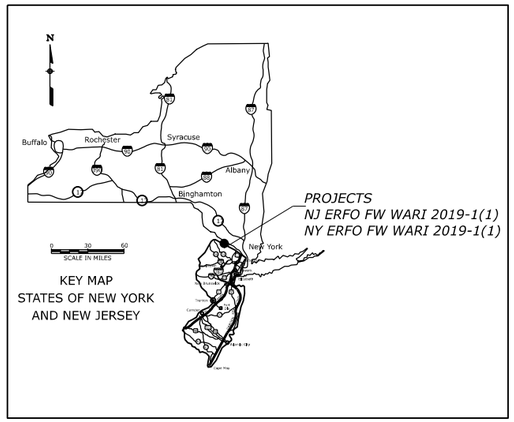
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	A01



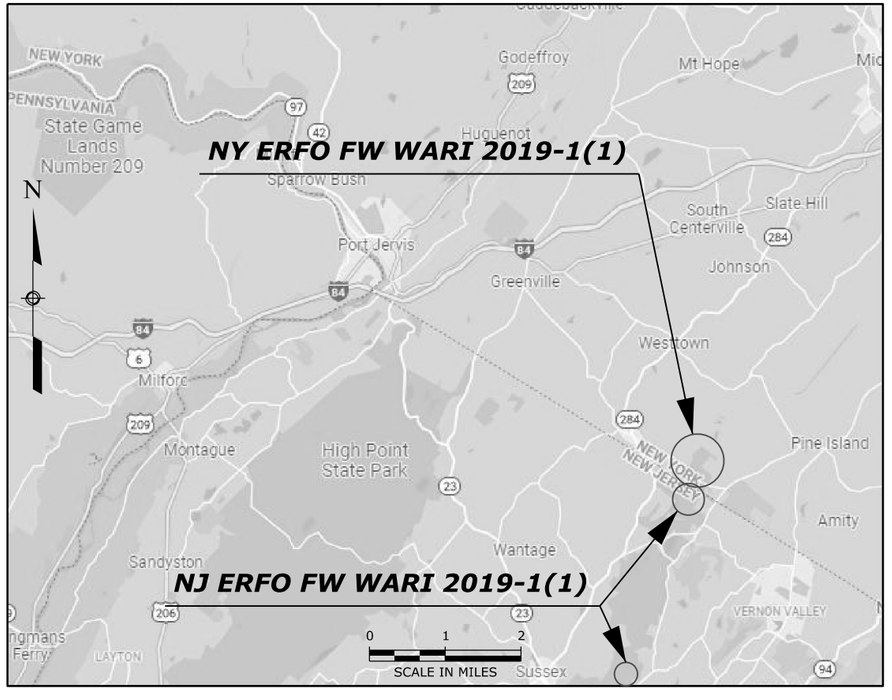
U.S. DEPARTMENT OF THE INTERIOR
U.S. FISH AND WILDLIFE SERVICE
WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PLANS FOR PROPOSED PROJECTS

NJ-NY ERFO FW WARI 2019-1(1)

REPAIR WINDING WATERS TRAIL
AND LIBERTY LOOP TRAIL (NE);
REPAIR LIBERTY LOOP TRAIL (SW);
REPAIR DAGMAR DALE NATURE TRAIL

SUSSEX COUNTY, NEW JERSEY
ORANGE COUNTY, NEW YORK



TRAIL:	WIDTH	SURFACE	BASE	SUBGRADE
Liberty Loop Trail (SW)	10' ±	Grass, 4" ATC	6" CCS with ABC	N/A
Winding Waters Trail Site #3	10' ±	N/A	N/A	N/A
Winding Waters Trail Site #240	12' ±	Grass, 4" ATC	6" ABC	N/A
Winding Waters Trail Site #324	6' ±	Grass, 4" ATC	6" ABC	N/A
Liberty Loop Trail (NE)	10' ±	Grass, 4" ATC	6" CCS with ABC	N/A
Dagmar Dale Nature Trail Site #1	9' ±	Grass, 4" ATC	6" CCS with ABC	N/A
Dagmar Dale Nature Trail Site #2	13' ±	Grass, 4" ATC	6" ABC	N/A
Dagmar Dale Nature Trail Site #3	9' ±	Grass, 4" ATC	6" ABC	N/A
Dagmar Dale Nature Trail Site #4	9' ±	Grass, 4" ATC	6" ABC	N/A
Dagmar Dale Nature Trail Site #5	10' ±	Grass, 4" ATC	6" CCS with ABC	N/A
Dagmar Dale Nature Trail Site #6	9' ±	Grass, 4" ATC	6" CCS with ABC	N/A

DESCRIPTION OF PROJECT

IMPROVEMENT:

Schedule A: Embankment repairs, trail reconstruction, and other miscellaneous work.

Schedule B: Trail reconstruction and other miscellaneous work.

Schedule C: Culvert replacements, trail reconstruction, stream regrading, headwall and wingwall installation, and other miscellaneous work.

SCHEDULE A LENGTH: 0.77 Miles

SCHEDULE B LENGTH: 0.88 Miles

SCHEDULE C LENGTH: 0.22 Miles

SPECIFICATIONS:

"Standard Specifications for Construction of Roads and Bridges on Federal Highway Projects", FP-14.

PLANS PREPARED BY
U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
EASTERN FEDERAL LANDS HIGHWAY DIVISION
ASHBURN, VIRGINIA
SEPTEMBER, 2026

Project Manager	Highway Design Manager	Lead Designer
Katerina Roman	Canisha Lyons	Reo Kunori

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PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	A02

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A01	Title Sheet
A02	Sheet Index
A03 to A04	Symbols And Abbreviations
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A06	Location Map - Sch B and C (NJ)
A07	Survey Information - Sch A (NY)
A08	Survey Information - Sch A and B (NY/NJ)
A09	Survey Information - Sch C (NJ)
A10 to A11	Horizontal Alignment Data - Sch A (NY)
A12	Horizontal Alignment Data - Sch A and B (NY/NJ)
A13 to A14	Horizontal Alignment Data - Sch C (NJ)
B01 to B07	Typical Sections - Sch A (NY)
B08	Typical Sections - Sch B (NJ)
B09 to B10	Typical Sections - Sch C (NJ)
C01 to C03	Tabulation Of Quantities
C04 to C07	Summaries And Schedules - Sch A (NY)
C08 to C09	Summaries And Schedules - Sch B (NJ)
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D01 to D17	Construction Plans And Profiles - Sch A (NY)
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K01 to K04	Drainage Profiles - Sch C (NJ)
M01	Erosion And Sediment Control Narrative - Sch A (NY)
M02 to M03	Erosion And Sediment Control Narrative - Sch B and C (NJ)
M04 to M07	Erosion And Sediment Control Plans (Winding Waters Trail) - Sch A (NY)
M08 to M10	Revegetation Plan (Winding Waters Trail) - Sch A (NY)
M11 to M17	Erosion And Sediment Control Plans (Liberty Loop Trail (NE)) - Sch A (NY)
M18 to M26	Erosion And Sediment Control Plans - Sch B (NJ)
M27 to M35	Impact Maps - Sch B (NJ)
M36 to M44	Proposed Contour Maps - Sch C (NJ)
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T01 to T03	Cross Sections (Winding Waters Site #3) - Sch A (NY)
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V01 to V07	Cross Sections (Winding Waters Site #324) - Sch A (NY)
W01 to W20	Cross Sections (Liberty Loop (NE)) - Sch A (NY)
X01 to X34	Cross Sections (Liberty Loop (SW)) - Sch B (NJ)
Y01 to Y05	Cross Sections (Dagmar Dale Site #1) - Sch C (NJ)
Z01 to Z02	Cross Sections (Dagmar Dale Site #2) - Sch C (NJ)
AA01 to AA03	Cross Sections (Dagmar Dale Site #3) - Sch C (NJ)
BB01 to BB04	Cross Sections (Dagmar Dale Site #4) - Sch C (NJ)
CC01 to CC05	Cross Sections (Dagmar Dale Site #5) - Sch C (NJ)
DD01 to DD05	Cross Sections (Dagmar Dale Site #6) - Sch C (NJ)

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PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	A03

Δ	total central angle
Δc	curve central angle
Ø	diameter
θs	spiral central angle
abut.	abutment
ACP	asphalt concrete pavement
ADT	average daily traffic
AH	ahead
appr.	approach
BK	back
BM	benck mark
BP	balance point
br.	bridge
brg.	bearing
cc or c. to c.	center to center
CL	centerline
CMP	corrugated metal pipe
CO	contracting officer
col.	column
conc.	concrete
conn.	connection
constr. jt.	construction joint
cont.	continuous
CS	curve to spiral
ctrs.	centers
D	diameter
DHV	design hourly volume
dia.	diameter
diag.	diagonal
diaph.	diaphragm
dist.	distance
drwg(s).	drawing(s)
DSY	double solid yellow
E	east
e	superelevation rate
elev.	elevation with number
elev.	elevation
emb.	embankment
EOP	edge of pavement
EOS	edge of shoulder
EOT	edge of travel way
EQ or eq.	equation
ER	edge of road
ESAL	equivalent single axle load
EW	edge of water
exc.	excavation
exp. jt.	expansion joint
fin.	finish
flg.	flange
ftg.	footing
ga.	gage (gauge)
galv.	galvanized
hdwl.	headwall
hex.	hexagon
HW	high water
ID	inside diameter
jt.	joint
L	length of curve
lam.	lamination
lat.	latitude
LOD	Limits of Disturbance
long.	longitudinal
LPSM	lump sum
Ls	length of spiral
lt. or LT	left
LW	low water
ML	main line
MP	mile post
max.	maximum
min.	minimum
mon.	monument
N	north
NC	normal crown
NMSA	nominal maximum size aggregate
o. c.	on center
ohwm	ordinary high water mark
o. to o.	out to out
OD	outside diameter
OG	original ground
PC	point of curve
PCC	point of compound curve
PCS	point of curve to spiral
PI	point of intersection
pl.	plate
POC	point on curve
POS	point on spiral
POT	point on tangent
PS	point of tangent to spiral
PSC	point of spiral to curve
PST	point of spiral to tangent
PT	point of tangent
pvmt.	pavement
R	radius
R.	range
R/W	right-of-way
rdwy.	roadway
RECP	rolled erosion control product
reinf.	reinforcement
reqd.	required
rt. or RT	right
rte.	route
S	south
SADT	seasonal average daily traffic
SC	point of spiral to curve
sec.	section
shldr.	shoulder
spa.	spacing, spaces or spaced
sqft	square foot
sqyd	square yard
SRS	point of spiral to reverse spiral
SS	point of spiral to spiral (no curve)
ST	point of spiral to tangent
Sta.	station
std.	standard
stgr.	stringer
stiff.	stiffener
struc.	structural
STS	point of spiral to tangent spiral
sym.	symmetrical
T	tangent distance
T.	township
TBM	temporary bench mark
thd.	thread
TS	point of tangent to spiral
Ts	tangent distance (spiraled curve)
typ.	typical
V	design speed
vph	vehicles per hour
VPI	vertical point of intersection
W	west

Control Point (Terrestrial and GPS); Jump Hub — RBAR 3000

National Boundary

State Boundary

County Boundary

City Boundary

Township or Range Line

Section Line

Section Corner (Found, Projected) — 36 31 1 6

¼ Section Line

¼ Section Corner (Found, Projected) — 15 22

1/16 Section Line

1/16 Section Corner (Found, Projected) — 1/6 SEC.

Property Line w/Found Property Corner — P/L

Parcel Number

National Park Boundary — NP

National Forest Boundary

National Wildlife Refuge Boundary — NWR

BLM Lands Boundary

Indian Reservation Boundary

Existing Roadway (Road, Paved, Gravel)

Railroad

Trail

Intermittent Drainage or Small Creek

Large Creek or River

Lake, Pond or Reservoir; Marshland

Spring or Seep

Treeline; Individual Trees, Pine

Material Source; Bore Hole; Test Pit

Spot Elevation; Coordinate Grid Tick — EL. 0.00 — N 0 E 1

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SYMBOLS AND ABBREVIATIONS

SHEET 1 OF 2

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2 August 2026 1:07 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	A04

North Arrow N

		EXISTING	PROPOSED
Slope Stake Limits	*Top of Cut*		
	Toe of Fill		
	Transition		
Fence			
Gate with Fence			
Cattleguard			
Guardrail			
Concrete Barrier			
Retaining Wall			
Signs (single, double post; portable)			
Delineators			
Pipe Culvert (arrow shows flow)			
Pipe Culvert with End Section			
Pipe Culvert with Headwall			
Pipe Culvert with Drop Inlet			DI
Box Culvert			
Underdrain		UD	UD
Overhead/Above Ground Utilities		P	P
Underground Utilities		W	W

FM = force main, FO = fiber optic, G = gas, IRR = irrigation, O = oil, P = power, SA = sanitary sewer, SD = storm drain, SS = storm sewer, STEAM = steam, T = telephone, TV = CATV, W = water

Poles (Power, Telephone, Joint Use, Light, Support w/Anchor)

Miscellaneous Utility Features

EM = electric meter, T = telephone pedestal, TV = CATV pedestal, UP = transformer or junction box, WF = water fountain

Building

Right-of-Way Line with Monument r/w R/W

Permanent Easement P/E P/E

Construction Easement TCE

Riprap

	AREA PATTERN
Pavement Removal / Roadway Obliteration	
Full Depth Pavement	
Sidewalk Asphalt/Concrete	
Mill and Overlay	
Overlay	

Silt Fence SF

Diversion Berm DB

Drainage Divide

Check Dam

Limits of Disturbance LOD LOD LOD

Fiber Roll or Wattle

PROJECT SPECIFIC SYMBOLS AND ABBREVIATIONS:

ABC	*Aggregate Base Course*
ATC	*Aggregate Topsoil Course*
CCS	*Cellular Confinement System*
HDPE	*High-Density Polyethylene*
RCP	*Reinforced Concrete Pipe*
TOE	*Toe of Embankment*
Tr.	*Transition*
WARI	*Wallkill River National Wildlife Refuge*
HP	*High Point*
LP	*Low Point*

	AREA PATTERN
Trail reconstruction (and Trail reconstruction with CCS)	
Embankment repair	
Clearing and grubbing	
Floating Turbidity Curtain	TC
Apparent TOE	
Ordinary high water	
Water level (typ.)	
Permanent Impacts	
Temporary Impacts	

NO.	DATE	BY	REVISIONS

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WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SYMBOLS AND ABBREVIATIONS

SHEET 2 OF 2

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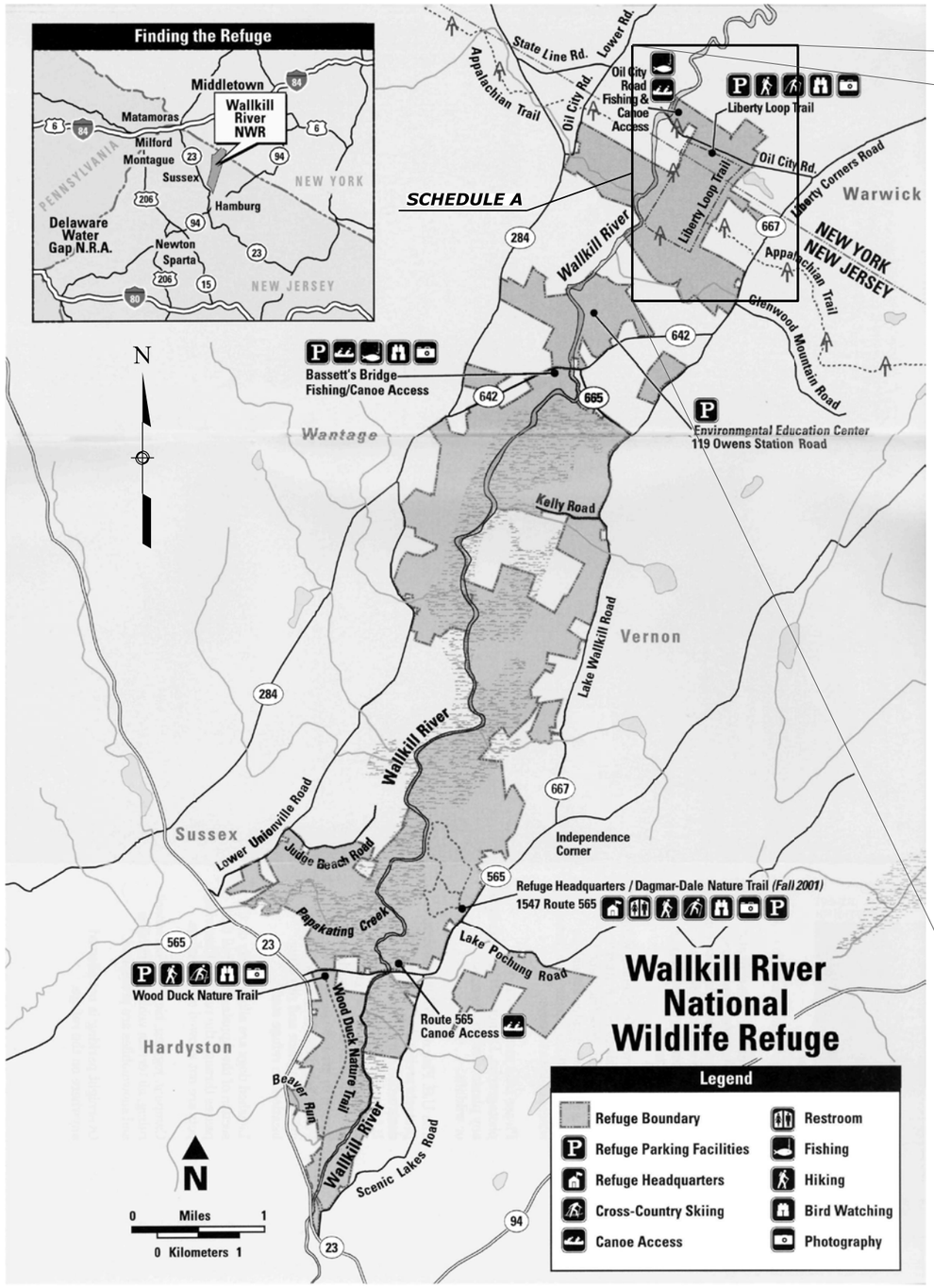

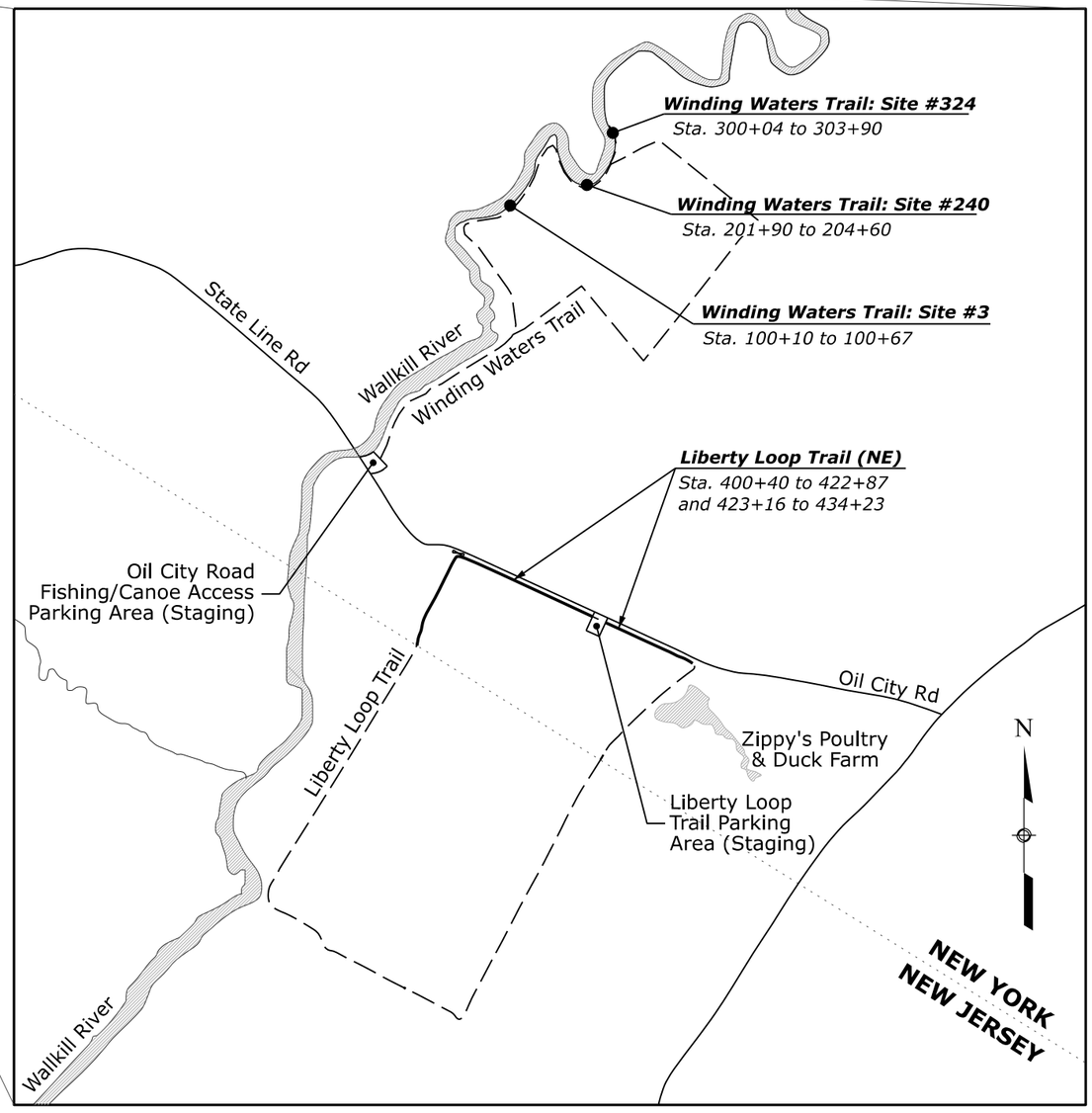

NOTE:

Obtain approval from the CO prior to setting up staging areas for the various sites.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

NOT TO SCALE

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

LOCATION MAP

SCHEDULE A: LIBERTY LOOP TRAIL (NE) & WINDING WATERS TRAIL

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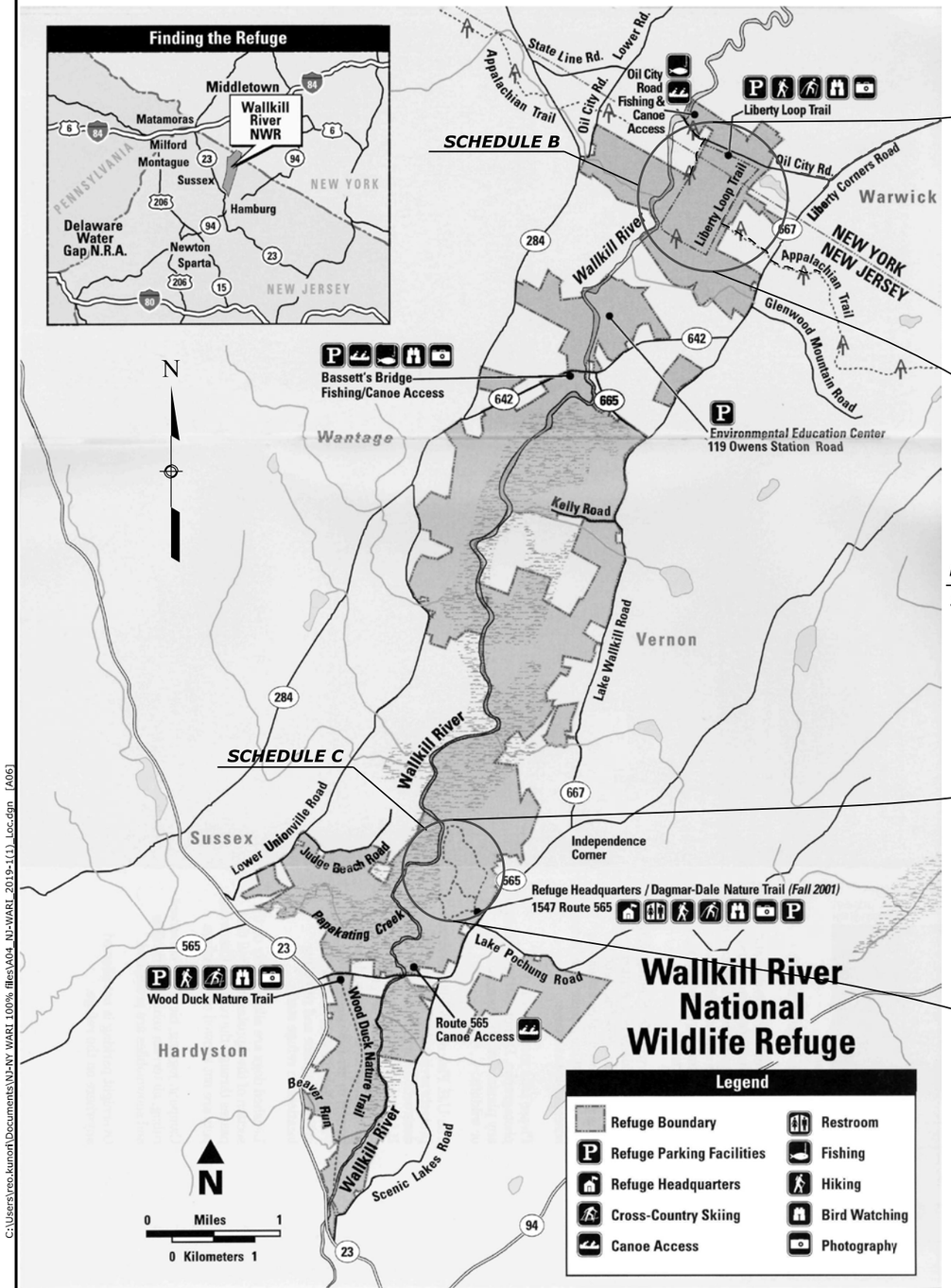
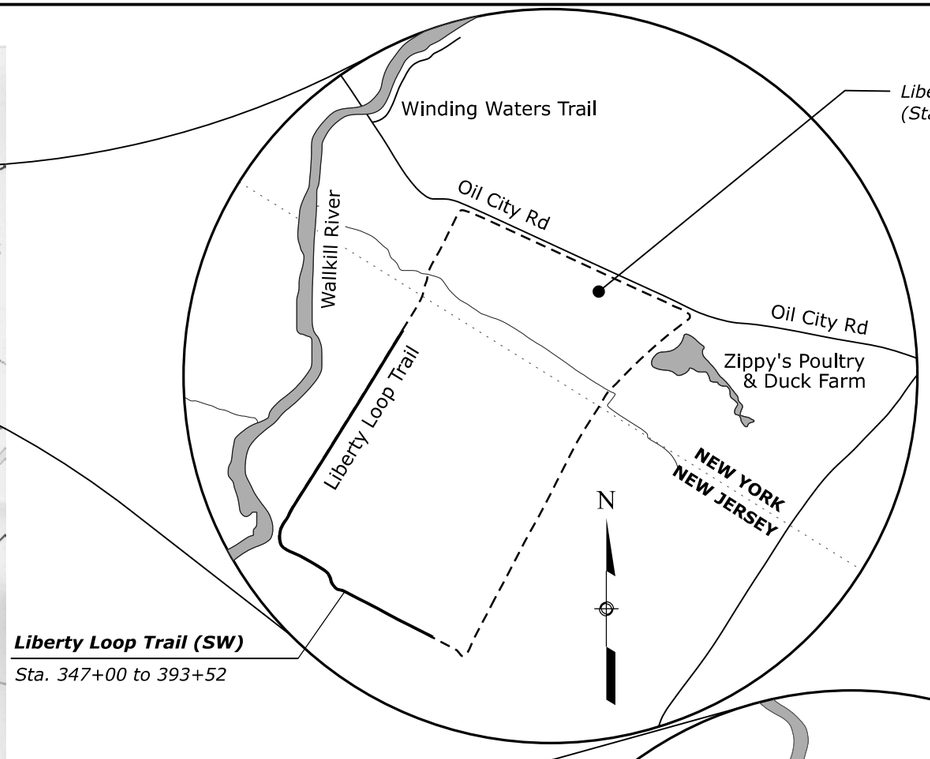
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

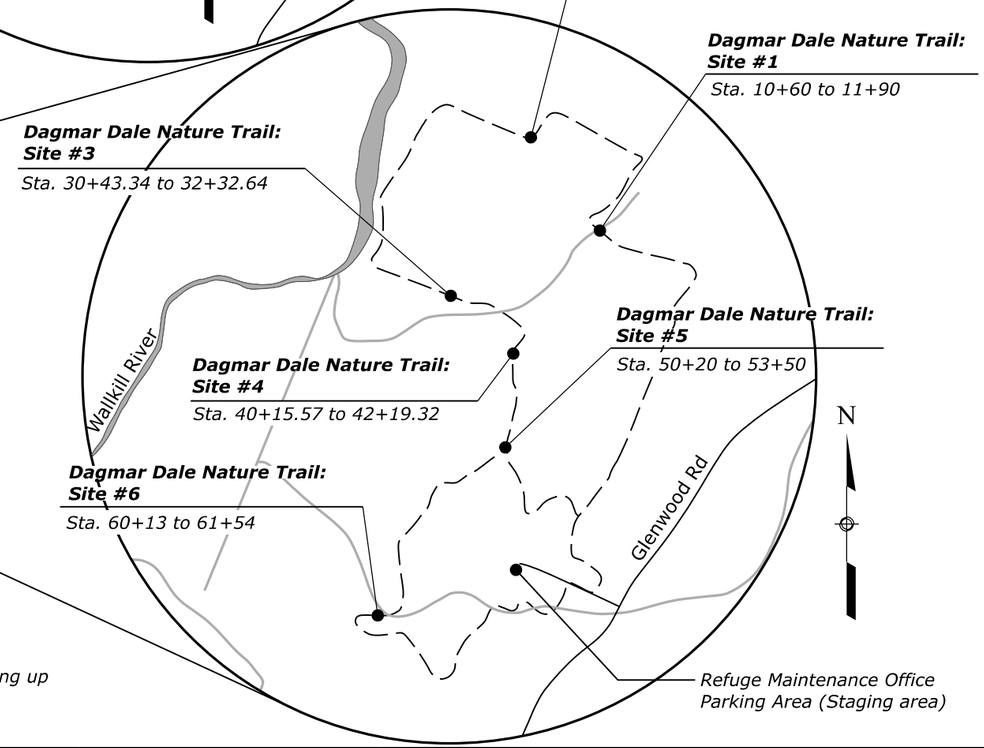
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	A06





Liberty Loop Trail (SW)

Sta. 347+00 to 393+52



Dagmar Dale Nature Trail: Site #1

Sta. 10+60 to 11+90

Dagmar Dale Nature Trail: Site #2

Sta. 21+37.68 to 22+91.23

Dagmar Dale Nature Trail: Site #3

Sta. 30+43.34 to 32+32.64

Dagmar Dale Nature Trail: Site #4

Sta. 40+15.57 to 42+19.32

Dagmar Dale Nature Trail: Site #5

Sta. 50+20 to 53+50

Dagmar Dale Nature Trail: Site #6

Sta. 60+13 to 61+54

NOTE:

Obtain approval from the CO prior to setting up staging areas for the various sites.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

NOT TO SCALE

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

LOCATION MAP

SCHEDULES B AND C: LIBERTY LOOP TRAIL (SW) & DAGMAR DALE NATURE TRAIL

SHEET 2 OF 2

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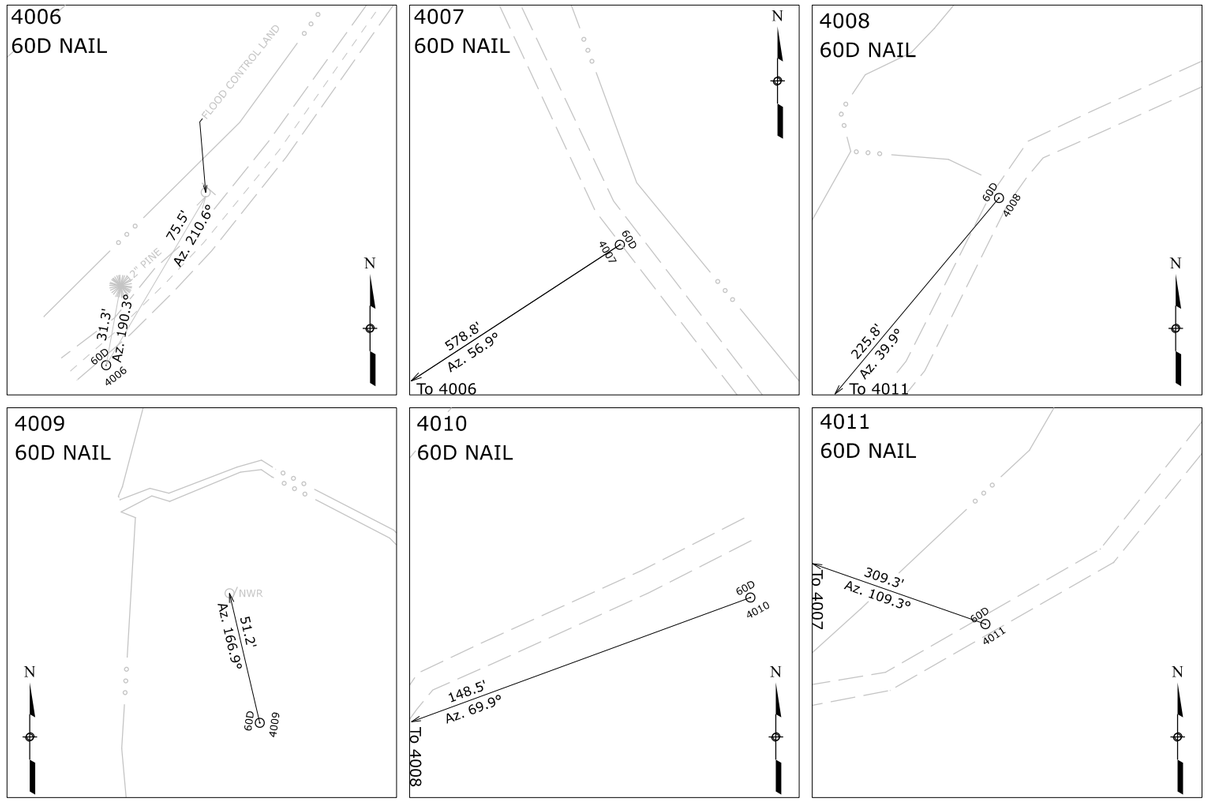
PROJECT CONTROL POINTS REFERENCES



NOTE:

Azimuth angle shown is based on north axis at clockwise rotation.

PROJECT COORDINATE SYSTEM

Coordinate System : US State Plane 1983
Zone : New Jersey 2900
Hor. Datum : NAD 1983 (Conus)
Vert. Datum : NAVD 1988 (Conus)
Geoid Model : GEIOD18 (Conus)

PROJECT CONTROL POINTS

NAME	NORTHING	EASTING	ELEV.	TYPE
4006	896480.762	484142.950	388.423	60D NAIL
4007	896797.103	484627.665	387.177	60D NAIL
4008	896868.211	485064.578	386.789	60D NAIL
4009	897505.439	485020.454	385.525	60D NAIL
4010	896919.231	485203.784	381.570	60D NAIL
4011	896695.095	484919.607	388.218	60D NAIL

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

NOT TO SCALE

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SURVEY INFORMATION

SCHEDULE A: WINDING WATERS TRAIL

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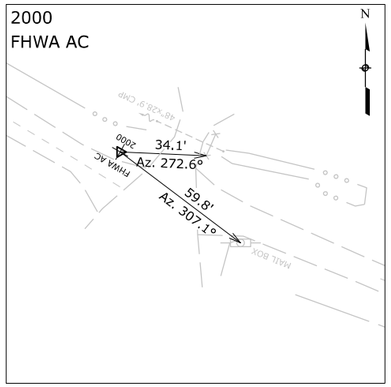
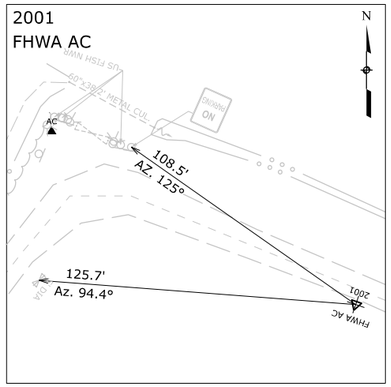
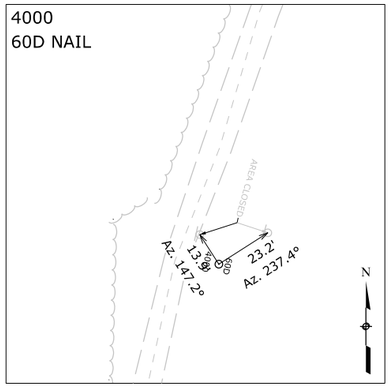
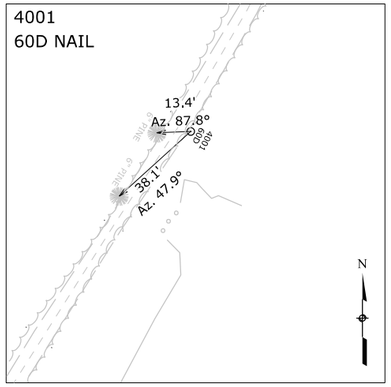
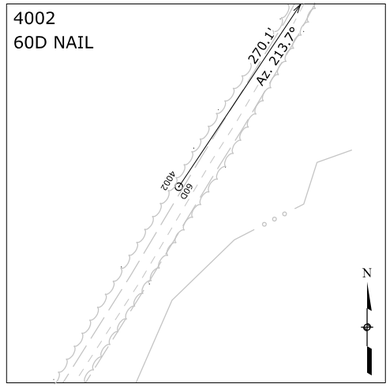
PROJECT CONTROL POINTS REFERENCES

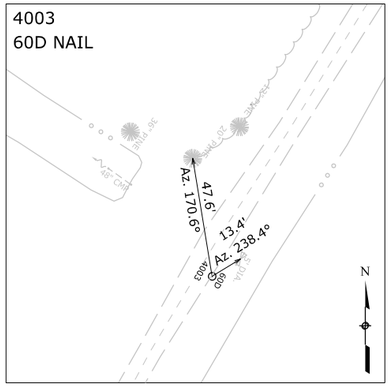


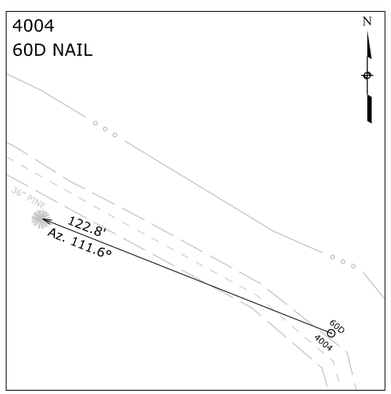


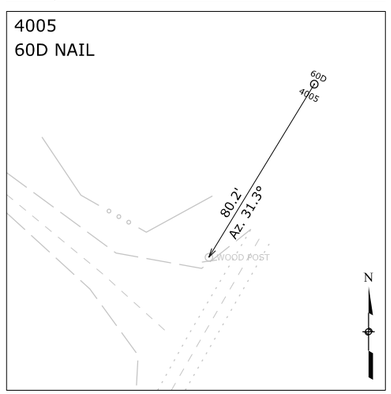












NOTE:

Azimuth angle shown is based on north axis at clockwise rotation.

PROJECT COORDINATE SYSTEM

Coordinate System : US State Plane 1983
Zone : New Jersey 2900
Hor. Datum : NAD 1983 (Conus)
Vert. Datum : NAVD 1988 (Conus)
Geoid Model : GEIOD18 (Conus)

PROJECT CONTROL POINTS

NAME	NORTHING	EASTING	ELEV.	TYPE
2000	892499.845	484945.047	387.418	FHWA AC
2001	893031.739	483752.910	384.893	FHWA AC
4000	892390.771	483299.285	387.614	60D NAIL
4001	892090.964	483178.841	387.350	60D NAIL
4002	891866.315	483028.846	388.431	60D NAIL
4003	891207.780	482626.454	386.657	60D NAIL
4004	889333.024	482259.914	387.310	60D NAIL
4005	888559.266	483714.080	391.872	60D NAIL

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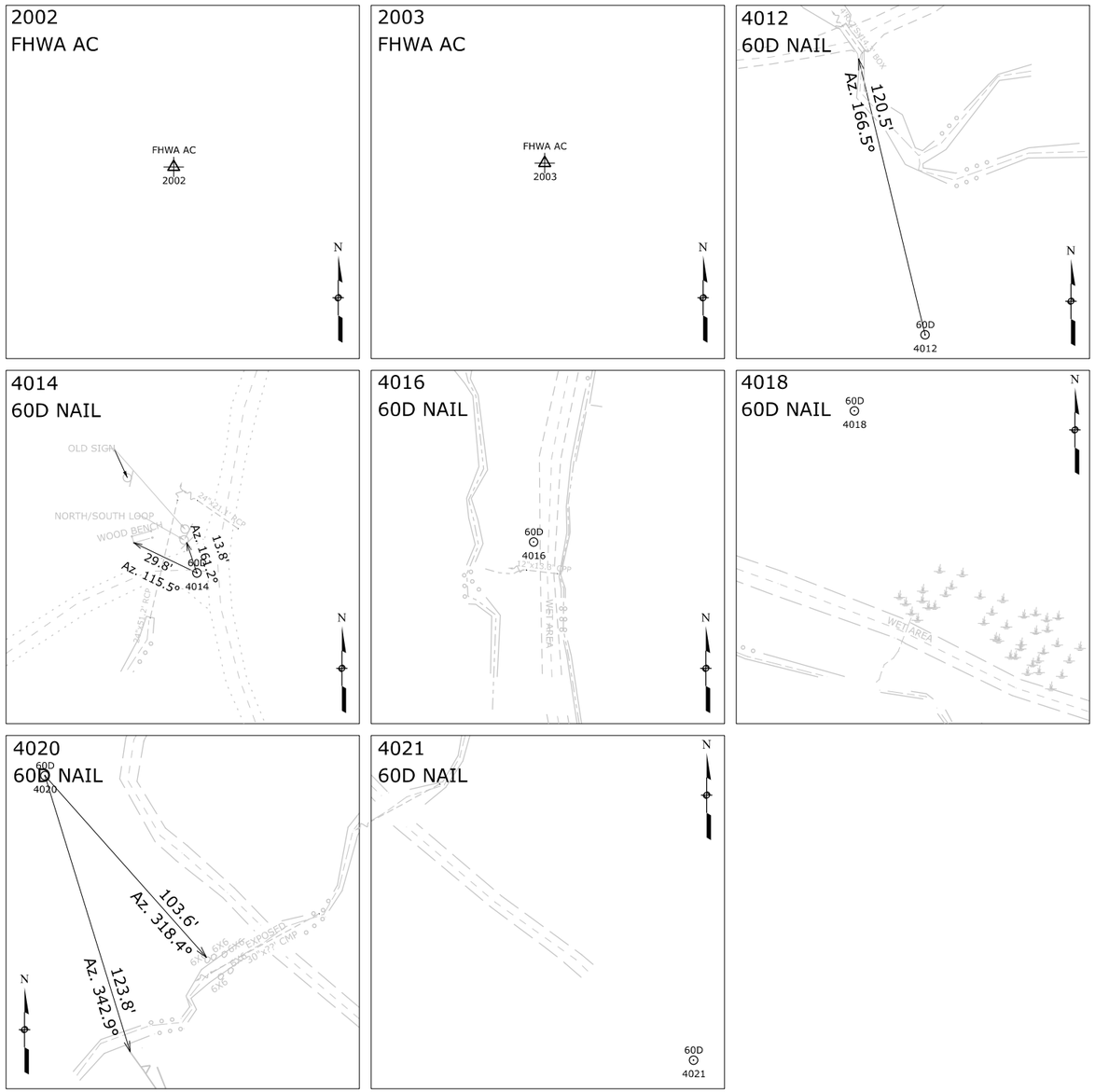
2 August 2026 5:00 PM

PROJECT CONTROL POINTS REFERENCES



NOTE:

Azimuth angle shown is based on north axis at clockwise rotation.

PROJECT COORDINATE SYSTEM

Coordinate System : US State Plane 1983
Zone : New Jersey 2900
Hor. Datum : NAD 1983 (Conus)
Vert. Datum : NAVD 1988 (Conus)
Geoid Model : GEOID18 (Conus)

PROJECT CONTROL POINTS

NAME	NORTHING	EASTING	ELEV.	TYPE
2002	862305.937	474137.599	533.929	101#FHWA AC
2003	862671.681	474093.237	528.396	101#FHWA AC
4012	861952.465	472995.070	409.581	103#60D SITE 6
4014	863167.809	473716.279	445.636	103#60D SITE 5
4016	863718.909	473768.185	417.232	103#60D SITE 4
4018	864298.924	473369.686	400.075	103#60D SITE 3
4020	864710.394	474249.050	444.599	103#60D SITE 2
4021	864534.558	474473.213	460.649	103#60D SITE 1

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

NOT TO SCALE

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SURVEY INFORMATION

SCHEDULE C: DAGMAR DALE NATURE TRAIL

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24 July 2026 3:45 PM

CONTROL LINE DATA - WINDING WATERS SITE #3

POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	N48.829°E	26.411 '	896478.562	484129.227		100+00.000					
WW-A1	N45.503°E	23.825 '	896495.949	484149.108		100+26.411					
WW-A2	N43.237°E	21.049 '	896512.647	484166.102		100+50.236					
WW-A3	N39.241°E	45.746 '	896527.982	484180.521		100+71.285					
WW-A4	N34.960°E	113.385 '	896563.412	484209.459		101+17.031					
WW-A5	N28.895°E	117.731 '	896656.337	484274.429		102+30.416					
END			896759.411	484331.317		103+48.147					

CONTROL LINE DATA - WINDING WATERS SITE #240

POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	S37.602°E	104.610 '	896811.761	484621.765		200+00.000					
WW-B1	S64.617°E	99.525 '	896713.656	484697.321	201+04.610	201+23.828	201+42.331	27.016°	-80.000 '	37.721 '	19.218 '
WW-B2	N54.912°E	132.658 '	896641.012	484850.427	202+22.638	202+92.578	203+49.287	60.471°	-120.000 '	126.649 '	69.941 '
WW-B3	N38.944°E	89.345 '	896723.718	484968.158	204+12.004	204+23.224	204+34.300	15.968°	-80.000 '	22.296 '	11.221 '
WW-B4	N26.490°E	76.957 '	896803.39	485032.547	205+12.424	205+25.518	205+38.508	12.454°	-120.000 '	26.084 '	13.094 '
WW-B5	N65.596°E	90.496 '	896884.346	485072.892	206+02.371	206+15.866	206+28.306	39.106°	37.998 '	25.935 '	13.495 '
WW-B6	N62.981°E	44.732 '	896925.508	485163.615	207+05.307	207+14.434	207+23.558	2.614°	-400.000 '	18.251 '	9.127 '
END			896945.829	485203.465		207+59.163					

CONTROL LINE DATA - WINDING WATERS SITE #324

POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	N21.277°W	0.100 '	896882.672	485074.506		300+00.000					
WW-C1	N26.026°E	44.711 '	896893.784	485070.179	300+00.100	300+11.925	300+22.391	47.303°	27.000 '	22.291 '	11.825 '
WW-C2	N18.368°E	62.512 '	896960.809	485102.907	300+55.278	300+85.155	301+14.943	7.658°	-446.403 '	59.666 '	29.877 '
WW-C3	N8.746°W	54.166 '	897041.881	485129.826	301+47.299	301+70.768	301+93.089	27.114°	-96.172 '	45.512 '	23.190 '
WW-C4	N28.225°W	127.369 '	897138.168	485115.012	302+24.065	302+67.319	303+09.737	19.479°	-252.000 '	85.672 '	43.253 '
END			897250.393	485054.775		303+93.853					

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2 August 2026 5:08 PM

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

HORIZONTAL ALIGNMENT DATA

SCHEDULE A: WINDING WATERS TRAIL

CONTROL LINE DATA - LIBERTY LOOP TRAIL (NE)

POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	N17.928°E	13.769 '	892317.967	483263.228		400+00.00					
LL-1	N12.332°E	37.791 '	892341.299	483270.777	400+13.77	400+24.52	400+35.26	5.597°	-220.000 '	21.490 '	10.754 '
LL-2	N19.472°E	107.796 '	892387.675	483280.915	400+62.30	400+71.98	400+81.63	7.140°	155.159 '	19.335 '	9.680 '
LL-T1	N19.472°E	22.105 '	892489.306	483316.848		401+79.75					
LL-3	N28.672°E	60.567 '	892525.32	483329.581	402+01.85	402+17.95	402+33.97	9.201°	200.000 '	32.117 '	16.093 '
LL-4	N26.109°E	140.444 '	892588.669	483364.224	402+78.44	402+90.08	403+01.71	2.564°	-520.000 '	23.268 '	11.636 '
LL-5	N30.896°E	154.810 '	892777.466	483456.75	404+30.52	405+00.33	405+70.05	4.787°	1670.022 '	139.532 '	69.807 '
LL-6	N27.094°E	77.597 '	892921.167	483542.739	406+55.06	406+67.71	406+80.35	3.801°	-381.297 '	25.297 '	12.653 '
LL-7	N38.385°E	70.732 '	893011.384	483588.894	407+45.30	407+69.04	407+92.63	11.290°	240.179 '	47.328 '	23.741 '
LL-8	S67.672°E	168.306 '	893082.443	483645.184	408+39.62	408+59.54	408+73.77	73.943°	26.463 '	34.152 '	19.921 '
LL-9	S64.690°E	131.244 '	893011.085	483818.927	410+22.15	410+41.67	410+61.18	2.982°	750.000 '	39.030 '	19.520 '
LL-10	S66.178°E	158.686 '	892948.313	483951.663	411+72.91	411+88.50	412+04.08	1.488°	-1200.000 '	31.172 '	15.587 '
LL-11	S64.555°E	46.248 '	892877.355	484112.384	413+47.18	413+64.18	413+81.18	1.624°	1200.000 '	34.003 '	17.003 '
LL-12	S65.191°E	150.354 '	892852.718	484164.164	414+10.43	414+21.52	414+32.62	0.636°	-2000.000 '	22.189 '	11.094 '
LL-13	S65.970°E	105.511 '	892783.352	484314.222	415+71.88	415+86.84	416+01.80	0.779°	-2200.000 '	29.922 '	14.961 '
LL-14	S65.327°E	263.808 '	892730.789	484432.112	416+92.35	417+15.91	417+39.48	0.643°	4200.000 '	47.132 '	23.566 '
LL-15	S65.093°E	126.904 '	892615.12	484683.905	419+79.72	419+93.00	420+06.28	0.234°	6500.000 '	26.565 '	13.282 '
LL-16	S64.166°E	86.341 '	892556.908	484809.271	421+19.91	421+31.22	421+42.54	0.926°	1400.000 '	22.636 '	11.318 '
LL-17	S58.506°E	43.970 '	892510.668	484904.78	422+17.56	422+37.34	422+57.08	5.660°	400.000 '	39.513 '	19.773 '
LL-18	S76.489°E	37.837 '	892484.963	484946.738	422+81.28	422+86.51	422+91.66	17.983°	-33.091 '	10.386 '	5.236 '
LL-19	S66.641°E	102.021 '	892474.342	484990.939	423+24.26	423+31.88	423+39.47	9.848°	88.472 '	15.207 '	7.622 '
LL-20	S65.272°E	113.150 '	892424.414	485106.545	424+33.87	424+57.77	424+81.68	1.370°	2000.000 '	47.809 '	23.906 '
LL-21	S64.267°E	45.023 '	892371.949	485220.464	425+70.92	425+83.19	425+95.46	1.004°	1400.000 '	24.539 '	12.270 '
LL-22	S65.332°E	77.853 '	892347.561	485271.063	426+28.21	426+39.36	426+50.51	1.064°	-1200.000 '	22.294 '	11.147 '
LL-23	S65.606°E	92.211 '	892310.874	485350.943	427+17.21	427+27.26	427+37.31	0.274°	-4200.000 '	20.097 '	10.049 '
LL-24	S64.565°E	242.136 '	892263.783	485454.783	428+19.47	428+41.28	428+63.09	1.041°	2400.000 '	43.614 '	21.808 '
LL-25	S65.878°E	263.875 '	892149.944	485694.147	430+83.41	431+06.34	431+29.25	1.313°	-2000.000 '	45.839 '	22.921 '
LL-26	S26.234°E	69.160 '	892027.203	485968.254	433+70.21	434+06.67	434+40.19	39.644°	101.143 '	69.983 '	36.458 '
END			891965.167	485998.825		434+72.89					

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2 August 2026 5:08 PM

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

HORIZONTAL ALIGNMENT DATA

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

CONTROL LINE DATA - LIBERTY LOOP TRAIL (SW)

POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	N47.866°W	54.873'	888452.182	483665.609		345+97.597					
LL-27	N62.216°W	315.939'	888536.288	483572.639	346+52.470	347+22.966	347+92.724	14.35°	-560.000'	140.254'	70.496'
LL-28	N62.836°W	195.811'	888713.856	483235.628	350+38.167	351+03.156	351+68.145	0.62°	-12006.948'	129.978'	64.990'
LL-29	N67.664°W	10.414'	888805.621	483056.799	352+98.967	353+04.155	353+09.337	4.828°	-123.076'	10.370'	5.188'
LL-30	N60.857°W	84.185'	888812.21	483040.761	353+14.562	353+21.487	353+28.395	6.806°	116.446'	13.832'	6.924'
LL-31	N64.072°W	55.034'	888864.539	482946.91	354+05.656	354+28.924	354+52.181	3.215°	-829.144'	46.525'	23.269'
LL-32	N61.716°W	30.129'	888890.714	482893.07	354+83.946	354+88.778	354+93.608	2.357°	234.908'	9.662'	4.831'
LL-33	N63.033°W	386.740'	888943.132	482795.655	355+18.905	355+99.399	356+79.887	1.318°	-6999.986'	160.982'	80.495'
LL-34	N60.962°W	118.467'	889141.252	482406.261	359+86.132	360+36.289	360+86.435	2.071°	2774.407'	100.302'	50.157'
LL-35	N29.137°W	43.835'	889201.522	482297.7	361+54.745	361+60.447	361+65.854	31.825°	20.000'	11.109'	5.702'
LL-36	N5.323°W	38.714'	889243.603	482274.242	362+03.987	362+08.330	362+12.548	23.814°	20.598'	8.561'	4.343'
LL-37	N14.888°W	29.963'	889287.579	482270.145	362+46.919	362+52.371	362+57.798	9.564°	-65.174'	10.879'	5.452'
LL-38	N49.873°W	36.875'	889322.206	482260.939	362+82.309	362+88.176	362+93.676	34.985°	-18.615'	11.366'	5.867'
LL-39	N61.178°W	114.557'	889348.267	482230.021	363+24.684	363+28.245	363+31.783	11.305°	-35.979'	7.099'	3.561'
LL-40	N61.766°W	229.320'	889468.034	482012.364	364+42.779	365+76.655	367+10.528	0.588°	-26078.935'	267.748'	133.875'
LL-41	N54.296°W	49.544'	889579.755	481804.299	368+05.973	368+12.815	368+19.637	7.47°	104.804'	13.665'	6.842'
LL-42	N21.457°W	46.019'	889611.779	481759.74	368+62.339	368+67.668	368+72.704	32.839°	18.084'	10.365'	5.329'
LL-43	N2.624°E	50.528'	889658.929	481741.208	369+13.394	369+18.036	369+22.540	24.082°	21.761'	9.146'	4.642'
LL-44	N10.662°E	99.482'	889713.431	481743.706	369+68.427	369+72.458	369+76.476	8.038°	57.383'	8.050'	4.031'
LL-45	N29.876°E	39.380'	889823.913	481764.506	370+71.927	370+84.867	370+97.565	19.214°	76.449'	25.638'	12.940'
LL-46	N32.547°E	152.814'	889884.454	481799.285	371+24.004	371+54.445	371+84.874	2.671°	1305.805'	60.870'	30.440'
LL-47	N30.086°E	90.566'	890027.311	481890.46	373+07.247	373+23.907	373+40.561	2.461°	-775.612'	33.313'	16.659'
LL-48	N32.069°E	335.993'	890129.569	481949.704	374+14.468	374+42.081	374+69.690	1.983°	1595.712'	55.222'	27.614'
LL-49	N34.410°E	13.719'	890419.007	482131.051	377+78.069	377+83.632	377+89.194	2.341°	272.307'	11.125'	5.563'
LL-50	N32.627°E	318.900'	890456.752	482156.905	377+97.350	378+29.382	378+61.409	1.782°	-2059.172'	64.059'	32.032'
LL-51	N32.066°E	257.780'	890738.942	482337.563	381+48.277	381+64.442	381+80.606	0.561°	-3299.993'	32.330'	16.165'
LL-52	N29.141°E	51.829'	890961.816	482477.187	384+22.221	384+27.439	384+32.654	2.925°	-204.353'	10.433'	5.218'
LL-53	N31.250°E	73.897'	891025.232	482512.544	384+79.265	385+00.043	385+20.817	2.109°	1128.829'	41.552'	20.778'
LL-54	N29.563°E	95.513'	891118.204	482568.961	385+73.935	386+08.789	386+43.638	1.687°	-2367.168'	69.702'	34.854'
LL-55	N32.345°E	355.293'	891214.56	482623.616	387+04.297	387+19.561	387+34.820	2.782°	628.703'	30.522'	15.264'
LL-56	N29.249°E	52.235'	891517.244	482815.294	390+74.848	390+77.826	390+80.804	3.095°	-110.243'	5.956'	2.979'
LL-57	N32.434°E	99.541'	891565.754	482842.46	391+30.060	391+33.424	391+36.786	3.185°	120.998'	6.726'	3.364'
LL-58	N36.717°E	12.775'	891658.293	482901.265	392+32.963	392+43.064	392+53.155	4.283°	270.107'	20.192'	10.101'
LL-59	N33.922°E	66.485'	891678.963	482916.681	392+55.830	392+68.840	392+81.845	2.796°	-533.126'	26.015'	13.010'
LL-60	N31.657°E	198.200'	891742.33	482959.297	393+35.320	393+45.200	393+55.078	2.264°	-499.999'	19.758'	9.880'
LL-61	N27.216°E	40.732'	891924.243	483071.462	395+43.397	395+58.910	395+74.407	4.442°	-399.999'	31.009'	15.512'
LL-62	N32.683°E	141.131'	891968.265	483094.101	395+99.626	396+08.397	396+17.154	5.467°	183.698'	17.528'	8.771'
LL-63	N26.722°E	62.411'	892121.217	483192.23	397+49.515	397+90.107	398+30.626	5.961°	-779.645'	81.111'	40.592'
LL-64	N16.173°E	116.452'	892183.542	483223.606	398+52.445	398+59.811	398+67.137	10.549°	-79.795'	14.692'	7.367'
LL-65	N17.928°E	33.753'	892298.954	483257.077	399+76.222	399+79.937	399+83.652	1.756°	242.486'	7.430'	3.715'
LL-66	N12.332°E	37.791'	892341.299	483270.777	400+13.689	400+24.443	400+35.179	5.597°	-220.000'	21.490'	10.754'
LL-67	N19.472°E	17.781'	892387.675	483280.915	400+62.217	400+71.897	400+81.552	7.14°	155.159'	19.335'	9.680'
END			892404.439	483286.842		400+89.653					

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2 August 2026 5:11 PM

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

HORIZONTAL ALIGNMENT DATA

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

CONTROL LINE DATA - DAGMAR DALE SITE #1											
POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	N53.810°W	38.443'	864576.63	474423.34		10+00.00					
DD-A1	N49.629°W	42.441'	864602.35	474388.19	10+38.44	10+43.55	10+48.66	4.181°	140.000'	10.216'	5.110'
DD-A2	N40.734°W	16.467'	864632.36	474352.89	10+85.99	10+89.88	10+93.75	8.895°	50.000'	7.762'	3.889'
DD-A3	N49.467°W	45.575'	864654.67	474333.67	11+06.33	11+19.31	11+32.24	8.733°	-170.000'	25.911'	12.980'
DD-A4	N30.409°W	24.901'	864685.38	474297.76	11+64.84	11+66.51	11+68.16	19.058°	10.000'	3.326'	1.679'
DD-A5	N8.294°E	12.783'	864708.37	474284.27	11+91.38	11+93.14	11+94.76	38.703°	5.000'	3.377'	1.756'
DD-A6	N29.525°E	15.740'	864725.65	474286.79	12+05.79	12+10.47	12+15.05	21.23°	25.000'	9.264'	4.686'
DD-A7	N67.876°E	78.305'	864749.03	474300.03	12+26.11	12+37.24	12+47.53	38.351°	32.000'	21.419'	11.128'
END			864778.52	474372.57		13+14.70					

CONTROL LINE DATA - DAGMAR DALE SITE #1 (STREAM)											
POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	S13.767°W	23.325'	864686.71	474379.75		1+00.00					
DDS-A1	S25.064°W	4.887'	864663.10	474373.96	1+23.33	1+24.31	1+25.30	11.296°	10.000'	1.972'	0.989'
DDS-A2	S55.355°W	45.774'	864656.22	474370.75	1+29.20	1+31.90	1+34.48	30.291°	10.000'	5.287'	2.707'
DDS-A3	S48.805°W	27.297'	864630.20	474333.09		1+77.55					
DDS-A4	S37.140°W	5.364'	864611.68	474311.93	2+04.85	2+05.66	2+06.47	11.666°	-8.000'	1.629'	0.817'
DDS-A5	S84.447°W	13.802'	864605.66	474307.37	2+11.02	2+13.21	2+15.15	47.307°	5.000'	4.128'	2.190'
END			864604.32	474293.64		2+26.76					

CONTROL LINE DATA - DAGMAR DALE SITE #2											
POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	S31.104°W	27.080'	865371.82	473986.68		20+00.00					
DD-B1	S35.366°W	54.358'	865347.35	473971.92	20+27.08	20+28.57	20+30.06	4.262°	40.000'	2.976'	1.488'
DD-B2	S44.298°W	27.748'	865299.2	473937.75	20+82.93	20+87.61	20+92.28	8.932°	60.000'	9.354'	4.686'
DD-B3	S67.609°W	21.396'	865276.39	473915.49	21+15.34	21+19.47	21+23.48	23.310°	20.000'	8.137'	4.125'
DD-B4	N86.940°W	20.613'	865266.52	473891.53	21+40.75	21+45.26	21+49.63	25.451°	20.000'	8.884'	4.517'
DD-B5	N69.726°W	145.395'	865267.94	473864.9	21+65.73	21+71.78	21+77.75	17.214°	40.000'	12.018'	6.055'
END			865318.33	473728.51		23+17.09					

CONTROL LINE DATA - DAGMAR DALE SITE #3											
POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	S70.336°E	50.437'	864244.288	473284.96		30+00.00					
DD-C1	S69.211°E	44.517'	864225.663	473337.08	30+50.44	30+55.35	30+60.26	1.126°	500.000'	9.823'	4.912'
DD-C2	S63.470°E	50.803'	864206.303	473388.07	30+99.87	31+09.89	31+19.90	5.741°	200.000'	20.040'	10.028'
DD-C3	S72.090°E	46.892'	864181.254	473438.25	31+60.68	31+65.96	31+71.21	8.621°	-70.000'	10.532'	5.276'
DD-C4	S81.989°E	46.420'	864164.171	473491.11	32+12.83	32+21.49	32+30.10	9.899°	-100.000'	17.277'	8.660'
END			864157.702	473537.07		32+67.86					

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2 August 2026 5:12 PM

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

HORIZONTAL ALIGNMENT DATA

SCHEDULE C: DAGMAR DALE NATURE TRAIL

CONTROL LINE DATA - DAGMAR DALE SITE #4												
POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T	
START	S11.183°W	68.080'	863825.29	473789.81		40+00.00						
DD-D1	S6.297°W	54.418'	863754.32	473775.78	40+68.08	40+72.35	40+76.61	4.886°	-100.000'	8.528'	4.267'	
DD-D2	S12.122°E	80.053'	863687.34	473768.39	41+26.76	41+39.73	41+52.48	18.419°	-80.000'	25.718'	12.971'	
END			863609.07	473785.2		42+19.56						

CONTROL LINE DATA - DAGMAR DALE SITE #5											
POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	S13.870°W	81.761'	863351.767	473767.298		50+00.00					
DD-E1	S18.885°W	27.948'	863268.139	473746.648	50+81.76	50+86.14	50+90.51	5.015°	100.000'	8.752'	4.379'
DD-E2	S10.154°W	57.749'	863238.084	473736.367	51+14.08	51+17.90	51+21.70	8.731°	-50.000'	7.620'	3.817'
DD-E3	S14.761°E	44.445'	863164.93	473723.265	51+75.63	51+92.20	52+08.25	24.915°	-75.000'	32.614'	16.569'
DD-E4	S22.657°E	36.576'	863113.943	473736.7	52+36.12	52+44.41	52+52.66	7.896°	-120.000'	16.538'	8.282'
DD-E5	S30.748°E	82.943'	863073.664	473753.514	52+80.96	52+88.03	52+95.08	8.090°	-100.000'	14.120'	7.072'
END			863002.381	473795.919		53+70.95					

CONTROL LINE DATA - DAGMAR DALE SITE #6											
POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	S52.434°W	18.405'	862106.669	473014.329		60+00.00					
DD-F1	S63.475°W	36.226'	862094.564	472998.591	60+18.41	60+19.86	60+21.30	11.041°	15.000'	2.891'	1.450'
DD-F2	S79.021°W	31.418'	862076.862	472963.125	60+56.07	60+59.48	60+62.86	15.546°	25.000'	6.783'	3.412'
DD-F3	S84.893°W	42.723'	862070.341	472929.513	60+90.86	60+93.68	60+96.50	5.872°	55.000'	5.637'	2.821'
DD-F4	S89.043°W	20.161'	862066.215	472883.351	61+36.40	61+40.02	61+43.64	4.151°	100.000'	7.244'	3.624'
END			862065.879	472863.192		61+60.18					

CONTROL LINE DATA - DAGMAR DALE SITE #6 (STREAM)											
POINT ID	BEARING	DISTANCE (FEET)	NORTHING (Y)	EASTING (X)	PC	PI	PT	DELTA	R	L	T
START	N43.464°W	21.395'	862028.61	472992.37		6+00.00					
DDS-F1	N27.304°W	45.003'	862046.20	472975.70	6+21.40	6+24.23	6+27.04	16.16°	20.000'	5.641'	2.839'
DDS-F2	N41.345°W	25.954'	862088.37	472953.93	6+69.20	6+71.66	6+74.10	14.041°	-20.000'	4.901'	2.463'
DDS-F3	N32.547°W	35.447'	862109.01	472935.77	6+97.59	6+99.13	7+00.66	8.798°	20.000'	3.071'	1.539'
END			862138.89	472916.70		7+34.57					

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2 August 2026 5:12 PM

NO.	DATE	BY	REVISIONS

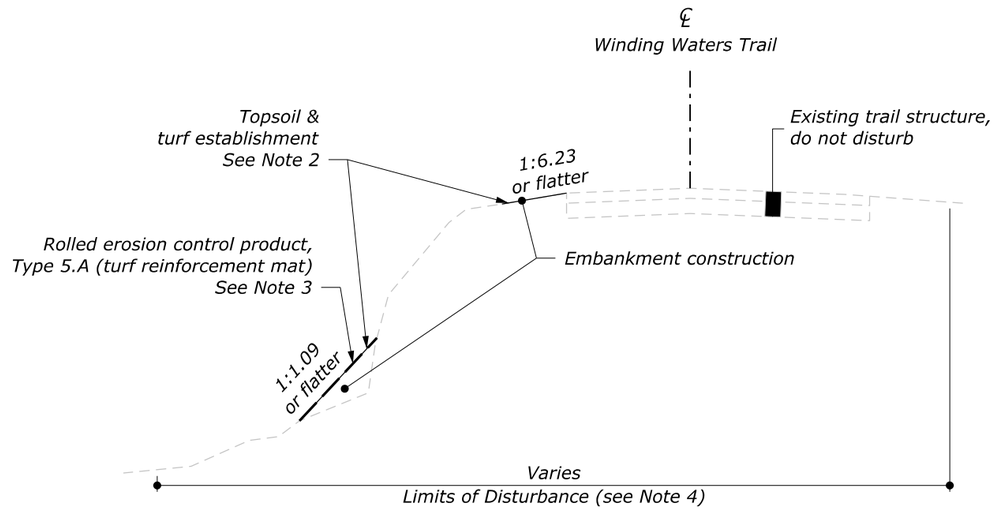
U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

HORIZONTAL ALIGNMENT DATA

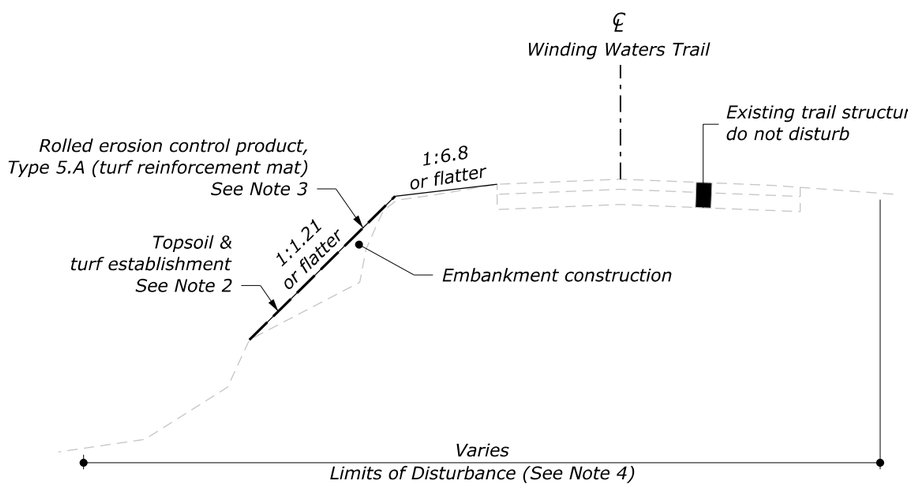
SCHEDULE C: DAGMAR DALE NATURE TRAIL

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	B01



EMBANKMENT CONSTRUCTION

Winding Waters Trail
Site 3: 100+10 to 100+20



EMBANKMENT CONSTRUCTION

Winding Waters Trail
Site 3: 100+40 to 100+67

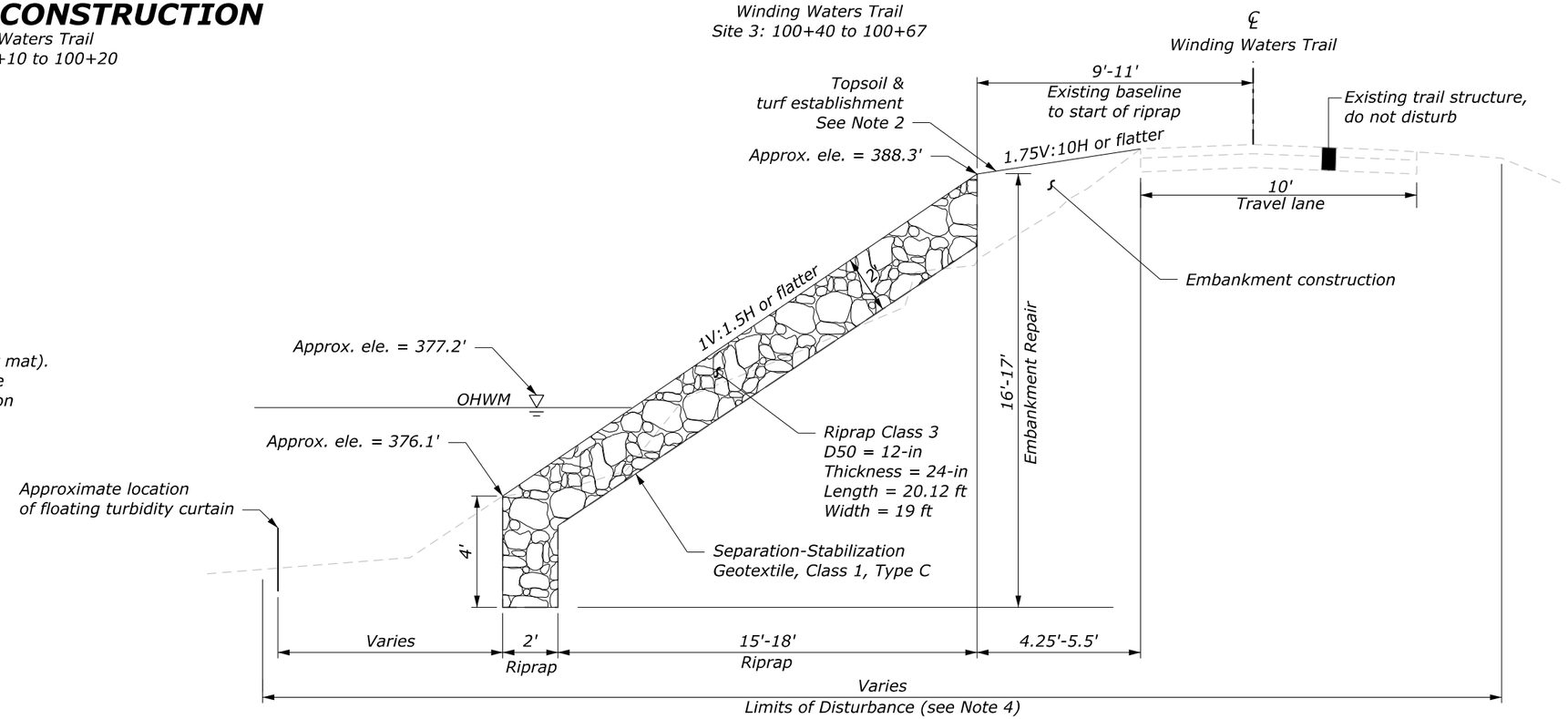
NOTES:

1. Change the construction limits to fit the field conditions as approved by the CO.

2. Place topsoil a minimum depth of 4 inches on all disturbed areas outside the limits of the trail reconstruction, in accordance with Section 624. Establish turf in accordance with Section 625.

3. Place rolled erosion control product, type 5.A. (turf reinforcement mat). Use bio-degradable stakes and stake the matting according to the manufacturer's recommendations. Key and stake the rolled erosion control product a minimum of 12 inches into the top of the bank. See Sections 629 and 713 for more information.

4. At Site #3, perform clearing and grubbing up to the limits disturbed ground only (see Construction Plan and Profile).



EMBANKMENT REPAIR

Winding Waters Trail
Site 3: 100+20 to 100+40

NO.	DATE	BY	REVISIONS

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FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

NOT TO SCALE

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TYPICAL SECTIONS

SCHEDULE A: WINDING WATERS TRAIL

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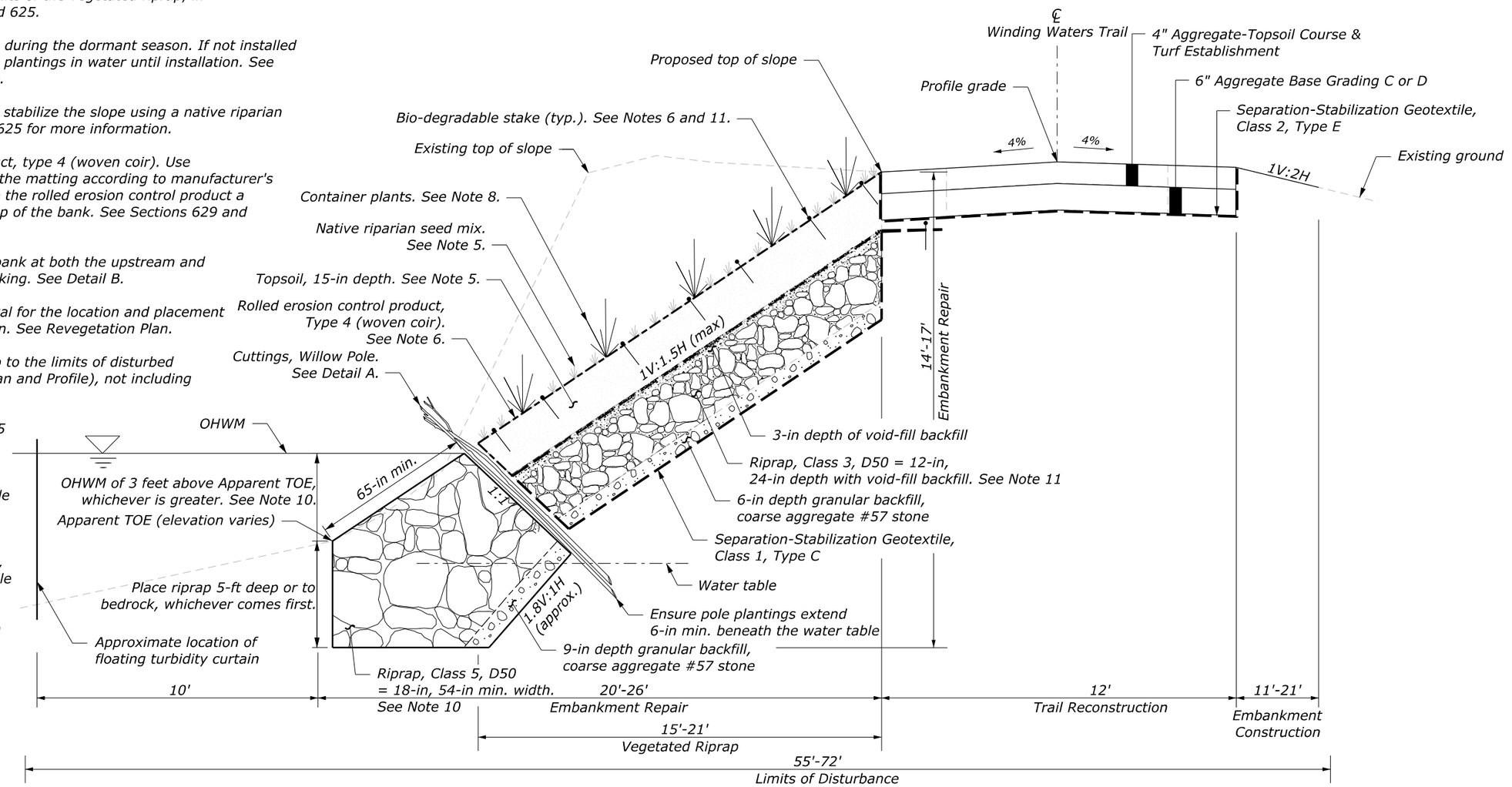
2 September 2026 9:30 AM

NOTES:

1. Stake the proposed trail, including all curvatures, for the CO's approval prior to the trail reconstruction.
2. Change the construction limits to fit the field conditions as approved by the CO.
3. Place topsoil a minimum depth of 4 inches and establish turf on all disturbed areas outside the limits of the vegetated riprap, in accordance with Sections 624 and 625.
4. Harvest and install pole plantings during the dormant season. If not installed immeditaely, soak and store pole plantings in water until installation. See Section 626 for more information.
5. Place topsoil (15-inch depth) and stabilize the slope using a native riparian seed mix. See Sections 624 and 625 for more information.
6. Place rolled erosion control product, type 4 (woven coir). Use bio-degradable stakes and stake the matting according to manufacturer's recommendations. Key and stake the rolled erosion control product a minimum of 12 inches into the top of the bank. See Sections 629 and 713 for more information.
7. Provide a key-in trench into the bank at both the upstream and downstream ends to prevent flanking. See Detail B.
8. Coordinate and obtain CO approval for the location and placement of all plantings prior to installation. See Revegetation Plan.
9. Perform clearing and grubbing up to the limits of disturbed ground only (see Construction Plan and Profile), not including areas of trail reconstruction.
10. Place granular backfill and Class 5 riprap up to the OHWM or to a 3 feet vertical rise (65 inches of slope length), whichever is greater. Place the largest available rocks at the toe of the slope as directed by the CO.
11. Place geotextile, granular backfill, and then place Class 3 riprap while simultaneously backfilling voids with the void-fill backfill material approved by the CO. Place 3-inch depth void-fill backfill before encapsulating the rock with geotextile. Key and stake the geotextile into the top of bank.



TRAIL RECONSTRUCTION AND EMBANKMENT REPAIR WITH VEGETATED RIPRAP

Winding Waters Trail
Site 240: 202+15 to 204+00

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TYPICAL SECTIONS

SCHEDULE A: WINDING WATERS TRAIL

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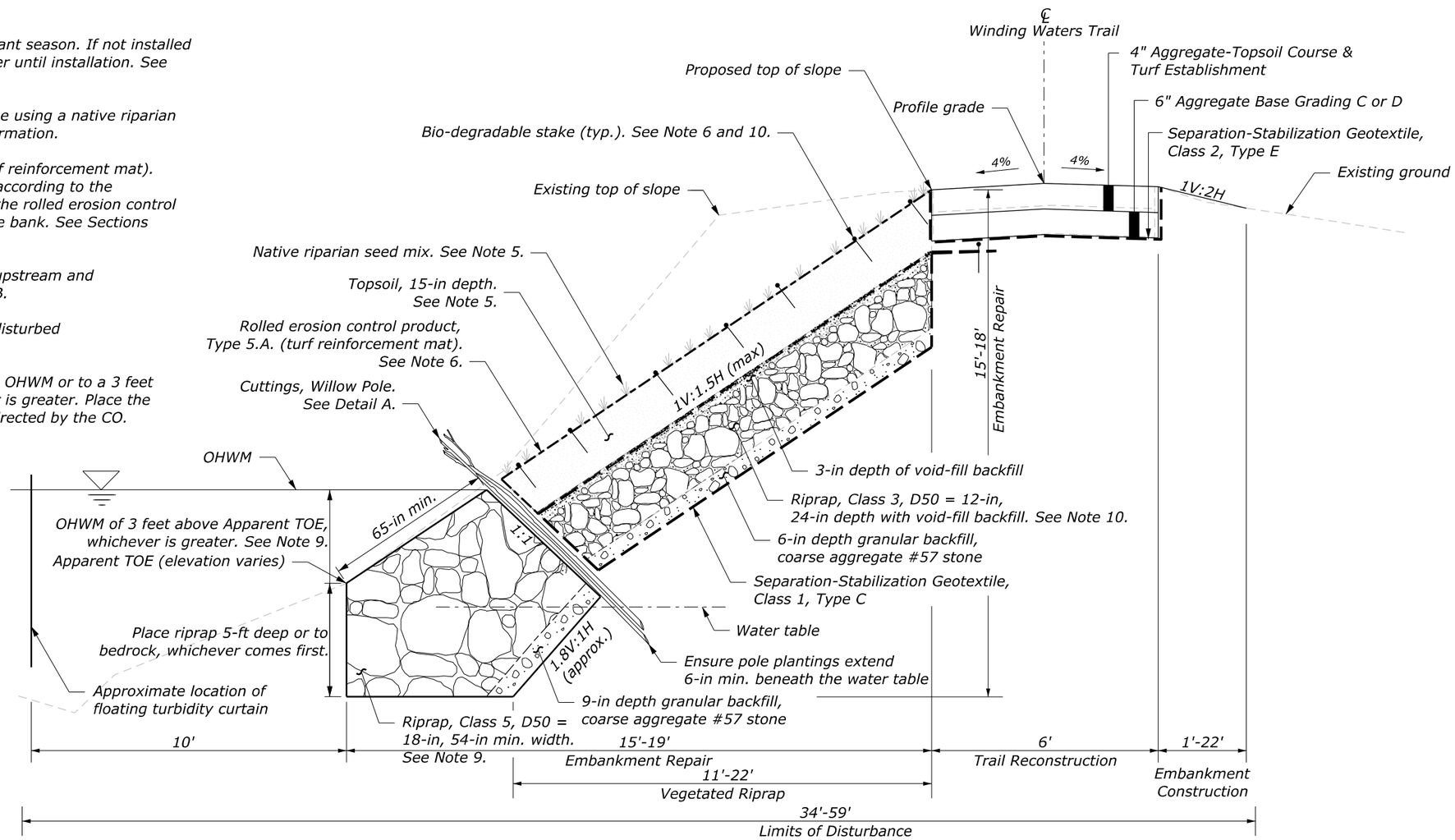
31 August 2026 9:24 AM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	B03

NOTES:

1. Stake the proposed trail, including all curvatures, for the CO's approval prior to the trail reconstruction.
2. Change the construction limits to fit the field conditions as approved by the CO.
3. Place topsoil a minimum depth of 4 inches and establish turf on all disturbed areas outside the limits of the vegetated riprap, in accordance with Sections 624 and 625.
4. Harvest and install pole plantings during the dormant season. If not installed immeditaely, soak and store pole plantings in water until installation. See Section 626 for more information.
5. Place topsoil (15-inch depth) and stabilize the slope using a native riparian seed mix. See Sections 624 and 625 for more information.
6. Place rolled erosion control product, type 5.A. (turf reinforcement mat). Use bio-degradable stakes and stake the matting according to the manufacturer's recommendations. Key and stake the rolled erosion control product a minimum of 12 inches into the top of the bank. See Sections 629 and 713 for more information.
7. Provide a key-in trench into the bank at both the upstream and downstream ends to prevent flanking. See Detail B.
8. Perform clearing and grubbing up to the limits of disturbed ground only (see Construction Plan and Profile).
9. Place granular backfill and Class 5 riprap up to the OHWM or to a 3 feet vertical rise (65 inches of slope length), whichever is greater. Place the largest available rocks at the toe of the slope as directed by the CO.
10. Place geotextile, granular backfill, and then place Class 3 riprap while simultaneously backfilling voids with the void-fill backfill material approved by the CO. Place 3-inch depth void-fill backfill before encapsulating the rock with geotextile. Key and stake the geotextile into the top of bank.



TRAIL RECONSTRUCTION AND EMBANKMENT REPAIR WITH VEGETATED RIPRAP

Winding Waters Trail
Site 324: 300+90 to 303+35

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
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WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TYPICAL SECTIONS

SCHEDULE A: WINDING WATERS TRAIL

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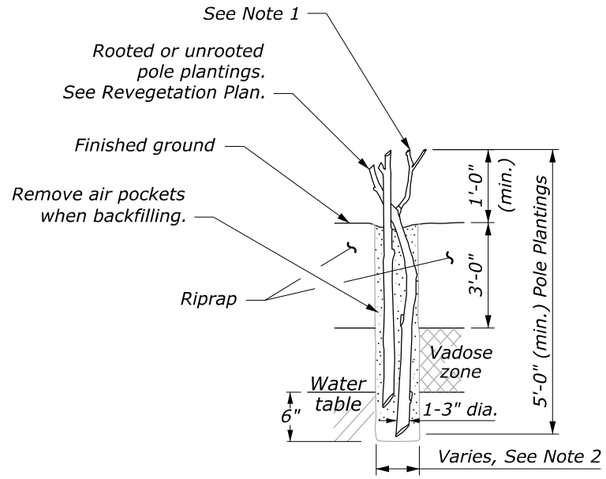
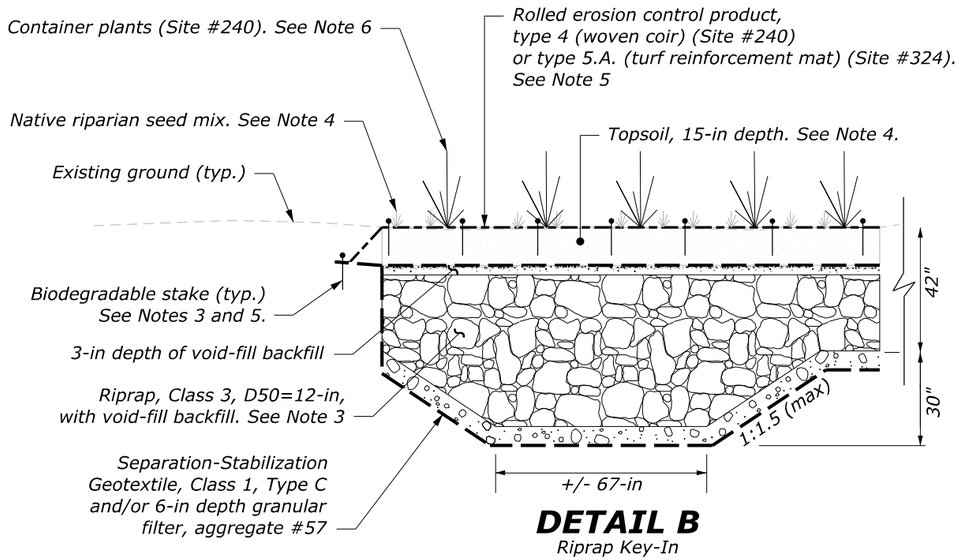
31 August 2026 9:24 AM



DETAIL A
Pole Plantings



DETAIL B
Riprap Key-In

VEGETATED RIPRAP DETAILS

NOTES:

1. *Harvest and install pole plantings during the dormant season. If not installed immeditaely, soak and store pole plantings in water until installation. See Section 626 for more information.*

2. *Ensure the width of excavation hole is at least twice the diameter of the live cutting (pole).*

3. *Place geotextile, granular backfill, and then place Class 3 riprap while simultaneously backfilling voids with the void-fill backfill material approved by the CO. Place 3-inch depth void-fill backfill before encapsulating the rock with geotextile. Key and stake the geotextile into the top of bank.*

4. *Place topsoil (15-inch depth) and stabilize the slope using a native riparian seed mix. See Sections 624 and 625 for more information.*

5. *Place rolled erosion control product, type 4 (woven coir) (Site #240) or rolled erosion control product, type 5.A. (turf reinforcement mat) (Site #324). Use bio-degradable stakes and stake the matting according to the manufacturer's recommendations. Key and stake the rolled erosion control product a minimum of 12 inches into the top of of the bank. See Sections 629 and 713 for more information.*

6. *Coordinate and obtain CO approval for the location and placement of all plantings prior to installation. See Revegetation Plan.*

NO.	DATE	BY	REVISIONS

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OFFICE OF FEDERAL LANDS HIGHWAY

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WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TYPICAL SECTIONS

SCHEDULE A: WINDING WATERS TRAIL

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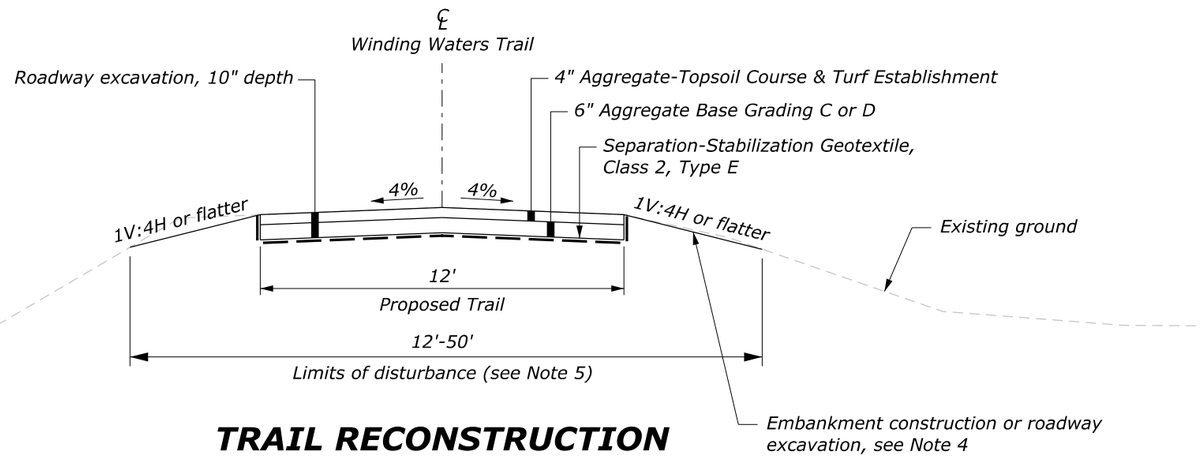
2 September 2026 9:15 AM

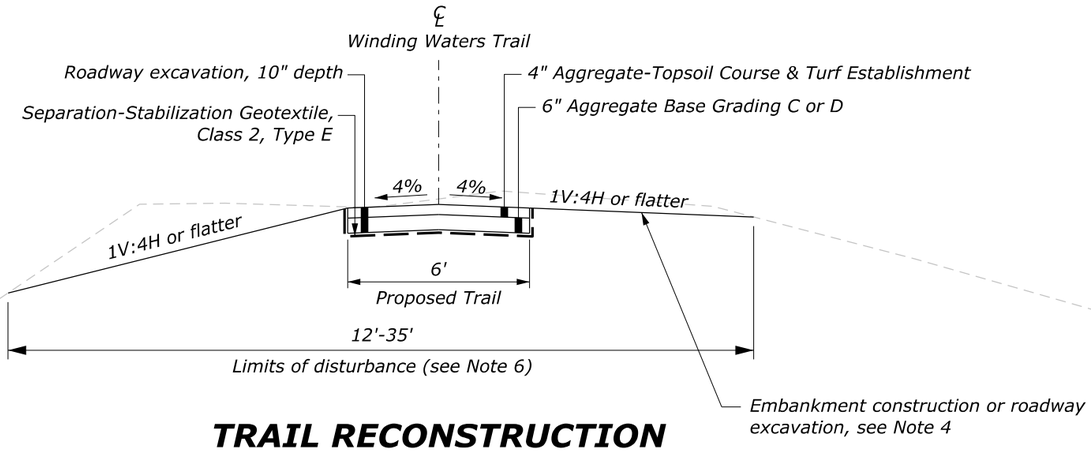
NOTES:

1. *Stake the proposed trail, including curvatures, for the CO's approval prior to the trail's construction.*
2. *Change the construction limits to fit the field conditions as approved by the CO.*
3. *Place topsoil a minimum depth of 4 inches on all disturbed areas outside the limits of the trail reconstruction, in accordance with Section 624. Establish turf in accordance with Section 625.*
4. *Regrade cut slopes and fill slopes to match the cross slope of the proposed trail reconstruction to the extent practicable. Cut and fill slopes that cannot match the cross slope of the proposed trail reconstruction should not exceed a slope of 1V:4H.*
5. *At Site #240, perform clearing and grubbing up to the limits of disturbed ground only (see Construction Plan and Profile), not including areas of trail reconstruction.*
6. *At Site #324, perform clearing and grubbing up to the limits of disturbed ground only (see Construction Plan and Profile).*



TRAIL RECONSTRUCTION

Winding Waters Trail
Site 240: 201+90 to 202+15 and 204+00 to 204+60



TRAIL RECONSTRUCTION

Winding Waters Trail
Site 324: 300+04 to 300+90 and 303+35 to 303+90

NO.	DATE	BY	REVISIONS

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FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TYPICAL SECTIONS

SCHEDULE A: WINDING WATERS TRAIL

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9 August 2026 6:25 PM

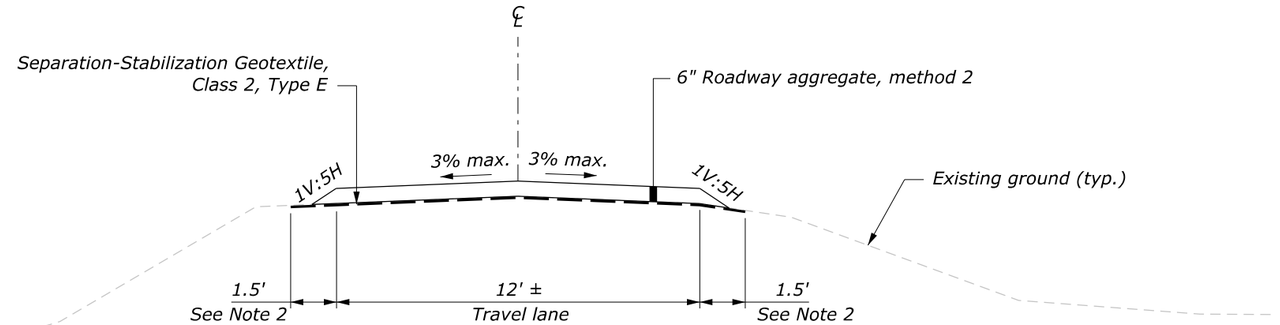
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	B06

NOTES:

1. *When construction activities are complete, remove the temporary access road, and regrade the area to the preconstruction contours as directed by the CO.*

2. *Place geotextile over the existing ground according to Section 207 and extend to a minimim of 18" beyond the limitis of placed roadway aggregate.*

3. *Construct, maintain, and remove the temproary access roads and temporary crossing according to Section 650.*



TEMPORARY ACCESS ROADWAY

Aggregate Overlay

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

NOT TO SCALE

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TYPICAL SECTIONS

SCHEDULE A: WINDING WATERS TRAIL

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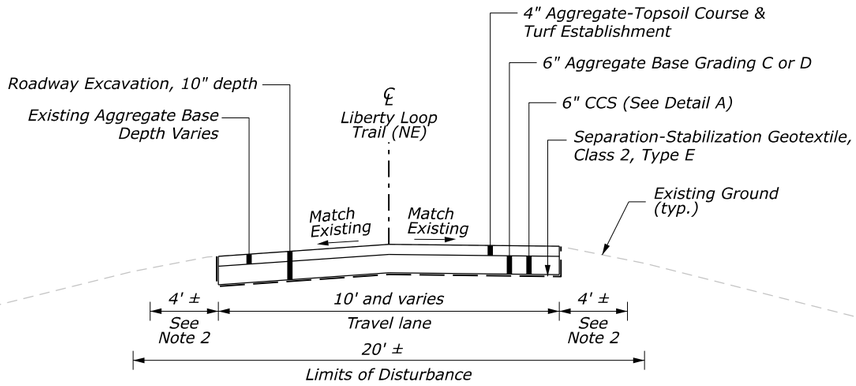
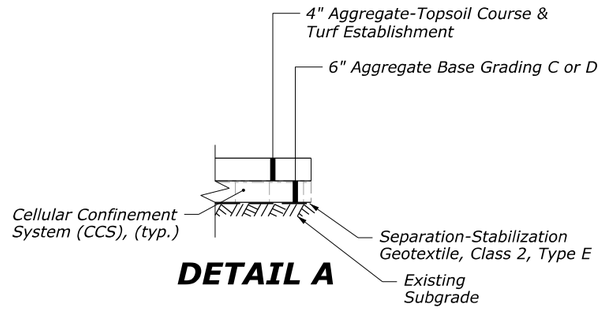
2 August 2026 6:27 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	B07



TRAIL RECONSTRUCTION WITH CCS

Liberty Loop Trail (NE)
Sta. 400+40 to 422+87 and 423+16 to 434+23



DETAIL A

NOTES:

1. *Reestablish the trail to its current alignment and elevation. Adjust final grading based on site conditions as directed by the CO.*
2. *Recondition shoulder as directed by CO. Locations to be verified in field.*
3. *Utilize the on-site material for the embankment construction, shoulder reconditioning, and other repairs. Reuse excavated material for this work to the most extent possible.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

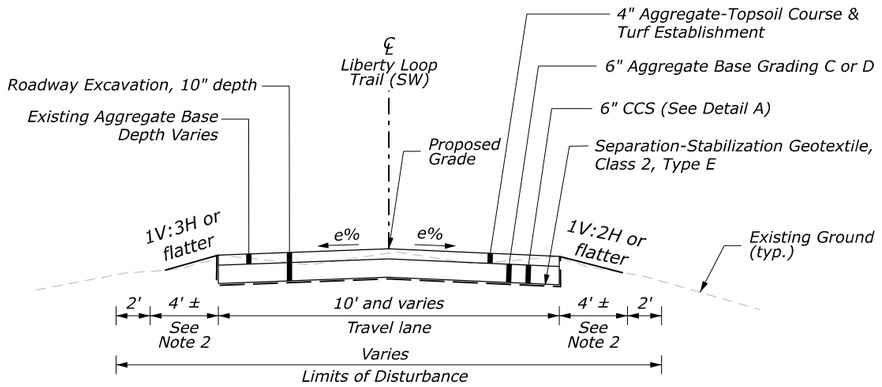
NOT TO SCALE

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TYPICAL SECTIONS

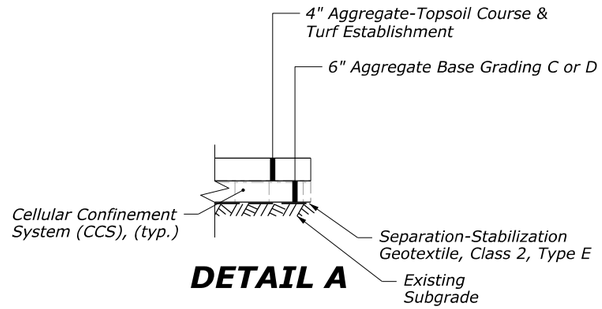
SCHEDULE A: LIBERTY LOOP TRAIL (NE)

C:\Users\rreo.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\B01_NY-WARI_2019-1(1)\LLTrail.dgn [B07]
2 August 2026 7:21 PM



TRAIL RECONSTRUCTION WITH CCS

Liberty Loop Trail (SW)
347+00 to 393+52



DETAIL A

NOTES:

1. *Reestablish the trail to its current alignment and elevation. Adjust final grading based on site conditions as directed by the CO.*
2. *Recondition shoulder as directed by the CO. Locations to be verified in field.*
3. *Utilize the on-site material for the embankment construction, shoulder reconditioning, and other repairs. Reuse excavated material for this work to the most extent possible.*
4. *Place topsoil a minimum depth of 4 inches on all disturbed areas outside the limits of the trail reconstruction, in accordance with Section 624. Establish turf in accordance with Section 625.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

NOT TO SCALE

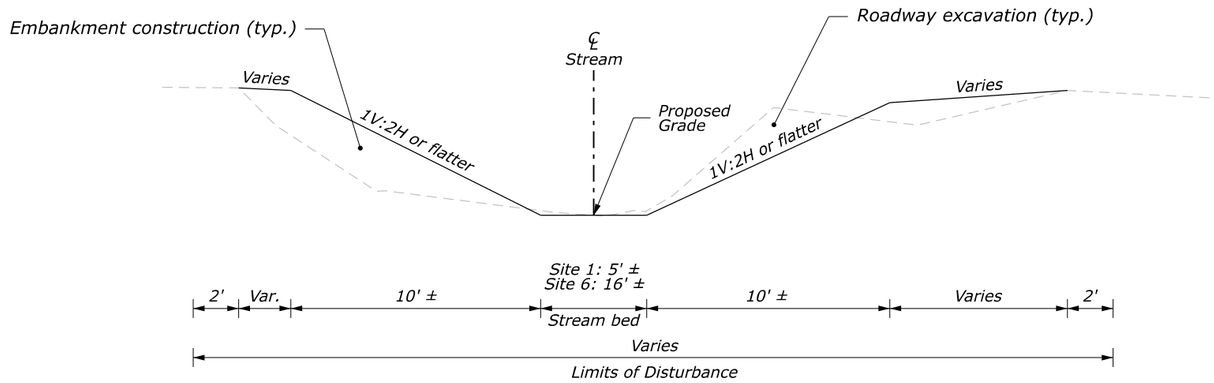
WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TYPICAL SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

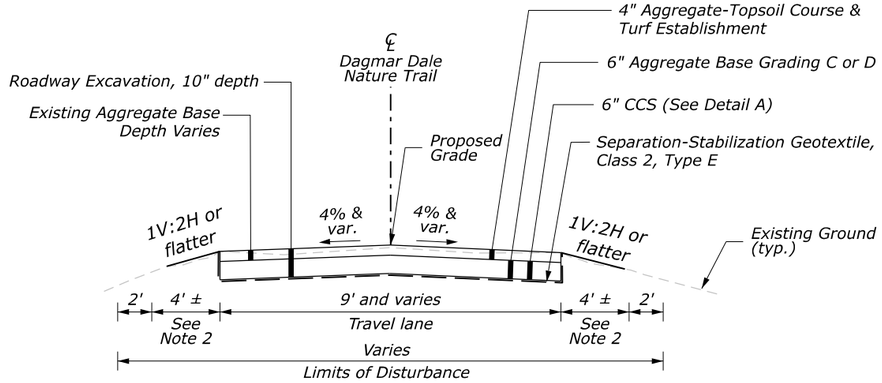
2 September 2026 9:26 AM \\fhf15fileserve.fhfd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\B01-B03_NJ-WARI_2019-1(1).dgn [B08]

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	B09



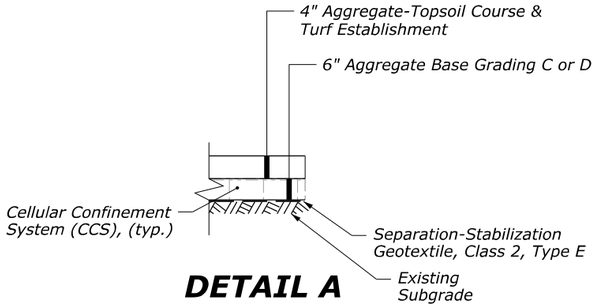
STREAM REGRADING

Dagmar Dale Nature Trail
Site 1: 1+15 to 2+10
Site 6: 0+30 to 1+10

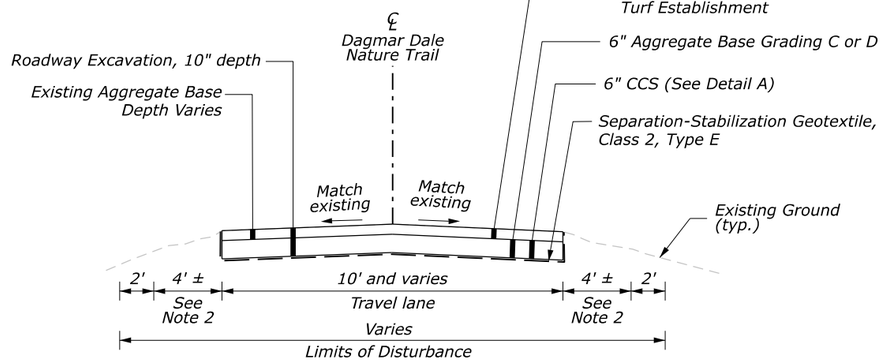


TRAIL RECONSTRUCTION WITH CCS

Dagmar Dale Nature Trail
Site 1: 10+60 to 11+90
Site 6: 60+13 to 61+54



DETAIL A



TRAIL RECONSTRUCTION WITH CCS

Dagmar Dale Nature Trail
Site 5: 50+20 to 53+50

NOTES:

1. *Adjust final grading based on site conditions as directed by the CO.*
2. *Recondition shoulder as directed by the CO. Locations to be verified in field.*
3. *Utilize the on-site material for the embankment construction, shoulder reconditioning, and other repairs. Reuse excavated material for this work to the most extent possible.*
4. *Place topsoil a minimum depth of 4 inches on all disturbed areas outside the limits of the trail reconstruction, in accordance with Section 624. Establish turf in accordance with Section 625.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

NOT TO SCALE

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TYPICAL SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL

2 September 2026 9:26 AM
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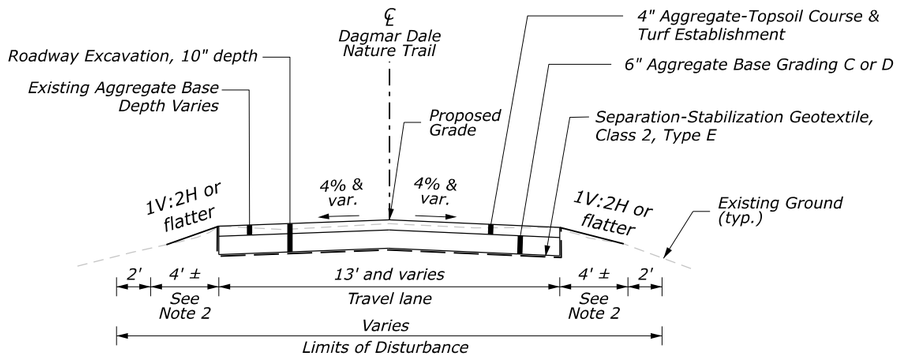
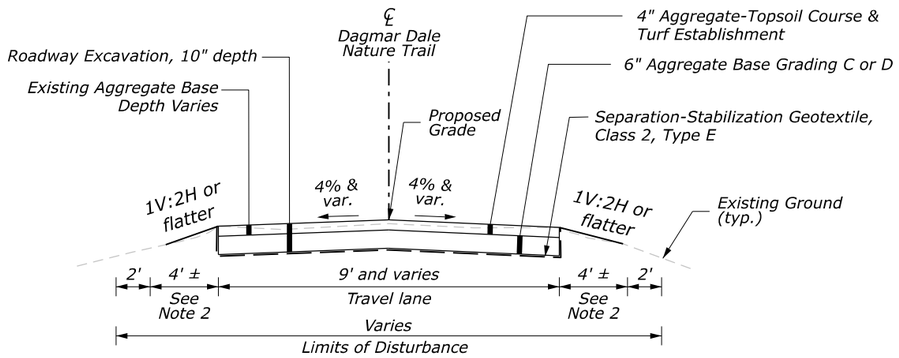
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	B10



TRAIL RECONSTRUCTION

Dagmar Dale Nature Trail
Site 2: 21+37.68 to 22+91.23



TRAIL RECONSTRUCTION

Dagmar Dale Nature Trail
Site 3: 30+43.34 to 32+32.64
Site 4: 40+15.57 to 42+19.32

NOTES:

1. *Adjust final grading based on site conditions as directed by the CO.*
2. *Recondition shoulder as directed by the CO. Locations to be verified in field.*
3. *Utilize the on-site material for the embankment construction, shoulder reconditioning, and other repairs. Reuse excavated material for this work to the most extent possible.*
4. *Place topsoil a minimum depth of 4 inches on all disturbed areas outside the limits of the trail reconstruction, in accordance with Section 624. Establish turf in accordance with Section 625.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TYPICAL SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL

SHEET 2 OF 2

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PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	C01

Line Item No.	Pay Item No.	Pay Item Description	Unit	Estimated Quantities
A0010	15101-0000	MOBILIZATION	LPSM	ALL
A0020	15201-0000	CONSTRUCTION SURVEY AND STAKING	LPSM	ALL
A0030	15401-0000	CONTRACTOR TESTING	LPSM	ALL
A0040	15705-0100	SOIL EROSION CONTROL, SILT FENCE	LNFT	1,500
A0050	15705-0100	SOIL EROSION CONTROL, SILT FENCE,(WILDLIFE EXCLUSION)	LNFT	7,500
A0060	15705-2000	SOIL EROSION CONTROL, FLOATING TURBIDITY CURTAIN	LNFT	540
A0070	15720-0000	STORM WATER POLLUTION PREVENTION PLAN	LPSM	ALL
A0080	20103-0000	CLEARING AND GRUBBING	SQYD	2,700
A0090	20220-1000	REMOVAL, INDIVIDUAL TREE	EACH	17
A0100	20401-0000	ROADWAY EXCAVATION	CUYD	3,900
A0110	20435-2000	BACKFILL, GRANULAR,(3-INCH VOID-FILL BACKFILL)	CUYD	215
A0120	20701-0300	SEPARATION-STABILIZATION GEOTEXTILE, CLASS 1, TYPE C	SQYD	1,900
A0130	20701-1000	SEPARATION-STABILIZATION GEOTEXTILE, CLASS 2, TYPE E	SQYD	5,500
A0140	25102-0300	PLACED RIPRAP, METHOD A, CLASS 3	TON	1,100
A0150	25102-0500	PLACED RIPRAP, METHOD A, CLASS 5	TON	1,900
A0160	25510-0000	SELECT GRANULAR BACKFILL,(COARSE AGGREGATE #57 STONE)	CUYD	200
A0170	30101-4000	AGGREGATE BASE GRADING C OR D	TON	1,700
A0180	30302-2000	SHOULDER RECONDITIONING	LNFT	370
A0190	31302-0400	AGGREGATE-TOPSOIL COURSE, 4-INCH DEPTH	SQYD	5,400
A0200	62303-1000	SPECIAL LABOR, HIRED TECHNICAL SERVICES,(ARBORIST)	LPSM	ALL
A0210	62303-1000	SPECIAL LABOR, HIRED TECHNICAL SERVICES,(BIOLOGICAL MONITOR)	LPSM	ALL
A0220	62303-1000	SPECIAL LABOR, HIRED TECHNICAL SERVICES,(HORTICULTURIST)	LPSM	ALL
A0230	62403-0000	FURNISHING AND PLACING TOPSOIL	CUYD	95
A0240	62403-0000	FURNISHING AND PLACING TOPSOIL,(15-INCH DEPTH)	CUYD	350
A0250	62501-0000	TURF ESTABLISHMENT	ACRE	1.45
A0260	62601-0600	ACER RUBRUM 'OCTOBER GLORY', OCTOBER GLORY RED MAPLE, 3 1/2-INCH TO 4-INCH CALIPER, BALLED AND BURLAPPED	EACH	6
A0270	62616-0500	PLATANUS OCCIDENTALIS, SYCAMORE, 60-INCH TO 72-INCH HEIGHT, BALLED AND BURLAPPED	EACH	2
A0280	62617-0700	QUERCUS PALUSTRUS, PIN OAK, 3 1/2-INCH TO 4-INCH CALIPER, BALLED AND BURLAPPED	EACH	3
A0290	62619-0300	SALIX NIGRA, BLACK WILLOW, 2 1/2-INCH TO 3 1/2-INCH CALIPER, CONTAINER GROWN	EACH	6
A0300	62630-0300	PLANTINGS, SEEDLINGS, CONTAINER GROWN	EACH	190
A0310	62635-3000	CUTTINGS, WILLOW POLE	EACH	165
A0320	62901-1100	ROLLED EROSION CONTROL PRODUCT, TYPE 4,(WOVEN COIR)	SQYD	430
A0330	62901-1200	ROLLED EROSION CONTROL PRODUCT, TYPE 5.A,(TURF REINFORCEMENT MAT)	SQYD	470
A0340	62903-0000	CELLULAR CONFINEMENT SYSTEM	SQYD	4,700
A0350	63501-0000	TEMPORARY TRAFFIC CONTROL	LPSM	ALL
A0360	63701-0000	FIELD OFFICE	EACH	1
A0370	65001-1000	CONSTRUCT AND MAINTAIN DIVERSION,(TEMPORARY ACCESS ROAD)	LPSM	ALL

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30 August 2026 9:07 PM

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TABULATION OF QUANTITIES

SCHEDULE A: LIBERTY LOOP TRAIL (NE) &
WINDING WATERS TRAIL

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	C02

Line Item No.	Pay Item No.	Pay Item Description	Unit	Estimated Quantities
B0010	15101-0000	MOBILIZATION	LPSM	ALL
B0020	15201-0000	CONSTRUCTION SURVEY AND STAKING	LPSM	ALL
B0030	15401-0000	CONTRACTOR TESTING	LPSM	ALL
B0040	15705-0100	SOIL EROSION CONTROL, SILT FENCE	LNFT	9,800
B0050	15720-0000	STORM WATER POLLUTION PREVENTION PLAN	LPSM	ALL
B0060	20401-0000	ROADWAY EXCAVATION	CUYD	1,200
B0070	20701-1000	SEPARATION-STABILIZATION GEOTEXTILE, CLASS 2, TYPE E	SQYD	5,300
B0080	30101-4000	AGGREGATE BASE GRADING C OR D	TON	1,650
B0090	31302-0400	AGGREGATE-TOPSOIL COURSE, 4-INCH DEPTH	SQYD	5,200
B0100	62303-1000	SPECIAL LABOR, HIRED TECHNICAL SERVICES,(BIOLOGICAL MONITOR)	LPSM	ALL
B0110	62403-0000	FURNISHING AND PLACING TOPSOIL	CUYD	88
B0120	62501-0000	TURF ESTABLISHMENT	ACRE	1.24
B0130	62903-0000	CELLULAR CONFINEMENT SYSTEM	SQYD	5,200
B0140	63501-0000	TEMPORARY TRAFFIC CONTROL	LPSM	ALL
B0150	64620-0300	REMOVE AND RESET BENCH	EACH	2

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24 August 2026 3:29 PM

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TABULATION OF QUANTITIES

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

Line Item No.	Pay Item No.	Pay Item Description	Unit	Estimated Quantities
C0010	15101-0000	MOBILIZATION	LPSM	ALL
C0020	15201-0000	CONSTRUCTION SURVEY AND STAKING	LPSM	ALL
C0030	15401-0000	CONTRACTOR TESTING	LPSM	ALL
C0040	15702-6000	SOIL EROSION CONTROL, TEMPORARY STREAM DIVERSION	LPSM	ALL
C0050	15705-0100	SOIL EROSION CONTROL, SILT FENCE	LNFT	3,000
C0060	15706-0200	SOIL EROSION CONTROL, CHECK DAM	EACH	2
C0070	20103-0000	CLEARING AND GRUBBING	SQYD	970
C0080	20220-1000	REMOVAL, INDIVIDUAL TREE	EACH	12
C0090	20301-0300	REMOVAL OF BOX CULVERT,(AND WINGWALLS)	EACH	1
C0100	20301-1900	REMOVAL OF PIPE CULVERT	EACH	2
C0110	20420-0000	EMBANKMENT CONSTRUCTION	CUYD	495
C0120	20701-1000	SEPARATION-STABILIZATION GEOTEXTILE, CLASS 2, TYPE E	SQYD	1,400
C0130	20702-0300	GEOTEXTILE FILTER, CLASS 1, TYPE C	SQYD	18
C0140	25102-0300	PLACED RIPRAP, METHOD A, CLASS 3	TON	12
C0150	25126-0000	REMOVE AND RESET BOULDER	EACH	5
C0160	25510-0000	SELECT GRANULAR BACKFILL,(COARSE AGGREGATE #57 STONE)	CUYD	4
C0170	30101-4000	AGGREGATE BASE GRADING C OR D	TON	400
C0180	30302-1000	DITCH RECONDITIONING	LNFT	310
C0190	31302-0400	AGGREGATE-TOPSOIL COURSE, 4-INCH DEPTH	SQYD	1,300
C0200	60103-0240	CONCRETE, HEADWALL FOR 54-INCH PIPE CULVERT	EACH	2
C0210	60103-0700	CONCRETE, HEADWALL FOR DOUBLE 48-INCH PIPE CULVERT	EACH	2
C0220	60103-1920	CONCRETE, HEADWALL FOR 24-INCH EQUIVALENT DIAMETER PIPE CULVERT	EACH	4
C0230	60201-0800	24-INCH PIPE CULVERT,(HDPE)	LNFT	37
C0240	60201-1200	48-INCH PIPE CULVERT,(HDPE)	LNFT	36
C0250	60201-1300	54-INCH PIPE CULVERT,(HDPE)	LNFT	32
C0260	62303-1000	SPECIAL LABOR, HIRED TECHNICAL SERVICES,(BIOLOGICAL MONITOR)	LPSM	ALL
C0270	62403-0000	FURNISHING AND PLACING TOPSOIL	CUYD	120
C0280	62501-0000	TURF ESTABLISHMENT	ACRE	0.48
C0290	62903-0000	CELLULAR CONFINEMENT SYSTEM	SQYD	650
C0300	63501-0000	TEMPORARY TRAFFIC CONTROL	LPSM	ALL

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24 August 2026 6:05 PM

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TABULATION OF QUANTITIES

SCHEDULE C: DAGMAR DALE NATURE TRAIL

SOIL EROSION CONTROL SUMMARY

Sheet	SITE	STATION TO STATION			SIDE	Schedule A Pay Item 15705-0100 SOIL EROSION CONTROL, SILT FENCE (WILDLIFE EXCLUSION) LNFT	Schedule A Pay Item 15705-0100 SOIL EROSION CONTROL, SILT FENCE LNFT	Schedule A Pay Item 15705-2000 SOIL EROSION CONTROL, FLOATING TURBIDITY CURTAIN LNFT	Schedule A Pay Item 62901-1100 ROLLED EROSION CONTROL PRODUCT, TYPE 4 (WOVEN COIR) SQYD	Schedule A Pay Item 62901-1200 ROLLED EROSION CONTROL PRODUCT, TYPE 5.A (TURF REINFORCEMENT MAT) SQYD	REMARKS
M12-M15	Liberty Loop Trail (NE)	400+42	to	422+83	LT	2,262					Liberty Loop Trail before parking
M12-M15	Liberty Loop Trail (NE)	400+38	to	422+71	RT	2,224					Liberty Loop Trail before parking
M11	Liberty Loop Trail Staging Entrance	422+47	to	422+83	LT	70					Liberty Loop Trail staging entrance
M11	Liberty Loop Trail Staging Entrance	423+23	to	423+42	LT	65					Liberty Loop Trail staging entrance
M11	Liberty Loop Trail Staging	422+71	to	423+32	RT	200					Liberty Loop Trail staging
M15-M17	Liberty Loop Trail (NE)	423+23	to	434+23	LT	1,107					Liberty Loop Trail after parking
M15-M17	Liberty Loop Trail (NE)	423+32	to	434+23	RT	1,087					Liberty Loop Trail after parking
M06	Winding Waters Trail Site #240	201+90	to	202+14	LT		35				
M06	Winding Waters Trail Site #240	201+90	to	204+60	RT		315				
M06	Winding Waters Trail Site #240	204+03	to	204+60	LT		59				
M07	Winding Waters Trail Site #324	300+00	to	300+88	LT		112				
M07	Winding Waters Trail Site #324	300+06	to	303+90	RT		399				
M07	Winding Waters Trail Site #324	303+42	to	303+90	LT		52				
M04	Winding Waters Trail Staging						178				
M04	Winding Waters Trail Staging						215				
M05	Winding Waters Trail Site #3	100+09	to	100+69	LT			79			
M06	Winding Waters Trail Site #240	202+05	to	204+10	LT			179			
M07	Winding Waters Trail Site #324	300+77	to	303+44	LT			258			
B02	Winding Waters Trail Site #240	202+15	to	204+00					404		
B01	Winding Waters Trail Site #3	100+10	to	100+67						30	Not needed in riprap section
B03	Winding Waters Trail Site #324	300+90	to	303+35						413	
				Subtotal this sheet		7,014	1,364	516	404	443	
				Rounded Total		7,500	1,500	540	430	470	

CLEARING AND REMOVALS SUMMARY

SHEET	SITE	STATION TO STATION			SIDE	Schedule A Pay Item 20103-0000 CLEARING AND GRUBBING SQYD	Schedule A Pay Item 20220-1000 REMOVAL, INDIVIDUAL TREE EACH	REMARKS
D01	Winding Waters Trail Site #3	100+10	to	100+67		123		LOD minus edge of water
D02	Winding Waters Trail Site #240	201+90	to	204+60		1162		LOD minus edge of water minus roadway reconditioning
D04	Winding Waters Trail Site #324	300+00	to	303+90		1311		LOD minus edge of water
M08	Winding Waters Trail Site #3	100+37	to	100+37	LT		1	
M09	Winding Waters Trail Site #240	201+90	to	204+60	RT & LT		12	
M10	Winding Waters Trail Site #324	300+04	to	303+90	RT & LT		4	
				Subtotal This sheet		2,596	17	
				Rounded Total		2,700	17	

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SUMMARIES AND SCHEDULES

SCHEDULE A: LIBERTY LOOP TRAIL (NE) &
WINDING WATERS TRAIL

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2 September 2026 9:36 AM

EARTHWORK SUMMARY

SHEET	SITE	STATION TO STATION	Schedule A Pay Item 20401-0000 ROADWAY EXCAVATION CUYD	Schedule A No Pay Section 204 EMBANKMENT CONSTRUCTION CUYD	Schedule A Pay Item 20435-2000 BACKFILL, GRANULAR (3-INCH VOID-FILL BACKFILL) CUYD	Schedule A Pay Item 20701-0300 SEPARATION-STABILIZATION GEOTEXTILE, CLASS 1, TYPE C SQYD	Schedule A Pay Item 20701-1000 SEPARATION-STABILIZATION GEOTEXTILE, CLASS 2, TYPE E SQYD	Schedule A Pay Item 25510-0000 SELECT GRANULAR BACKFILL (COARSE AGGREGATE #57 STONE) CUYD	Schedule A Pay Item 30101-4000 AGGREGATE BASE GRADING C OR D TON	Schedule A Pay Item 30302-2000 SHOULDER RECONDITIONING LNFT	Schedule A Pay Item 31302-0400 AGGREGATE-TOPSOIL COURSE, 4-INCH DEPTH SQYD	Schedule A Pay Item 62403-0000 FURNISHING AND PLACING TOPSOIL CUYD	Schedule A Pay Item 62403-0000 FURNISHING AND PLACING TOPSOIL (15-INCH DEPTH) CUYD	Schedule A Pay Item 62501-0000 TURF ESTABLISHMENT ACRE	Schedule A Pay Item 62903-0000 CELLULAR CONFINEMENT SYSTEM SQYD	REMARKS
D06-D17	Liberty Loop Trail (NE)	400+40 to 434+23	1,215	0.03												
D01	Winding Waters Trail Site #3	100+10 to 100+67	15	8												
D02	Winding Waters Trail Site #240	201+90 to 204+60	1,032	211												
D03	Winding Waters Trail Site #324	300+04 to 303+90	1,421	4												
B02	Winding Waters Trail Site #240	202+15 to 204+00			98			84					164			
B03	Winding Waters Trail Site #324	300+90 to 303+35			98			104					166			
B01	Winding Waters Trail Site #3	100+20 to 100+40				70										
B02	Winding WatersTrail Site #240	202+15 to 204+00				838										
B03	Winding Waters Trail Site #324	300+90 to 303+35				865										
B07	Liberty Loop Trail (NE) before staging area	202+15 to 204+00					2,827									Roadway reconditioning area
B07	Liberty Loop Trail (NE) after staging area	202+15 to 204+00					1,693									Roadway reconditioning area
B05	Winding Waters Trail Site #240	300+90 to 303+35					348									Roadway reconditioning area
B05	Winding Waters Trail Site #324	300+90 to 303+35					257									Roadway reconditioning area
D02	Winding Waters Trail Site #240	201+90 to 204+60							109		348					
D04	Winding Waters Trail Site #324	300+04 to 303+90							81		257					
D06-D13	Liberty Loop Trail (NE) before staging area	400+40 to 422+87							890		2,827				2,827	
D13-D17	Liberty Loop Trail (NE) after staging area	423+16 to 434+23							533		1,693				1,693	
B07	Liberty Loop Trail (NE) before staging area	400+40 to 422+87								225				0.58		
B07	Liberty Loop Trail after staging area	423+16 to 434+23								111				0.35		
B01	Winding Waters Trail Site #3	100+10 to 100+67										6		0.01		Disturbed area (Cut/fill region)
B02	Winding Waters Trail Site #240	201+90 to 204+60												0.07		Trail top
B02	Winding Waters Trail Site #240	201+90 to 204+60										49		0.09		Disturbed area (Cut/fill region)
B03	Winding Waters Trail Site #324	300+04 to 303+90												0.05		Trail top
B03	Winding Waters Trail Site #324	300+04 to 303+90										34		0.06		Disturbed area (Cut/fill region)
B02	Winding Waters Trail Site #240	202+15 to 204+00												0.08		Seeding over vegetated riprap
B03	Winding Waters Site #324	300+90 to 303+35												0.09		Seeding over vegetated riprap
		Subtotal this Sheet	3,683	223	196	1,773	5,125	188	1,614	336	5,125	89	330	1.39	4,520	
		Rounded Total	3,900	230	215	1,900	5,500	200	1,700	370	5,400	95	350	1.45	4,700	

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SUMMARIES AND SCHEDULES

SCHEDULE A: LIBERTY LOOP TRAIL (NE) & WINDING WATERS TRAIL

\\fhfl15fileserve.flhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\C01-C05_NY-WARI_2019-1(1)_qty.dgn [Earthwork]
2 September 2026 9:40 AM

DRAINAGE SUMMARY

SHEET	SITE	STATION	SIDE	Schedule A Pay Item 25102-0300 PLACED RIPRAP, METHOD A, CLASS 3 TON	Schedule A Pay Item 25102-0500 PLACED RIPRAP, METHOD A, CLASS 5 TON	REMARKS
D13	Winding Waters Trail Site #3	100+23 to 100+29	LT	21		
D14	Winding Waters Trail Site #240	202+15 to 204+00	LT	482	599	
D16	Winding Waters Trail Site #324	300+90 to 303+35	LT	477	1,155	
			Subtotal this Sheet	980	1754	
			Rounded Total	1,100	1,900	

LANDSCAPING SUMMARY

SHEET	SITE	STATION TO STATION			SIDE	Schedule A Pay Item 62601-0200 ACER RUBRUM, RED MAPLE, 2 1/2-INCH TO 3 1/2-INCH CALIPER, BALLED AND BURLAPPED EACH	Schedule A Pay Item 62616-0500 PLATANUS OCCIDENTALIS, SYCAMORE, 60-INCH TO 72-INCH HEIGHT, BALLED AND BURLAPPED EACH	Schedule A Pay Item 62617-0700 QUERCUS PALUSTRUS, PIN OAK, 3 1/2-INCH TO 4-INCH CALIPER, BALLED AND BURLAPPED EACH	Schedule A Pay Item 62619-0300 SALIX NIGRA, BLACK WILLOW, 2 1/2-INCH TO 3 1/2-INCH CALIPER, CONTAINER GROWN EACH	Schedule A Pay Item 62630-0300 PLANTINGS, SEEDLINGS, CONTAINER GROWN EACH	Schedule A Pay Item 62635-3000 CUTTINGS, WILLOW POLE EACH	REMARKS
M09	Winding Waters Trail Site #240	202+24	to	204+24	RT & LT	5						Revegetation Site #240
M10	Winding Waters Trail Site #324	300+51	to	301+68	RT	1		1				Revegetation Site #324
M09	Winding Waters Trail Site #240	202+24	to	204+24	RT		1	2	4			Revegetation Site #240
M08	Winding Waters Trail Site #3	100+65	to	100+65	RT		1					Revegetation Site #324
M10	Winding Waters Trail Site #324	300+51	to	301+68	LT				2			Revegetation Site #324
B02/M09	Winding Waters Trail Site #240	202+15	to	204+00	LT					164	61	Quantities obatained from most recent HYD report
B03/M10	Winding Waters Trail Site #324	300+90	to	303+35	LT						81	
					Subtotal This sheet	6	2	3	6	164	142	
					Rounded Total	6	2	3	6	190	165	

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FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SUMMARIES AND SCHEDULES

SCHEDULE A: LIBERTY LOOP TRAIL (NE) &
WINDING WATERS TRAIL

2 September 2026 9:54 AM \\fhfl15fileserve.flhd.fhwa.dot.gov\data\PROJECTS_FWSWARI(NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\C01-C05_NY-WARI_2019-1(1)_qty.dgn [Drainage - Landscaping]

MISCELLANEOUS SUMMARY

Sheet	Schedule A Pay Item 15720-0000 STORM WATER POLLUTION PLAN LPSM	Schedule A Pay Item 62303-1000 SPECIAL LABOR, HIRED TECHNICAL SERVICES (ARBORIST) LPSM	Schedule A Pay Item 62303-1000 SPECIAL LABOR, HIRED TECHNICAL SERVICES (BIOLOGICAL MONITOR) LPSM	Schedule A Pay Item 62303-1000 SPECIAL LABOR, HIRED TECHNICAL SERVICES (HORTICULTURIST) LPSM	Schedule A Pay Item 65001-1000 CONSTRUCTION AND MAINTAIN DIVERSION (TEMPORARY ACCESS ROAD) LPSM	REMARKS
	1					Refer to SCRs for information on SWPPP
		1				Refer to SCRs for information regarding arborist requirements
			1			Refer to SCRs for information regarding biological monitor requirements
				1		Refer to SCRs for information regarding horticulturist requirements
B06					1	Refer to Temporary Access Roadway Typical Section
Subtotal this Sheet	1	1	1	1	1	
Rounded Total	1	1	1	1	1	

TEMPORARY TRAFFIC CONTROL SUMMARY

MUTCD NO.	SIGN TEXT	PANEL SIZE WIDTH (in)	HEIGHT (in)	AREA (sqft)	COLOR COMBINATION	SITE	QUANTITY	Schedule A Pay Item 63501-0000 TEMPORARY TRAFFIC CONTROL LPSM	Schedule A Section 635 TEMPORARY TRAFFIC CONTROL, CONSTRUCTION SIGN SQFT	Schedule A Section 635 TEMPORARY TRAFFIC CONTROL, BARRICADE TYPE 3 EACH	REMARKS
R11-2 (Mod.)	PATH CLOSED	48	30	10	Black on White		5		50		Mount on barricade or gate
R11-2 (Mod.)	PARKING CLOSED	48	30	10	Black on White		2		20		Mount on barricade or gate
W20-3 (Mod.)	PARKING CLOSED AHEAD	36	36	9	Black on Orange		4		36		
						Liberty Loop Trail (NE)		1		10	
						Subtotal this Sheet		1	106	10	
						Rounded Total		*	*	*	

* For information only

NOTE:

Construction signs locations are approximate and final locations to be identified in the field per CO's review and approval.

NO.	DATE	BY	REVISIONS

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OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SUMMARIES AND SCHEDULES

SCHEDULE A: LIBERTY LOOP TRAIL (NE) &
WINDING WATERS TRAIL

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2 September 2026 9:44 AM

SOIL EROSION CONTROL SUMMARY

SITE	STATION TO STATION			SIDE	Schedule B Pay Item 15705-0100 SOIL EROSION CONTROL, SILT FENCE LNFT	REMARKS
Liberty Loop Trail (SW)	347+00	to	393+52	RT	4,654	
	347+00	to	393+52	LT	4,667	
					335	For Liberty Loop staging area
			Subtotal this sheet		9,656	
			Rounded Total		9,800	

EARTHWORK SUMMARY

SITE	STATION TO STATION			Schedule B Pay Item 20401-0000 ROADWAY EXCAVATION CUYD	Schedule B No Pay Section 204 EMBANKMENT CONSTRUCTION CUYD	Schedule B Pay Item 20701-1000 SEPARATION-STABILIZATION GEOTEXTILE, CLASS 2, TYPE E (COARSE AGGREGATE #57 STONE) SQYD	Schedule B Pay Item 30101-4000 AGGREGATE BASE GRADING C OR D TON	Schedule B Pay Item 31302-0400 AGGREGATE-TOPSOIL COURSE, 4-INCH DEPTH SQYD	Schedule B Pay Item 62403-0000 FURNISHING AND PLACING TOPSOIL CUYD	Schedule B Pay Item 62501-0000 TURF ESTABLISHMENT ACRE	Schedule B Pay Item 62903-0000 CELLULAR CONFINEMENT SYSTEM SQYD	REMARKS
Liberty Loop Trail (SW)	347+00	to	393+52	1,165	32	5,101	1,607	5,100	86	1.05	5,100	Turf establishment for trail top
Liberty Loop Trail (SW)	347+00	to	393+52							0.16		Turf establishment for cut/fill region
			Subtotal this Sheet	1,165	32	5,101	1,607	5,100	86	1.21	5,100	
			Rounded Total	1,200	35	5,300	1,650	5,200	88	1.24	5,200	

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SUMMARIES AND SCHEDULES

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

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24 August 2026 6:11 PM

CONSTRUCTION SIGN SUMMARY

MUTCD NO.	SIGN TEXT	PANEL SIZE WIDTH (in)	PANEL SIZE HEIGHT (in)	PANEL SIZE AREA (sqft)	COLOR COMBINATION	SITE	QUANTITY	Schedule B Pay Item 63501-0000 TEMPORARY TRAFFIC CONTROL LPSM	Schedule B Section 635 TEMPORARY TRAFFIC CONTROL, CONSTRUCTION SIGN SQFT	Schedule B Section 635 TEMPORARY TRAFFIC CONTROL, BARRICADE TYPE 3 EACH	REMARKS
						Liberty Loop Trail (SW)		1		2	
R11-2 (mod.)	TRAIL CLOSED	48	30	10	Black on Orange	Liberty Loop Trail (SW)	2		20		Mounted on Barricade Type 3
W20-3 (mod.)	TRAIL CLOSED AHEAD	36	36	9	Black on Orange	Liberty Loop Trail (SW)	2		18		
M4-9	DETOUR ARROW	30	24	5	Black on Orange	Liberty Loop Trail (SW)	4		20		LT
M4-9	DETOUR ARROW	30	24	5	Black on Orange	Liberty Loop Trail (SW)	4		20		RT
M4-8a	END DETOUR	24	18	3	Black on Orange	Liberty Loop Trail (SW)	2		6		
						Subtotal this Sheet		1	84	2	
						Rounded Total		*	*	*	

* For information only

NOTE:

Construction signs locations are approximate and final locations to be identified in the field per CO's review and approval.

MISCELLANEOUS SUMMARY

Sheet	SITE	STATION TO STATION	SIDE	Schedule B Pay Item 15720-0000 STORM WATER POLLUTION PREVENTION PLAN LPSM	Schedule B Pay Item 62303-1000 SPECIAL LABOR HIRED TECHNICAL SERVICES (BIOLOGICAL MONITOR) LPSM	Schedule B Pay Item 64620-0300 REMOVE AND RESET BENCH EACH	REMARKS
				1	1		Refer to SCRs for information on SWPPP and biological monitor requirements for New Jersey
E02	Liberty Loop Trail (SW)	349+53 to 349+53	RT			1	
			Subtotal this Sheet	1	1	1	
			Rounded Total	1	1	2	

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SUMMARIES AND SCHEDULES

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

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24 August 2026 6:13 PM

SOIL EROSION CONTROL SUMMARY

SITE	STATION TO STATION			SIDE	Schedule C Pay Item 15702-6000 SOIL EROSION CONTROL, TEMPORARY STREAM DIVERSION LPSM	Schedule C Pay Item 15705-0100 SOIL EROSION CONTROL, SILT FENCE LNFT	Schedule C Pay Item 15706-0200 SOIL EROSION CONTROL, CHECK DAM EACH	REMARKS
Dagmar Dale Nature Trail Sites #1, 3, 4, 6					1			Diversion needed to dewater site for culverts installation
Dagmar Dale Nature Trail Site #1	10+60	to	11+07	RT		66		
	11+11	to	11+90	RT		103		
	10+60	to	10+98	LT		72		
	11+05	to	11+90	LT		104		
Dagmar Dale Nature Trail Site #2	21+38	to	22+91	RT		149		
	21+38	to	22+91	LT		162		
Dagmar Dale Nature Trail Site #3	30+43	to	31+07	RT		74		
	31+13	to	32+33	RT		129		
	30+43	to	31+09	LT		74		
	31+16	to	32+33	LT		123		
Dagmar Dale Nature Trail Site #4	40+16	to	41+20	RT		108		
	41+32	to	42+19	RT		94		
	40+16	to	42+20	LT		206		
Dagmar Dale Staging Area						474		
Dagmar Dale Nature Trail Site #5	50+20	to	51+66	RT		145		
	51+70	to	53+50	RT		177		
	50+20	to	51+77	LT		157		
	52+03	to	53+50	LT		149		
Dagmar Dale Nature Trail Site #6	60+13	to	60+77	RT		79		
	60+98	to	61+54	RT		100		
	60+13	to	60+55	LT		58		
	60+62	to	61+54	LT		103		
Dagmar Dale Nature Trail Site #4	40+55		40+55	LT			1	Drainage ditch after passing the culvert
Subtotal this sheet					1	2,906	1	
Rounded Total					1	3,000	2	

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2 September 2026 10:03 AM

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U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SUMMARIES AND SCHEDULES

SCHEDULE C: DAGMAR DALE NATURE TRAIL

CLEARING AND REMOVALS SUMMARY

SITE	STATION TO STATION			SIDE	Schedule C Pay Item 20103-0000 CLEARING AND GRUBBING SQYD	Schedule C Pay Item 20220-1000 REMOVAL, INDIVIDUAL TREE EACH	Schedule C Pay Item 20301-0300 REMOVAL OF BOX CULVERT (AND WINGWALLS) EACH	Schedule C Pay Item 20301-1900 REMOVAL OF PIPE CULVERT EACH	REMARKS
Dagmar Dale Nature Trail Site #1	10+75	to	11+43	LT	233				
Dagmar Dale Nature Trail Site #1	10+67	to	11+36	RT	211				
Dagmar Dale Nature Trail Site #3	30+96	to	31+26	LT	46				
Dagmar Dale Nature Trail Site #3	30+99	to	31+22	RT	42				
Dagmar Dale Nature Trail Site #4	41+09	to	41+35	LT	24				
Dagmar Dale Nature Trail Site #4	41+09	to	41+36	RT	68				
Dagmar Dale Nature Trail Site #6	60+25	to	61+00	LT	149				
Dagmar Dale Nature Trail Site #6	60+32	to	61+00	RT	147				
Dagmar Dale Nature Trail Site #1	11+06		11+06	RT		1			
Dagmar Dale Nature Trail Site #6	60+49		60+49	LT		1			
Dagmar Dale Nature Trail Site #6	60+62		60+62	LT		1			
Dagmar Dale Nature Trail Site #6	60+89		60+89	RT		1			
Dagmar Dale Nature Trail Site #6	60+54	to	60+64				1		4'R x 2'S x 14.3' box culvert
Dagmar Dale Nature Trail Site #1	11+01		11+01					1	Exposed 30" CMP from washout
Dagmar Dale Nature Trail Site #4	40+21		40+21					1	12" x 13.3' CPP
Subtotal this Sheet					920	4	1	2	
Rounded Total					970	12	1	2	

EARTHWORK SUMMARY

SITE	STATION TO STATION			Schedule C No Pay Section 204 ROADWAY EXCAVATION CUYD	Schedule C Pay Item 20420-0000 EMBANKMENT CONSTRUCTION CUYD	Schedule C Pay Item 20701-1000 SEPARATION-STABILIZATION GEOTEXTILE, CLASS 2, TYPE E SQYD	Schedule C Pay Item 20702-0300 GEOTEXTILE FILTER, CLASS 1, TYPE C SQYD	Schedule C Pay Item 30101-4000 AGGREGATE BASE GRADING C OR D TON	Schedule C Pay Item 31302-0400 AGGREGATE-TOPSOIL COURSE, 4-INCH DEPTH SQYD	Schedule C Pay Item 62403-0000 FURNISHING AND PLACING TOPSOIL CUYD	Schedule C Pay Item 62501-0000 TURF ESTABLISHMENT ACRE	Schedule C Pay Item 62903-0000 CELLULAR CONFINEMENT SYSTEM SQYD	REMARKS
Dagmar Dale Nature Trail Site #1	10+60	to	11+90	63	200	125	17	40	125	47	0.03	125	
	10+60	to	11+90								0.09		Turf for cut/fill region
Dagmar Dale Nature Trail Site #2	21+38	to	22+91	24	4	231		73	231	3	0.05		
	21+38	to	22+91								0.01		Turf for cut/fill region
Dagmar Dale Nature Trail Site #3	30+43	to	32+33	14	71	190		60	190	13	0.04		
	30+43	to	32+33								0.02		Turf for cut/fill region
Dagmar Dale Nature Trail Site #4	40+16	to	42+19	7	157	202		64	202	25	0.04		
	40+16	to	42+19								0.05		Turf for cut/fill region
Dagmar Dale Nature Trail Site #5	50+20	to	53+50	89	49	321		101	321		0.07	321	
Dagmar Dale Nature Trail Site #6	60+13	to	61+54	49		141		44	141	28	0.03	141	
	60+13	to	61+54								0.05		Turf for cut/fill region
Subtotal this Sheet				246	481	1,211	17	381	1,211	116	0.47	588	
Rounded Total				255	495	1,400	18	400	1,300	120	0.48	650	

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U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

SUMMARIES AND SCHEDULES

SCHEDULE C: DAGMAR DALE NATURE TRAIL

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2 September 2026 10:04 AM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	C12

DRAINAGE SUMMARY

SITE	STATION		STATION	Schedule C Pay Item 25102-0300 PLACED RIPRAP, METHOD A, CLASS 3 TON	Schedule C Pay Item 25510-0000 SELECT GRANULAR BACKFILL (COARSE AGGREGATE #57 STONE) CUYD	Schedule C Pay Item 30302-1000 DITCH RECONDITIONING LNFT	Schedule C Pay Item 60103-0240 CONCRETE, HEADWALL FOR 54-INCH PIPE CULVERT EACH	Schedule C Pay Item 60103-0700 CONCRETE, HEADWALL FOR DOUBLE 48-INCH PIPE CULVERT EACH	Schedule C Pay Item 60103-1920 CONCRETE, HEADWALL FOR 24-INCH EQUIVALENT DIAMETER PIPE CULVERT EACH	Schedule C Pay Item 60201-0800 24-INCH PIPE CULVERT (HDPE) LNFT	Schedule C Pay Item 60201-1200 48-INCH PIPE CULERT (HDPE) LNFT	Schedule C Pay Item 60201-1300 54-INCH PIPE CULVERT (HDPE) LNFT	REMARKS
Dagmar Dale Nature Trail Site #4	41+18	to	41+27	11	3								
Dagmar Dale Nature Trail Site #1	10+99	to	10+99				2					32	
Dagmar Dale Nature Trail Site #6	60+59	to	60+66					2					
Dagmar Dale Nature Trail Site #3	31+13	to	31+13						2	20			
Dagmar Dale Nature Trail Site #4	41+22	to	41+22						2	17			
Dagmar Dale Nature Trail Site #4	40+63	to	41+97			129							
Dagmar Dale Nature Trail Site #5	50+20	to	51+51			131							
Dagmar Dale Nature Trail Site #5	51+73	to	52+11			34							
Dagmar Dale Nature Trail Site #6	60+59	to	60+59								18		Double barrel
Dagmar Dale Nature Trail Site #6	60+66	to	60+66								18		Double barrel
Subtotal this Sheet				11	3	294	2	2	4	37	36	32	
Rounded Total				12	4	310	2	2	4	37	36	32	

CONSTRUCTION SIGN SUMMARY

MUTCD NO.	SIGN TEXT	PANEL SIZE WIDTH (in)	HEIGHT (in)	AREA (sqft)	COLOR COMBINATION	SITE	QUANTITY	Schedule C Pay Item 63501-0000 TEMPORARY TRAFFIC CONTROL LPSM	Schedule C Section 635 TEMPORARY TRAFFIC CONTROL, CONSTRUCTION SIGN SQFT	Schedule C Section 635 TEMPORARY TRAFFIC CONTROL, BARRICADE TYPE 3 EACH	REMARKS
						Dagmar Dale Nature Trail Sites		1		2	Dagmar dale Nature Trail is expected to be closed during construction See Sheet N03 for temporary traffic control plan
R11-2 (mod.)	TRAIL CLOSED	48	30	10	Black on Orange	Dagmar Dale Nature Trail Sites	2		20		Mounted on Barricade Type 3
Subtotal this Sheet								1	20	2	
Rounded Total								*	*	*	

* For information only

NOTE:

Construction signs locations are approximate and final locations to be identified in the field per CO's review and approval.

MISCELLANEOUS SUMMARY

Sheet	SITE	STATION TO STATION	SIDE	Schedule C Pay Item 25126-0000 REMOVE AND RESET BOULDER EACH	Schedule C Pay Item 62303-1000 SPECIAL LABOR, HIRED TECHNICAL SERVICES (BIOLOGICAL MONITOR) LPSM	REMARKS
F01	Dagmar Dale Nature Trail Site #1	111+03 to 11+14	LT	5		Outlet side 6'x6' size boulders reported along the waterway
					1	Refer to SCRs for information on biological monitor requirements for New Jersey
Subtotal this Sheet				5	1	
Rounded Total				5	1	

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

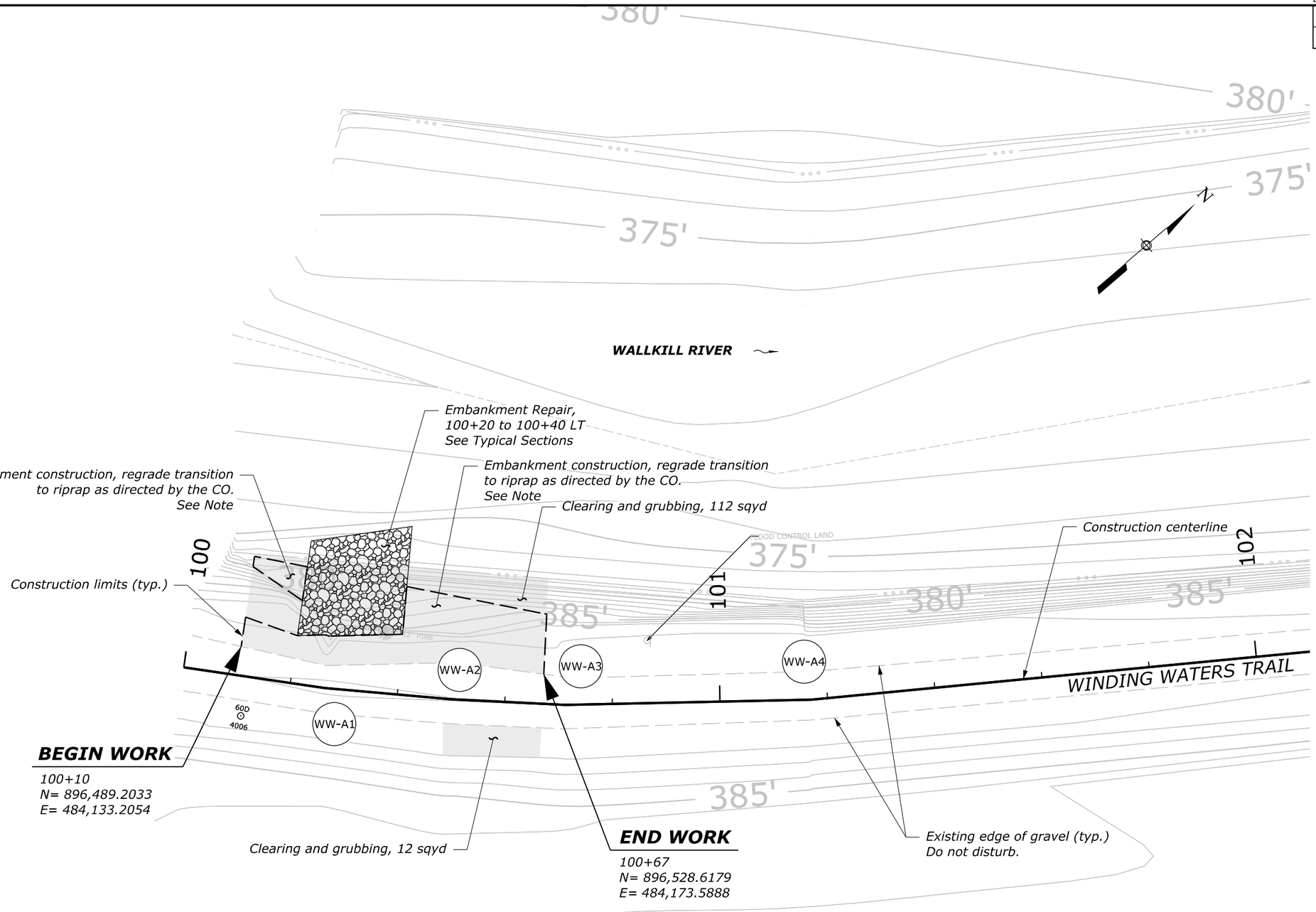
SUMMARIES AND SCHEDULES

SCHEDULE C: DAGMAR DALE NATURE TRAIL

2 September 2026 10:00 AM \\fhf15fileserve.flhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\C01-CXX_NJ-WARI_2019-1(1).dgn [Drainage-CON Signs-Miscellaneous Sch.C (C12)]

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D01



NOTE:

Do not perform embankment construction within the limits of the Wallkill River.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN

SCHEDULE A: WINDING WATERS TRAIL SITE #3
100+10 TO 100+67

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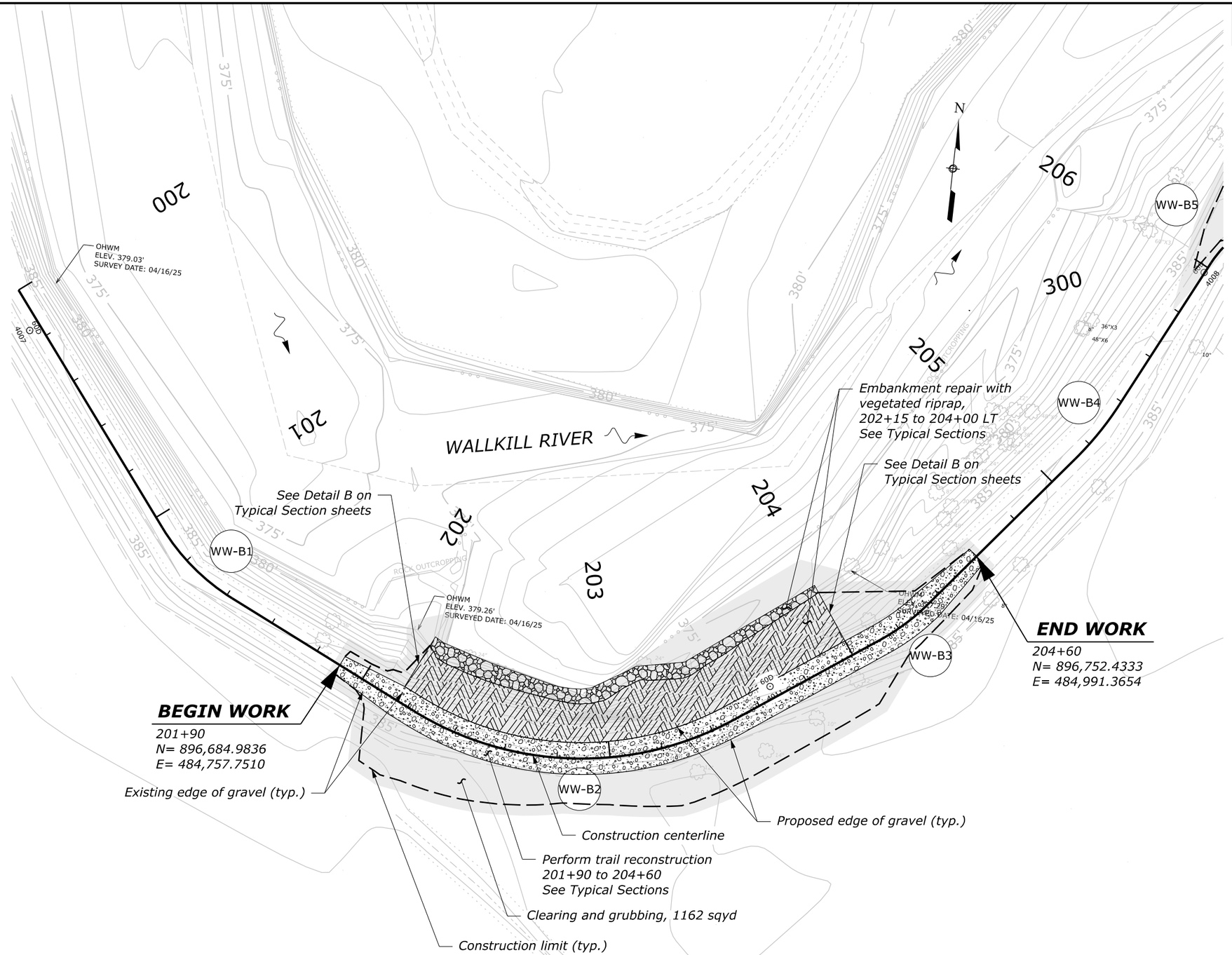
3 September 2026 8:43 AM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D02



NO.	DATE	BY	REVISIONS

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OFFICE OF FEDERAL LANDS HIGHWAY

0 20 40
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN

SCHEDULE A: WINDING WATERS TRAIL SITE #240

201+90 TO 204+60

SHEET 2 OF 5

2 September 2026 10:36 AM
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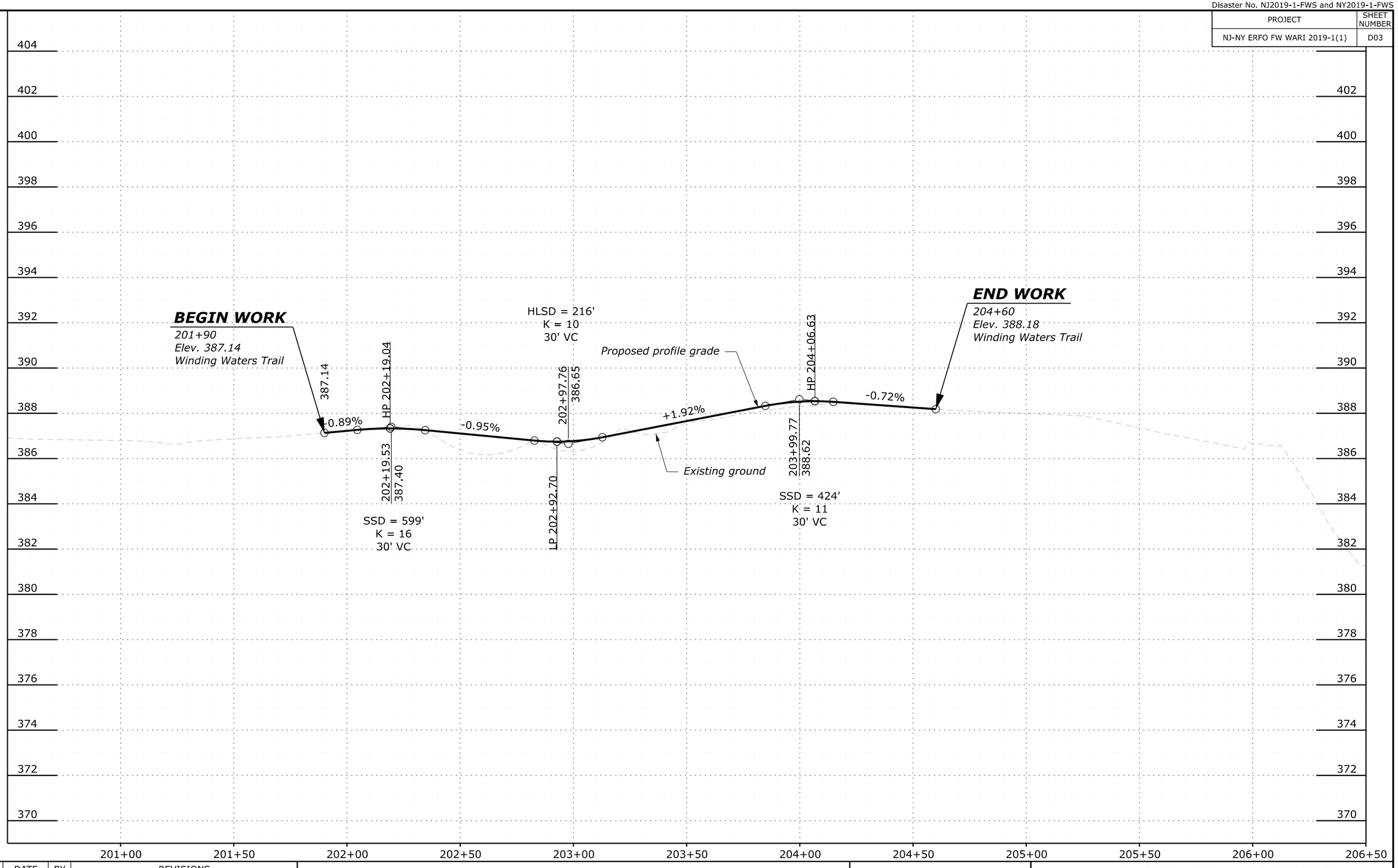
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D03



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 20 40
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PROFILE

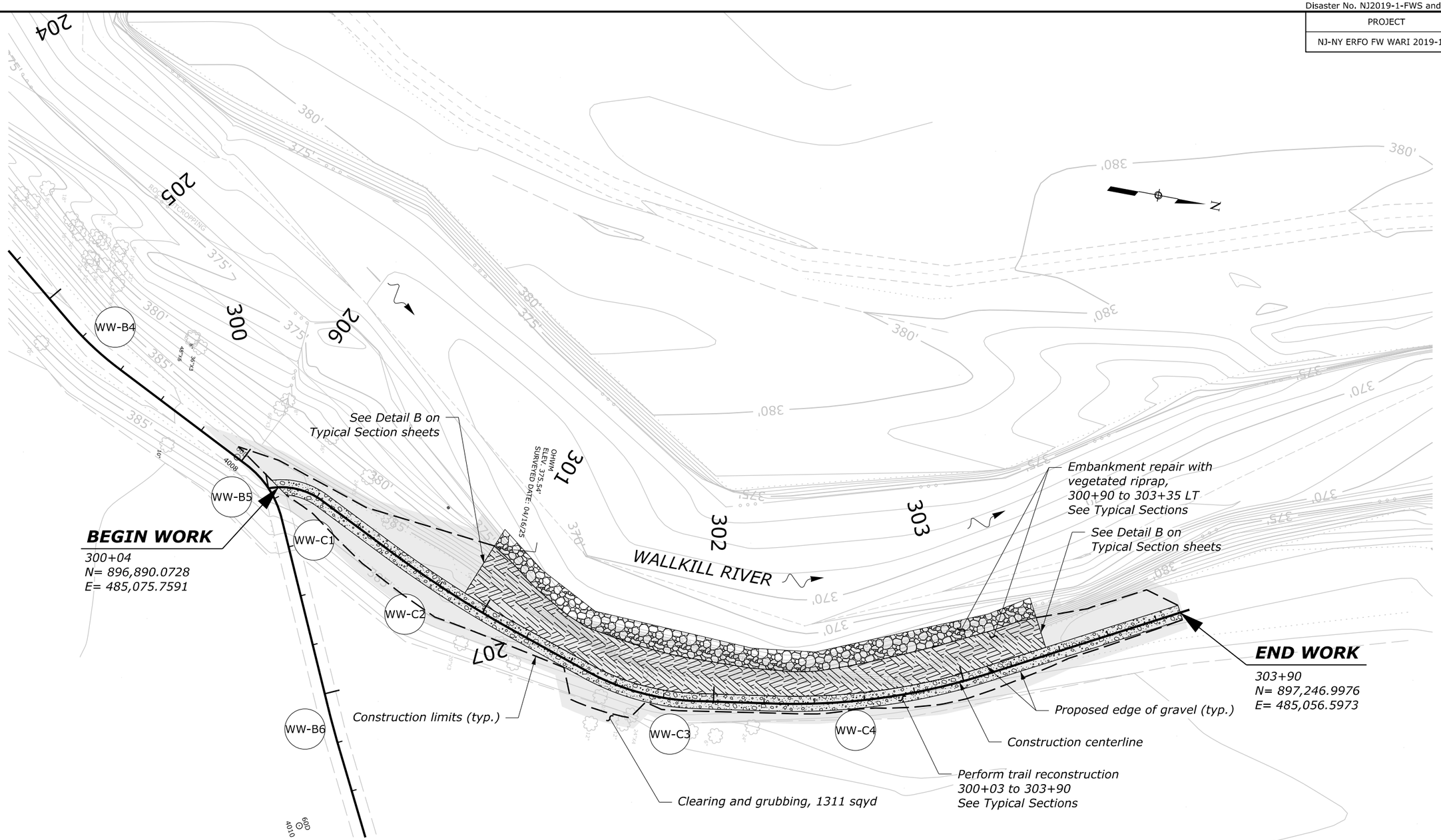
SCHEDULE A: WINDING WATERS TRAIL SITE #240
201+90 TO 204+60

SHEET 3 OF 5

3 August 2026 10:38 AM
\\fhfil5fileserve.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\D08-D11-NY-WARI_2019-1(1)Trail_PLAN - Sites 240 and 324.dgn [Pro_Site #240 - Profile_Site#240]

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D04



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 20 40
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN

SCHEDULE A: WINDING WATERS TRAIL SITE #324

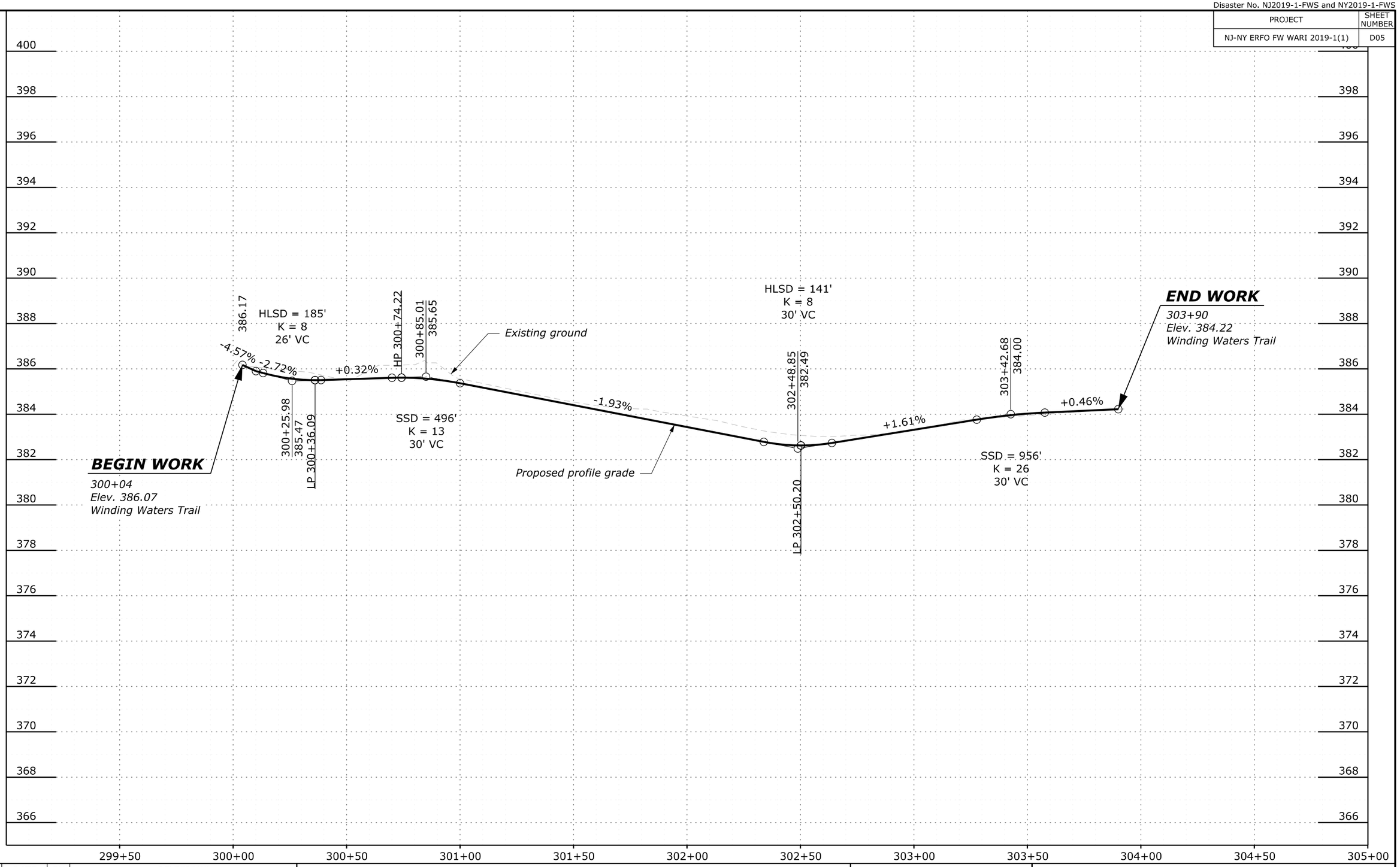
300+04 TO 303+90

SHEET 4 OF 5

2 September 2026 10:41 AM \\fhf15fileserve.flhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\D08-D11-NY-WARI_2019-1(1)Trail_PLAN - Sites 240 and 324.dgn [Site #324 (Sheet)]

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D05



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 20 40
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PROFILE

SCHEDULE A: WINDING WATERS TRAIL SITE #324
300+04 TO 303+90

SHEET 5 OF 5

3 August 2026 10:49 AM

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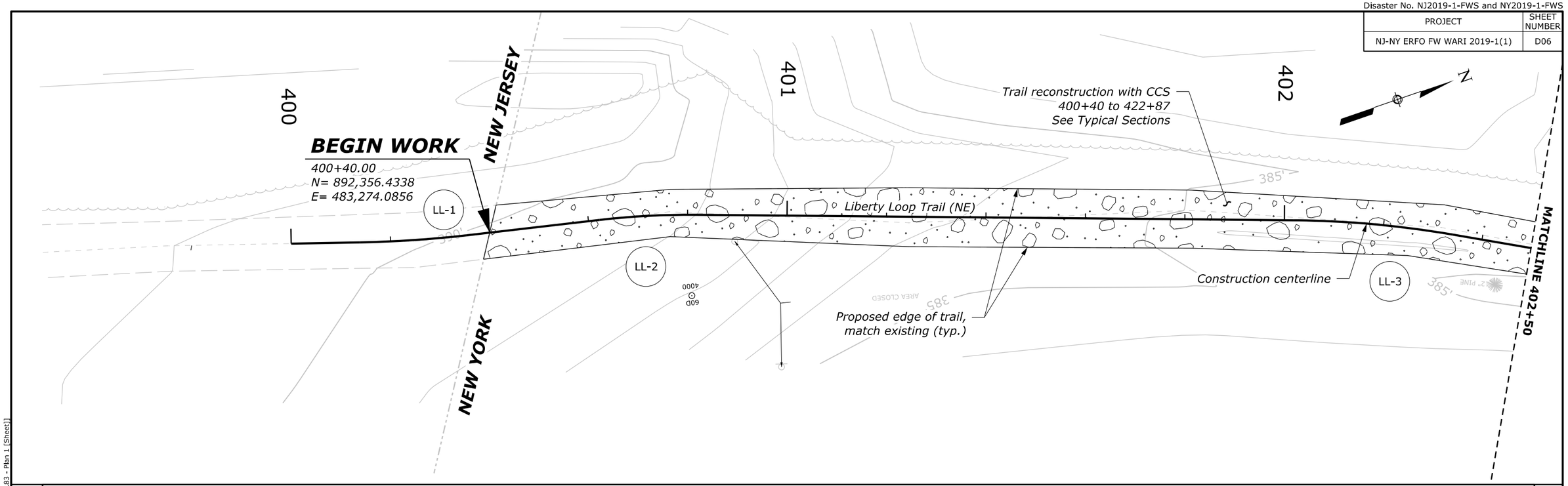
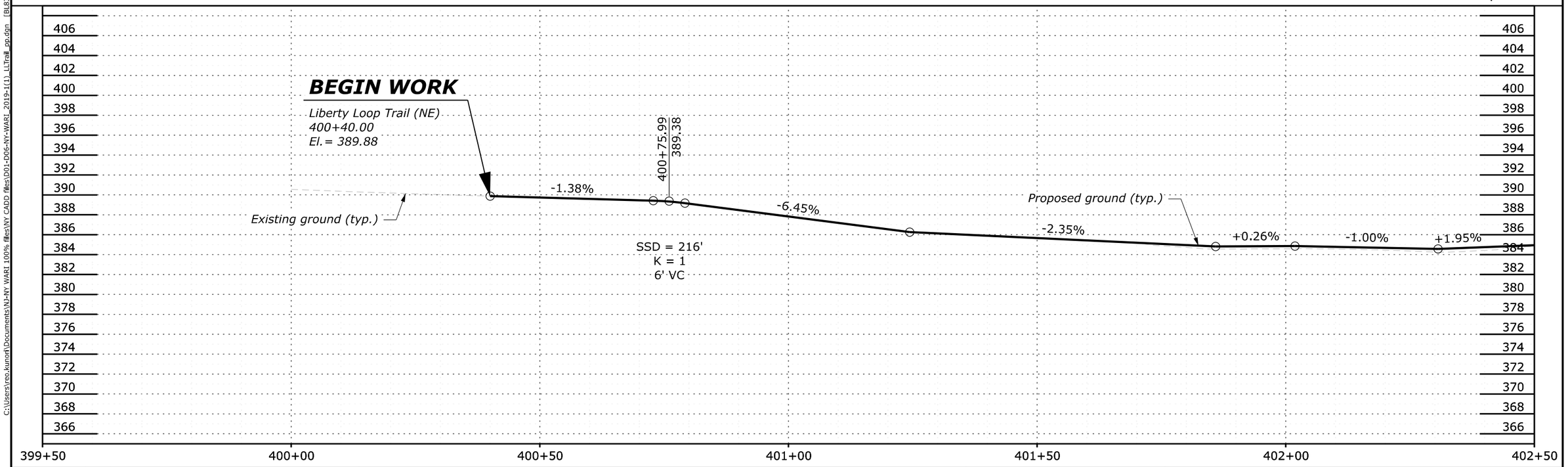
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D06





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

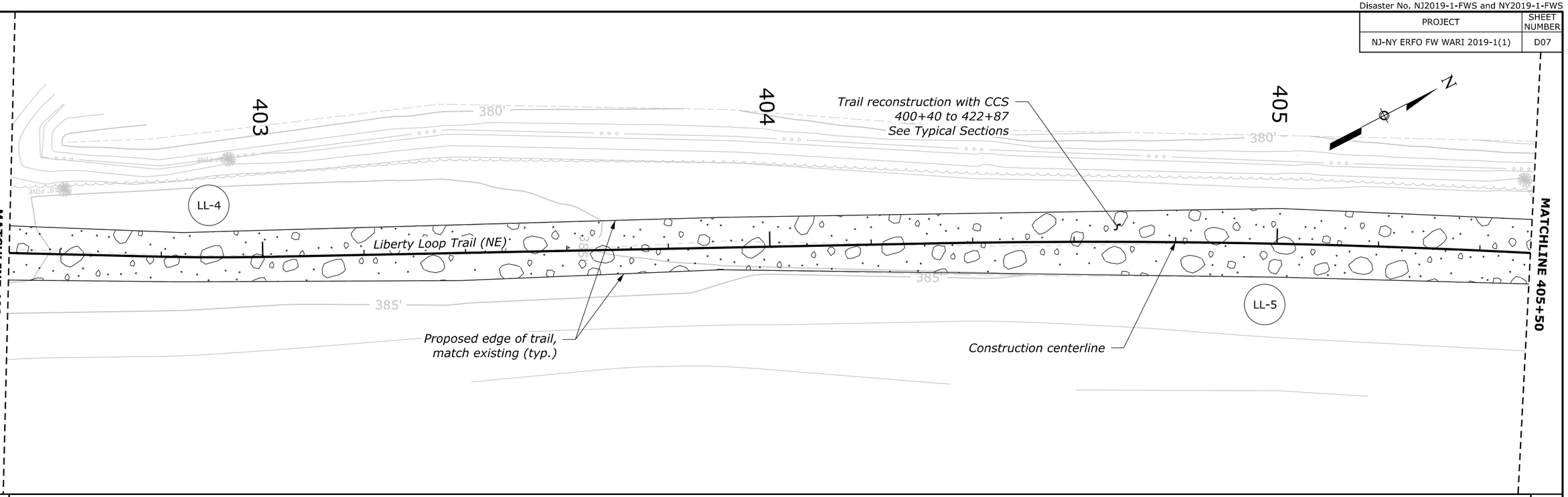
400+40 TO 402+50

SHEET 1 OF 12

C:\Users\rea.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\D01-D06-NY-WARI_2019-1(1)_LLTrail_pp.dgn [BL83 - Plan 1 [Sheet]]
2 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D07



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

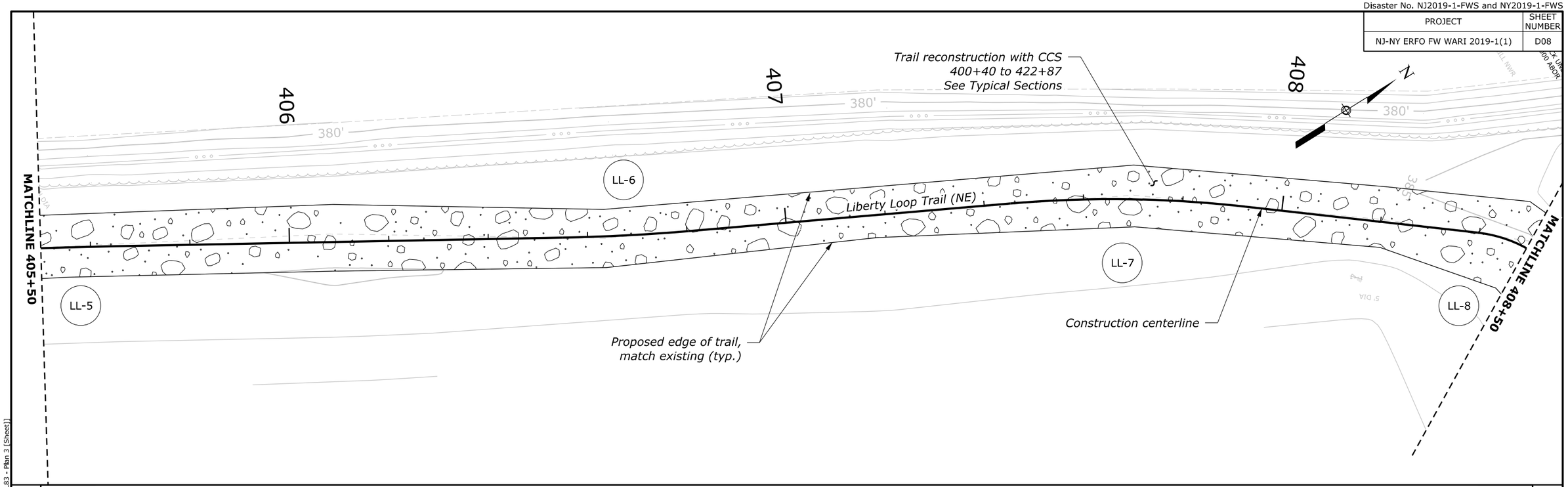
402+50 TO 405+50

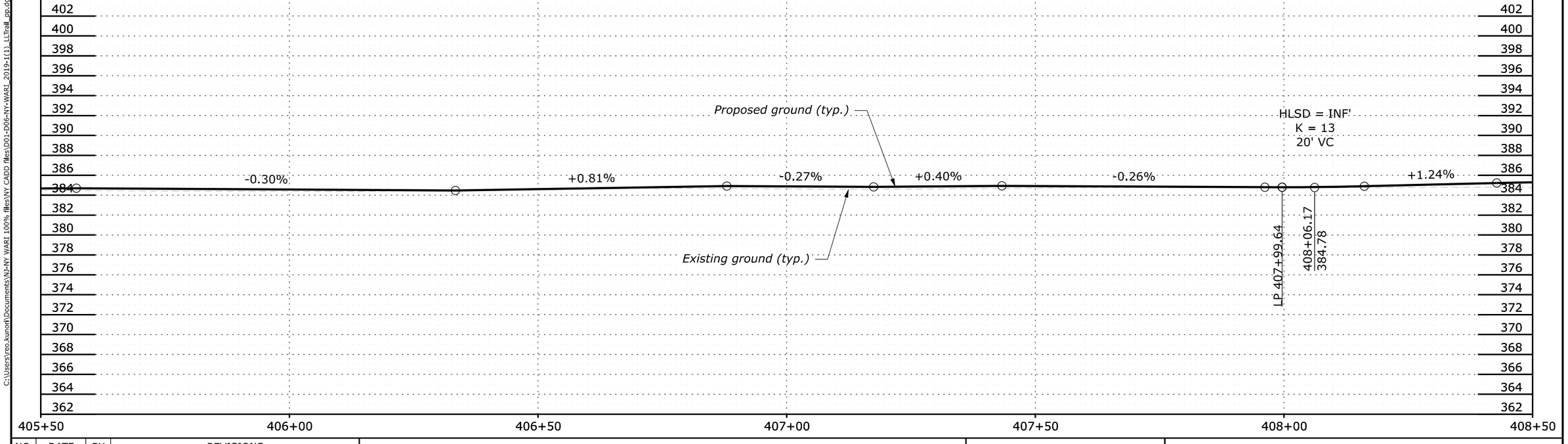
SHEET 2 OF 12

C:\Users\rea.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\D01-D06-NY-WARI_2019-1(1)_LLTrail_pp.dgn [BL83 - Plan 2 [Sheet]]
2 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D08





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

405+50 TO 408+50

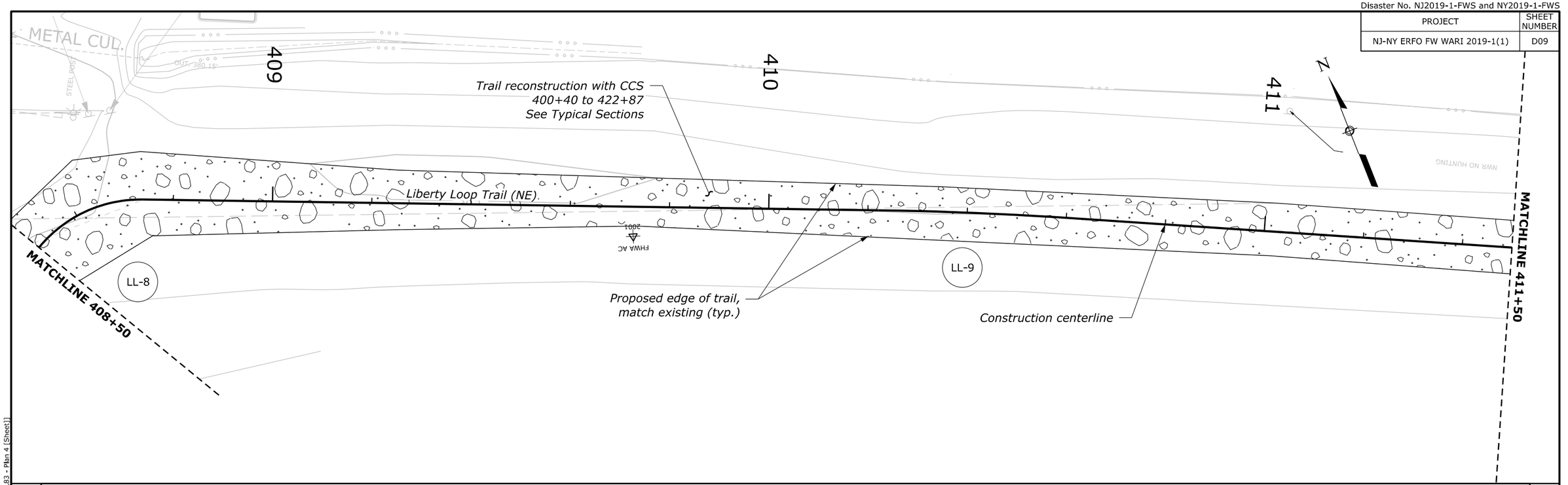
SHEET 3 OF 12

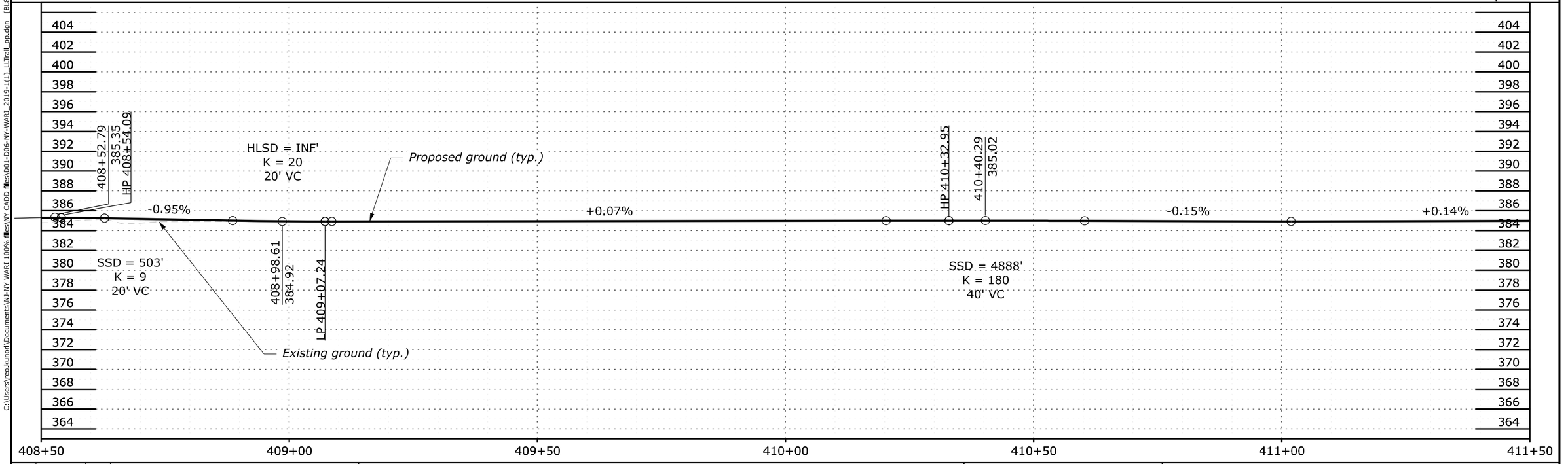
C:\Users\vreo.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\D01-D06-NY-WARI_2019-1(1)_LLTrail_pp.dgn [BL83 - Plan 3 [Sheet]]

2 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D09





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

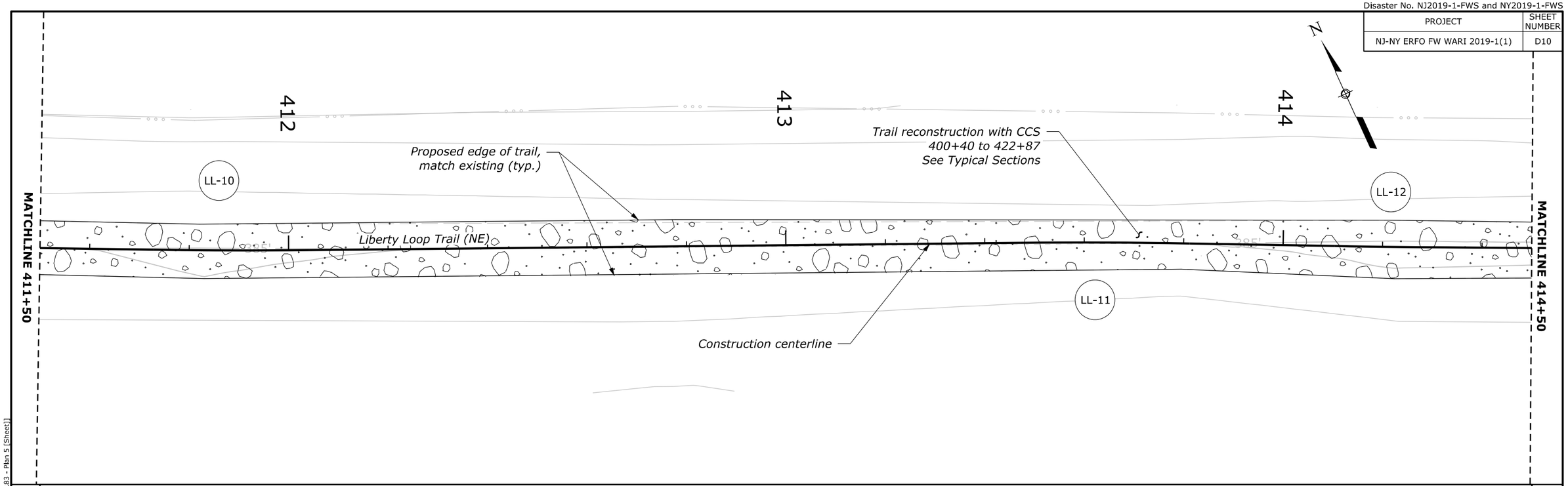
408+50 TO 411+50

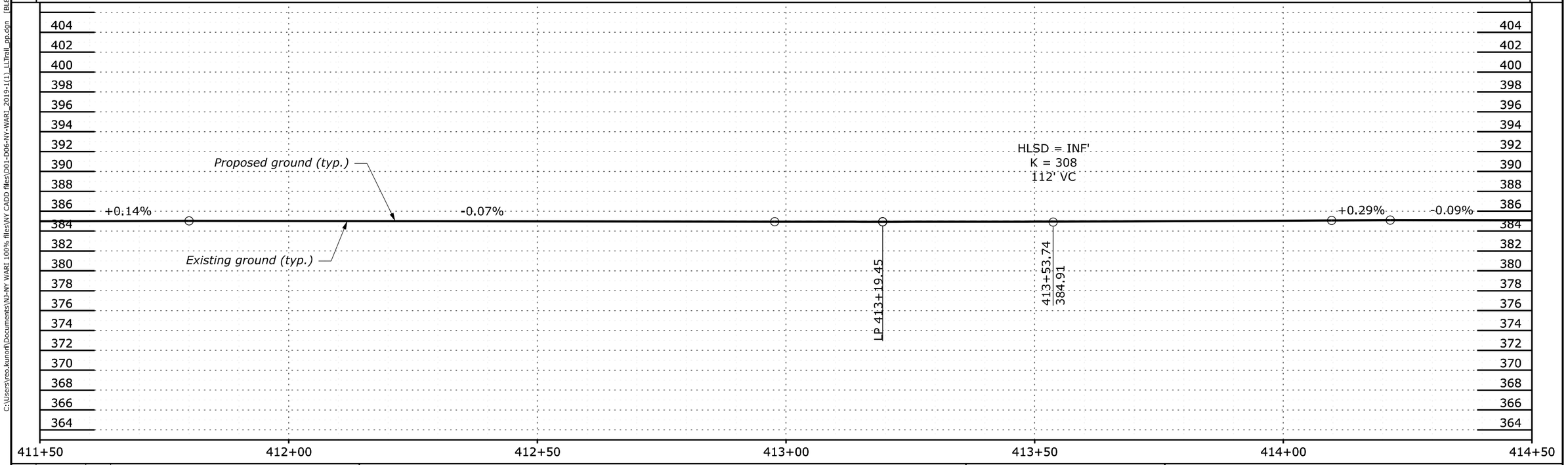
SHEET 4 OF 12

C:\Users\rea.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\D01-D06-NY-WARI_2019-1(1)_LLTrail_pp.dgn [BL83 - Plan 4 [Sheet]]
2 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D10





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

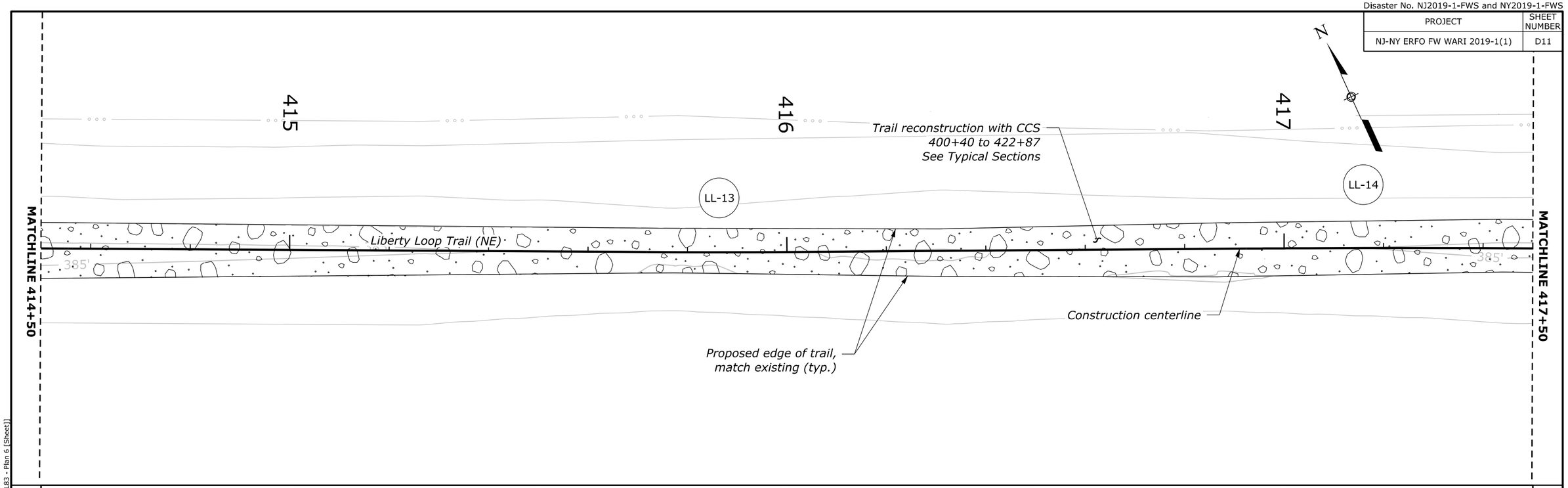
411+50 TO 414+50

SHEET 5 OF 12

C:\Users\rea.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\D01-D06-NY-WARI_2019-1(1)_LLTrail_pp.dgn [BL83 - Plan 5 [Sheet]]
2 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D11



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

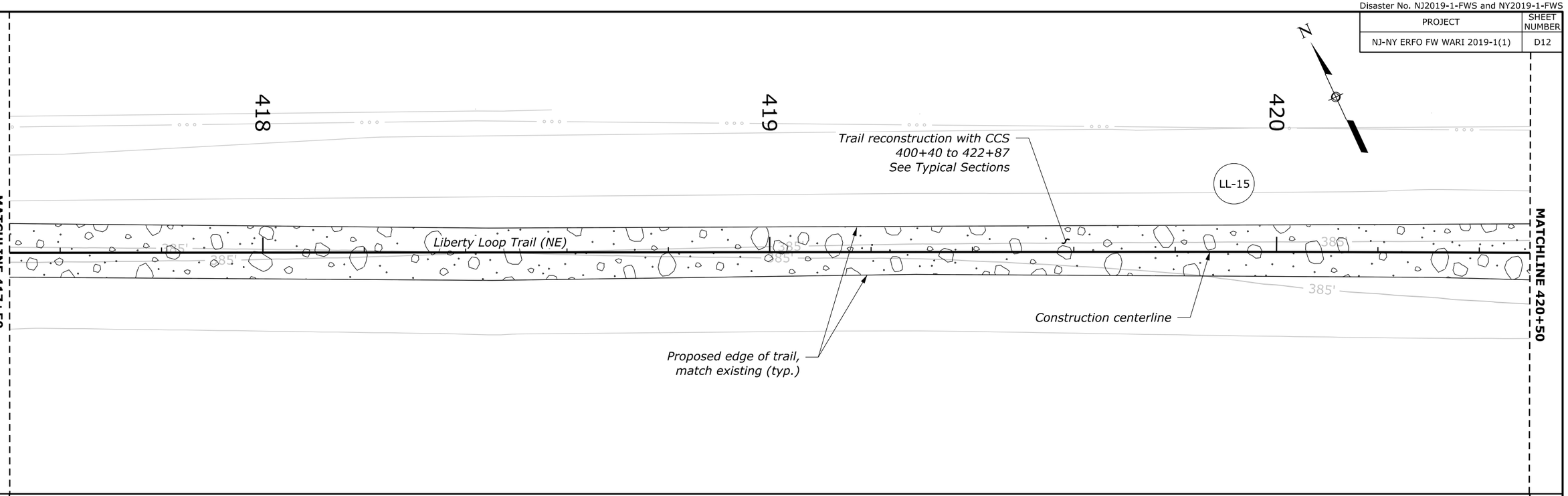
414+50 TO 417+50

SHEET 6 OF 12

C:\Users\rea.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\D01-D06-NY-WARI_2019-1(1)_LLTrail_pp.dgn [BL83 - Plan 6 [Sheet]]
2 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D12



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

417+50 TO 420+50

SHEET 7 OF 12

2 August 2026 7:50 PM
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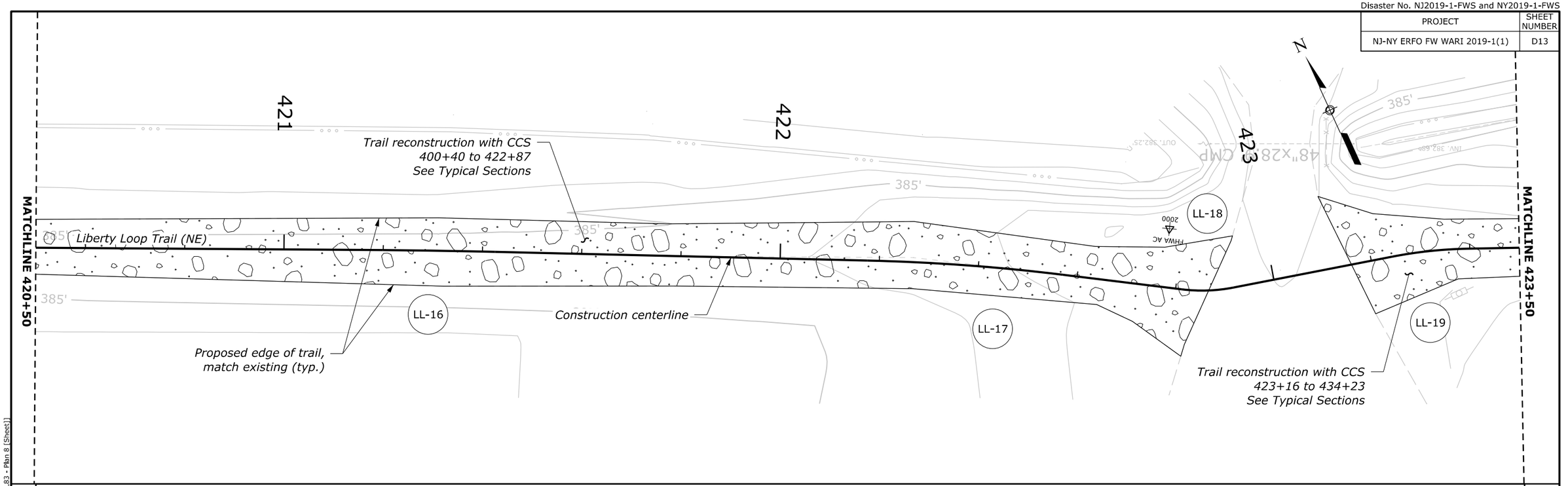
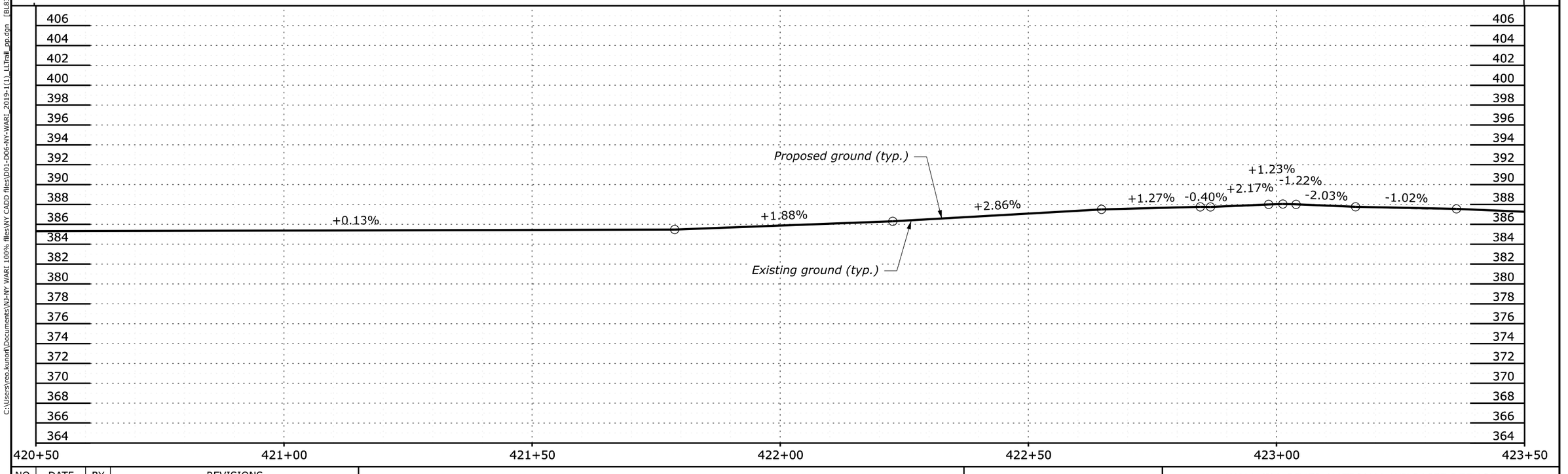
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D13





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

420+50 TO 423+50

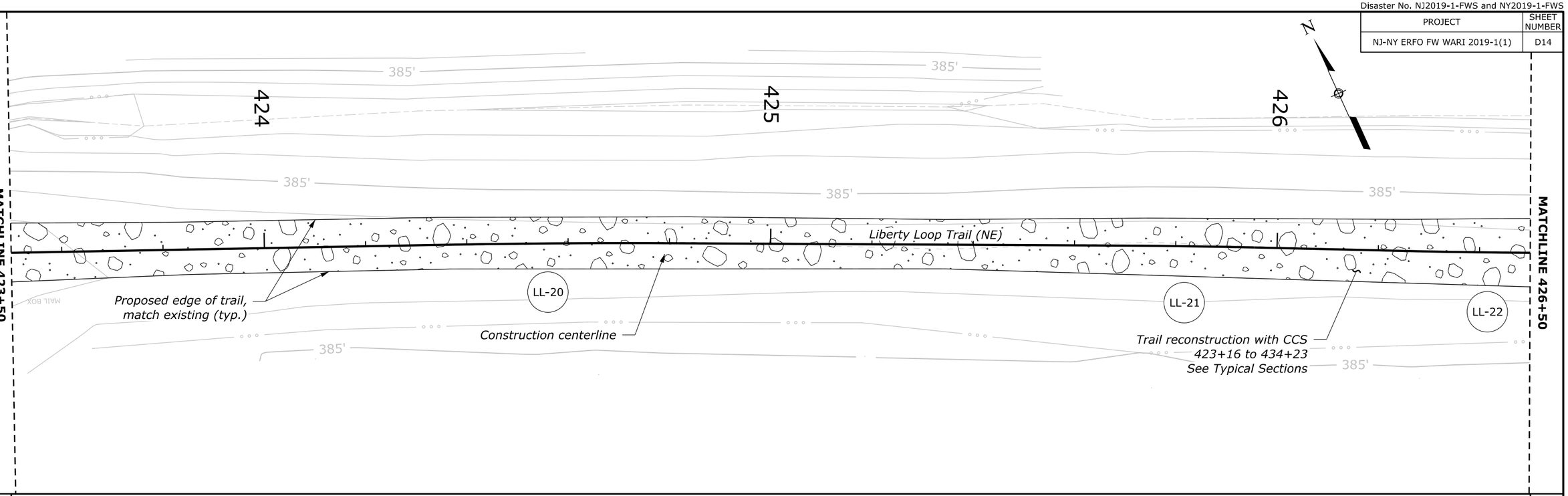
SHEET 8 OF 12

C:\Users\vreo.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\D01-D06-NY-WARI_2019-1(1)_LLTrail_pp.dgn [BL83 - Plan 8 [Sheet]]

2 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D14



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

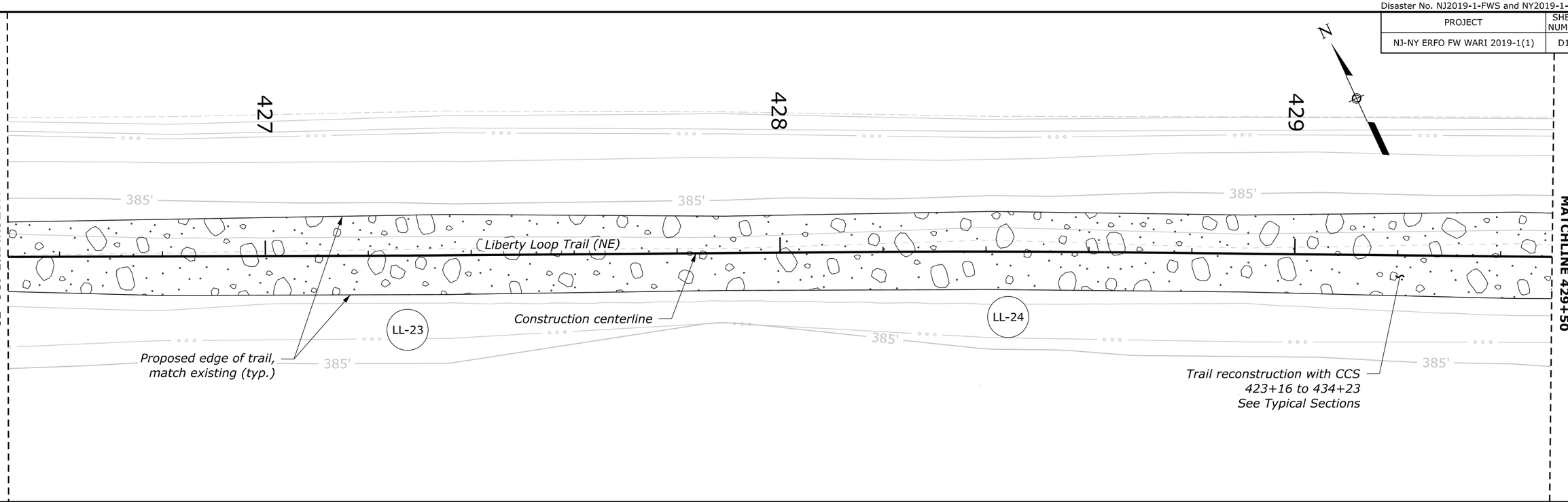
SCHEDULE A: LIBERTY LOOP TRAIL (NE)

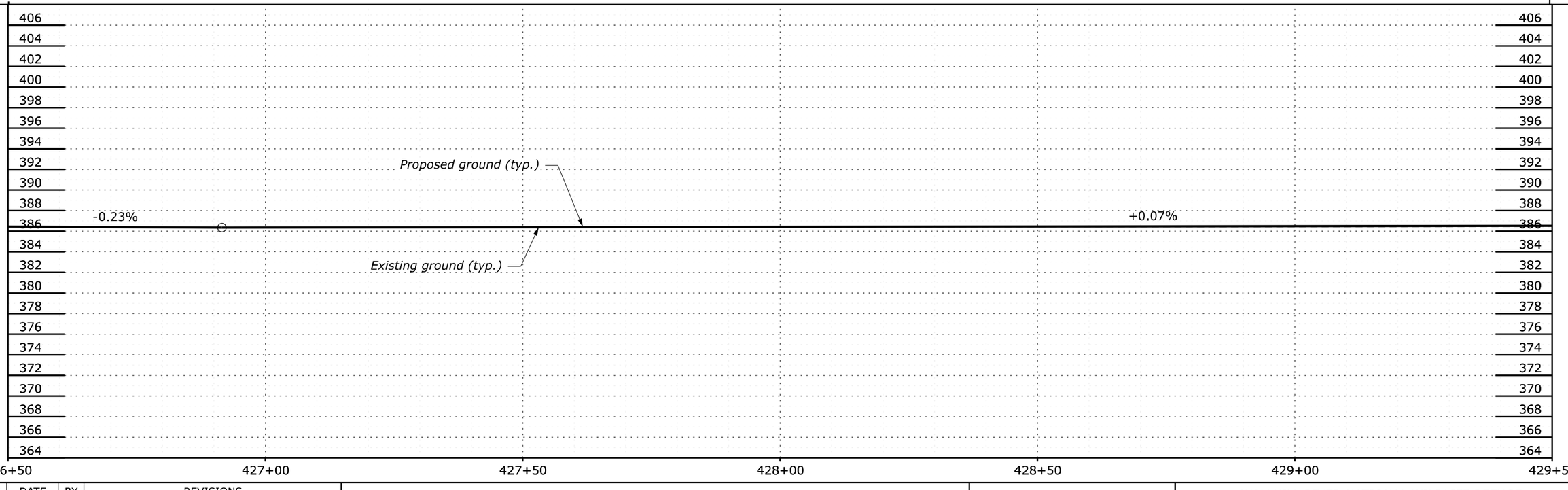
423+50 TO 426+50

C:\Users\vreo.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\D01-D06-NY-WARI_2019-1(1)_LLTrail_pp.dgn [BL83 - Plan 9 [Sheet]]
2 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D15





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

426+50 TO 429+50

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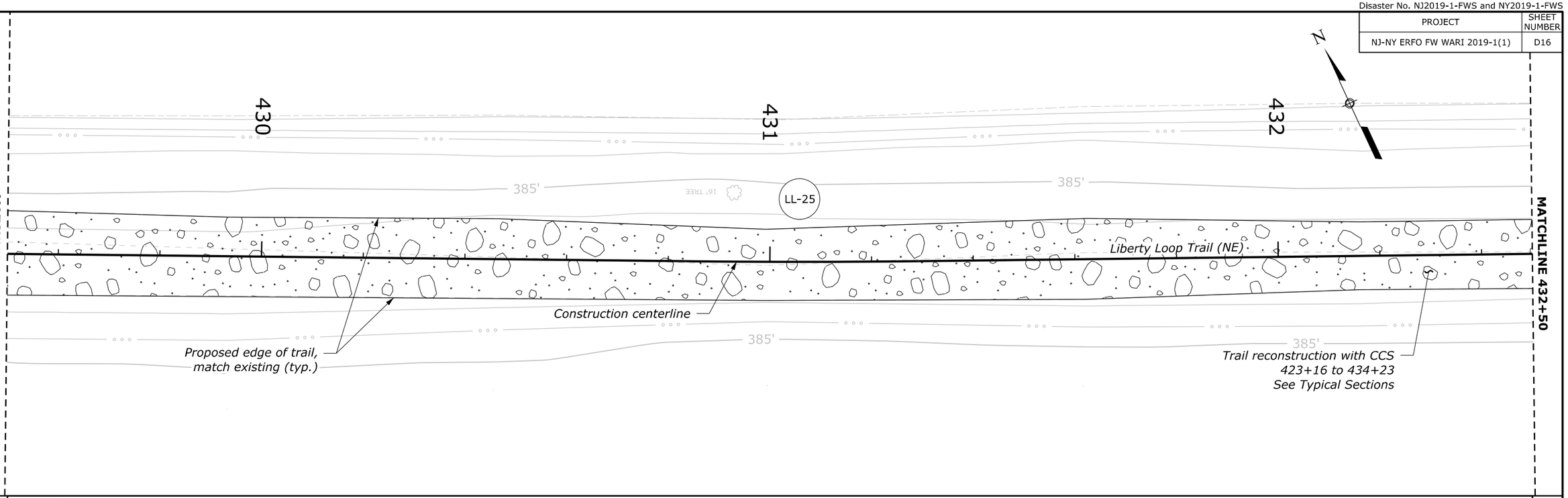
2 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D16



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

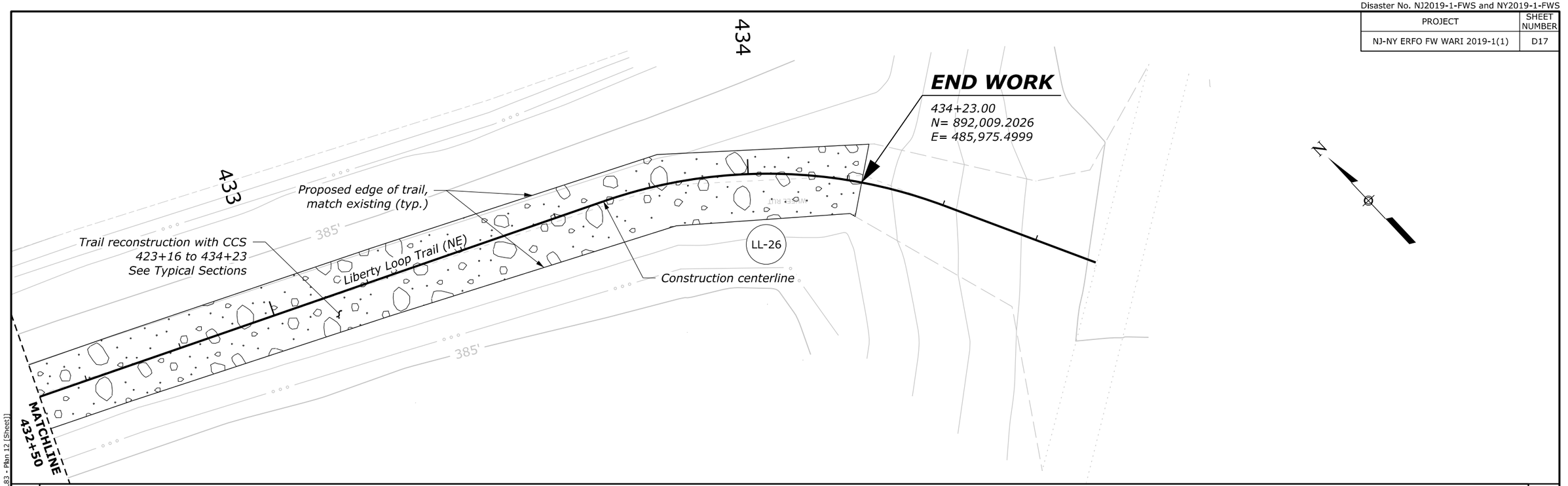
429+50 TO 432+50

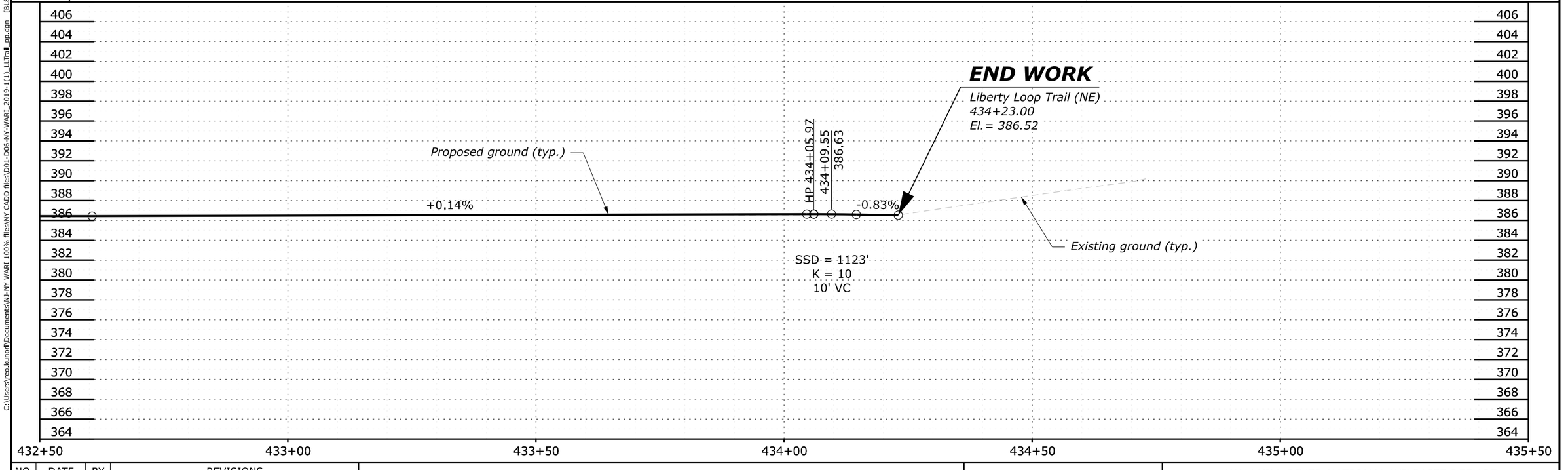
SHEET 11 OF 12

2 August 2026 7:50 PM C:\Users\rex.kunoh\Documents\NJ-NY WARI 100% files\NY CADD files\D01-D06-NY-WARI_2019-1(1)_LLTrail_pp.dgn [BL83 - Plan 11 [Sheet]]

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	D17





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

432+50 TO 434+23

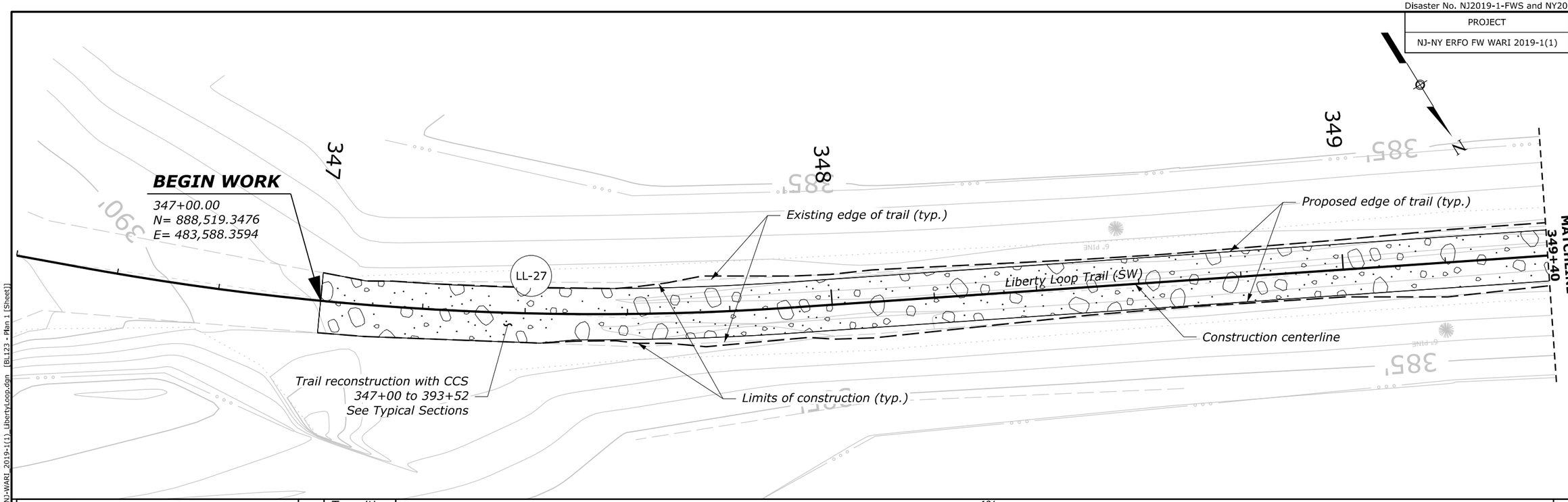
SHEET 12 OF 12

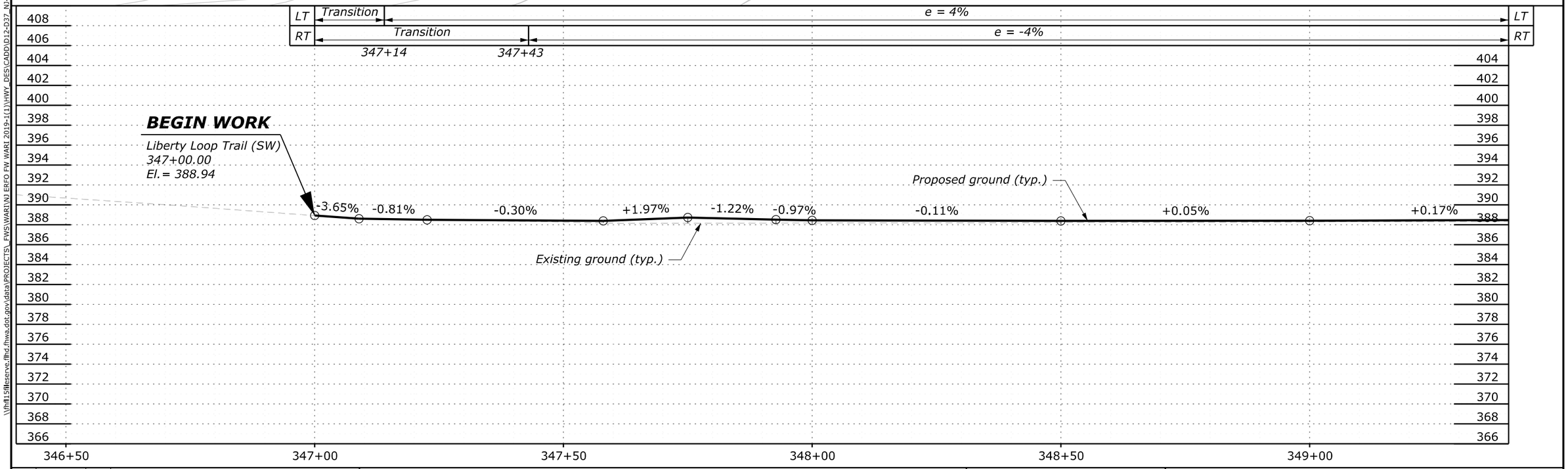
C:\Users\vreo.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\D01-D06-NY-WARI_2019-1(1)_LLTrail_pp.dgn [BL83 - Plan 12 (Sheet)]

2 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E01





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

347+00 TO 349+40

SHEET 1 OF 17

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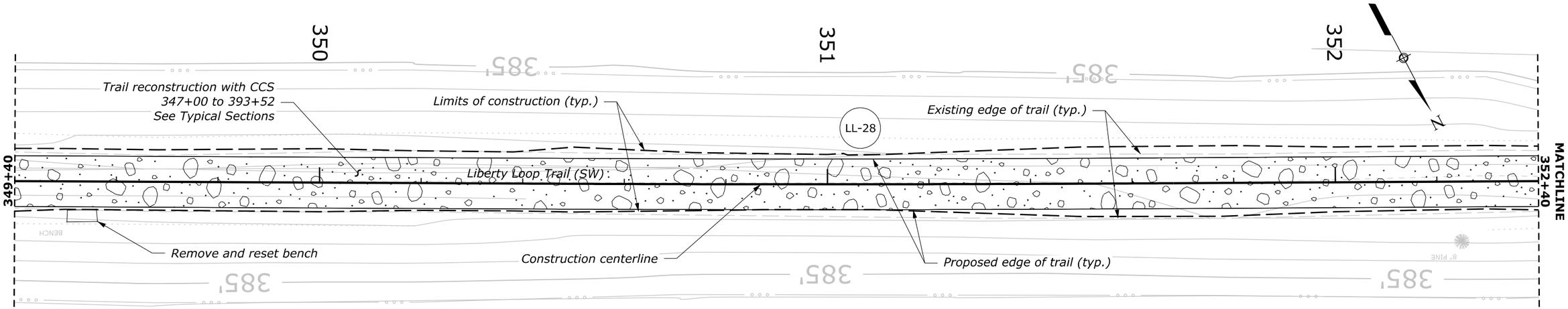
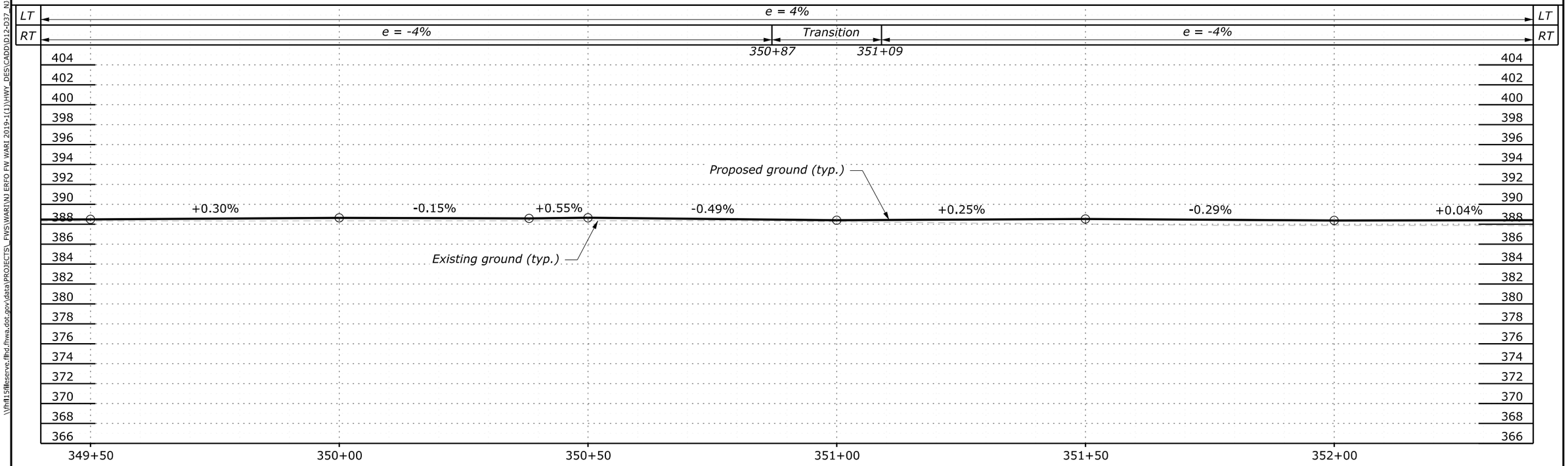
3 August 2026 2:18 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E02





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

349+40 TO 352+40

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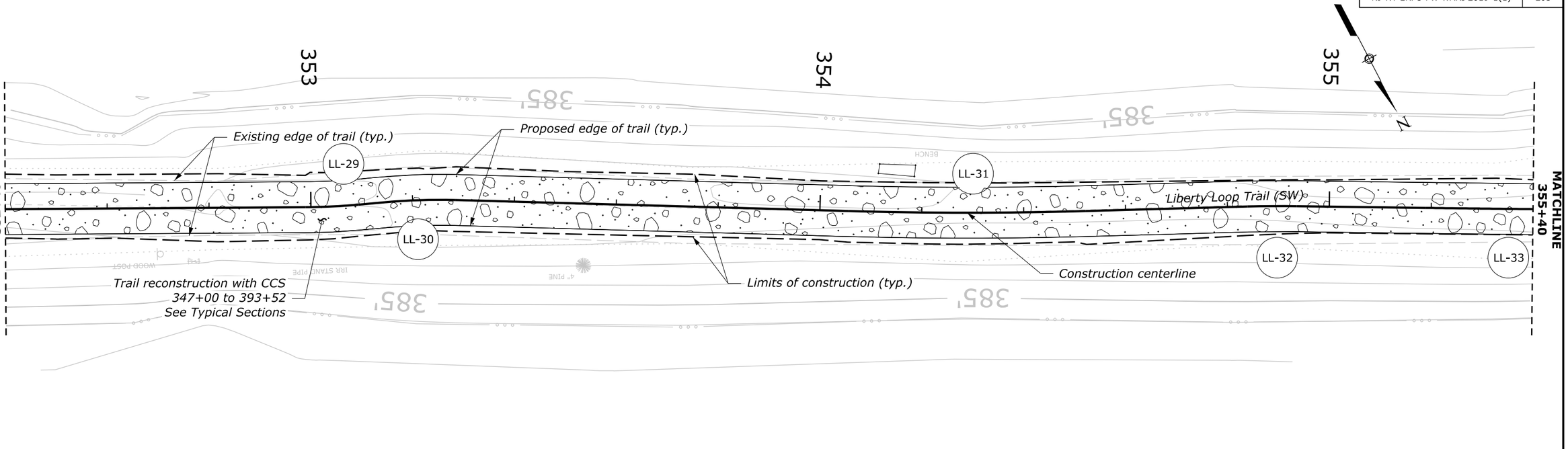
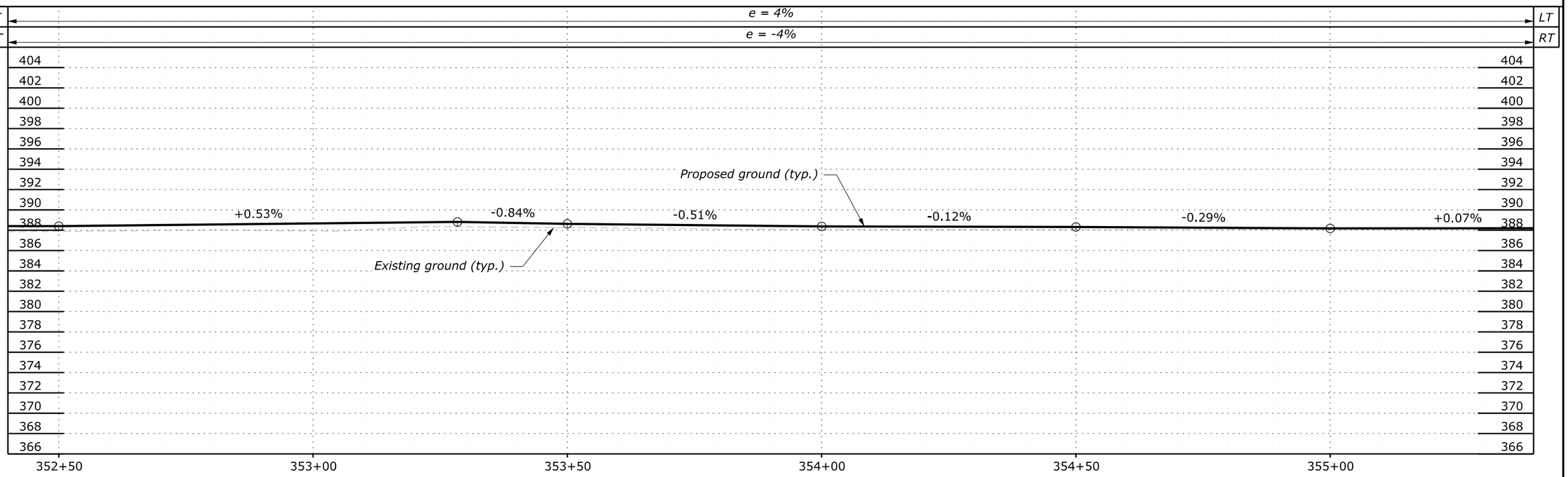
3 August 2026 2:18 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E03





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
352+40 TO 355+40

SHEET 3 OF 17

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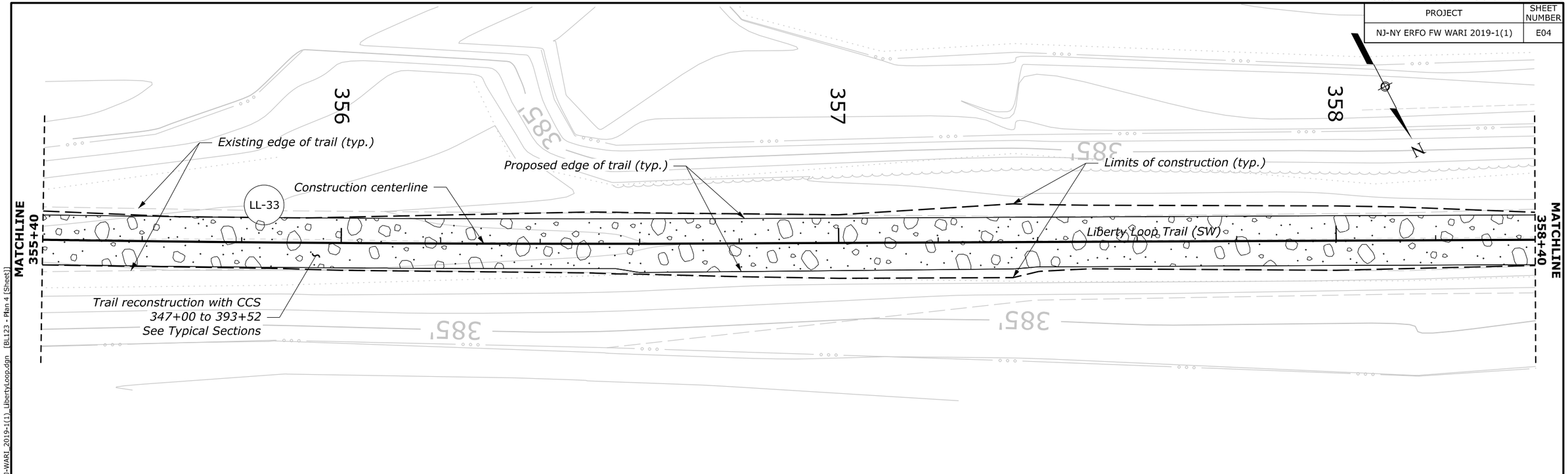
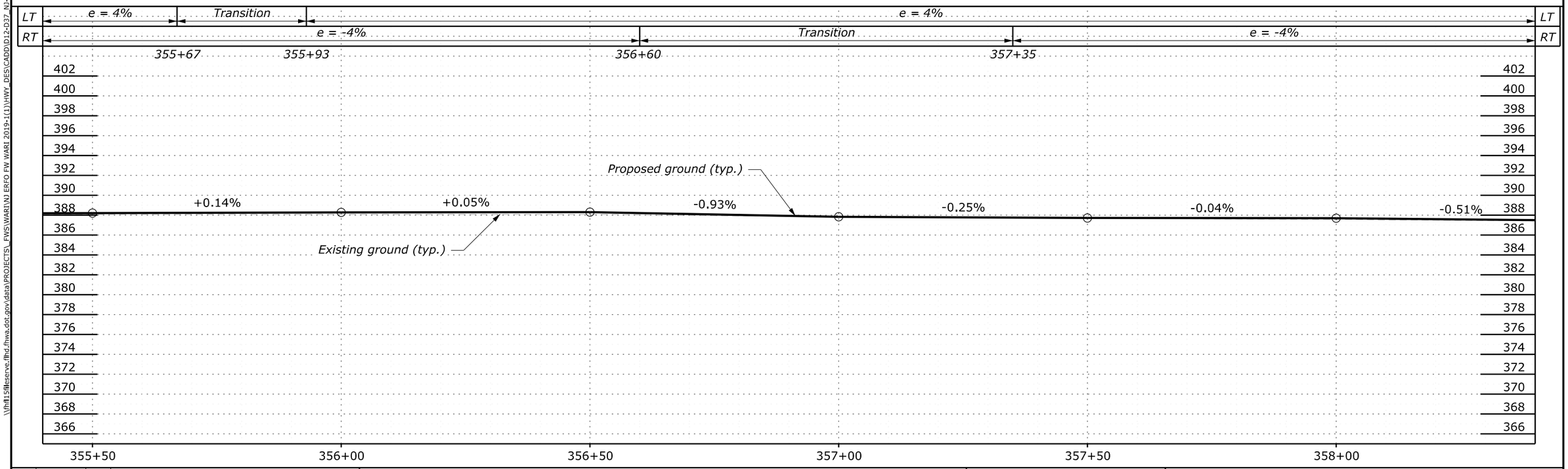
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E04





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

355+40 TO 358+40

SHEET 4 OF 17

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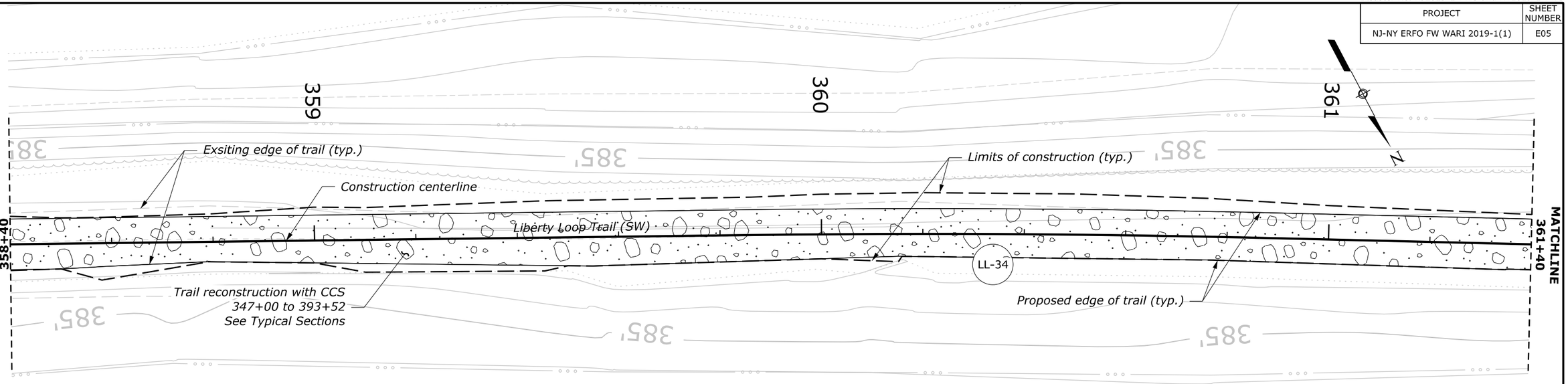
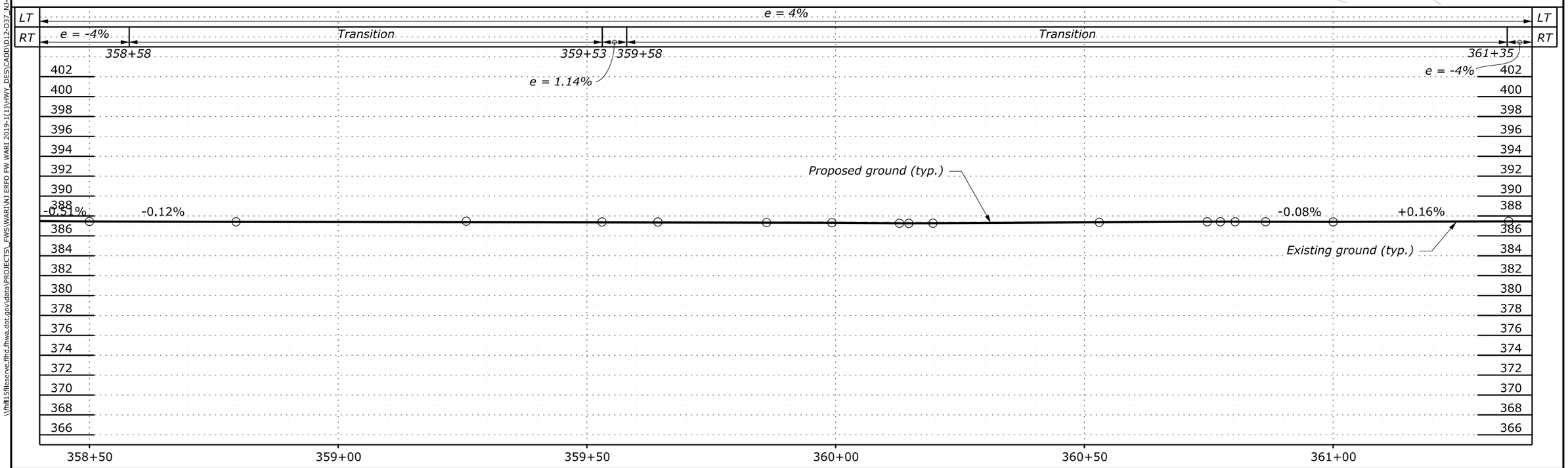
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E05





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
358+40 TO 361+40

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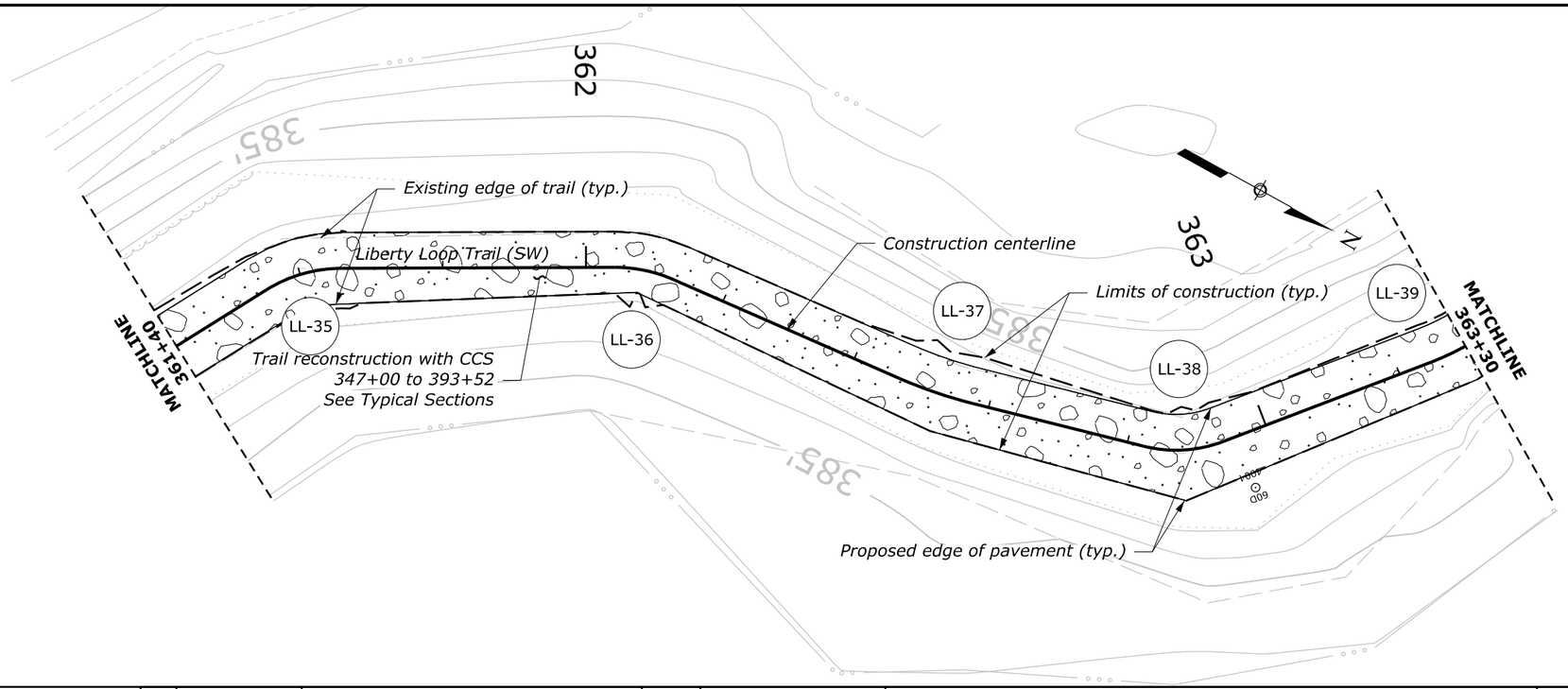
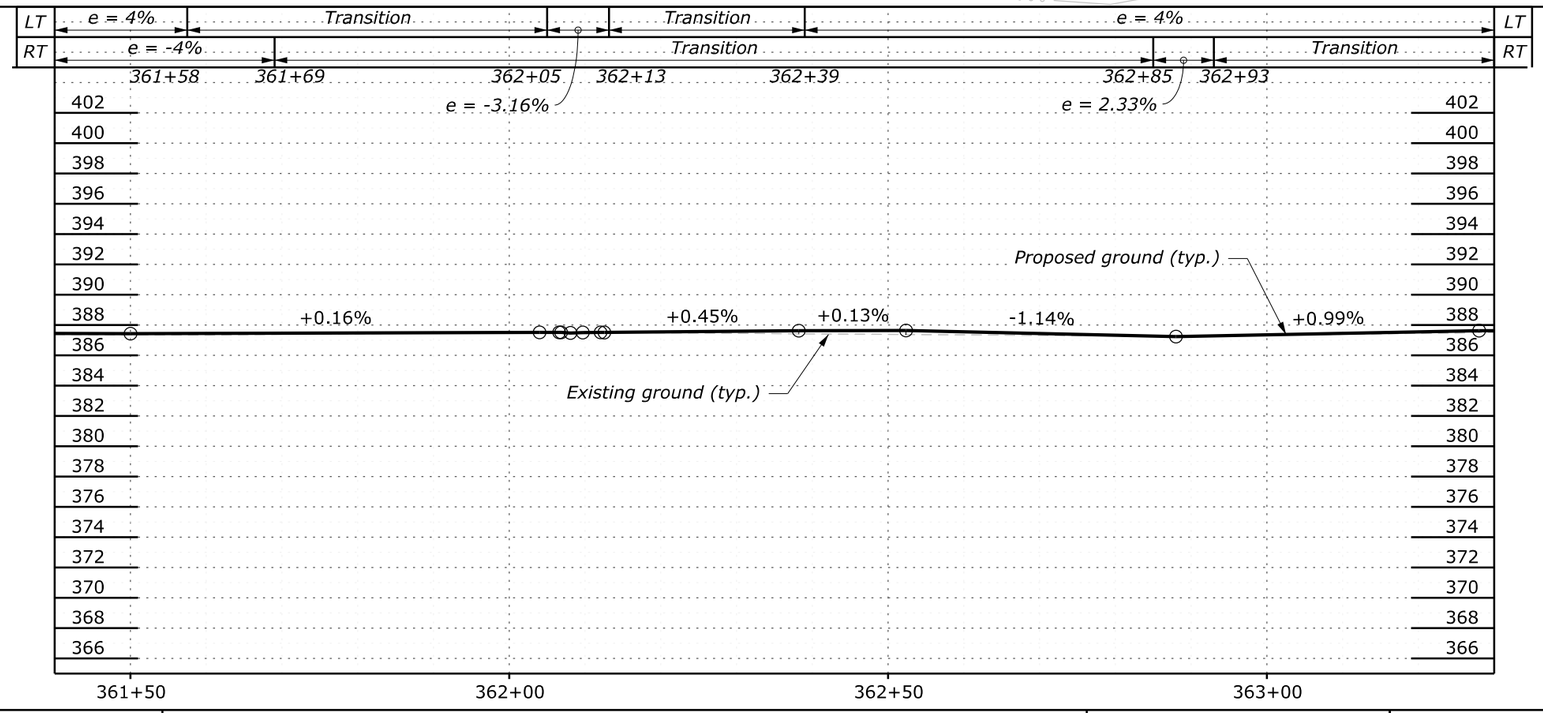
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E06





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

361+40 TO 363+30

SHEET 6 OF 17

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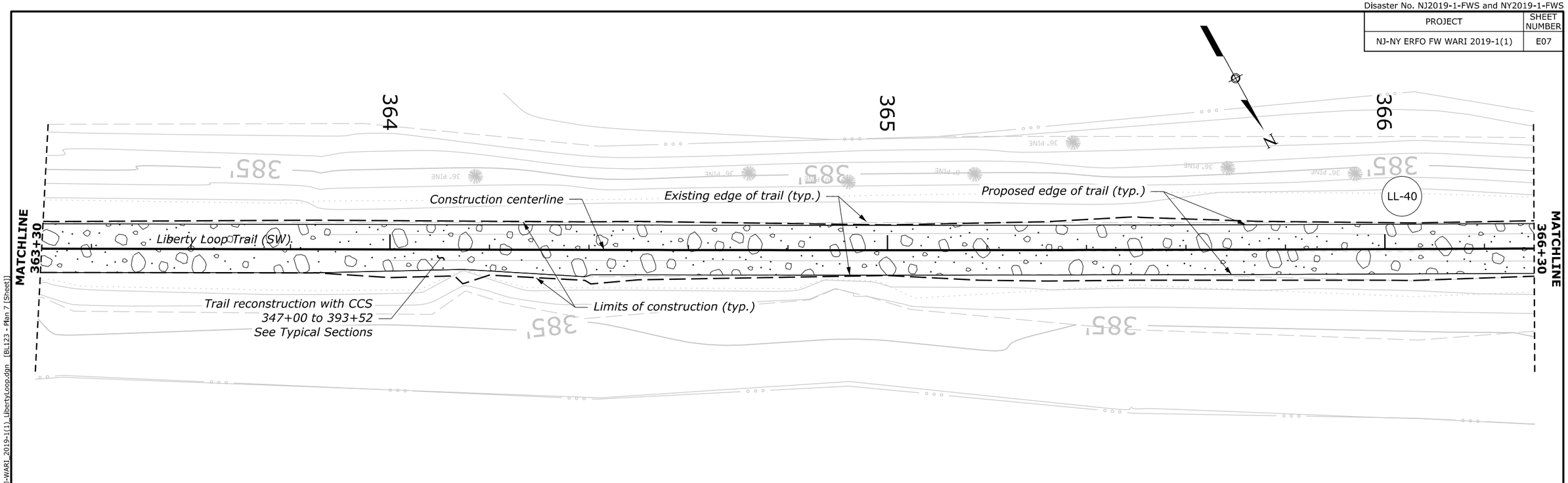
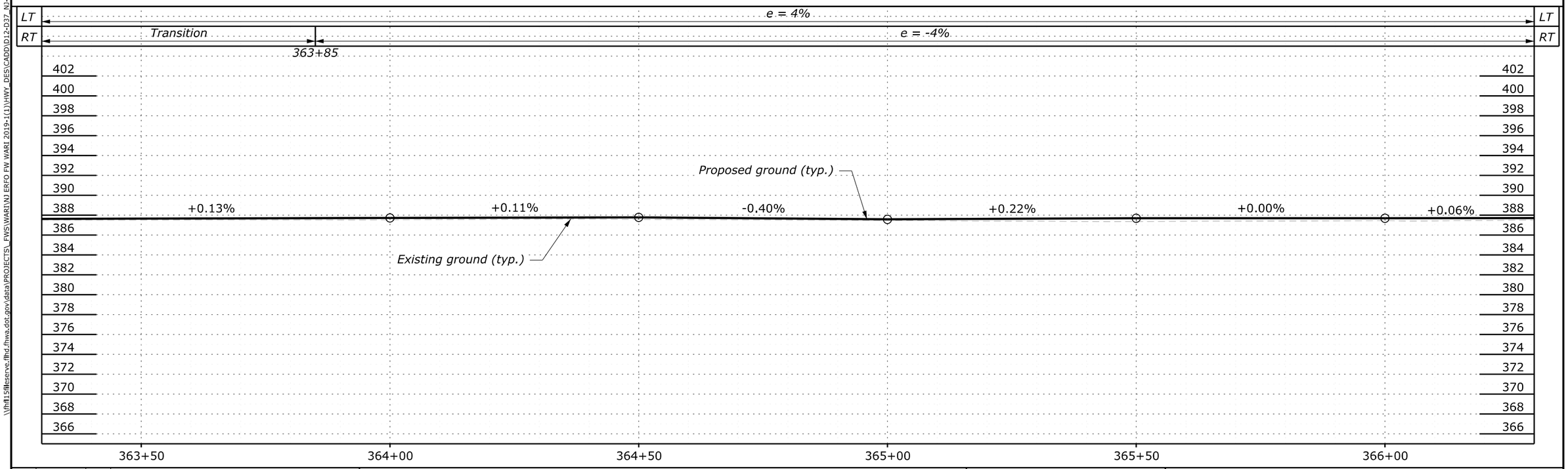
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E07





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

363+30 TO 366+30

SHEET 7 OF 17

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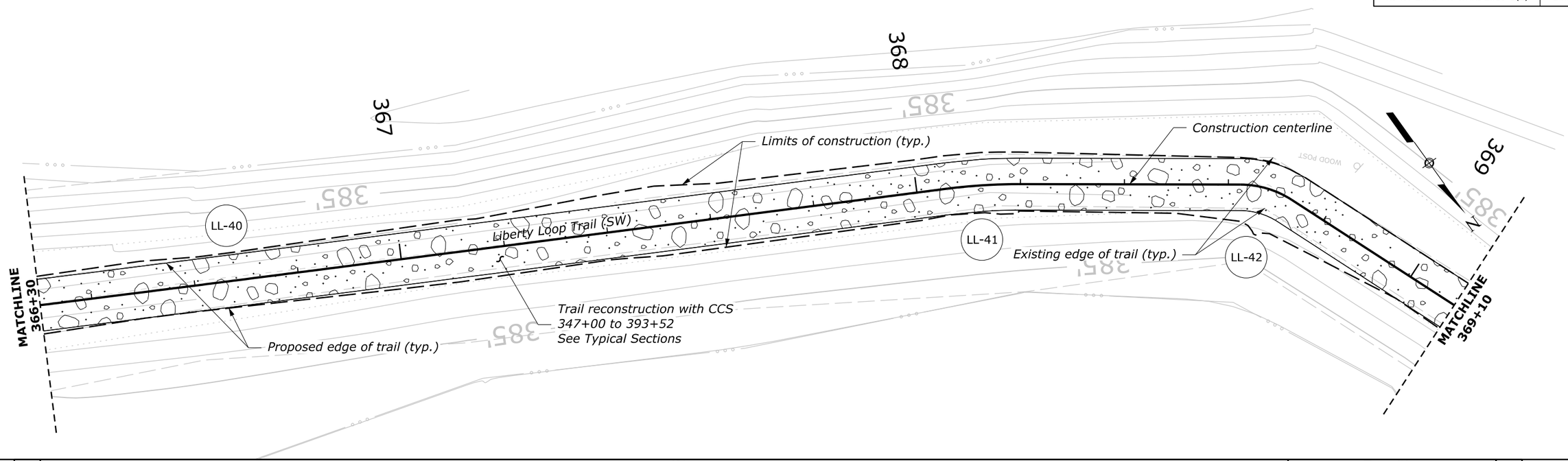
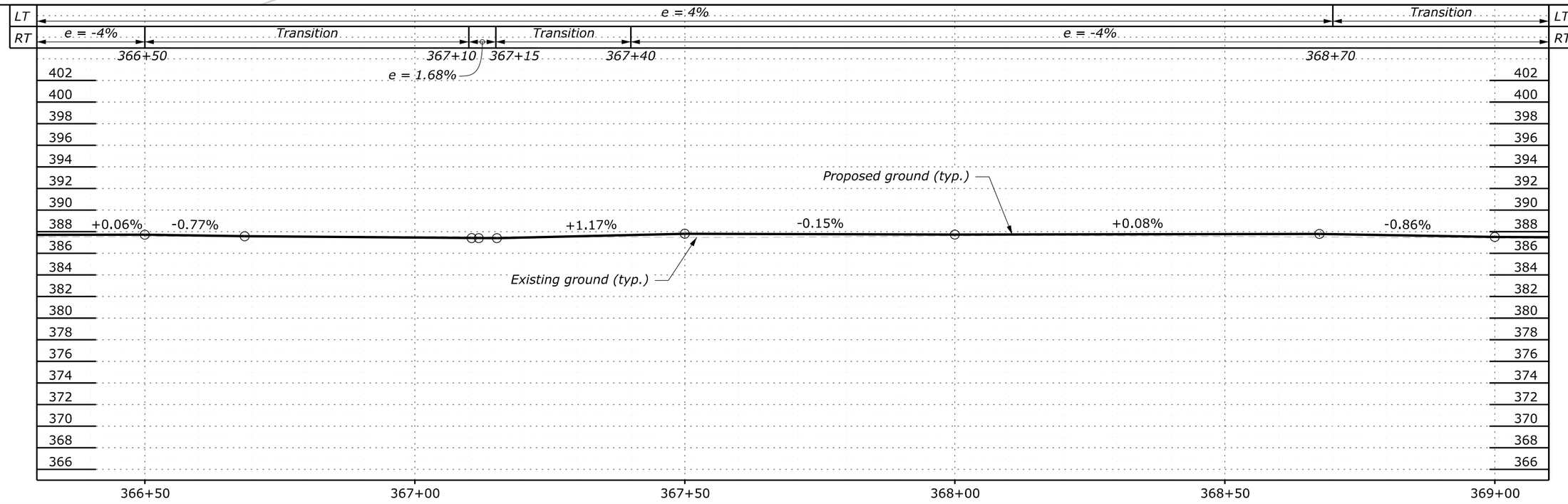
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E08





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
366+30 TO 369+10

SHEET 8 OF 17

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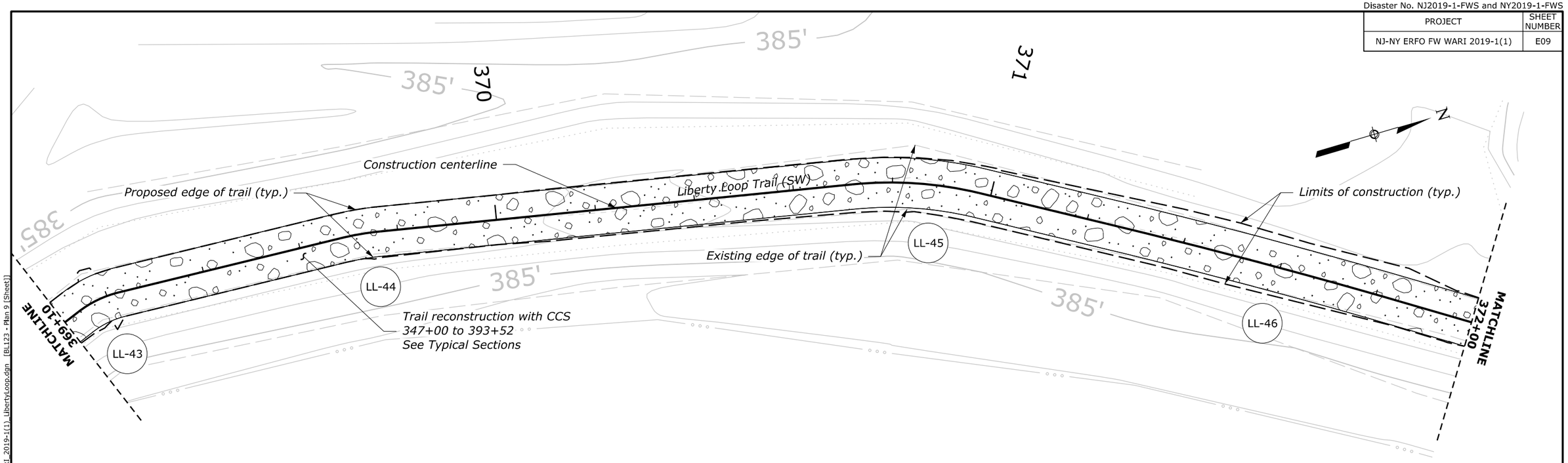
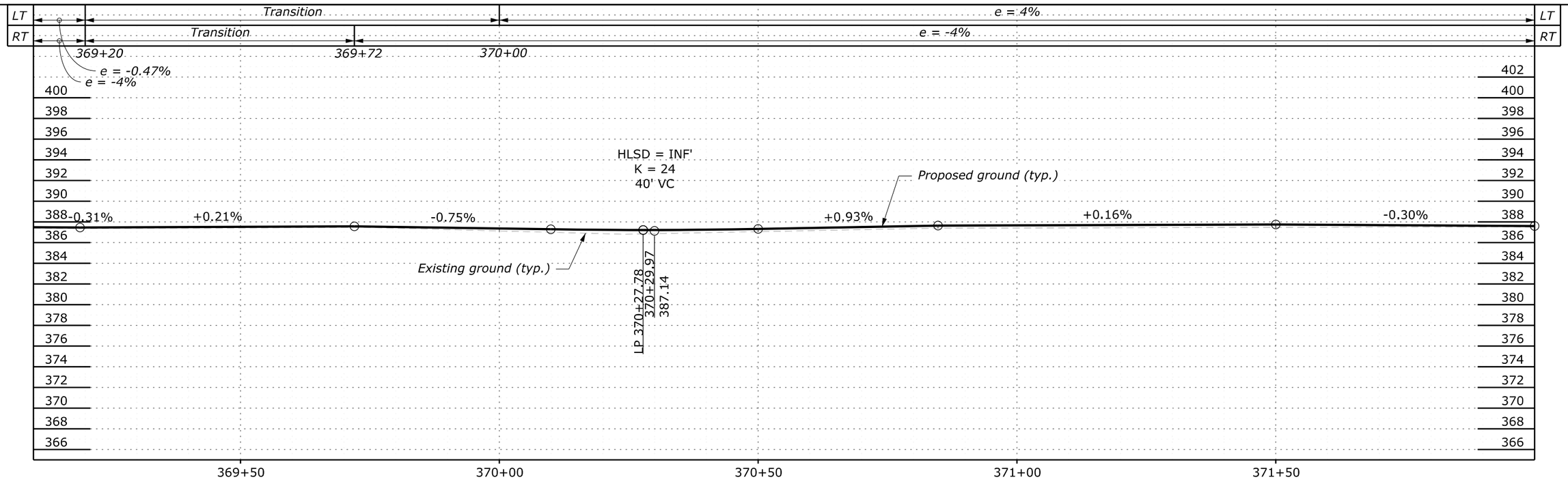
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E09





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

369+10 TO 372+00

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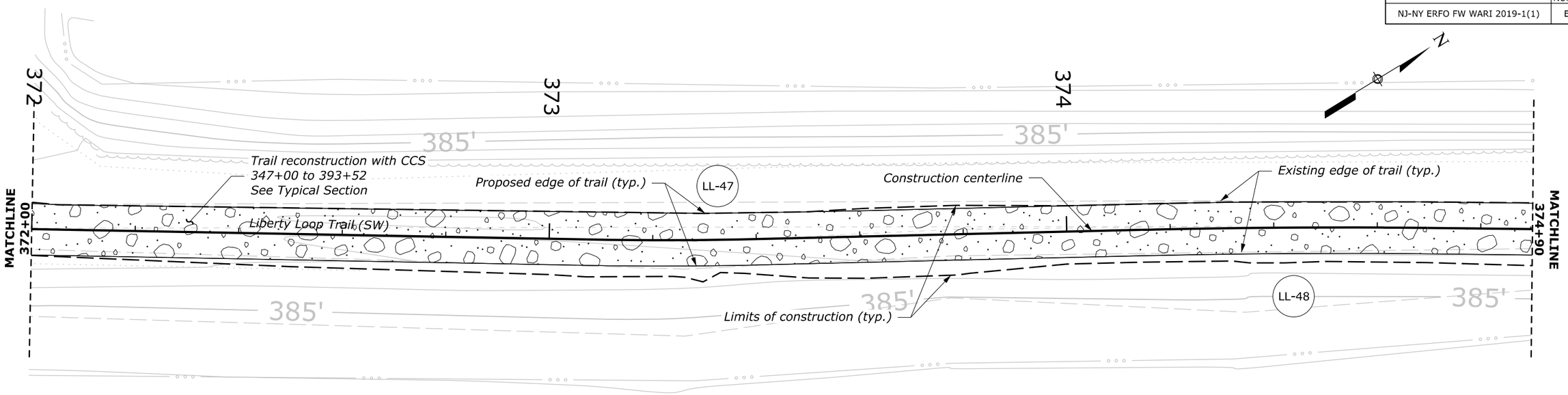
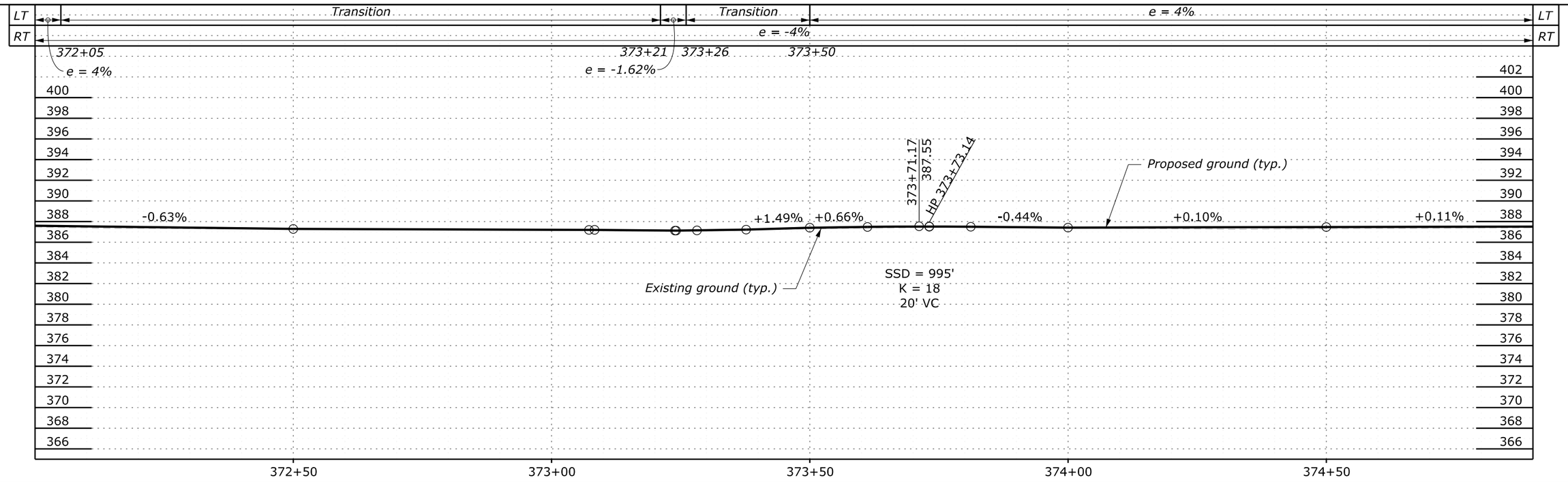
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E10





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

372+00 TO 374+90

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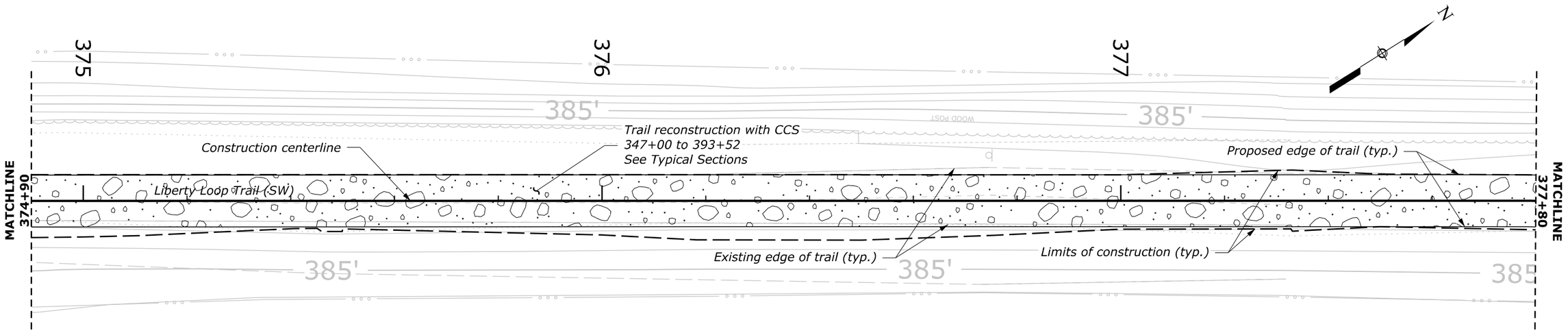
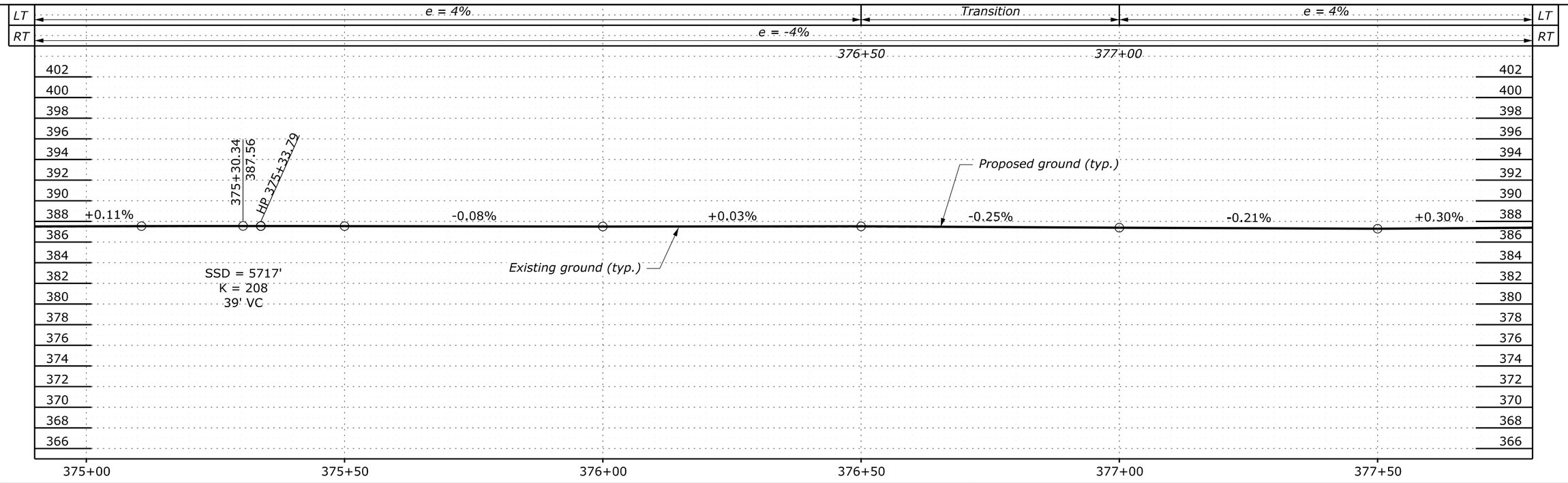
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E11





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

374+90 TO 377+80

SHEET 11 OF 17

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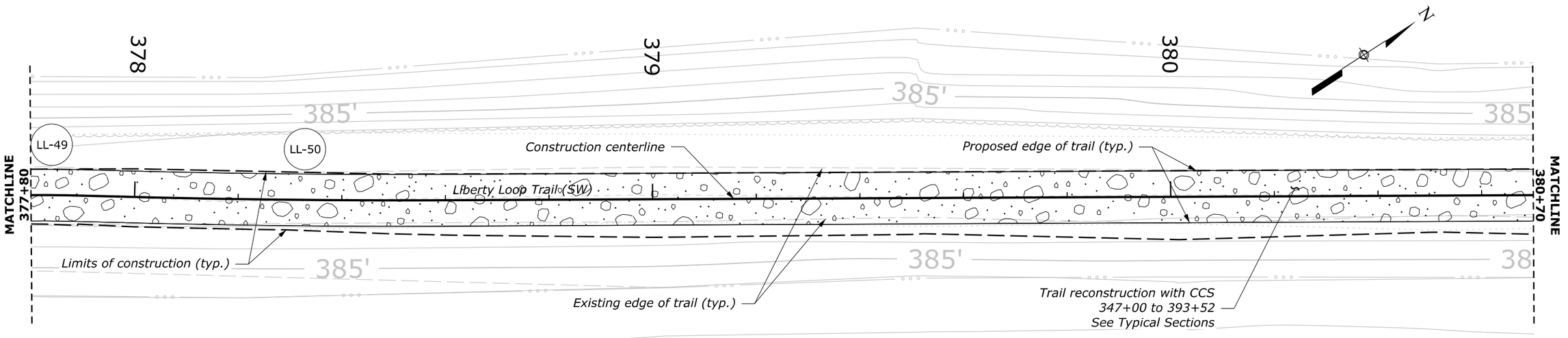
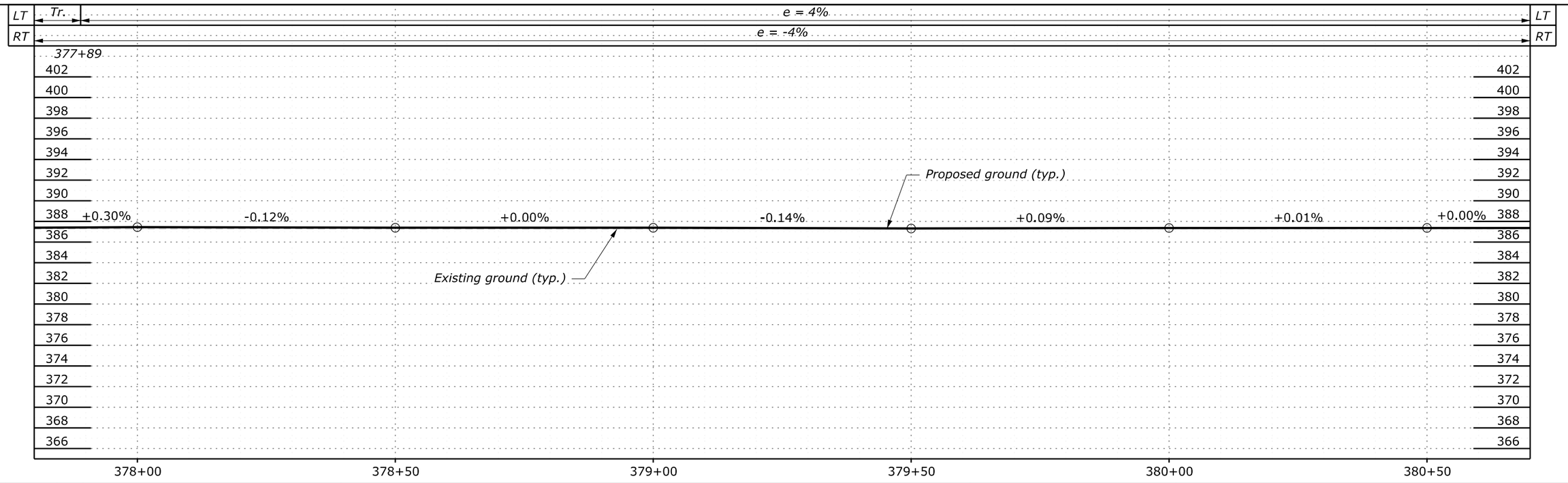
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E12





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

377+80 TO 380+70

SHEET 12 OF 17

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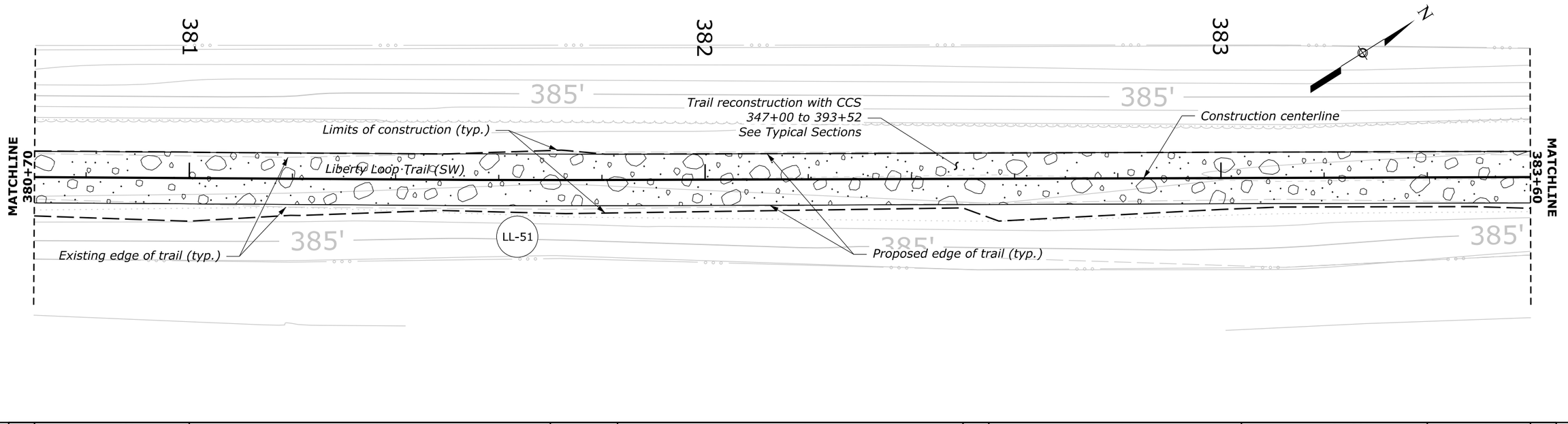
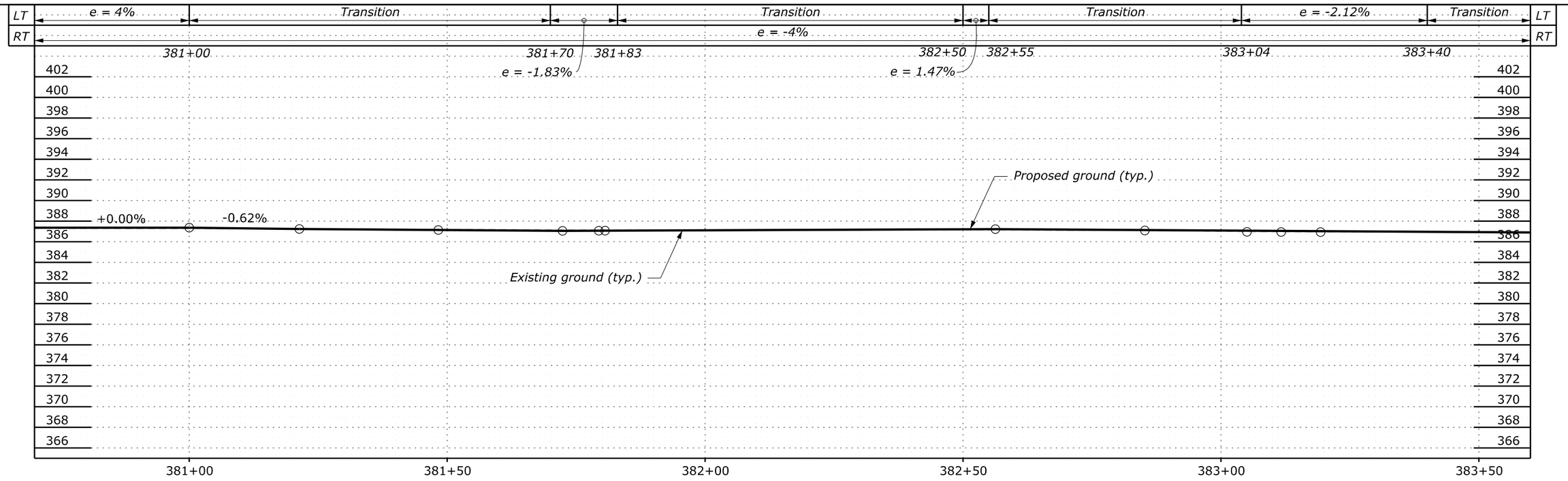
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E13





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

380+70 TO 383+60

SHEET 13 OF 17

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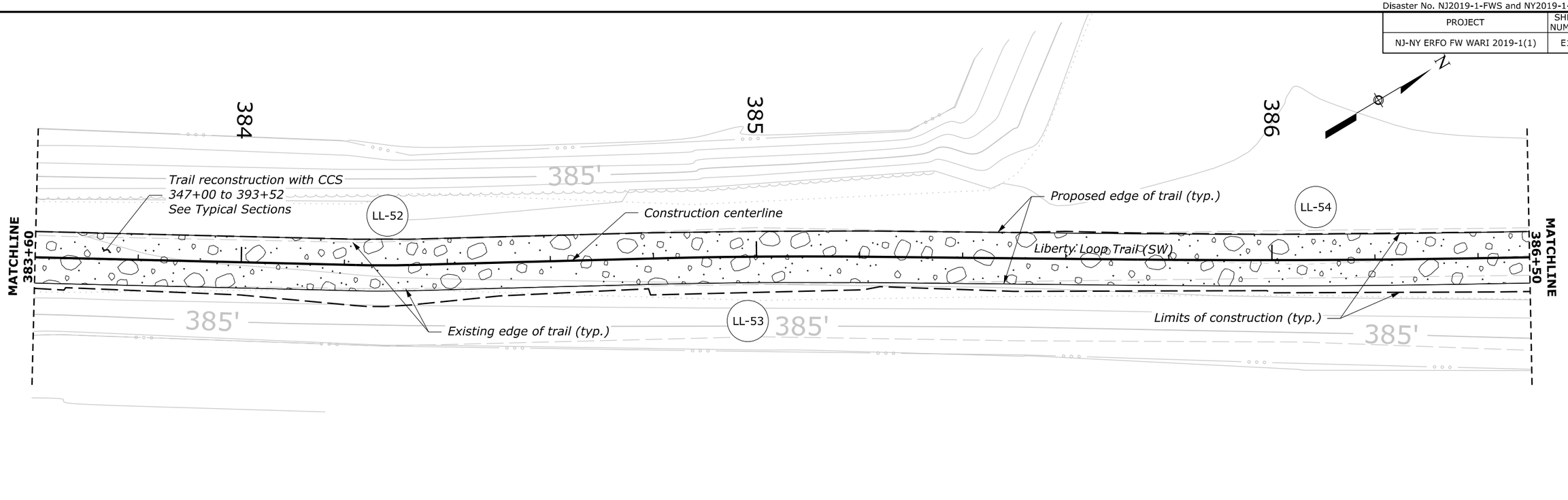
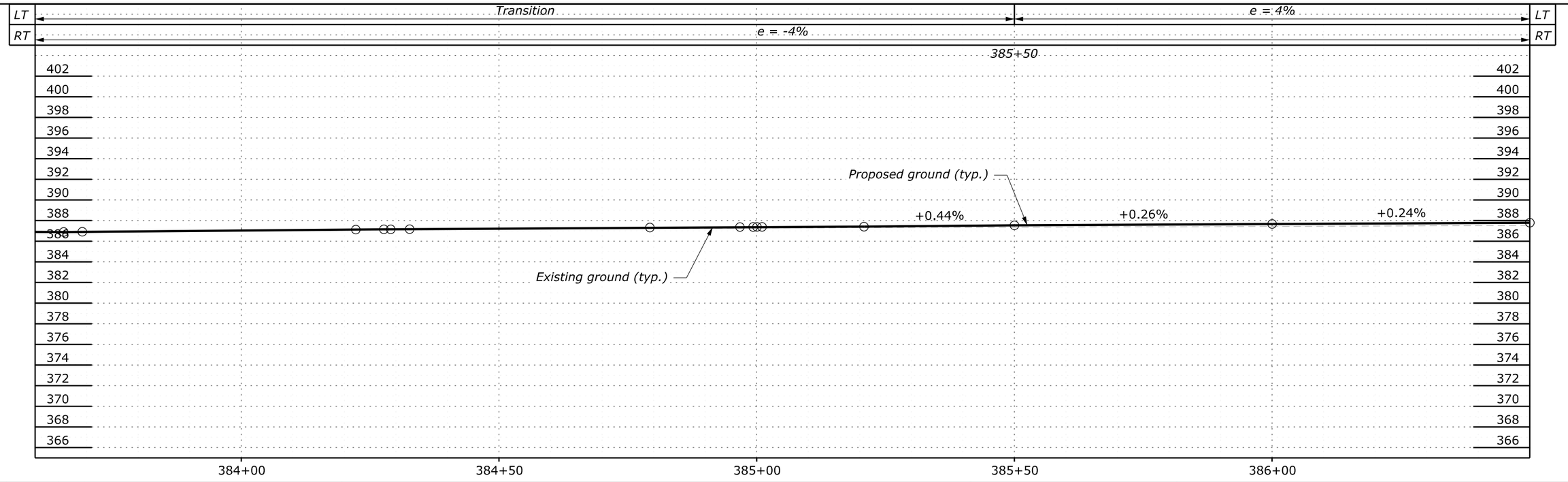
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E14





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

383+60 TO 386+50

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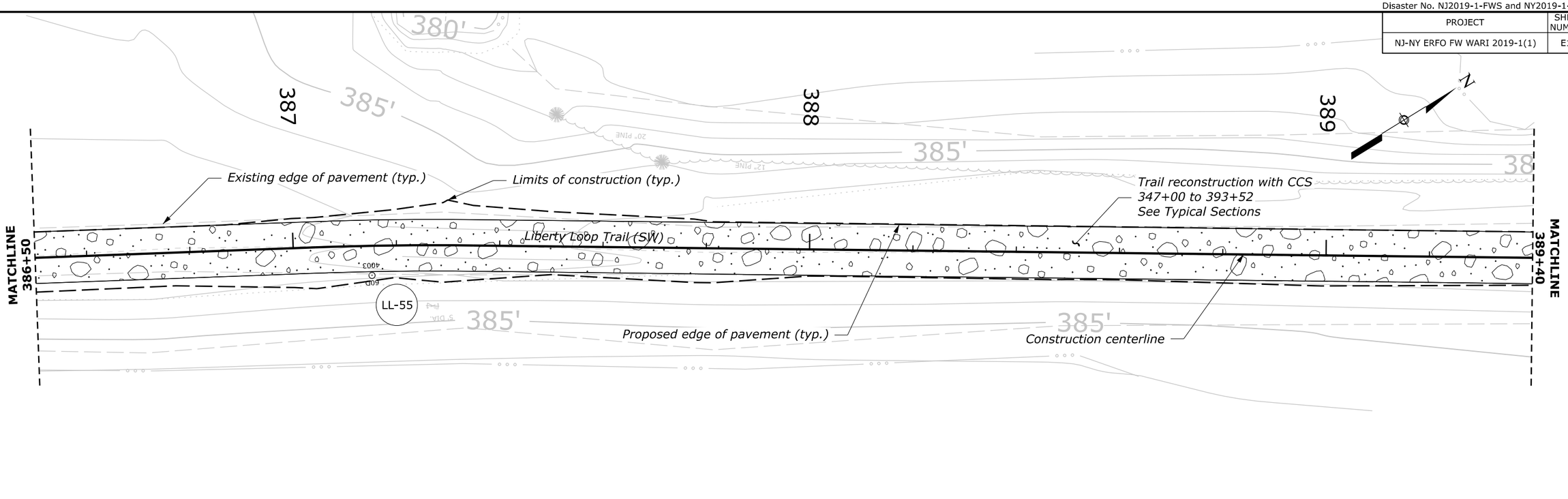
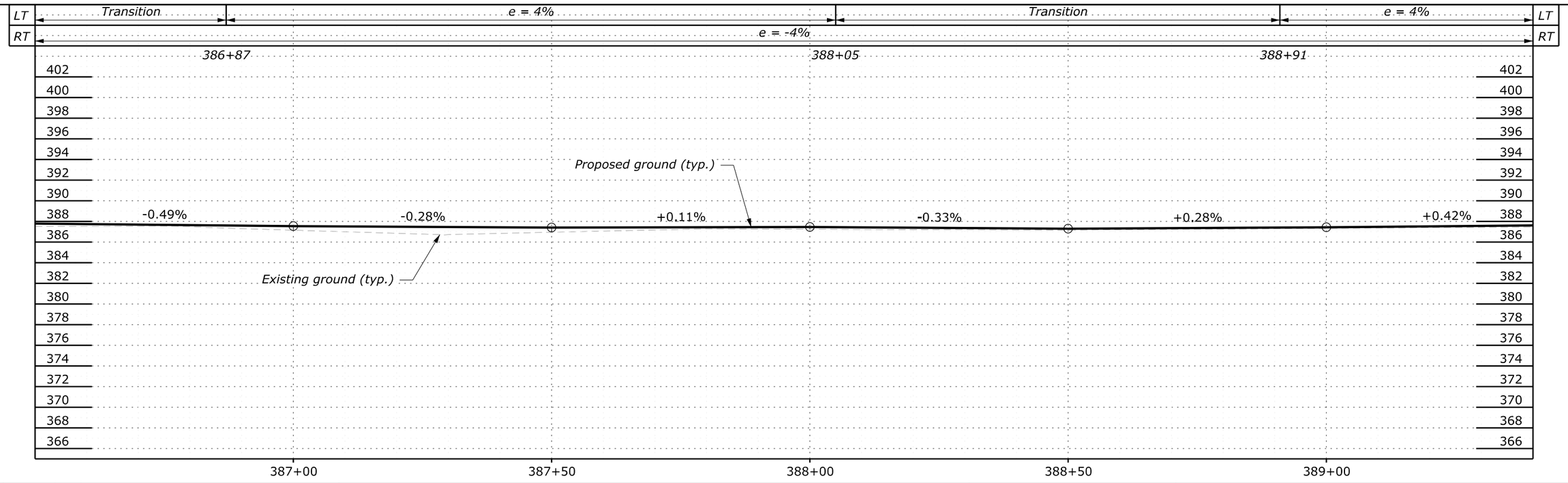
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E15





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
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OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
386+50 TO 389+40

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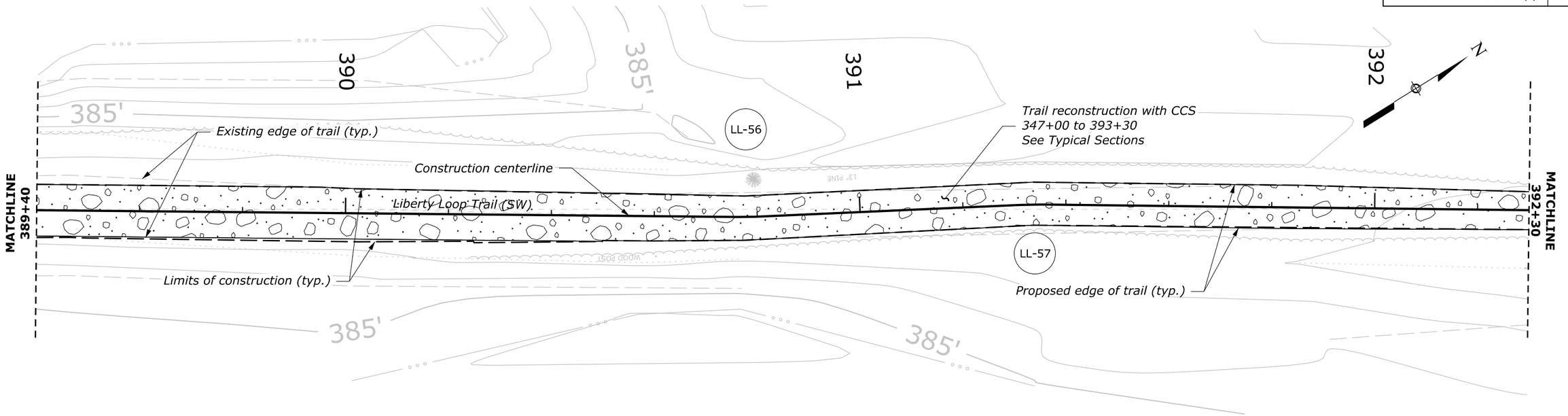
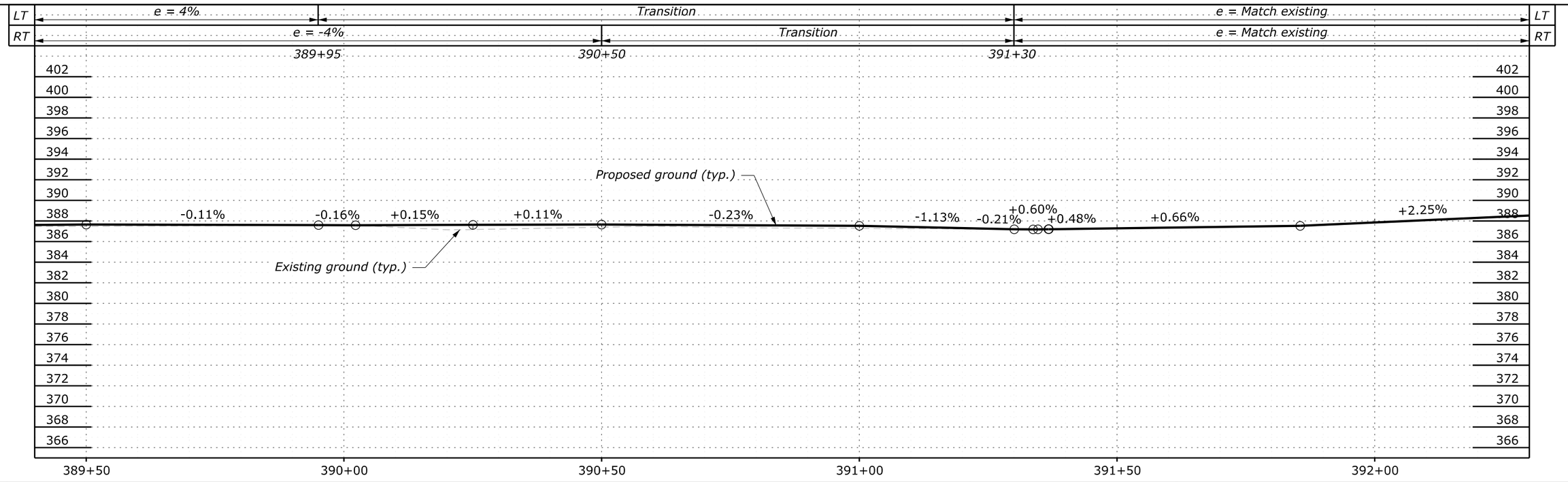
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	E16





NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

389+40 TO 392+30

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Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
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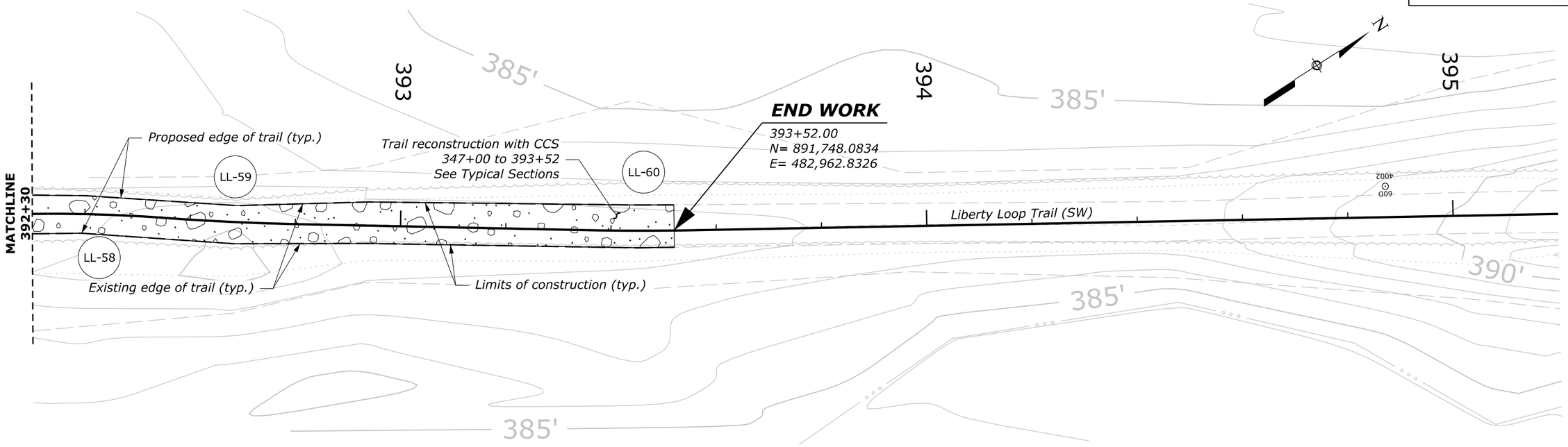

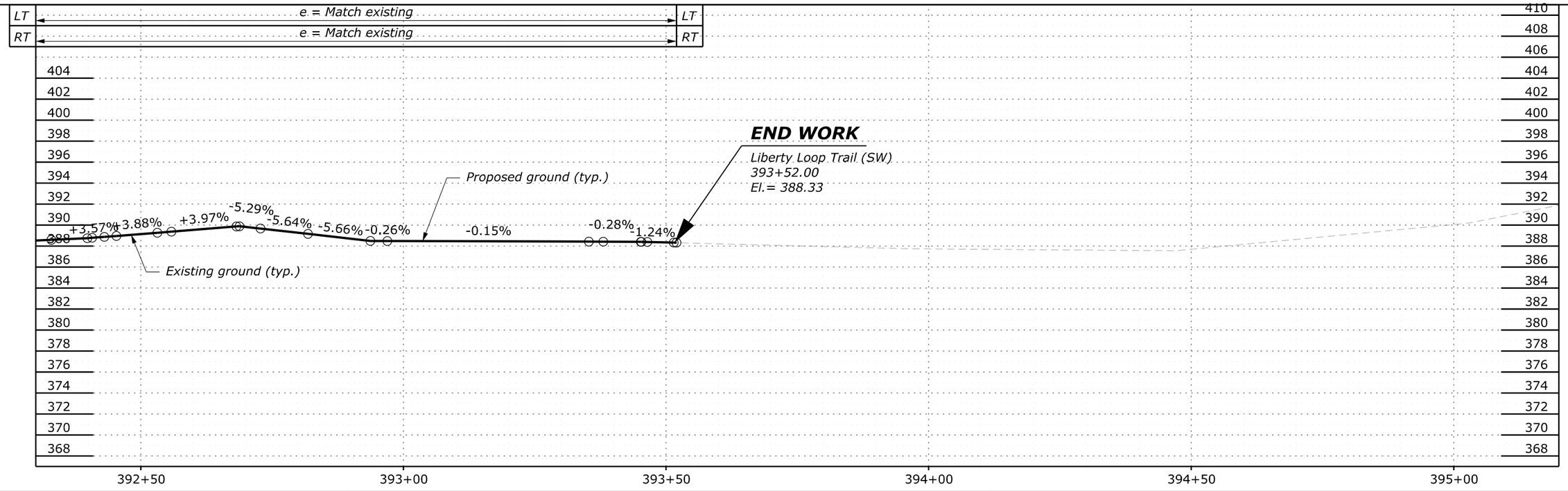

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OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN & PROFILE

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

392+30 TO 393+52

SHEET 17 OF 17

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Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F01

END STREAM REGRADING
2+25.00
N= 864,604.5111
E= 474,295.5671
See Typical Sections

Perform clearing and grubbing, LT
233 sqyd

Remove existing 30" CMP culvert.
11+01.17

Remove and reset boulders (typ.)

West headwall

NW wingwall

SW wingwall

Trail reconstruction with CCS
10+60 to 11+90
See Typical Sections

END WORK
11+90.00
N= 864,705.6598
E= 474,285.8555

Dagmar Dale Nature Trail

BEGIN WORK
10+60.00
N= 864,613.0015
E= 474,375.6532

Limits of construction (typ.)

Proposed edge of trail (typ.)

Existing edge of trail (typ.)

NE wingwall

Remove individual tree

East headwall

Install 54" HDPE culvert with concrete headwalls and wingwalls.
See Detail E601-5A and Standard 602-5
10+99.40

SE wingwall

Stream

Construction centerline

BEGIN STREAM REGRADING
1+02.00
N= 864,684.9168
E= 474,379.3093
See Typical Sections

Perform clearing and grubbing, RT
211 sqyd

DDS-A5 DDS-A4 DDS-A3 DDS-A2 DDS-A1
DD-A1 DD-A2 DD-A3 DD-A4 DD-A5 DD-A6 DD-A7

435' 430' 440' 445' 450'

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1
10+60 TO 11+90

SHEET 1 OF 11

9 August 2026 8:04 PM C:\Users\rseo.kunnri\Documents\NJ-NY WARI 100% files\D01-D02_NJ-WARI_2019-1(1)_DagmarDale_Site1.dgn [DagmarDale_Site1 [Sheet1]

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F02

464
462
460
458
456
454
452
450
448
446
444
442
440
438
436
434
432
430
428
426
424
422
420

HLSD = 133'
K = 3
10' VC

HLSD = 79'
K = 3
15' VC

HLSD = 23'
K = 0
5' VC

-7.82%
-4.44%
+0.30%
+11.95%

Proposed profile grade

10+66.20
444.54

10+99.84
443.04

LP 11+06.39

11+78.50
443.28

BEGIN WORK
Dagmar Dale Nature Trail
10+60.00
El.= 445.02

Existing ground

END WORK
Dagmar Dale Nature Trail
11+90.00
El.= 423.16

2' Embedded depth
(stream material)

Install 54" HDPE pipe culvert
10+99.40

Remove existing 30" CMP culvert.
Elevation unknown.
11+01.17

10+00 10+50 11+00 11+50 12+00 12+50

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9 August 2026 7:15 PM

NO.	DATE	BY	REVISIONS

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FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

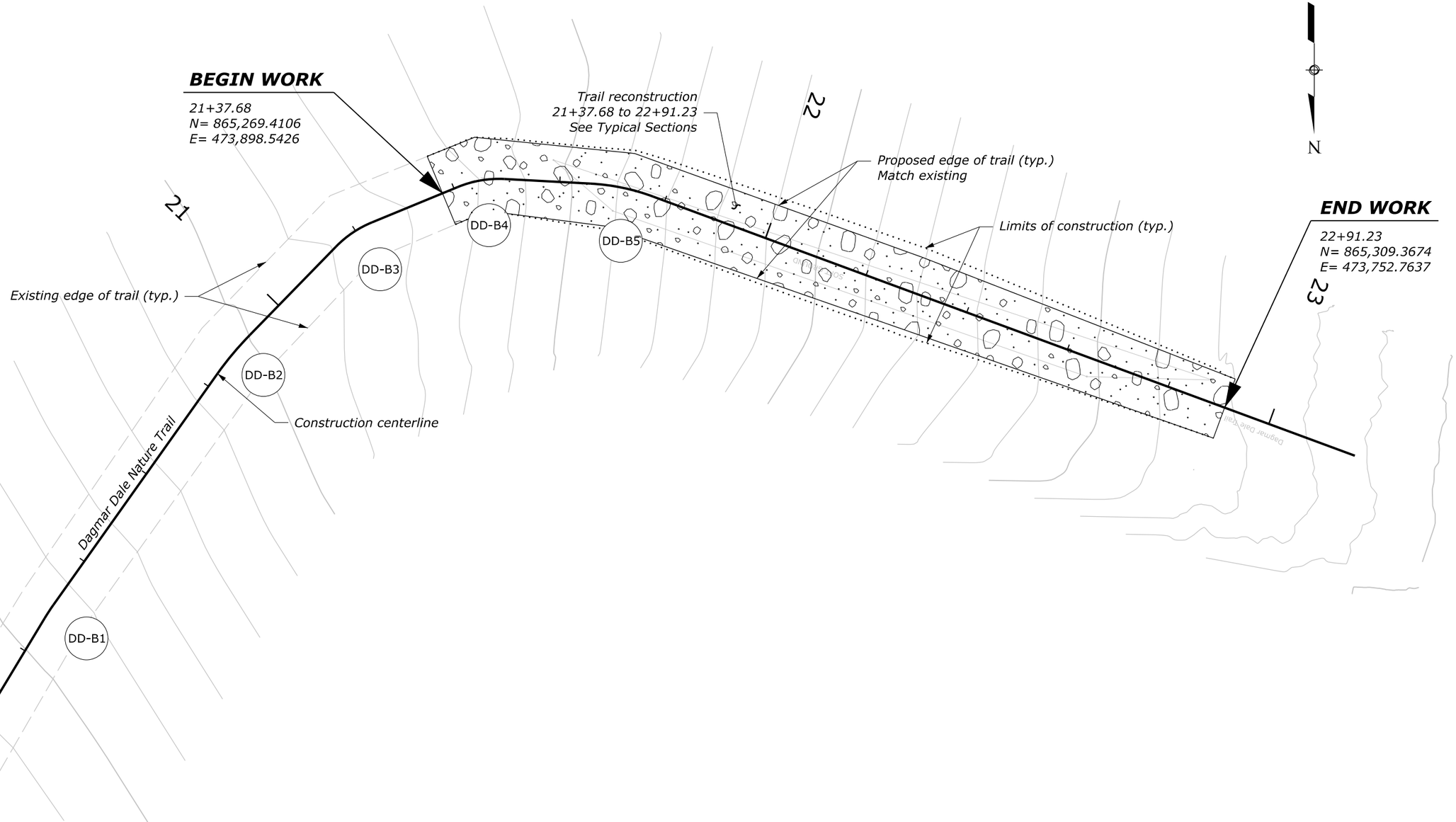
CONSTRUCTION PROFILE

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1
10+60 TO 11+90

SHEET 2 OF 11

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F03



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #2

21+37.68 TO 22+91.23

SHEET 3 OF 11

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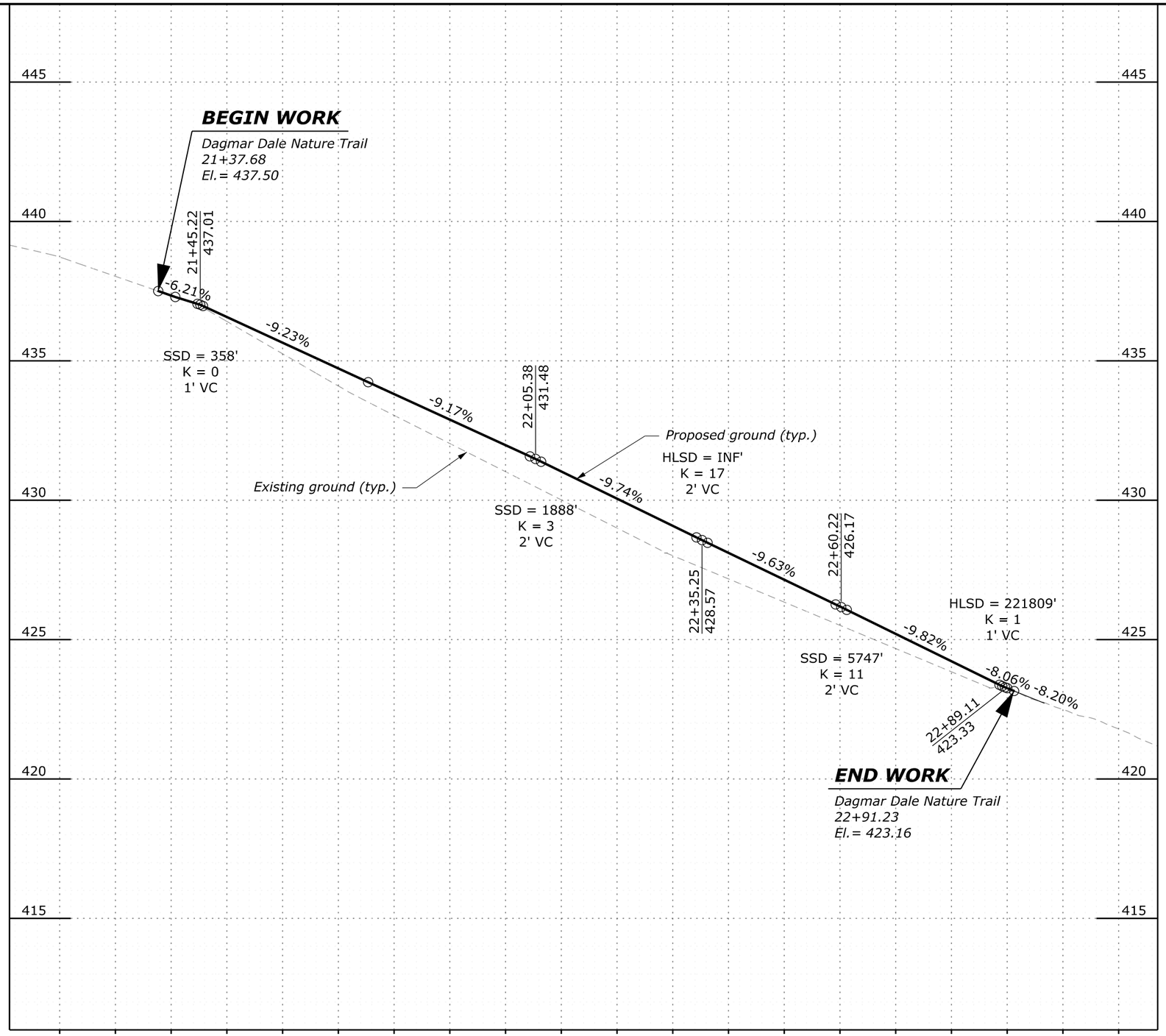
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F04



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PROFILE

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #2

21+37.68 TO 22+91.23

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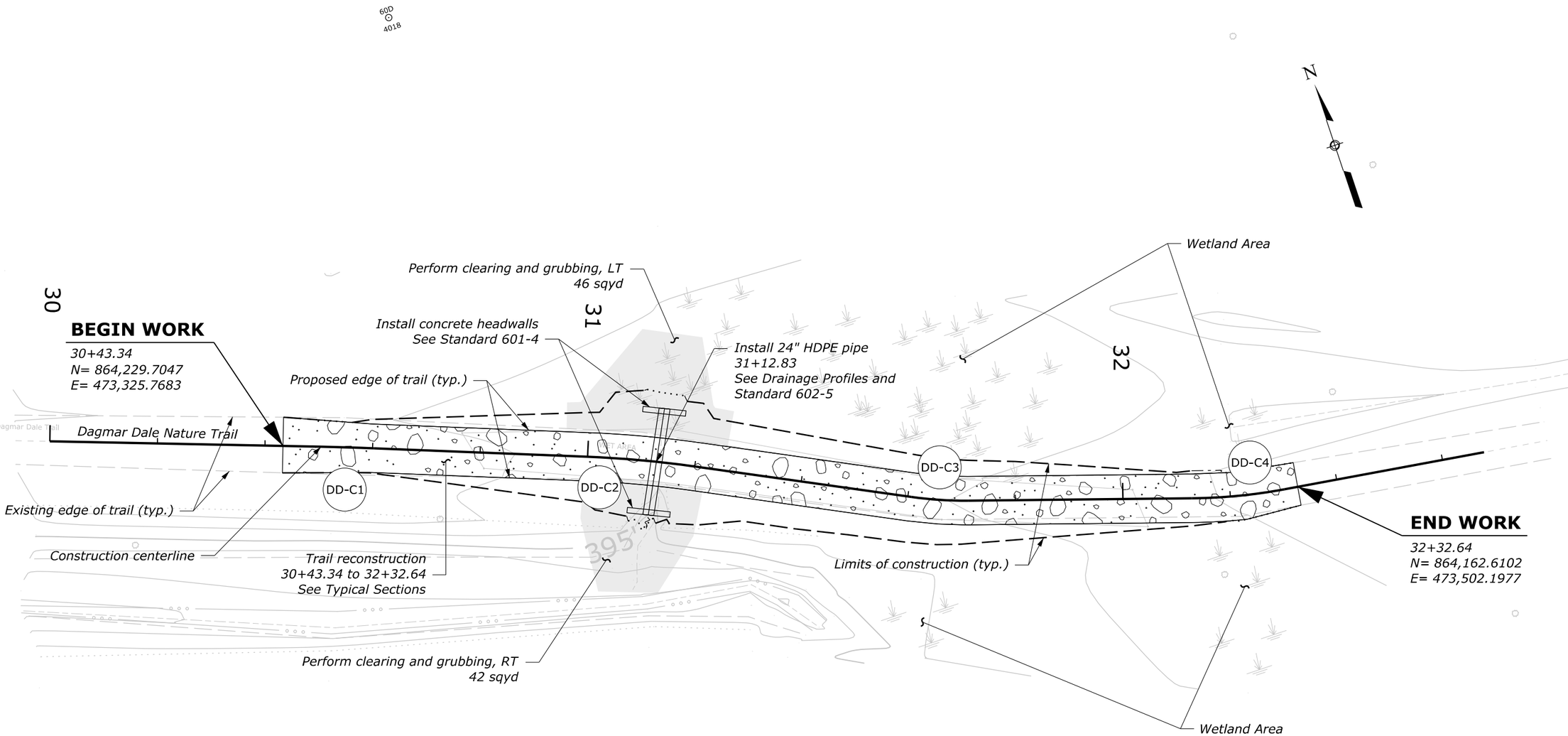
9 August 2026 7:30 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F05



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #3
30+43.34 TO 32+32.64

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9 August 2026 7:44 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F06

415
410
405
400
395
390
385
380

HLSD = 1547'
K = 5
10' VC

+0.63%

+2.52%

HP 31+13.84

31+12.84
399.00

SSD = 324'
K = 1
4' VC

-0.84%

Proposed profile grade

HLSD = 446'
K = 2
5' VC

+1.37%

30+54.67
397.53

Existing ground

32+21.45
398.09

LP 32+20.84

BEGIN WORK
Dagmar Dale Nature Trail
30+43.34
El.= 397.69

Install 24" HDPE pipe
31+13.91
See Drainage Profiles

END WORK
Dagmar Dale Nature Trail
32+32.64
El.= 398.24

30+00 30+10 30+20 30+30 30+40 30+50 30+60 30+70 30+80 30+90 31+00 31+10 31+20 31+30 31+40 31+50 31+60 31+70 31+80 31+90 32+00 32+10 32+20 32+30 32+40 32+50 32+60 32+70

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PROFILE

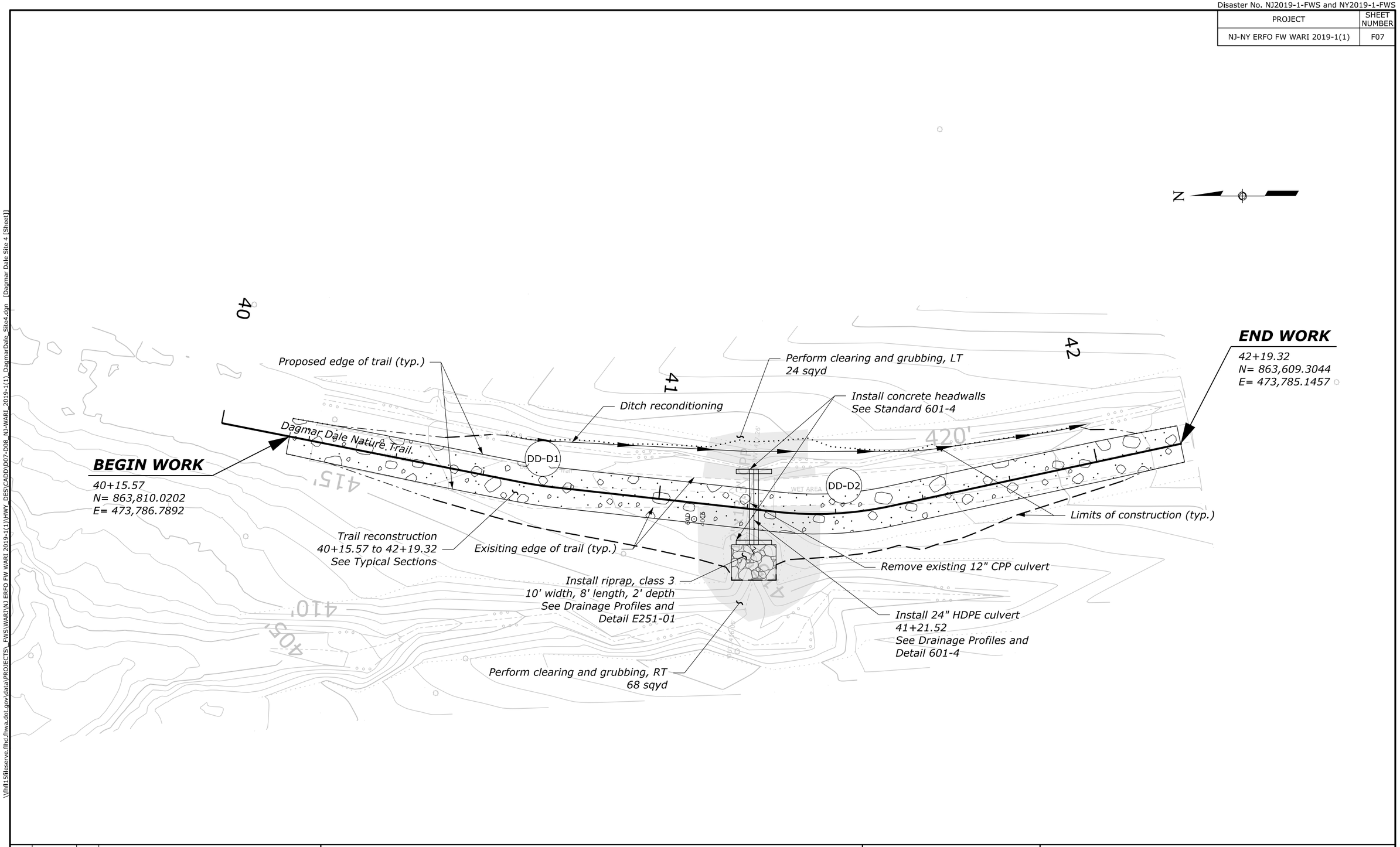
SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #3
30+43.34 TO 32+32.64

SHEET 6 OF 11

C:\Users\reo.kunorl\Documents\NJ-NY WARI 100% files\D05-D06_NJ-WARI_2019-1(1)_DagmarDale_Site3.dgn [Dagmar Dale Site 3 Profile (Sheet)]
9 August 2026 7:41 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F07



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #4

40+15.57 TO 42+19.32

10 August 2026 2:44 PM
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Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F08

435
430
425
420
415
410
405
400

BEGIN WORK
Dagmar Dale Nature Trail
40+15.57
El.= 415.49

HLSD = 52'
K = 0
2' VC

+5.65%

Proposed profile grade

SSD = 190'
K = 1
5' VC

41+21.51
421.35

HP 41+23.92

-0.10%

END WORK
Dagmar Dale Nature Trail
42+19.32
El.= 421.32

HLSD = 69'
K = 0
2' VC

LP 42+16.77
42+17.73
421.25

Install 24" HDPE culvert
See Drainage Profile

Remove existing 12" CPP culvert

Existing ground

LP 40+16.75
40+17.73
415.49

39+80 39+90 40+00 40+10 40+20 40+30 40+40 40+50 40+60 40+70 40+80 40+90 41+00 41+10 41+20 41+30 41+40 41+50 41+60 41+70 41+80 41+90 42+00 42+10 42+20

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #4
40+15.57 TO 42+19.32

SHEET 8 OF 11

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9 August 2026 7:50 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F09

N

BEGIN WORK
50+20.00
N= 863,332.3502
E= 473,762.5037

END WORK
53+50.00
N= 863,020.3827
E= 473,785.2101

50
51
52
53

Proposed edge of trail (typ.)
Match existing

Ditch reconditioning

DD-E1
DD-E2
DD-E3
DD-E4
DD-E5

Dagmar Dale Nature Trail

Construction centerline

Trail reconstruction with CCS
50+20 to 53+50
See Typical Sections

60D
4014

OUT 430.20
DITCH 437.85
24"X51.2' RCP
INV 440.30
NORTH SOUTH LOOP
WOOD BENCH
OLD SIGN

430'
440'
445'

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NO.	DATE	BY	REVISIONS

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OFFICE OF FEDERAL LANDS HIGHWAY

0 15 30
SCALE IN FEET

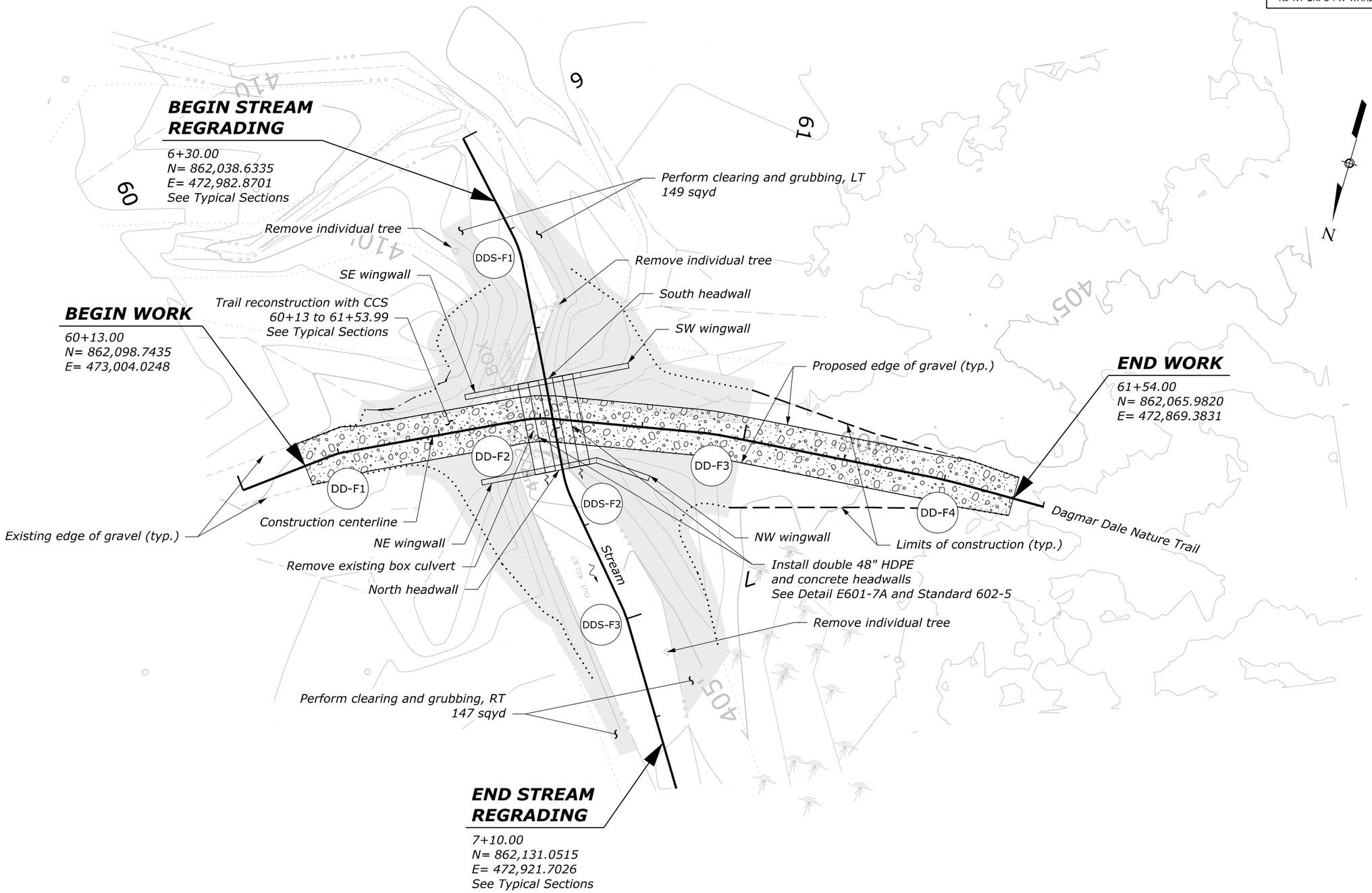
WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #5
50+20 TO 53+50

SHEET 9 OF 11

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F10



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PLAN

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

60+13 TO 61+53.99

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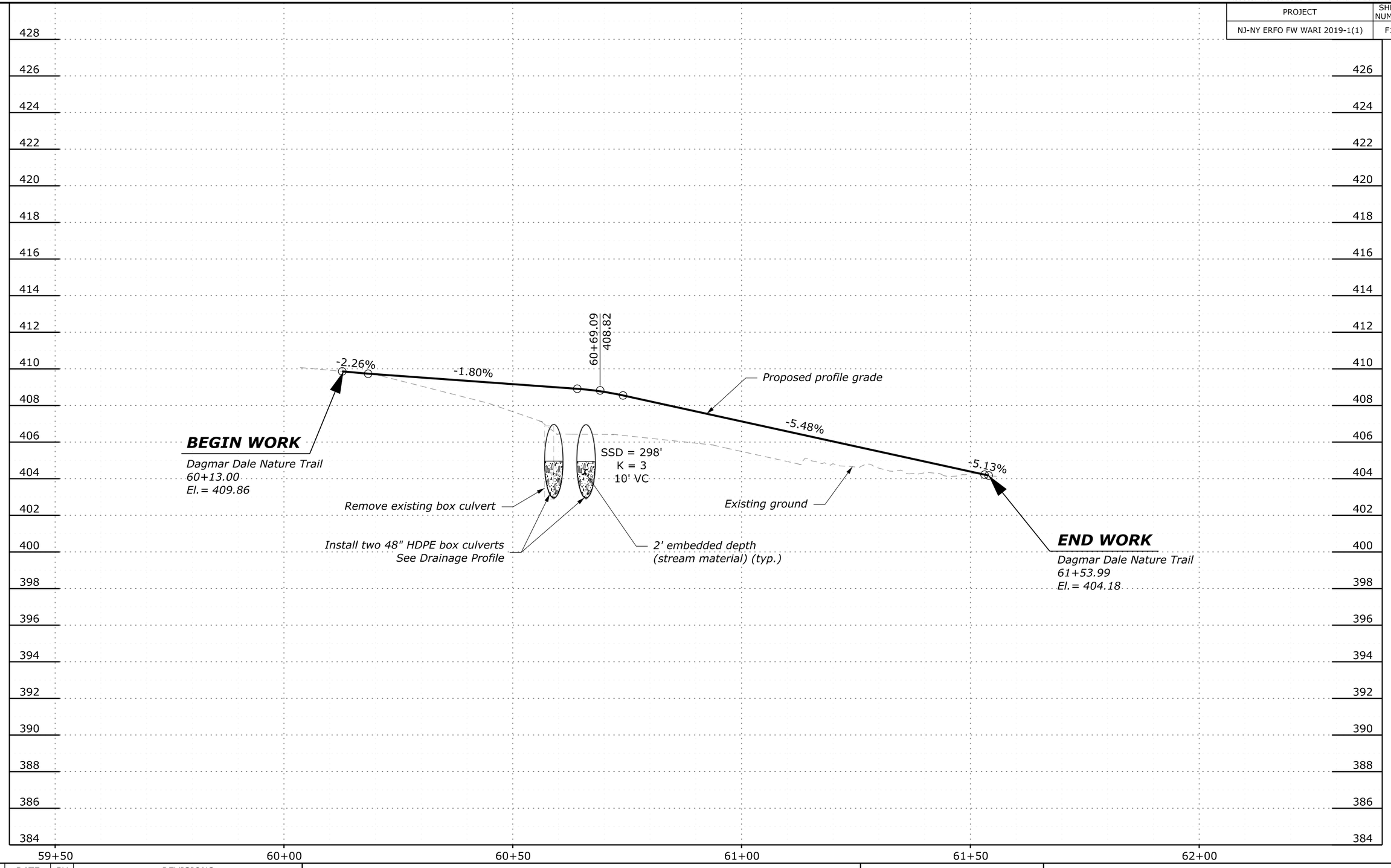
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	F11



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CONSTRUCTION PROFILE

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

60+13 TO 61+53.99

SHEET 11 OF 11

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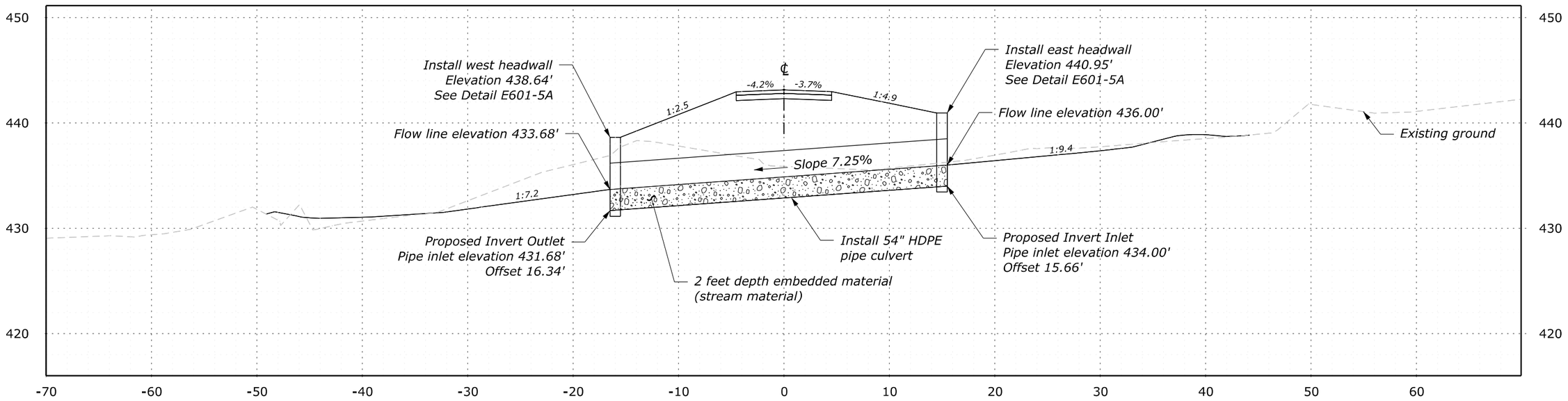
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	K01



10+99.40
Des Grade = 443.14
Subgrade = 442.31
Original Grnd = 435.99
54-Inch x 32 Feet HDPE Pipe Culvert
Skew = 6.09 °

NOTE:

Verify a minimum allowable soil bearing capacity of 4000psf immediately prior to casting the foundation.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
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OFFICE OF FEDERAL LANDS HIGHWAY

0 5 10
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

DRAINAGE PROFILE

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1
10+99.40

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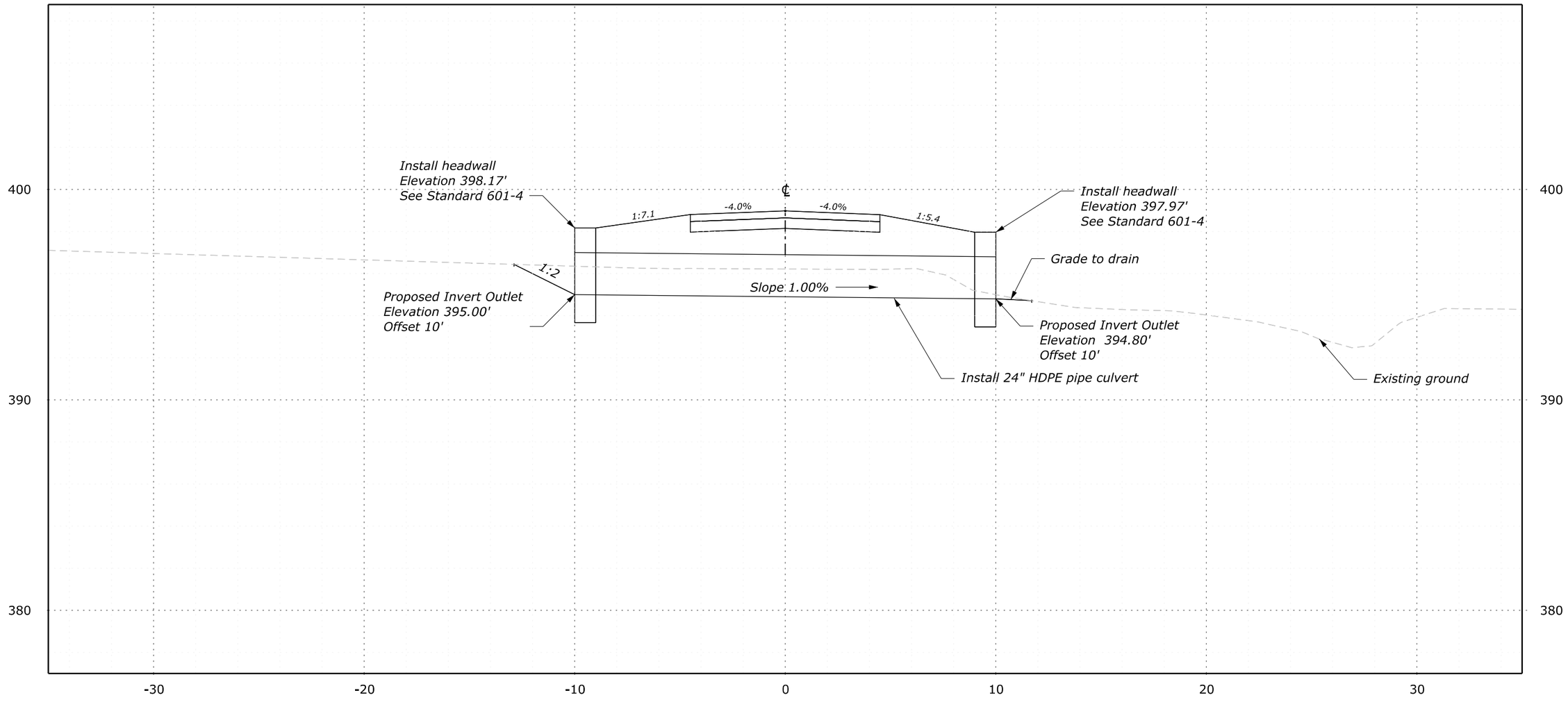
21 July 2026 1:38 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	K02



31+12.83
Des Grade = 398.98
Subgrade = 398.15
Original Grnd = 396.25
24-Inch x 20 Feet HDPE Pipe Culvert
Skew = 2.72°

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 2.5 5
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

DRAINAGE PROFILE

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #3
31+12.83

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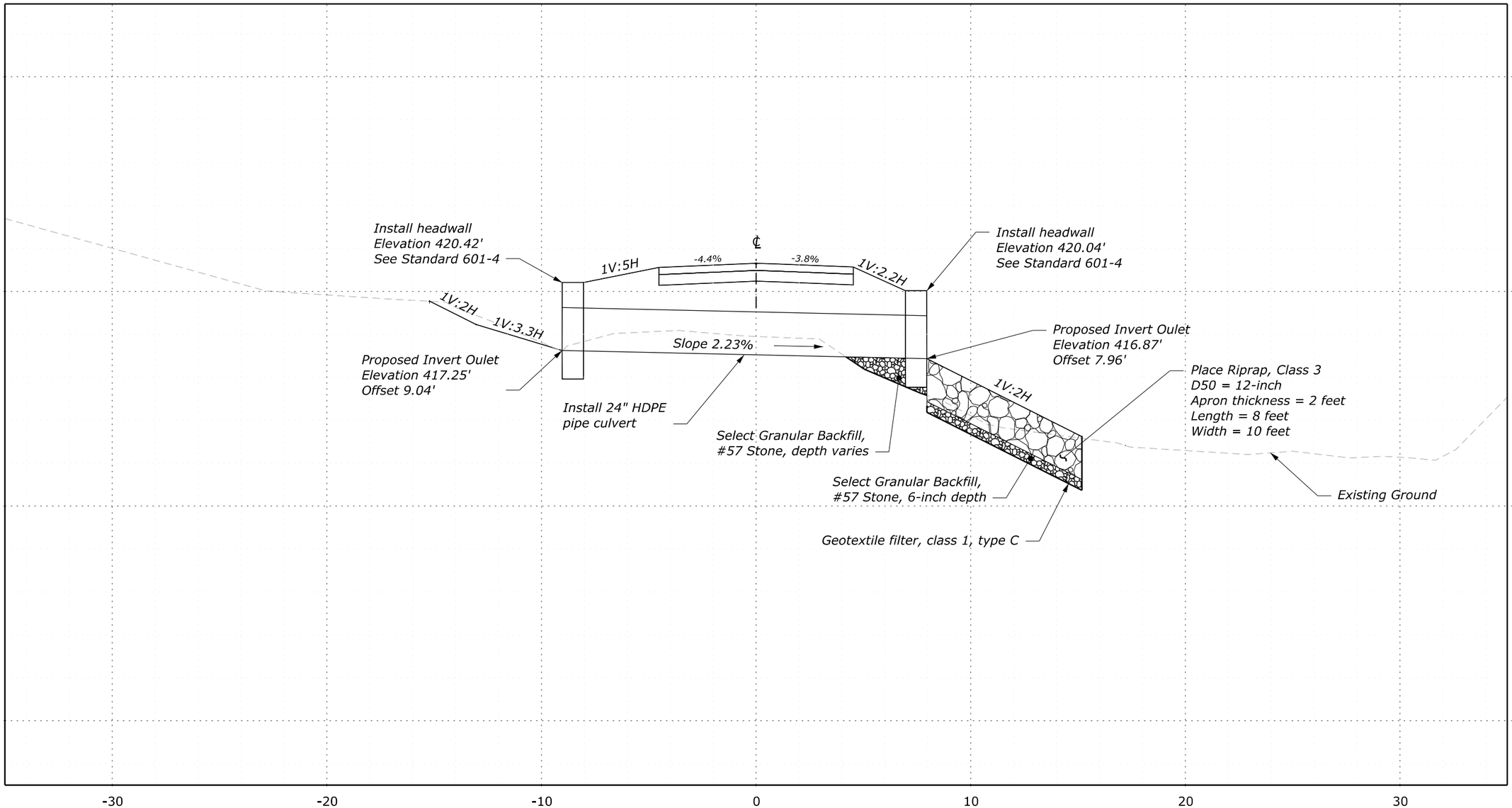
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	K03



41+21.52
Des Grade = 421.31
Subgrade = 420.48
Original Grnd = 417.90
24-Inch x 17 Feet HDPE Pipe Culvert
Skew = 6.3°

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 2.5 5
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

DRAINAGE PROFILE

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #4
41+21.52

3 August 2026 6:08 PM
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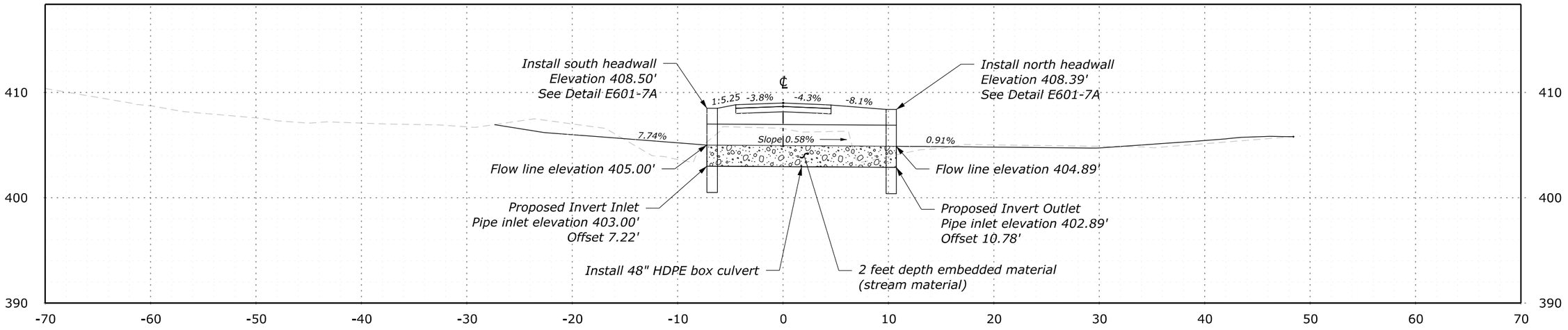

60+58.97
Des Grade = 409.00
Subgrade = 408.17
Original Grnd = 406.57
48-Inch x 18 Feet HDPE Box Culvert
Skew 7.41 °

60+66.01
Des Grade = 408.87
Subgrade = 408.03
Original Grnd = 406.42
48-Inch x 18 Feet HDPE Box Culvert
Skew = 16.33 °

NOTE:

Verify a minimum allowable soil bearing capacity of 4000psf immediately prior to casting the foundation.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 5 10
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

DRAINAGE PROFILE

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6
60+58.97 AND 60+66.01

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21 July 2026 4:43 PM

Project Description

This project consists of embankment repairs, trail reconstruction, and other miscellaneous work.

Soil disturbing activities include clearing and grubbing, tree removal, revegetation, trail reconstruction, grading, excavation, embankment construction, and riprap installation.

The total disturbed area for the project is approximately 2.29 acres. The receiving water is the Wallkill River.

The Runoff Coefficient prior to and after construction is 0.43.

DISTURBANCE SUMMARY

Activity	Include in Calculation? Yes or No	Disturbed Area (SQFT)	Stream Bed Impacts		Wetland Impacts	
			Temporary (SQFT)	Permanent (SQFT)	Temporary (SQFT)	Permanent (SQFT)
Liberty Loop Trail (NE) (incl. staging)	Yes	64,164				
Winding Waters Staging Area	Yes	7,745				
Winding Waters Trail Site #3	Yes	2,330	0	100		
Winding Waters Trail Site #240	Yes	12,786	0	41	815	2,568
Winding Waters Trail Site #324	Yes	12,723	0	726	1,116	1,140
TOTAL (SQFT) =		99,748	0	866	1,931	3,709
TOTAL (ACRE) =		**2.2899**	**0.0000**	**0.0199**	**0.0443**	**0.0851**

Prohibited Discharges

The following discharges are prohibited:
Wash-water from concrete, paint, curing compounds, and other construction materials
Fuels, oils, equipment-related compounds
Soaps, solvents used for vehicle washing
Waste, garbage, sanitary waste

Inspect and maintain on a regular basis, all mechanized equipment used in or near surface water to prevent contamination from fuels, lubricants, hydraulic fluids, or other toxic materials.

Solid waste generated from the project will consist of construction debris, garbage, and empty containers. Collect and store all waste in dumpsters, or in metal or plastic drums, as appropriate.

Hazardous waste will not be generated from normal construction activities. Equipment fueling and maintenance could generate spills, leaks, and hazardous wastes like motor oil, diesel, gasoline, and battery fluid. If feasible, conduct these activities in a covered area to avoid contact with storm water. Store all hazardous waste materials in appropriate and clearly marked containers away from other non-waste materials. Do not dispose of hazardous waste materials into the on-site dumpsters. Dispose of material according to Federal, State, and local regulations.

Develop and implement a Spill Prevention Control and Countermeasures (SPCC) plan following the requirements under 40 CFR 112. Report spills large enough to discharge to surface waters to the National Response Center at 1-800-424-8802.

General Guidelines

The Erosion & Sediment Control Narrative is meant as a guideline for preventing erosion and controlling sediment. The work consists of applying measures throughout the life of the project to control erosion and to minimize the sedimentation of rivers, streams, and impoundments such as lakes, reservoirs, bays, and coastal waters. The measures consist of soil erosion control measures which are also defined and outlined in the Standard Specifications for Construction of Roads and Bridges on Federal Highway Projects, FP-14, and the Special Contract Requirements.

Do not modify the type, size, or location of any control or practice without prior approval from the Contracting Officer (CO).
No construction access will be permitted through a wetland or waterway.
Do not allow construction vehicles to track sediment outside the project limits.
Do not allow any construction equipment to operate on or access the down-slope side of the perimeter control measures.
Direct storm water to vegetated buffer areas and do not discharge directly into surface waters.

Sequence of Construction

Phase I Establish Perimeter Controls

At all sites, prior to any clearing, grubbing, or excavation, construct perimeter controls to ensure that disturbed sediment does not leave the project site. Perimeter controls include silt fence and other specified measures outside the construction limits.

Phase II Intermediate Controls

At all sites, apply intermediate controls during rough grading operations. Install silt fence in areas surrounding the culverts as called out in the Erosion and Sediment Control plans.

Apply temporary turf establishment in disturbed areas that will remain exposed for over 14 days within 7 days. Apply permanent turf establishment to the finished slopes according to Section 625.

At the end of each day's grading operations, shape earthwork to minimize and control erosion from storm runoff.

Provide silt fence around all stockpiled excavated roadway material. Apply temporary turf establishment to stockpiles remaining in place longer than 14 days within 7 days of stockpiling.

Provide watering for dust control within the construction limits, on active haul roads, and in pits and staging areas.

At Sites #240 and #324, excavate the areas for vegetative riprap, followed by placing the Class 5 riprap key-in. Place the Class 3 riprap, pole plantings and soil mix.

Phase III Final Construction / Stabilization

At Sites #240 and #324, after placing the vegetative riprap, reconstruct the trail as shown in the plans.

For all sites, after completion of roadway construction, do the following as directed by the CO:
Finish grading, place riprap, and apply permanent turf establishment to any remaining disturbed areas.
Where necessary, replace eroded topsoil and re-apply permanent turf establishment to disturbed areas where vegetation has not established.
Inspect, clean, and repair all riprap basins, and stabilized channels.
Remove all devices used for dewatering.
Remove silt fence only after all upslope areas are stabilized and vegetation is well established.

Remove all other perimeter controls when directed by the CO.

Maintenance and Inspection Procedures

Unless stated otherwise, construct and maintain all vegetated and structural erosion control practices according to Section 157, the details shown in the plans, and the individual permitting requirements. Check and maintain erosion control measures once every 7 days and within 24 hours after a rain of 0.25 inches or more, and daily during wet weather. Repair or replace any damaged measures by the end of the day.

Silt fence - inspect for buildup of excess sediment, undercutting, sags, and other failures. If the fabric becomes damaged, repair or replace as necessary. Remove sediment from behind the silt fence when it becomes 0.5 feet deep at the fence.

Floating turbidity curtain - inspect daily and repair if necessary. Remove any floating construction or natural debris immediately to prevent any damage. If necessary, remove sediment deposited behid the curtain by hand prior to removal. Leave the curtain in place until work is completed in the enclosed area and remove curtain by carefully pulling it toward the construction site to minimize the release of attached sediment.

Record the inspection date and summary of findings within 24 hours of completing a site inspection.

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Project Description

This project consists of culvert replacements, trail reconstruction, stream regrading, headwall and wingwall installation, and other miscellaneous work.

Soil disturbing activity includes trail reconstruction, culvert removal and replacement, excavation, grading, embankment construction, clearing and grubbing, and riprap installation.

The total disturbed area for the project is approximately 2.67 acres. Refer to the table "Disturbance Calculations Summary" for a breakdown by site. The receiving water is Wallkill River.

The Runoff Coefficient prior to construction is 0.82. The Runoff Coefficient after construction will be 0.82.

Prohibited Discharges

The following discharges are prohibited:
- *Wash-water from concrete, paint, curing compounds, and other construction materials*
- *Fuels, oils, equipment-related compounds*
- *Soaps, solvents used for vehicle washing*
- *Waste, garbage, sanitary waste*

Inspect and maintain on a regular basis, all mechanized equipment used in or near surface water to prevent contamination from fuels, lubricants, hydraulic fluids, or other toxic materials.

Solid waste generated from the project will consist of construction debris, garbage, and empty containers. Collect and store all waste in dumpsters, or in metal or plastic drums, as appropriate.

Hazardous waste will not be generated from normal construction activities. Equipment fueling and maintenance could generate spills, leaks, and hazardous wastes like motor oil, diesel, gasoline, and battery fluid. If feasible, conduct these activities in a covered area to avoid contact with storm water. Store all hazardous waste materials in appropriate and clearly marked containers away from other non-waste materials. Do not dispose of hazardous waste materials into the on-site dumpsters. Dispose of material according to Federal, State, and local regulations.

Develop and implement a Spill Prevention Control and Countermeasures (SPCC) plan following the requirements under 40 CFR 112. Report spills large enough to discharge to surface waters to the National Response Center at 1-800-424-8802.

General Guidelines

The Erosion & Sediment Control Narrative is meant as a guideline for preventing erosion and controlling sediment. The work consists of applying measures throughout the life of the project to control erosion and to minimize the sedimentation of rivers, streams, and impoundments such as lakes, reservoirs, bays, and coastal waters. The measures consist of soil erosion control measures which are also defined and outlined in the Standard Specifications for Construction of Roads and Bridges on Federal Highway Projects, FP-14, and the Special Contract Requirements.

Do not modify the type, size, or location of any control or practice without prior approval from the Contracting Officer (CO).
No construction access will be permitted through a wetland or waterway.
Do not allow construction vehicles to track sediment outside the project limits.
Do not allow any construction equipment to operate on or access the down-slope side of the perimeter control measures.
Direct storm water to vegetated buffer areas and do not discharge directly into surface waters.

Sequence of Construction

Phase I Establish Perimeter Controls

Prior to any excavation, construct perimeter controls to ensure that disturbed sediment does not leave the project site. Perimeter controls include silt fence and other specified measures outside the construction limits.

Phase II Intermediate Controls

Apply intermediate controls during rough grading operations. Install silt fence in areas as called out in the Erosion and Sediment Control plans.

Apply temporary turf establishment in disturbed areas that will remain exposed for over 14 days within 7 days. Apply permanent turf establishment to the finished slopes according to Section 625.

At the end of each day's grading operations, shape earthwork to minimize and control erosion from storm runoff.

Upon completion of culverts, ensure that culvert entrances, outlets, and outlet channels are at final grade and are stabilized (with vegetation, riprap, or pavement) before routing drainage through completed culverts.

Provide silt fence around all stockpiled excavated roadway material. Apply temporary turf establishment to stockpiles remaining in place longer than 14 days within 7 days of stockpiling.

Provide watering for dust control within the construction limits, on active haul roads, and in pits and staging areas.

Phase III Final Construction / Stabilization

After completion of roadway construction, do the following as directed by the CO:
- *Finish grading, place riprap, and apply permanent turf establishment to any remaining disturbed areas.*
- *Where necessary, replace eroded topsoil and re-apply permanent turf establishment to disturbed areas where vegetation has not established.*
- *Inspect, clean, and repair all culvert outlet protection, riprap basins, and stabilized channels.*
- *Remove all devices used for dewatering.*
- *Remove silt fence only after all upslope areas are stabilized and vegetation is well established.*

Remove all other perimeter controls when directed by the CO.

Maintenance and Inspection Procedures

Unless stated otherwise, construct and maintain all vegetated and structural erosion control practices according to Section 157, the details shown in the plans, and the individual permitting requirements. Check and maintain erosion control measures once every 7 days and within 24 hours after a rain of 0.25 inches or more, and daily during wet weather. Repair or replace any damaged measures by the end of the day.

Silt fence - Inspect for sediment buildup, undercutting, overtopping, and damaged or sagging fabric. Repair or replace as needed and maintain proper anchoring. Remove sediment when it reaches 0.5 feet in depth or one-half the height of the fence.

Check Dam with Rolled Erosion Control Product (RECP) - Inspect for displacement, undercutting, and sediment buildup after storm events. Ensure RECP remains properly anchored. Repair as needed and remove sediment when it reaches one-half the height of the check dam.

Temporary In-Stream Diversion Berm - Inspect for erosion, breaching, settlement, and overtopping. Repair and recompact as needed to maintain function. Remove sediment to preserve capacity.

Bypass Pumping for Temporary In-Stream Diversions - Inspect pumps, hoses, and connections daily and during storm events. Ensure proper operation and capacity. Maintain backup equipment and stabilize discharge areas.

Record the inspection date and summary of findings within 24 hours of completing a site inspection.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL NARRATIVE

SCHEDULES B AND C: LIBERTY LOOP TRAIL (SW) AND DAGMAR DALE NATURE TRAIL

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UPPER DELAWARE CONSERVATION DISTRICT
SOIL EROSION AND SEDIMENT CONTROL
GENERAL NOTES

1. The District shall be represented at the project preconstruction meeting with the township engineer, contractors, and utility representatives. If the township engineer does not schedule a preconstruction meeting, it is the responsibility of the owner/applicant to schedule one before any land disturbance. Two weeks notice must be given for scheduling preconstruction meetings.

2. Failure of the aforementioned plan shall not relieve the applicant of any of its responsibilities relevant to the appropriate statutes. Additional erosion and sediment control measures may be required as deemed necessary by the District in the event of any unforeseen problems incurred during construction.

3. Any changes of approved plans shall require an additional submittal to the District including appropriate re-review fees.

4. A 48 hour Start of Land Disturbance Notification shall be given.

5. In that *N.J.S.A. 4-24-39 et seq* requires that no Certificates of Occupancy be issued by the municipality before the provisions of the certified plan for soil erosion and sediment control have been complied with for permanent measures. All site work relative to approved plans and all work around individual lots in subdivisions will be completed before the District issues a Certificate of Compliance. Two weeks notice must be given to the District to schedule inspection for Certificate of Compliance release.

6. Final stabilization of all land disturbances associated with underground utilities, irrespective of phasing, is the ultimate responsibility of the owner.

7. A cash bond of not less than $2,500. (Per disturbed acre or part thereof, or a lot) will be posted with the Upper Delaware Conservation District during the non-growing season if a Certificate of Compliance is needed and soil erosion and sediment control measures for permanent stabilization are not completed.

8. Sediment tracked onto public right-of-ways shall be swept at the end of each working day.

9. No building permits will be released until all soil erosion and sediment control measures as showed on approved plans are installed.

10. Dust to be controlled with water, calcium chloride or other method approved by the Soil Conservation District.

11. Tracking pad to be kept clean and repaired as necessary.

12. Soil Erosion and Sediment Control measures shall be in accordance with *Standards for Soil Erosion and Sediment Control in New Jersey, February 2014.*

13. See Detail Sheets for Additional Soil and Sediment Control Details.

Disturbance Calculations Summary

	Temporary Disturbance (acre)	Permanent Disturbance (acre)	Total Disturbance (acre)
Dagmar Dale Staging Area	0.35	0.00	0.35
Dagmar Dale Site #1	0.04	0.11	0.15
Dagmar Dale Site #2	0.06	0.01	0.07
Dagmar Dale Site #3	0.06	0.03	0.09
Dagmar Dale Site #4	0.07	0.05	0.12
Dagmar Dale Site #5	0.11	0.00	0.11
Dagmar Dale Site #6	0.06	0.07	0.13
Liberty Loop Trail (SW)	0.42	1.22	1.64
Total	**1.19**	**1.48**	**2.67**

Impact Calculations Summary

	State Open Water		Wetland		Transition Area		Riparian Zone	
	Temporary Disturbance (acre)	Permanent Disturbance (acre)	Temporary Disturbance (acre)	Permanent Disturbance (acre)	Temporary Disturbance (acre)	Permanent Disturbance (acre)	Temporary Disturbance (acre)	Permanent Disturbance (acre)
Dagmar Dale Staging Area	-	-	-	-	-	-	-	-
Dagmar Dale Site #1	0.013	0.063	-	-	-	-	0.023	0.104
Dagmar Dale Site #2	-	-	-	-	-	-	-	-
Dagmar Dale Site #3	0.005	0.00001	0.007	0.005	0.014	0.021	0.026	0.026
Dagmar Dale Site #4	-	-	-	-	0.021	0.036	0.032	0.050
Dagmar Dale Site #5	-	-	-	-	-	-	-	-
Dagmar Dale Site #6	0.015	0.036	-	-	0.014	0.035	0.029	0.070
Liberty Loop Trail (SW)	-	-	-	-	-	-	-	-
Total	**0.034**	**0.098**	**0.007**	**0.005**	**0.049**	**0.091**	**0.110**	**0.250**
	0.132		**0.011**		**0.140**		**0.360**	

Impact Calculations Summary (continued)

	Stream		Floodway	
	Temporary Disturbance (acre)	Permanent Disturbance (acre)	Temporary Disturbance (acre)	Permanent Disturbance (acre)
Dagmar Dale Staging Area	-	-	-	-
Dagmar Dale Site #1	0.013	0.063	-	-
Dagmar Dale Site #2	-	-	-	-
Dagmar Dale Site #3	0.005	0.000001	-	-
Dagmar Dale Site #4	-	-	-	-
Dagmar Dale Site #5	-	-	-	-
Dagmar Dale Site #6	0.014	0.035	-	-
Liberty Loop Trail (SW)	-	-	0.428	1.214
Total	**0.032**	**0.098**	**0.428**	**1.214**
	0.130		**1.642**	

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4 August 2026 8:05 AM

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL NARRATIVE

SCHEDULES B AND C: LIBERTY LOOP TRAIL (SW) AND DAGMAR DALE NATURE TRAIL

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M04

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U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE A: WINDING WATERS TRAIL STAGING AREA

SHEET 1 OF 4

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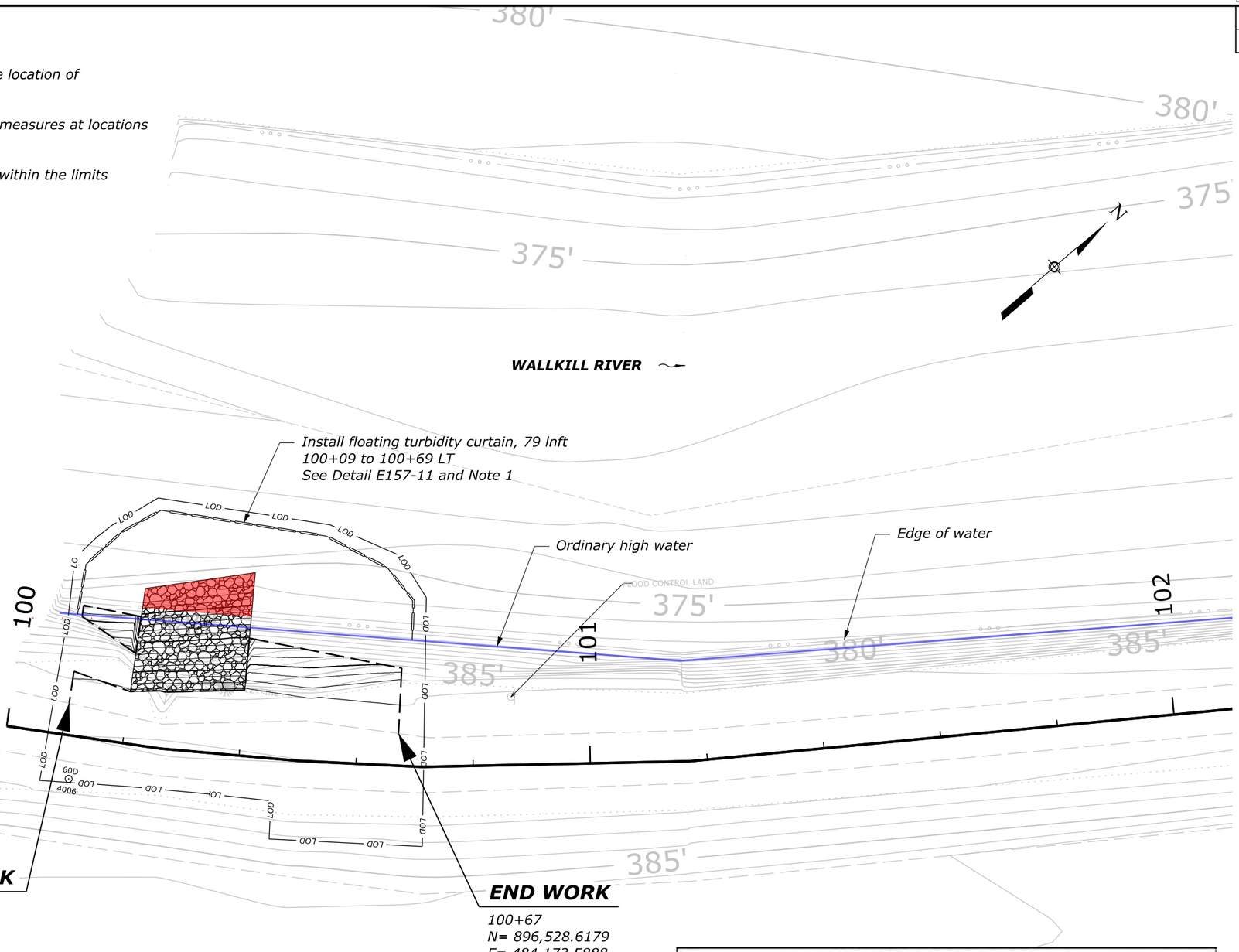
NOTES:

1. *Submit a plan for approval for the approximate location of floating turbidity curtains.*
2. *Place temporary erosion and sediment control measures at locations approved by the CO.*
3. *See Plan and Profile sheets for proposed work within the limits of disturbance.*



STREAM BED IMPACTS - SITE #3			
Impact Area	**Area (SQFT)**	**Area (ACRE)**	**Length (LNFT)**
Disturbed area	*2,330*	*0.053*	
Permanent Stream Bed Impacts	*100*	*0.002*	*19*

NO.	DATE	BY	REVISIONS

LEGEND

Limits of Disturbance

Permanent Stream Bed Impacts

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE A: WINDING WATERS TRAIL SITE #3
STREAM BED IMPACTS

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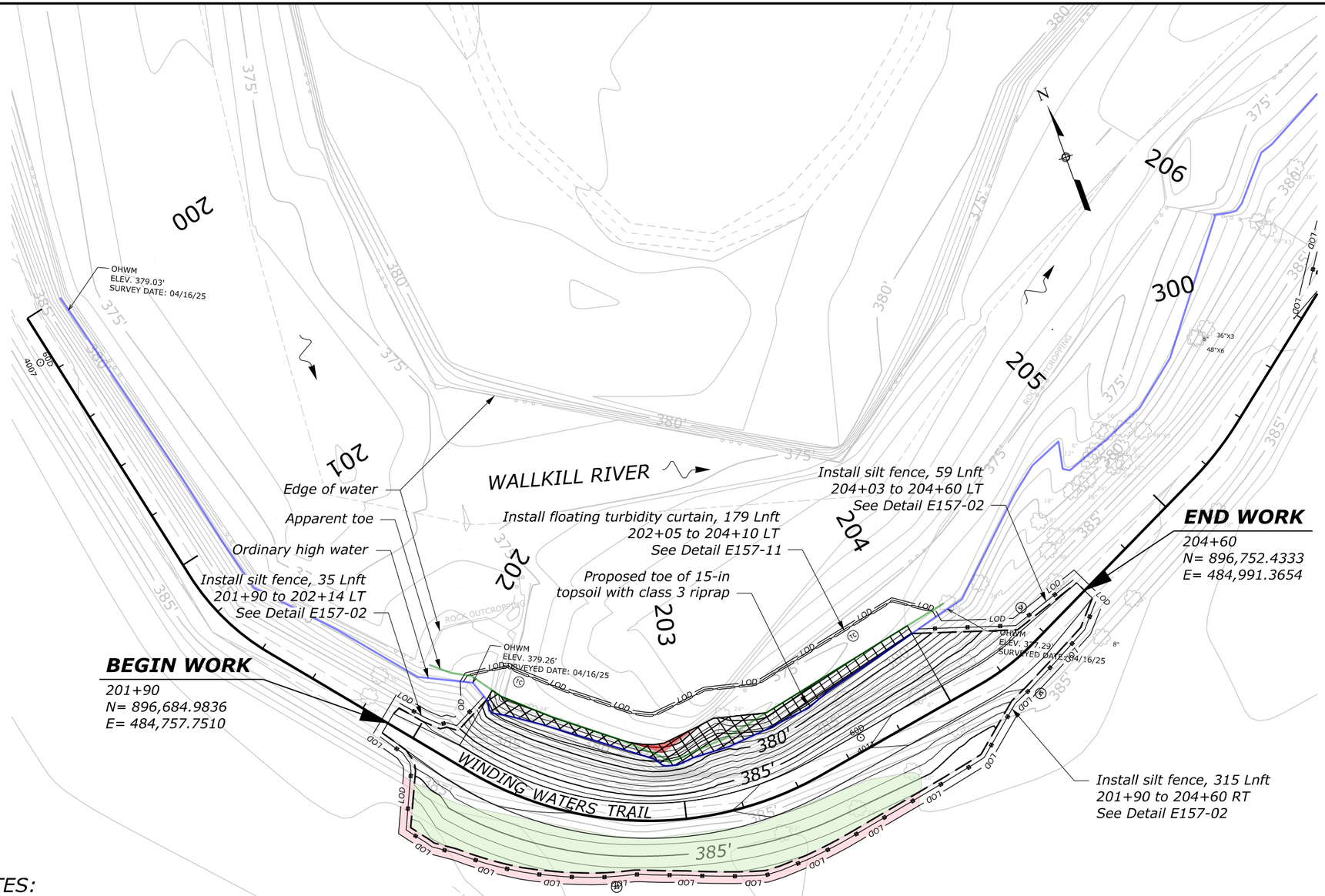
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M06



BEGIN WORK

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N= 896,684.9836
E= 484,757.7510

END WORK

204+60
N= 896,752.4333
E= 484,991.3654

NOTES:

1. Submit a plan for approval for the approximate location of floating turbidity curtains.
2. Place temporary erosion and sediment control measures at locations approved by the CO.
3. See Plan and Profile sheets for proposed work within the limits of disturbance.

WETLAND AND STREAM BED IMPACTS - SITE #240

Impact Area	Area (SQFT)	Area (ACRE)	Length (LNFT)
Disturbed area	12,786	0.294	
Permanent Wetland Impacts	2,568	0.059	
Temporary Wetland Impacts	815	0.019	
Permanent Stream Bed Impacts	41	0.001	42
Permanent Impacts Below OHWM	850	0.020	

NO.	DATE	BY	REVISIONS

LEGEND

- Limits of Disturbance
- Permanent Wetland Impacts
- Temporary Wetland Impacts
- Permanent Stream Bed Impacts
- Permanent Impacts Below OHWM

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE A: WINDING WATERS TRAIL SITE #240
WETLAND & STREAM BED IMPACTS

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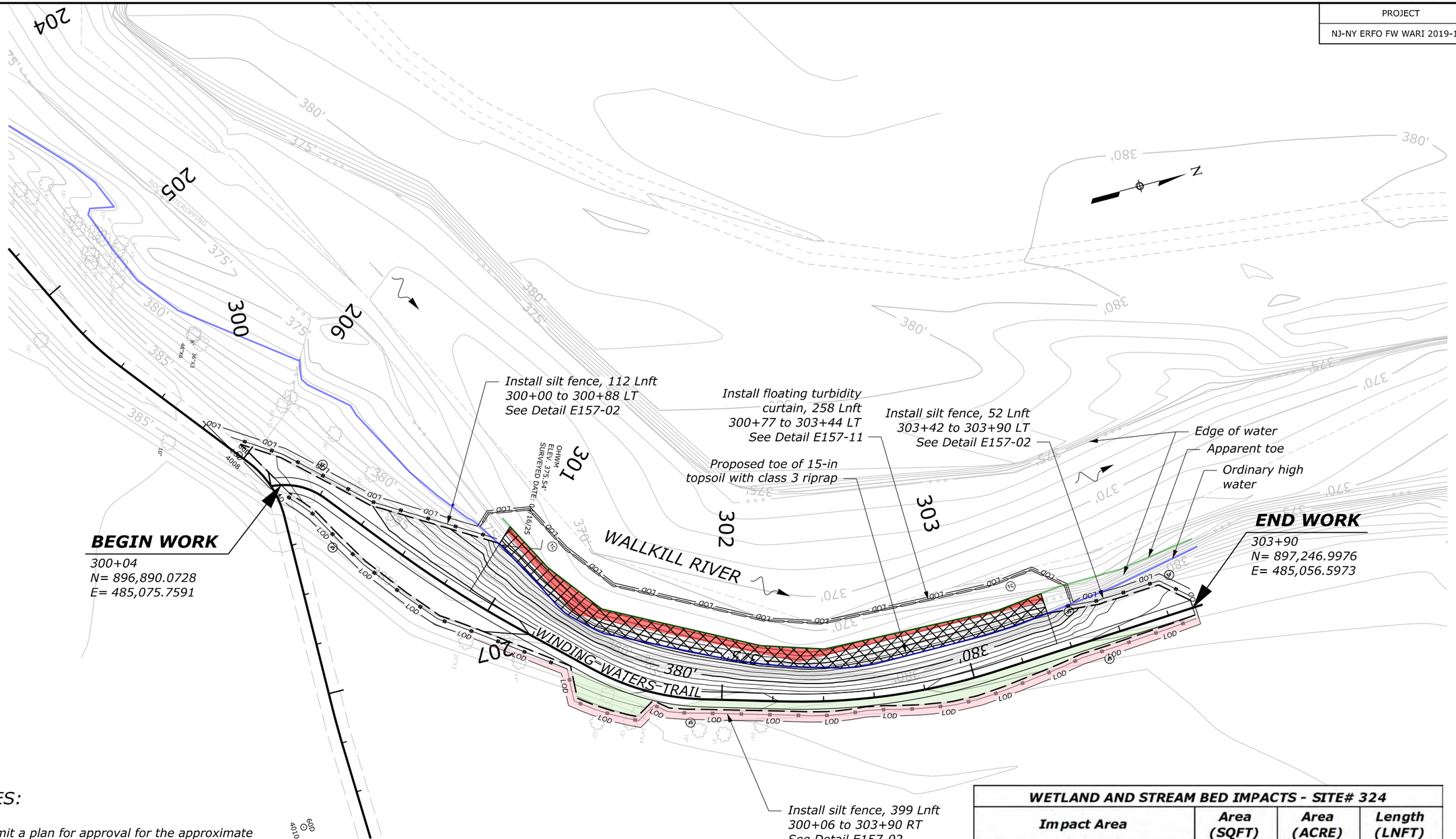
7 August 2026 11:21 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M07



NOTES:

1. Submit a plan for approval for the approximate location of floating turbidity curtains.
2. Place temporary erosion and sediment control measures at locations approved by the CO.
3. See Plan and Profile sheets for proposed work within the limits of disturbance.

WETLAND AND STREAM BED IMPACTS - SITE# 324

Impact Area	Area (SQFT)	Area (ACRE)	Length (LNFT)
Disturbed area	12,723	0.292	
Permanent Wetland Impacts	1,140	0.026	
Temporary Wetland Impacts	1,116	0.026	
Permanent Stream Bed Impacts	726	0.017	230
Permanent Impacts Below OHWM	1,947	0.045	

NO.	DATE	BY	REVISIONS

LEGEND

- Limits of Disturbance
- Permanent Stream Bed Impacts
- Permanent Wetland Impacts
- Permanent Impacts Below OHWM
- Temporary Wetland Impacts

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE A: WINDING WATERS TRAIL SITE #324
WETLAND & STREAM BED IMPACTS

SHEET 4 OF 4

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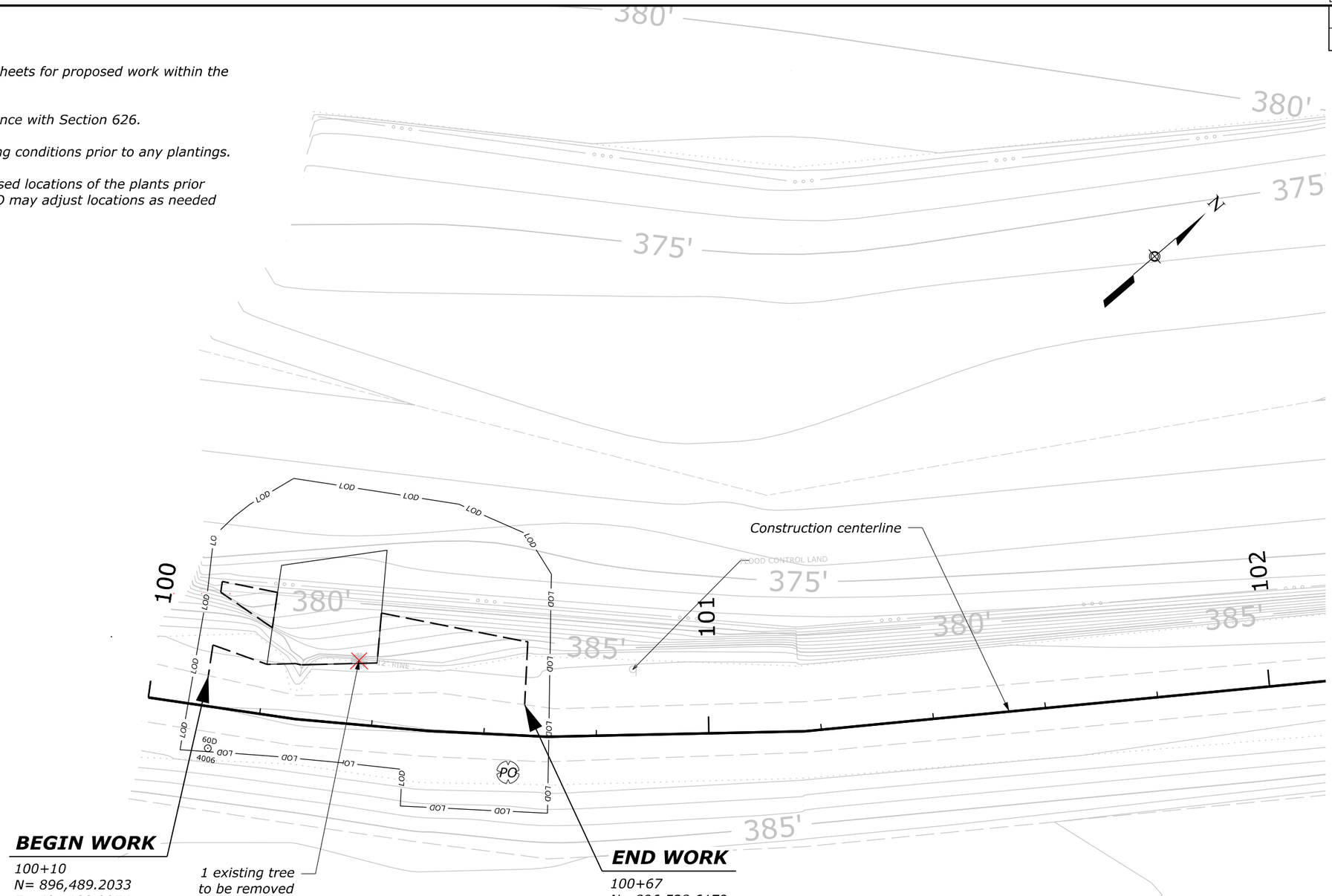
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M08

NOTES:

1. *See Plan and Profile sheets for proposed work within the limits of disturbance.*
2. *Plant trees in accordance with Section 626.*
3. *Field verify the existing conditions prior to any plantings.*
4. *Field verify the proposed locations of the plants prior to installation. The CO may adjust locations as needed in the field.*



BEGIN WORK

100+10
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E= 484,133.2054

END WORK

100+67
N= 896,528.6179
E= 484,173.5888

PLANT SCHEDULE - SITE #3

Symbol	*Scientific Name*	*Common Name*	*Type*	*Quantity*
PO	*Platinus Occidentalis*	*American Sycamore*	*Tree*	*1*

NO.	DATE	BY	REVISIONS

LEGEND:

Proposed new tree

Removed tree

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

REVEGETATION PLAN

SCHEDULE A: WINDING WATERS TRAIL SITE #3

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Disaster No. NJ2019-1-FWS and NY2019-1-FWS

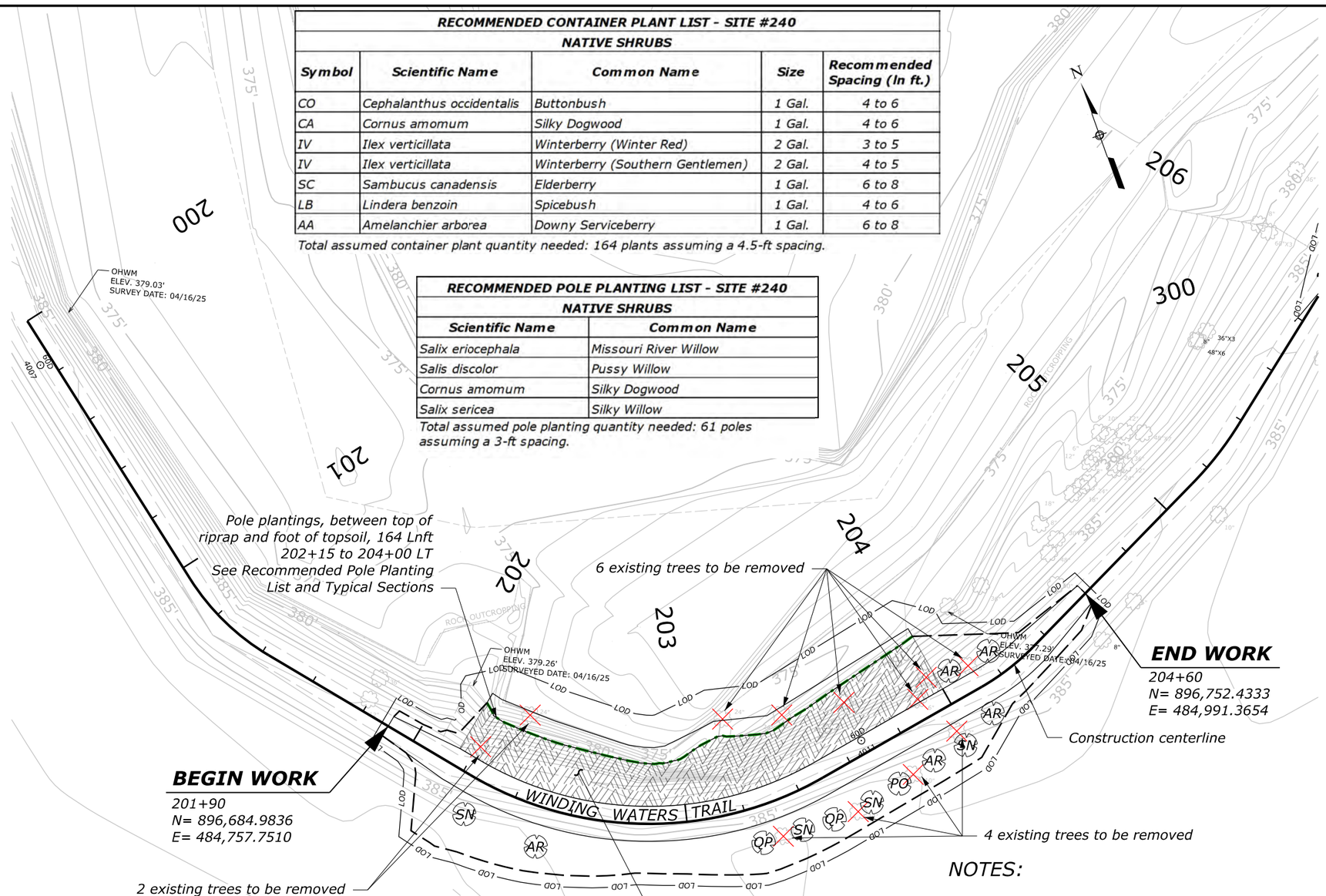
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M09

RECOMMENDED CONTAINER PLANT LIST - SITE #240				
NATIVE SHRUBS				
Symbol	**Scientific Name**	**Common Name**	**Size**	**Recommended Spacing (In ft.)**
CO	Cephalanthus occidentalis	Buttonbush	1 Gal.	4 to 6
CA	Cornus amomum	Silky Dogwood	1 Gal.	4 to 6
IV	Ilex verticillata	Winterberry (Winter Red)	2 Gal.	3 to 5
IV	Ilex verticillata	Winterberry (Southern Gentlemen)	2 Gal.	4 to 5
SC	Sambucus canadensis	Elderberry	1 Gal.	6 to 8
LB	Lindera benzoin	Spicebush	1 Gal.	4 to 6
AA	Amelanchier arborea	Downy Serviceberry	1 Gal.	6 to 8

Total assumed container plant quantity needed: 164 plants assuming a 4.5-ft spacing.

RECOMMENDED POLE PLANTING LIST - SITE #240	
NATIVE SHRUBS	
Scientific Name	**Common Name**
Salix eriocephala	Missouri River Willow
Salis discolor	Pussy Willow
Cornus amomum	Silky Dogwood
Salix sericea	Silky Willow

Total assumed pole planting quantity needed: 61 poles assuming a 3-ft spacing.



PLANT SCHEDULE - SITE #240				
Symbol	**Scientific Name**	**Common Name**	**Type**	**Quantity**
AR	Acer Rubrum	Red Maple	Tree	5
PO	Platinus Occidentalis	American Sycamore	Tree	1
SN	Salix Nigra	Black Willow	Tree	4
QP	Quercus Palustris	Pin Oak	Tree	2

NOTES:

1. See Plan and Profile sheets for proposed work within the limits of disturbance.
2. Plant trees in accordance with Section 626.
3. Field verify the existing conditions prior to any plantings.
4. Field verify the proposed locations of the plants prior to installation. The CO may adjust locations as needed in the field.

NO.	DATE	BY	REVISIONS

LEGEND:

Proposed new tree

Removed tree

Container plants

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 20 40
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

REVEGETATION PLAN

SCHEDULE A: WINDING WATERS TRAIL SITE #240

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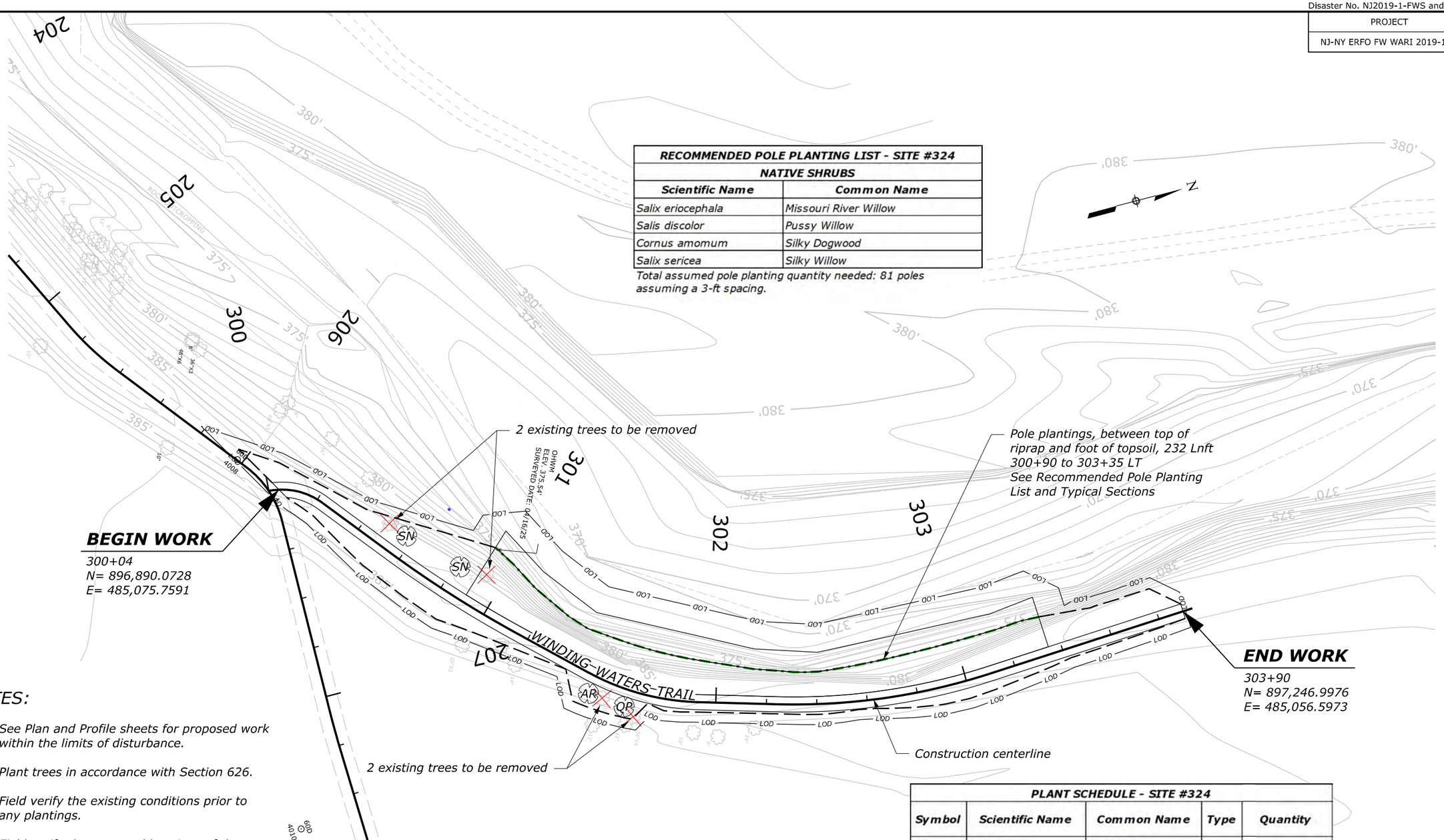
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M10

RECOMMENDED POLE PLANTING LIST - SITE #324	
NATIVE SHRUBS	
Scientific Name	**Common Name**
Salix eriocephala	*Missouri River Willow*
Salis discolor	*Pussy Willow*
Cornus amomum	*Silky Dogwood*
Salix sericea	*Silky Willow*

Total assumed pole planting quantity needed: 81 poles assuming a 3-ft spacing.



NOTES:

1. *See Plan and Profile sheets for proposed work within the limits of disturbance.*
2. *Plant trees in accordance with Section 626.*
3. *Field verify the existing conditions prior to any plantings.*
4. *Field verify the proposed locations of the plants prior to installation. The CO may adjust locations as needed in the field.*
5. *Live stakes/ pole plantings may be used as an alternative to the turf reinforcement mat as approved by the CO.*

PLANT SCHEDULE - SITE #324				
Symbol	**Scientific Name**	**Common Name**	**Type**	**Quantity**
AR	*Acer Rubrum*	*Red Maple*	*Tree*	1
SN	*Salix Nigra*	*Black Willow*	*Tree*	2
QP	*Quercus Palustris*	*Pin Oak*	*Tree*	1

NO.	DATE	BY	REVISIONS

LEGEND:

Proposed new tree

Removed tree

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 20 40
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

REVEGETATION PLAN

SCHEDULE A: WINDING WATERS TRAIL SITE #324

SHEET 3 OF 3

C:\Users\reo.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\M13-M14-NY-WARI_2019-1(1)_Revegetation_Sites 2408324.dgn [Revegetation - Site 324 [Sheet]]
7 August 2026 10:44 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M11

NOTES:

1. See Plan and Profile sheets for proposed work within the limits of disturbance.
2. Provide biological monitoring for the presence of Atlantic Coast leopard frog (ACLF) and other State-listed species in accordance to Sections 107 and 623.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULES A AND B: LIBERTY LOOP TRAIL STAGING AREA

SHEET 1 OF 7

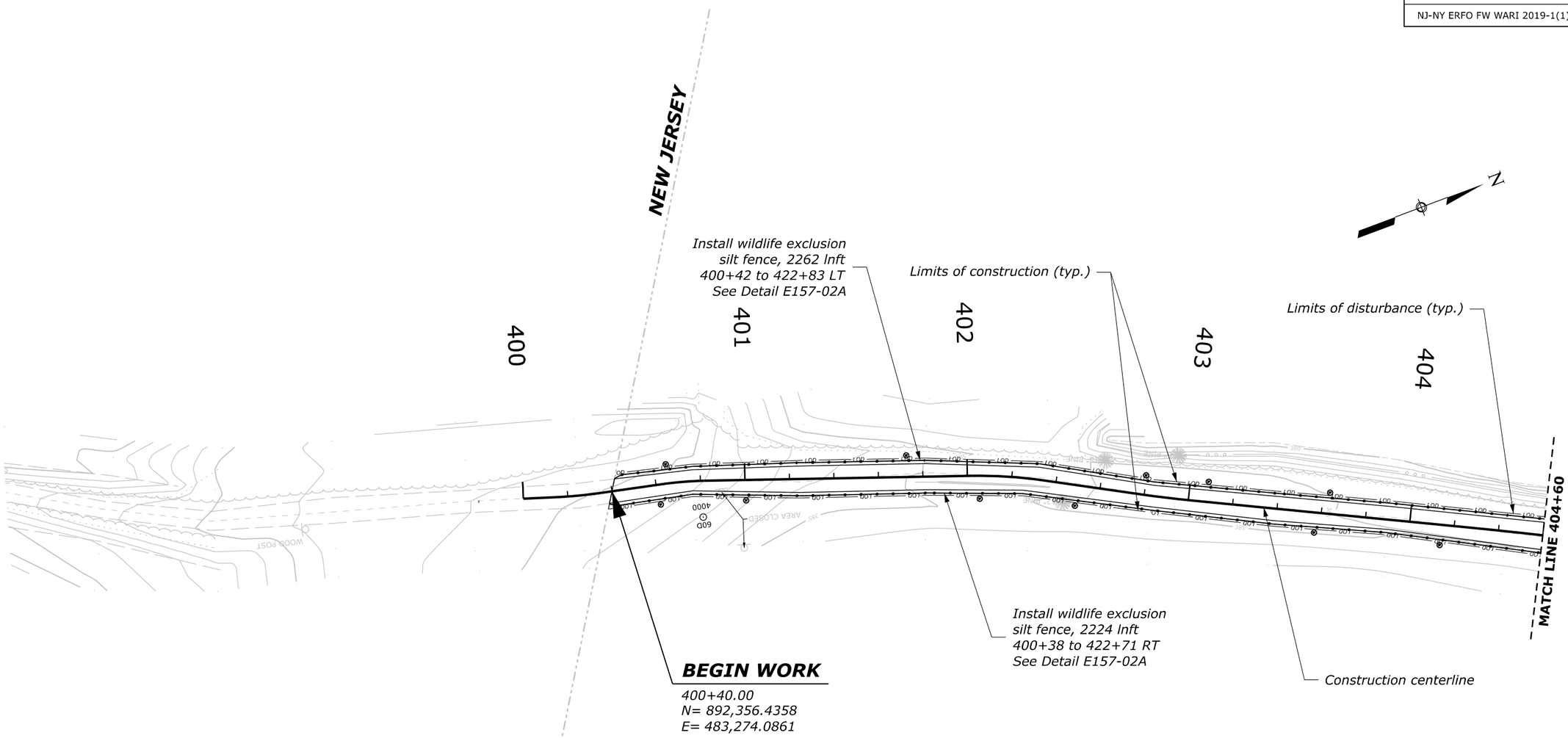
22 October 2026 3:52 PM

\\fhfl15filesrve.flhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\N1 ERFO FW WARI 2019-1(1)\HWY-DES\CADD\(100%) NY ERFO CADD Files\M02-NY-WARI_2019-1(1)_LLStagingArea_esc.dgn [BL03 - Plan 1 [Sheet]]

© 2026 Microsoft Corporation © Maxar ©CNES (2026) Distribution Airbus DS © Mapbox © 2026 TomTom

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M12



NOTES:

1. *See Plan and Profile sheets for proposed work within the limits of disturbance.*
2. *Provide biological monitoring for the presence of Atlantic Coast leopard frog (ACLF) and other State-listed species in accordance to Sections 107 and 623.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 25 50
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)
400+40 TO 404+60

SHEET 2 OF 7

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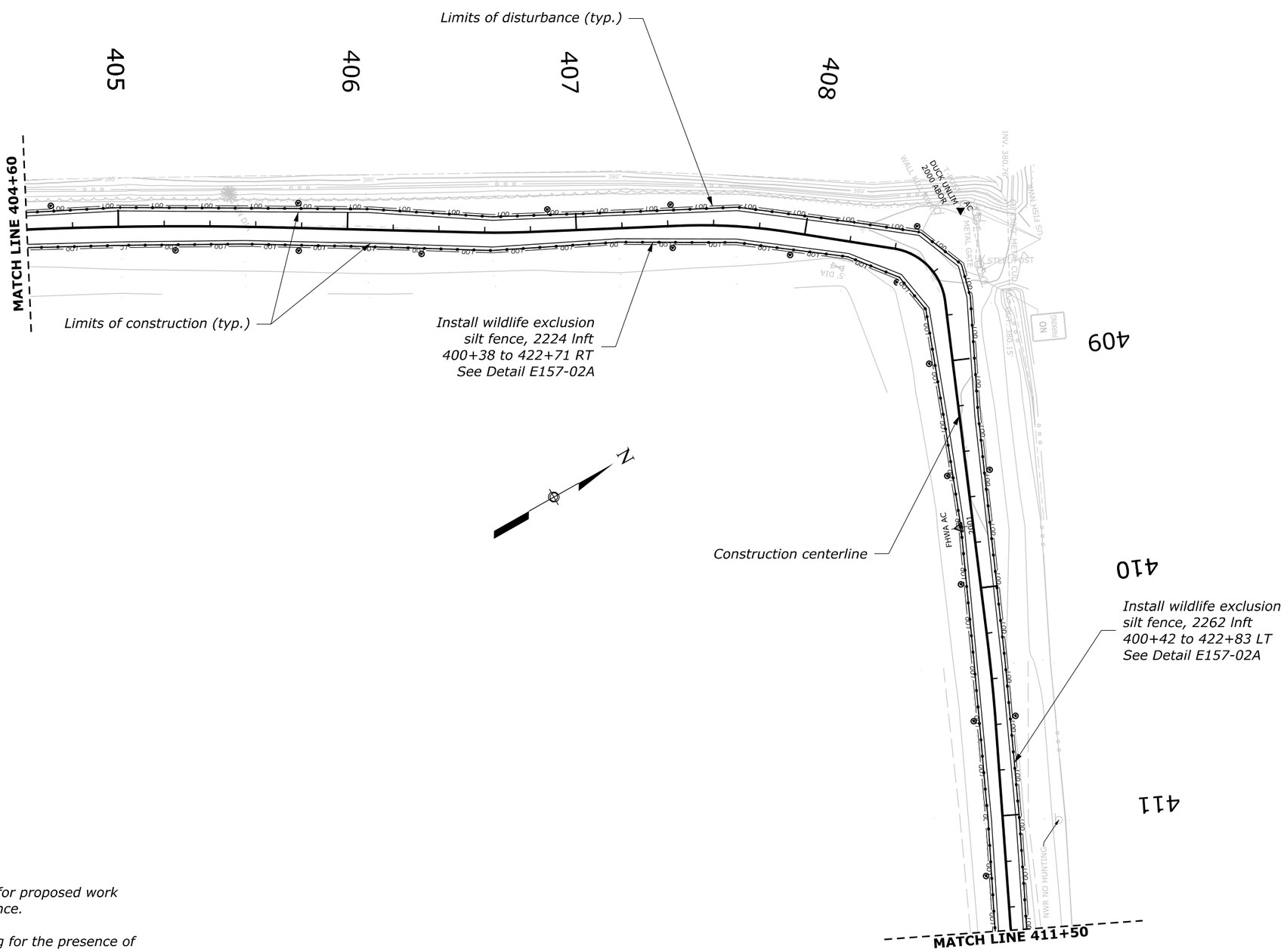
10 August 2026 3:21 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M13



NOTES:

1. *See Plan and Profile sheets for proposed work within the limits of disturbance.*

2. *Provide biological monitoring for the presence of Atlantic Coast leopard frog (ACLF) and other State-listed species in accordance to Sections 107 and 623.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 25 50
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)
404+60 TO 411+50

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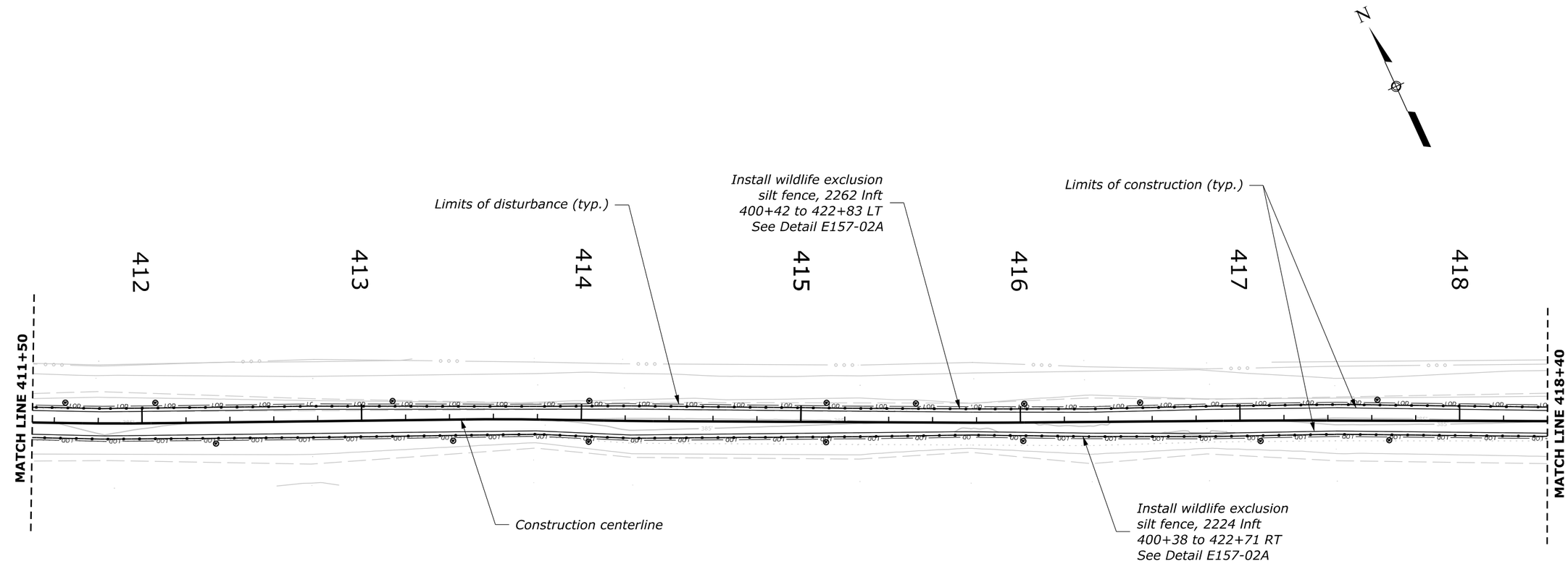
10 August 2026 3:21 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M14



NOTES:

1. See Plan and Profile sheets for proposed work within the limits of disturbance.

2. Provide biological monitoring for the presence of Atlantic Coast leopard frog (ACLF) and other State-listed species in accordance to Sections 107 and 623.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 25 50
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)
411+50 TO 418+40

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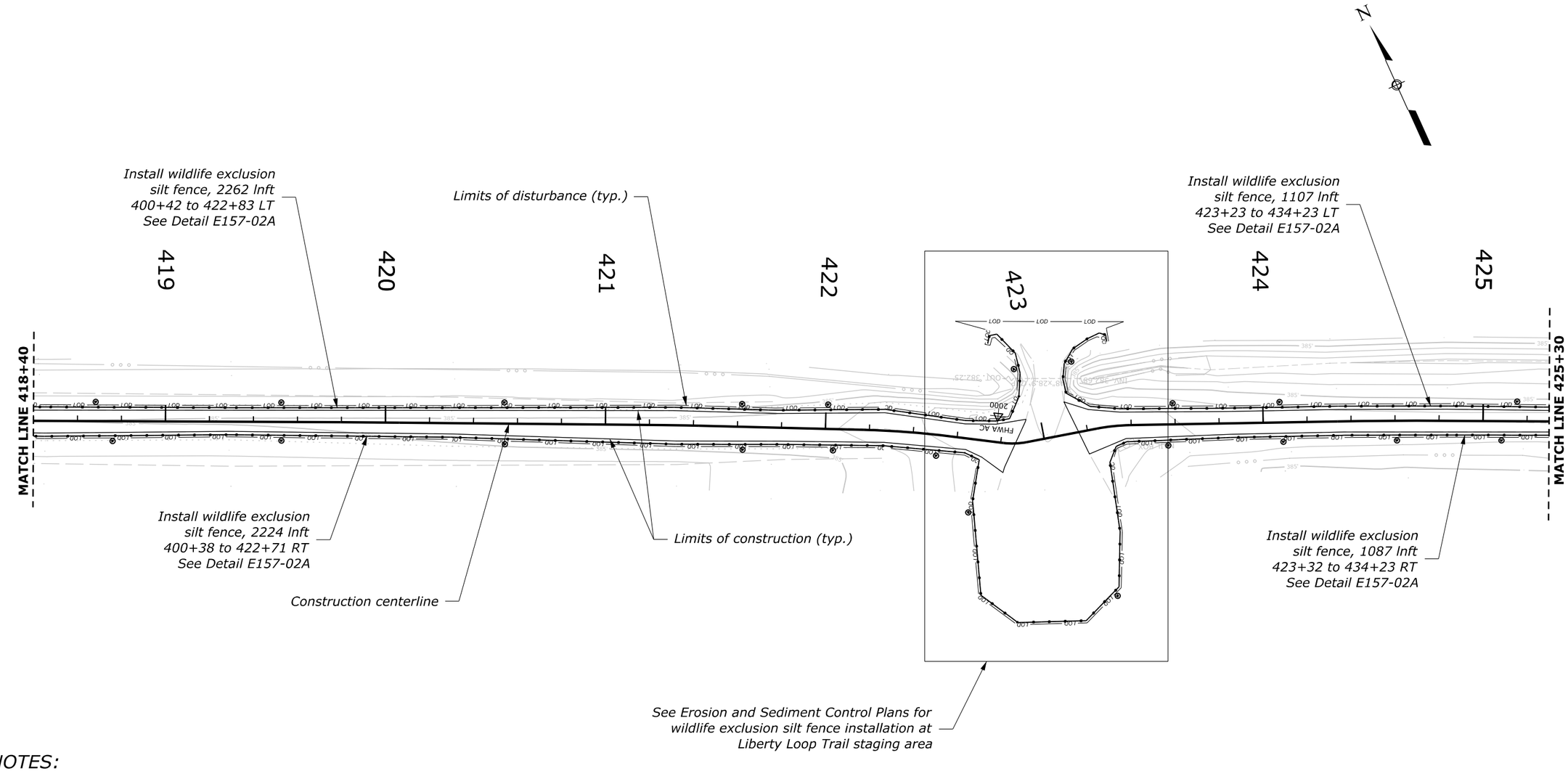
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M15



NOTES:

1. See Plan and Profile sheets for proposed work within the limits of disturbance.
2. Provide biological monitoring for the presence of Atlantic Coast leopard frog (ACLF) and other State-listed species in accordance to Sections 107 and 623.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 25 50
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)
418+40 TO 425+30

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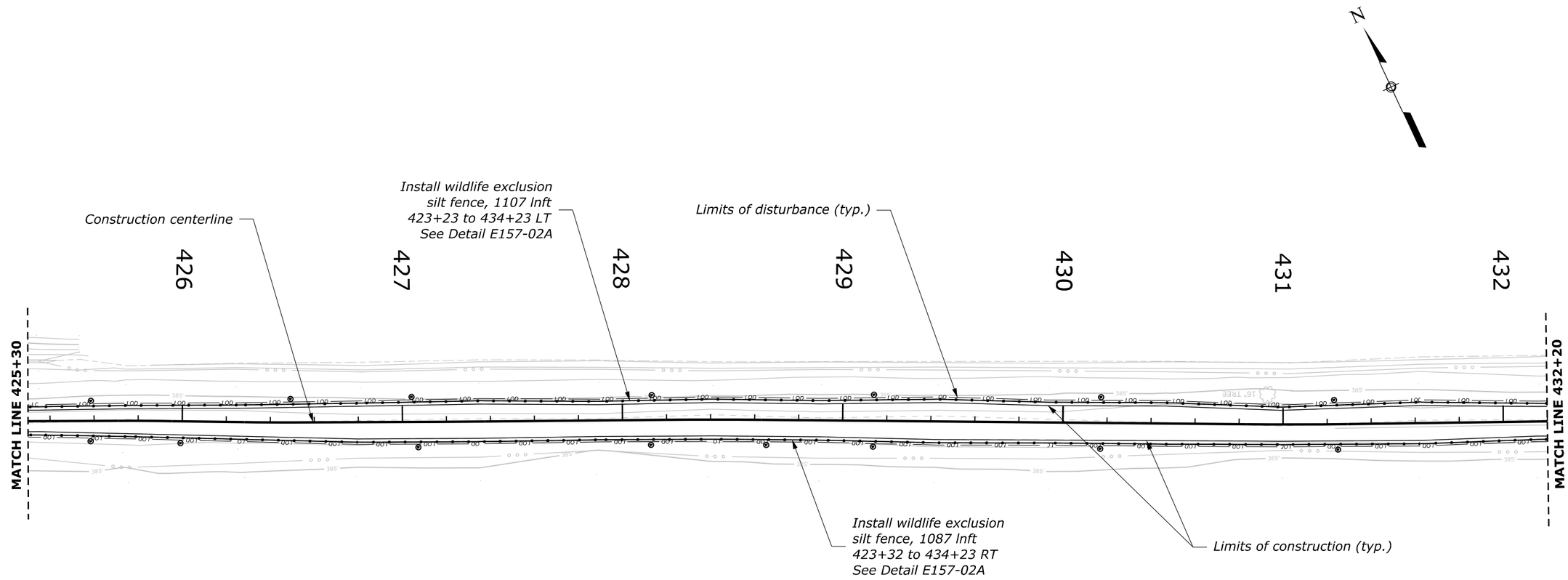
22 October 2026 4:03 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M16



NOTES:

1. See Plan and Profile sheets for proposed work within the limits of disturbance.

2. Provide biological monitoring for the presence of Atlantic Coast leopard frog (ACLF) and other State-listed species in accordance to Sections 107 and 623.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 25 50
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)
425+30 TO 432+20

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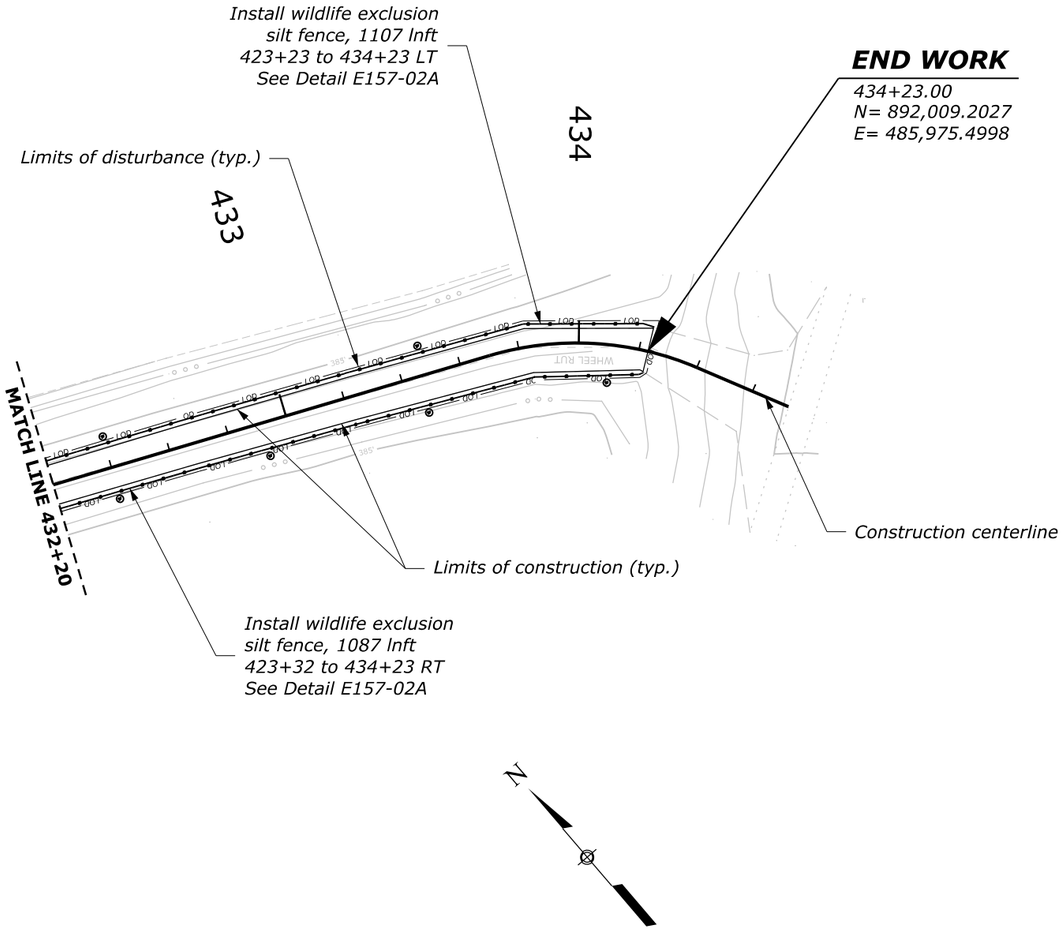
10 August 2026 3:21 PM



NOTES:

1. See Plan and Profile sheets for proposed work within the limits of disturbance.
2. Provide biological monitoring for the presence of Atlantic Coast leopard frog (ACLF) and other State-listed species in accordance to Sections 107 and 623.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 25 50
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)
432+20 TO 434+23

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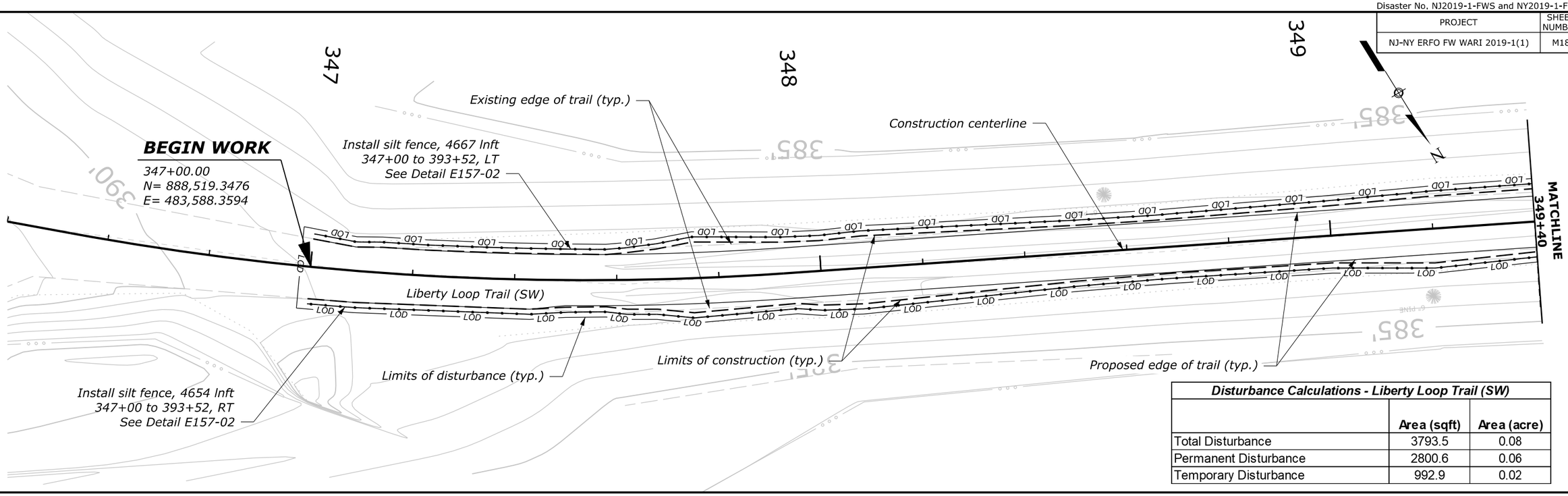
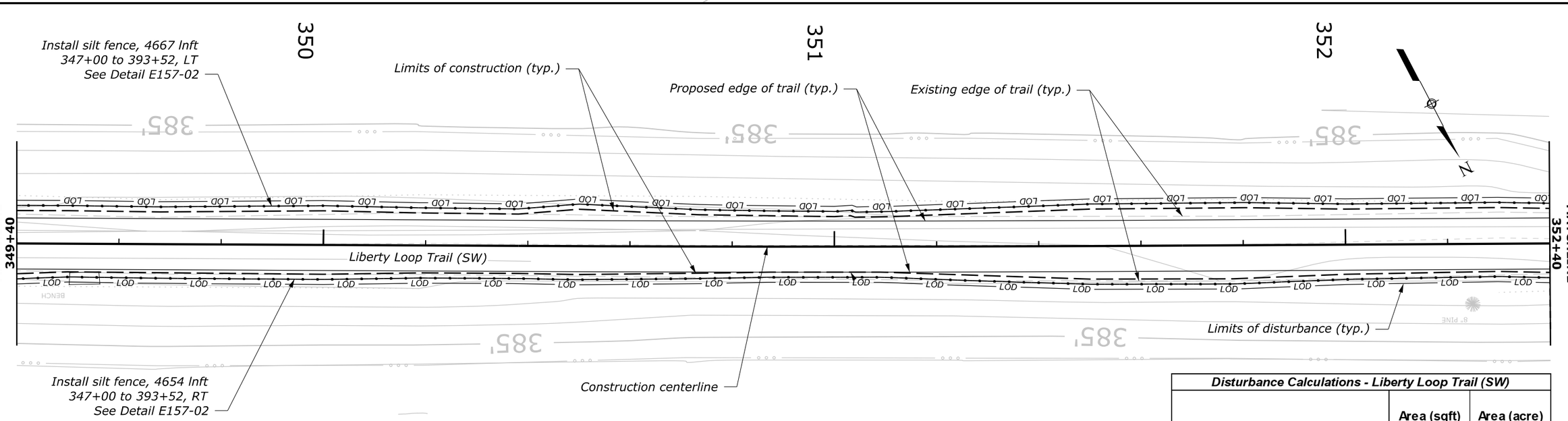
10 August 2026 3:21 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M18



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	3793.5	0.08
Permanent Disturbance	2800.6	0.06
Temporary Disturbance	992.9	0.02



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	4889.7	0.11
Permanent Disturbance	3689.2	0.08
Temporary Disturbance	1200.5	0.03

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
347+00 TO 352+40

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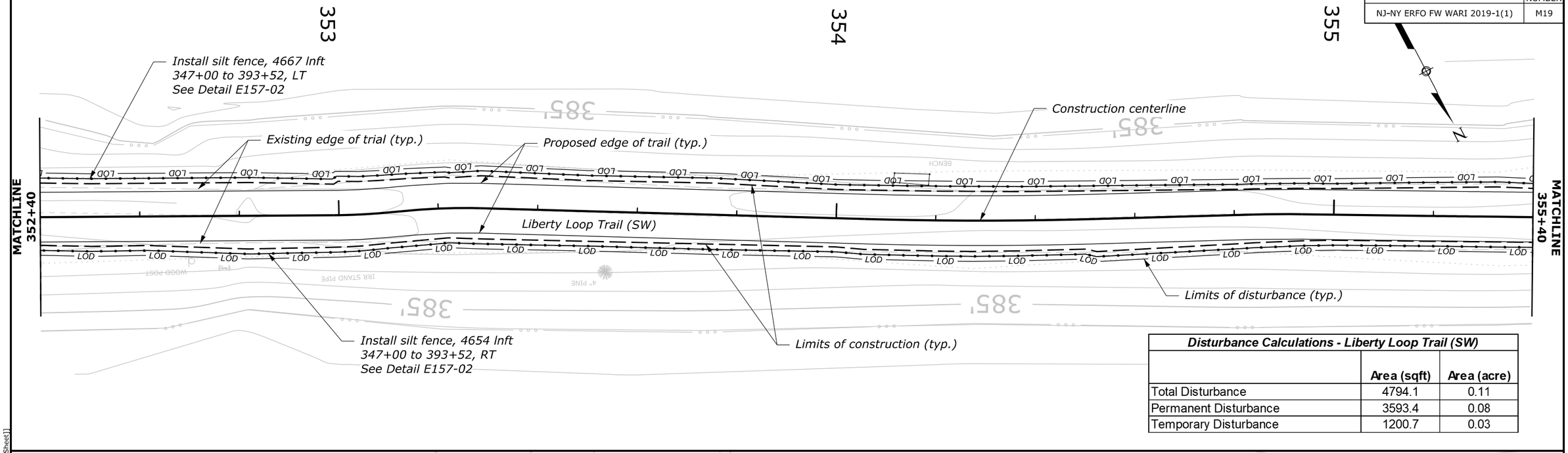
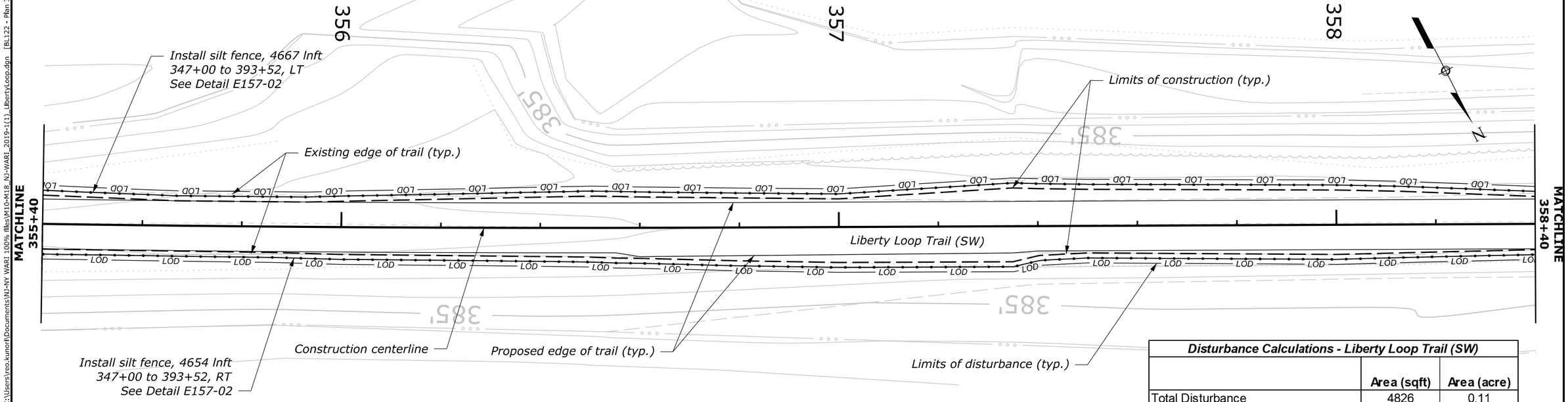
8 August 2026 2:07 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M19



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	4794.1	0.11
Permanent Disturbance	3593.4	0.08
Temporary Disturbance	1200.7	0.03



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	4826	0.11
Permanent Disturbance	3625.3	0.08
Temporary Disturbance	1200.7	0.03

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
352+40 TO 358+40

SHEET 2 OF 9

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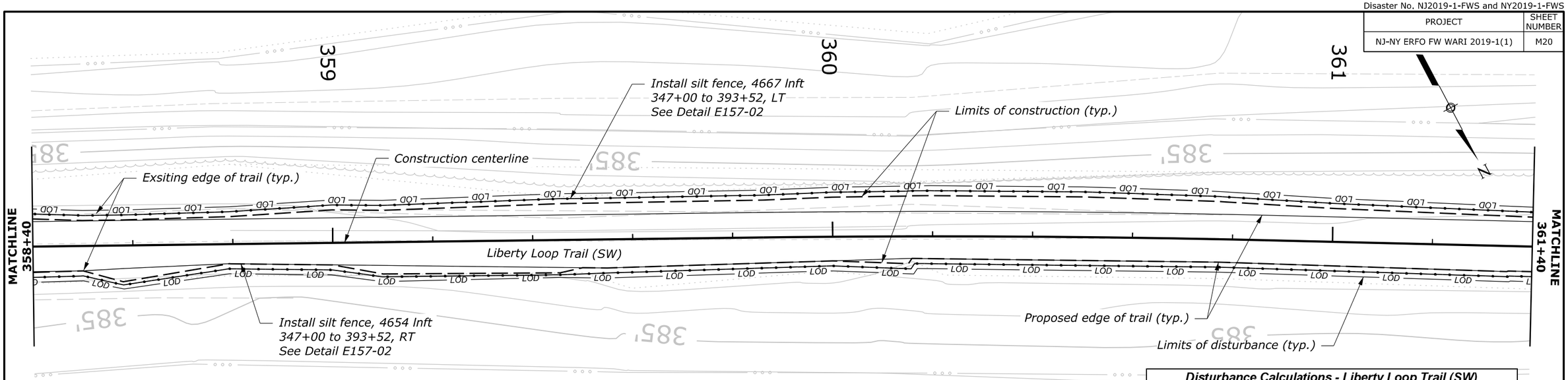
8 August 2026 2:07 PM

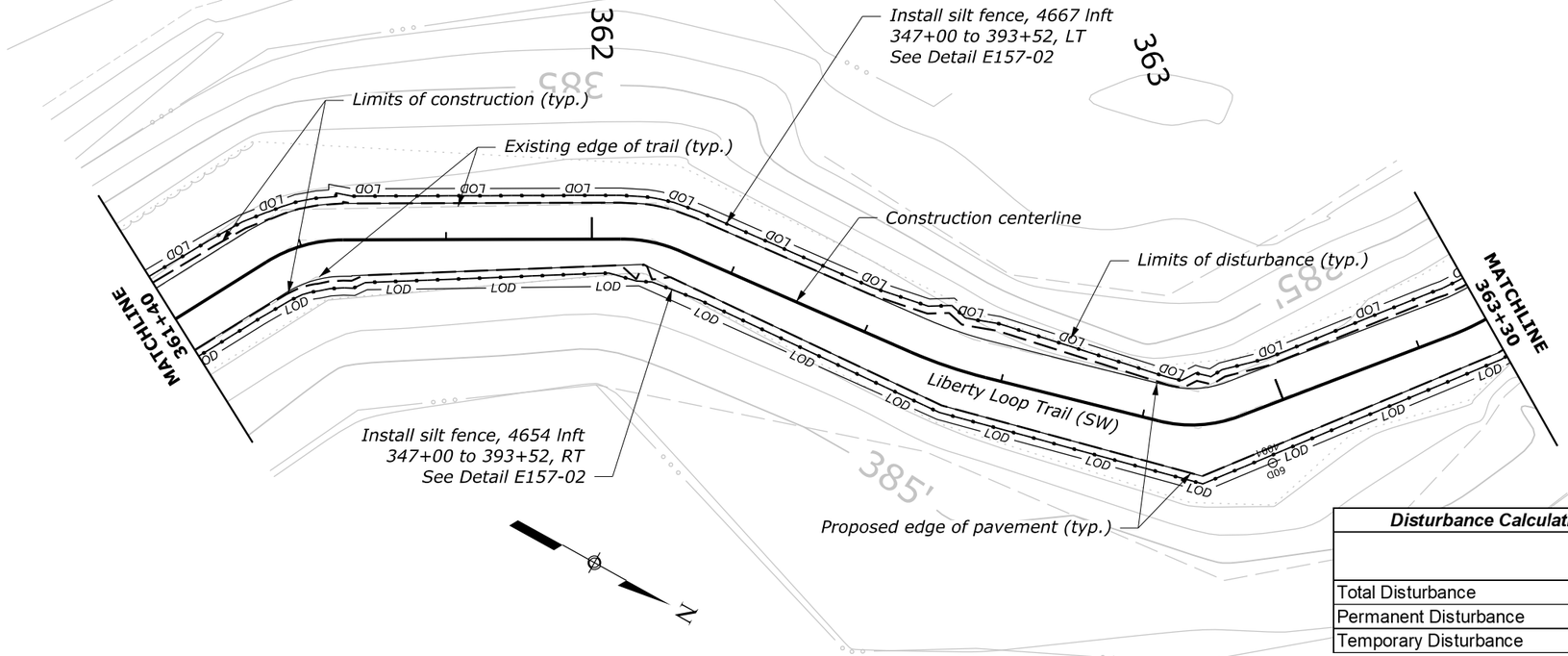
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M20



Disturbance Calculations - Liberty Loop Trail (SW)

	Area (sqft)	Area (acre)
Total Disturbance	4786.41	0.11
Permanent Disturbance	3625.26	0.08
Temporary Disturbance	1161.15	0.03



Disturbance Calculations - Liberty Loop Trail (SW)

	Area (sqft)	Area (acre)
Total Disturbance	2749.98	0.06
Permanent Disturbance	1951.52	0.04
Temporary Disturbance	798.46	0.02

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
358+40 TO 363+30

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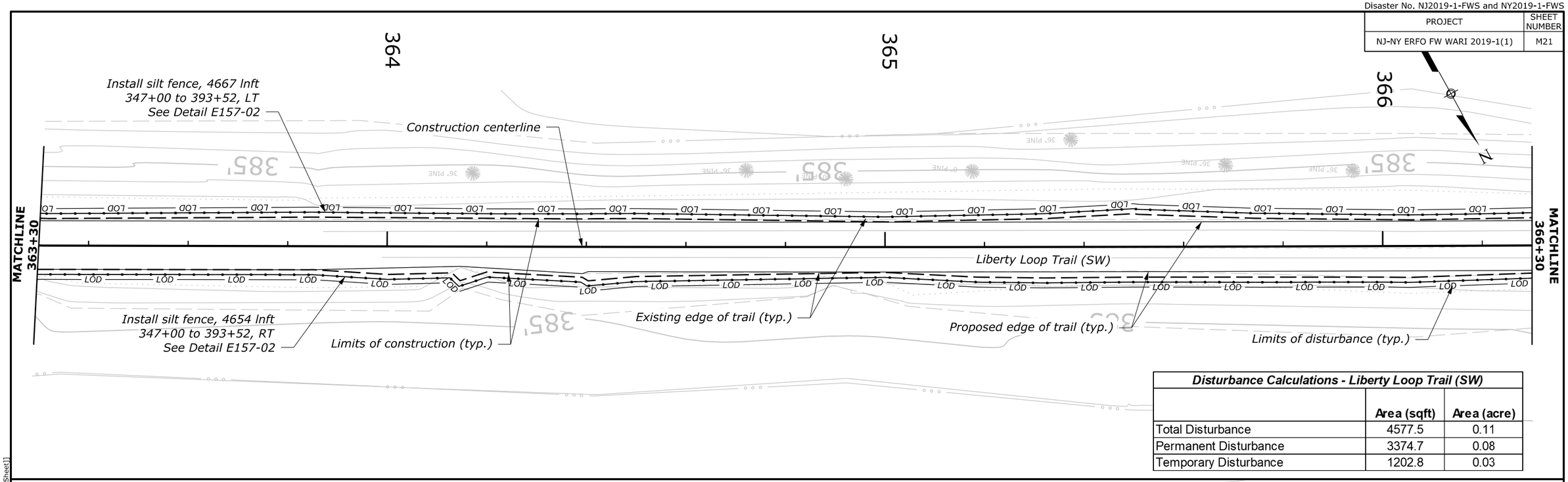
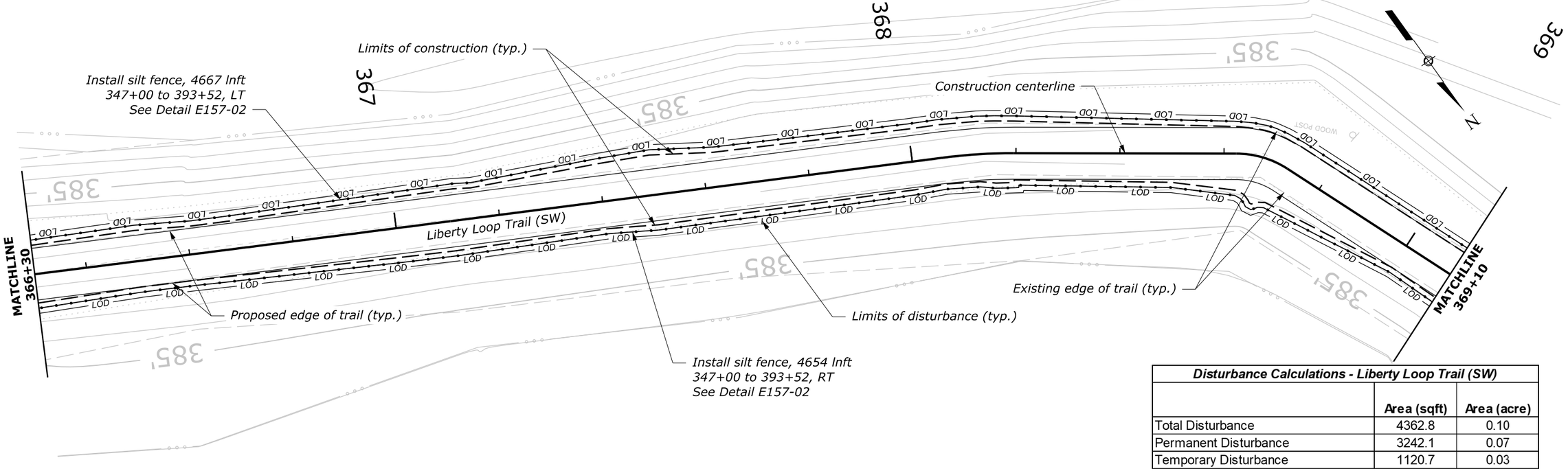
8 August 2026 2:07 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M21



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	4577.5	0.11
Permanent Disturbance	3374.7	0.08
Temporary Disturbance	1202.8	0.03



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	4362.8	0.10
Permanent Disturbance	3242.1	0.07
Temporary Disturbance	1120.7	0.03

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
363+30 TO 369+10

SHEET 4 OF 9

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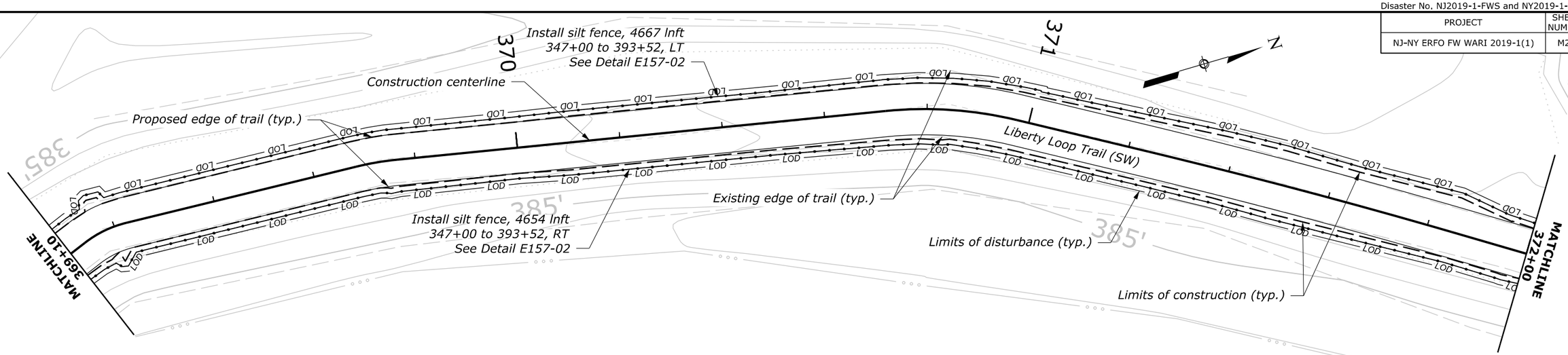
8 August 2026 2:07 PM

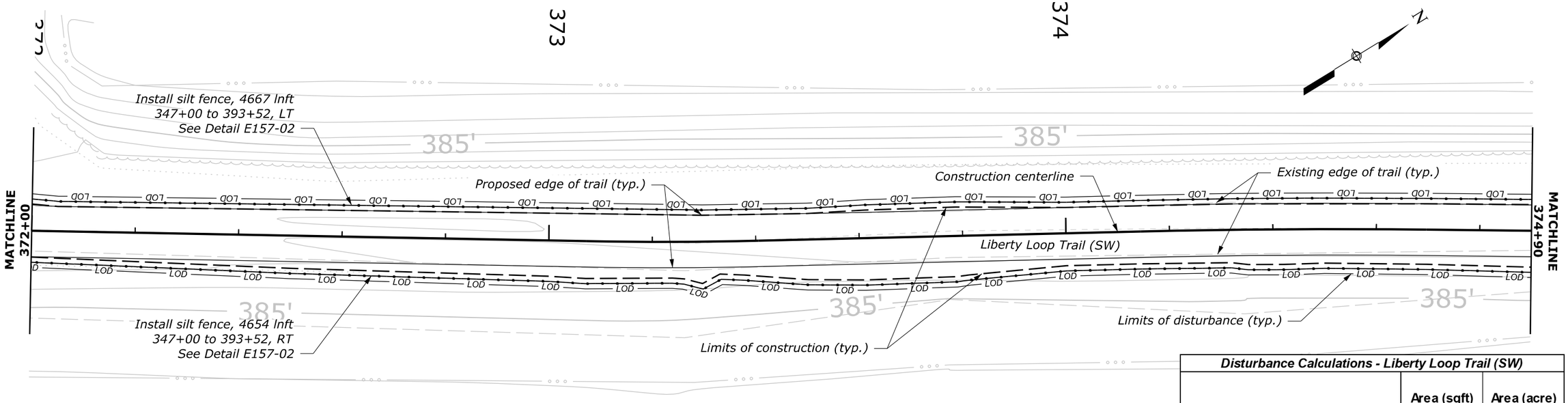
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M22



Disturbance Calculations - Liberty Loop Trail (SW)

	Area (sqft)	Area (acre)
Total Disturbance	4333.4	0.10
Permanent Disturbance	3165.8	0.07
Temporary Disturbance	1167.6	0.03



Disturbance Calculations - Liberty Loop Trail (SW)

	Area (sqft)	Area (acre)
Total Disturbance	4581.6	0.11
Permanent Disturbance	3419.9	0.08
Temporary Disturbance	1161.7	0.03

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
369+10 TO 374+90

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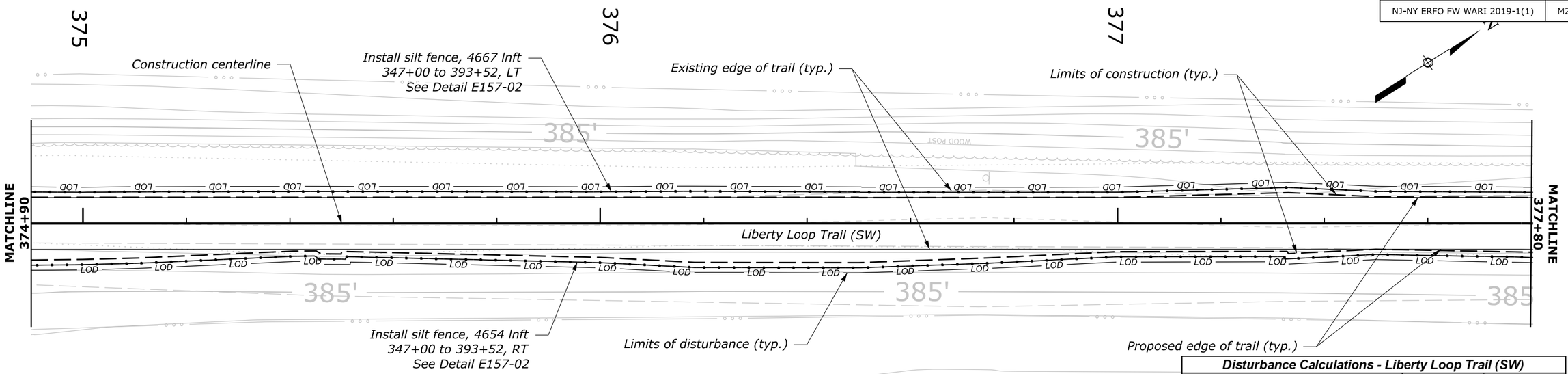
8 August 2026 2:07 PM

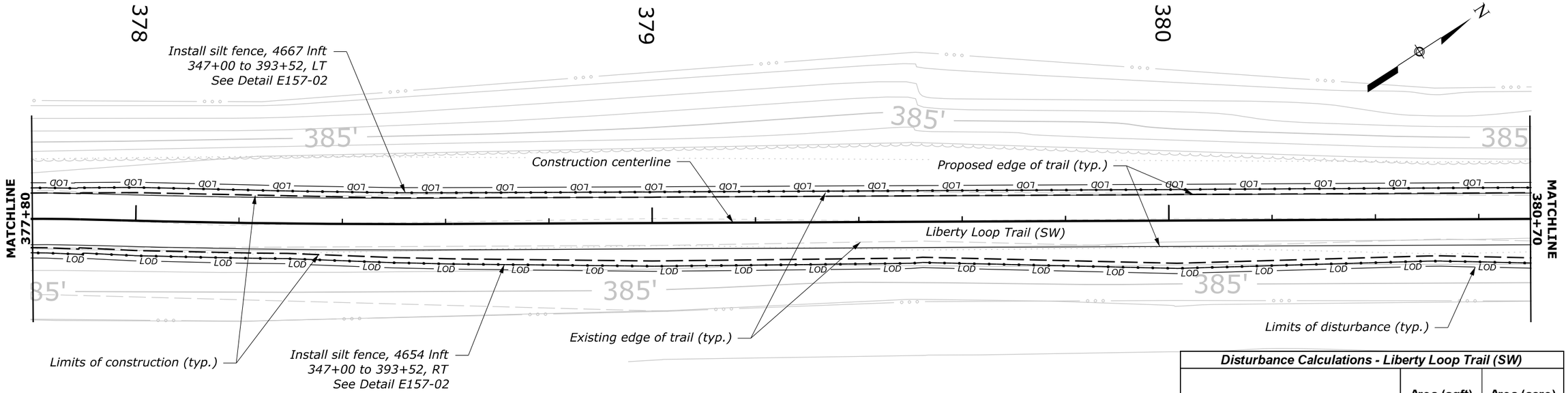
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M23



Disturbance Calculations - Liberty Loop Trail (SW)

	Area (sqft)	Area (acre)
Total Disturbance	4459.8	0.11
Permanent Disturbance	3297.6	0.08
Temporary Disturbance	1162.2	0.03



Disturbance Calculations - Liberty Loop Trail (SW)

	Area (sqft)	Area (acre)
Total Disturbance	4657.3	0.11
Permanent Disturbance	3946.6	0.09
Temporary Disturbance	710.7	0.02

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
374+90 TO 380+70

SHEET 6 OF 9

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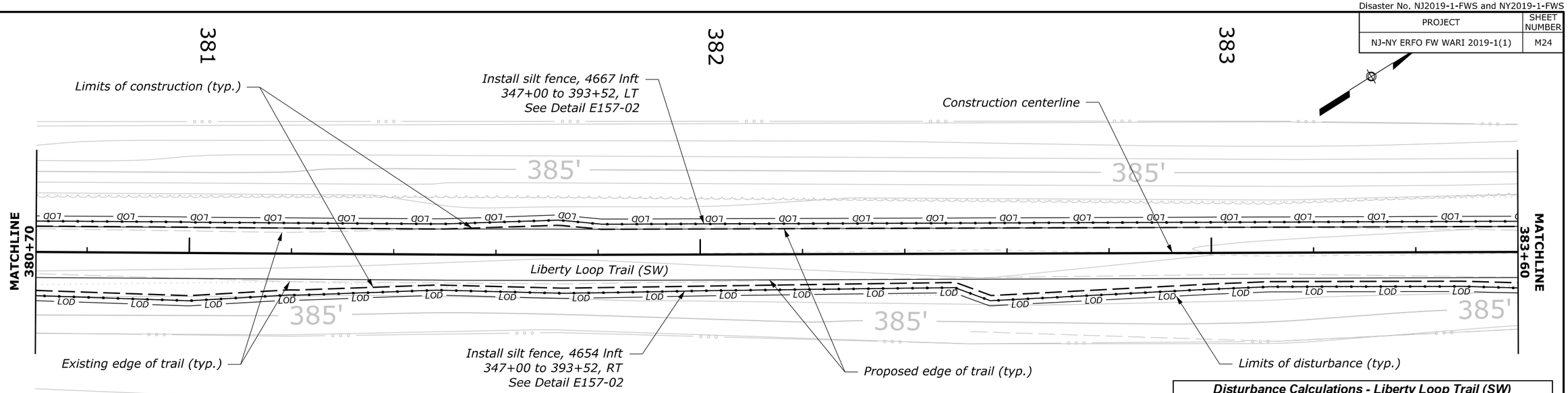
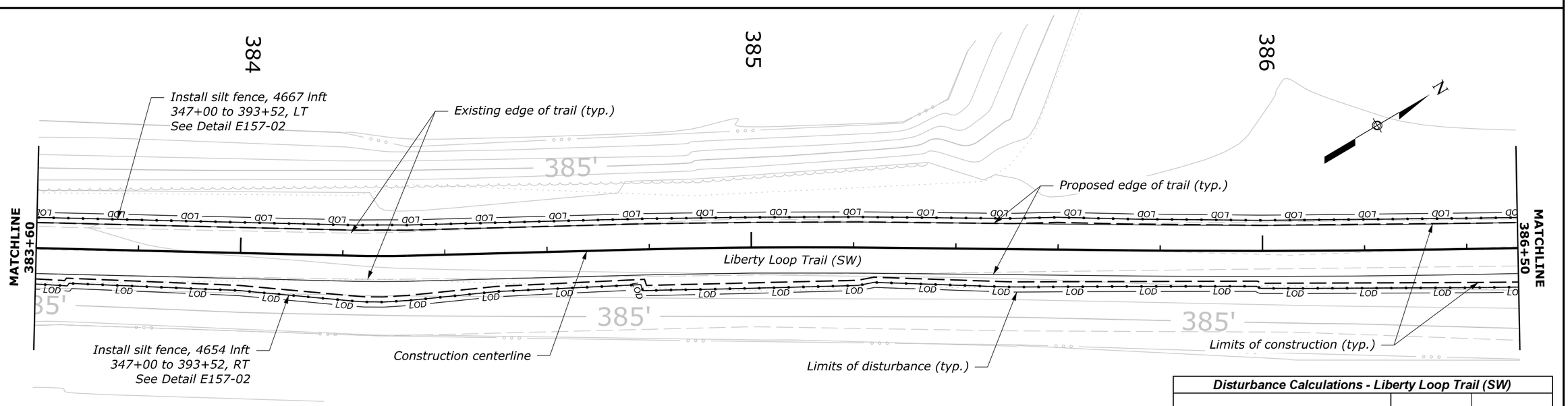
8 August 2026 2:07 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M24



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	4524.7	0.11
Permanent Disturbance	3363.4	0.08
Temporary Disturbance	1161.3	0.03



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	4500.1	0.11
Permanent Disturbance	3337	0.08
Temporary Disturbance	1163.1	0.03

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

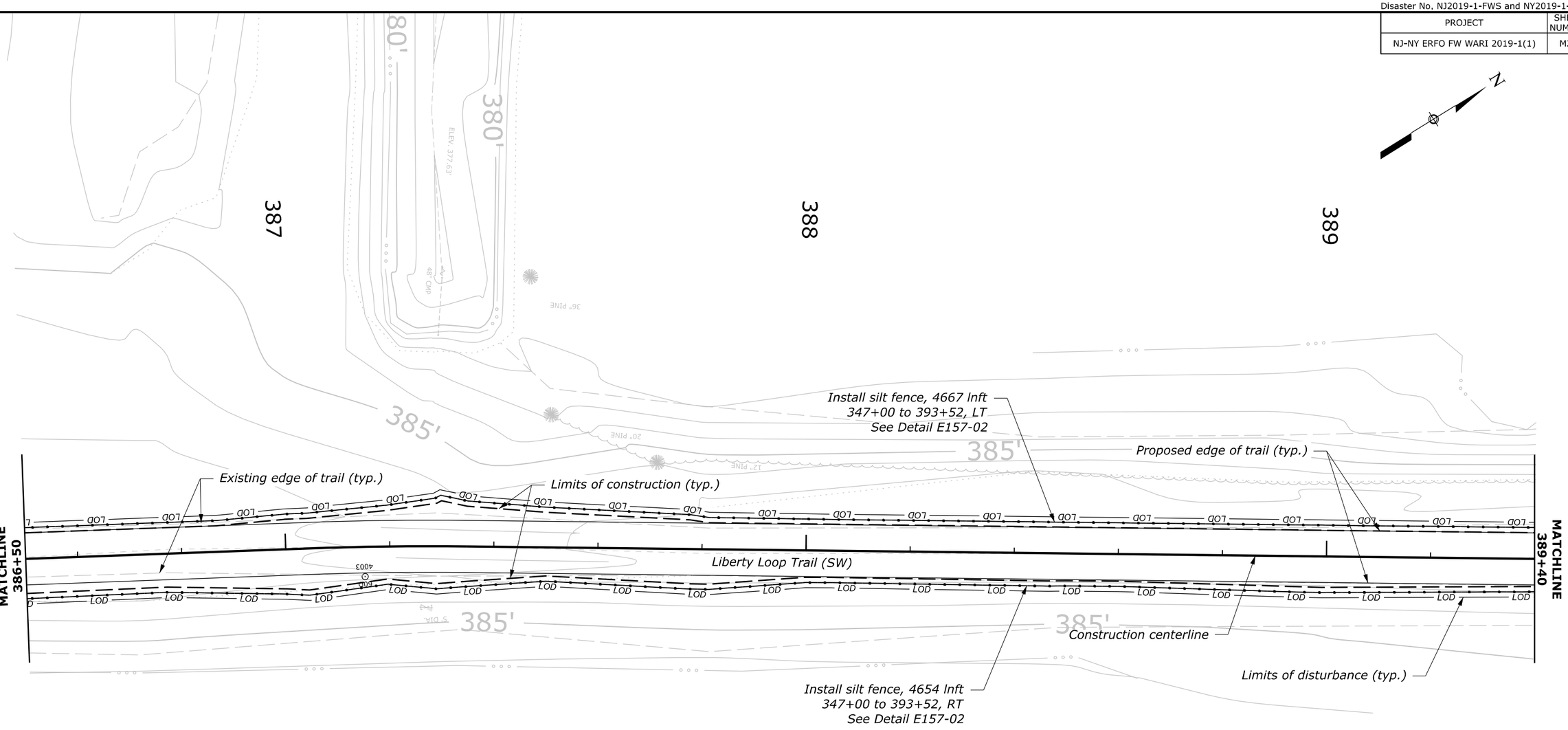
SCHEDULE B: LIBERTY LOOP TRAIL (SW)
380+70 TO 386+50

SHEET 7 OF 9

C:\Users\reo.kunorh\Documents\NJ-NY WARI 100% files\M10-M18_NJ-WARI_2019-1(1)_LIbertyLoop.dgn [BL124 - Plan 13 (Sheet)]
8 August 2026 2:07 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M25



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	4495.6	0.11
Permanent Disturbance	3333.9	0.08
Temporary Disturbance	1161.7	0.03

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

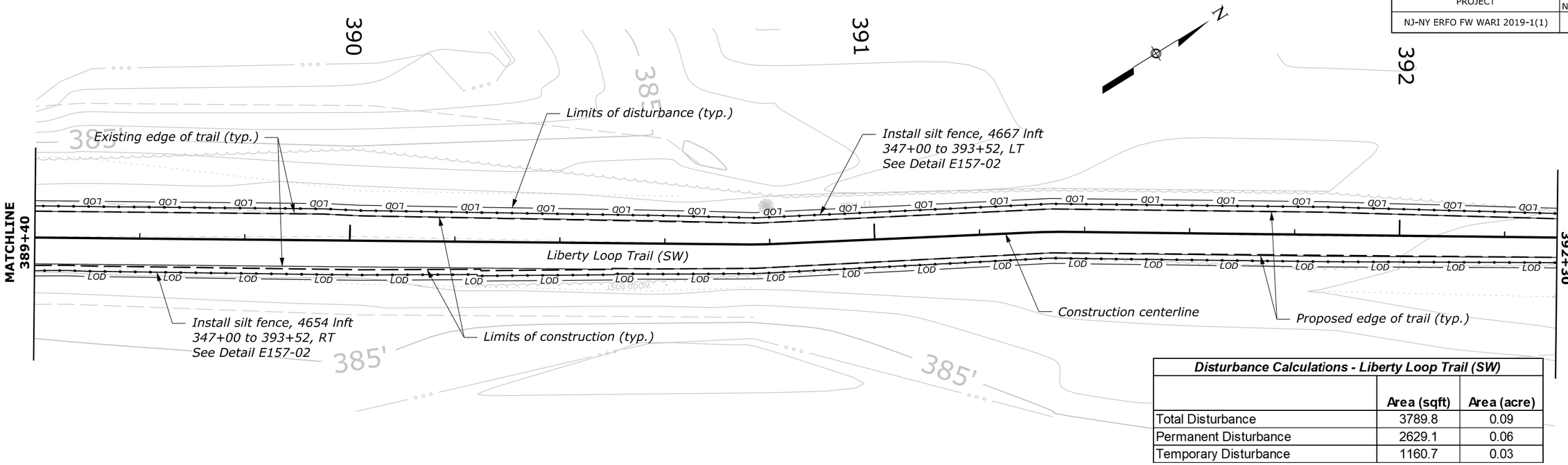
EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
386+50 TO 389+40

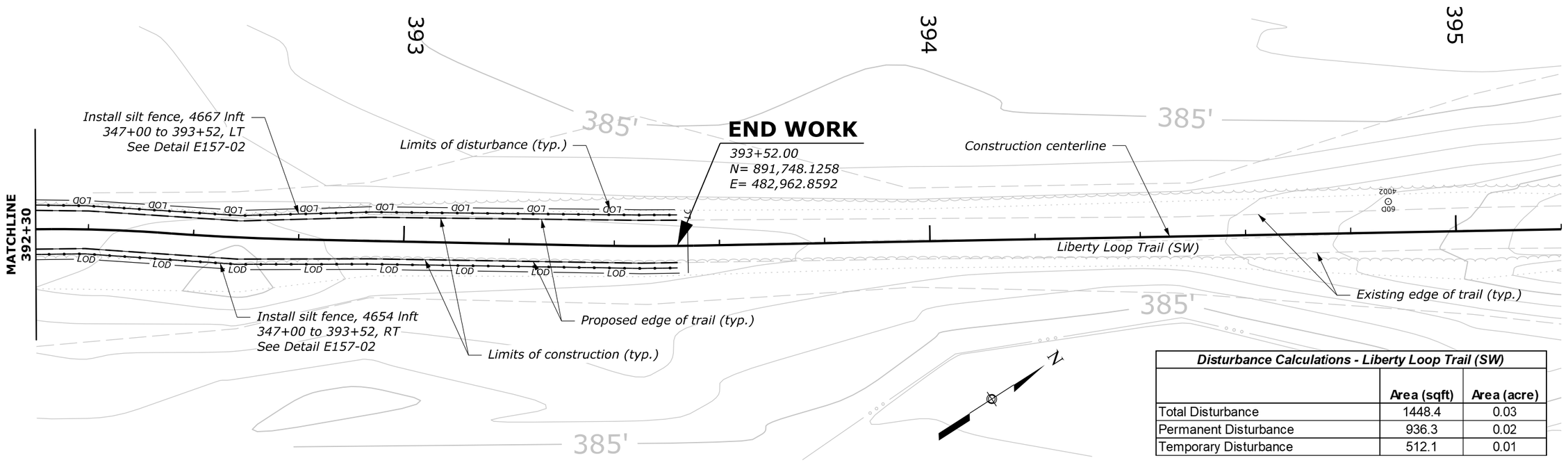
C:\Users\reo.kunorh\Documents\NJ-NY WARI 100% files\M10-M18_NJ-WARI_2019-1(1)_LibertyLoop.dgn [BL124 - Plan 15 [Sheet]]
8 August 2026 2:07 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M26



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	3789.8	0.09
Permanent Disturbance	2629.1	0.06
Temporary Disturbance	1160.7	0.03



Disturbance Calculations - Liberty Loop Trail (SW)		
	Area (sqft)	Area (acre)
Total Disturbance	1448.4	0.03
Permanent Disturbance	936.3	0.02
Temporary Disturbance	512.1	0.01

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
389+40 TO 393+52

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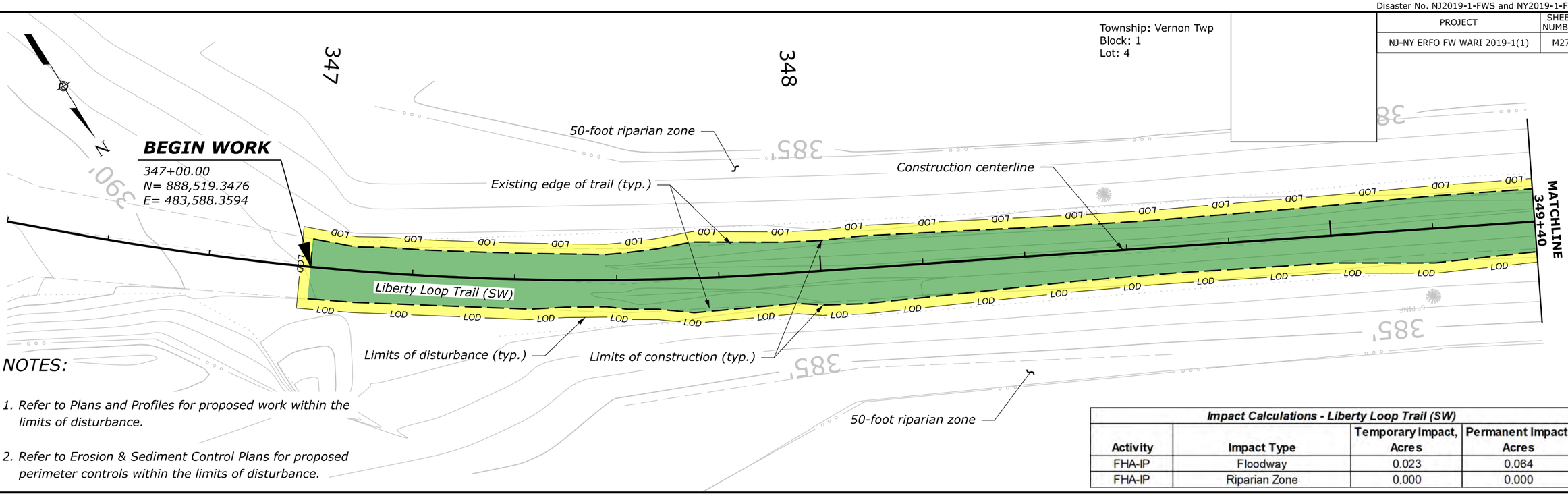
8 August 2026 2:07 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M27

Township: Vernon Twp
Block: 1
Lot: 4

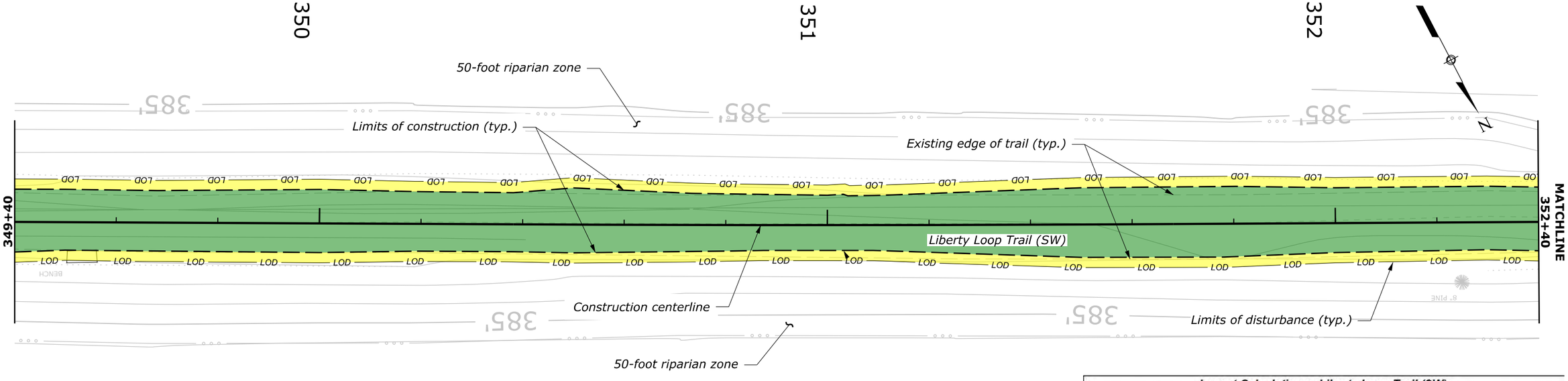


NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.

Impact Calculations - Liberty Loop Trail (SW)

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.023	0.064
FHA-IP	Riparian Zone	0.000	0.000



Township: Vernon Twp
Block: 1
Lot: 4

Impact Calculations - Liberty Loop Trail (SW)

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.028	0.085
FHA-IP	Riparian Zone	0.000	0.000

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
347+00 TO 352+40

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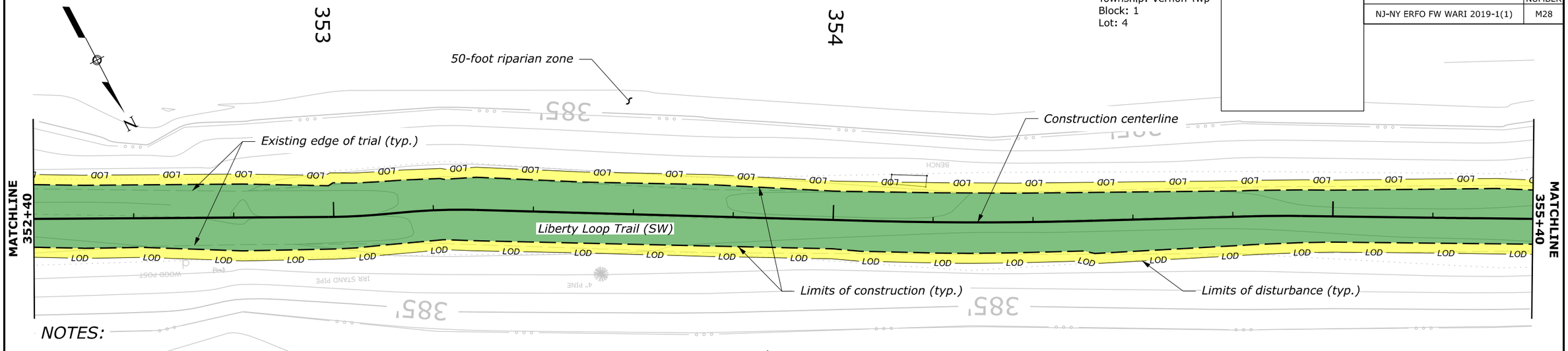
8 August 2026 2:52 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M28

Township: Vernon Twp
Block: 1
Lot: 4

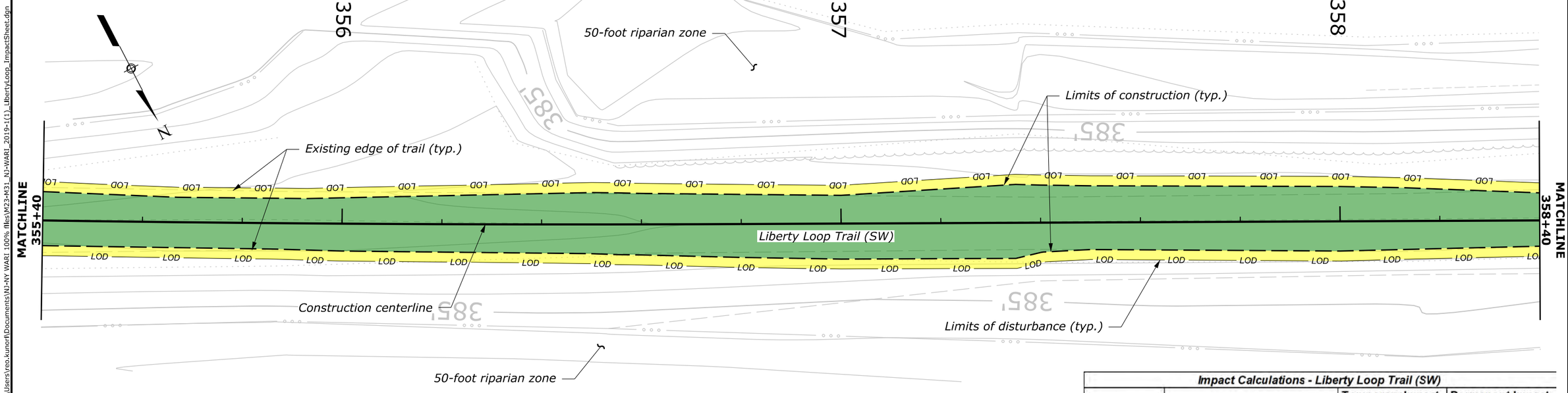


NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.

Impact Calculations - Liberty Loop Trail (SW)

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.028	0.082
FHA-IP	Riparian Zone	0.000	0.000



Township: Vernon Twp
Block: 1
Lot: 4

Impact Calculations - Liberty Loop Trail (SW)

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.028	0.083
FHA-IP	Riparian Zone	0.000	0.000

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
352+40 TO 358+40

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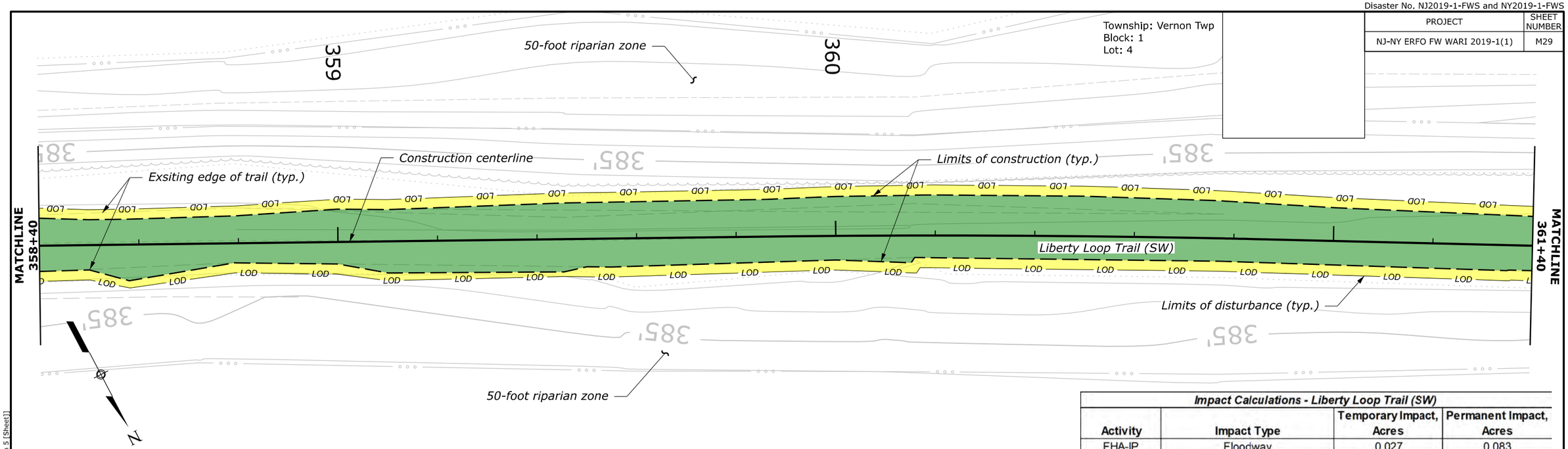
8 August 2026 2:52 PM

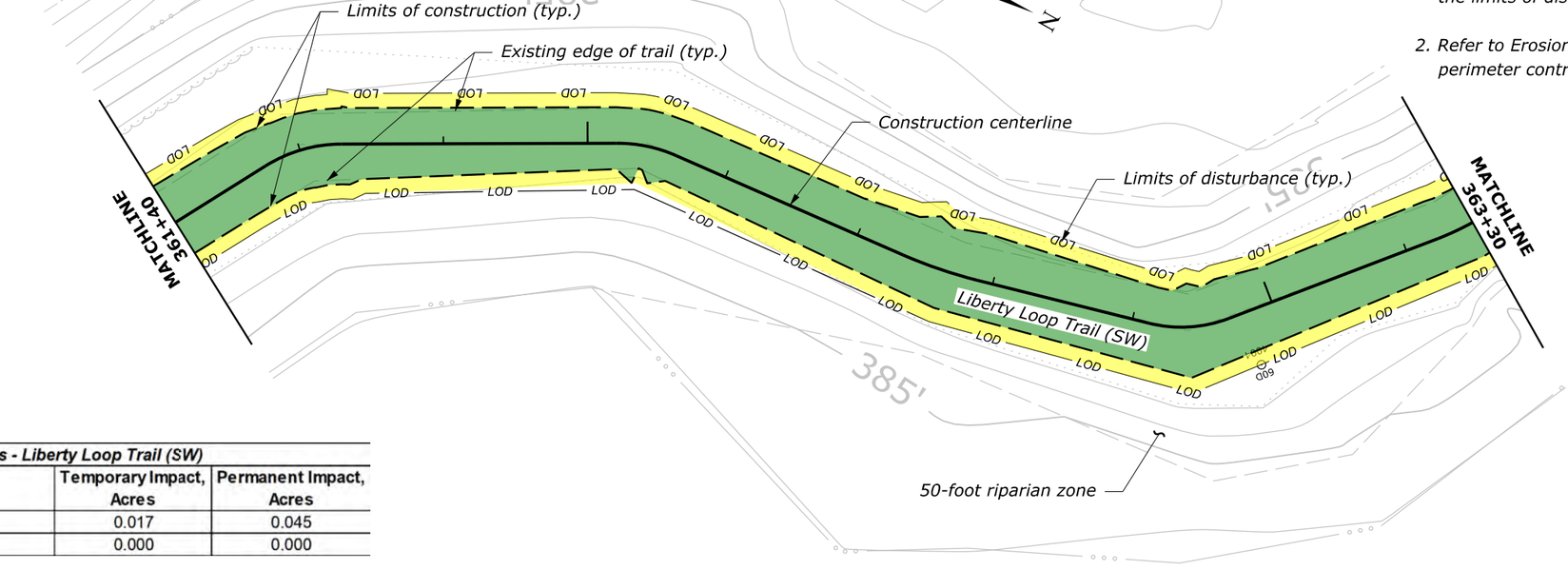
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M29

Township: Vernon Twp
Block: 1
Lot: 4



Impact Calculations - Liberty Loop Trail (SW)			
Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.027	0.083
FHA-IP	Riparian Zone	0.000	0.000

Township: Vernon Twp
Block: 1
Lot: 4



Impact Calculations - Liberty Loop Trail (SW)			
Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.017	0.045
FHA-IP	Riparian Zone	0.000	0.000

NOTES:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
358+40 TO 363+30

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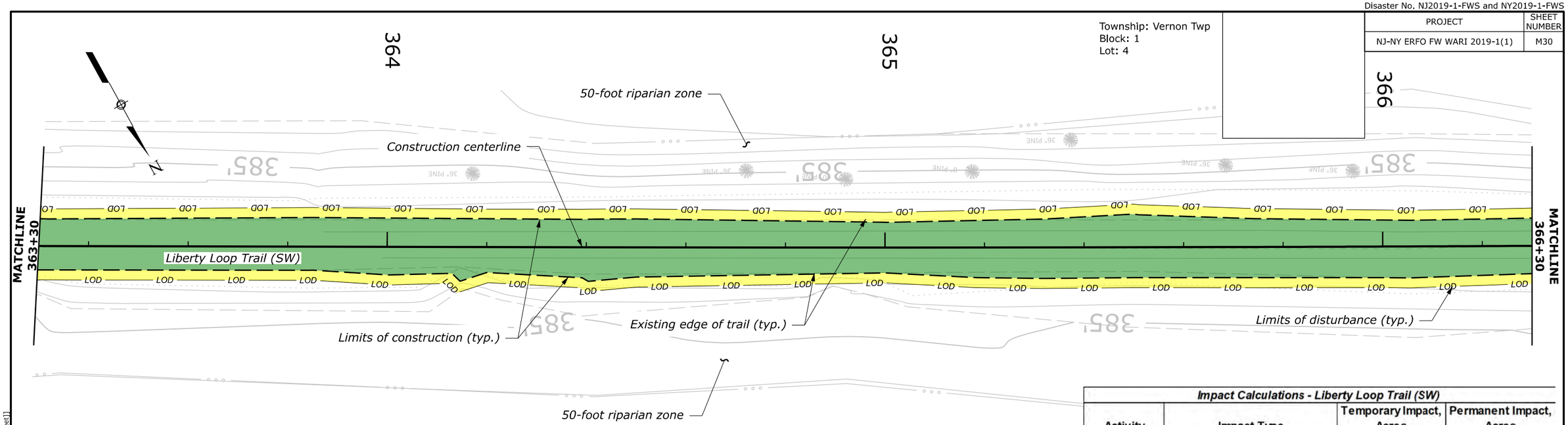
8 August 2026 2:52 PM

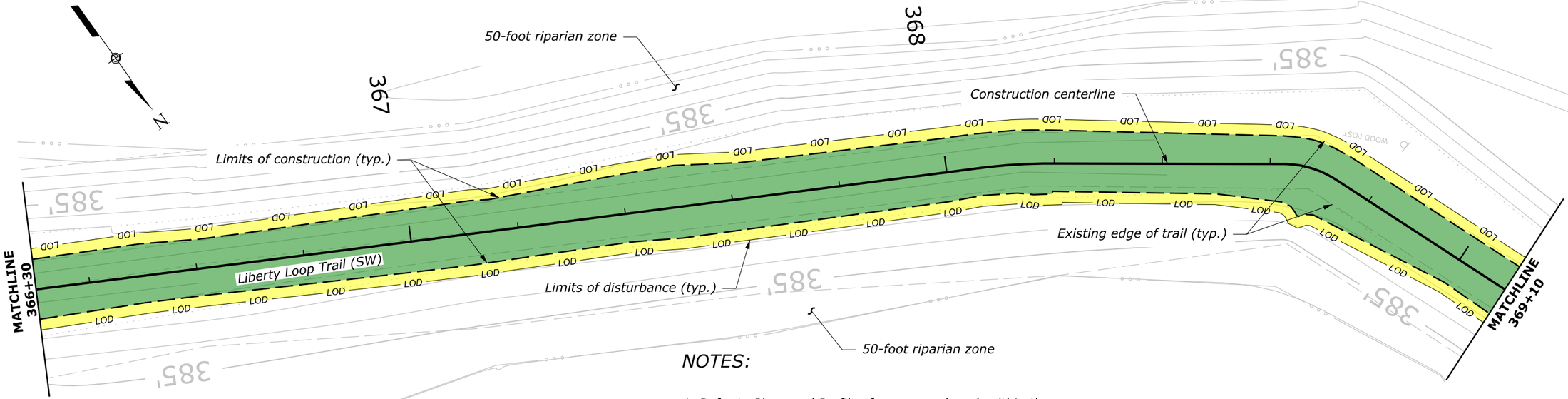
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M30

Township: Vernon Twp
Block: 1
Lot: 4



Impact Calculations - Liberty Loop Trail (SW)			
Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.028	0.077
FHA-IP	Riparian Zone	0.000	0.000



NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.

Township: Vernon Twp
Block: 1
Lot: 4

Impact Calculations - Liberty Loop Trail (SW)			
Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.026	0.074
FHA-IP	Riparian Zone	0.000	0.000

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
363+30 TO 369+10

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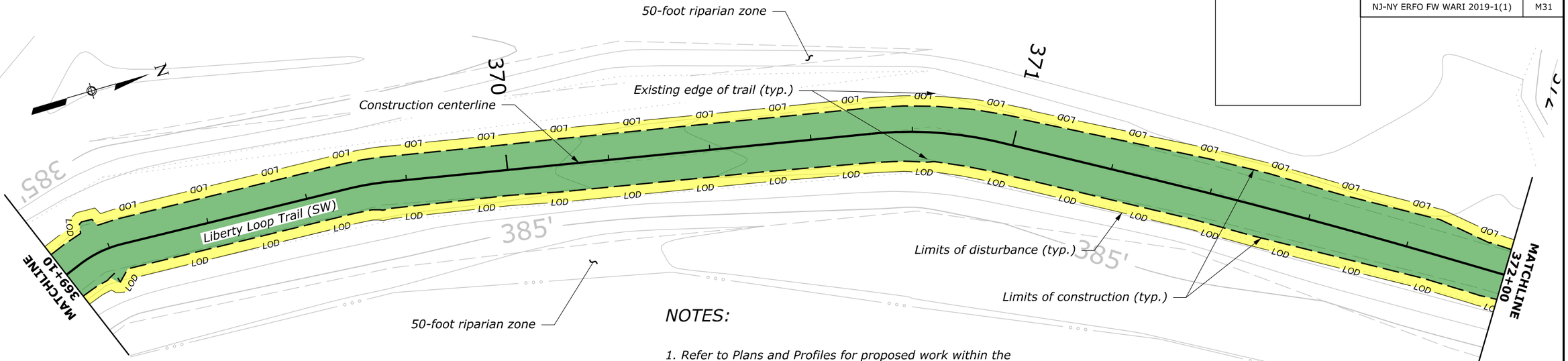
8 August 2026 2:52 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M31



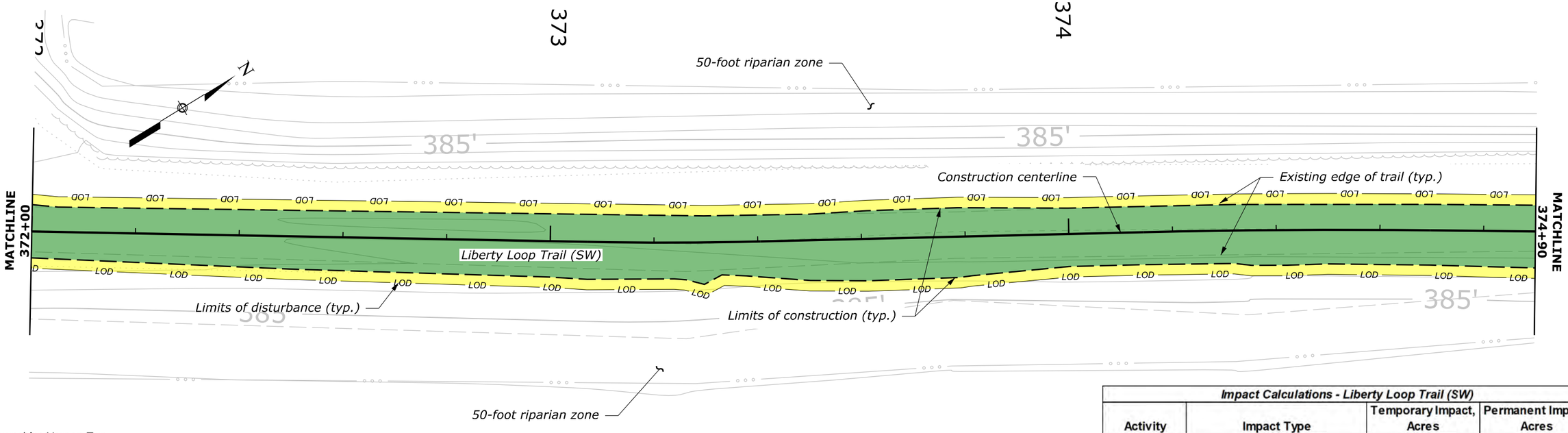
NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.

Impact Calculations - Liberty Loop Trail (SW)

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.027	0.073
FHA-IP	Riparian Zone	0.000	0.000

Township: Vernon Twp
Block: 1
Lot: 4



Impact Calculations - Liberty Loop Trail (SW)

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.027	0.079
FHA-IP	Riparian Zone	0.000	0.000

Township: Vernon Twp
Block: 1
Lot: 4

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
369+10 TO 374+90

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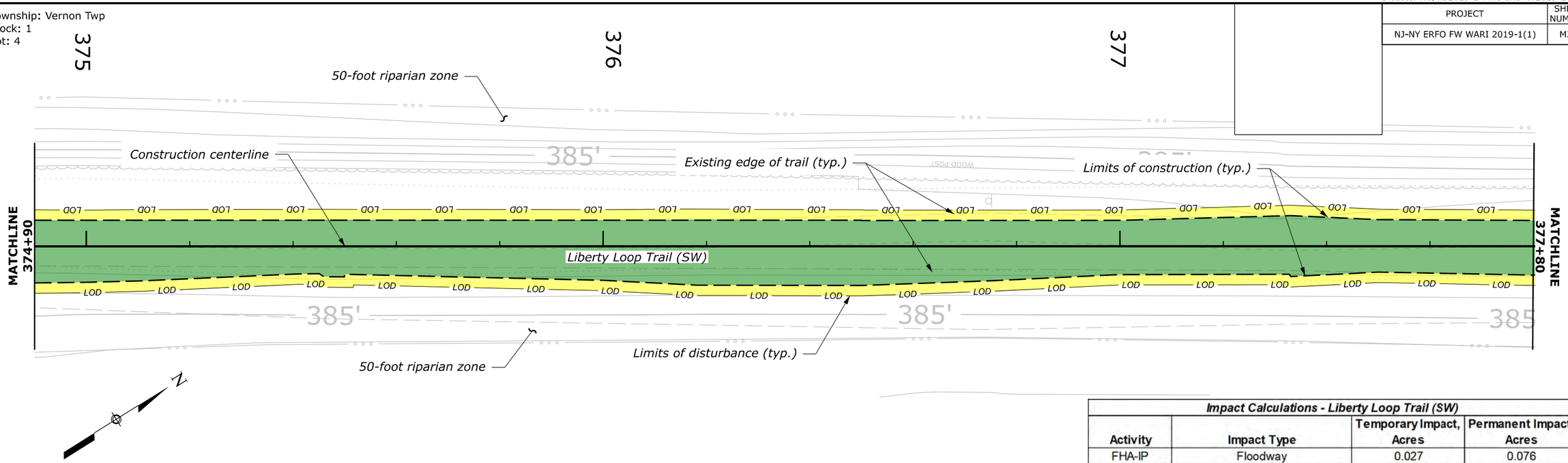
8 August 2026 2:52 PM

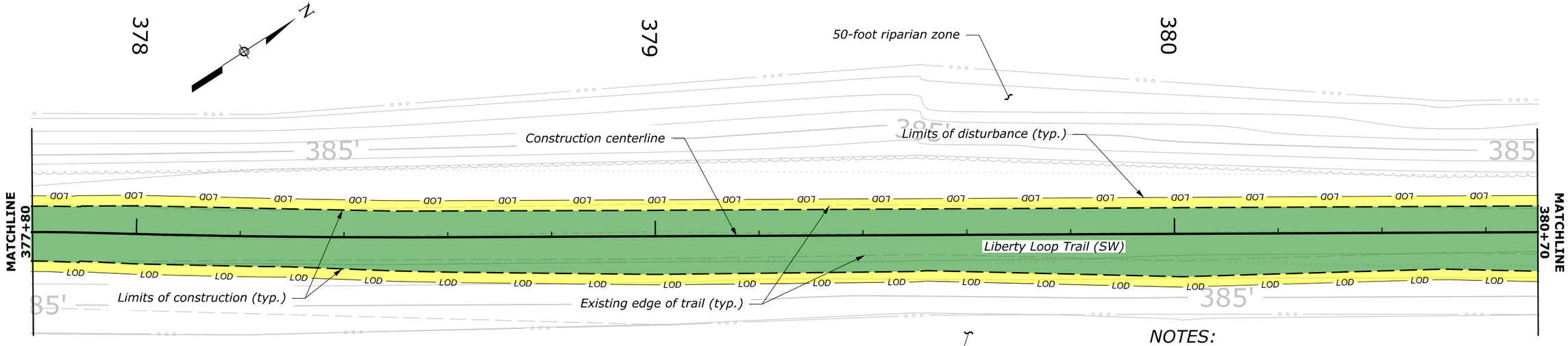
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M32

Township: Vernon Twp
Block: 1
Lot: 4



Impact Calculations - Liberty Loop Trail (SW)

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.027	0.076
FHA-IP	Riparian Zone	0.000	0.000



Township: Vernon Twp
Block: 1
Lot: 4

Impact Calculations - Liberty Loop Trail (SW)

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.027	0.080
FHA-IP	Riparian Zone	0.000	0.000

NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
374+90 TO 380+70

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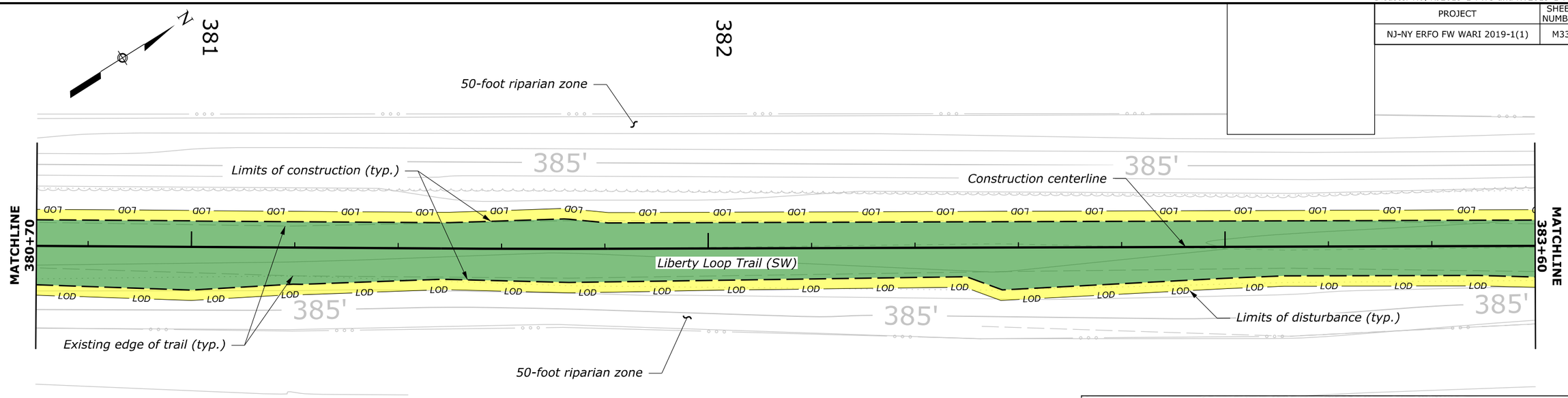
8 August 2026 2:52 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M33

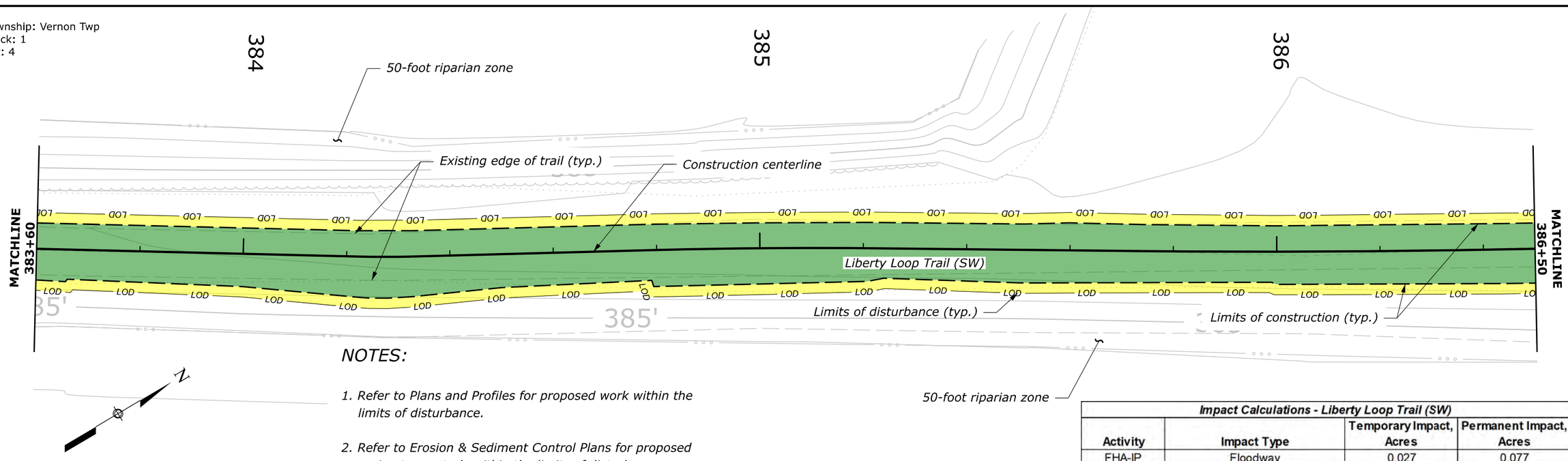


Township: Vernon Twp
Block: 1
Lot: 4

Impact Calculations - Liberty Loop Trail (SW)

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.027	0.077
FHA-IP	Riparian Zone	0.000	0.000

Township: Vernon Twp
Block: 1
Lot: 4



NOTES:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*

Impact Calculations - Liberty Loop Trail (SW)

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.027	0.077
FHA-IP	Riparian Zone	0.000	0.000

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
380+70 TO 386+50

SHEET 7 OF 9

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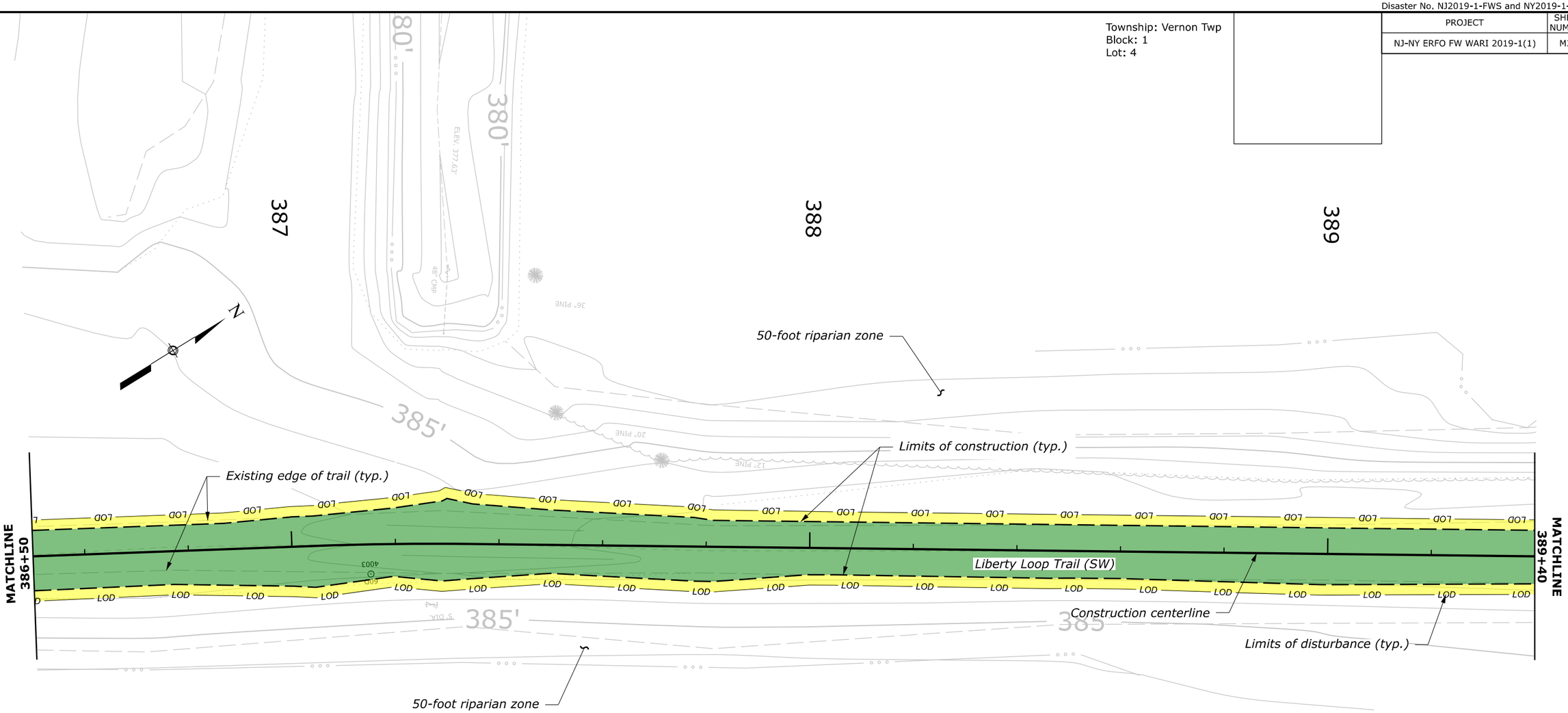
8 August 2026 2:52 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M34

Township: Vernon Twp
Block: 1
Lot: 4



NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.

Impact Calculations - Liberty Loop Trail (SW)			
Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.027	0.077
FHA-IP	Riparian Zone	0.000	0.000

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
386+50 TO 389+40

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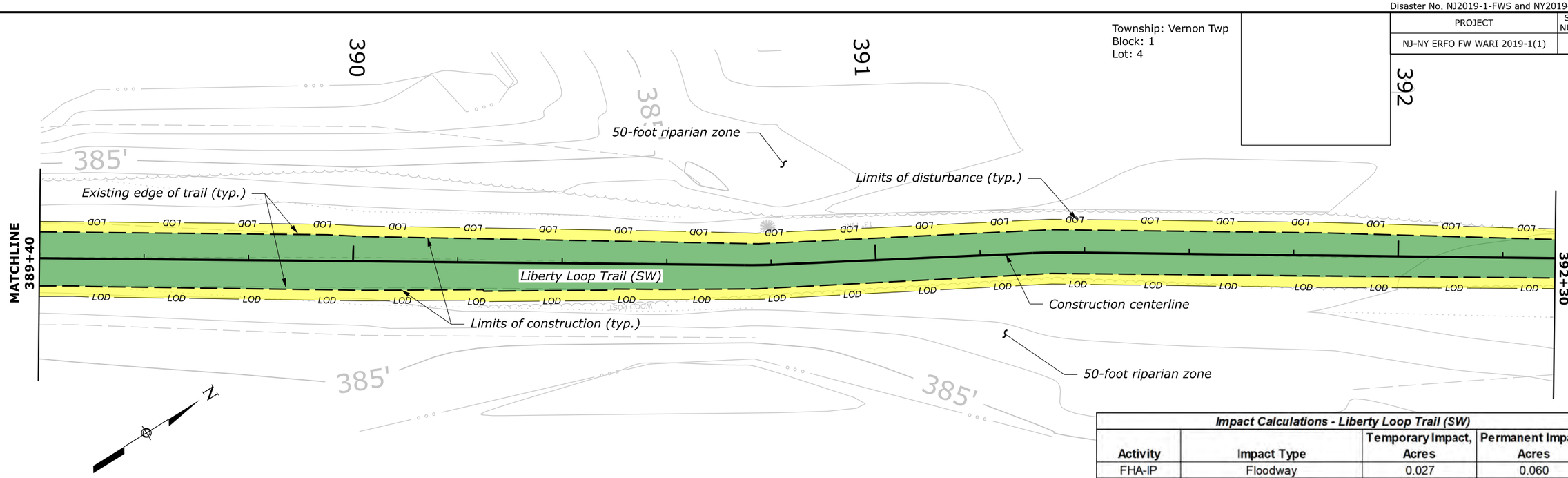
8 August 2026 2:52 PM

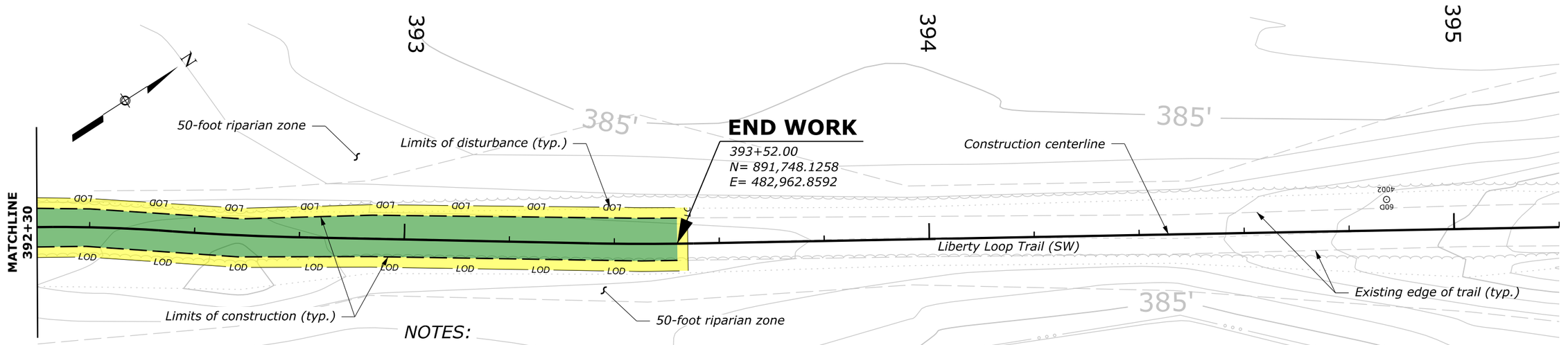
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M35

Township: Vernon Township
Block: 1
Lot: 4



Impact Calculations - Liberty Loop Trail (SW)			
Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.027	0.060
FHA-IP	Riparian Zone	0.000	0.000



NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.

Township: Vernon Twp
Block: 1
Lot: 4

Impact Calculations - Liberty Loop Trail (SW)			
Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FHA-IP	Floodway	0.012	0.021
FHA-IP	Riparian Zone	0.000	0.000

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
389+40 TO 393+52

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8 August 2026 2:52 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M36

Township: Vernon Twp
Block: 1
Lot: 4

347

348

N

BEGIN WORK

347+00.00
N= 888,519.3476
E= 483,588.3594

Existing edge of trail (typ.)

Construction centerline

MATCHLINE 349+40

Liberty Loop Trail (SW)

NOTES:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*

350

351

352

N

MATCHLINE 349+40

Existing edge of trail (typ.)

388'

Liberty Loop Trail (SW)

Construction centerline

MATCHLINE 352+40

Township: Vernon Twp
Block: 1
Lot: 4

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

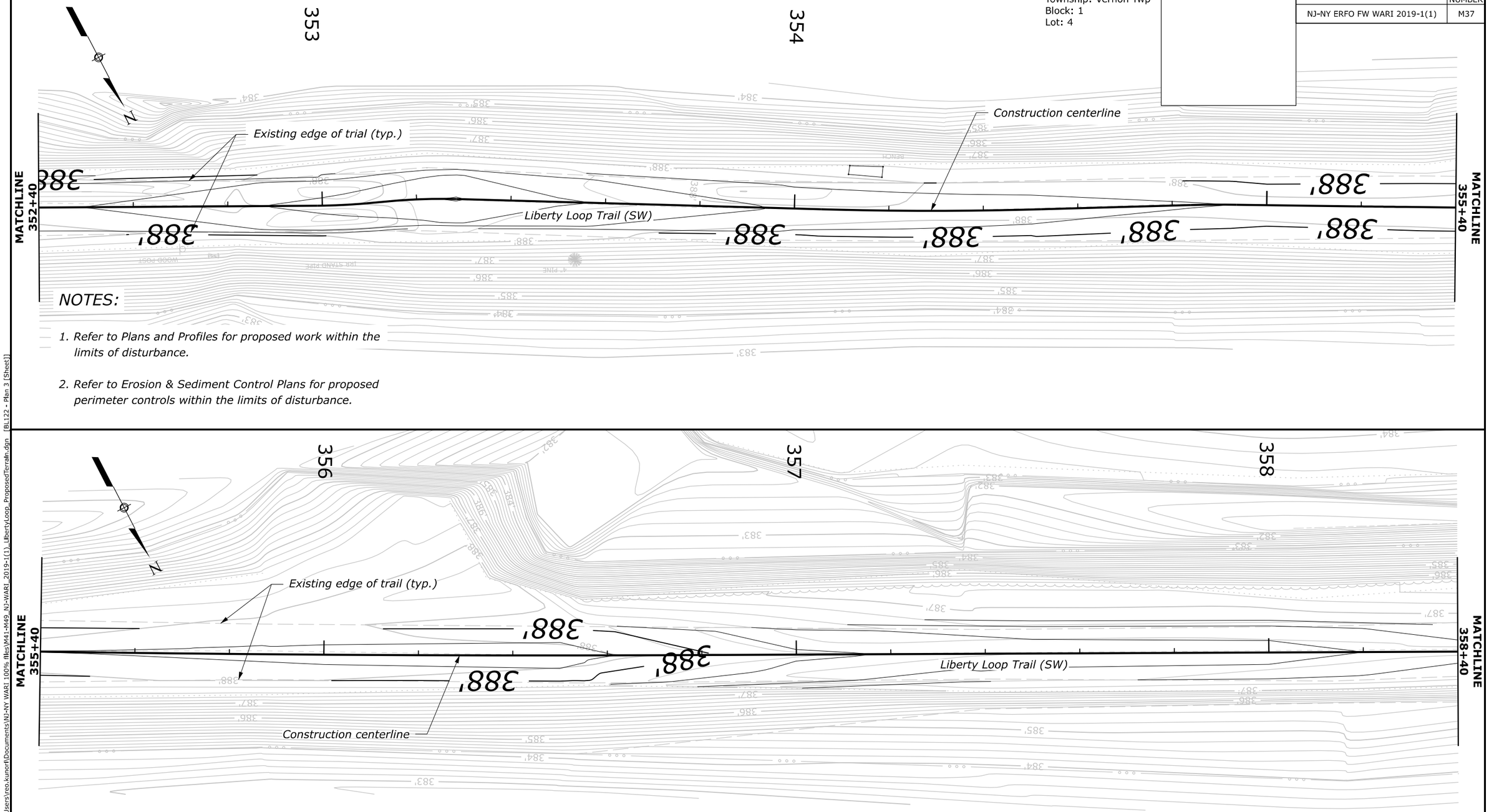
SCHEDULE B: LIBERTY LOOP TRAIL (SW)
347+00 TO 352+40

SHEET 1 OF 9

C:\Users\reo.kuner\Documents\NJ-NY WARI 100% files\M41-M49_NJ-WARI_2019-1(1)_LibertyLoop_ProposedTerrain.dgn [BL122 - Plan 1 [Sheet]]
8 August 2026 3:17 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M37

Township: Vernon Twp
Block: 1
Lot: 4



Township: Vernon Twp
Block: 1
Lot: 4

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
352+40 TO 358+40

C:\Users\vreo.kunori\Documents\NJ-NY WARI 100% files\M41-M49_NJ-WARI_2019-1(1)_LibertyLoop_ProposedTerrain.dgn [BL122 - Plan 3 [Sheet]]
8 August 2026 3:17 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M38

Township: Vernon Twp
Block: 1
Lot: 4

359

360

Construction centerline

Exsiting edge of trail (typ.)

387'

MATCHLINE
358+40

MATCHLINE
361+40

Liberty Loop Trail (SW)

N

Township: Vernon Twp
Block: 1
Lot: 4

362

363

Existing edge of trail (typ.)

Construction centerline

MATCHLINE
361+40

MATCHLINE
363+30

Liberty Loop Trail (SW)

N

NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.

2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

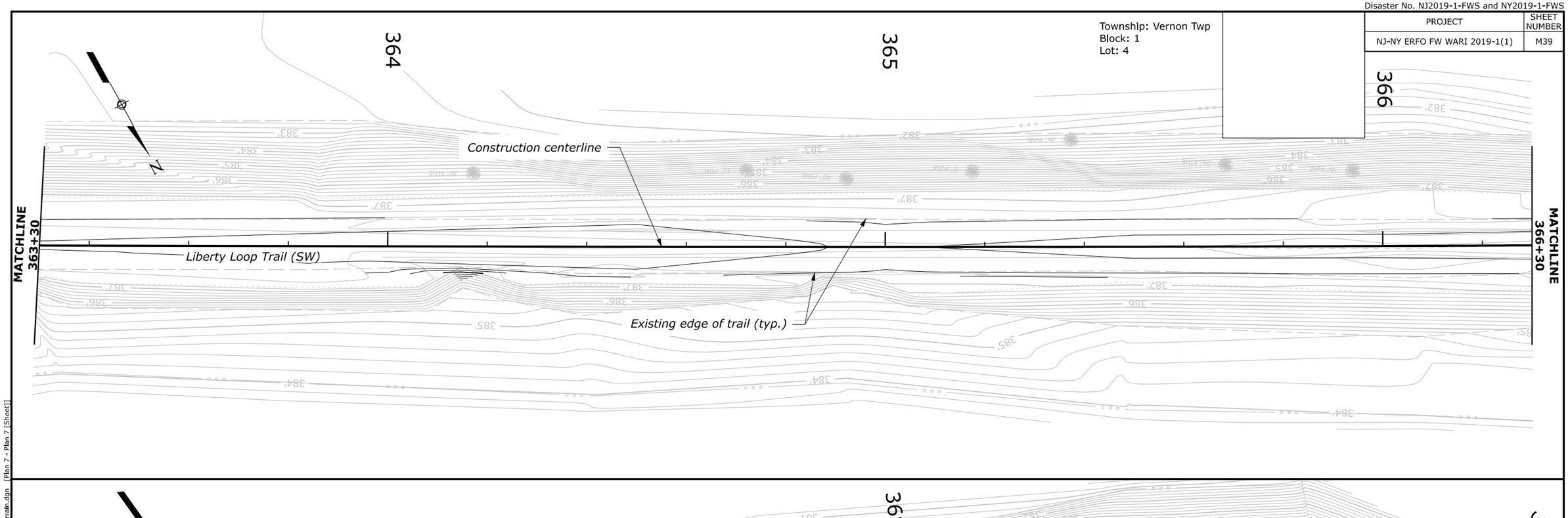
SCHEDULE B: LIBERTY LOOP TRAIL (SW)
358+40 TO 363+30

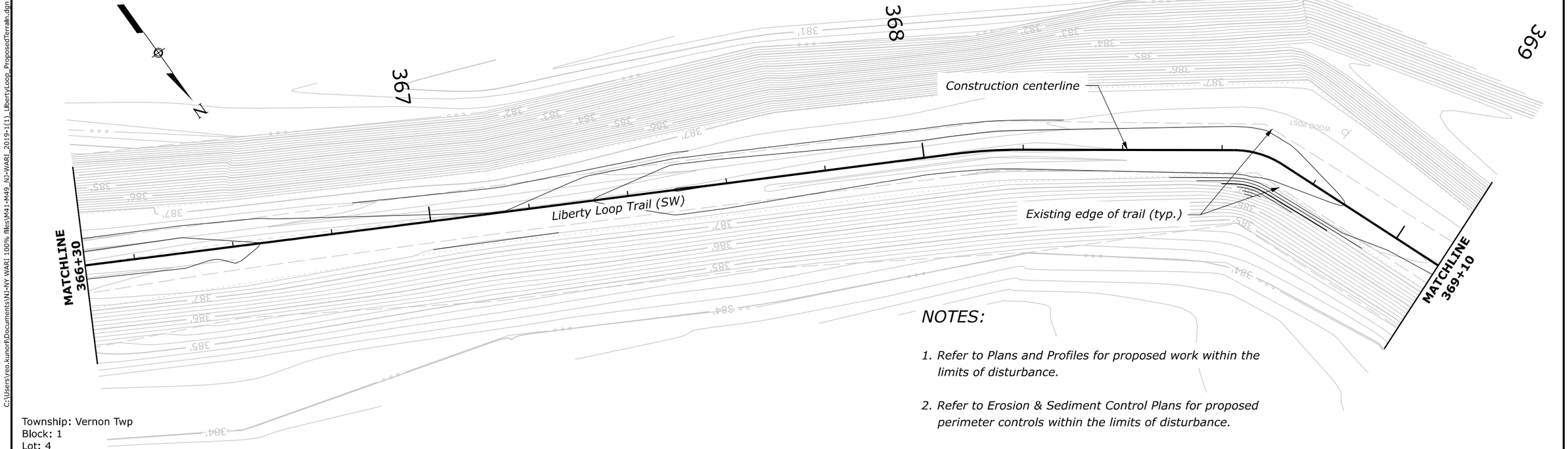
C:\Users\reo.kunorh\Documents\NJ-NY WARI 100% files\M41-M49_NJ-WARI_2019-1(1)_LibertyLoop_ProposedTerrain.dgn [BL122 - Plan 5 [Sheet]]
8 August 2026 3:17 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M39

Township: Vernon Twp
Block: 1
Lot: 4





Township: Vernon Twp
Block: 1
Lot: 4

NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

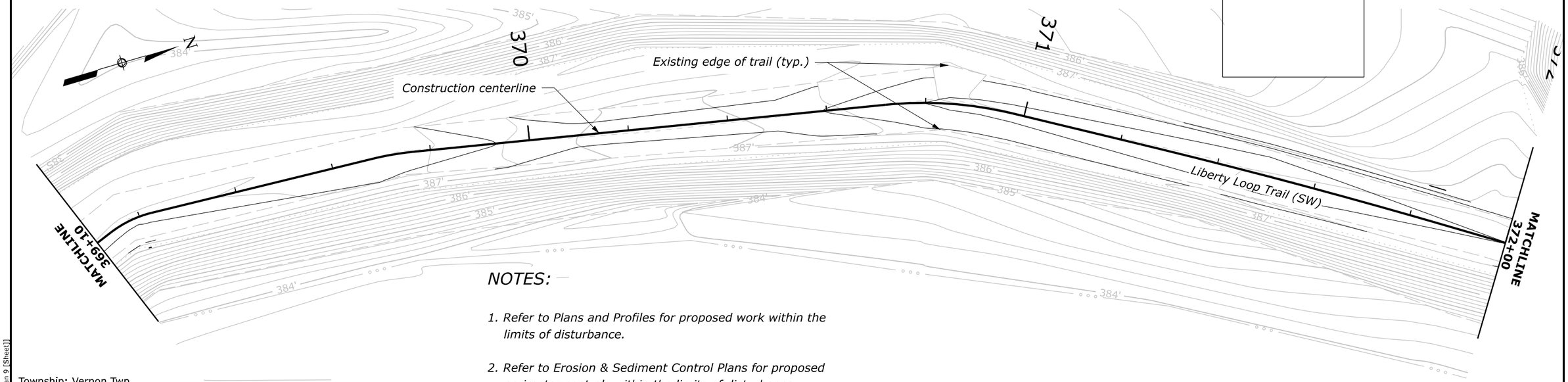
PROPOSED CONTOUR MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
363+30 TO 369+10

C:\Users\reo.kunorl\Documents\NJ-NY WARI 100% files\M41-M49_NJ-WARI_2019-1(1)_LibertyLoop_ProposedTerrain.dgn [Plan 7 - Plan 7 [Sheet]]
8 August 2026 3:17 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

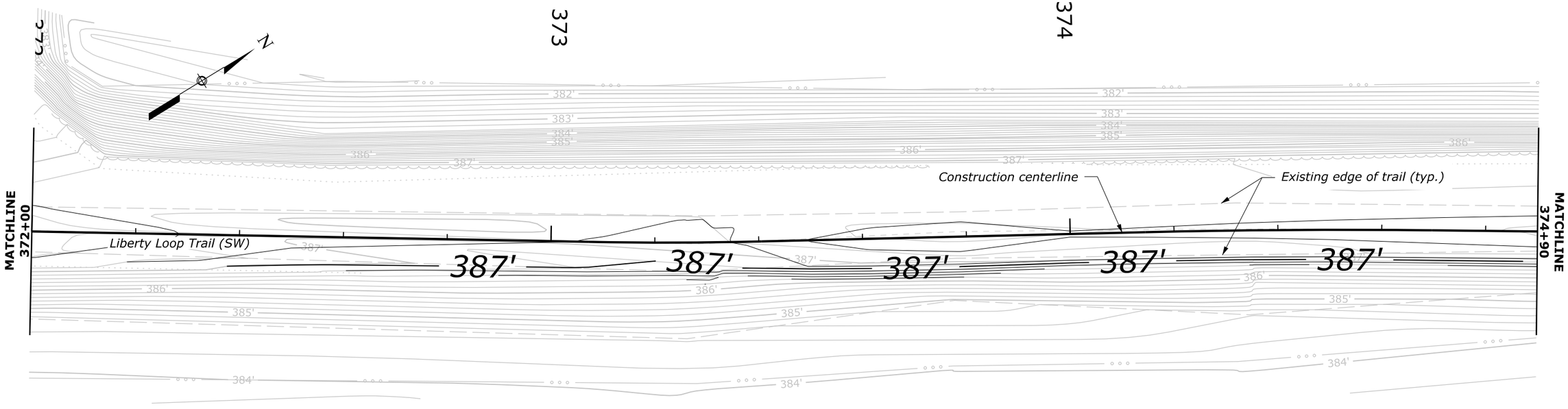
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M40



NOTES:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*

Township: Vernon Twp
Block: 1
Lot: 4



Township: Vernon Twp
Block: 1
Lot: 4

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
369+10 TO 374+90

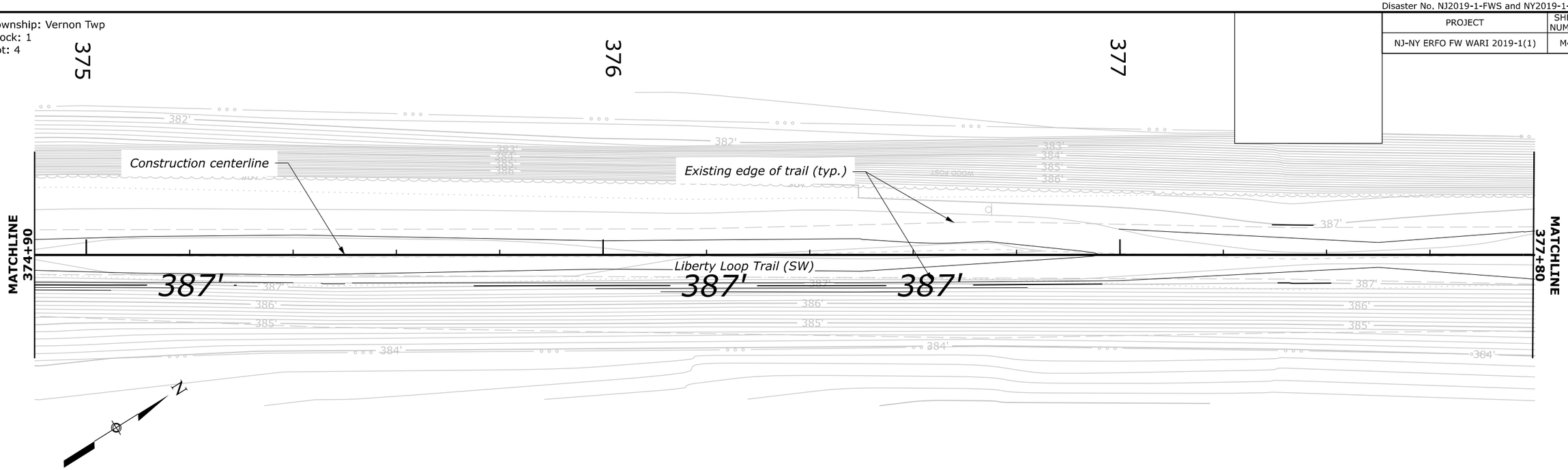
SHEET 5 OF 9

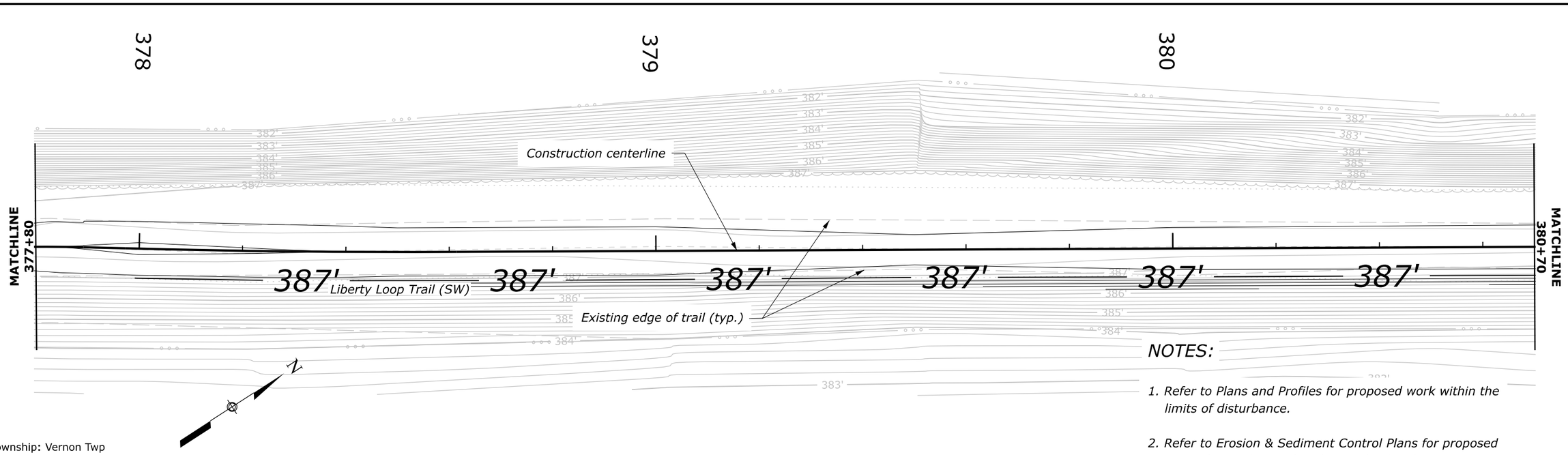
C:\Users\reb.kunori\Documents\NJ-NY WARI 100% files\M41-M49_NJ-WARI_2019-1(1)_LibertyLoop_ProposedTerrain.dgn [BL124 - Plan 9 [Sheet]]
8 August 2026 3:17 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M41

Township: Vernon Twp
Block: 1
Lot: 4





Township: Vernon Twp
Block: 1
Lot: 4

NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

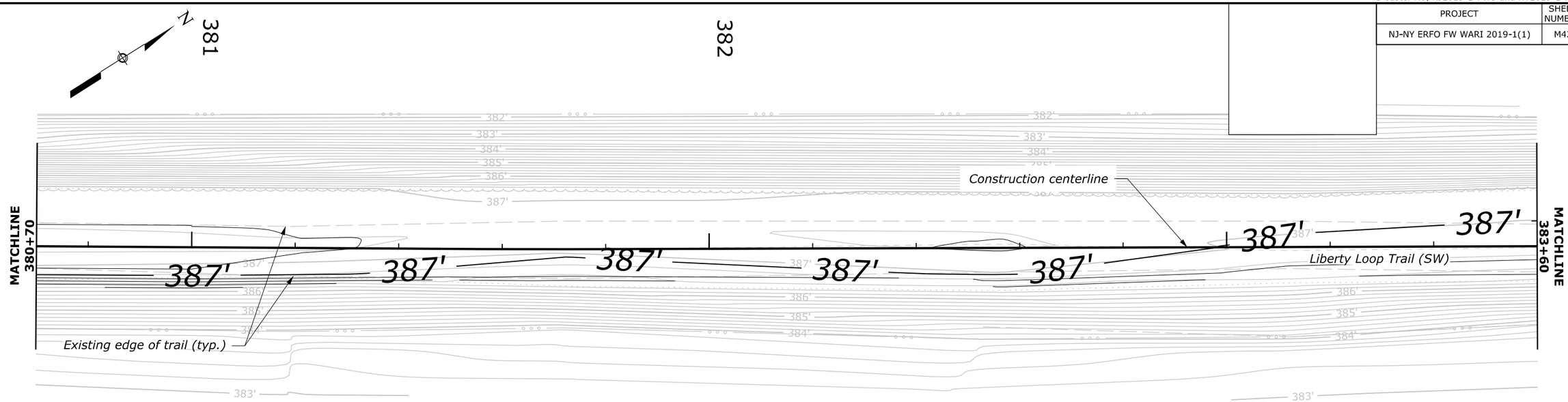
SCHEDULE B: LIBERTY LOOP TRAIL (SW)
374+90 TO 380+70

SHEET 6 OF 9

C:\Users\reo.kunorl\Documents\NJ-NY WARI 100% files\M41-M49_NJ-WARI_2019-1(1)_LibertyLoop_ProposedTerrain.dgn [BL124 - Plan 11 (Sheet)]
8 August 2026 3:17 PM

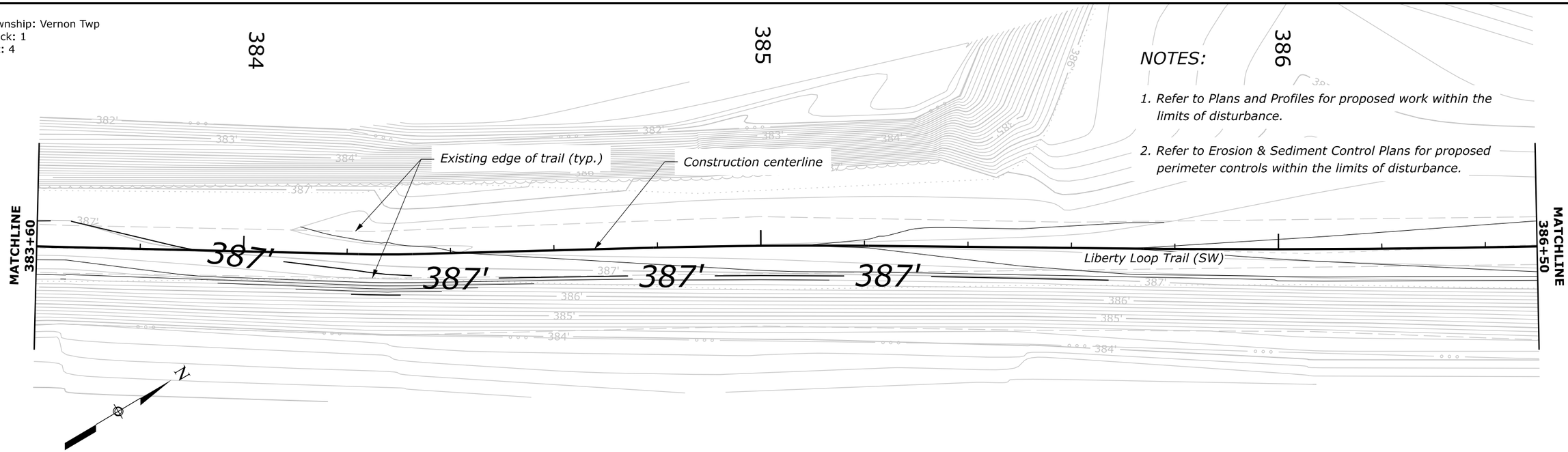
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M42



Township: Vernon Twp
Block: 1
Lot: 4

Township: Vernon Twp
Block: 1
Lot: 4



NOTES:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
380+70 TO 386+50

SHEET 7 OF 9

C:\Users\vreo.kunori\Documents\NJ-NY WARI 100% files\M41-M49_NJ-WARI_2019-1(1)_LibertyLoop_ProposedTerrain.dgn [BL124 - Plan 13 [Sheet]]
8 August 2026 3:17 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

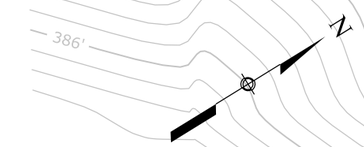
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M43

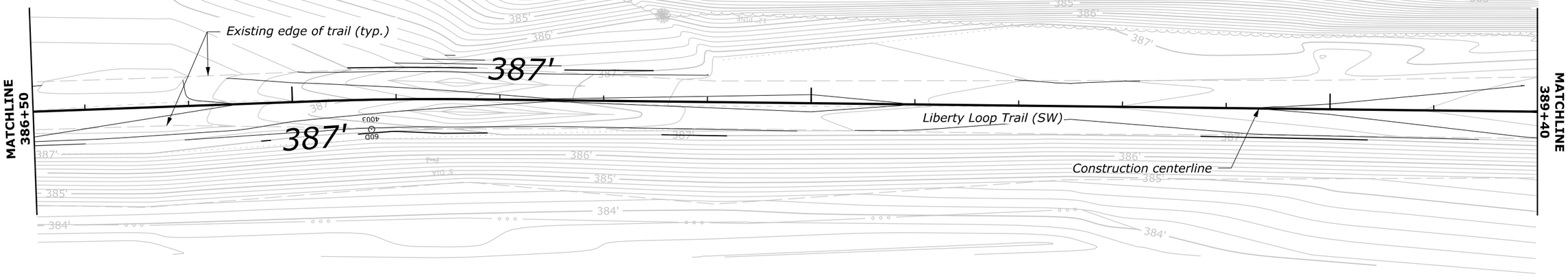
Township: Vernon Twp
Block: 1
Lot: 4

387

388

389





NOTES:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*

2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
386+50 TO 389+40

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8 August 2026 3:17 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M44

Township: Vernon Twp
Block: 1
Lot: 4

390 391 392

MATCHLINE 389+40

MATCHLINE 392+30

Existing edge of trail (typ.)

Liberty Loop Trail (SW)

Construction centerline

NOTES:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*

393 394 395

MATCHLINE 392+30

END WORK

393+52.00
N= 891,748.1258
E= 482,962.8592

Construction centerline

Liberty Loop Trail (SW)

Existing edge of trail (typ.)

Township: Vernon Twp
Block: 1
Lot: 4

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

SCHEDULE B: LIBERTY LOOP TRAIL (SW)
389+40 TO 393+52

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8 August 2026 3:17 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M45

Disturbance Calculations - Dagmar Dale Staging Area		
	Area (sqft)	Area (acre)
Total Disturbance	15425.5	0.35
Permanent Disturbance	0.00	0.00
Temporary Disturbance	15425.5	0.35

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 15 30
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE C: DAGMAR DALE NATURE TRAIL STAGING AREA

SHEET 1 OF 7

C:\Users\rco.kunofi\Documents\NJ-NY WARI 100% files\M03_NJ-WARI_2019-1(1)_DagmarDale_StagingArea.dgn [DD Staging Area [Sheet]]
8 August 2026 3:28 PM

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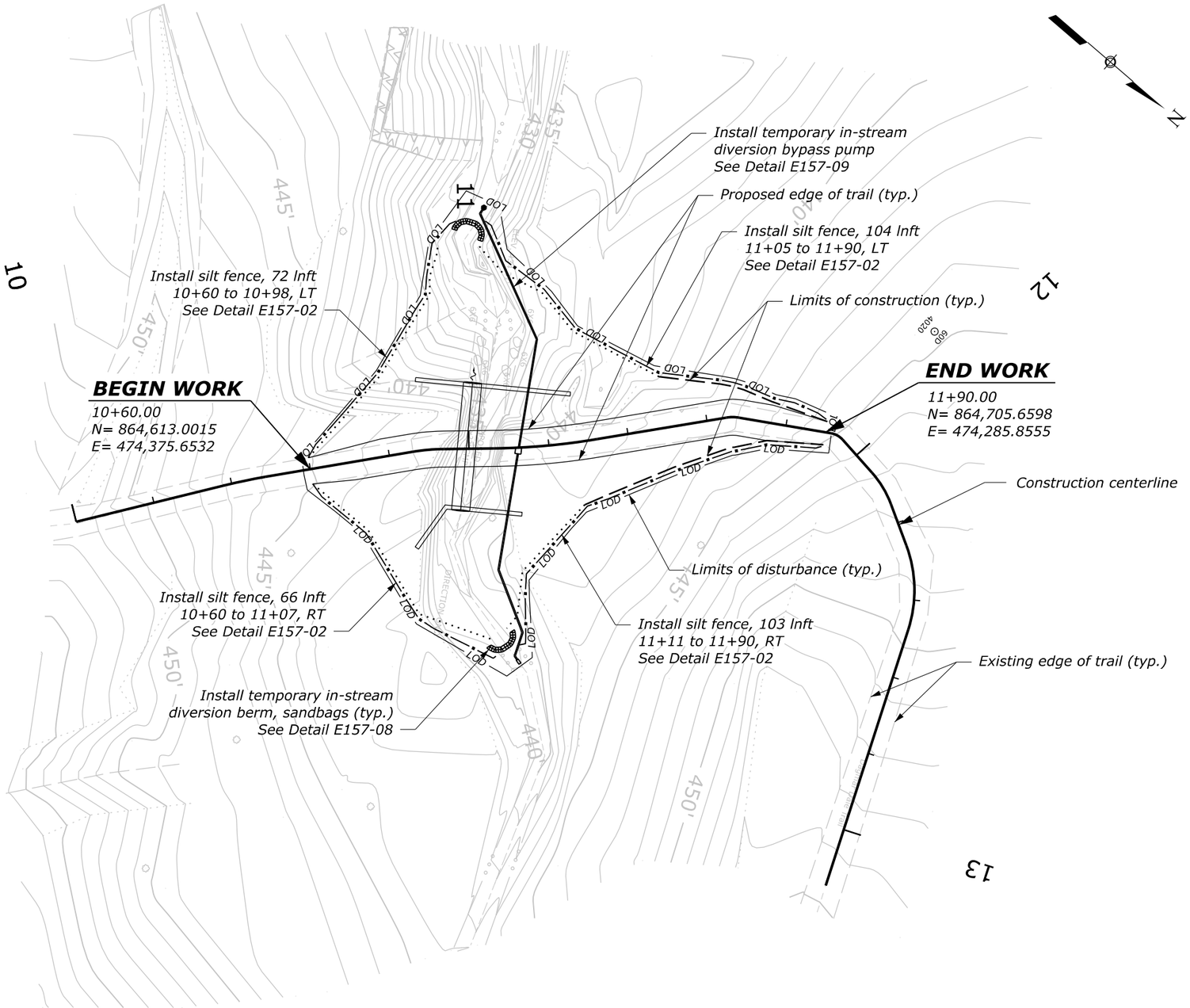
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M46



NOTES:

1. Refer to Plans and Profiles for proposed work within the limits of disturbance.

2. Refer to Drainage Profiles for cross sections of proposed pipe culverts.

Disturbance Calculations - Dagmar Dale Site #1		
	Area (sqft)	Area (acre)
Total Disturbance	6445.9	0.15
Permanent Disturbance	4548.3	0.11
Temporary Disturbance	1897.6	0.04

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 15 30
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1

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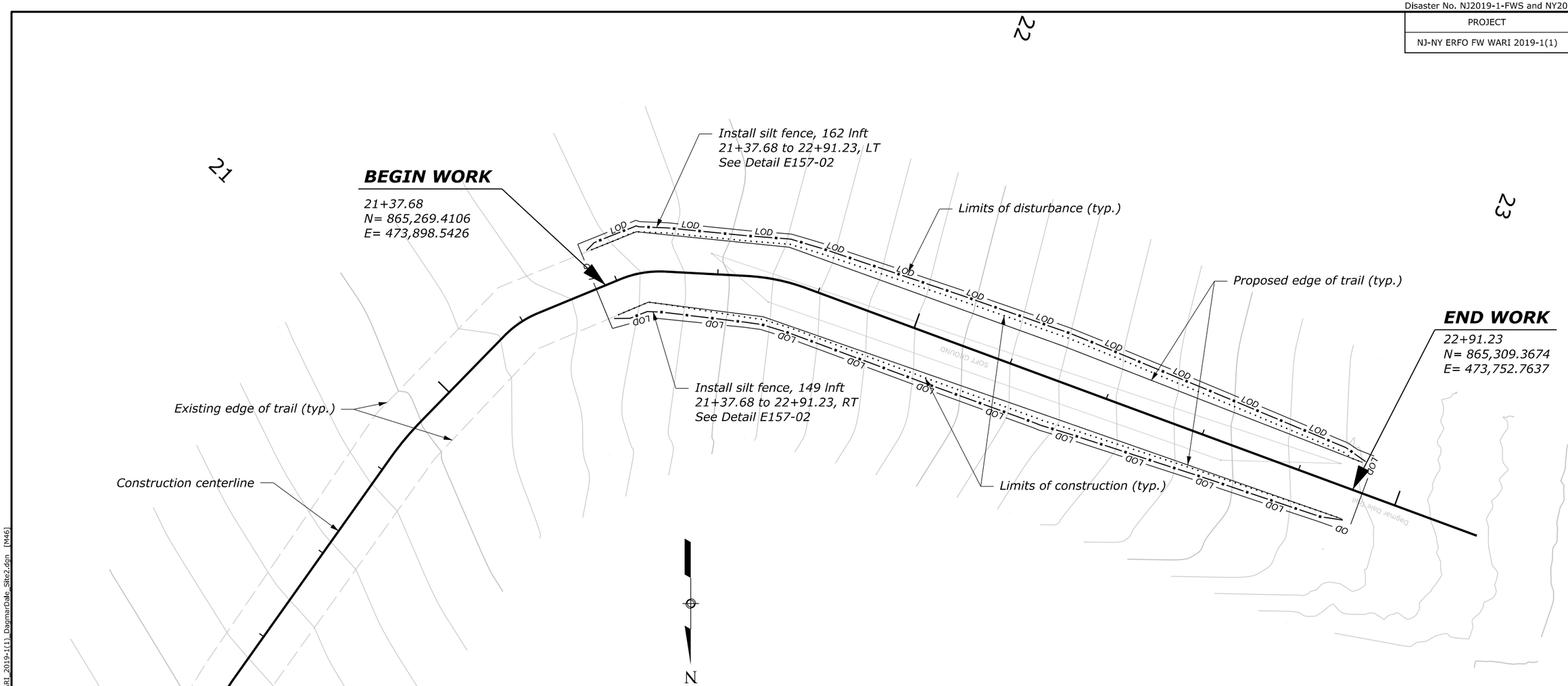
8 August 2026 3:35 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M47



NOTE:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*

Disturbance Calculations - Dagmar Dale Site #2		
	Area (sqft)	**Area (acre)**
Total Disturbance	3012.8	0.07
Permanent Disturbance	245.8	0.01
Temporary Disturbance	2767.0	0.06

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #2

SHEET 3 OF 7

8 August 2026 3:44 PM

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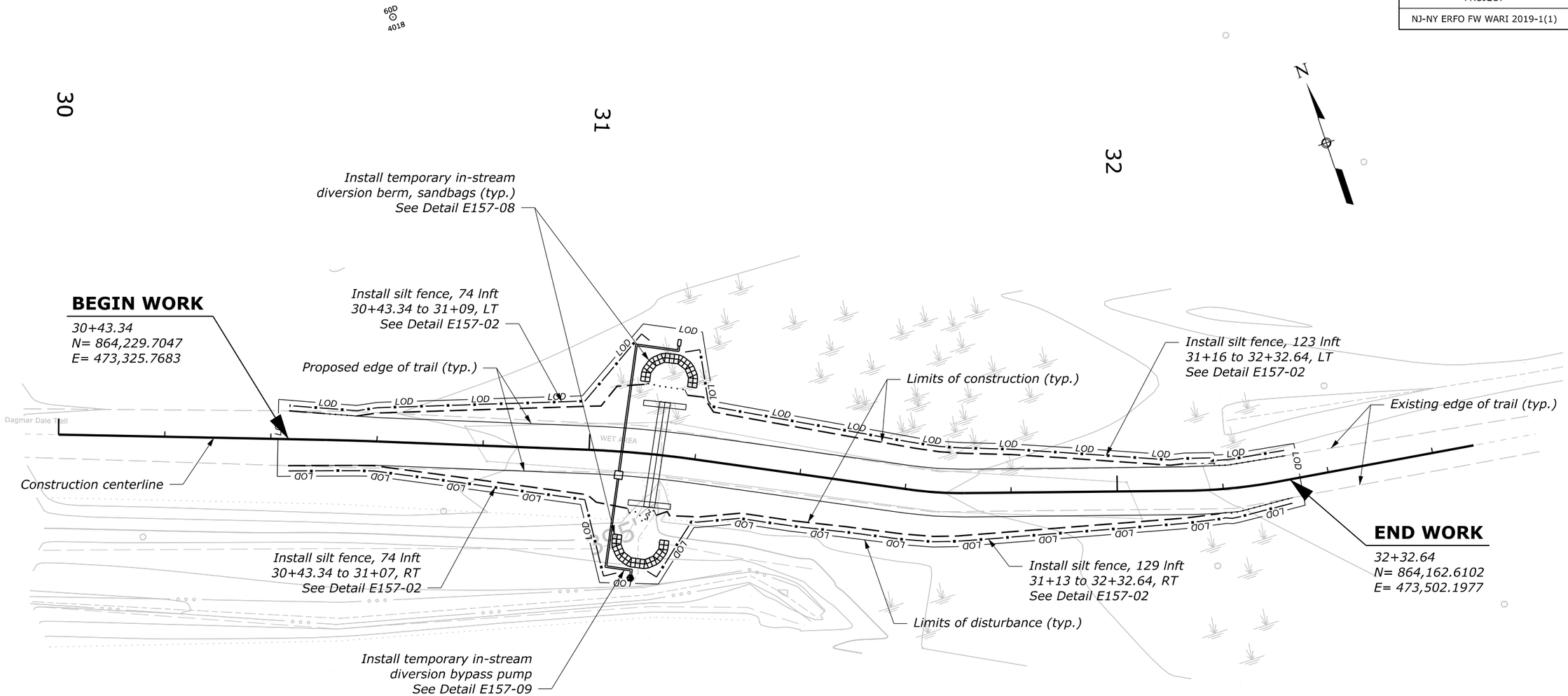
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M48



Disturbance Calculations - Dagmar Dale Site #3		
	Area (sqft)	Area (acre)
Total Disturbance	3920.8	0.09
Permanent Disturbance	1112.3	0.03
Temporary Disturbance	2808.5	0.06

NOTES:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*
2. *Refer to Drainage Profiles for cross sections of proposed pipe culverts.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #3

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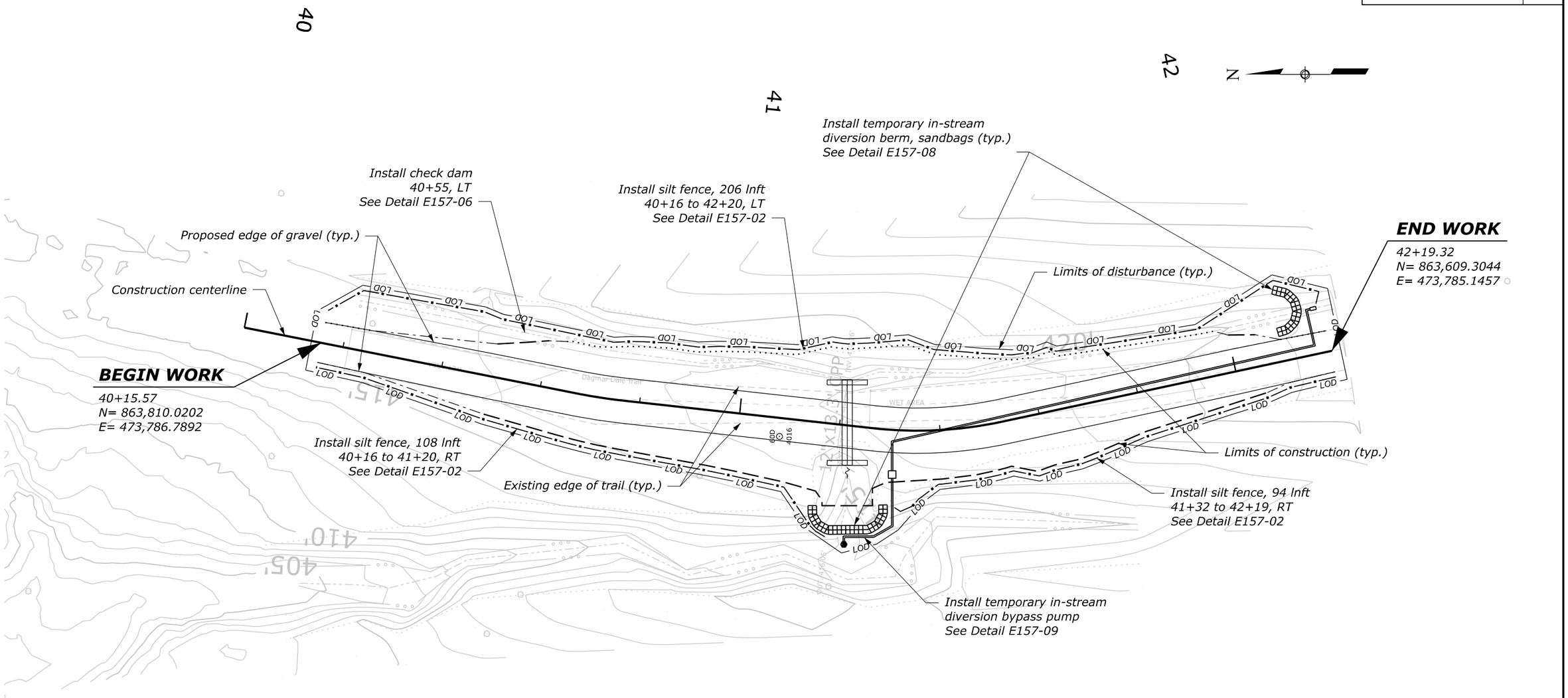
8 August 2026 3:47 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M49



NOTES:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*

2. *Refer to Drainage Profiles for cross sections of proposed pipe culverts.*

Disturbance Calculations - Dagmar Dale Site #4		
	Area (sqft)	**Area (acre)**
Total Disturbance	5240.1	0.12
Permanent Disturbance	1991.0	0.05
Temporary Disturbance	3249.1	0.07

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #4

SHEET 5 OF 7

10 August 2026 2:20 PM
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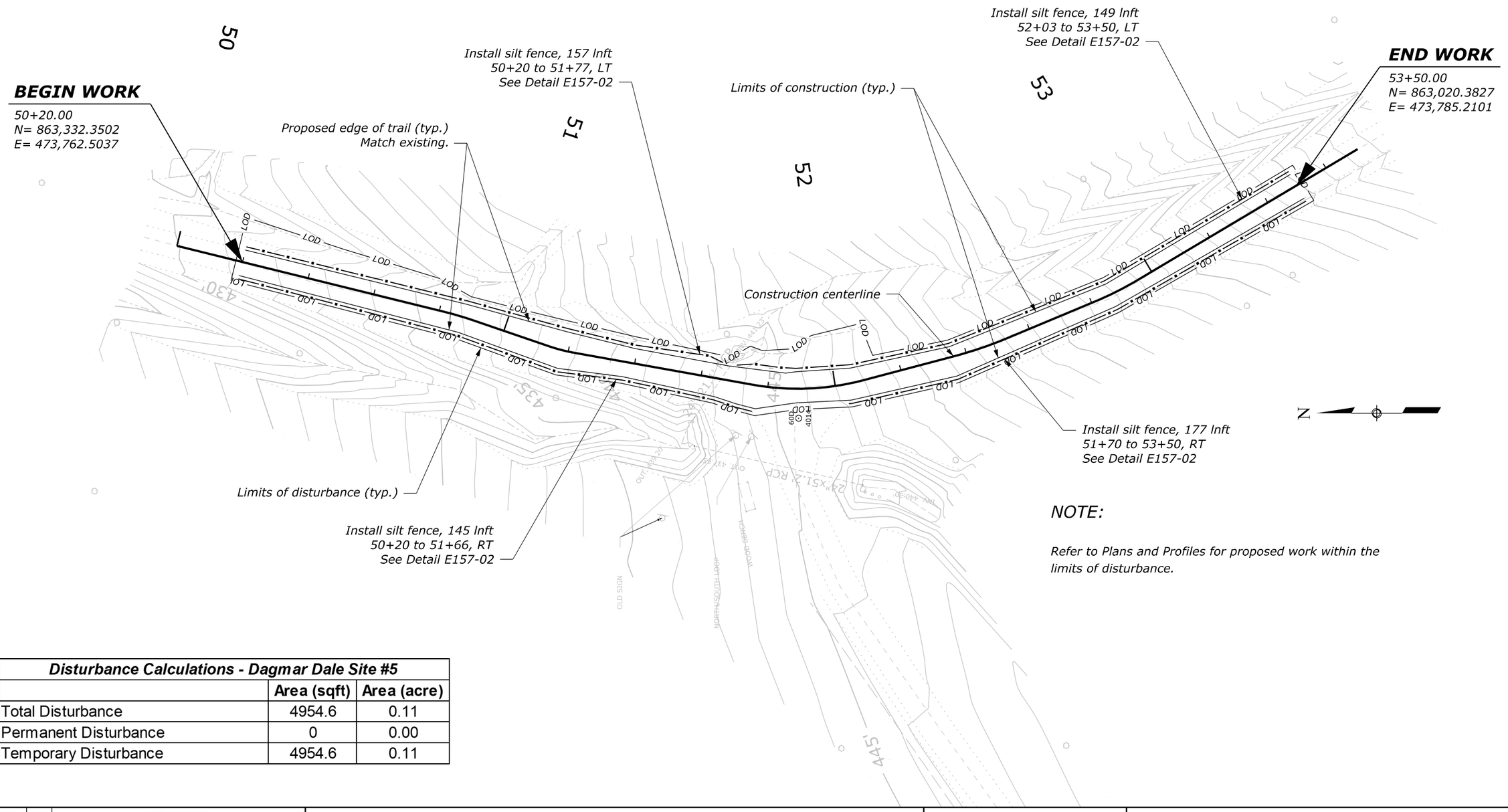

NOTE:

Refer to Plans and Profiles for proposed work within the limits of disturbance.

Disturbance Calculations - Dagmar Dale Site #5		
	Area (sqft)	Area (acre)
Total Disturbance	4954.6	0.11
Permanent Disturbance	0	0.00
Temporary Disturbance	4954.6	0.11

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #5

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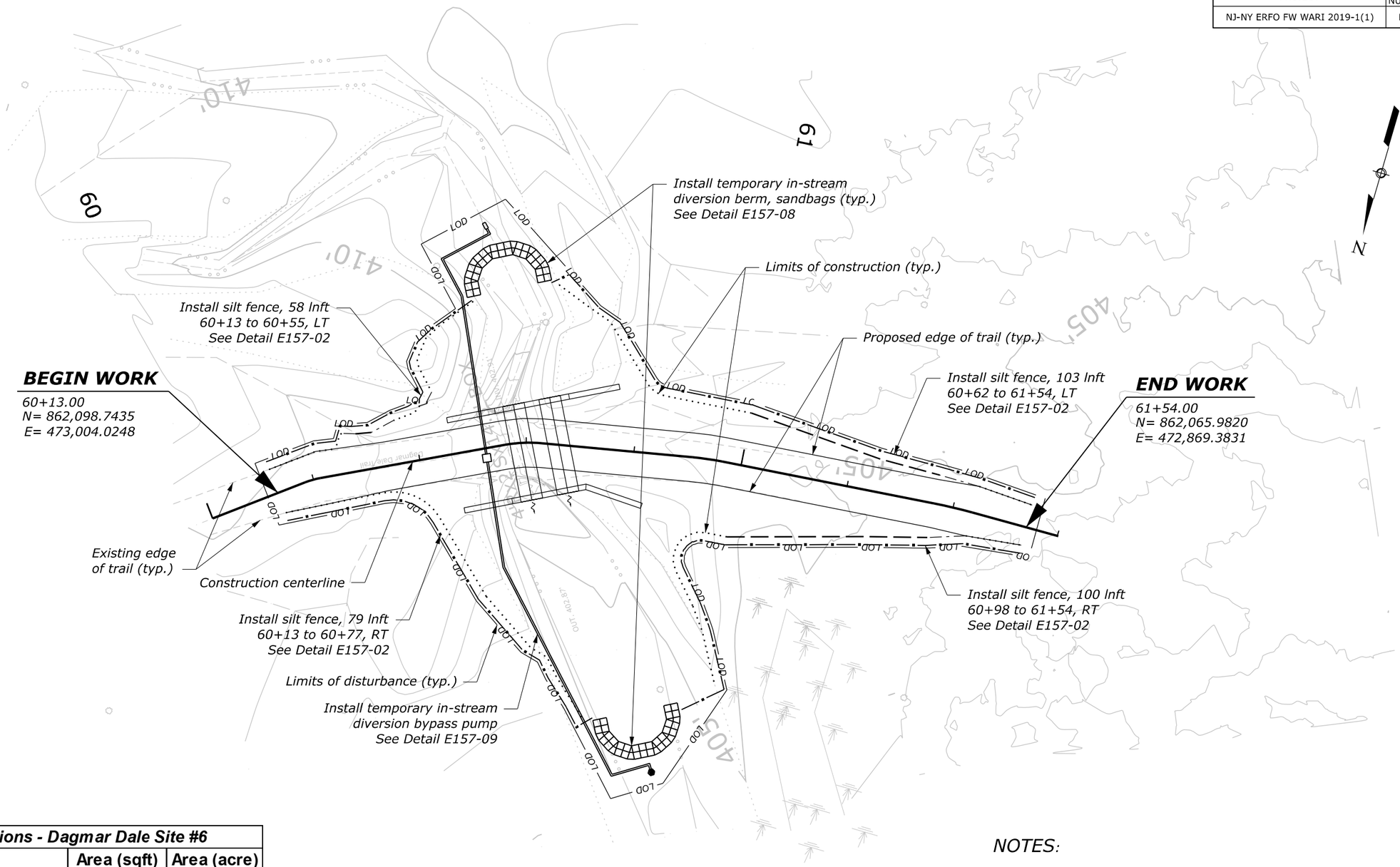
10 August 2026 2:21 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M51



Disturbance Calculations - Dagmar Dale Site #6		
	Area (sqft)	Area (acre)
Total Disturbance	5597	0.13
Permanent Disturbance	3051	0.07
Temporary Disturbance	2546	0.06

NOTES:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*
2. *Refer to Drainage Profiles for cross sections of proposed pipe culverts.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT CONTROL PLANS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

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10 August 2026 2:23 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M52

Township: Vernon Rwp
Block: 321
Lot: 4

NOTES:

1. Refer to Plans and Profiles for proposed work within limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.
3. Refer to Drainage Profiles for cross sections of proposed pipe culverts.

Impact Calculations - Dagmar Dale Site #1

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FWW-GP10	State Open Water	0.013	0.063
FWW-GP10	Wetlands	0	0
FWW-GP10	Transition Area	0	0
FHA-IP	Riparian Zone (300-ft buffer)	0.023	0.104

NO.	DATE	BY	REVISIONS

LEGEND

- *Permanent Impacts*
- *Temporary Impacts*
- *State Open Waters Impacts*

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 15 30
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

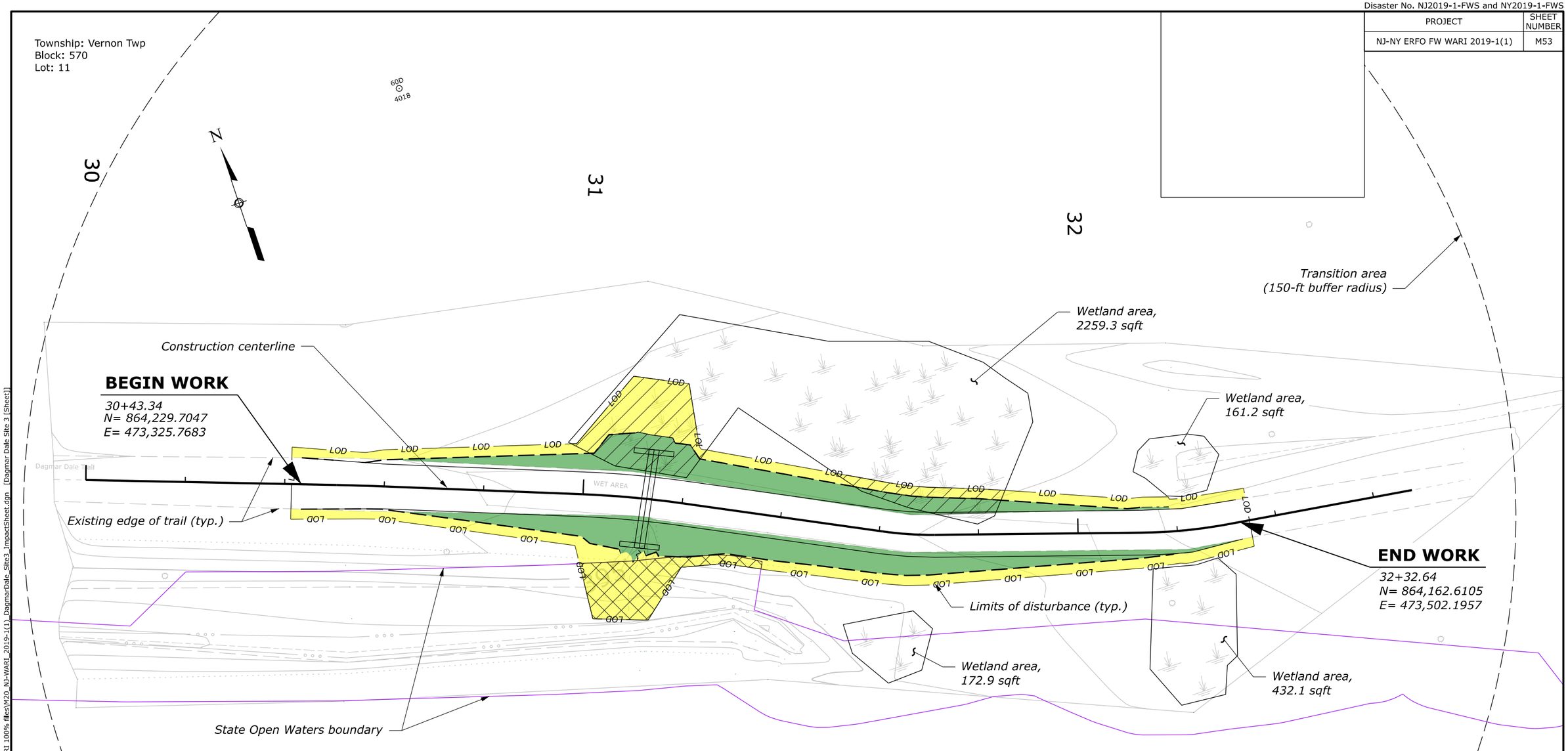
SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1

C:\Users\rreo.kunori\Documents\NJ-NY WARI 100% files\M19_NJ-WARI_2019-1(1)_DagmarDale_Site1_ImpactSheet.dgn [M18 [Sheet]]
8 August 2026 4:15 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M53

Township: Vernon Twp
Block: 570
Lot: 11



Impact Calculations - Dagmar Dale Site #3			
Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FWW-GP10	State Open Water	0.005	0.00001
FWW-GP10	Wetlands	0.007	0.005
FWW-GP10	Transition Area (150-ft radius)	0.014	0.021
FHA-IP	Riparian Zone (300-ft buffer)	0.026	0.026

NOTES:

1. Refer to Plans and Profiles for proposed work within limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.
3. Refer to Drainage Profiles for cross sections of proposed pipe culverts.

NO.	DATE	BY	REVISIONS

LEGEND

- Permanent Impacts
- Temporary Impacts
- Wetland Impacts
- State Open Water Impacts
- Wetland Area

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

IMPACT MAP

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #3

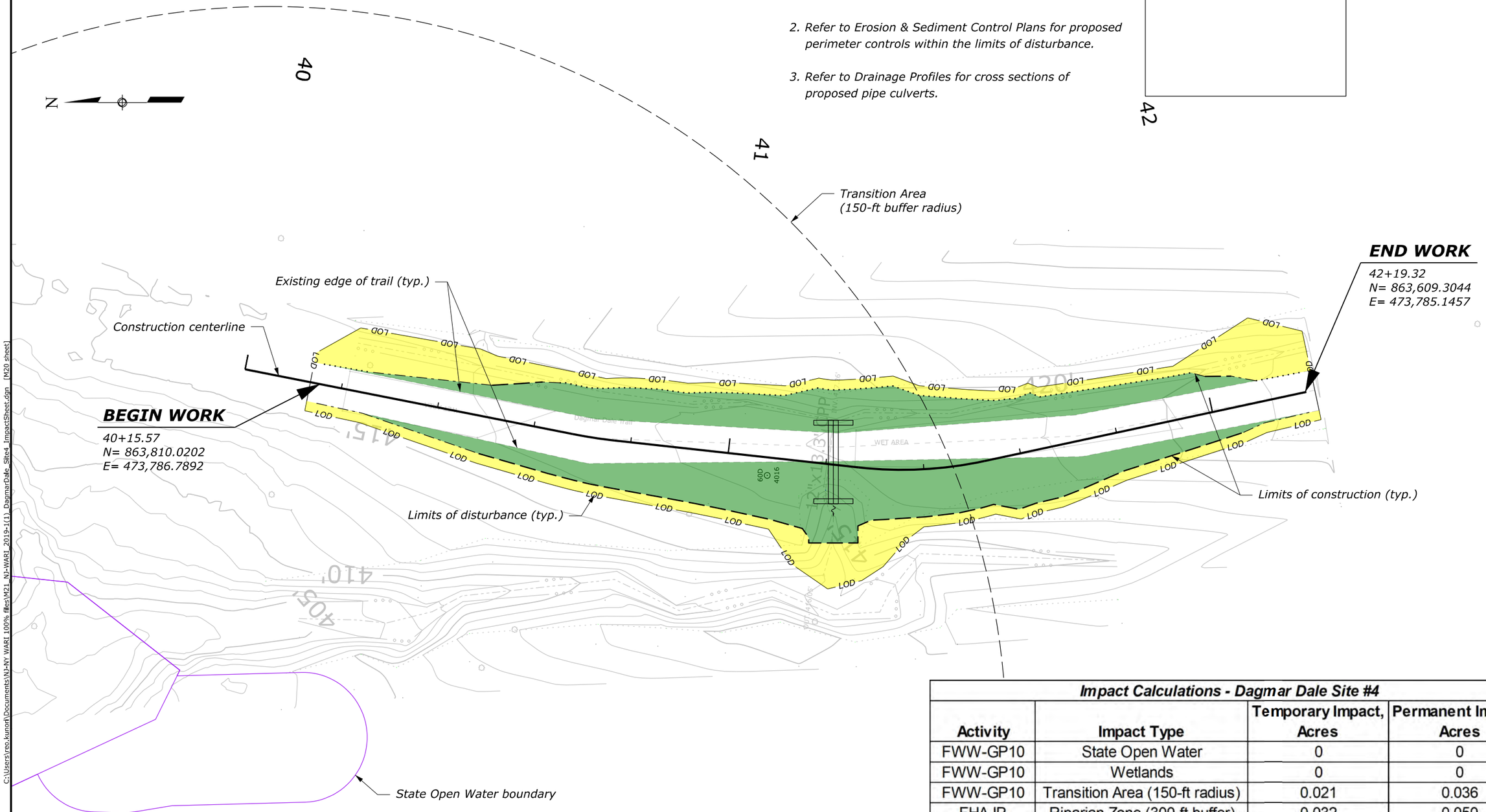
C:\Users\reo.kumorl\Documents\NJ-NY WARI 100% files\M20_NJ-WARI_2019-1(1)_DagmarDale_Site3_ImpactSheet.dgn [Dagmar Dale Site 3 [Sheet]]
8 August 2026 4:18 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M54

Township: Vernon Twp
Block: 570
Lot: 11

NOTES:

1. *Refer to Plans and Profiles for proposed work within limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*
3. *Refer to Drainage Profiles for cross sections of proposed pipe culverts.*



Impact Calculations - Dagmar Dale Site #4

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FWW-GP10	State Open Water	0	0
FWW-GP10	Wetlands	0	0
FWW-GP10	Transition Area (150-ft radius)	0.021	0.036
FHA-IP	Riparian Zone (300-ft buffer)	0.032	0.050

NO.	DATE	BY	REVISIONS

LEGEND

- *Permanent Impacts*
- *Temporary Impacts*

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

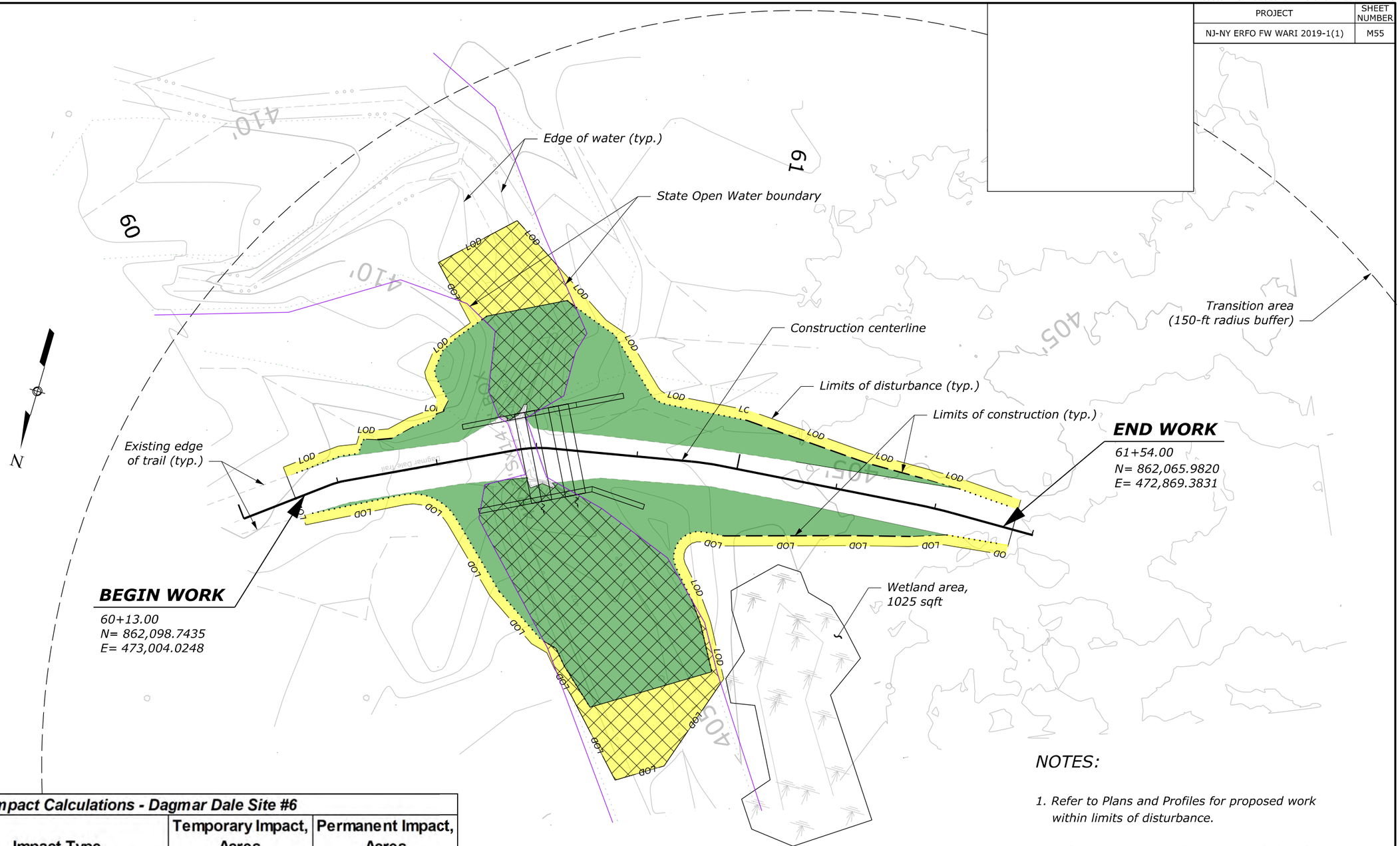
IMPACT MAP

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #4

C:\Users\reo.kunofi\Documents\NJ-NY WARI 100% files\M2s_NJ-WARI_2019-1(1)_DagmarDale_Site4_ImpactSheet.dgn [M20 sheet]
8 August 2026 4:24 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M55

Township: Vernon Twp
Block: 570
Lot: 9



Impact Calculations - Dagmar Dale Site #6

Activity	Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
FWW-GP10	State Open Water	0.015	0.036
FWW-GP10	Transition Area (150-ft radius)	0.014	0.035
FWW-GP10	Wetlands	0	0
FHA-IP	Riparian Zone (300-ft buffer)	0.029	0.070

NOTES:

1. Refer to Plans and Profiles for proposed work within limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.
3. Refer to Drainage Profiles for cross sections of proposed pipe culverts.

NO.	DATE	BY	REVISIONS

LEGEND

- Permanent Impacts
- Temporary Impacts
- State Open Waters Impacts
- Wetland Area

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

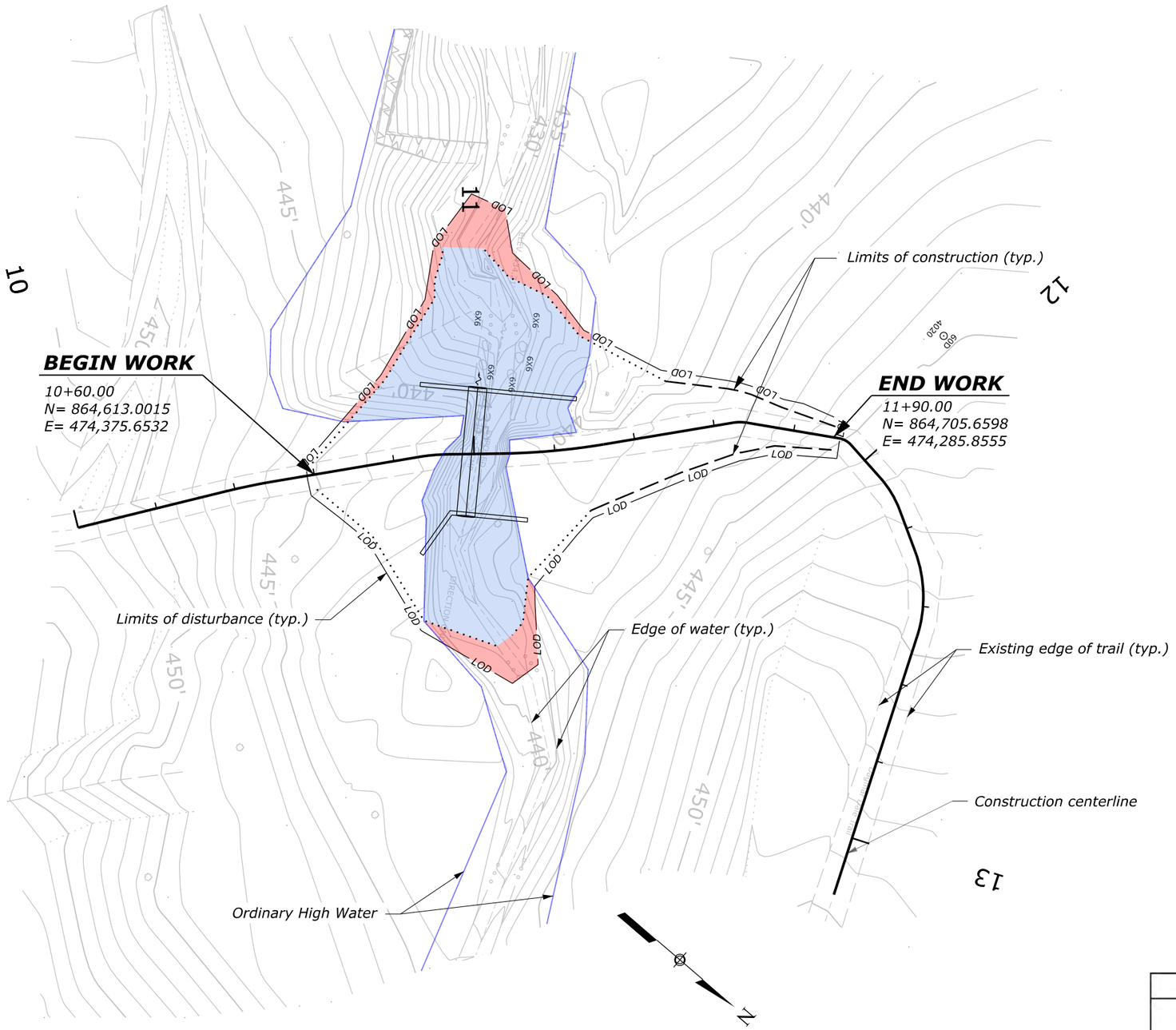
IMPACT MAP

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

C:\Users\rso.kunnh\Documents\NJ-NY WARI 100% files\M22_NJ-WARI_2019-1(1)_DagmarDale_Site6_ImpactSheet.dgn [DagmarDale_Site6 (Sheet)]
8 August 2026 4:25 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M56

Township: Vernon Rwp
Block: 321
Lot: 4



NOTES:

1. *Refer to Plans and Profiles for proposed work within limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*
3. *Refer to Drainage Profiles for cross sections of proposed pipe culverts.*
4. *Ordinary High Water Mark (OHWM) is based on the existing field conditions at the time of the survey.*

Impact Calculations - Dagmar Dale Site #1

Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
Stream	0.013	0.063

NO.	DATE	BY	REVISIONS

LEGEND

- *Permanent Stream Impacts*
- *Temporary Stream Impacts*

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 15 30
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

STREAM IMPACT MAP

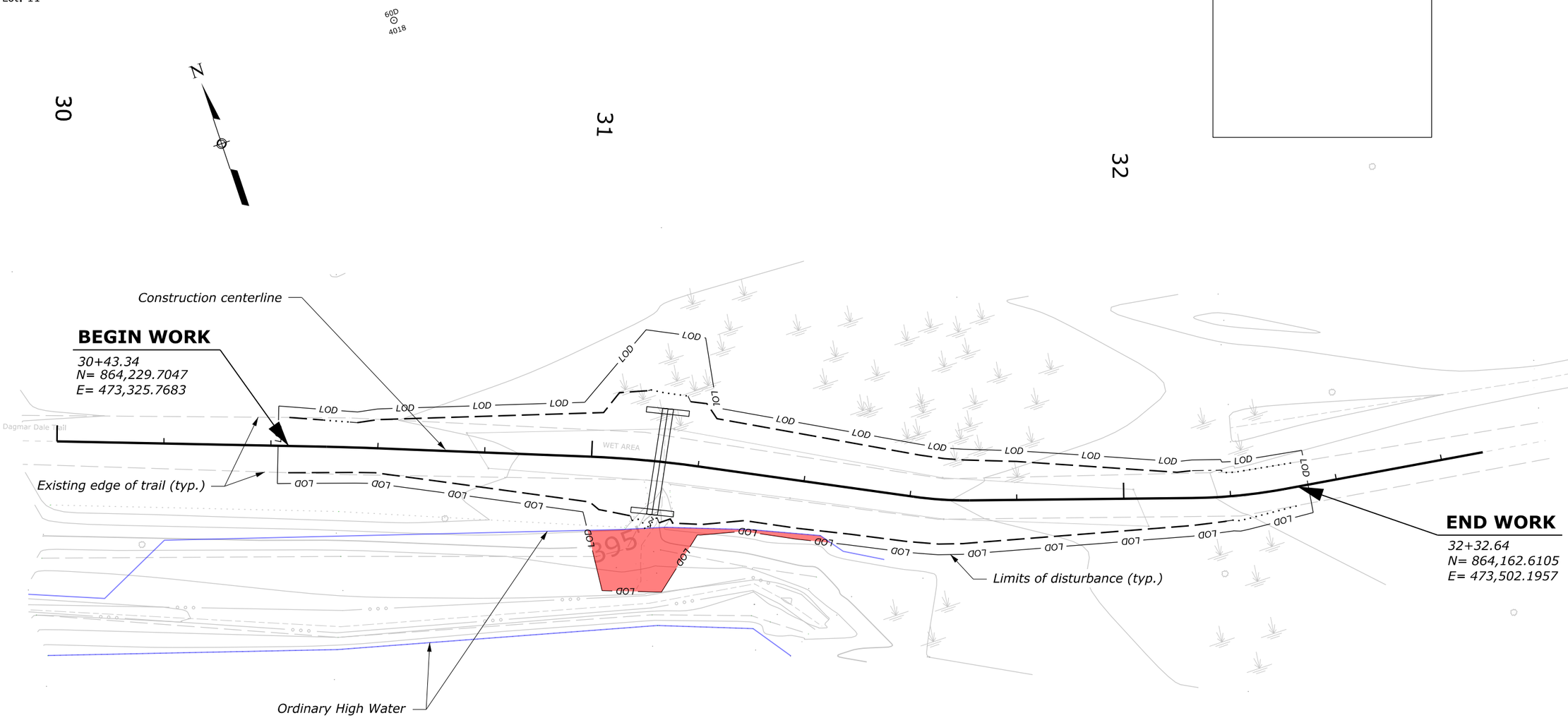
SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1

SHEET 1 OF 3

C:\Users\reo.kumofi\Documents\NJ-NY WARI 100% files\M32_NJ-WARI_2019-1(1)_DagmarDale_Site1_StreamImpact.dgn [M31 (Sheet)]
8 August 2026 4:27 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M57

Township: Vernon Twp
Block: 570
Lot: 11



Impact Calculations - Dagmar Dale Site #3		
Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
Stream	0.005	0.000001

NOTES:

1. *Refer to Plans and Profiles for proposed work within limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*
3. *Refer to Drainage Profiles for cross sections of proposed pipe culverts.*

NO.	DATE	BY	REVISIONS

LEGEND

- *Permanent Stream Impacts*
- *Temporary Stream Impacts*

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

STREAM IMPACT MAP

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #3

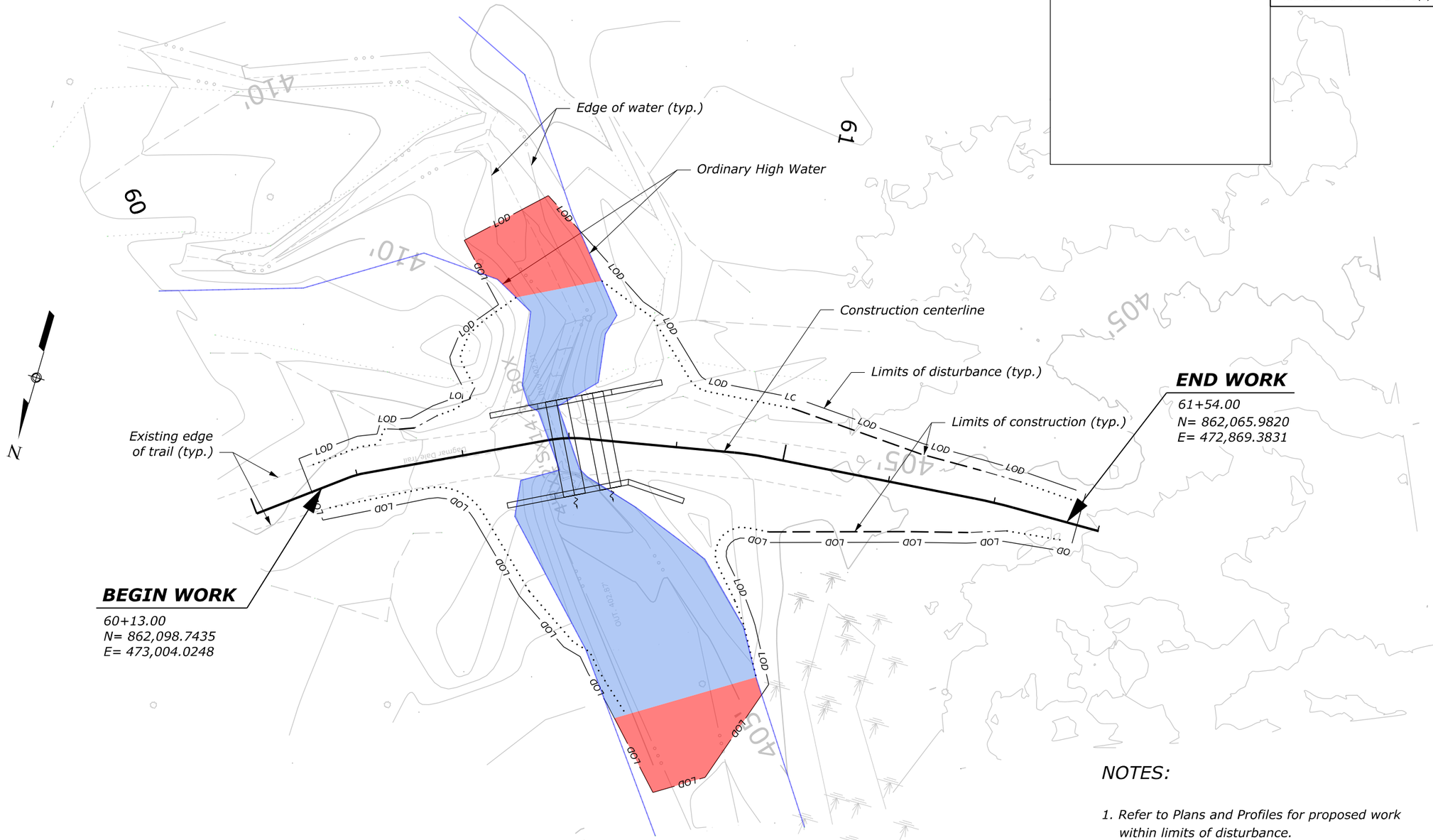
SHEET 2 OF 3

C:\Users\reo.kunori\Documents\NJ-NY WARI 100% files\M33_NJ-WARI_2019-1(1)_DagmarDale_Site3_StreamImpact.dgn [Dagmar Dale Site 3 [Sheet]]
8 August 2026 4:35 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M58

Township: Vernon Twp
Block: 570
Lot: 9



BEGIN WORK
60+13.00
N= 862,098.7435
E= 473,004.0248

END WORK
61+54.00
N= 862,065.9820
E= 472,869.3831

Impact Calculations - Dagmar Dale Site #6		
Impact Type	Temporary Impact, Acres	Permanent Impact, Acres
Stream	0.014	0.035

NOTES:

1. *Refer to Plans and Profiles for proposed work within limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*
3. *Refer to Drainage Profiles for cross sections of proposed pipe culverts.*

NO.	DATE	BY	REVISIONS

LEGEND

- *Permanent Stream Impacts*
- *Temporary Stream Impacts*

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

STREAM IMPACT MAP

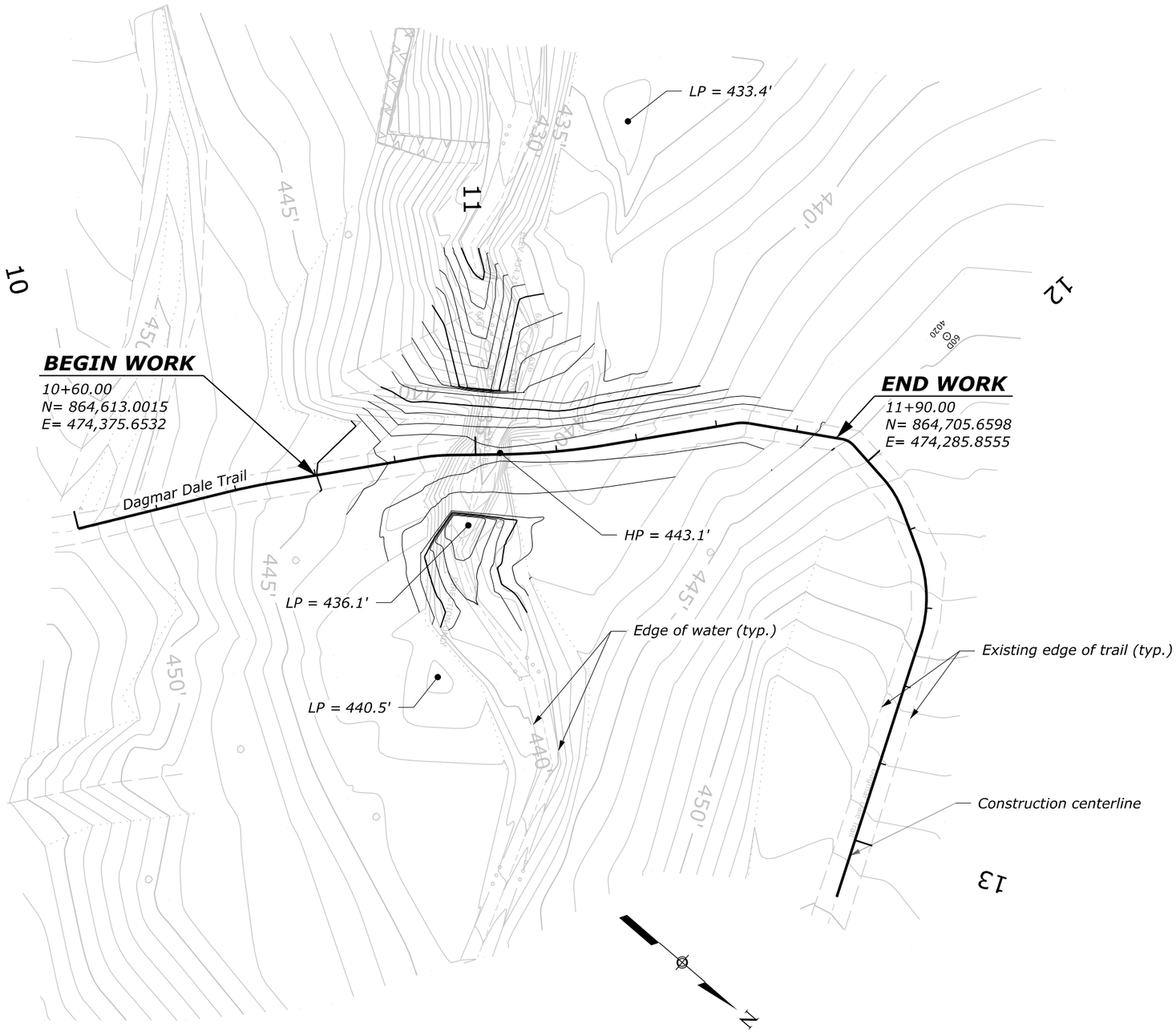
SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

SHEET 3 OF 3

C:\Users\rreo.kunori\Documents\NJ-NY WARI 100% files\M34_NJ-WARI_2019-1(1)_DagmarDale_Site6_StreamImpact.dgn [DagmarDale_Site6 [Sheet1]]
8 August 2026 4:37 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M59

Township: Vernon Rwp
Block: 321
Lot: 4



NOTES:

1. Refer to Plans and Profiles for proposed work within limits of disturbance.
2. Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.
3. Refer to Drainage Profiles for cross sections of proposed pipe culverts.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 15 30
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

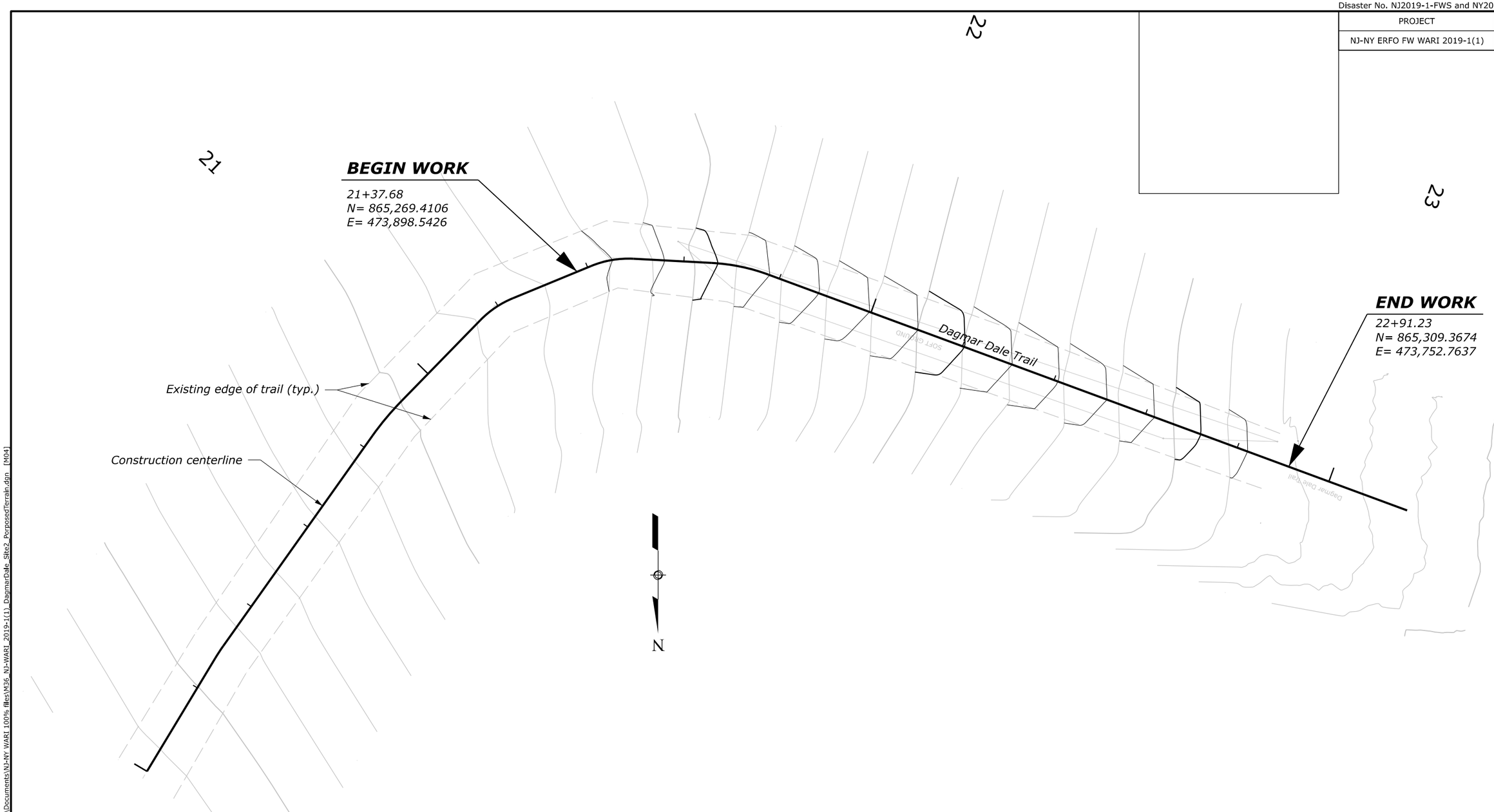
SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1

SHEET 1 OF 5

C:\Users\reo.kumoh\Documents\NJ-NY WARI 100% files\M35_NJ-WARI_2019-1(1)_DagmarDale_Site1_PorposedTerrain.dgn [M59 (Sheet)]
8 August 2026 4:47 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M60



NOTE:

1. *Refer to Plans and Profiles for proposed work within the limits of disturbance.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #2

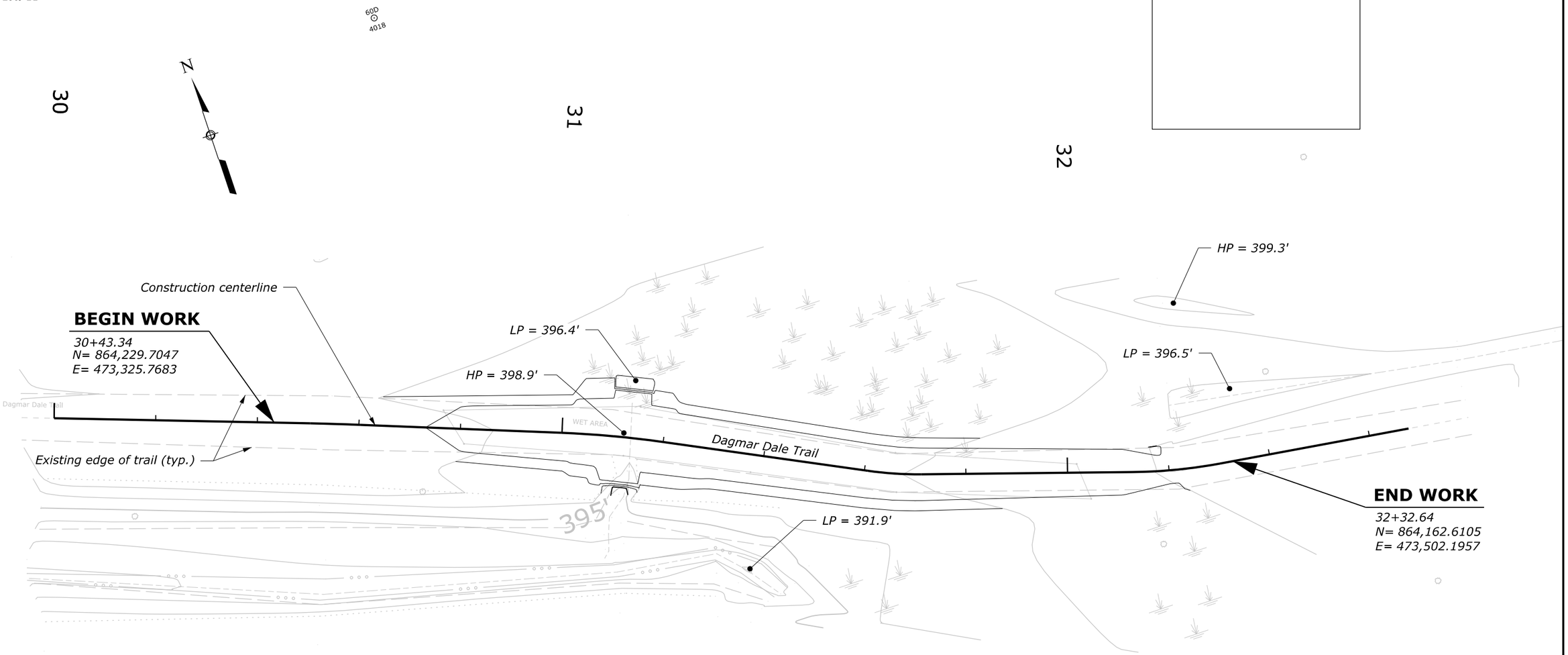
SHEET 2 OF 5

C:\Users\vreo.kunori\Documents\NJ-NY WARI 100% files\M36_NJ-WARI_2019-1(1)_DagmarDale_Site2_ProposedTerrain.dgn [M04]
8 August 2026 4:48 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M61

Township: Vernon Twp
Block: 570
Lot: 11



NOTES:

1. *Refer to Plans and Profiles for proposed work within limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*
3. *Refer to Drainage Profiles for cross sections of proposed pipe culverts.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #3

C:\Users\rseo.kunnri\Documents\NJ-NY WARI 100% files\M37_NJ-WARI_2019-1(1)_DagmarDale_Site3_PorposedTerrain.dgn [Dagmar Dale Site 3 [Sheet]]
20 July 2026 11:21 AM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M62

Township: Vernon Twp
Block: 570
Lot: 11

NOTES:

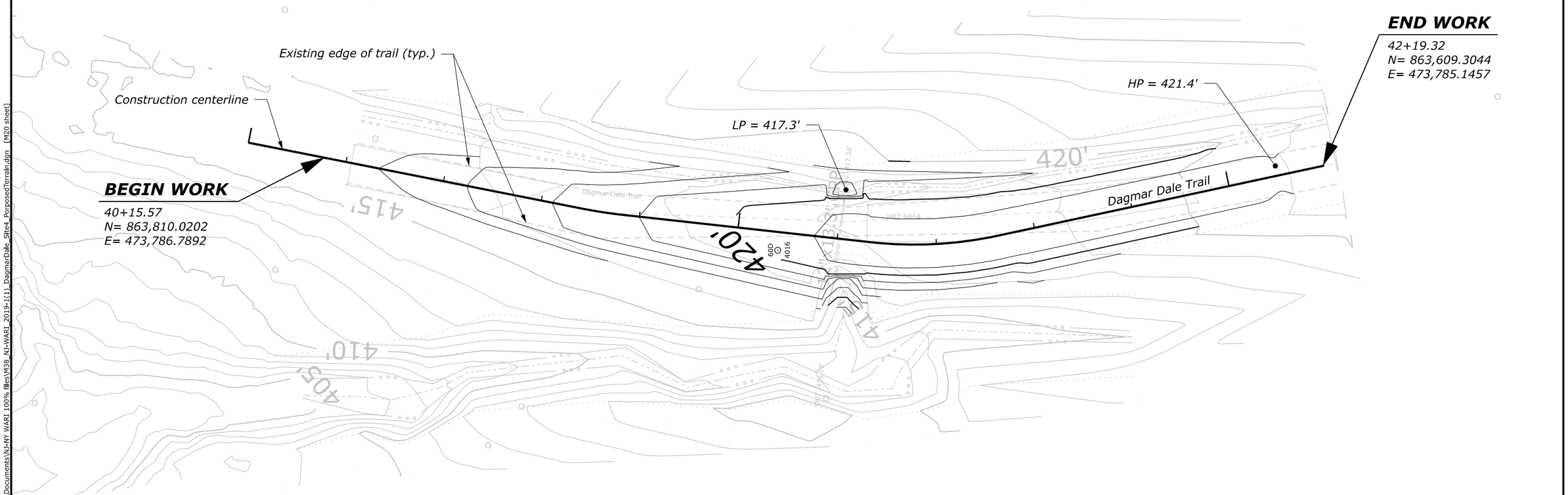
1. *Refer to Plans and Profiles for proposed work within limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*
3. *Refer to Drainage Profiles for cross sections of proposed pipe culverts.*

40

41

42

N



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #4

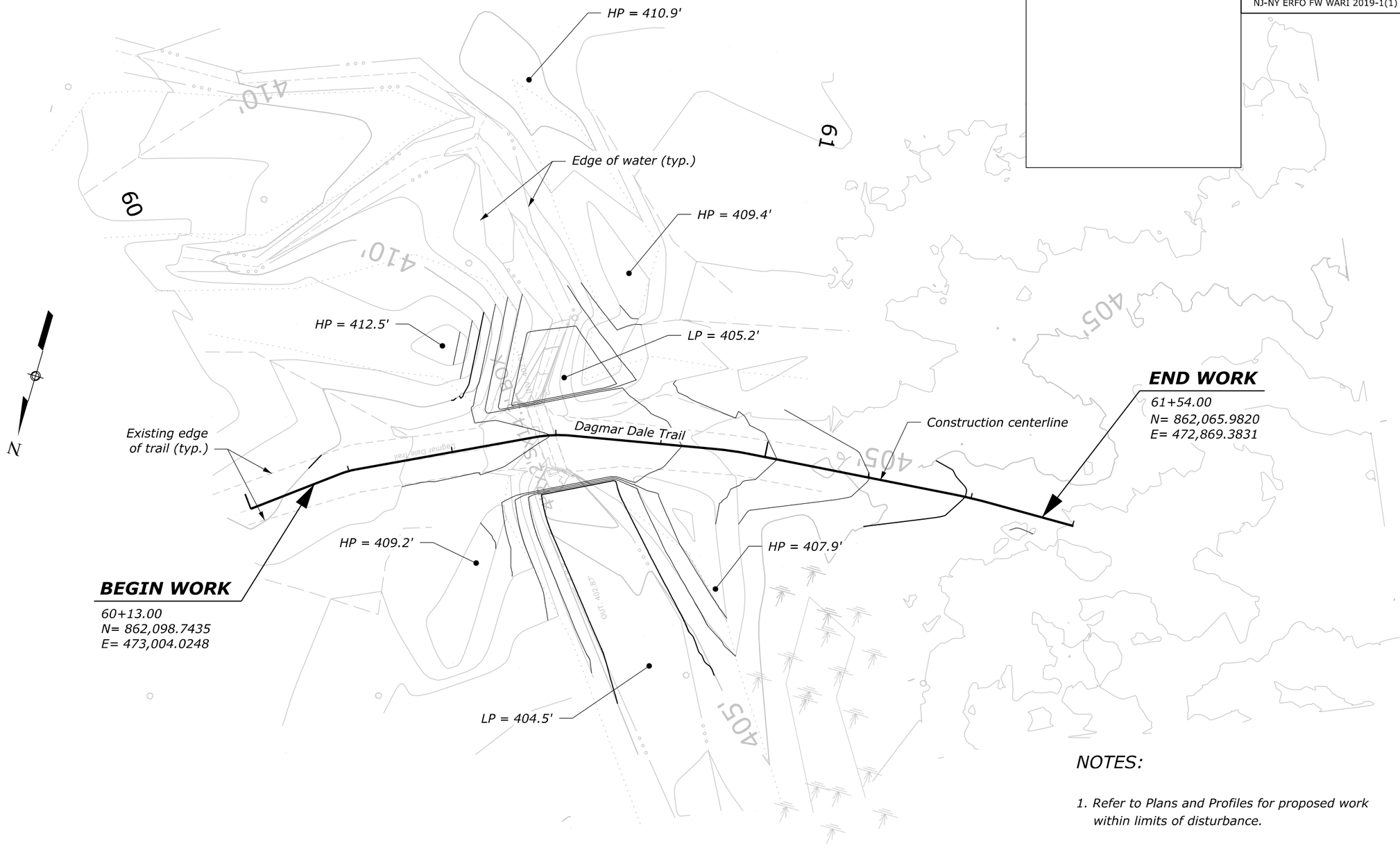
SHEET 4 OF 5

C:\Users\rec.kunori\Documents\NJ-NY WARI 100% files\M38_NJ-WARI_2019-1(1)_DagmarDale_Site4_PorposedTerrain.dgn [M20 sheet]
20 July 2026 11:45 AM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	M63

Township: Vernon Twp
Block: 570
Lot: 9



BEGIN WORK
60+13.00
N= 862,098.7435
E= 473,004.0248

END WORK
61+54.00
N= 862,065.9820
E= 472,869.3831

NOTES:

1. *Refer to Plans and Profiles for proposed work within limits of disturbance.*
2. *Refer to Erosion & Sediment Control Plans for proposed perimeter controls within the limits of disturbance.*
3. *Refer to Drainage Profiles for cross sections of proposed pipe culverts.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 10 20
SCALE IN FEET

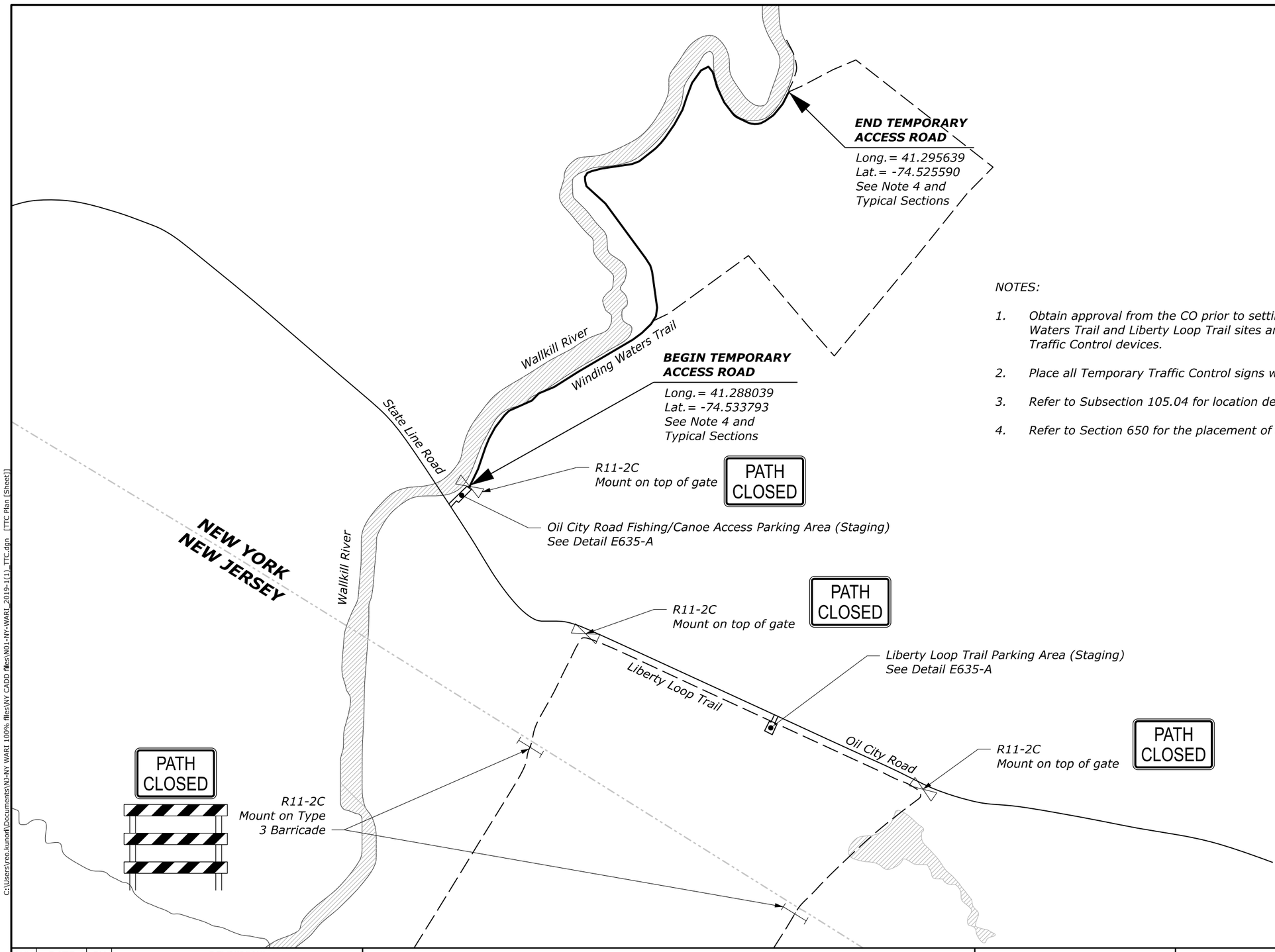
WALLKILL RIVER NATIONAL WILDLIFE REFUGE

PROPOSED CONTOUR MAP

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

C:\Users\vreo.kunori\Documents\NJ-NY WARI 100% files\M40_NJ-WARI_2019-1(1)_DagmarDale_Site6_PorposedTerrain.dgn [DagmarDale_Site6 [Sheet]]
8 August 2026 4:54 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	N01



NOTES:

1. *Obtain approval from the CO prior to setting up staging areas for the Winding Waters Trail and Liberty Loop Trail sites and before placement of Temporary Traffic Control devices.*
2. *Place all Temporary Traffic Control signs within Fish and Wildlife Services right-of-way.*
3. *Refer to Subsection 105.04 for location designated for the staging areas.*
4. *Refer to Section 650 for the placement of the temporary access road.*

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 300 600
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TEMPORARY TRAFFIC CONTROL PLAN

SCHEDULE A: LIBERTY LOOP TRAIL (NE) & WINDING WATERS TRAIL

SHEET 1 OF 3

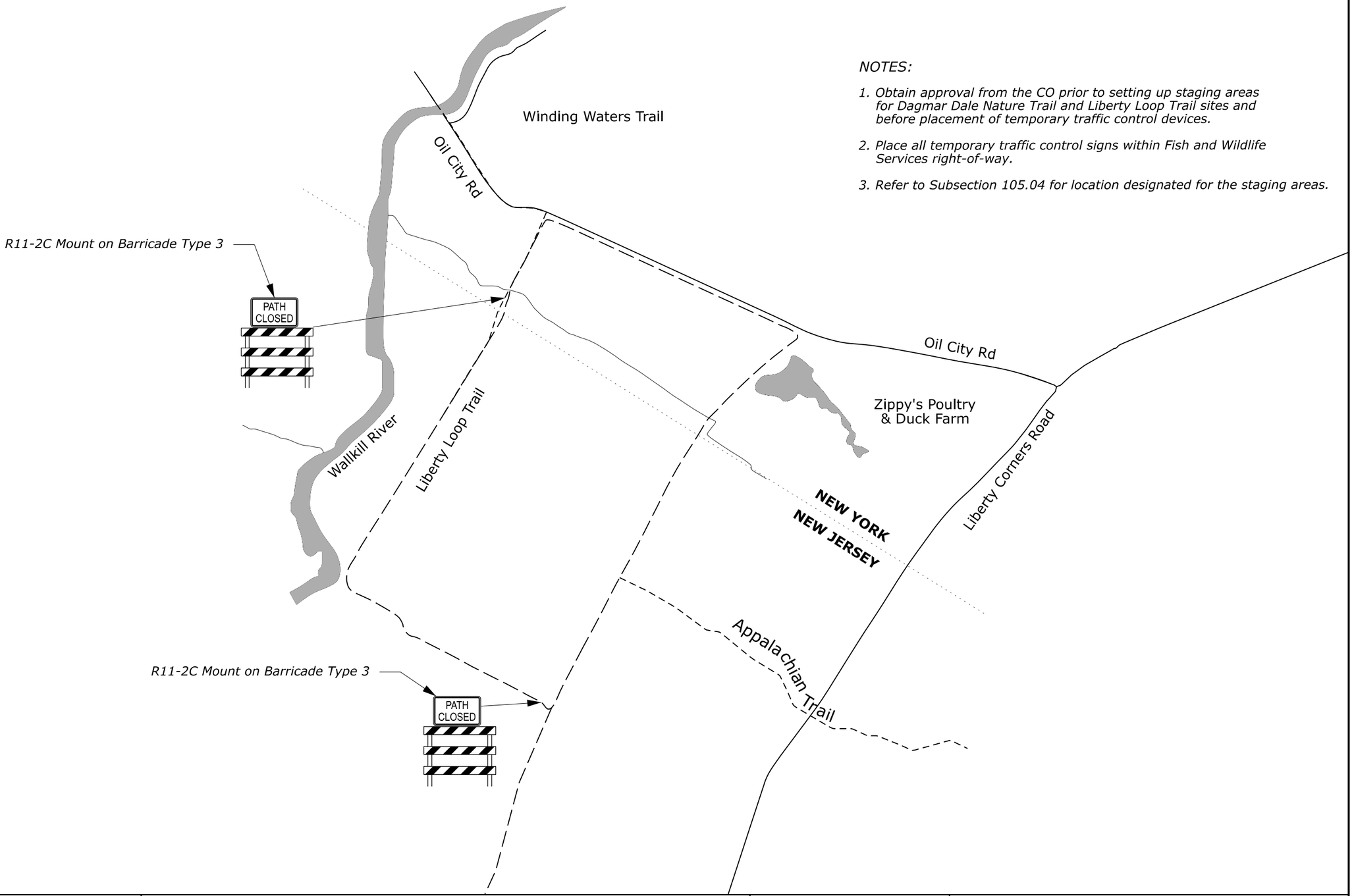
C:\Users\rcox.kunori\Documents\NJ-NY WARI 100% files\NY CADD files\N01-NY-WARI_2019-1(1)_TTC.dgn [TTC Plan [Sheet]]
5 August 2026 8:47 AM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	N02

NOTES:

1. *Obtain approval from the CO prior to setting up staging areas for Dagmar Dale Nature Trail and Liberty Loop Trail sites and before placement of temporary traffic control devices.*
2. *Place all temporary traffic control signs within Fish and Wildlife Services right-of-way.*
3. *Refer to Subsection 105.04 for location designated for the staging areas.*



C:\Users\vreo.kumori\Documents\NJ-NY WARI 100% files\N01-detour.dgn [TEMPORARY TRAFFIC CONTROL PLAN]
8 August 2026 10:05 PM

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 250 500
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

TEMPORARY TRAFFIC CONTROL PLAN

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

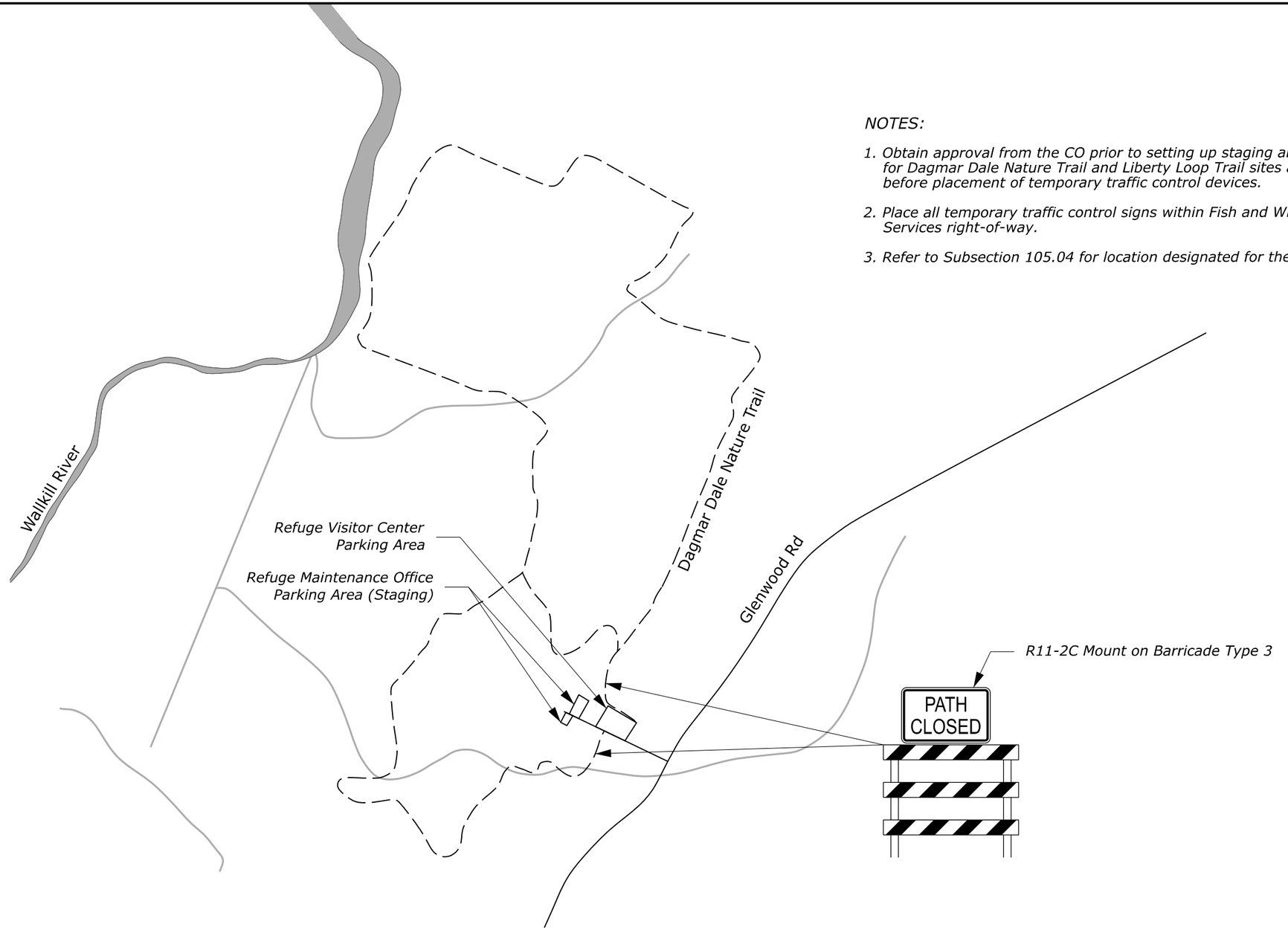
SHEET 2 OF 3

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	N03

NOTES:

1. *Obtain approval from the CO prior to setting up staging areas for Dagmar Dale Nature Trail and Liberty Loop Trail sites and before placement of temporary traffic control devices.*
2. *Place all temporary traffic control signs within Fish and Wildlife Services right-of-way.*
3. *Refer to Subsection 105.04 for location designated for the staging areas.*



NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

0 250 500
SCALE IN FEET

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

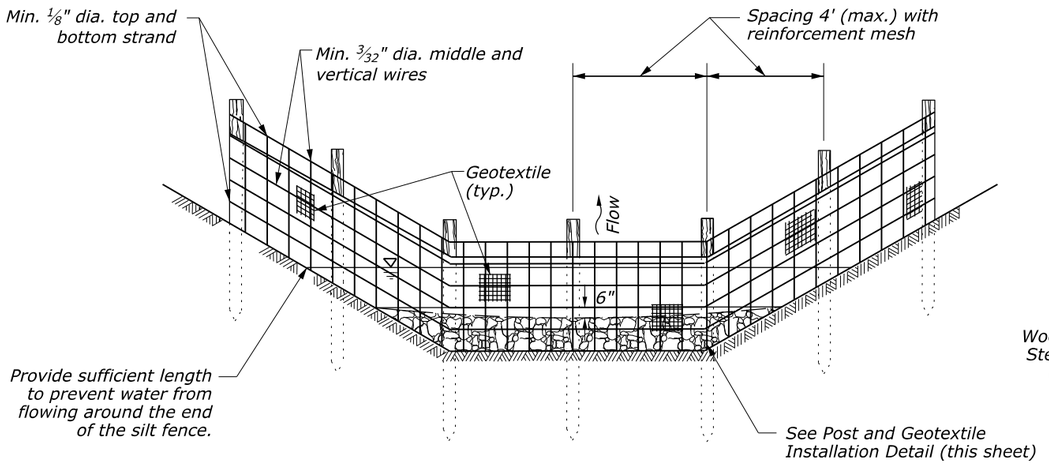
TEMPORARY TRAFFIC CONTROL PLAN

SCHEDULE C: DAGMAR DALE NATURE TRAIL

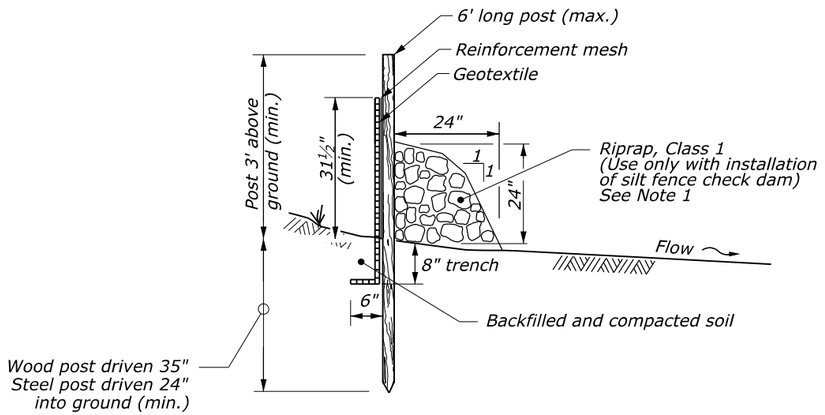
SHEET 3 OF 3

C:\Users\rreo.kunori\Documents\NJ-NY WARI 100% files\N01-detour.dgn [TEMPORARY TRAFFIC CONTROL PLAN (02)]
8 August 2026 10:04 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	S01



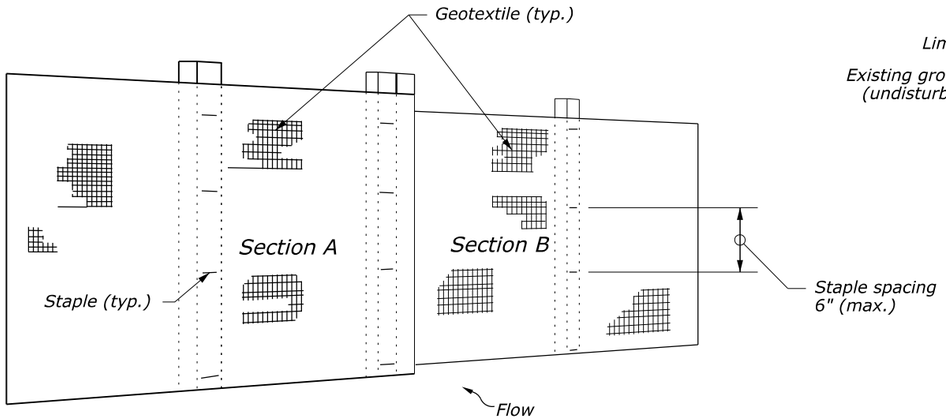
WIRE-BACKED SILT FENCE WITH CHECK DAM



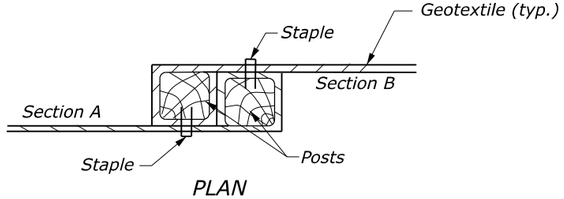
POST AND GEOTEXTILE INSTALLATION

NOTES:

1. *Install silt fence with Class 1 riprap only in low-flow drainage ditches where shown on the Erosion and Sediment Control Plan.*
2. *Install pre-assembled silt fence according to the manufacturer's recommendations.*
3. *Install silt fence along ground contours. Curve ends of silt fence upgrade to prevent water from running around the ends.*
4. *Attach geotextile and reinforcement mesh so they do not slide down posts. Provide detail for attaching to steel posts for approval.*
5. *Use reinforcement mesh that is a minimum of 32 inches in width and has a minimum of 6 line wires with 12-inch stay spacing.*
6. *Use geotextile that is a minimum of 45 inches in width and fasten adequately to the reinforcement mesh as directed.*
7. *Use 60-inch minimum height steel posts of the self-fastener angle steel type.*
8. *Use 70-inch minimum height by 3-inch diameter wood posts.*
9. *Extend reinforcement mesh and geotextile into trench.*



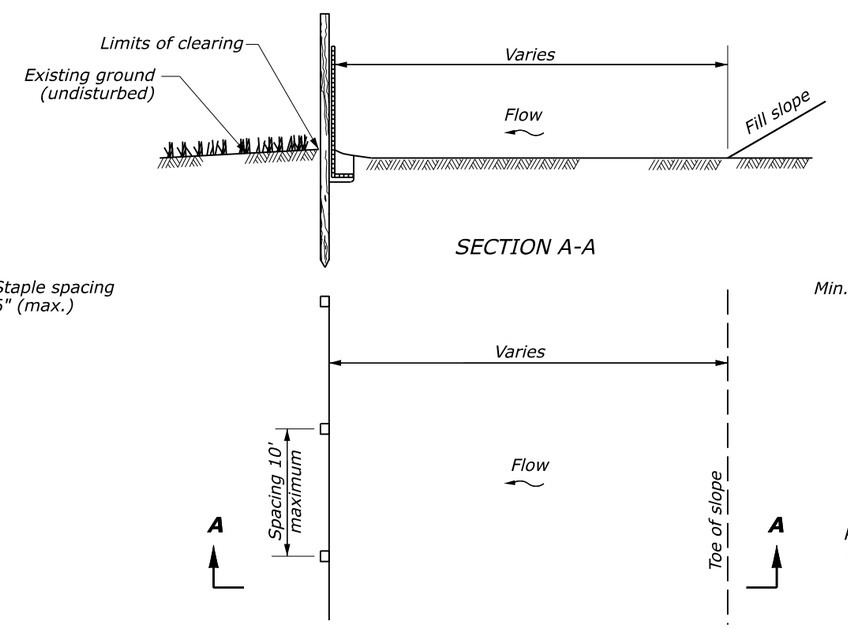
ELEVATION



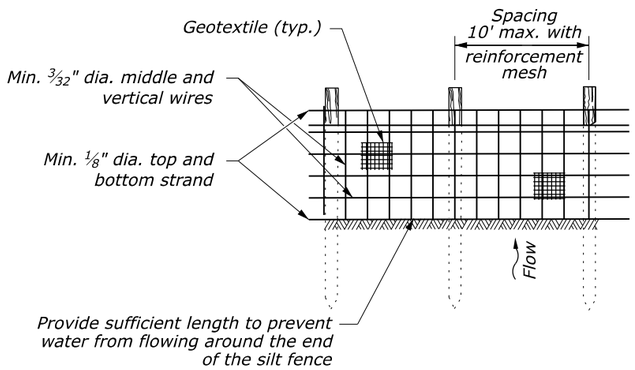
PLAN

JOINING TWO ADJACENT SILT FENCE SECTIONS

(See Note 4)



PLAN



ELEVATION

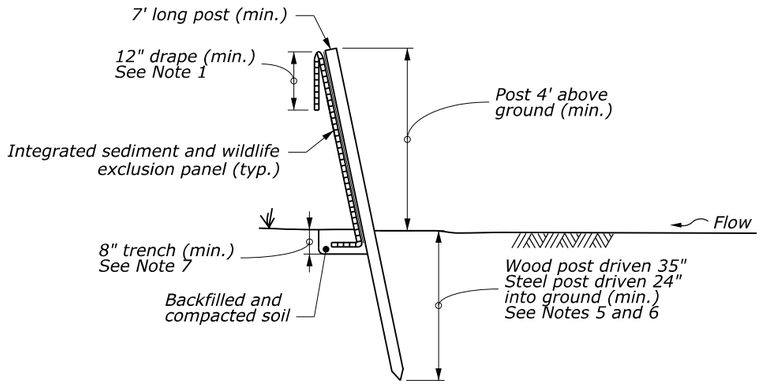
WIRE-BACKED SILT FENCE INSTALLATION AT TOE OF FILL

NO SCALE

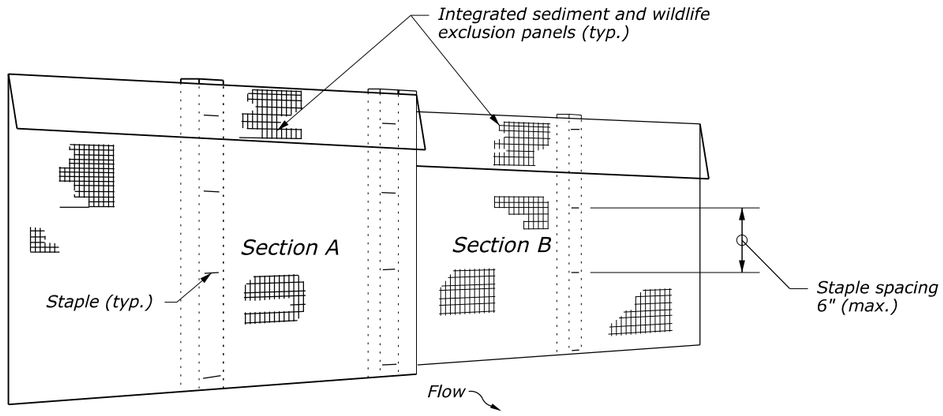
U.S. DEPARTMENT OF TRANSPORTATION, FHWA OFFICE OF FEDERAL LANDS HIGHWAY	EFLHD DETAIL E157-02
WIRE-BACKED SILT FENCE	SPECIFICATION FP-24, FP-14
	APPROVED FOR USE 05/2024

\\fhfil5\fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) Combined NJ-NY\Std-Det\S01-E157-02.dgn [S01]
28 July 2026 10:29 AM

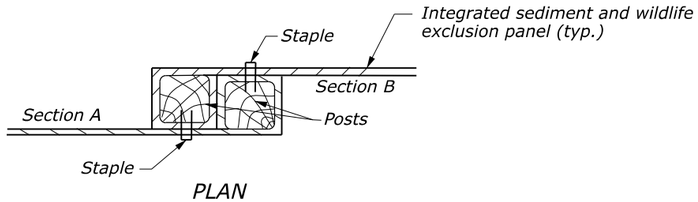
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	S02



POST AND INTEGRATED SEDIMENT AND WILDLIFE EXCLUSION PANEL INSTALLATION



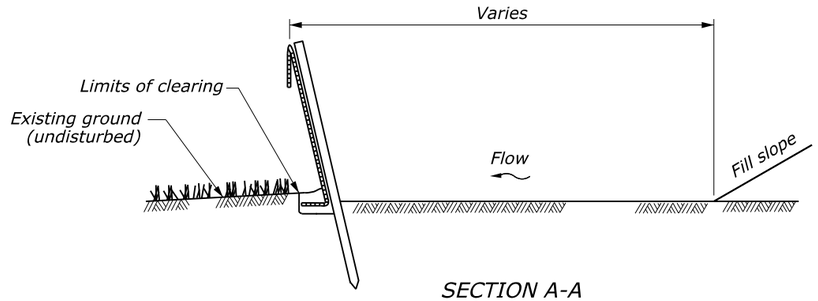
ELEVATION



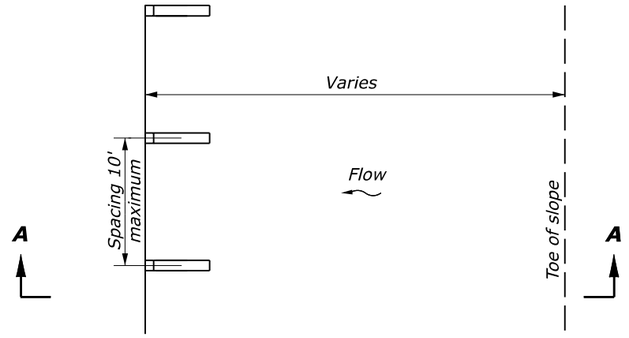
PLAN

JOINING TWO ADJACENT INTEGRATED SEDIMENT AND WILDLIFE EXCLUSION BARRIER SECTIONS

(See Note 4)

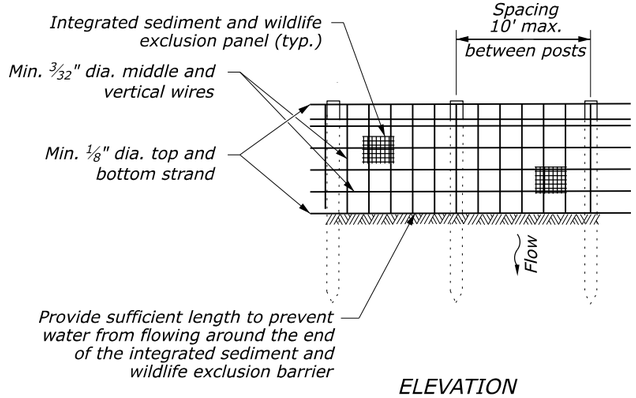


SECTION A-A



PLAN

INTEGRATED SEDIMENT AND WILDLIFE EXCLUSION BARRIER INSTALLATION AT TOE OF FILL



ELEVATION

NOTES:

1. Install integrated sediment and wildlife exclusion barrier at an obtuse angle to allow a minimum of 12 inches of material to drape at the top and create a vertical barrier.
2. Install pre-assembled integrated sediment and wildlife exclusion barrier system according to the manufacturer's recommendations. Ensure material used has a smooth, non-woven surface which inhibits climbing by amphibians.
3. Install integrated sediment and wildlife exclusion barrier system along ground contours. Curve ends of barrier upgrade to prevent water from running around the ends, unless curving upgrade will direct wildlife into the project area or other hazards. Curve ends away from wildlife hazard areas.
4. Attach integrated sediment and wildlife exclusion panels so they do not slide down posts. Provide detail for attaching to steel posts for approval.
5. Use 72-inch minimum height steel posts of the self-fastener angle steel type to anchor to the ground and keep taut.
6. Use 83-inch minimum height by 3-inch diameter wood posts to anchor to the ground and keep taut.
7. Extend integrated sediment and wildlife exclusion barrier a minimum of 8 inches into trench to secure the base of the fence.

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

WILDLIFE EXCLUSION SILT FENCE

EFLHD DETAIL	
E157-02A	
SPECIFICATION	FP-14
APPROVED FOR USE	05/2024

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22 October 2026 4:33 PM

PROJECT	SHEET NUMBER
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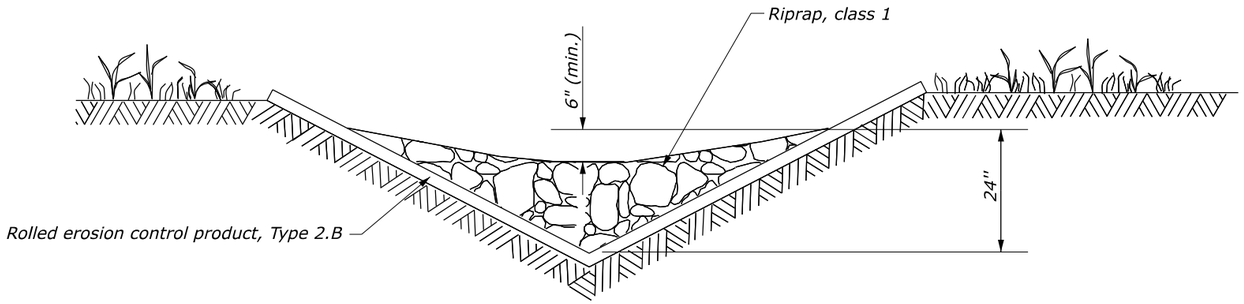
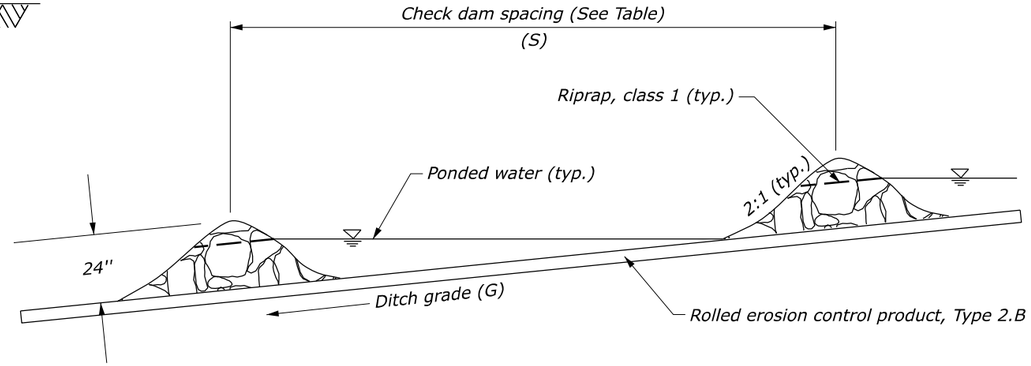
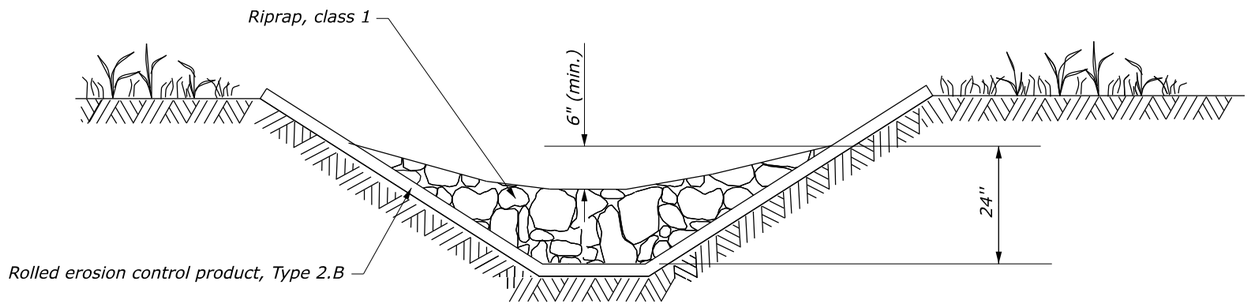


V-DITCH



DITCH PROFILE VIEW



TRAPEZOIDAL DITCH

DITCH CROSS-SECTION VIEW

CHECK DAM SPACING TABLE	
*DITCH GRADE (G)**	*SPACING (S) LNFT*
2%	*75*
3%	*50*
4%	*40*
5%	*30*
6%	*25*

** Do not use Check Dams below 2% or above 6% ditch grades.*

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

CHECK DAM WITH ROLLED EROSION CONTROL PRODUCT

EFLHD DETAIL
E157-06

SPECIFICATION
FP-24, FP-14

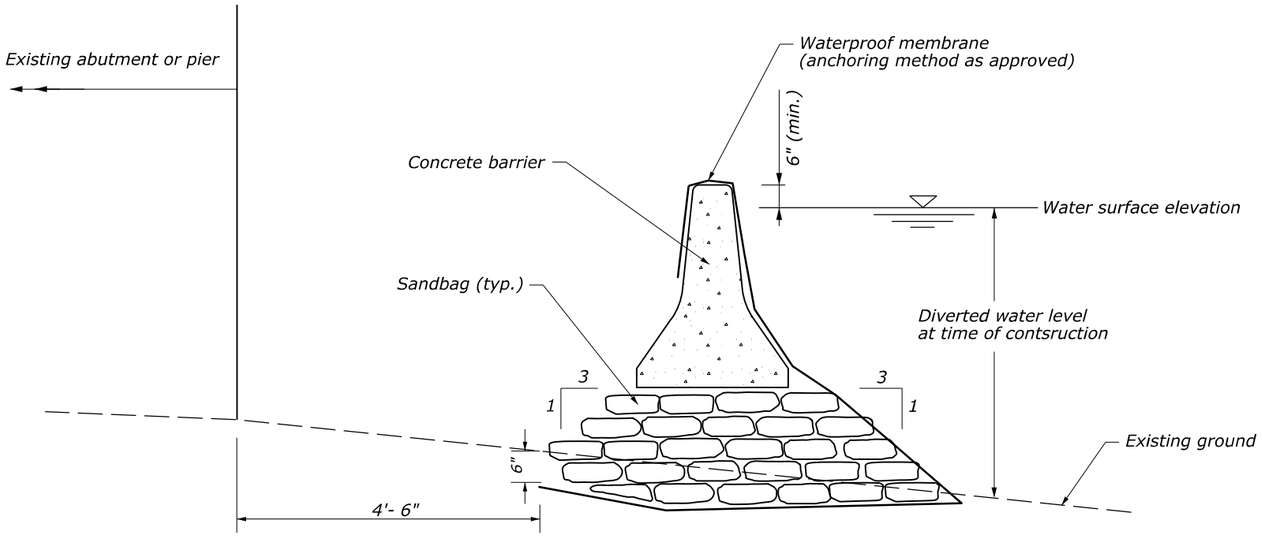
APPROVED FOR USE
05/2024

28 July 2026 10:29 AM \\fhl15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) Combined NJ-NY\Std-Det\S03-E157-06.dgn [Sheet]

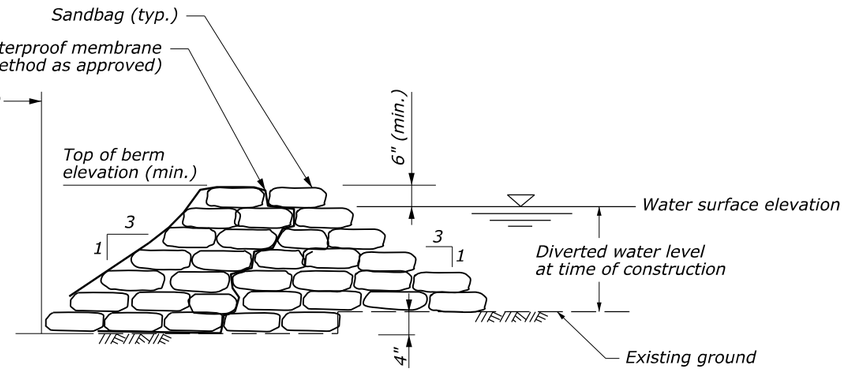
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	S04

NOTES:

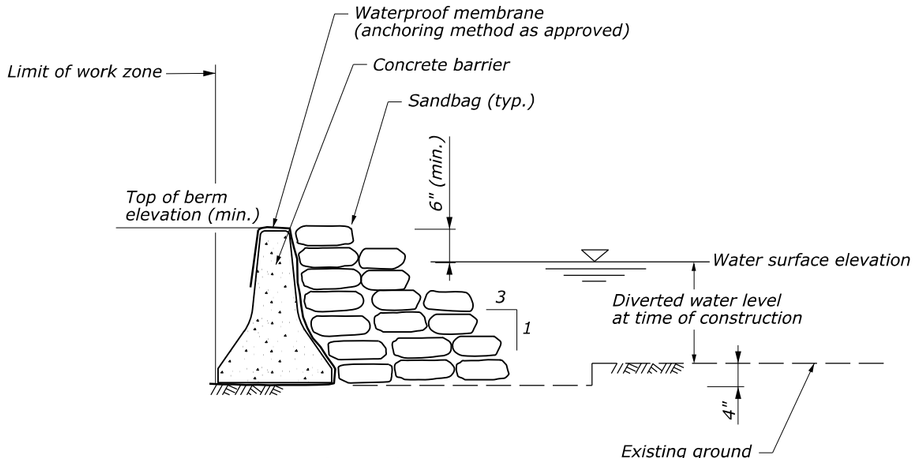
1. *These options suggest configurations for diverting a stream during in-water construction activities. Alternate stream diversion methods may be chosen (including any approved prefabricated or portable diversion berms, dams, etc.). As a minimum, provide a temporary diversion berm with a minimum height equivalent to the water surface elevation with 6-inch minimum freeboard. Submit temporary stream diversion drawings for approval prior to installation.*
2. *Construct diversion berms according to Subsection 157.10.*
3. *Place sandbags to form a pyramid by laying an equal number of bottom rows as there are vertical course. Overlap upper rows of sandbags above the joints in ower rows.*
4. *Place no more than one temporary diversion berm in the stream at any given time.*
5. *While in use, inspect and maintain in-stream diversion berms daily. Repair as needed after rainfall events or as directed. Remove sediment when deposits reach one-half the height of the sandbag barrier.*



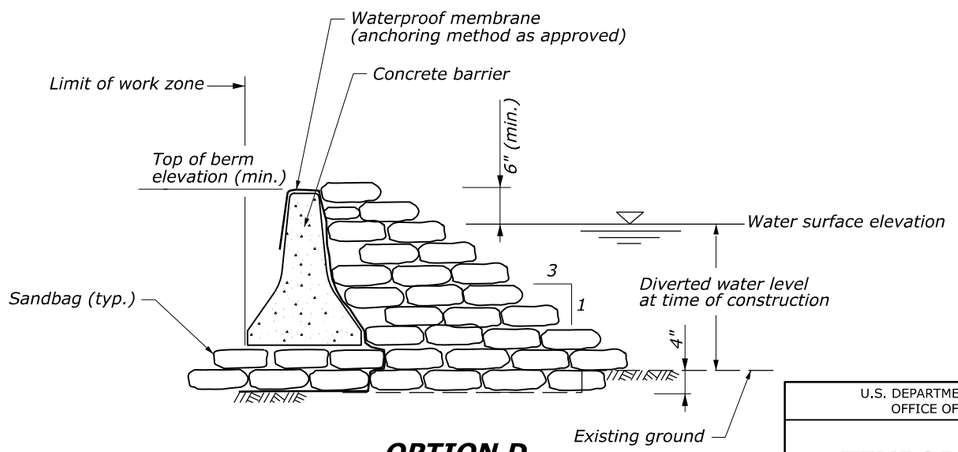
OPTION A



OPTION B



OPTION C



OPTION D

NO SCALE

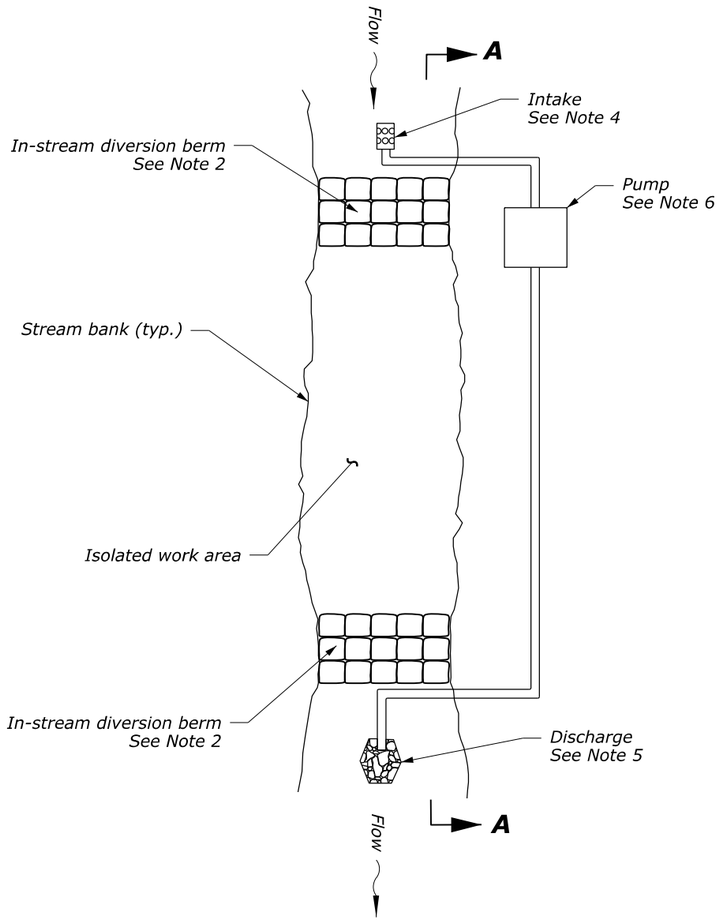
U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

TEMPORARY IN-STREAM DIVERSION BERM METHODS

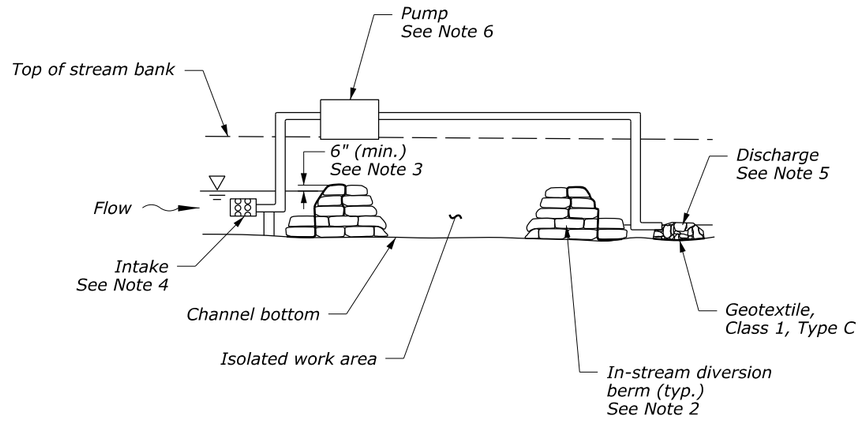
EFLHD DETAIL	E157-08
SPECIFICATION	FP-14
APPROVED FOR USE	05/2024

28 July 2026 10:30 AM \\fhl15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) Combined NJ-NY\Std-Det\S04-E157-08fp14.dgn [Sheet]

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	S05



PLAN VIEW



SECTION A-A

NO SCALE

NOTES:

1. *Provide a stream diversion system for expected flow during in-water work activities that accommodate at least a 2-year peak flood event.*
2. *Construct in-stream diversion berms according to Detail E157-08. Install the upstream barrier first, followed by the downstream barrier.*
3. *For stream diversion systems in place longer than one workday, ensure the height of the diversion berm is 1 foot higher than the 2-year peak flood event.*
4. *Suspend the intake above the channel bottom to prevent sediment from entering the intake. Place a mesh screen over the intake with a maximum mesh size of $^{3}/_{32}$ inch. Enclose the intake with a secondary screen box 3 feet on each side if velocities at the intake are greater than 0.4 feet per second.*
5. *Stabilize the discharge point at the downstream end to prevent erosion.*
6. *Have spare pumps on site in the event of pump failures or higher than expected flow rates. Place the pump on firm ground with secondary containment measures.*
7. *Remove water from the isolated work area as necessary to perform the work. Do not discharge sediment laden water directly or indirectly to wetlands or waters. Pump water from the isolated work area to a temporary storage and treatment site or into upland areas and allow water to infiltrate.*
8. *Upon completion of in-water work activities, restore isolated work area to match original or planned contours. Slowly restore stream flow to prevent loss of surface water downstream.*
9. *Remove the stream diversion system within 5 days after completion of in-water work or when directed. Restore existing stream bank and areas used for stream diversion to match original or planned contours.*

ADAPTED FROM:
WEST VIRGINIA DEPARTMENT OF ENVIRONMENT PROTECTION
EROSION AND SEDIMENT CONTROL BEST MANAGEMENT PRACTICE MANUAL,
FIGURE 3.21.1, PAGE 3.21-12

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

BYPASS PUMPING DIVERSION FOR TEMPORARY IN-STREAM DIVERSIONS

EFLHD DETAIL
E157-09

SPECIFICATION
FP-14

APPROVED FOR USE
05/2024

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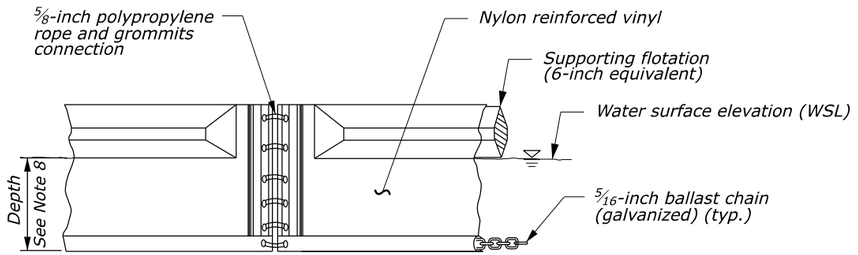
28 July 2026 10:32 AM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	S06



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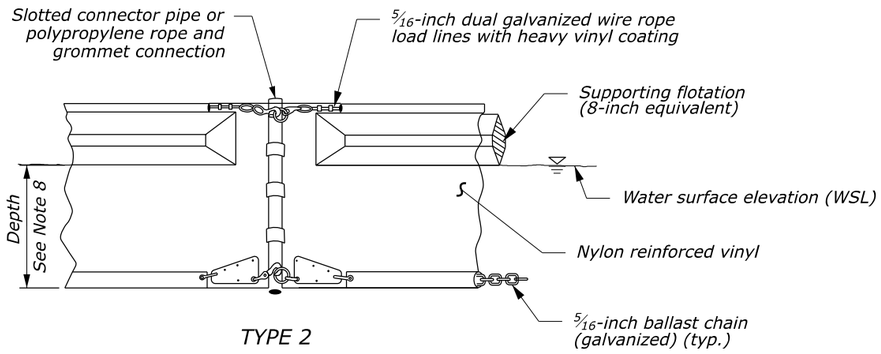

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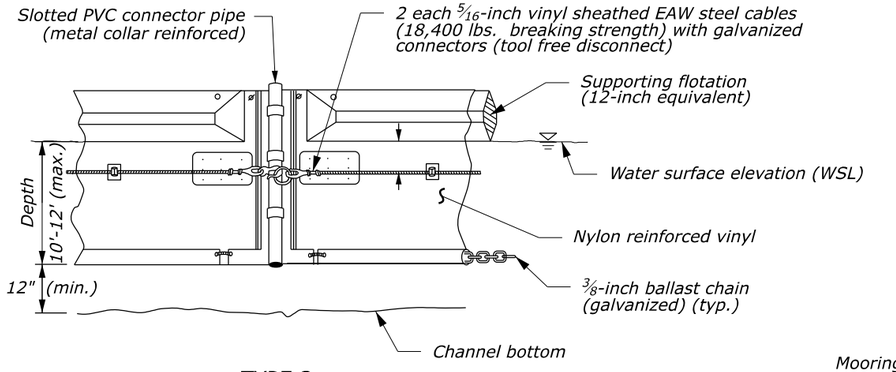
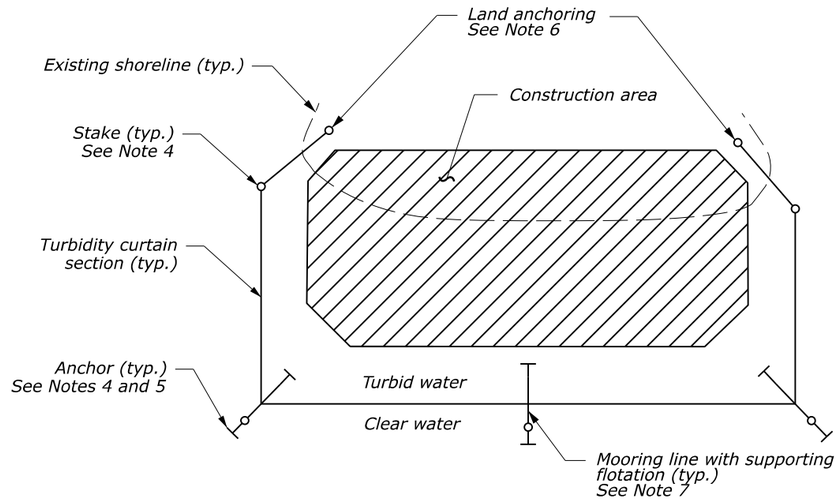


TYPE 3
(See Note 9)

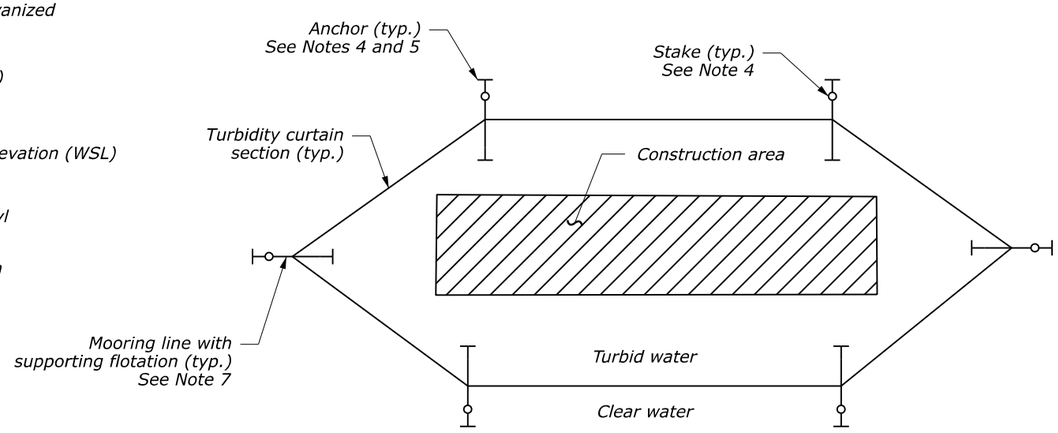
FLOATING TURBIDITY CURTAINS

USAGE CHART

TYPE	MAXIMUM CURRENT (FT/S)	ADDITIONAL CONDITIONS
1	0.0	No Wave/Wind/Tidal Action
2	3.5	Moving Water (One Direction)
3	5.0	Tidal Action or Subject to Wind and Wave Actions



PLAN VIEW - INSTALLATION FOR SHORELINE WORK



PLAN VIEW - INSTALLATION FOR PILE WORK

NOTES:

1. Select a floating turbidity curtain type to suit the site conditions and meet erosion control and water quality requirements. Submit details and specifications of the selected curtain type for approval prior to installation.
2. Install floating turbidity curtains before the start of any construction activities that could impact the area of concern.
3. Install floating turbidity curtains parallel to the direction of flow of a moving water body. Do not install curtains across the main flow of a significant moving water body. Do not install curtains where the flow of water will remove accumulated sediment or significantly move the curtain.
4. Install floating turbidity curtain stakes in a vertical position unless otherwise directed. Install anchors and stakes at no more than 100-foot spacing.
5. For Type 1 floating turbidity curtains, install anchors made of wooden stakes (2- by 4-inch rectangular or 2½-inch minimum diameter) or metal stakes (1⅓ pounds per linear foot). For Type 2 and 3 curtains, install anchors that are either weighted or that dig into the channel bottom.
6. Extend the ends of floating turbidity curtains well up into the shoreline as approved, especially if high water conditions are expected. Secure the ends firmly to the shoreline (preferably to rigid bodies such as trees or piles) to fully enclose the area where sediment may enter into the water body.
7. Provide minimum ½-inch nylon rope for adjustment line for mooring system. Ensure the mooring line is securely attached and is sufficiently buoyant to remain afloat under normal load conditions.
8. For Type 1 and 2 floating turbidity curtains, extend the nylon reinforced vinyl fabric through the entire depth of the water (to the channel bottom) unless tidal actions are present.
9. In areas subject to wind and wave actions or where tidal actions are present, use Type 3 floating turbidity curtains as directed. Install Type 3 curtains according to the manufacturer's recommendations and as approved.
10. Remove captured sediment behind floating turbidity curtains prior to removing the curtains. Do not disperse sediment into adjacent water body.

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

FLOATING TURBIDITY CURTAINS

EFLHD DETAIL	E157-11
SPECIFICATION	FP-14
APPROVED FOR USE	05/2024

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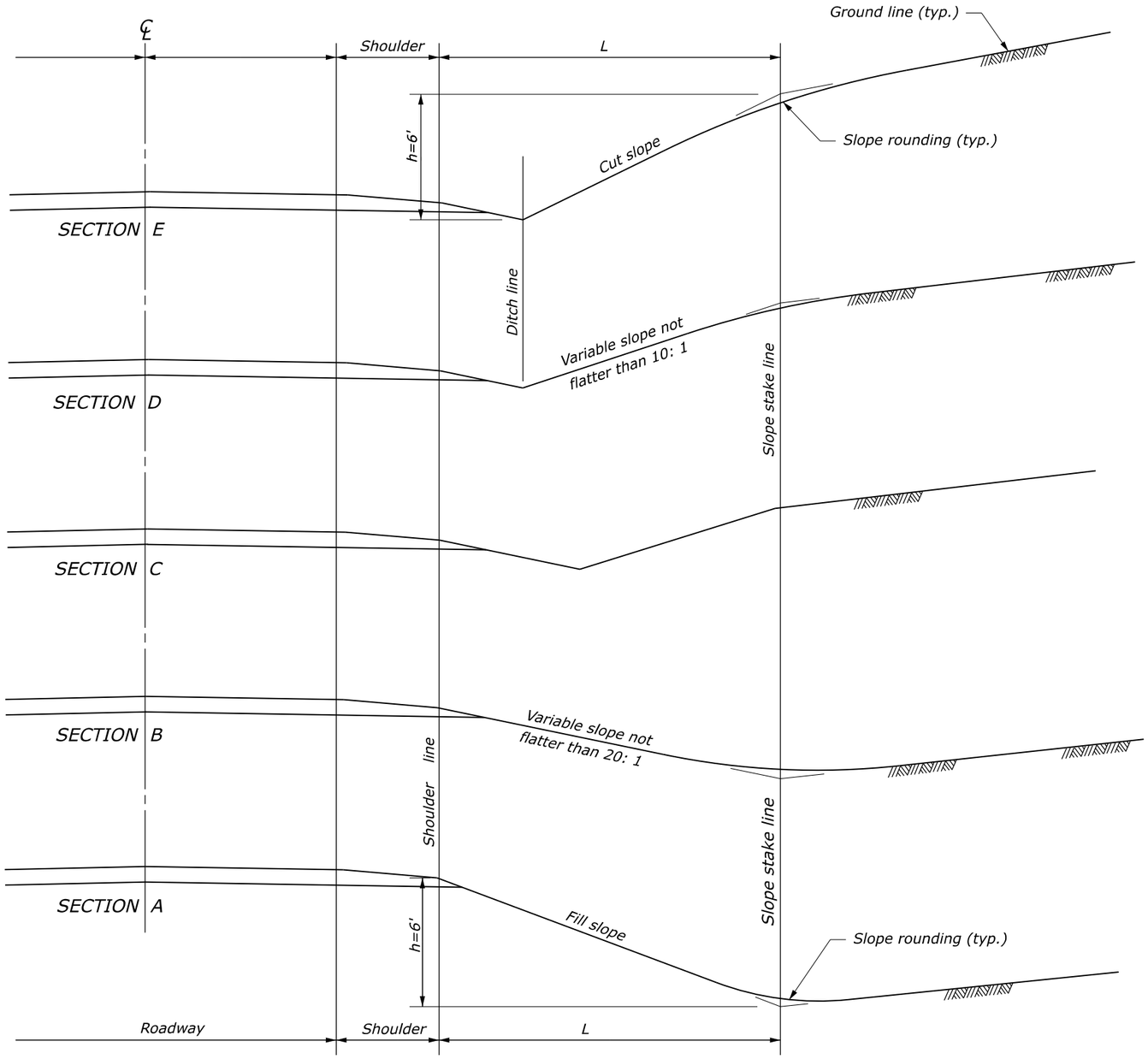
28 July 2026 10:33 AM

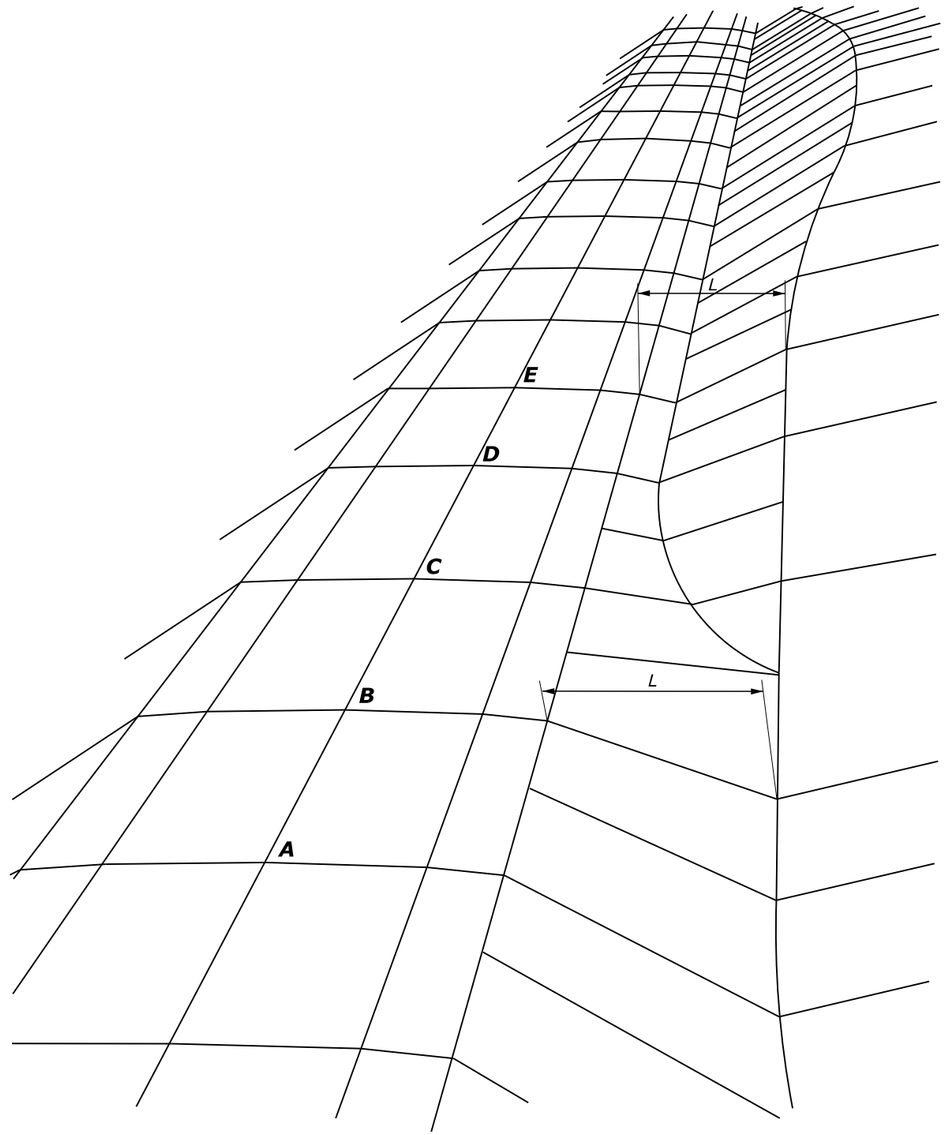
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	S07



NOTE:

Maintain distance L as the distance to slope stakes to establish the blend of cut and fill slopes into the original ground.



NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

CUT AND FILL SLOPE TRANSITIONS

EFLHD DETAIL
E204-01
SPECIFICATION
FP-24, FP-14
APPROVED FOR USE
05/2024

28 July 2026 10:33 AM \\fhfl1sfileserve.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) Combined NJ-NY\Std-Det\S07-E204-01.dgn [Sheet]

PROJECT	SHEET NUMBER
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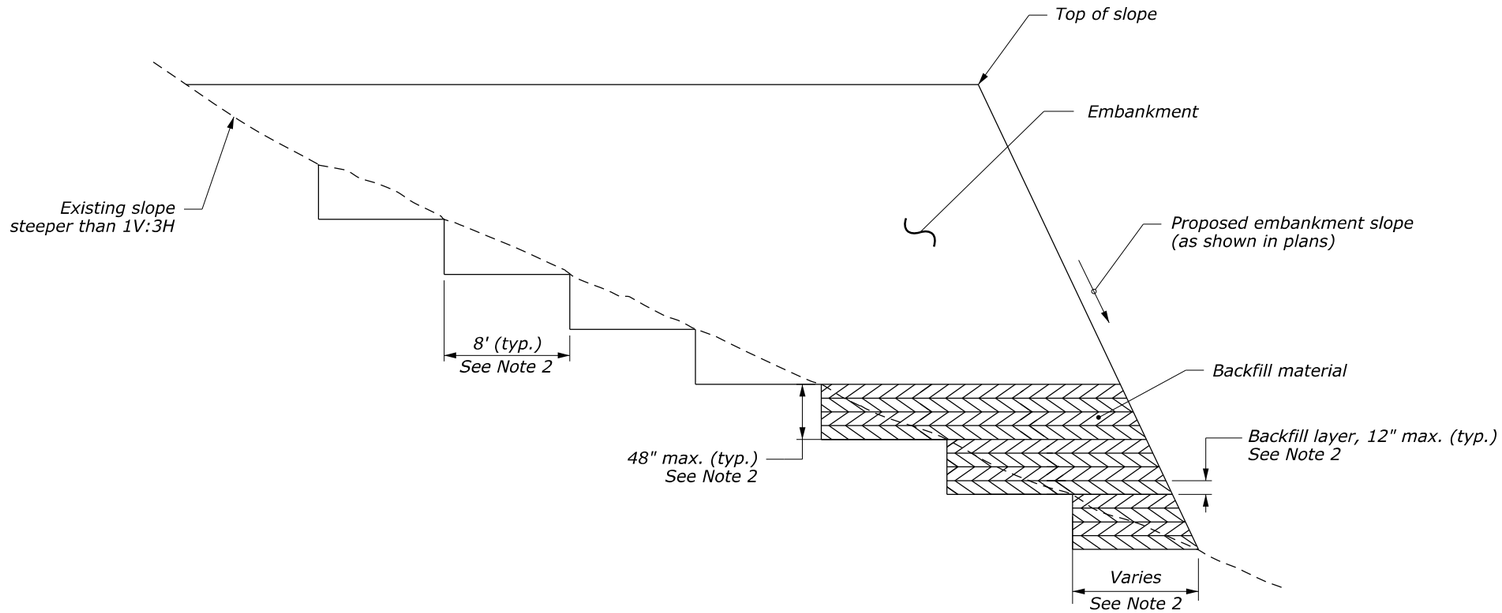
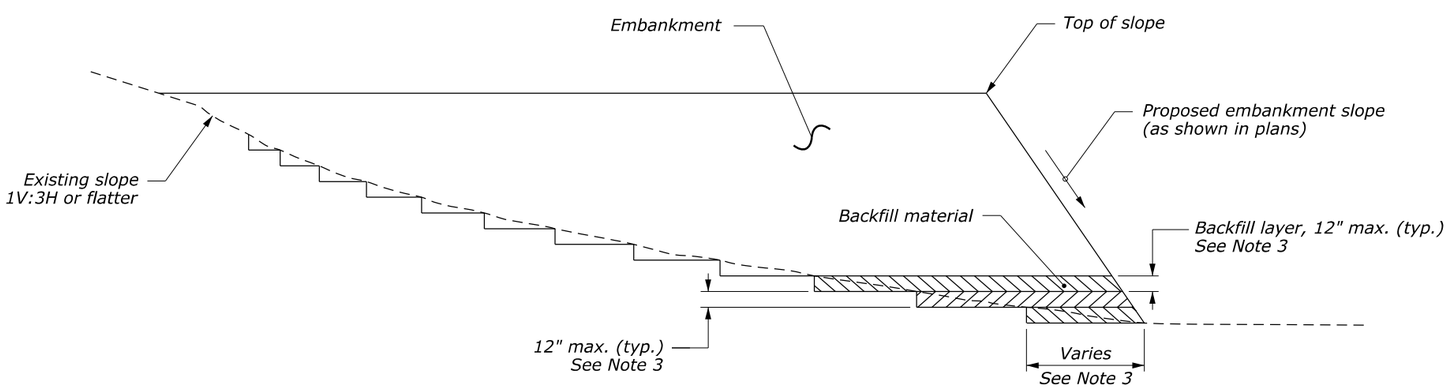


TYPICAL SECTION A

SLOPES STEEPER THAN 1V:3H



TYPICAL SECTION B

SLOPES 1V:3H OR FLATTER

NOTES:

1. *Where the embankment foundation is on side-hill or another existing embankment, bench the foundation while the embankment is being constructed.*
2. *For slopes steeper than 1V:3H, refer to Typical Section A. Cut 8-foot wide horizontal benches in the existing slope. Bench the slope as the embankment is placed and compacted in 1-foot thick layers until a backfill height of 4 feet is reached. Excavate and backfill in a similar manner until the top of slope is reached.*
3. *For slopes 1V:3H or flatter, refer to Typical Section B. Cut into the existing slope until a maximum height of 1 foot is reached, allowing the horizontal distance to vary. Place embankment material. Excavate and backfill in a similar manner until the top of slope is reached.*

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA OFFICE OF FEDERAL LANDS HIGHWAY	EFLHD DETAIL E204-02
BENCHING FOR EMBANKMENT	SPECIFICATION FP-24, FP-14
	APPROVED FOR USE 05/2024

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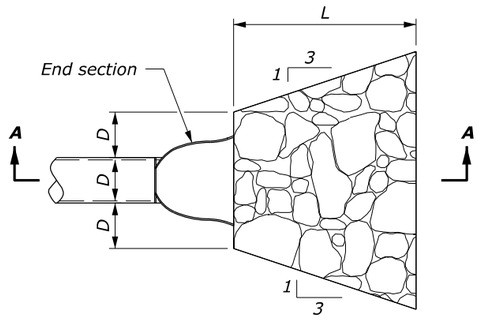
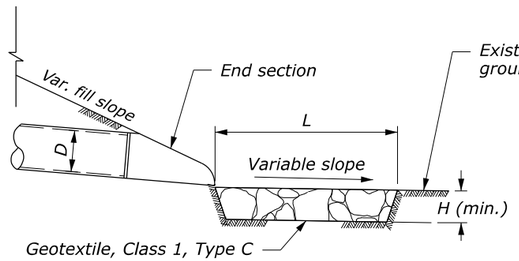
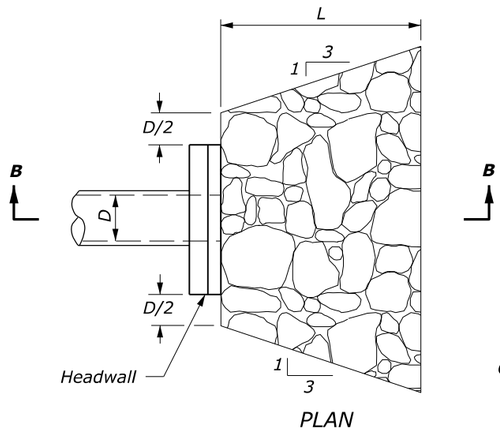


PLAN

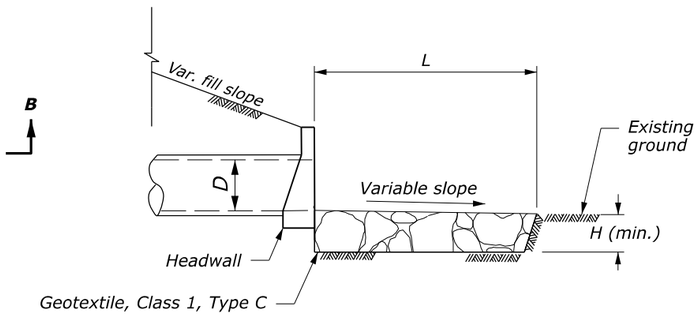


SECTION A-A

CULVERT WITH END SECTION



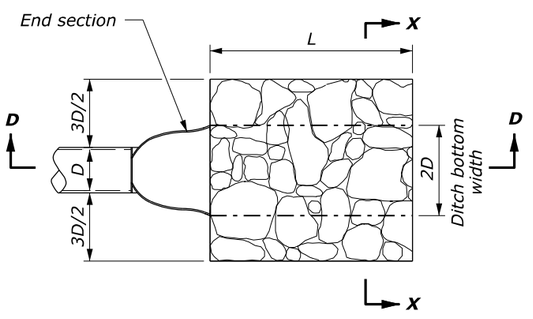
PLAN



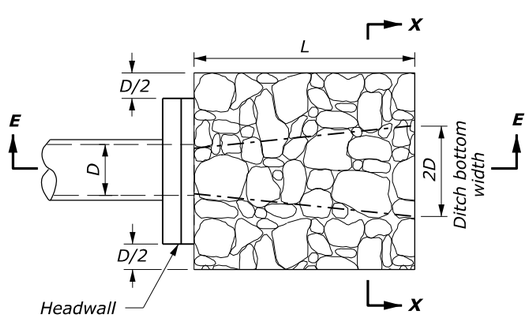
SECTION B-B

CULVERT WITH HEADWALL

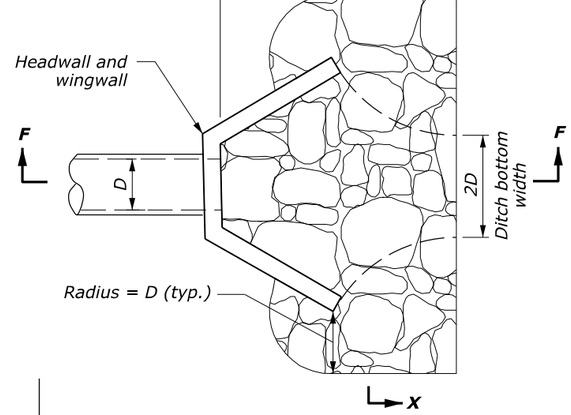
PROTECTIVE APRON AT CULVERT OUTLET WITHOUT DITCH



PLAN



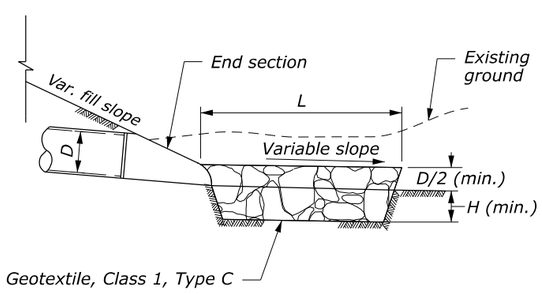
PLAN



PLAN

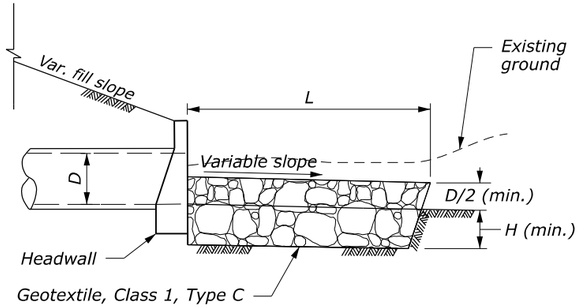
NOTES:

1. *Use for aprons serving culverts with slopes of less than 10%.*
2. *For arch or elliptical pipes, use equivalent diameter for (D) dimension.*
3. *See Sheet 2 of 2 for dimensions and estimated quantities.*



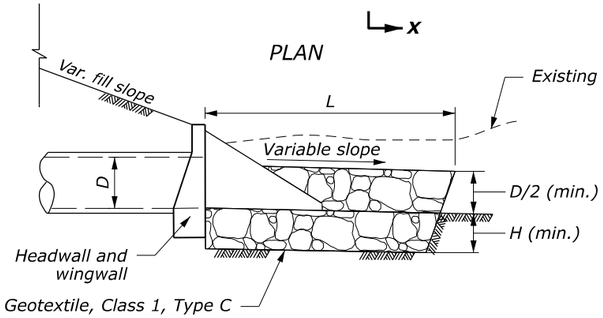
SECTION D-D

CULVERT WITH END SECTION



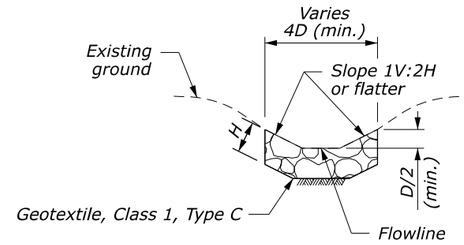
SECTION E-E

CULVERT WITH HEADWALL



SECTION F-F

CULVERT WITH HEADWALL AND WINGWALL



SECTION X-X

PROTECTIVE APRON AT CULVERT OUTLET WITH DITCH

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

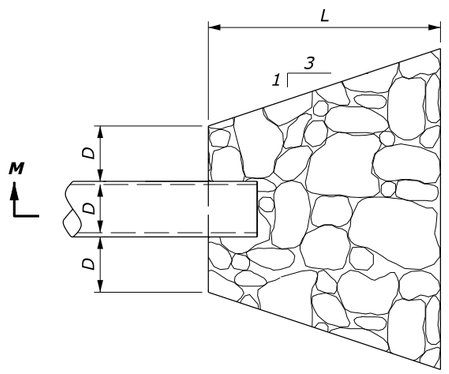
PLACED RIPRAP AT CULVERT OUTLETS
Sheet 1 of 2

EFLHD DETAIL
E251-01

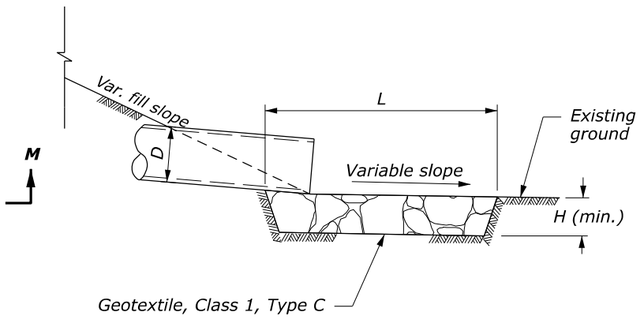
SPECIFICATION
FP-14

APPROVED FOR USE
05/2024

\\fhf15fileserve.flhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) Combined NJ-NY\Std-Det\S09-S10-E251-01.dgn [Sheet 1 of 2]
28 July 2026 10:33 AM

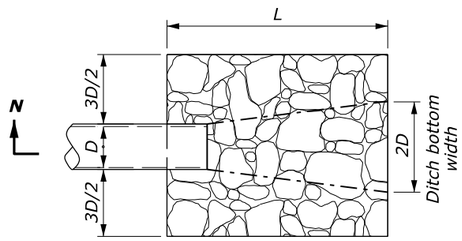


PLAN

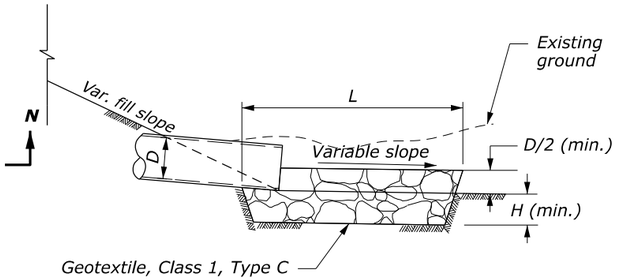


SECTION M-M

CULVERT OUTLET WITHOUT DITCH



PLAN



SECTION N-N

CULVERT OUTLET WITH DITCH

PROTECTIVE APRON AT CULVERT OUTLET WITH NO END TREATMENT

PROTECTIVE APRON DIMENSIONS AND ESTIMATED QUANTITIES

					OUTLET WITHOUT DITCH		OUTLET WITH DITCH	
	CULVERT SIZE (D) INCH	RIPRAP CLASS	LENGTH OF APRON (L) FT	DEPTH OF APRON (H) FT	ESTIMATED RIPRAP QUANTITY CUYD	ESTIMATED GEOTEXTILE QUANTITY SQYD	ESTIMATED RIPRAP QUANTITY CUYD	ESTIMATED GEOTEXTILE QUANTITY SQYD
WITH END SECTION	12	2	4	1.5	1	5	0.9	4
	18	2	6	1.5	2.2	9	2	8
	24	2	8	1.5	3.9	13	3.6	12
	30	3	12.5	2	10.8	27	9.3	24
	36	3	15	2	15.6	37	13.3	32
	42	4	21	2.5	34	63	27.2	52
	48	4	24	2.5	44.4	78	35.6	65
WITH HEADWALL	12	2	4	1.5	1.6	6	1.1	5
	18	2	6	1.5	3	10	2	8
	24	2	8	1.5	5.3	16	3.6	12
	30	3	12.5	2	15	32	9.3	24
	36	3	15	2	21.7	43	13.3	32
	42	4	21	2.5	47.6	73	27.2	52
	48	4	24	2.5	62.2	91	35.6	65
WITH HEADWALL AND WINGWALL	12	2	4	1.5	1.6	6	1.1	6.3
	18	2	6	1.5	3	10	2.5	11.1
	24	2	8	1.5	5.3	16	4.5	17
	30	3	12.5	2	15	32	11.6	32
	36	3	15	2	21.7	43	16.7	42.6
	42	4	21	2.5	47.6	73	34.1	68.4
	48	4	24	2.5	62.2	91	44.5	84.9
WITH NO END TREATMENT	12	2	6	1.5	1.7	7	1.3	6
	18	2	8	1.5	3.2	12	2.7	10
	24	2	10	1.5	5.2	17	4.4	15
	30	3	14.5	2	13.2	32	10.7	27
	36	3	17	2	18.5	42	15.1	36
	42	4	23	2.5	38.7	70	29.8	56
	48	4	26	2.5	49.8	86	38.5	70

NOTES:

1. *Use for aprons serving culverts with slopes of less than 10%.*
2. *For arch or elliptical pipes, use equivalent diameter for (D) dimension.*

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

PLACED RIPRAP AT CULVERT OUTLETS
Sheet 2 of 2

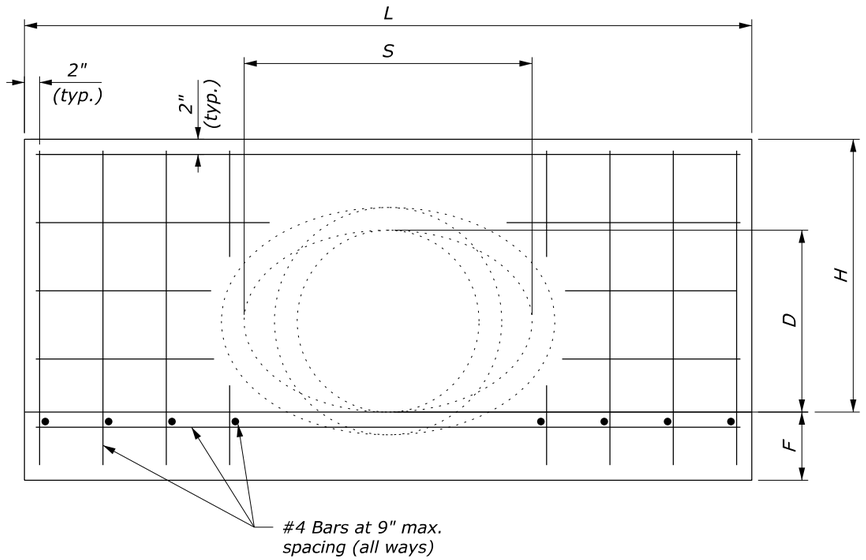
EFLHD DETAIL
E251-01

SPECIFICATION
FP-14

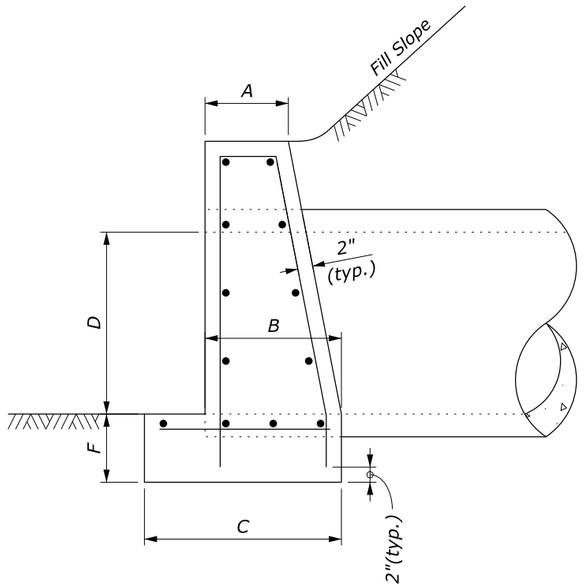
APPROVED FOR USE
05/2024

28 July 2026 10:33 AM \\fhf115fileserve.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) Combined NJ-NY\Std-Det\S09-S10-E251-01.dgn [Sheet 2 of 2]

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	S11



FRONT ELEVATION



SIDE ELEVATION

NOTE:

1. *Prepare foundation according to Section 209. Place headwalls on 6 inches of foundation fill.*
2. *Orient all headwalls parallel to the roadway centerline unless otherwise shown in the plans or as directed.*
3. *When pipes are on a skew, adapt and lengthen headwalls as directed.*
4. *Chamfer all exposed corners not rounded to ¾ inch.*
5. *Quantities shown are for one headwall with pipe at right angles.*

HEADWALL FOR ELLIPTICAL PIPE

SIZE OF ELLIPTICAL PIPE CULVERT (SPAN x RISE)										
	23" x 14"	30" x 19"	34" x 22"	38" x 24"	42" x 27"	45" x 29"	49" x 32"	53" x 34"	60" x 38"	68" x 43"
A	0'-8"	0'-9"	0'-10"	0'-10"	0'-11"	1'-0"	1'-0"	1'-0"	1'-0"	1'-0"
B	1'-2"	1'-5"	1'-6"	1'-8"	1'-9"	1'-10"	1'-11"	1'-11"	1'-11"	2'-0"
C	1'-8"	1'-11"	2'-1"	2'-4"	2'-5"	2'-7"	2'-8"	2'-9"	3'-3"	3'-6"
D	1'-2"	1'-7"	1'-10"	2'-0"	2'-3"	2'-5"	2'-8"	2'-10"	3'-2"	3'-7"
F	0'-8"	0'-8"	0'-9"	0'-9"	0'-9"	0'-9"	0'-9"	0'-9"	0'-9"	0'-9"
H	2'-10"	3'-3"	3'-7"	3'-9"	4'-0"	4'-2"	4'-5"	4'-7"	4'-11"	5'-4"
L	5'-5"	7'-2"	8'-6"	9'-2"	10'-2"	10'-11"	12'-1"	12'-11"	13'-0"	13'-0"
S	1'-11"	2'-6"	2'-10"	3'-2"	3'-6"	3'-9"	4'-1"	4'-5"	5'-0"	5'-8"
CUBIC YARDS OF CONCRETE										
Conc. Pipe	0.502	0.855	1.236	1.500	1.811	2.101	2.512	2.801	2.969	2.904

HEADWALL FOR CIRCULAR PIPE

DIAMETER OF PIPE CULVERT (D)						
	6"	15"	18"	21" or 24"	27" or 30"	33" or 36"
A	0'-6"	0'-8"	0'-9"	0'-11"	1'-0"	1'-0"
B	0'-9"	1'-1"	1'-3"	1'-6"	1'-9"	2'-0"
C	1'-2"	1'-7"	1'-9"	2'-2"	2'-6"	2'-9"
F	0'-6"	0'-8"	0'-8"	0'-9"	0'-9"	0'-9"
H	2'-0"	2'-11"	3'-2"	3'-9"	4'-3"	4'-9"
L	3'-8"	5'-0"	6'-0"	8'-0"	10'-0"	12'-0"
CUBIC YARDS OF CONCRETE						
Conc. Pipe	0.241	0.492	0.697	1.319	2.067	2.947
C.M. Pipe	0.257	0.521	0.739	1.398	2.198	3.145

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

CONCRETE HEADWALL FOR SMALL PIPE CULVERT

FLH STANDARD 601-4

SPECIFICATION FP-24, FP-14

APPROVED FOR USE 2/2024

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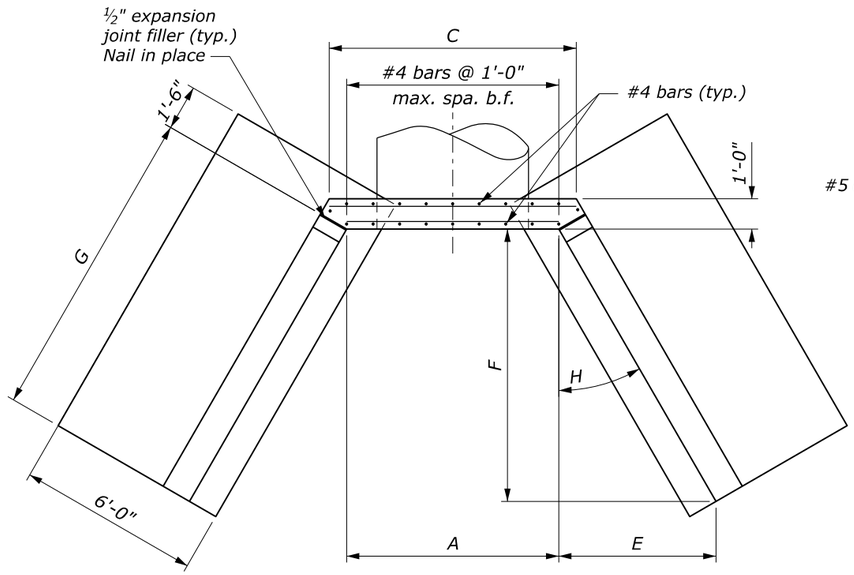
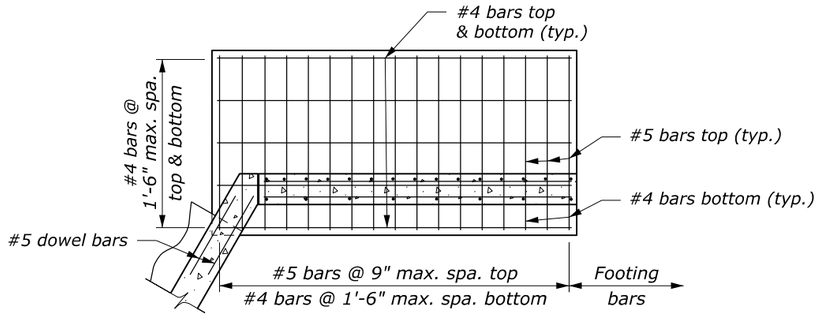
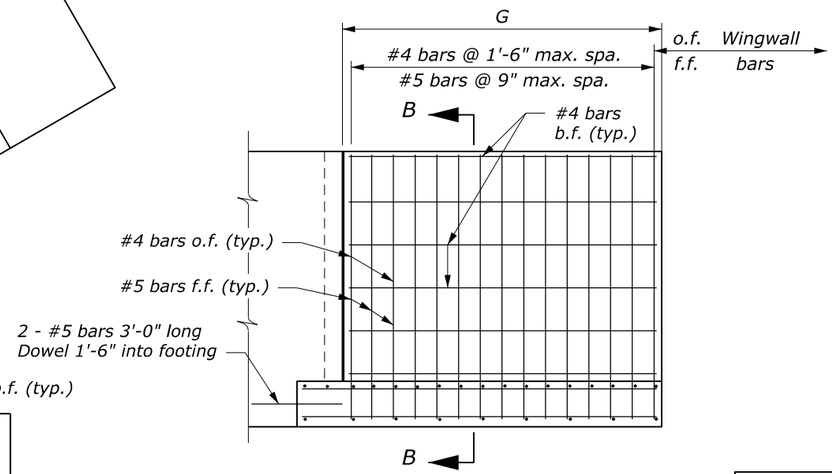


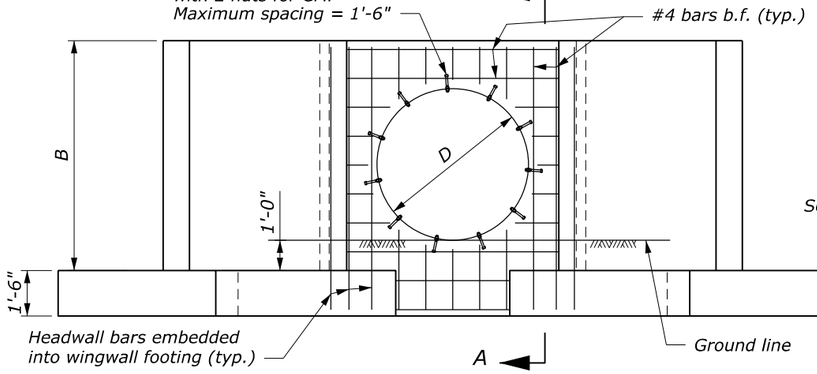
PLAN



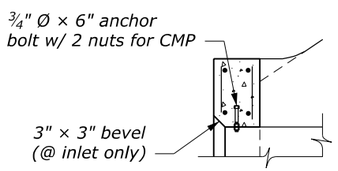
WINGWALL PLAN



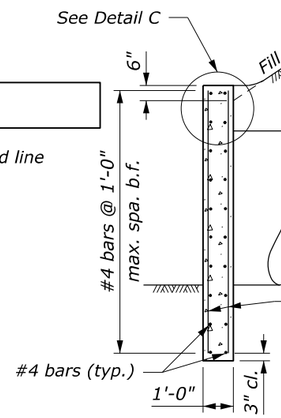
WINGWALL ELEVATION



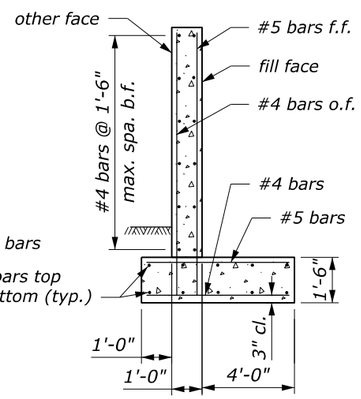
FRONT ELEVATION
(Showing headwall reinforcement)



DETAIL C
(For Metal Pipes)



SECTION A-A



SECTION B-B

NOTES:

1. *This drawing applies for normal crossings and skews up to 15°.*
2. *Prepare foundation according to Section 209. Place headwall/wingwalls on 6 inches of foundation fill.*
3. *On shallow fill where headwall is 2 feet or less below shoulder line, construct the headwall parallel to line and grade of the shoulder.*
4. *Do not allow top of wingwall to project above fill slope, ditch slope or shoulder.*
5. *Chamfer all exposed edges ¾ inch and finish all exposed surfaces with a Class 1 ordinary surface finish. Provide joint filler conforming to AASHTO M 213.*
6. *Bell end of concrete pipe may replace bevel at inlet headwall.*
7. *Quantities shown in table are for one headwall and two wingwalls and are based on CMP. Concrete and steel quantities shown will be used as basis for final payment for headwall/wingwalls constructed according to this drawing.*
8. *Reinforcing steel clearance is 2 inches unless shown otherwise.*
9. *Provide anchor bolts conforming to ASTM A307. Galvanize bolts and nuts according to ASTM A153.*
10. *Orientations shown in table are based on north orientation shown in Plan sheets.*

DIMENSIONS AND QUANTITIES (SITE 1)

Diameter of Pipe Culvert (D) = 54"

		West Headwall		East Headwall	
		NW Wingwall	SW Wingwall	NE Wingwall	SE Wingwall
1V:2H fill slope	A	6'-6"		6'-6"	
	B	6'-11½"	6'-11½"	6'-11½"	6'-11½"
	C	6'-6"		7'-1" (A + 7" towards SE wingwall)	
	E	12'-7¾"	12'-7¾"	12'-7¾"	6'-3¾"
	F	N/A	N/A	N/A	10'-11½"
	G	12'-7¾"	12'-7¾"	12'-7¾"	12'-7¾"
	H	90°	90°	90°	30°
Conc. (CUYD)		17.4		17.5	
Steel (LB)		1502		1502	

Abbreviations:
f.f. = Fill face
o.f. = Other face
b.f. = Both faces

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

CONCRETE HEADWALL/WINGWALL FOR SINGLE NORMAL 54" PIPE CULVERT

EFLHD DETAIL
E601-5A

SPECIFICATION
FP-14

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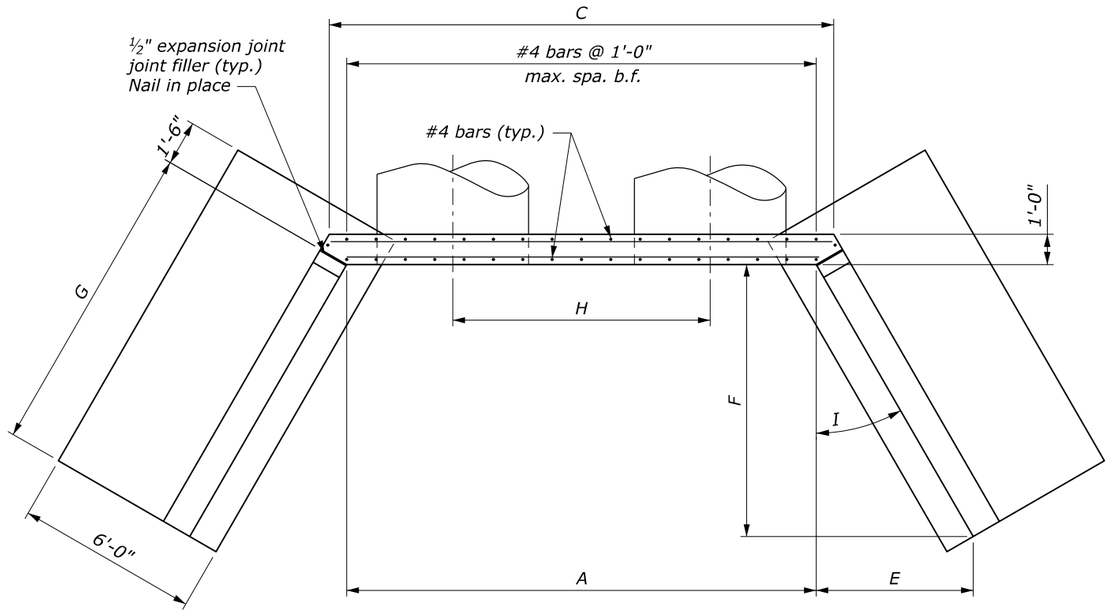
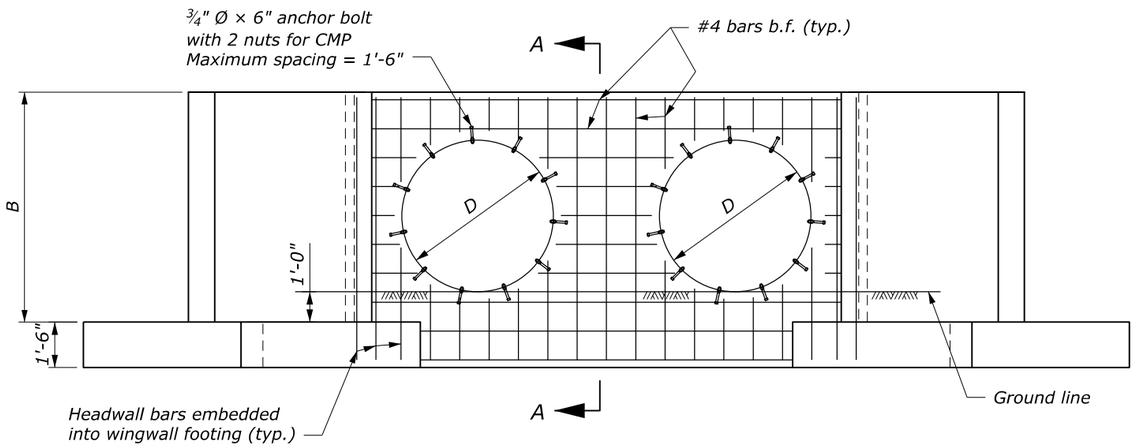
3 August 2026 6:11 PM



PLAN



FRONT ELEVATION
(Showing headwall reinforcement)

NOTES:

1. *This drawing applies for normal crossings and skews up to 15°.*
2. *Prepare foundation according to Section 209. Place headwall/wingwalls on 6 inches of foundation fill.*
3. *Quantities shown in table are for one headwall and two wingwalls and are based on CMP. Concrete and steel quantities shown will be used as basis for final payment for headwall/wingwalls constructed according to this drawing.*
4. *For views, notes, and reinforcing details not shown, see Detail E601-5A.*
5. *Orientations shown in table are based on north orientation shown in Plan sheets.*

DIMENSIONS AND QUANTITIES (SITE 6)
Diameter of Pipe Culvert (D) = 48"

		North Headwall		South Headwall	
		NW Wingwall	NE Wingwall	SW Wingwall	SE Wingwall
1V:2H fill slope	A	12'-10¼"		12'-10¼"	
	B	6'-5½"	6'-5½"	6'-5½"	6'-5½"
	C	13'-5¼" (A + 7" towards NW wingwall)		12'-10¼"	
	E	7'-5¾"	8'-7½"	8'-7½"	8'-7½"
	F	4'-3¾"	N/A	N/A	N/A
	G	8'-7½"	8'-7½"	8'-7½"	8'-7½"
	H	6'-10"		6'-10"	
	I	60°	90°	90°	90°
Conc. (CUYD)		14.0		14.0	
Steel (LB)		1535		1535	

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

CONCRETE HEADWALL/WINGWALL FOR MULTIPLE NORMAL 48" PIPE CULVERT

EFLHD DETAIL
E601-7A

SPECIFICATION
FP-14

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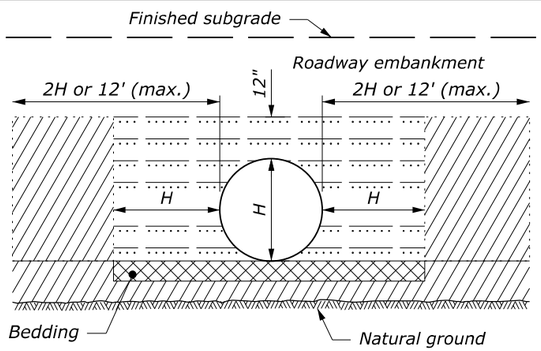
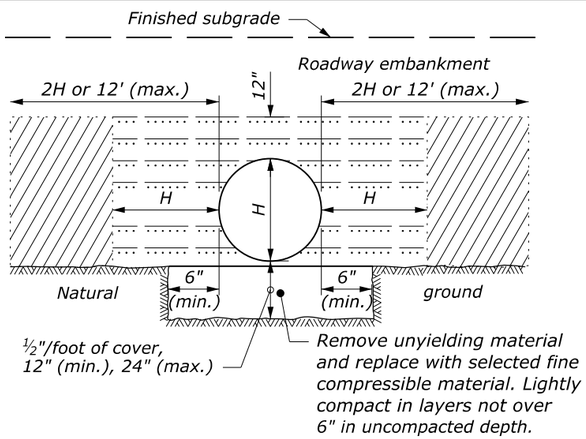
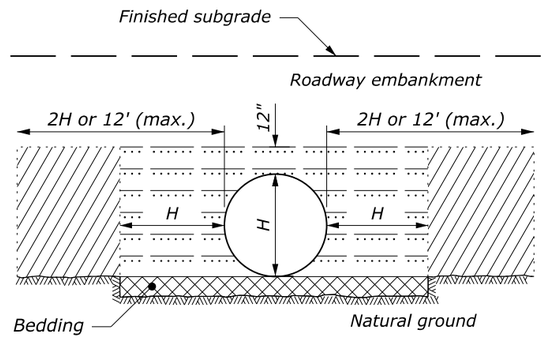


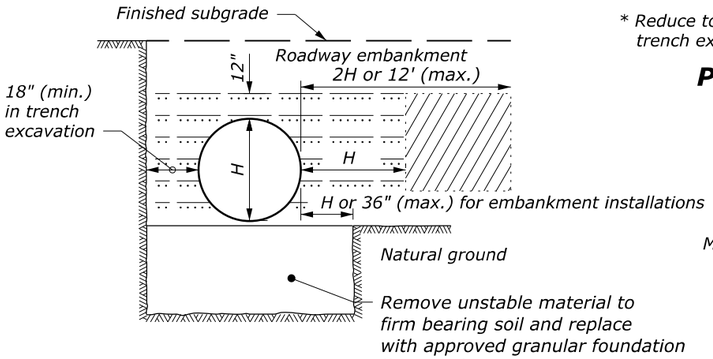
ABOVE NATURAL GROUND



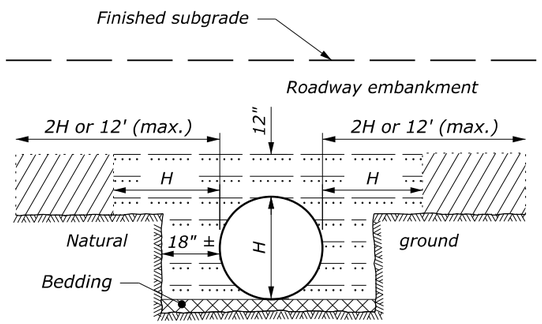
ON UNYIELDING MATERIAL



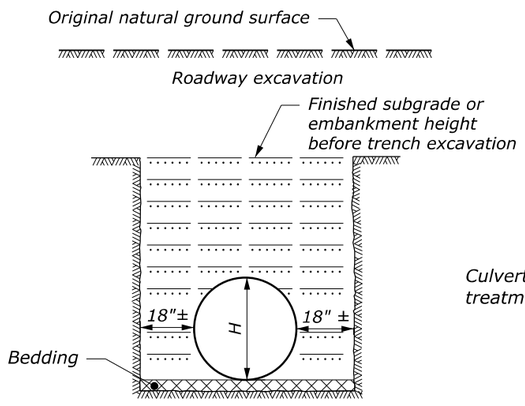
ON NATURAL GROUND



ON UNSTABLE MATERIAL



ABOVE AND BELOW NATURAL GROUND

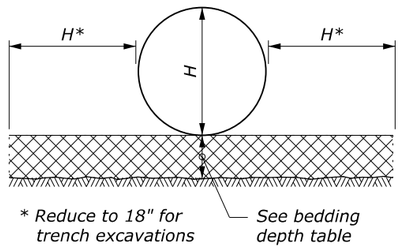


BELOW NATURAL GROUND OR TRENCH EXCAVATION IN EMBANKMENT

LEGEND:

- Bedding material (uncompacted)
- Embankment material placed in layers not exceeding 6" compacted depth.
- Compacted backfill material placed in layers not exceeding 6" compacted depth; or lean concrete backfill according to Section 614.
- Impermeable backfill material.

BEDDING DEPTH	
PIPE SIZE (H)	DEPTH
12" to 54"	4"
> 54"	6"

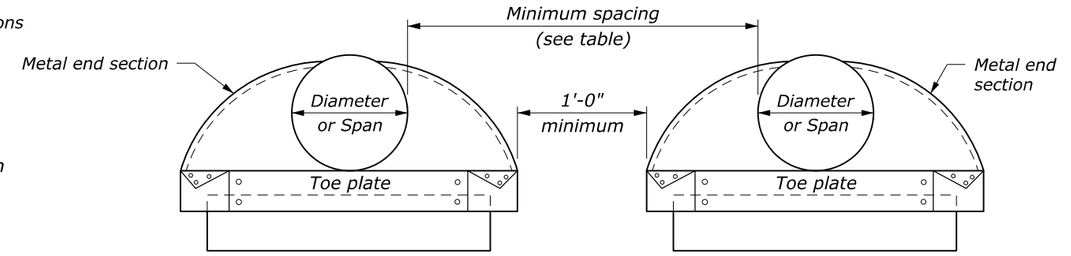


PIPE BEDDING

NOTE:

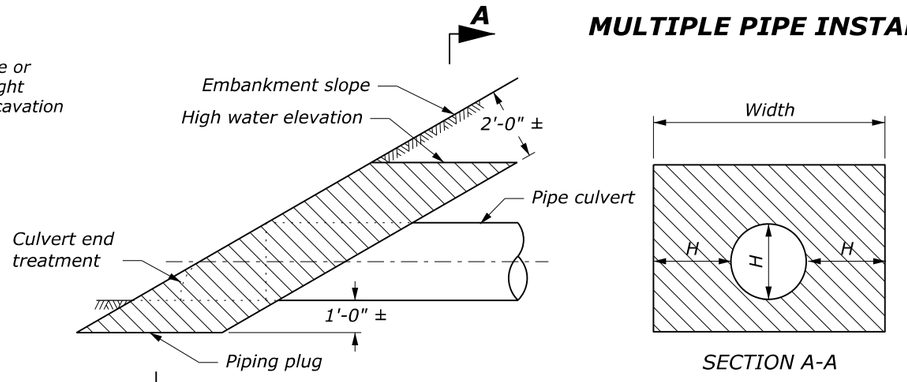
1. When directed, camber pipe culverts upward from a chord through the inlet and outlet inverts an ordinate amount equal to 1% of the pipe length. Develop camber on a parabolic curve. If the midpoint elevation on the parabolic curve as designed exceeds the elevation of the inlet invert, reduce the amount of camber or increase the pipe culvert gradient.
2. H equals the diameter of all round pipe culverts or the rise dimension of all pipe arch culverts.
3. See Section 704 for bedding and backfill requirements.

MINIMUM SPACING	
DIAMETER or SPAN	SPACING
UP to 48"	24"
48" and UP	Half diameter or span or 36", whichever is less



ELEVATION

MULTIPLE PIPE INSTALLATION



SECTION A-A

Construct piping plug of impermeable backfill material at the pipe culvert inlet where granular material is used for backfill. Width may be adjusted to tie into impervious material.

PIPING PLUG

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

METAL AND PLASTIC PIPE CULVERT BEDDING

FLH STANDARD 602-3
SPECIFICATION FP-14
APPROVED FOR USE 1/2024

\\fhl15filesrve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) Combined NJ-NY\Std-Det\S14-Std602-3.dgn [Std 602-3]
28 July 2026 10:35 AM

POLYETHYLENE (PE) PLASTIC ROUND PIPE CULVERT & STORM DRAIN

FILL HEIGHT TABLE AND MINIMUM CELL CLASSIFICATION NUMBER PER ASTM D 3350

SMOOTH WALL (SOLID WALL)

PIPE SIZE DIAMETER INCHES	MINIMUM COVER INCHES	CELL CLASSIFICATION NUMBER 335434C — MINIMUM WALL THICKNESS (INCHES) 0.607	0.857	0.923	1.154	1.385	1.292	1.477
		MAXIMUM FILL HEIGHT (FEET)						
12	12	57						
18	12		52					
24	12			38				
30	12				38			
36	12					38		
42	12						27	
48	12							27

CORRUGATED

PIPE SIZE DIAMETER INCHES	MINIMUM COVER INCHES	AASHTO M294 MAXIMUM FILL HEIGHT (FEET)
12	12	15
15	12	13
18	12	13
24	12	12
30	12	12
36	12	10

RIBBED

PIPE SIZE DIAMETER INCHES	MINIMUM COVER INCHES	CELL CLASS. NO. 334433C MAXIMUM FILL HEIGHT (FEET)	CELL CLASS. NO. 335434C MAXIMUM FILL HEIGHT (FEET)
18	12	18	24
24	12	22	28
30	12	22	28
36	12	25	31
42	12	21	27
48	12	21	26

NOTE:

1. When directed, camber pipe culverts upward from a chord through the inlet and outlet inverts an ordinate amount equal to 1% of the pipe length. Develop camber on a parabolic curve. If the midpoint elevation on the parabolic curve as designed exceeds the elevation of the inlet invert, reduce the amount of camber or increase the pipe culvert gradient.

2. For flexible pavement and aggregate surface roadways, measure minimum cover from the top of the pipe culvert to the bottom of the roadway subgrade. For rigid pavements, measure minimum cover from the top of the pipe culvert to the top of the pavement. For all roadway surface types, measure maximum fill height from the top of the pipe culvert to the top of the pavement.

POLYVINYL CHLORIDE (PVC) PLASTIC ROUND PIPE CULVERT

FILL HEIGHT TABLE AND MINIMUM CELL CLASSIFICATION NUMBER PER ASTM D 1784

SMOOTH WALL (SOLID WALL)

PIPE SIZE DIAMETER INCHES	MINIMUM COVER INCHES	CELL CLASS. NO. 12454 — MINIMUM WALL THICKNESS (INCHES) 0.358	0.438	CELL CLASS. NO. 12364 — MINIMUM WALL THICKNESS (INCHES) 0.358	0.438
		MAXIMUM FILL HEIGHT (FEET)			
12	12	65		69	
15	12		62		66

RIBBED

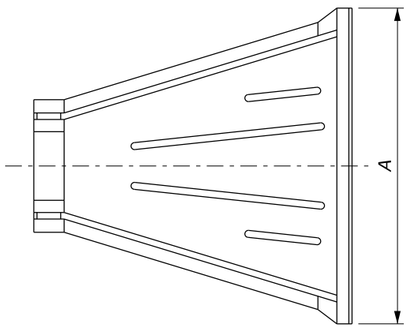
PIPE SIZE DIAMETER INCHES	MINIMUM COVER INCHES	CELL CLASS. NO. 12454C MAXIMUM FILL HEIGHT (FEET)	CELL CLASS. NO. 12364C MAXIMUM FILL HEIGHT (FEET)
12	12	37	26
15	12	32	22
18	12	33	23
24	12	29	21
30	12	28	20
36	12	27	19
42	12	26	18
48	12	24	17

POLYPROPYLENE (PP) PLASTIC ROUND PIPE CULVERT AND STORM DRAIN

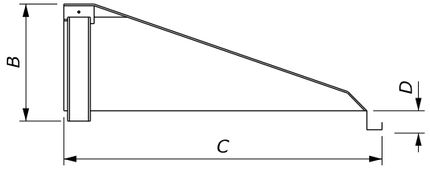
FILL HEIGHT TABLE AND MINIMUM CELL CLASSIFICATION NUMBER PER ASTM D 3350

CORRUGATED

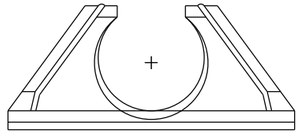
PIPE SIZE DIAMETER INCHES	MINIMUM COVER INCHES	AASHTO M330 MAXIMUM FILL HEIGHT (FEET)
12	12	17
15	12	17
18	12	17
24	12	16
30	12	14
36	12	12



TOP



SIDE



FRONT

END SECTION DIMENSIONS

PIPE SIZE DIAMETER INCHES	DIMENSIONS INCHES A	B	C	D
12	42	14.5	33	6
15	46	24.5	45.5	6
18	54	29	55	6
24	64	37	65	6
30	88	36	63.5	6
36	88	43	66.5	6

PLASTIC PIPE END SECTION

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

PLASTIC PIPE CULVERT

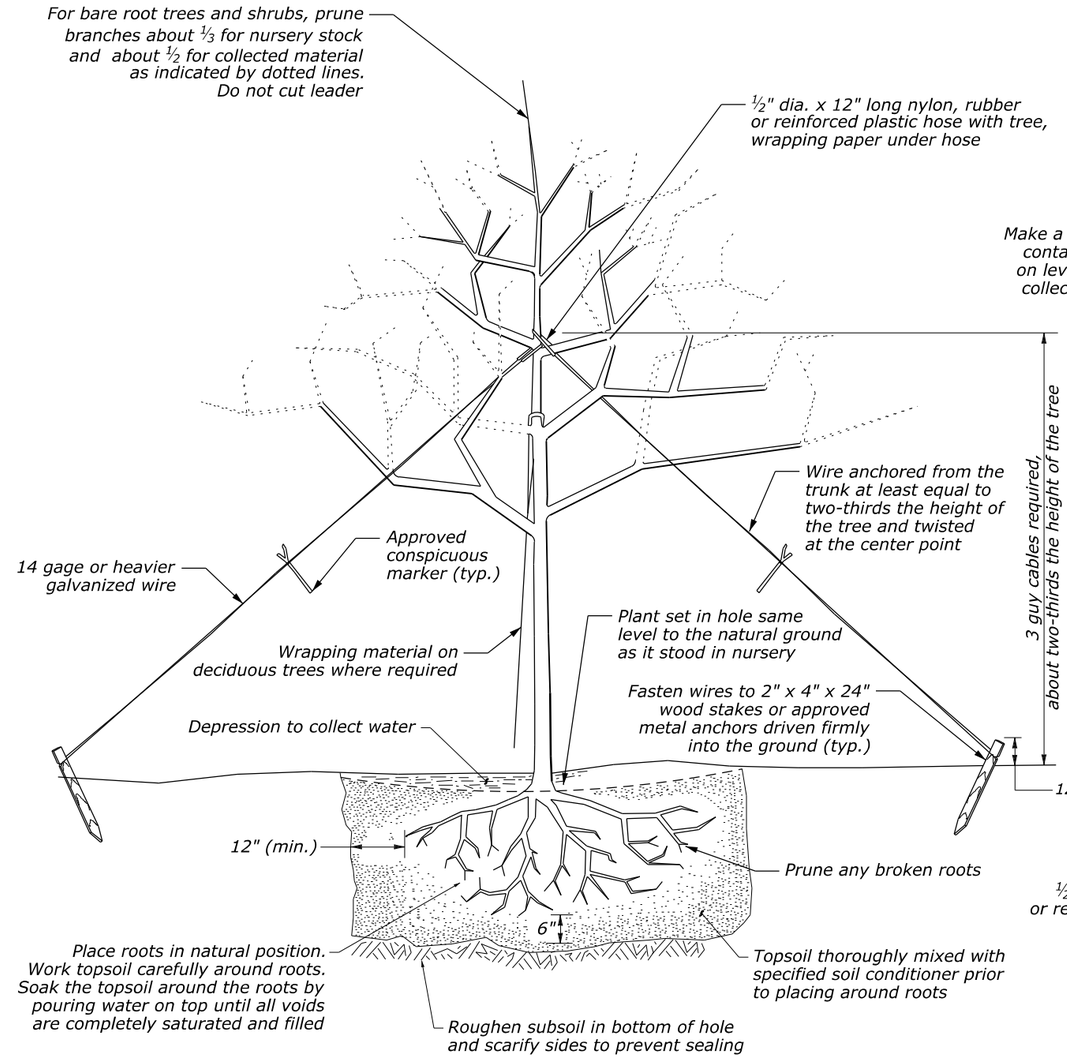
FLH STANDARD 602-5

SPECIFICATION FP-24, FP-14

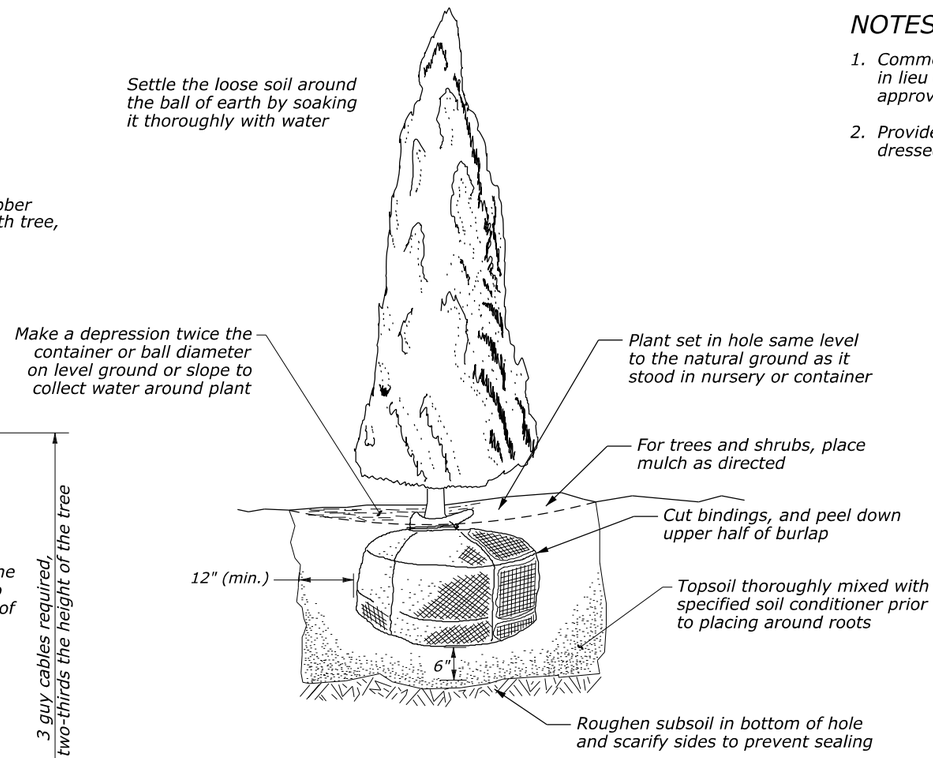
APPROVED FOR USE 5/2025

28 July 2026 10:35 AM \\fhf15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) Combined NJ-NY\Std-Det\S15-Std602-5.dgn [Std 602-5]

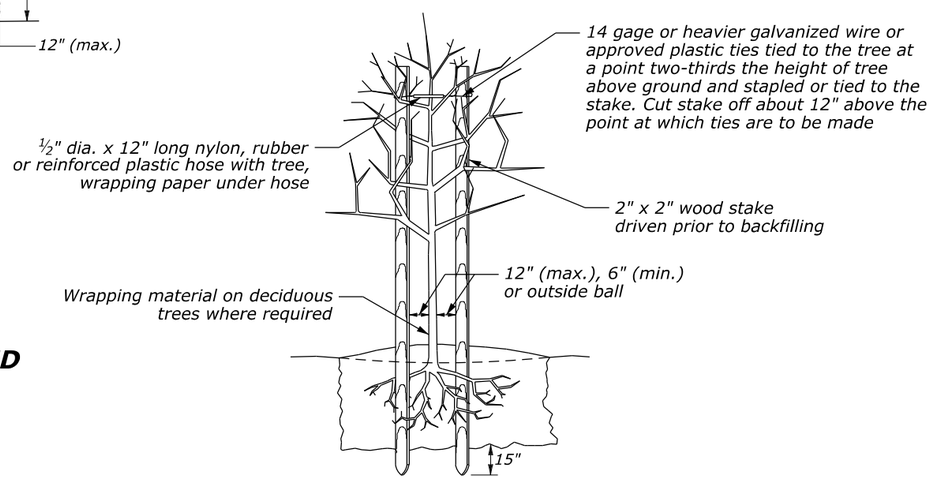
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	S16



METHOD OF PLANTING BARE ROOT TREES AND SHRUBS AND METHOD OF GUYING DECIDUOUS TREES OVER 12' AND CONIFERS OVER 4'



METHOD OF PLANTING CONTAINER OR BALLED AND BURLAPPED TREES AND SHRUBS



METHOD OF STAKING DECIDUOUS TREES UNDER 12'

NO SCALE

NOTES:

1. *Commercial plant ties may be used in lieu of hose and wire guying as approved.*
2. *Provide construction grade, rough or dressed Douglas Fir or Pine stakes.*

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

TREES AND SHRUBS PLANTING METHODS

EFLHD DETAIL	E626-01
SPECIFICATION	FP-24, FP-14
APPROVED FOR USE	06/2024

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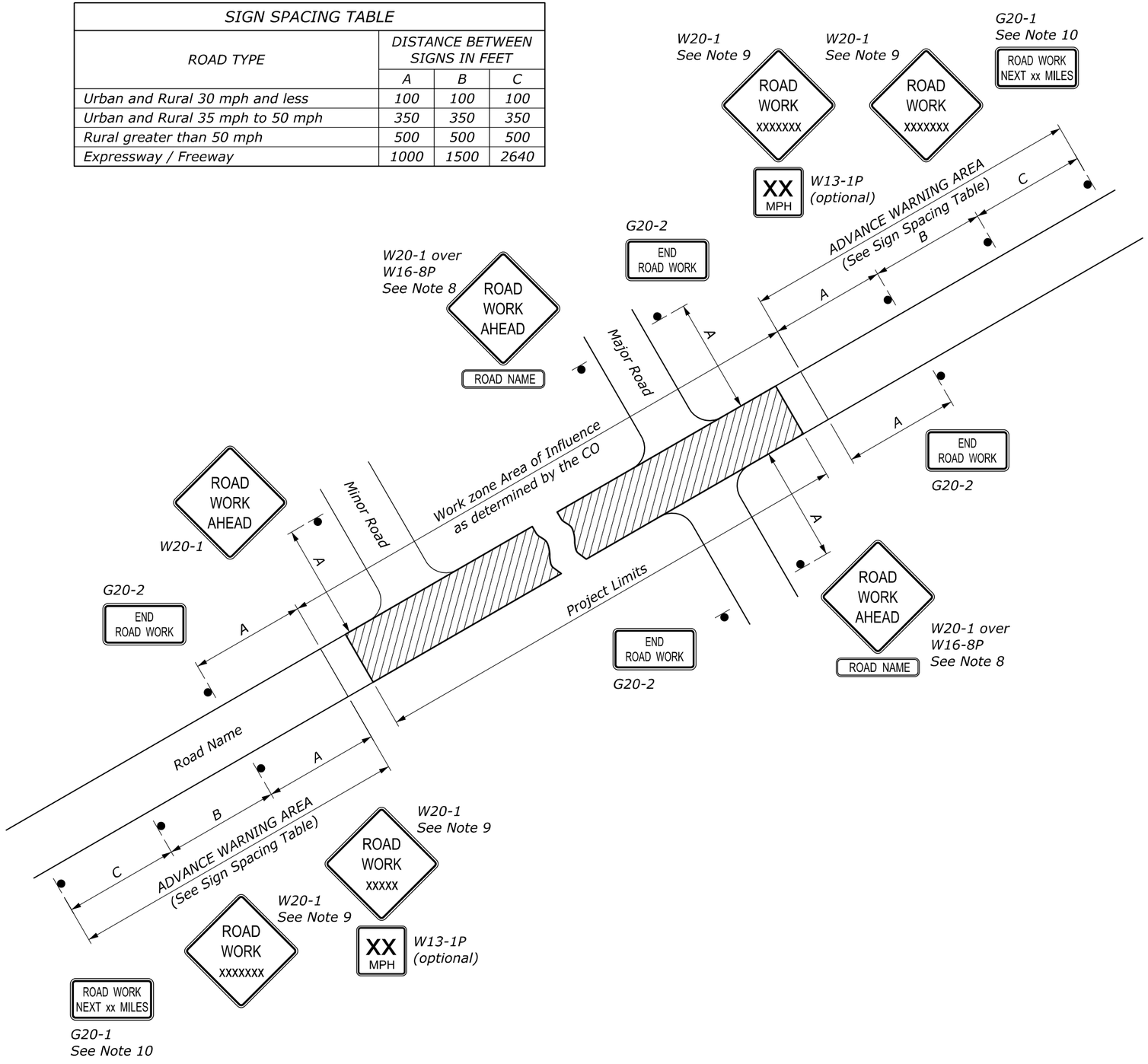
28 July 2026 10:35 AM

SIGN SPACING TABLE			
ROAD TYPE	DISTANCE BETWEEN SIGNS IN FEET		
	A	B	C
Urban and Rural 30 mph and less	100	100	100
Urban and Rural 35 mph to 50 mph	350	350	350
Rural greater than 50 mph	500	500	500
Expressway / Freeway	1000	1500	2640



NOTE:

1. *Erect all project advance warning signs before starting construction work.*
2. *Not all details shown on the temporary traffic control sheets may be applicable to this project. The Contractor may add or delete information and details in this traffic control plan as necessary to accommodate actual operations.*
3. *Where advance warning signs, placed as shown, interfere with permanent signs, locate the warning signs to fit field conditions as approved. Vary messages as required.*
4. *Install advisory speed plaques under the W20 series warning signs as needed to indicate a maximum recommended speed through the construction area.*
5. *Ensure all sign supports exposed to traffic are crashworthy.*
6. *Maintain two-way traffic during all non-work hours except as approved.*
7. *Do not store traffic control devices along the roadway when not in use. Cover post-mounted signs when not applicable.*
8. *If W20-1 is placed on a roadway other than that on which the actual construction work occurs, include a supplementary plaque indicating the name of the road on which the construction does occur (applies to major roads only).*
9. *The message on the W20-1 signs may be ROAD WORK AHEAD or may specify the distance to the work area in feet or in miles. Install an additional W20-1 sign when approach speeds exceed 50 mph. When used, place the two W20-1 signs "B" feet apart according to the Sign Spacing Table.*
10. *For work zones that are 2 miles or more in length, install G20-1 signs at each end of the project. Show the distance on the G20-1 sign to the nearest whole mile.*
11. *If signing on a roadway under a jurisdiction other than the client agency, verify that an encroachment permit has been obtained.*
12. *State standards may be used as an alternative if approved.*

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

TEMPORARY TRAFFIC CONTROL ADVANCE SIGNING

FLH STANDARD	635-1
SPECIFICATION	FP-24, FP-14
APPROVED FOR USE	2/2024

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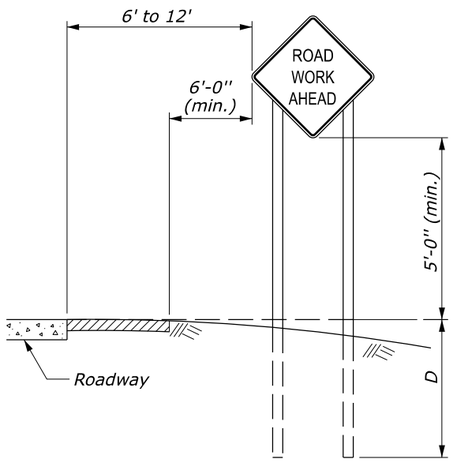
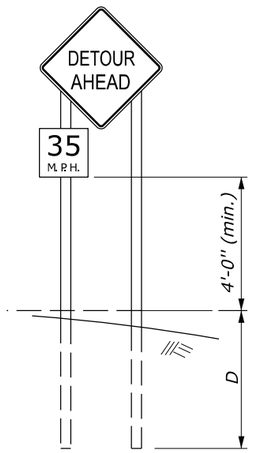
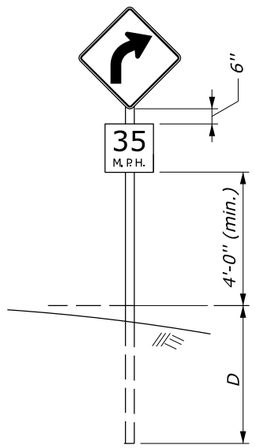
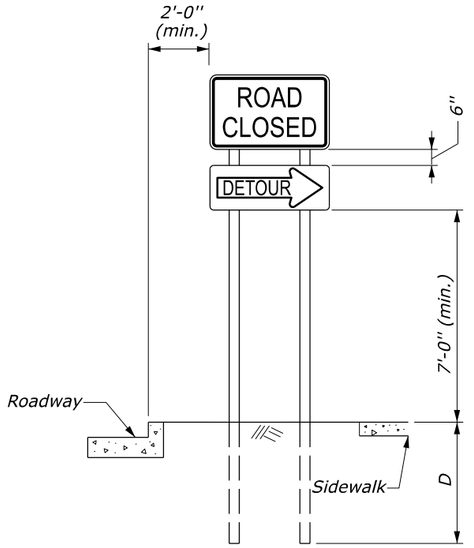
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	S18







RURAL AREA

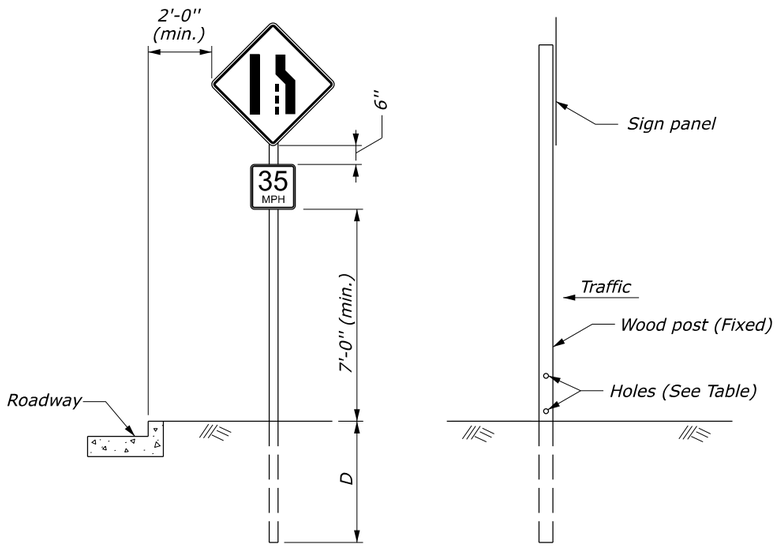


URBAN AREA

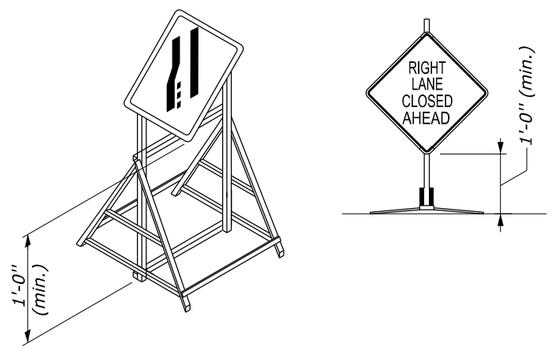
NOTES:

1. *Mount signs that are wider than 3 feet or larger than 10 square feet on double posts.*
2. *All lumber dimensions are nominal.*
3. *Submit alternate details for portable signs. Ensure sign mounts hold the sign face in a vertical plane. Portable signs may be mounted lower than fixed signs when approved. Ensure all portable sign supports are crashworthy.*
4. *When parking is permitted within 200 feet of the sign, mount the sign a minimum of 7 feet above the pavement surface.*
5. *When approved by the CO and the Utility Company, utility poles may be used for sign mounting.*
6. *For 4- by 6-inch and greater posts, see the Breakaway Sign Support View. If breakaway design cannot be used due to post spacing, place the sign outside the clearzone or shield with a barrier. Do not place holes in posts of non-breakaway signs.*
7. *Signs requiring 6- by 6-inch and greater posts are considered non-breakaway if multiple posts are required and the posts cannot be spaced a minimum of 7 feet apart.*

FIXED ROADWAY SIGNS



BREAKAWAY SIGN SUPPORT
(FIXED SIGNS 4" X 6" AND GREATER POSTS)
See Notes 6 and 7



PORTABLE SIGNS
See Notes 3 and 4

POST SIZE TABLE

POST SIZE	D	HOLE DIAMETER	MAXIMUM SIGN AREA (SQFT)			
			1 Post	2 Post	3 Post	4 Post
4" x 4"	4'	None Required	10	20	X	X
4" x 6"	4'	1.5"	X	35	50	70
6" x 6"	5'	2"	X	50	75	100
6" x 8"	5'	3"	X	85	125	165

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

TEMPORARY TRAFFIC CONTROL SIGN MOUNTING

EFLHD DETAIL	E635-01
SPECIFICATION	FP-24, FP-14
APPROVED FOR USE	06/2024

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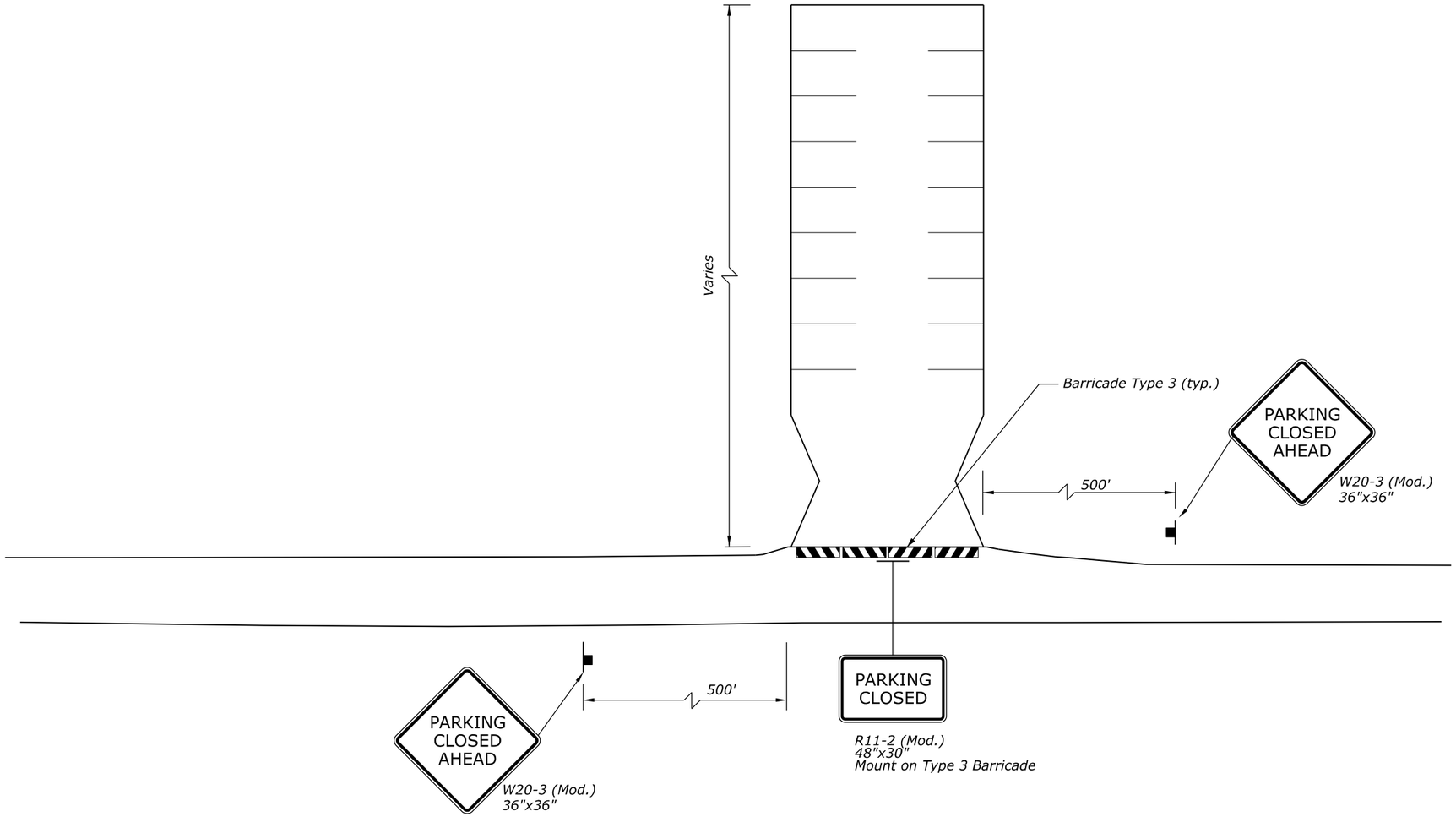
28 July 2026 10:36 AM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	S19



TYPICAL PARKING AREA CLOSURE

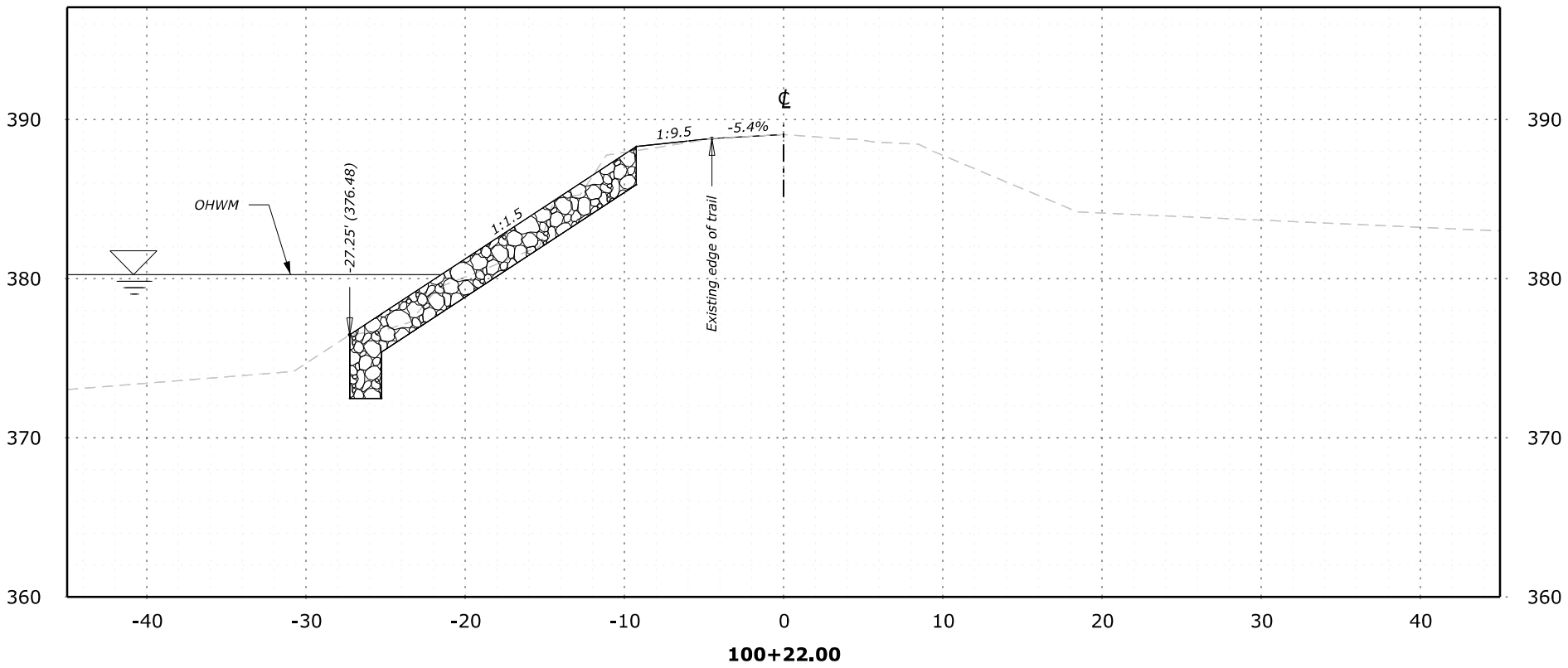
NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA OFFICE OF FEDERAL LANDS HIGHWAY	EFLHD DETAIL E635-A
TEMPORARY TRAFFIC CONTROL PARKING AREA CLOSURE	SPECIFICATION FP-14

28 July 2026 10:37 AM \\fhfil5\fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) Combined NJ-NY\Std-Det\S18-E635-A_Parking Area Closed.dgn [BDS]

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	T01



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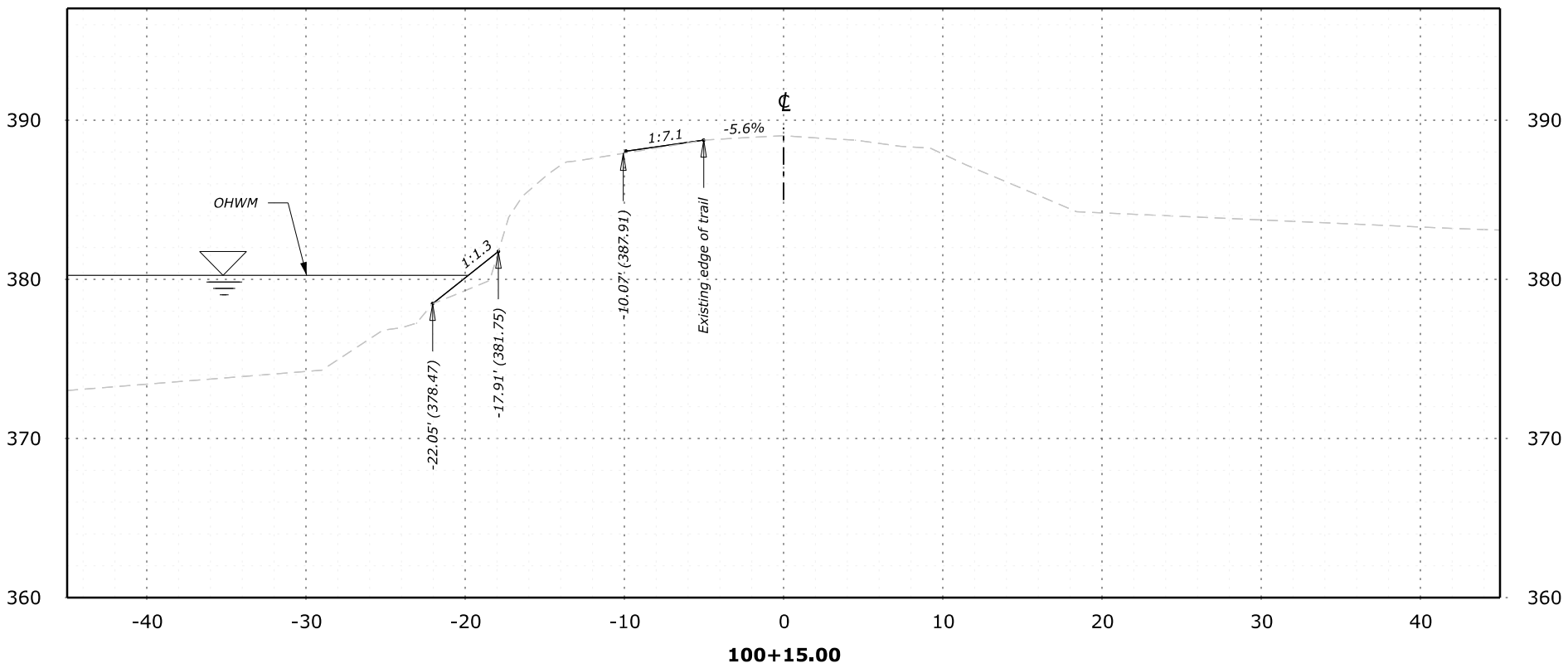

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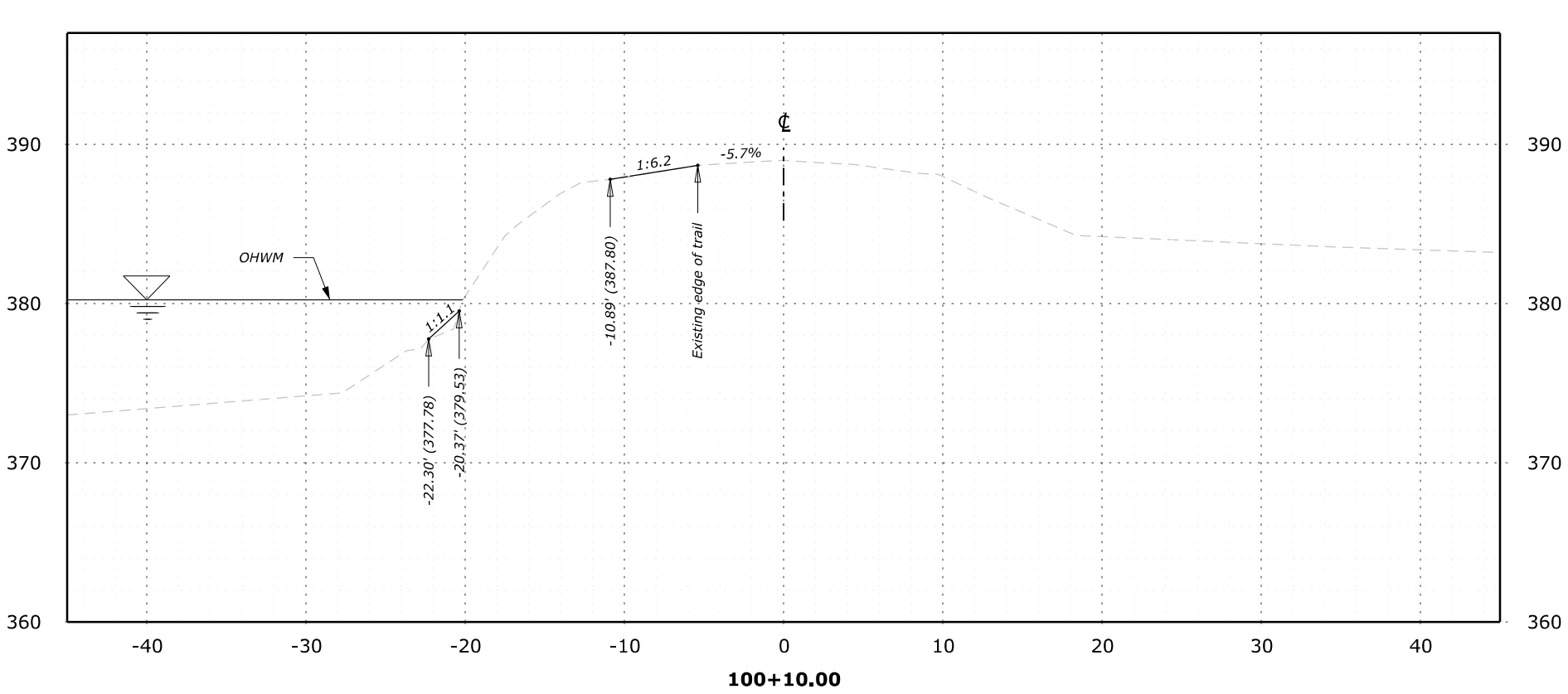

100+10.00

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #3

100+10 TO 100+22

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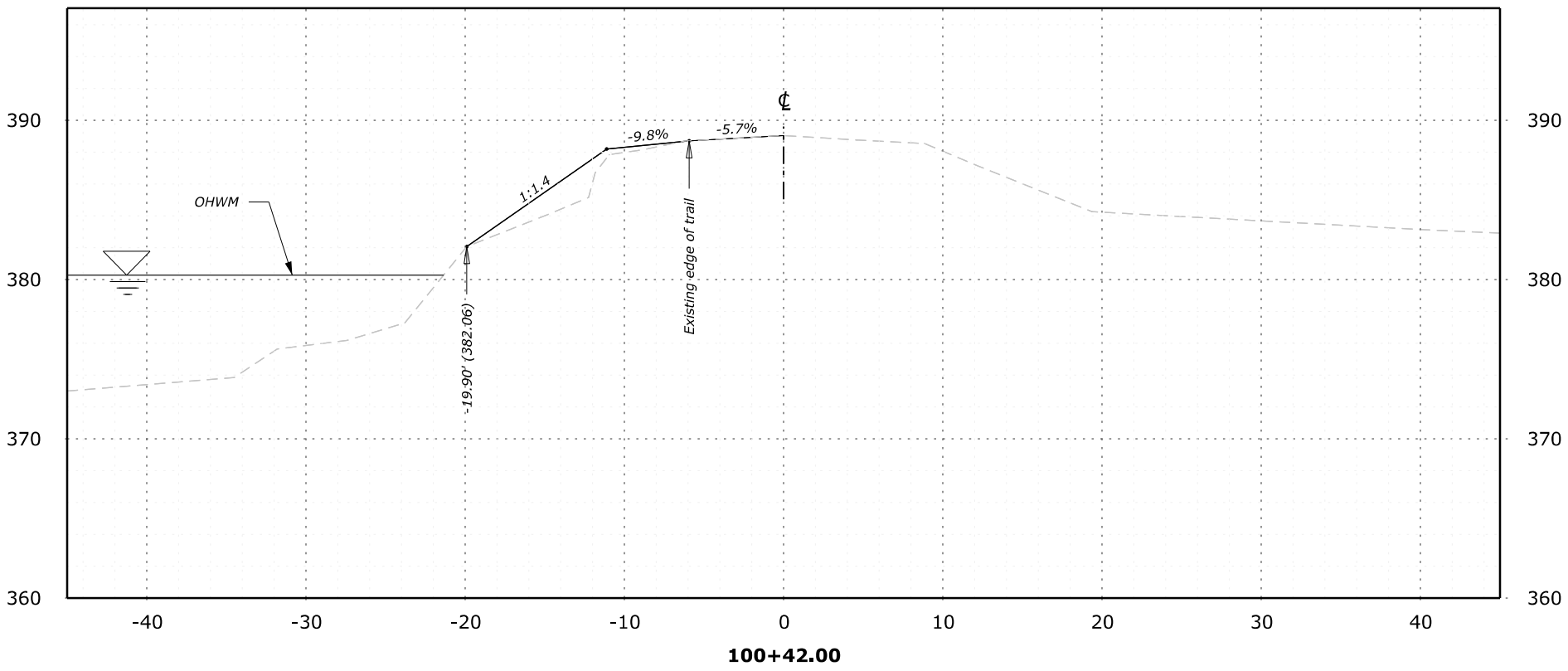
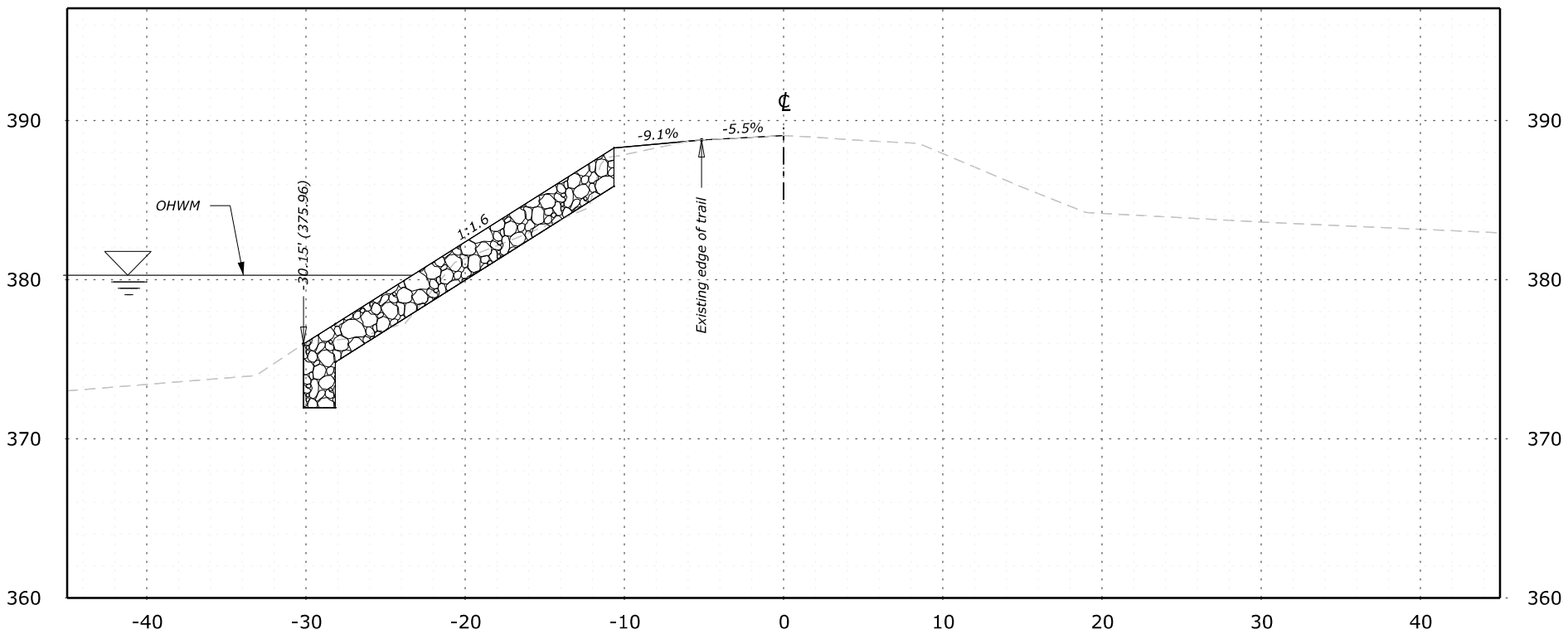
7 August 2026 6:40 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	T02



100+42.00



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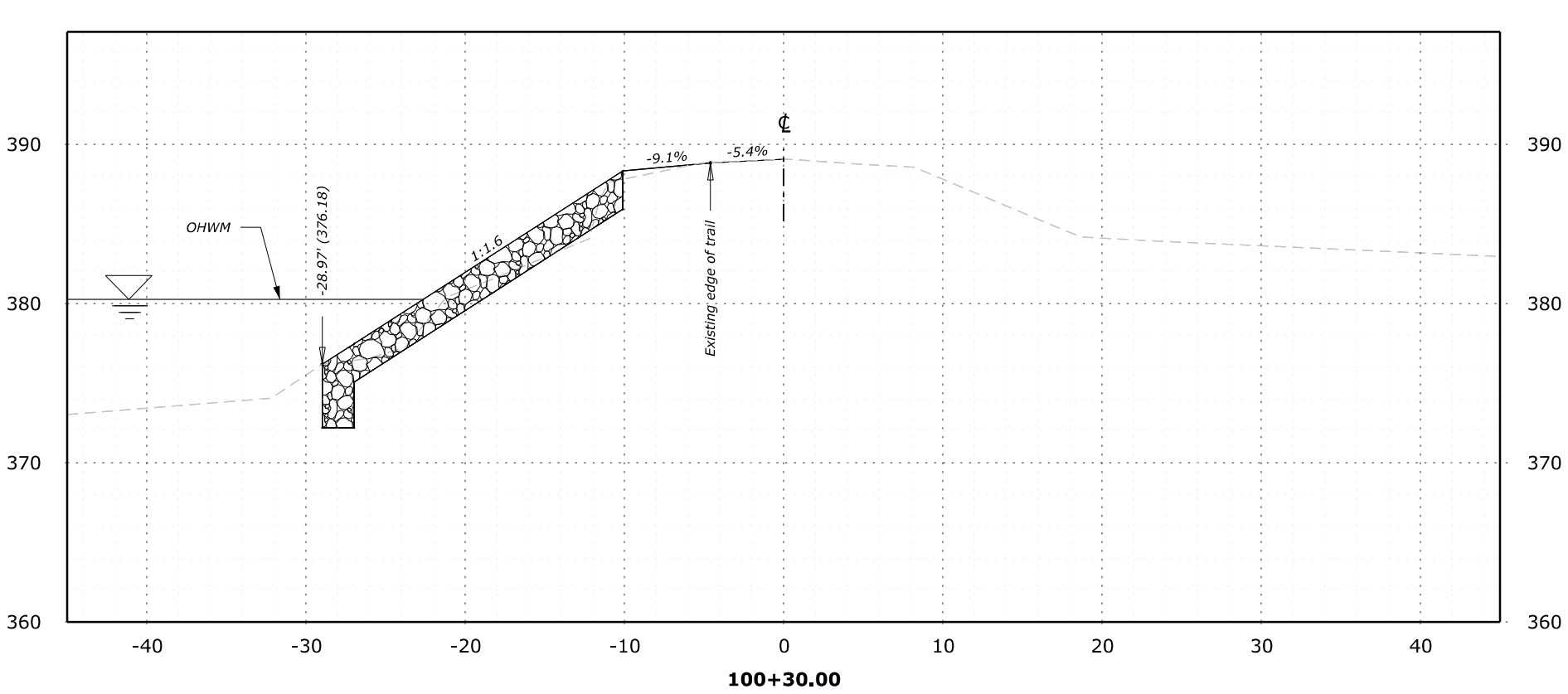

100+30.00

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7 August 2026 6:40 PM

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

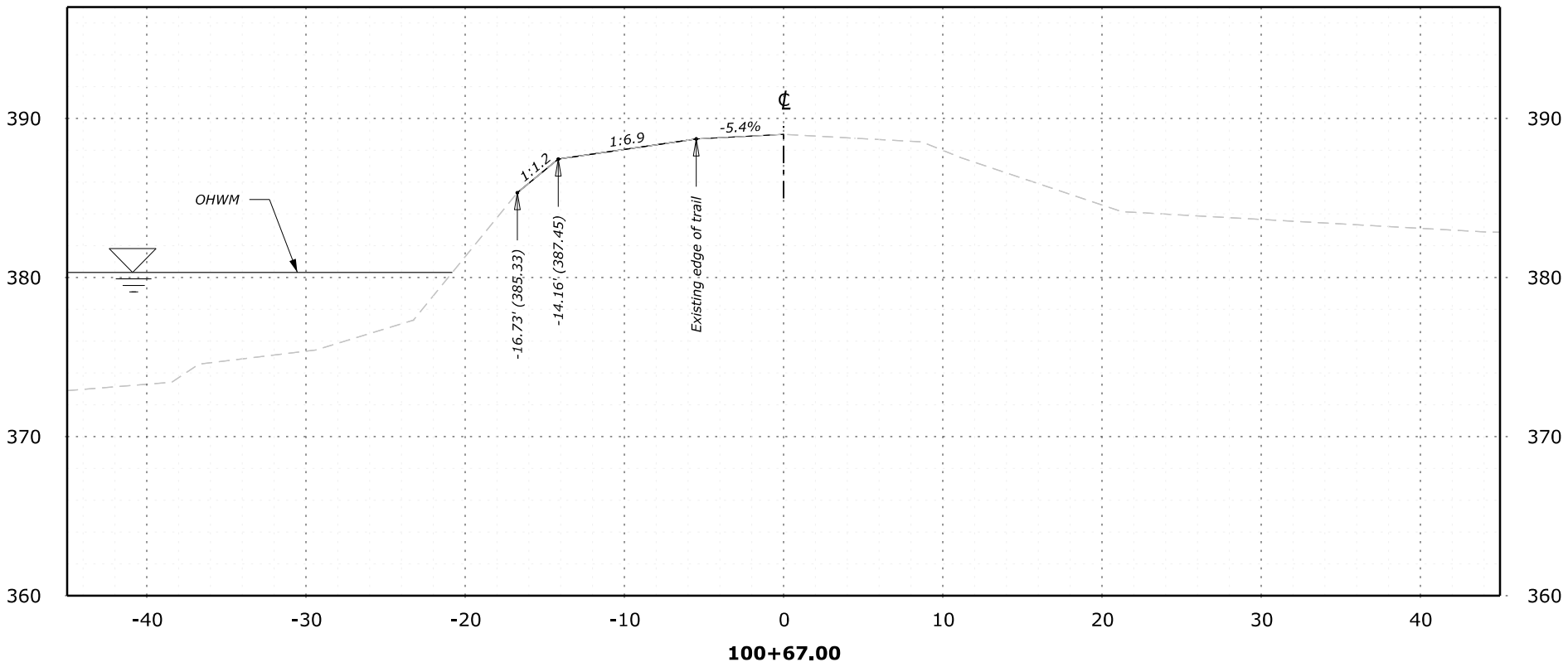
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #3

100+30 TO 100+42

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	T03



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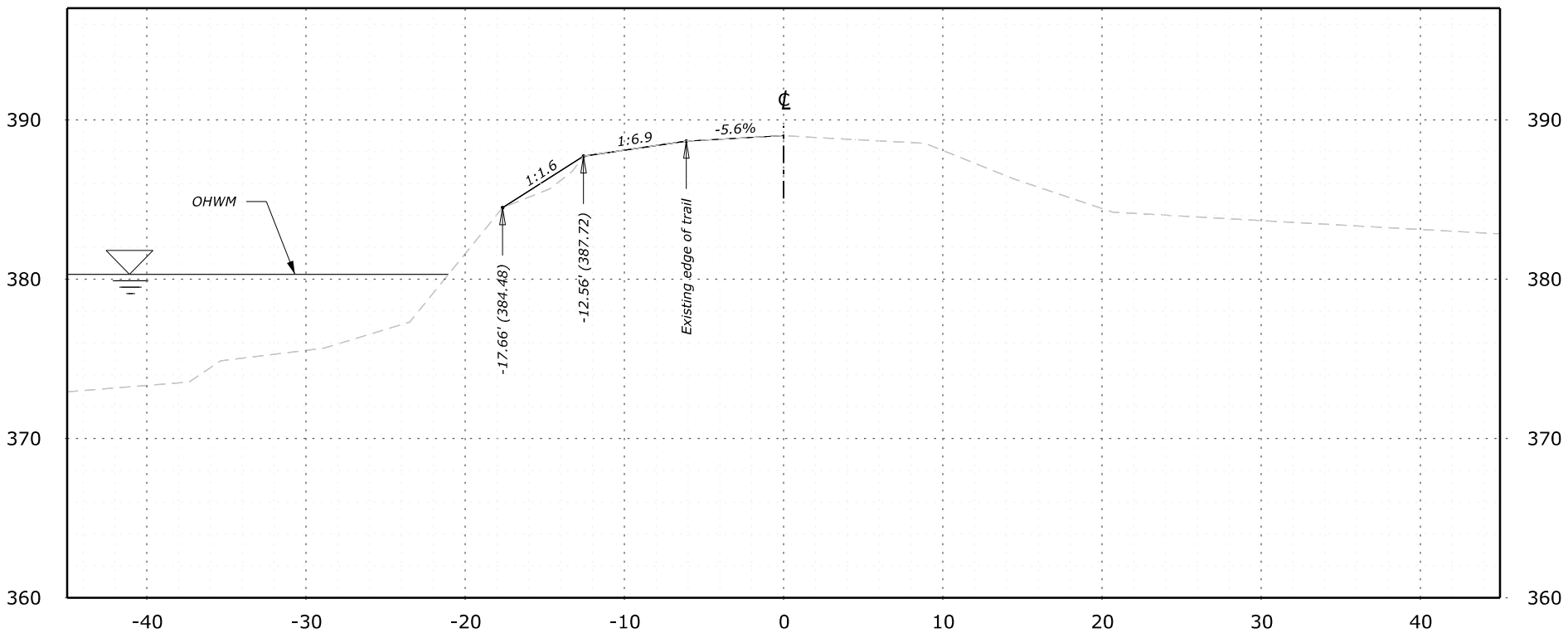

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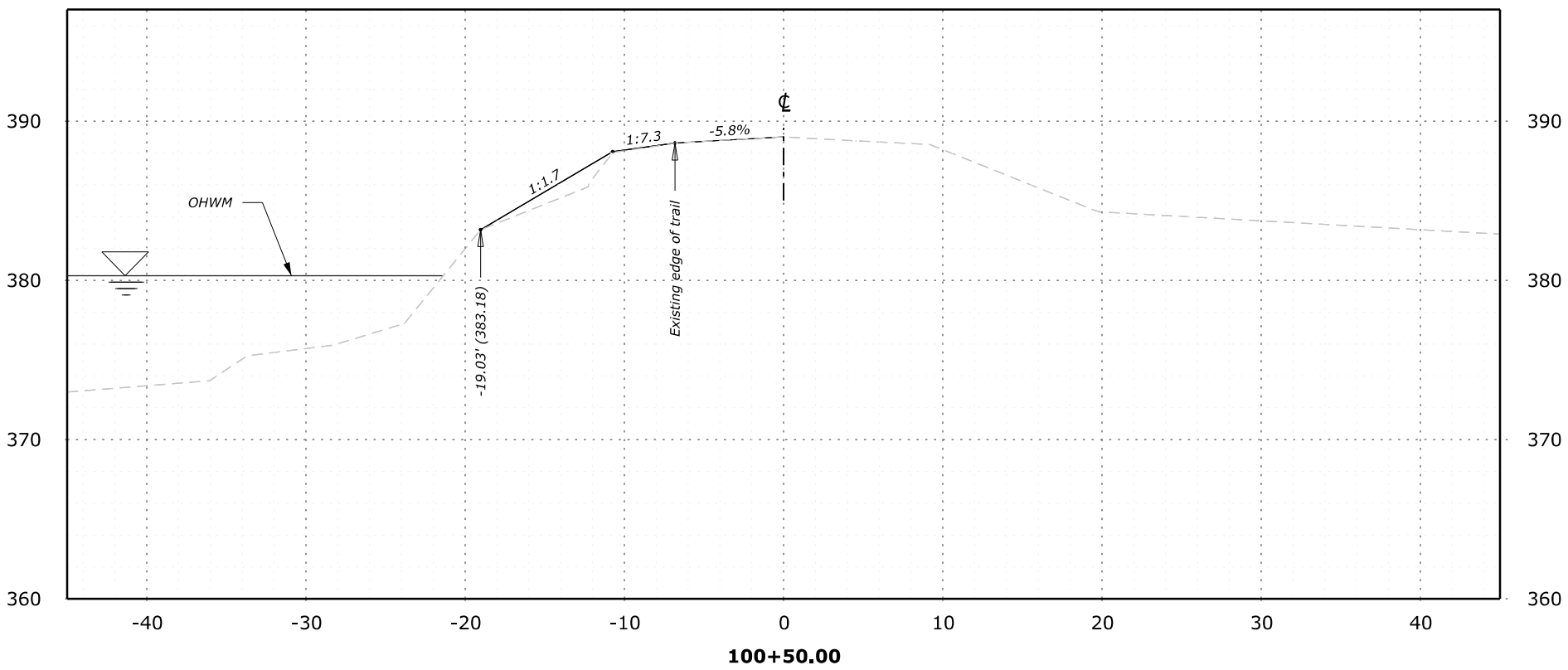

100+50.00

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #3

100+50 TO 100+67

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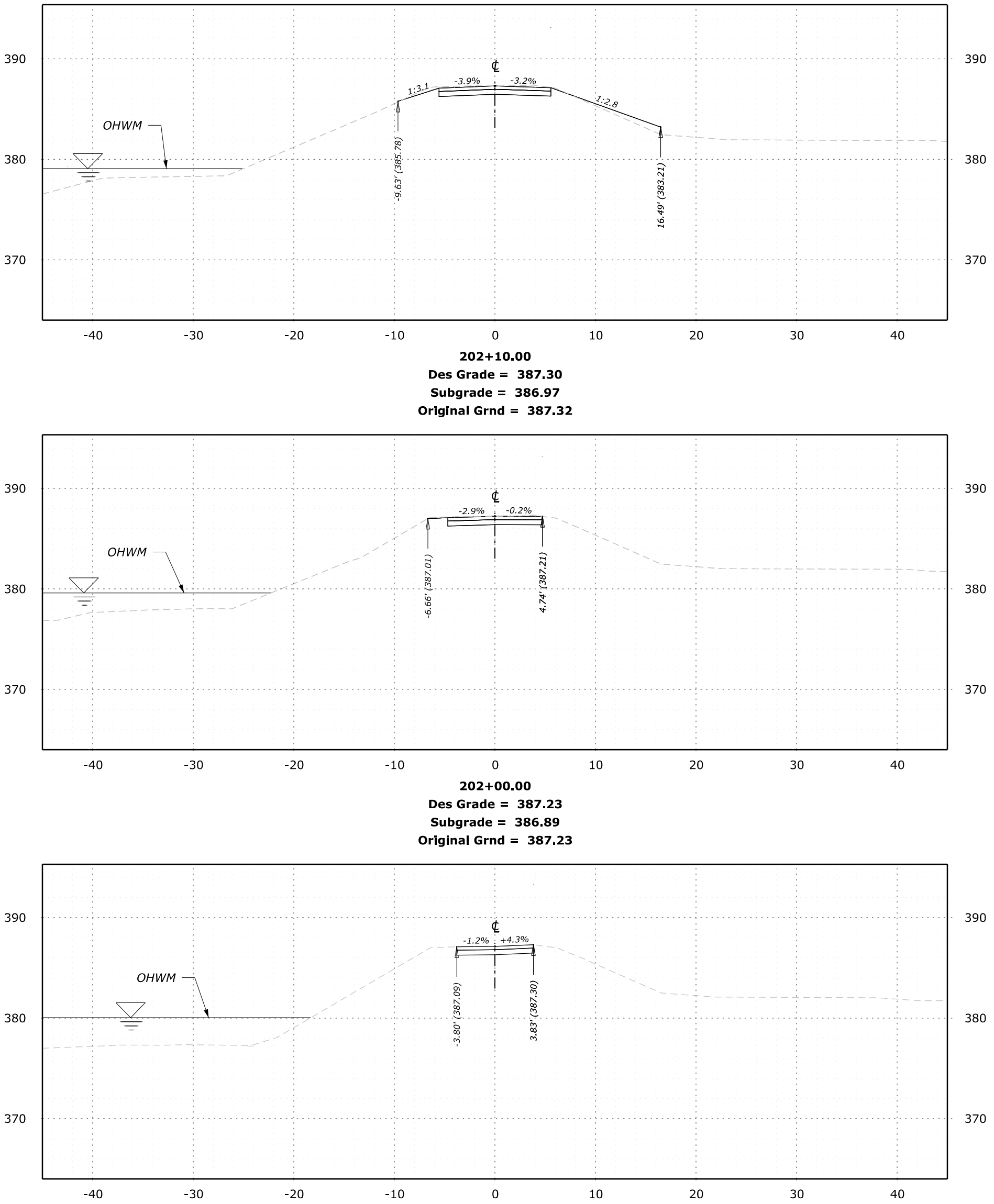
7 August 2026 6:40 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	U01



202+10.00
Des Grade = 387.30
Subgrade = 386.97
Original Grnd = 387.32

202+00.00
Des Grade = 387.23
Subgrade = 386.89
Original Grnd = 387.23

201+90.00
Des Grade = 387.14
Subgrade = 386.80
Original Grnd = 387.14

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

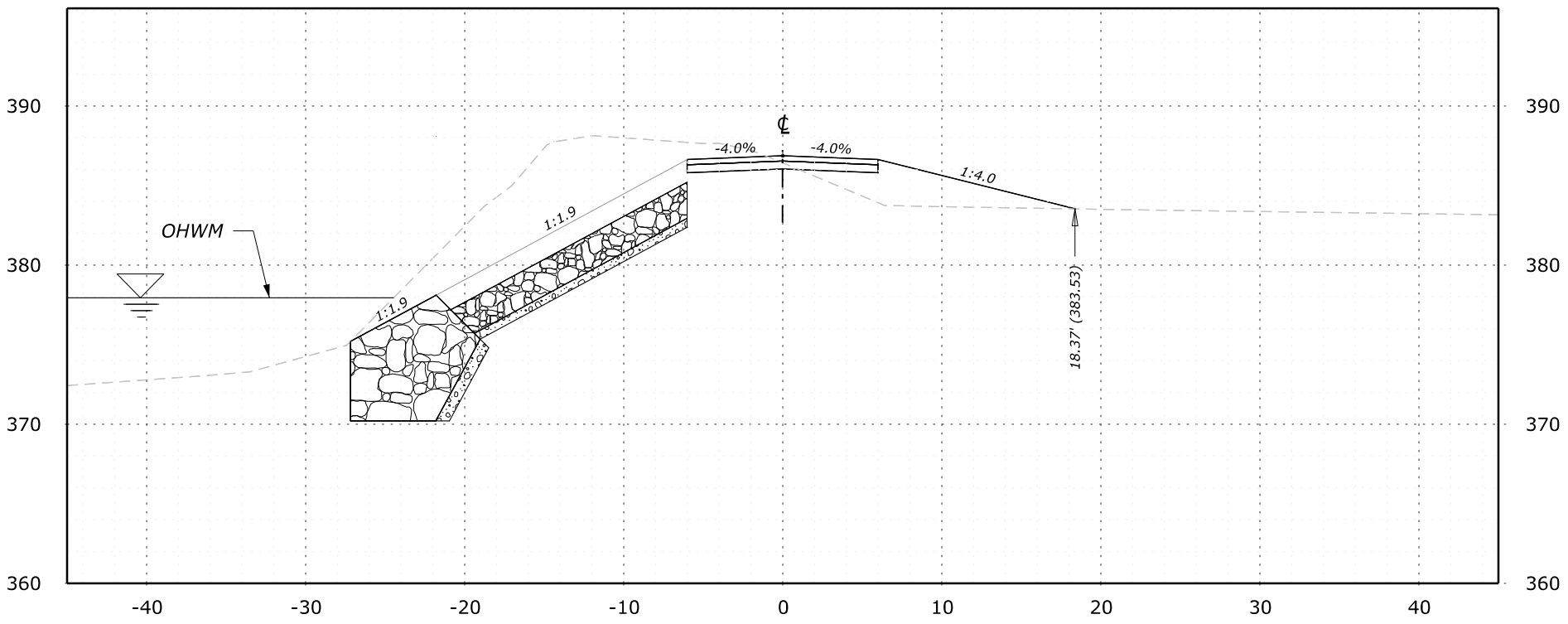
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #240

201+90 TO 202+10

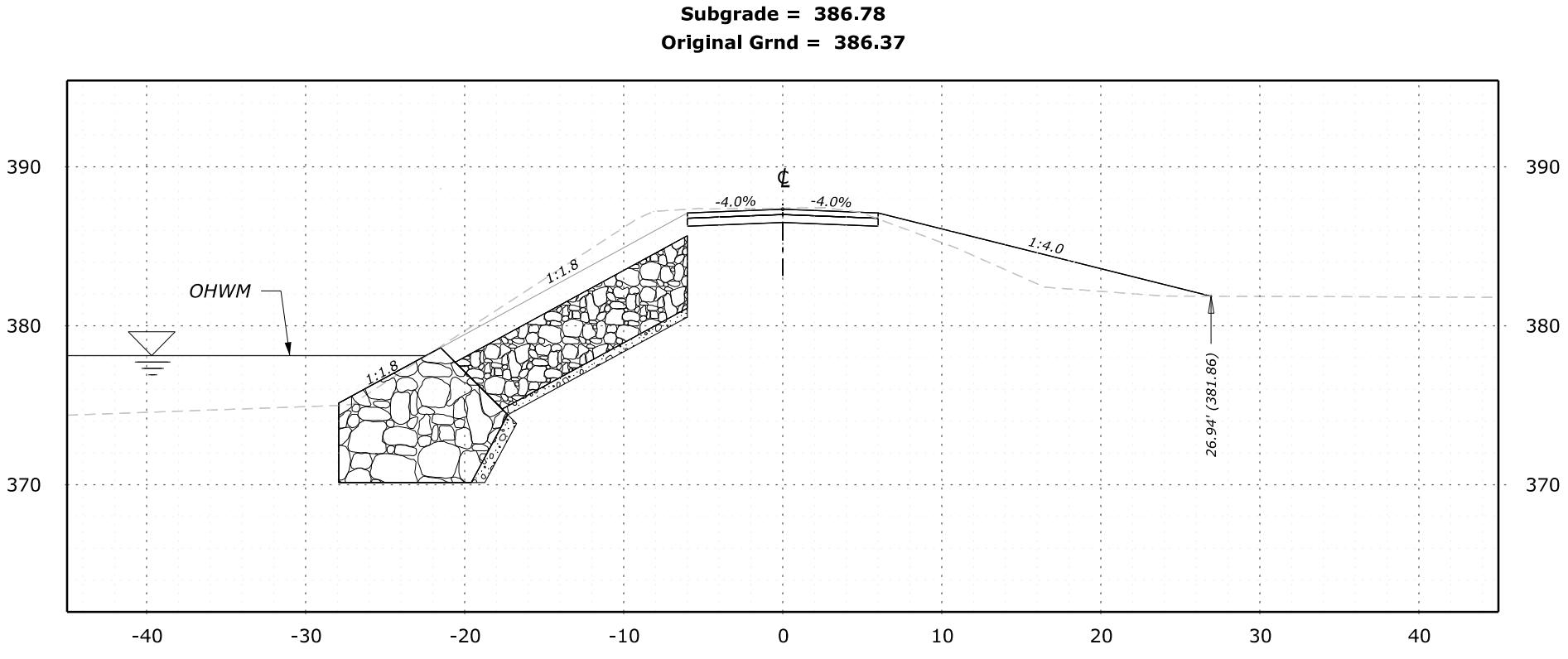
2 September 2026 11:15 AM \\fhfil5\fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\U01-U05-NY-WARI_2019-1(1)_xs_Site240.dgn [Bl_Site #240 - 201+90.00 [Sheet]]

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	U02



202+75.00
Des Grade = 386.87
Subgrade = 386.54
Original Grnd = 386.43

202+50.00
Des Grade = 387.11
Subgrade = 386.78
Original Grnd = 386.37



202+20.00
Des Grade = 387.33
Subgrade = 387.00
Original Grnd = 387.40

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

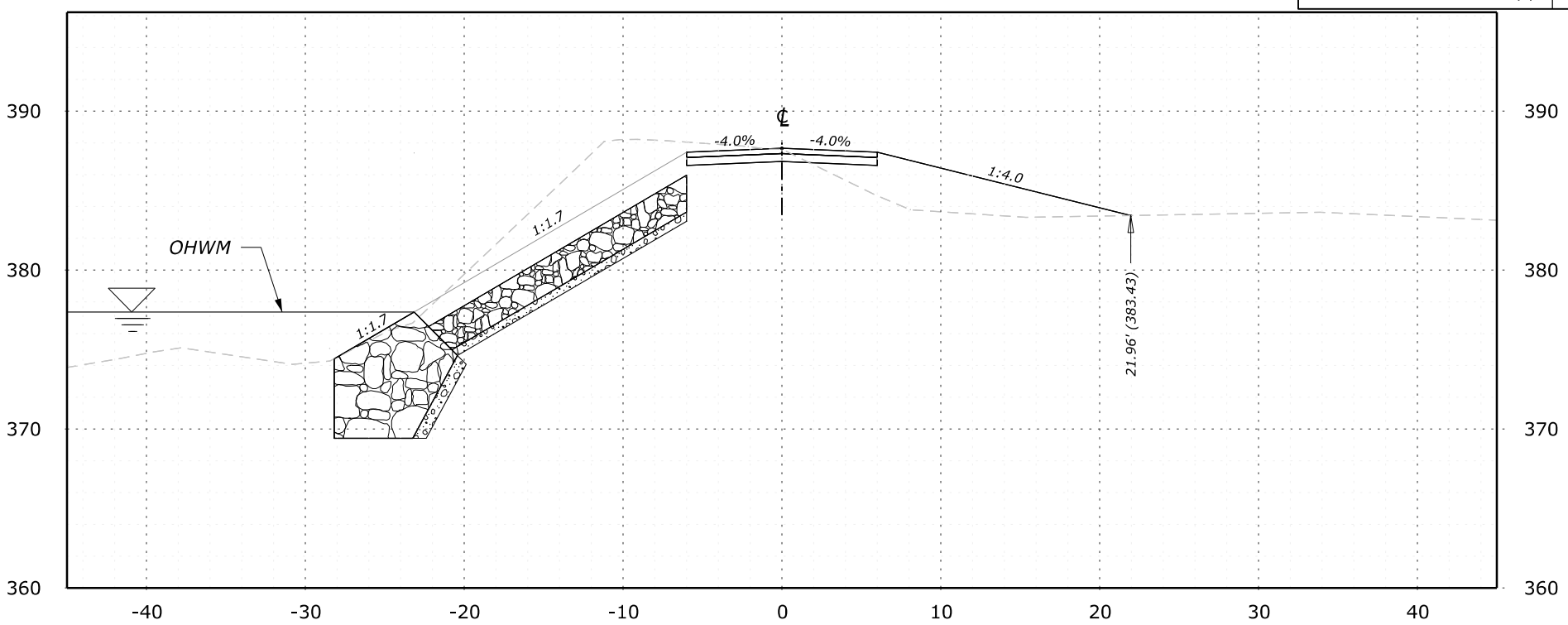
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #240

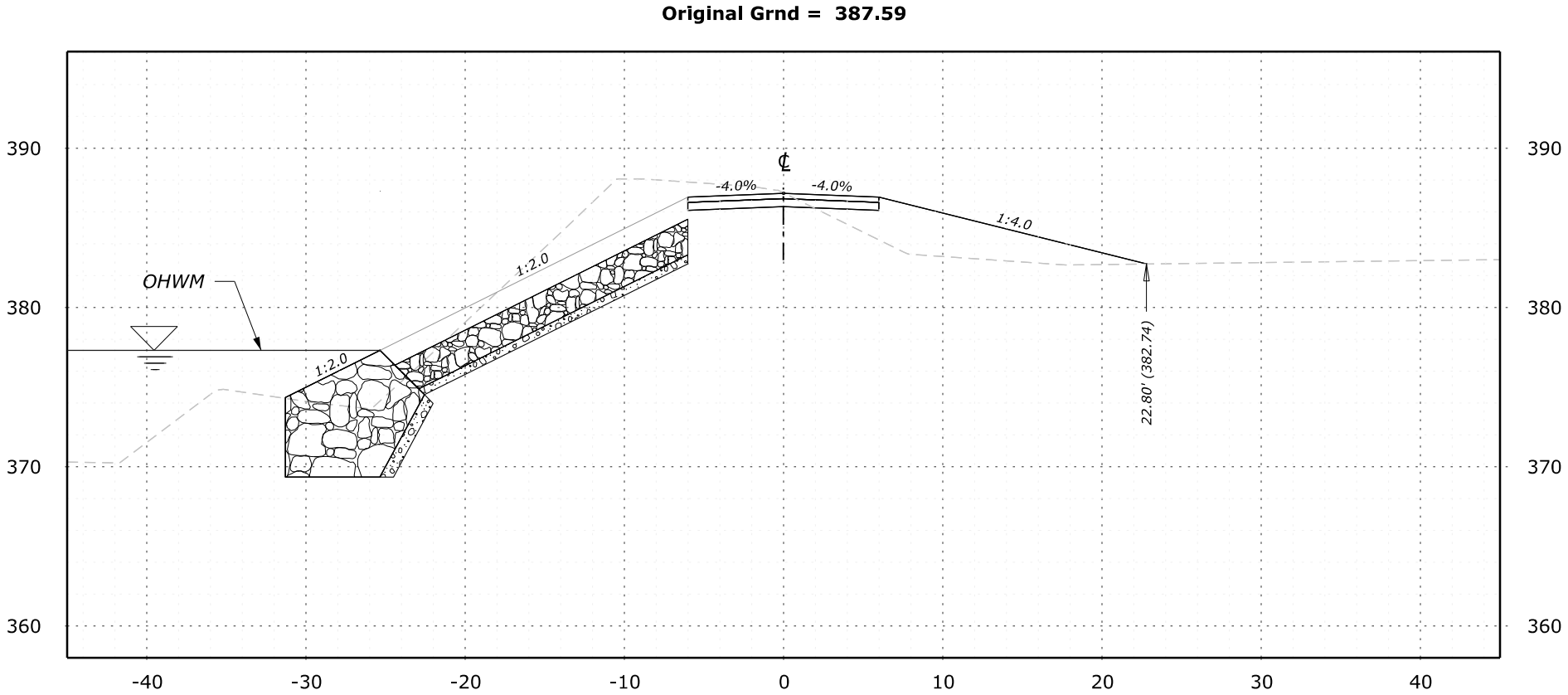
202+20 TO 202+75

2 September 2026 11:15 AM \\fhf1.fhfileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\U01-U05-NY-WARI_2019-1(1)_xs_Site240.dgn [BL_Site #240 - 202+20.00 [Sheet]]

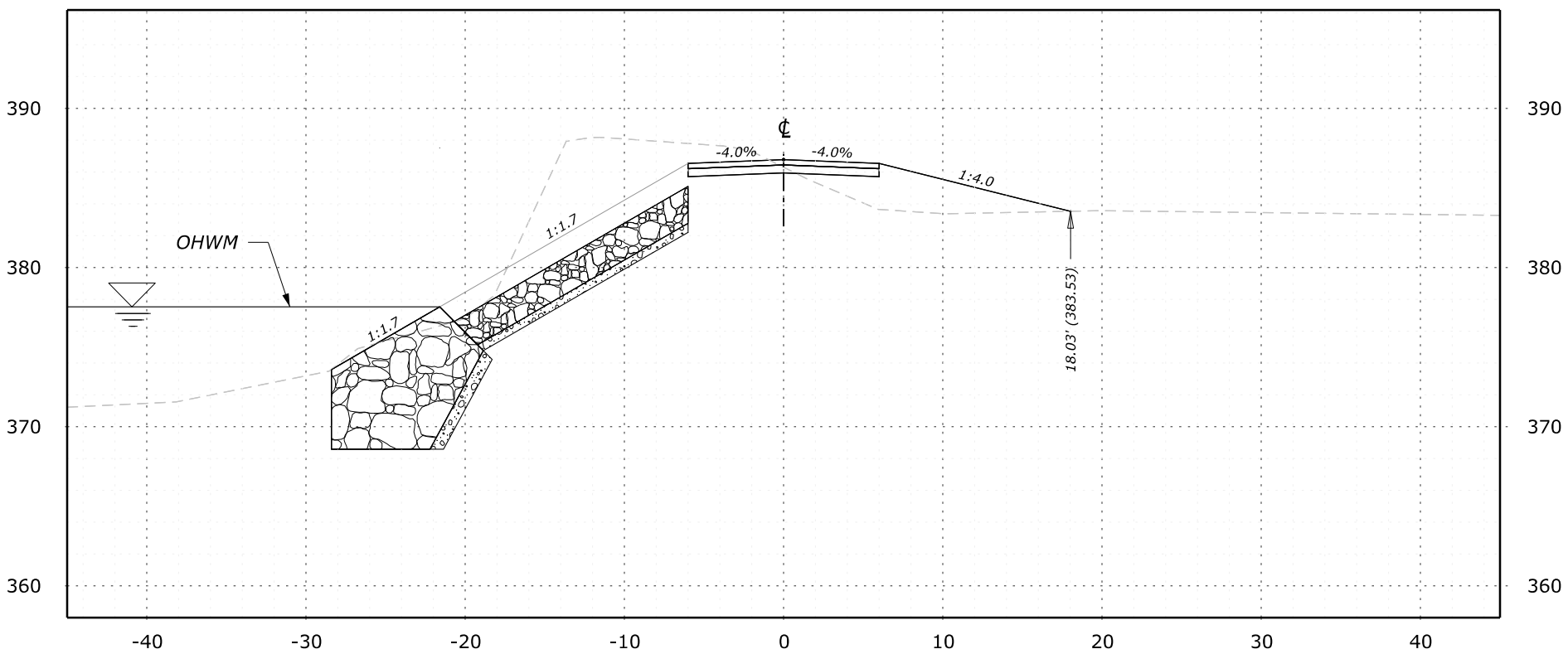
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	U03



203+50.00
Des Grade = 387.66
Subgrade = 387.33
Original Grnd = 387.59



203+25.00
Des Grade = 387.18
Subgrade = 386.84
Original Grnd = 387.20



203+00.00
Des Grade = 386.77
Subgrade = 386.44
Original Grnd = 386.30

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

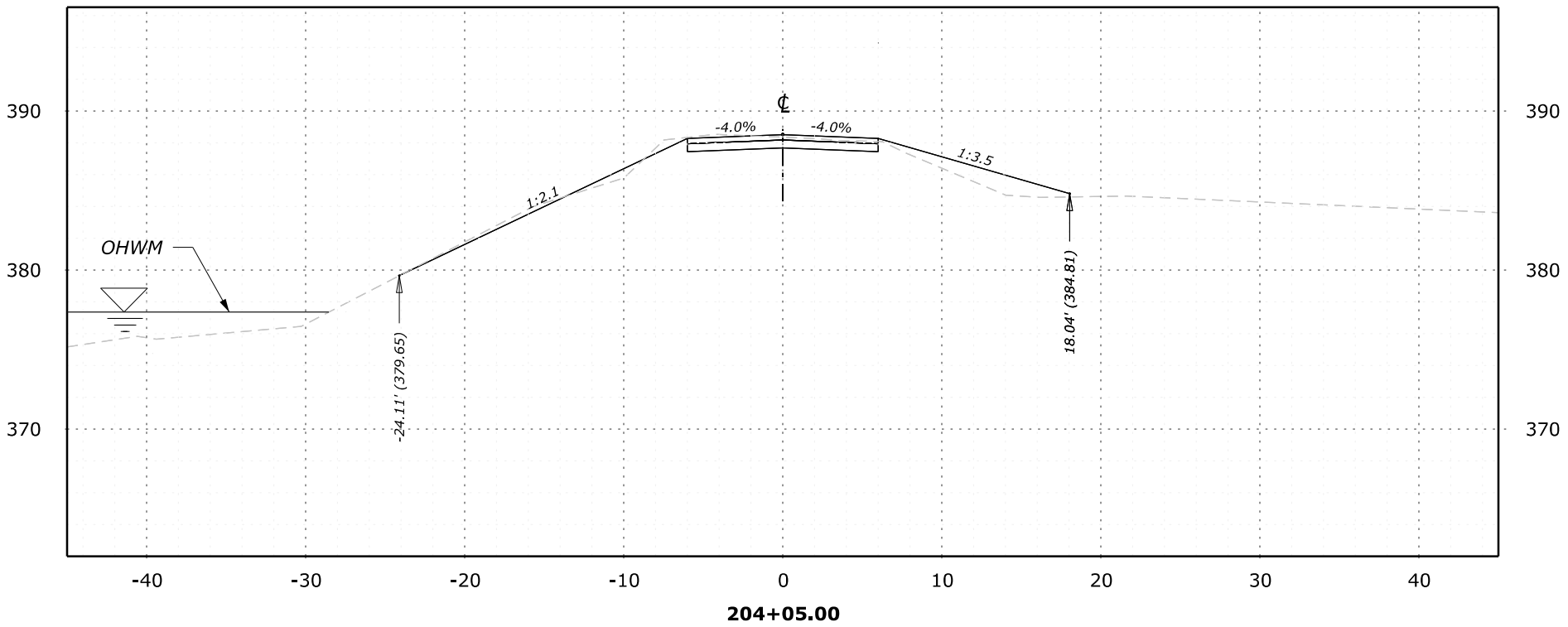
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #240

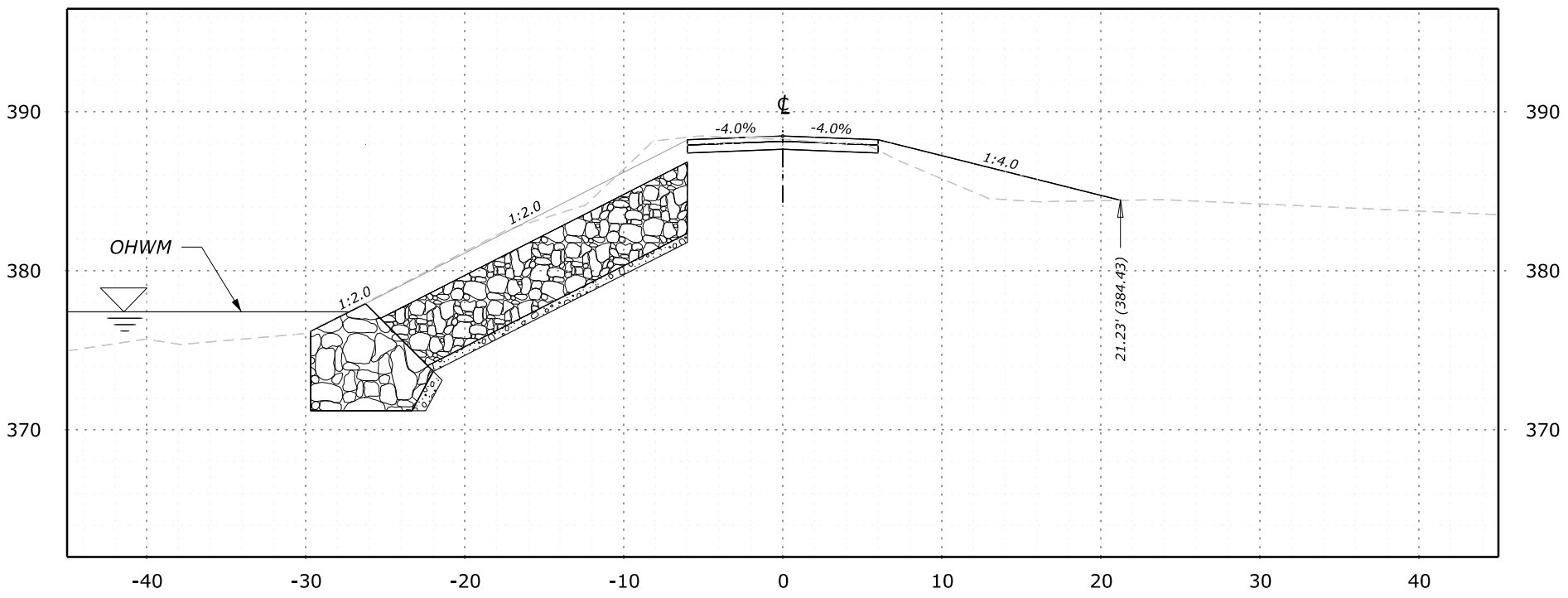
203+00 TO 203+50

2 September 2026 11:15 AM \\fhfl15\fleserve.flhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\U01-U05-NY-WARI_2019-1(1)_xs_Site240.dgn [BL_Site #240 - 203+00.00 [Sheet]]

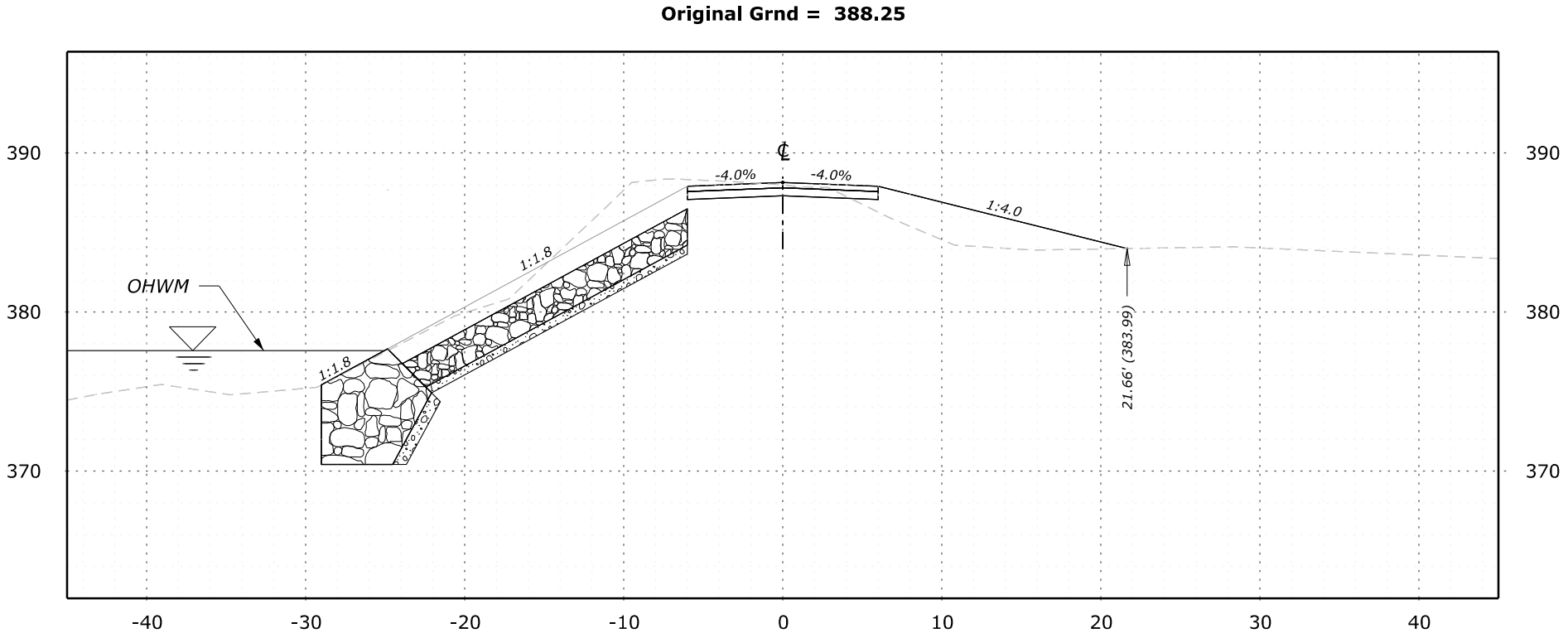
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	U04



204+05.00
Des Grade = 388.52
Subgrade = 388.19
Original Grnd = 388.35



203+95.00
Des Grade = 388.48
Subgrade = 388.14
Original Grnd = 388.25



203+75.00
Des Grade = 388.14
Subgrade = 387.81
Original Grnd = 387.98

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

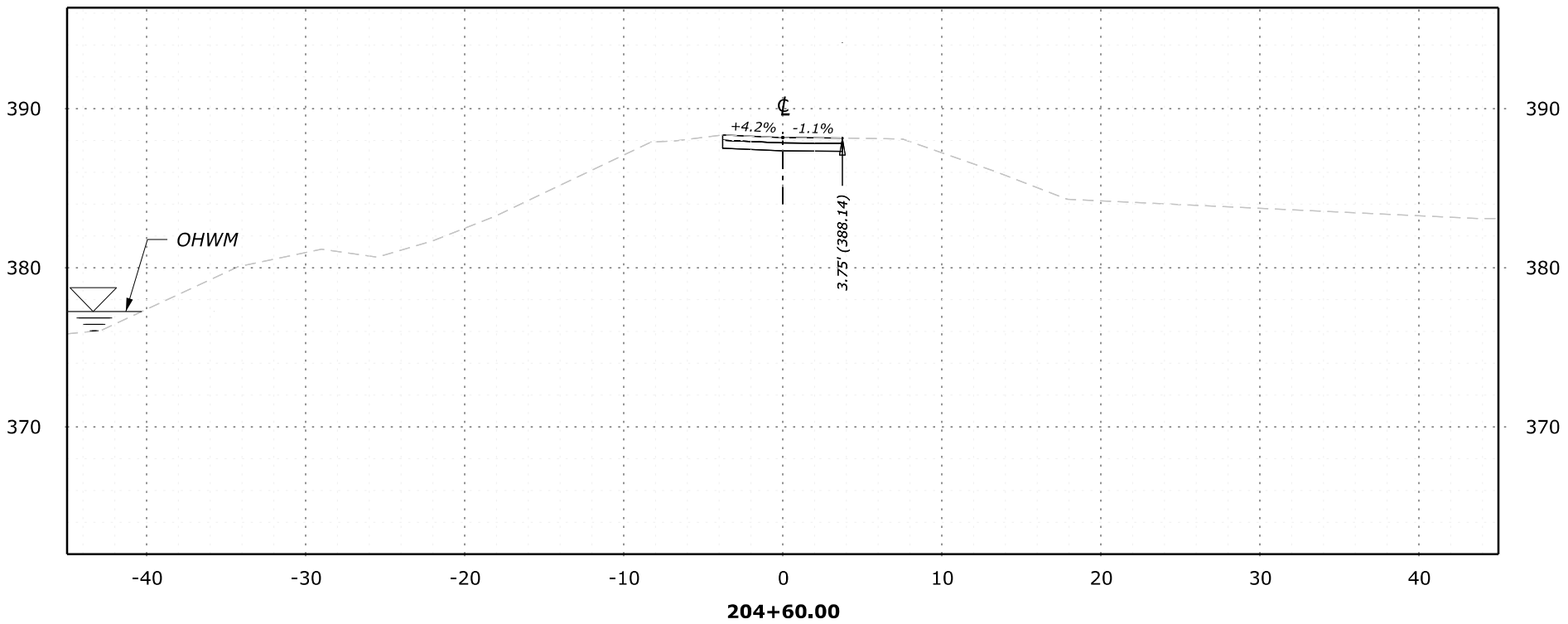
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #240

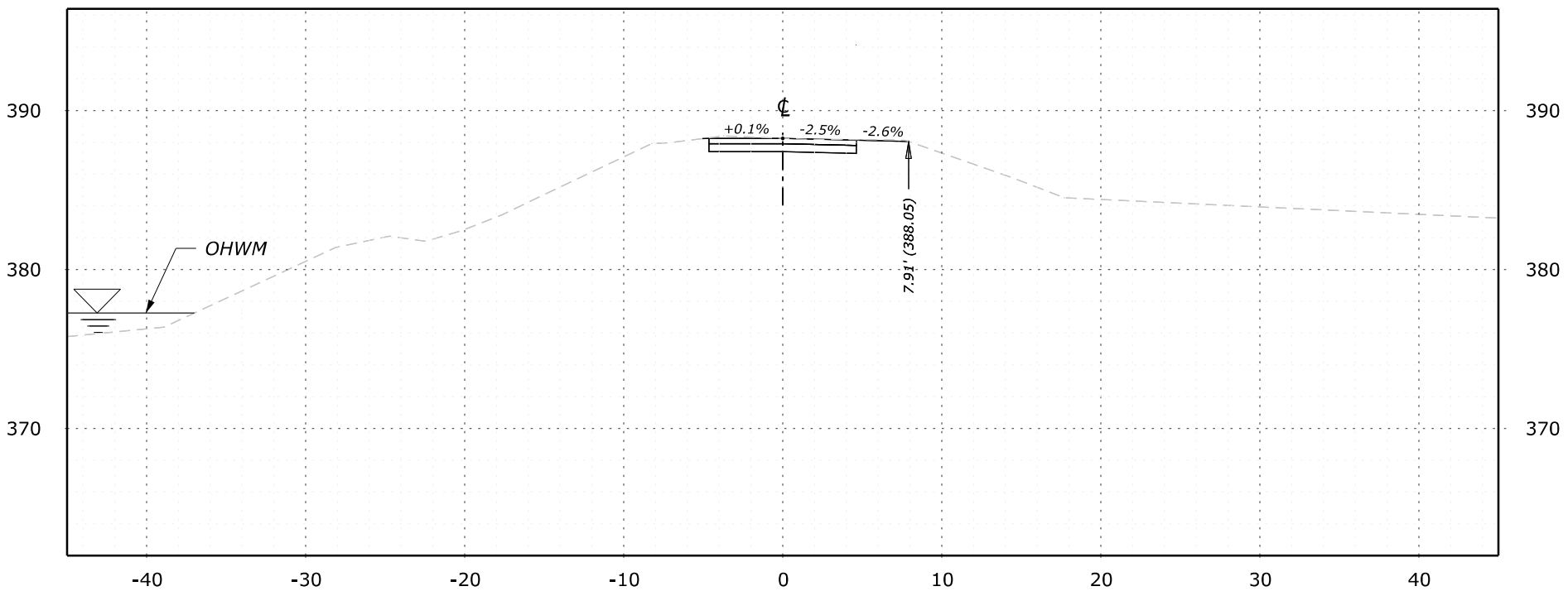
203+75 TO 204+05

2 September 2026 11:15 AM \\fhfl15\fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\U01-U05-NY-WARI_2019-1(1)_xs_Site240.dgn [Bl_Site #240 - 203+75.00 [Sheet]]

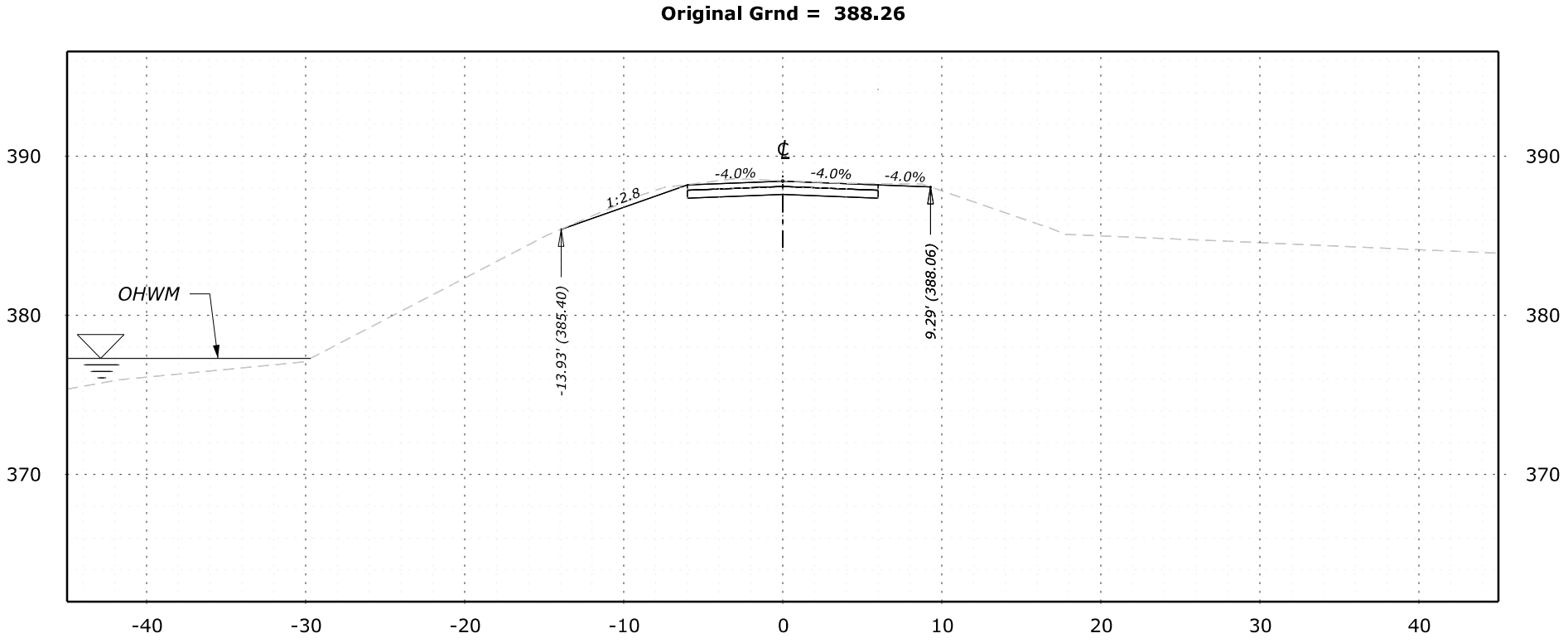
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	U05



204+60.00
Des Grade = 388.18
Subgrade = 387.85
Original Grnd = 388.18



204+50.00
Des Grade = 388.26
Subgrade = 387.92
Original Grnd = 388.26



204+25.00
Des Grade = 388.43
Subgrade = 388.10
Original Grnd = 388.46

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

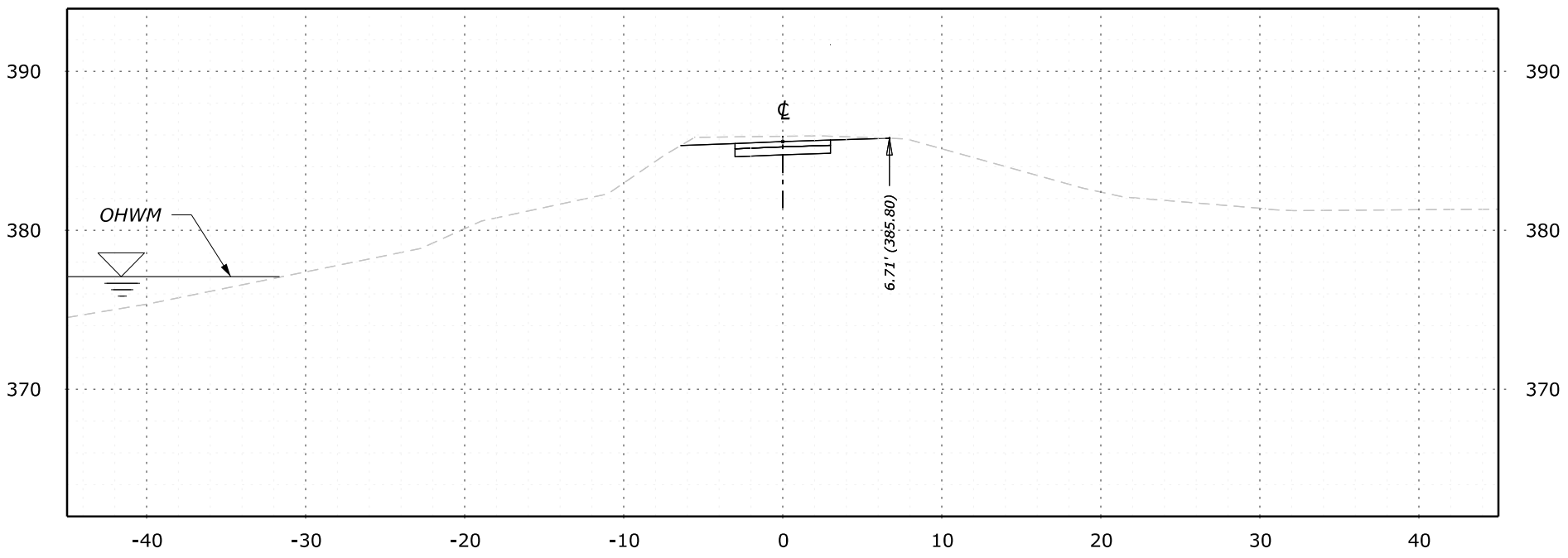
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #240

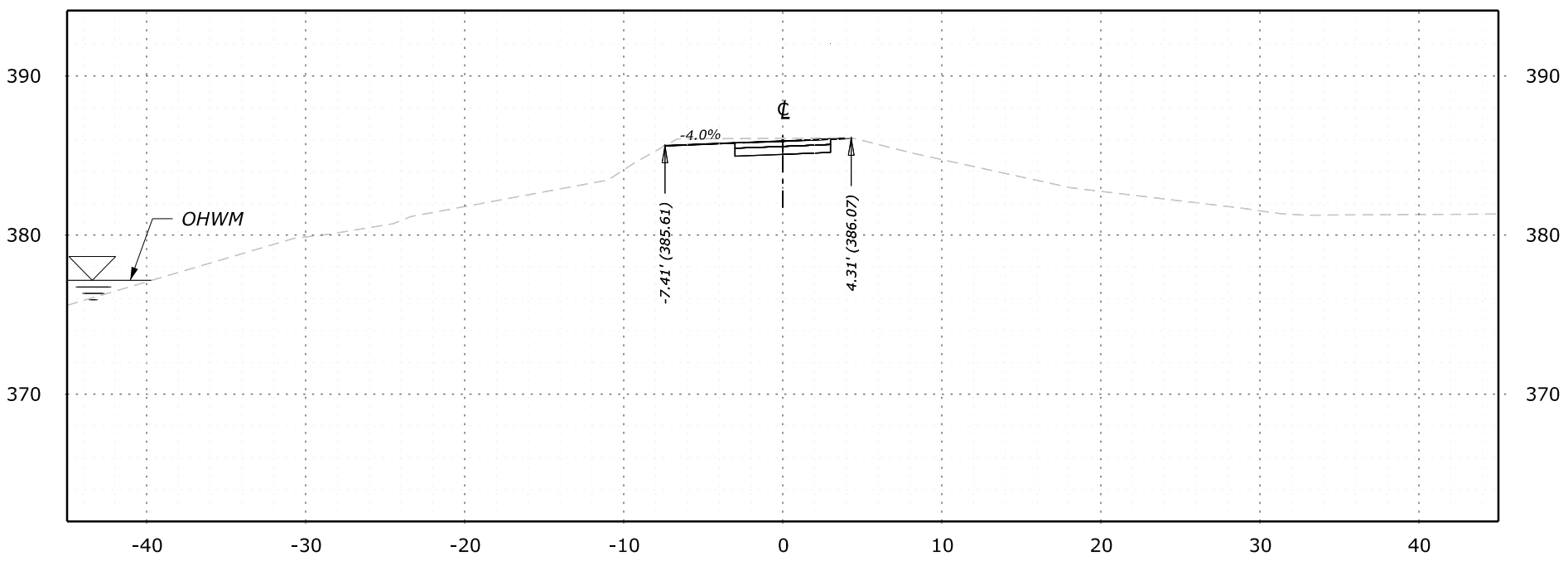
204+25 TO 204+60

2 September 2026 11:15 AM \\fhfl1s1fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\U01-U05-NY-WARI_2019-1(1)_xs_Site240.dgn [BL_Site #240 - 204+25.00 [Sheet]]

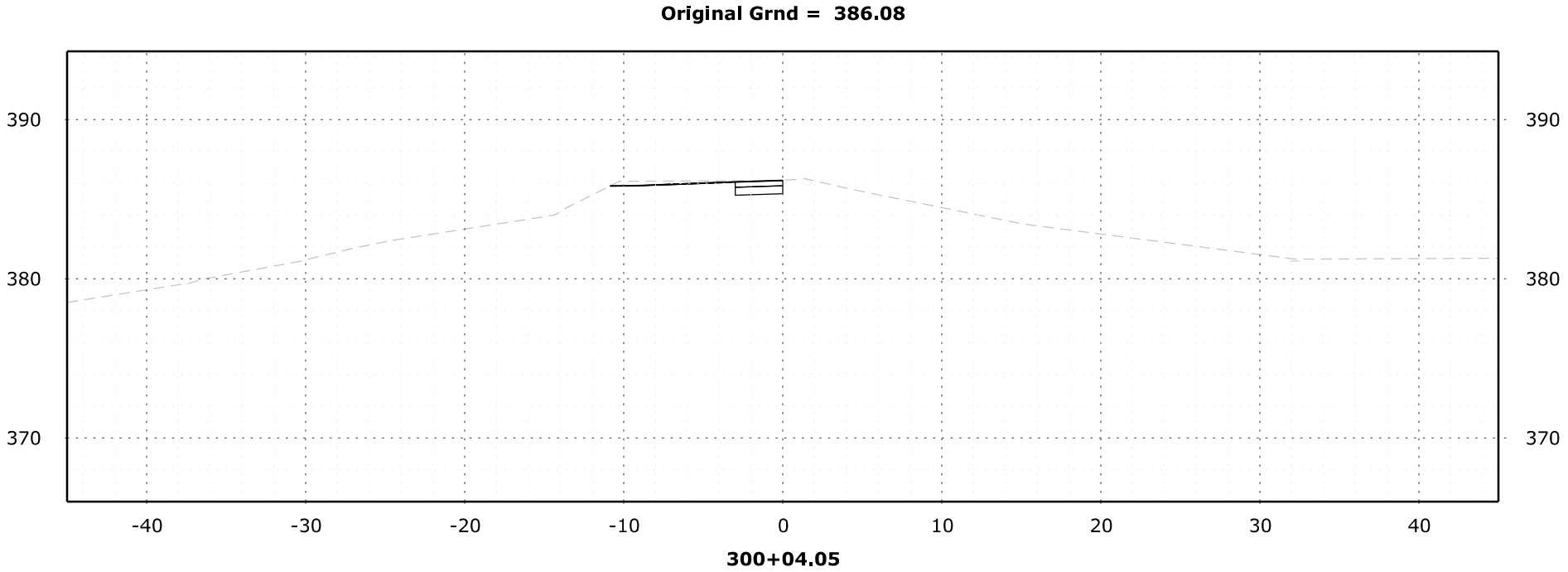
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	V01



300+25.00
Des Grade = 385.59
Subgrade = 385.26
Original Grnd = 385.92



300+10.00
Des Grade = 385.90
Subgrade = 385.57
Original Grnd = 386.08



300+04.05

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

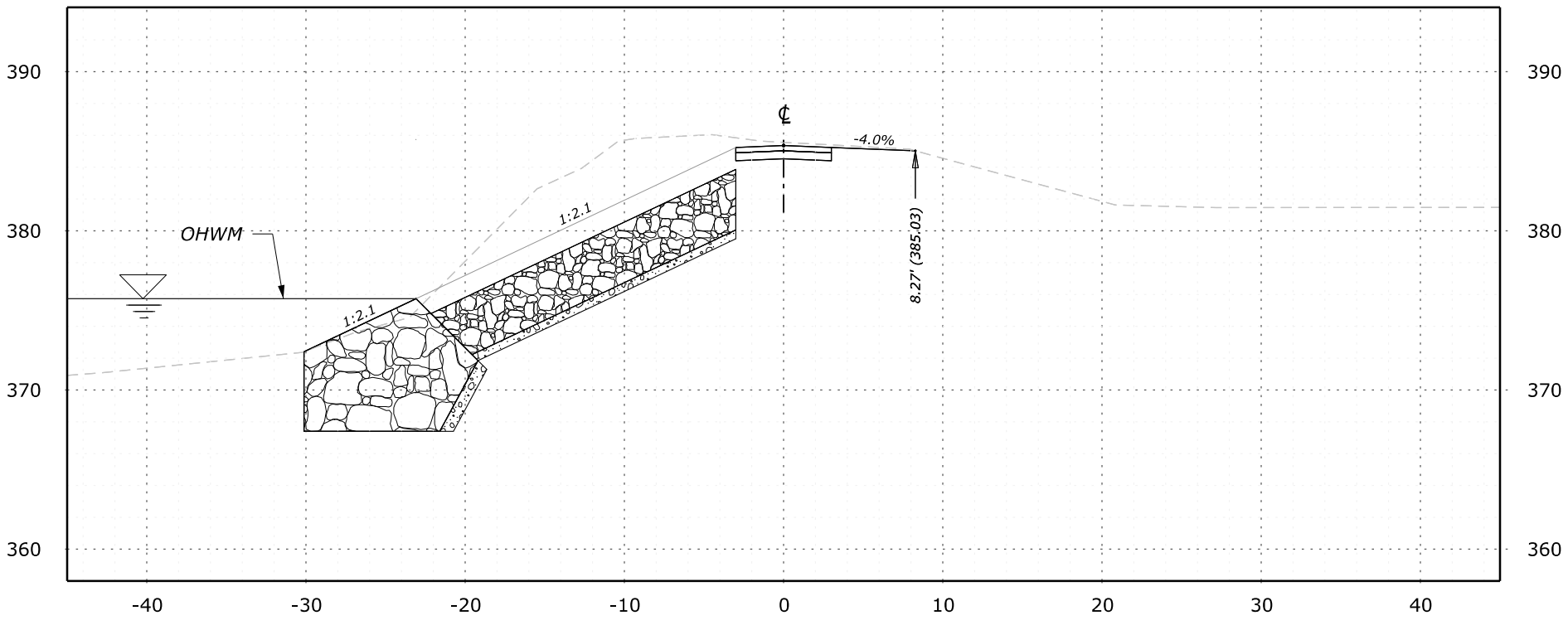
SCHEDULE A: WINDING WATERS TRAIL SITE #324

300+04.05 TO 300+25

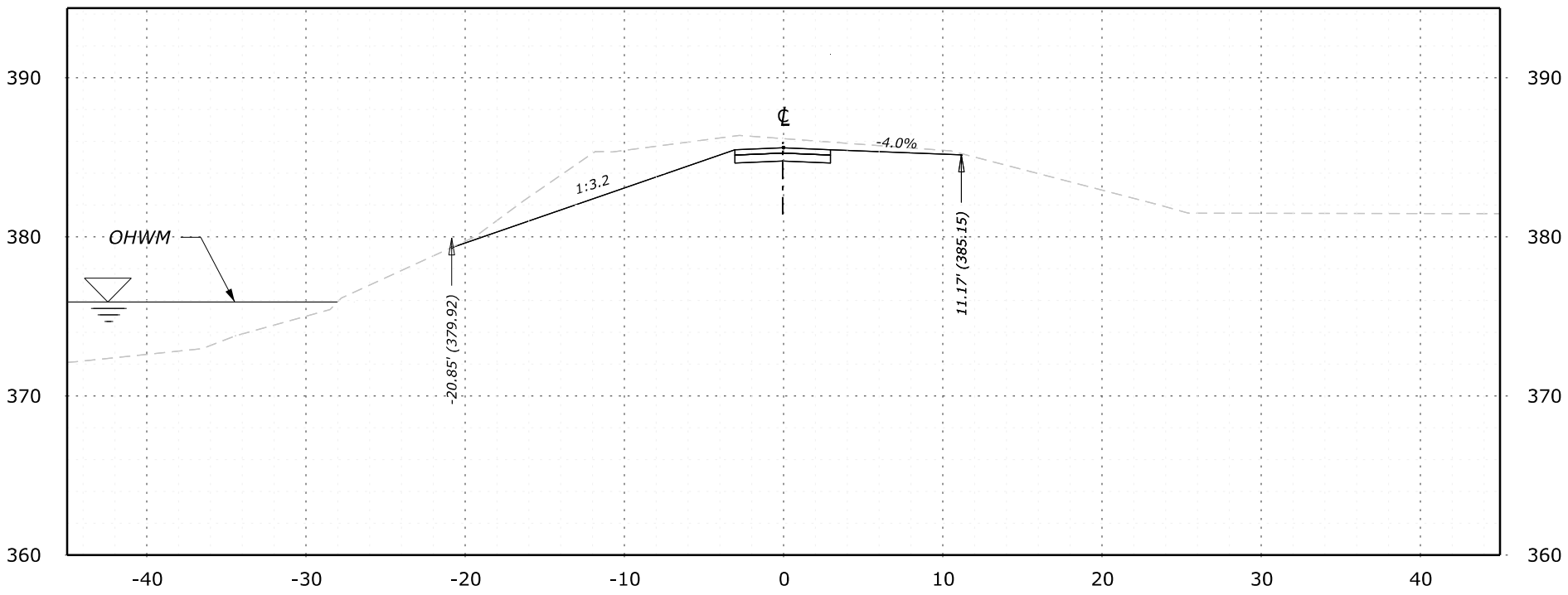
2 September 2026 11:55 AM \\fhfl15\fleserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\V01-V07-NY-WARI_2019-1(1)_xs_Site324.dgn [BL_Site #324 - 300+04.05 [Sheet1]]

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

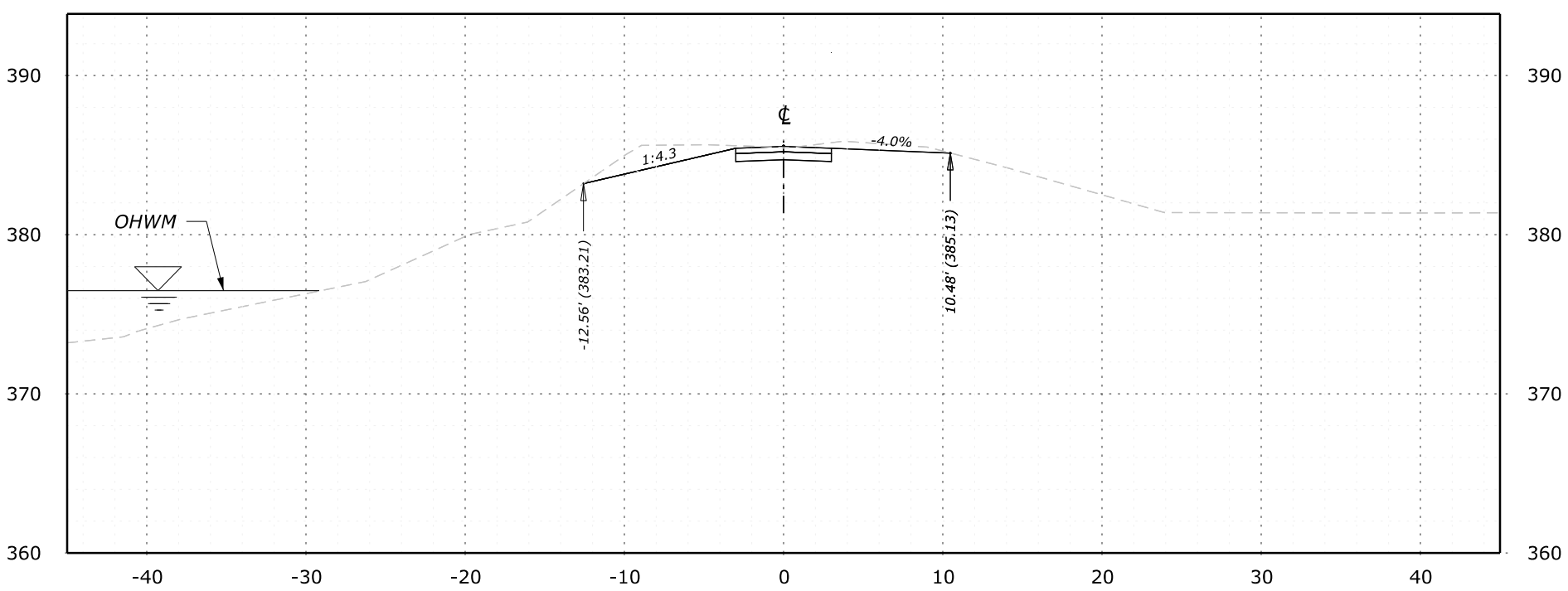
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	V02



301+00.00
Des Grade = 385.36
Subgrade = 385.03
Original Grnd = 385.57



300+75.00
Des Grade = 385.60
Subgrade = 385.26
Original Grnd = 386.20



300+50.00
Des Grade = 385.54
Subgrade = 385.21
Original Grnd = 385.48

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

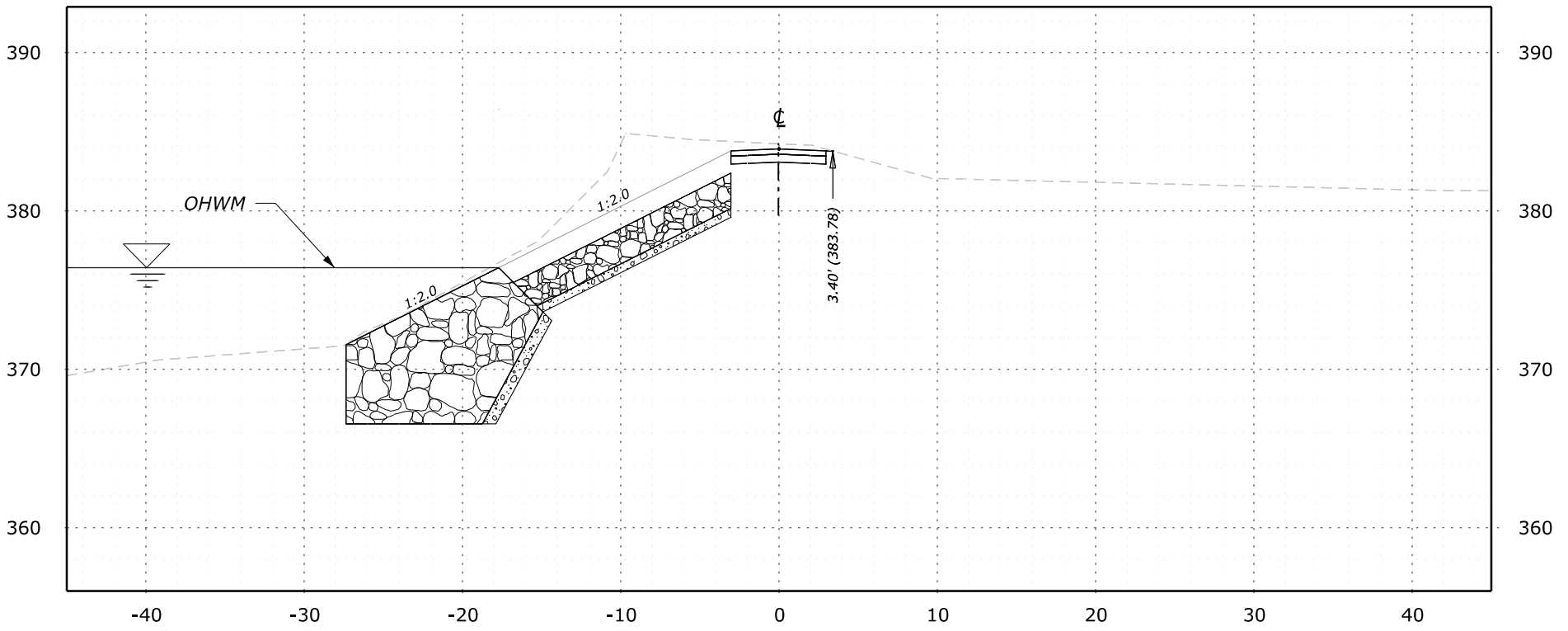
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #324

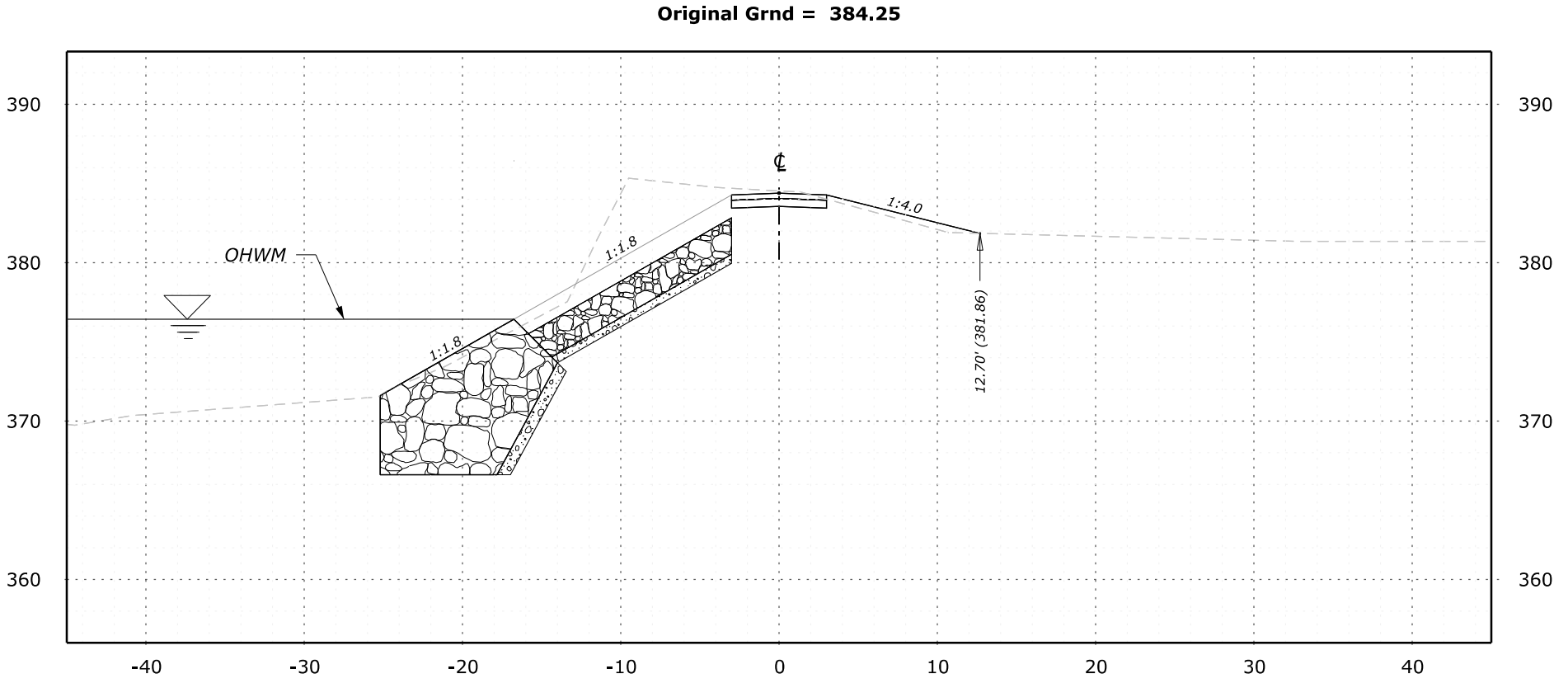
300+50 TO 301+00

2 September 2026 11:55 AM \\fhf15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WAR1\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\V01-V07-NY-WARI_2019-1(1)_xs_Site324.dgn [BL_Site #324 - 300+50.00 [Sheet1]]

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	V03



301+75.00
Des Grade = 383.92
Subgrade = 383.58
Original Grnd = 384.25



301+50.00
Des Grade = 384.40
Subgrade = 384.07
Original Grnd = 384.53

301+25.00
Des Grade = 384.88
Subgrade = 384.55
Original Grnd = 385.04

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

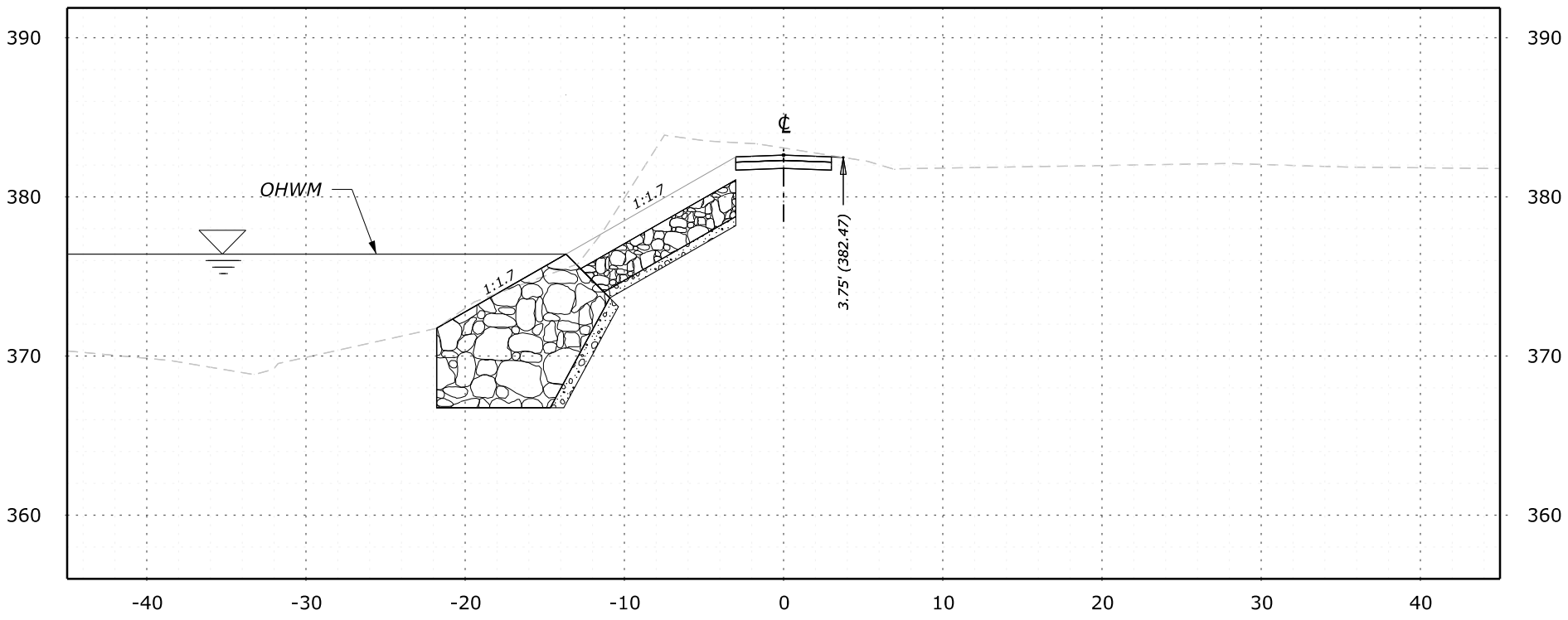
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #324

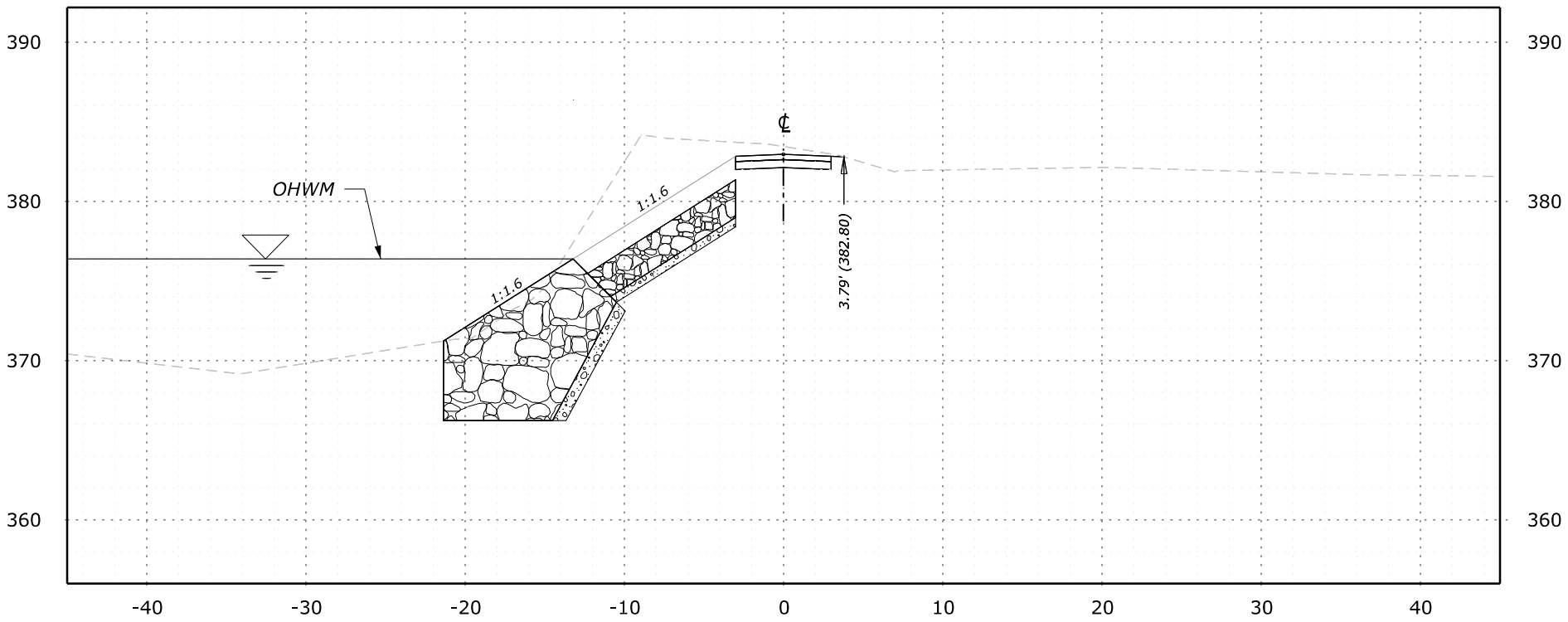
301+25 TO 301+75

2 September 2026 11:55 AM \\fhfl15\fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\V01-V07-NY-WARI_2019-1(1)_xs_Site324.dgn [BL_Site #324 - 301+25.00 [Sheet]]

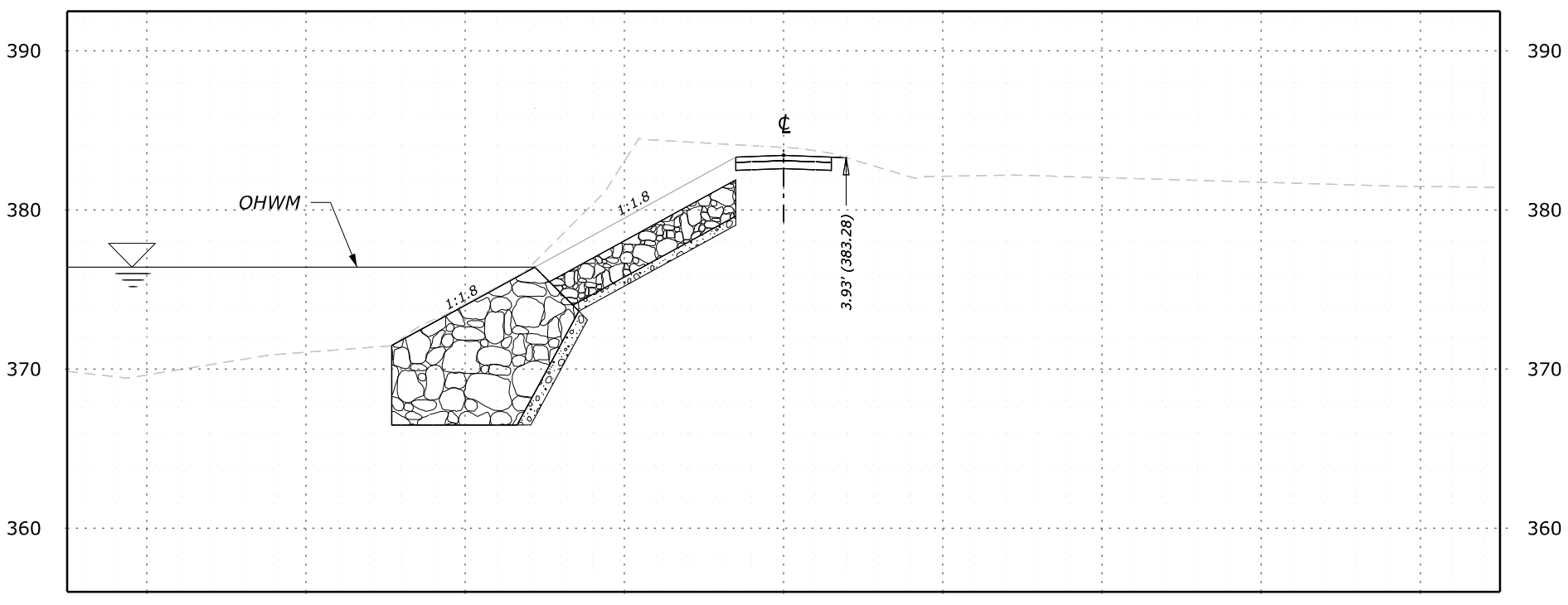
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	V04



302+50.00
Des Grade = 382.62
Subgrade = 382.29
Original Grnd = 383.06



302+25.00
Des Grade = 382.95
Subgrade = 382.62
Original Grnd = 383.44



302+00.00
Des Grade = 383.43
Subgrade = 383.10
Original Grnd = 383.92

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

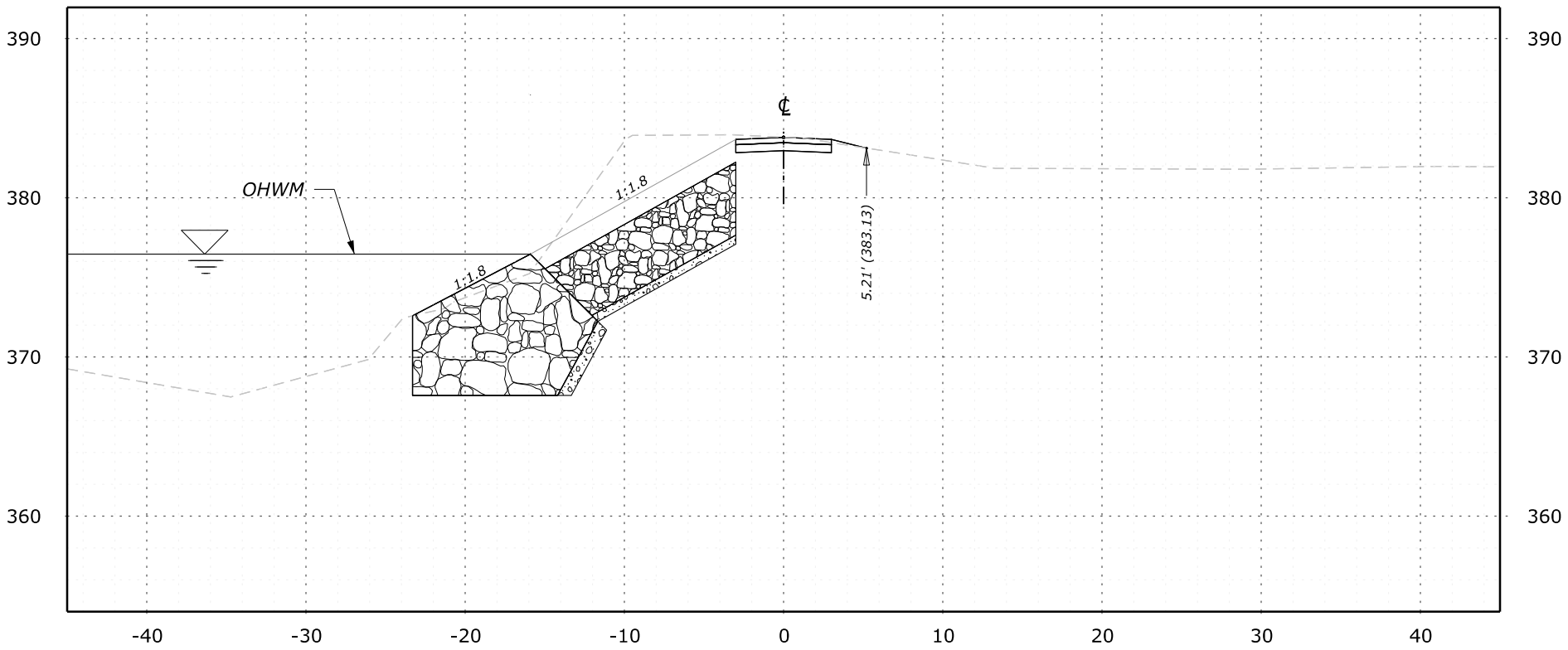
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #324

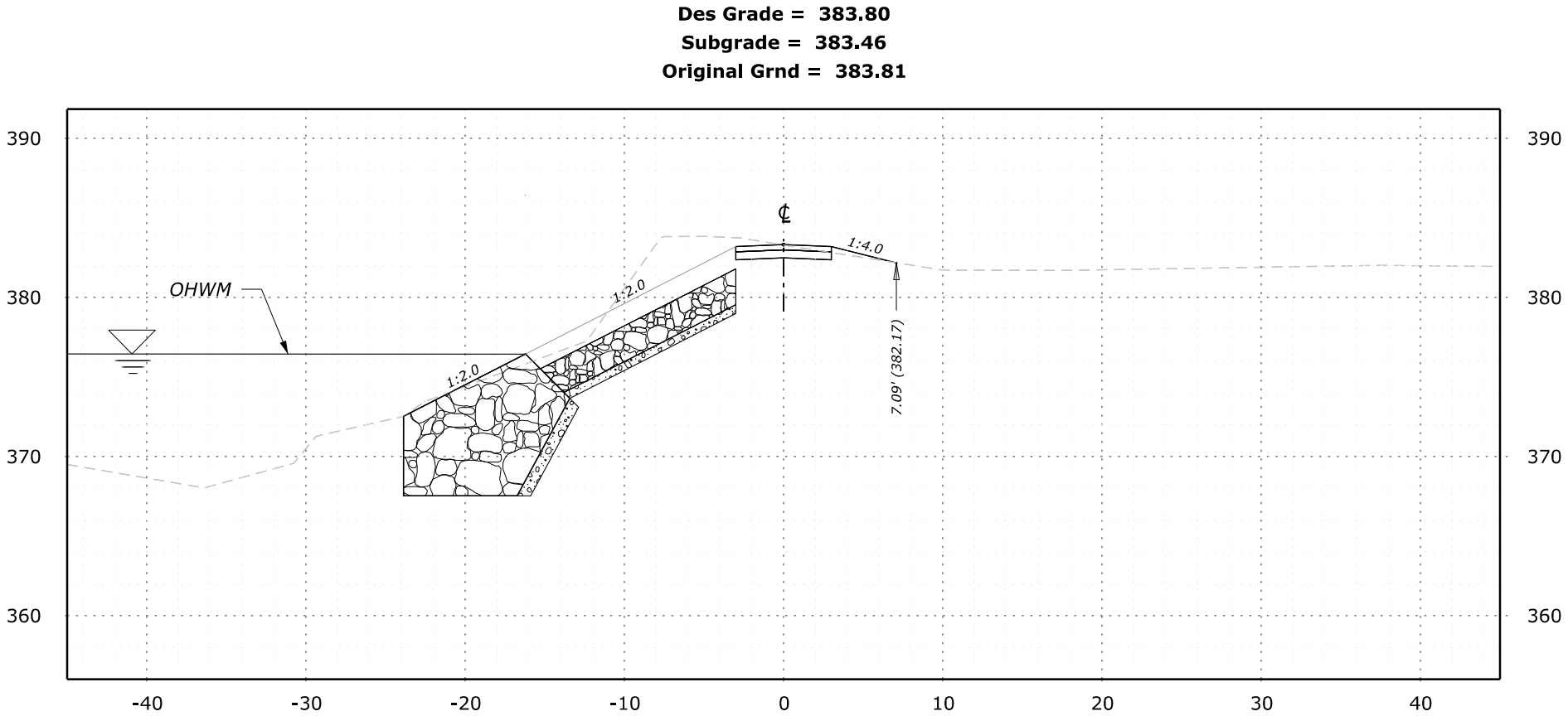
302+00 TO 302+50

2 September 2026 11:55 AM \\fhfl15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\V01-V07-NY-WARI_2019-1(1)_xs_Site324.dgn [BL_Site #324 - 302+00.00 [Sheet]]

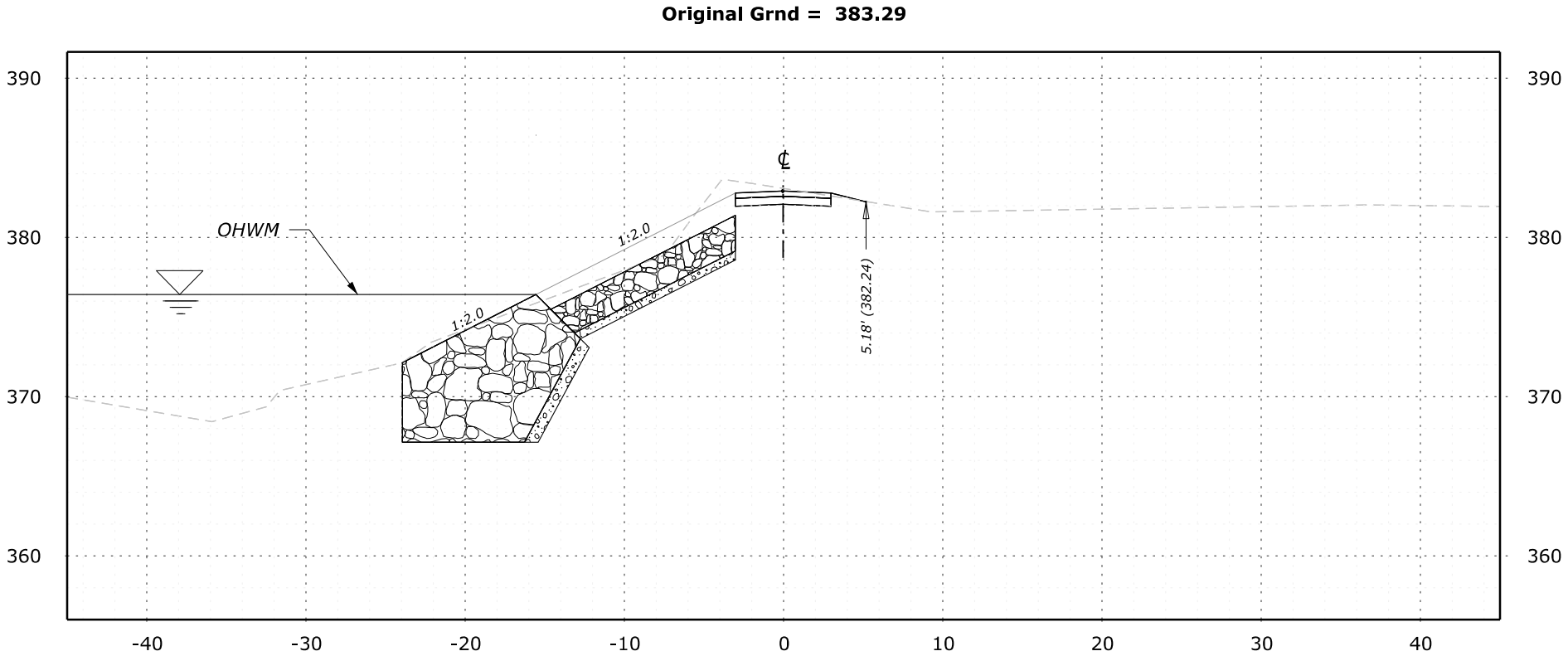
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	V05



303+30.00
Des Grade = 383.80
Subgrade = 383.46
Original Grnd = 383.81



303+00.00
Des Grade = 383.32
Subgrade = 382.98
Original Grnd = 383.29



302+75.00
Des Grade = 382.91
Subgrade = 382.58
Original Grnd = 383.07

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

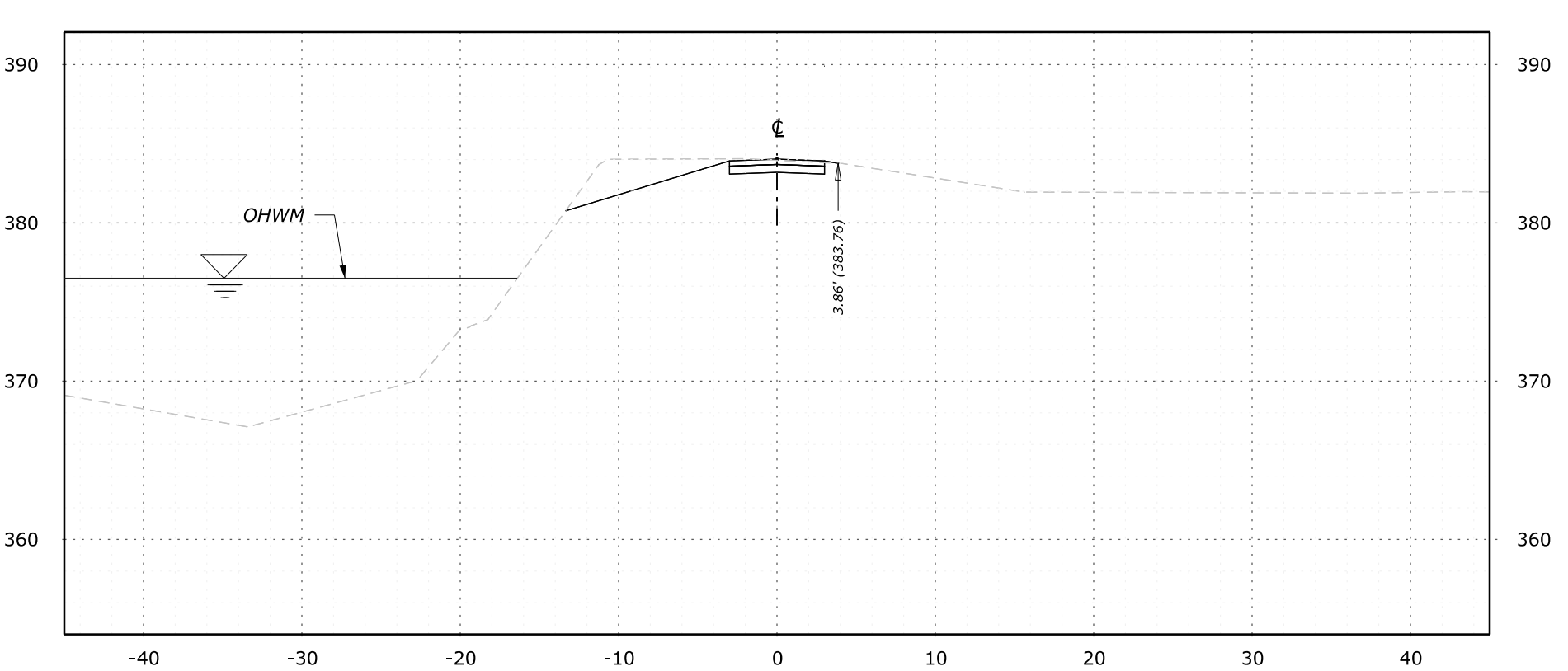
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #324

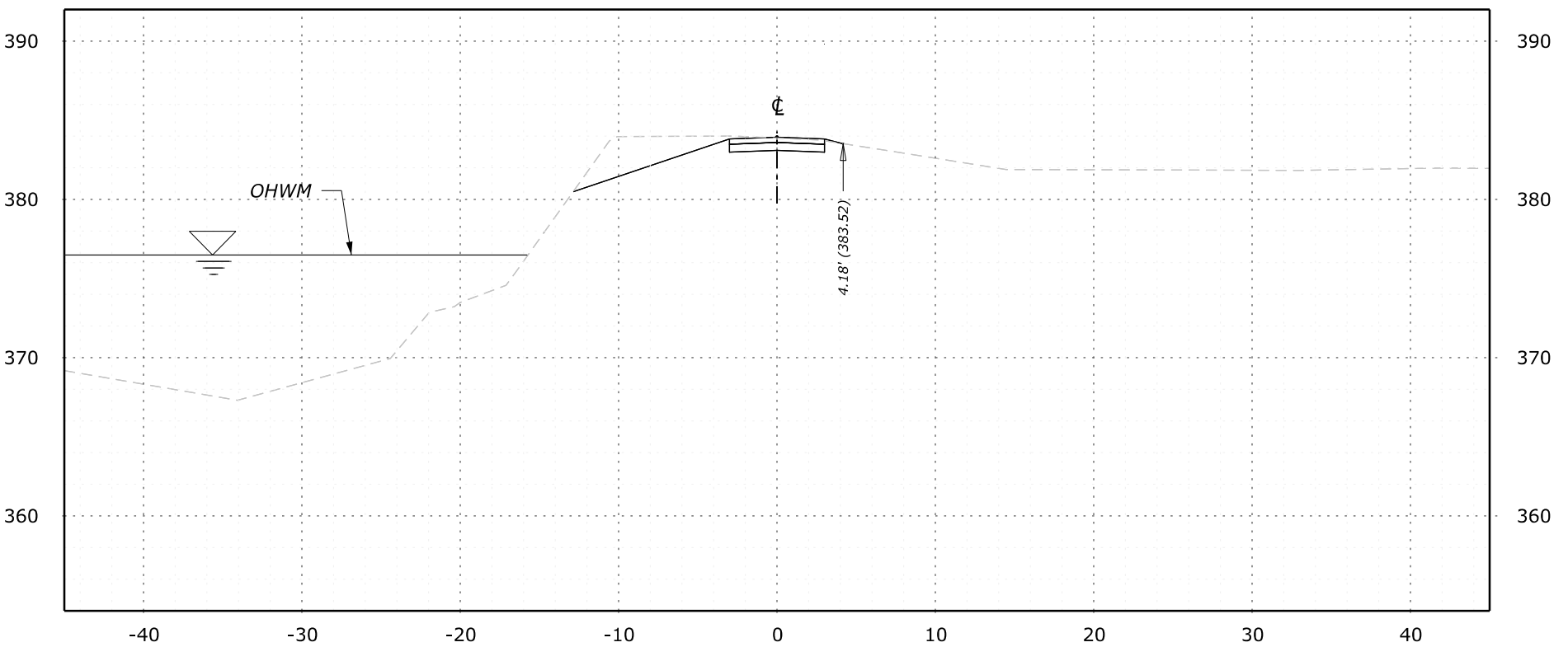
302+75 TO 303+30

2 September 2026 11:55 AM \\fhf15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WAR1\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\V01-V07-NY-WARI_2019-1(1)_xs_Site324.dgn [BL_Site #324 - 302+75.00 [Sheet]]

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	V06



303+50.00
Des Grade = 384.03
Subgrade = 383.69
Original Grnd = 383.95



303+40.00
Des Grade = 383.93
Subgrade = 383.60
Original Grnd = 383.88

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #324

303+40 TO 303+50

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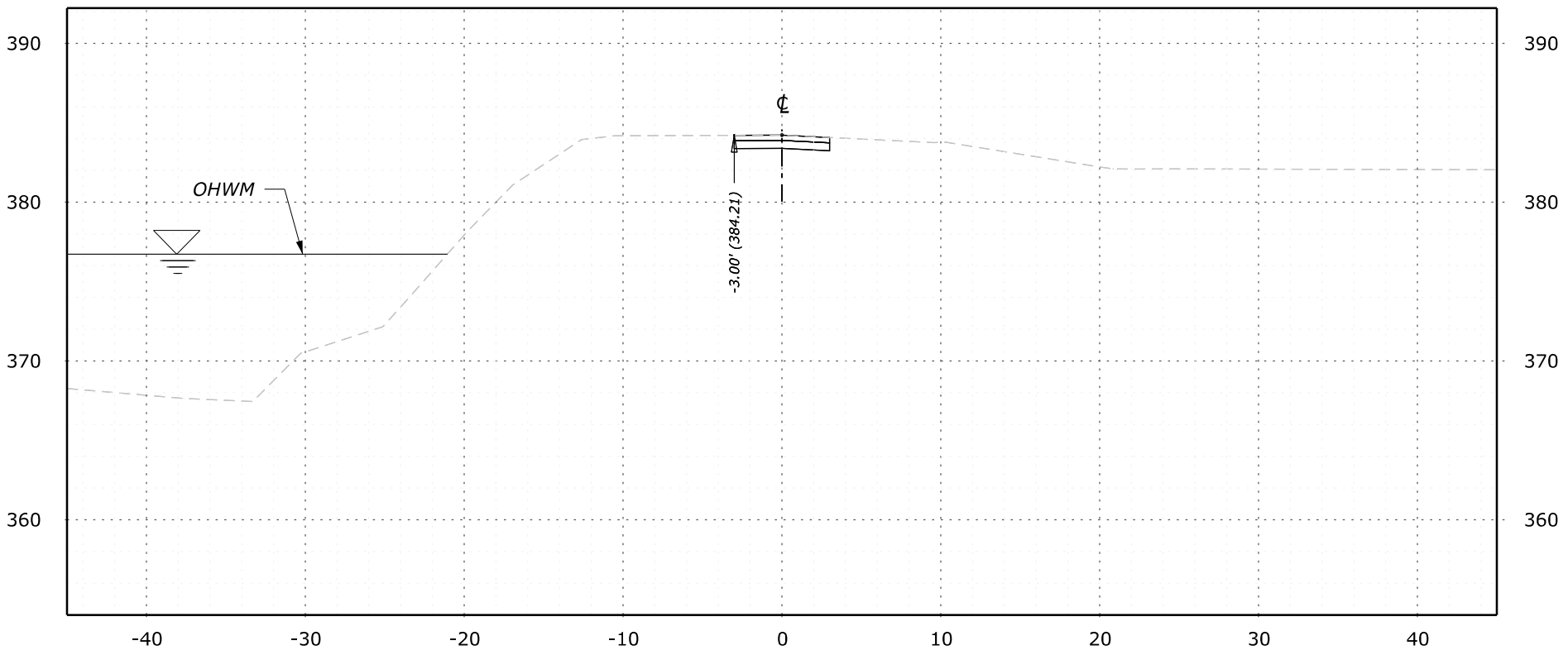
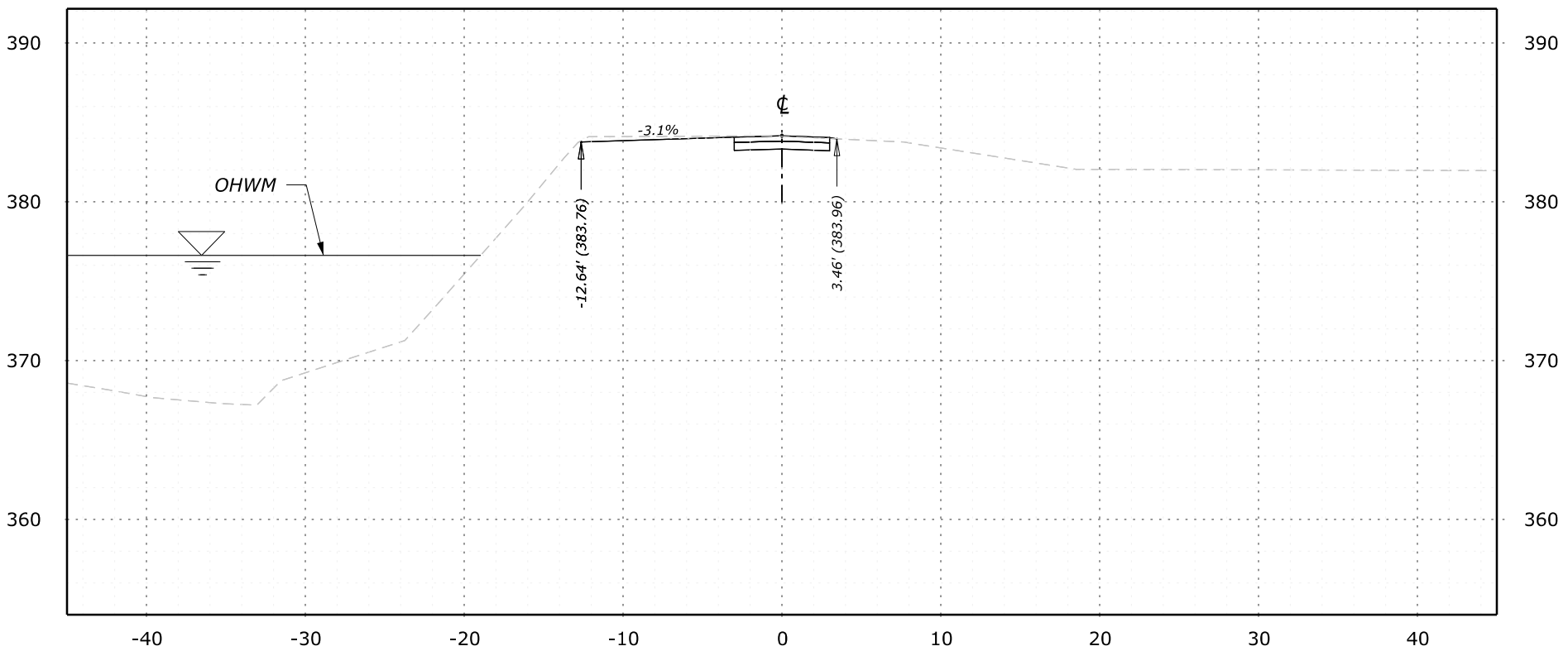
2 September 2026 11:55 AM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	V07



303+90.00
Des Grade = 384.22
Subgrade = 383.89
Original Grnd = 384.22



303+75.00
Des Grade = 384.15
Subgrade = 383.82
Original Grnd = 384.12

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

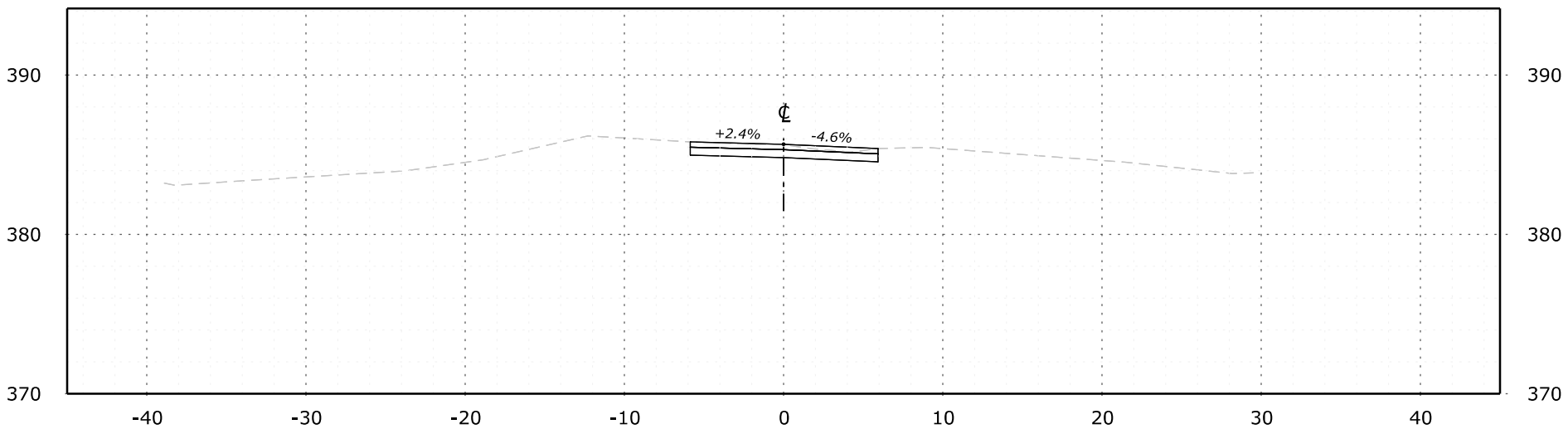
CROSS SECTIONS

SCHEDULE A: WINDING WATERS TRAIL SITE #324

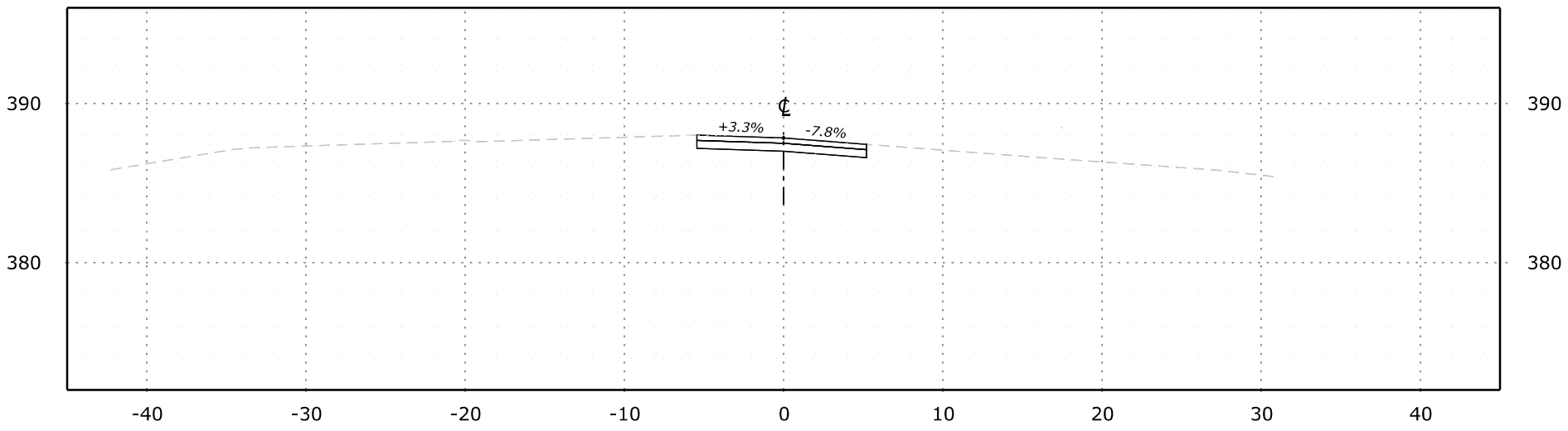
303+75 TO 303+90

2 September 2026 11:55 AM \\fhfl15\fleserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\(100%) NY ERFO CADD Files\V01-V07-NY-WARI_2019-1(1)_xs_SIte324.dgn [BL_SIte #324 - 303+75.00 [Sheet]]

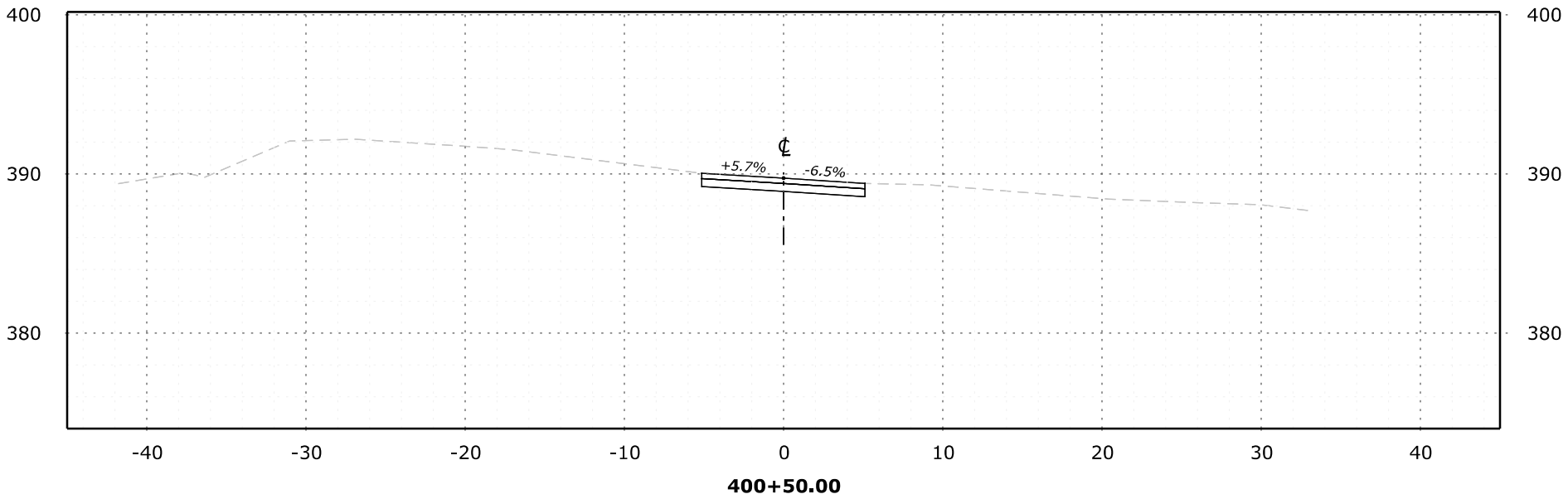
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W01



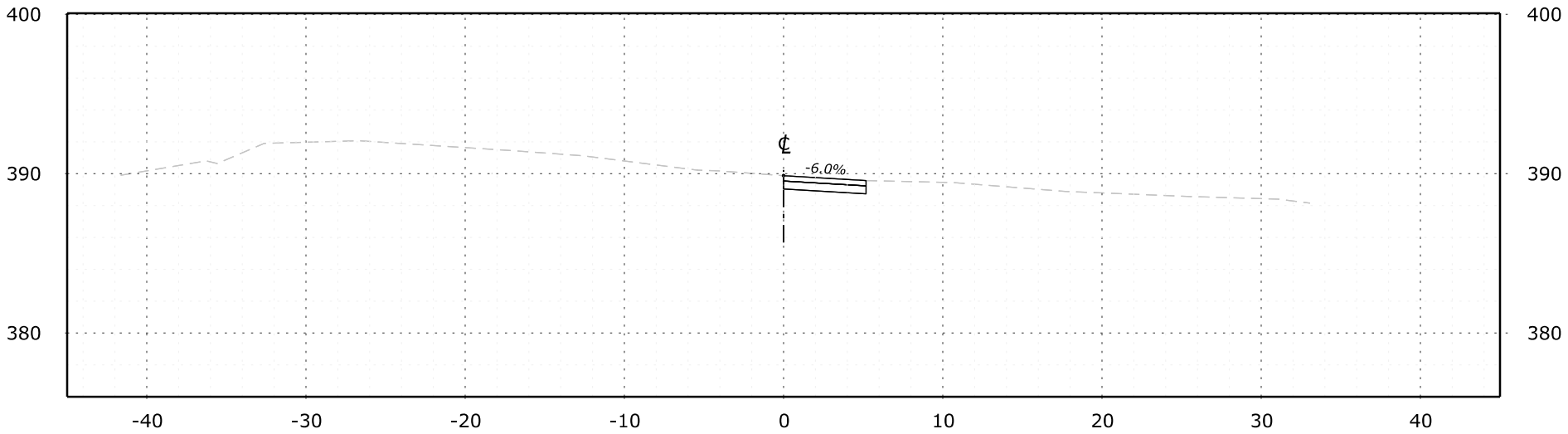
401+50.00
Des Grade = 385.66
Subgrade = 385.33
Original Grnd = 385.63



401+00.00
Des Grade = 387.83
Subgrade = 387.50
Original Grnd = 387.83



400+50.00
Des Grade = 389.74
Subgrade = 389.40
Original Grnd = 389.74



400+40.00
Des Grade = 389.88
Subgrade = 389.54
Original Grnd = 389.88

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

400+40 TO 401+50

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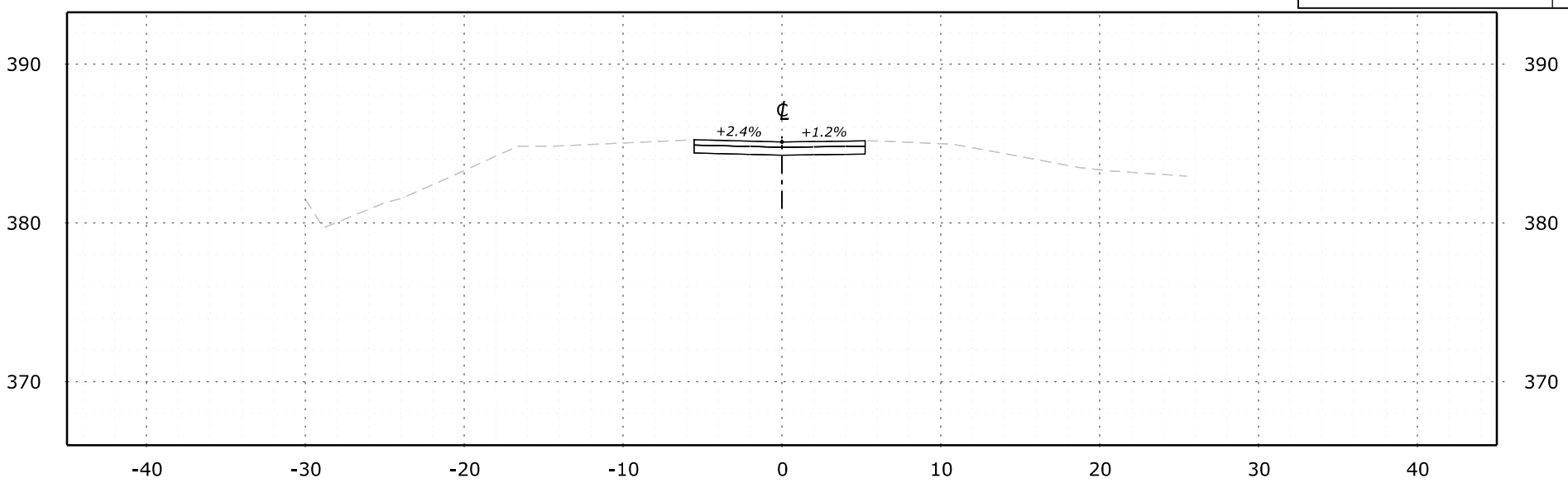
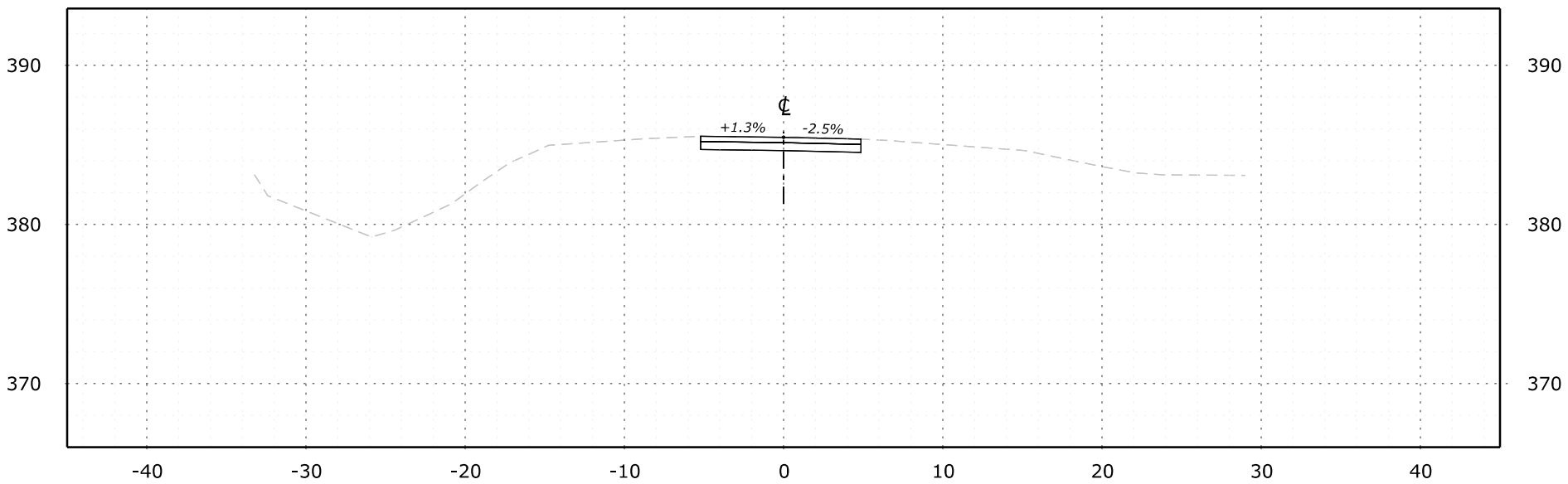
3 August 2026 5:11 PM

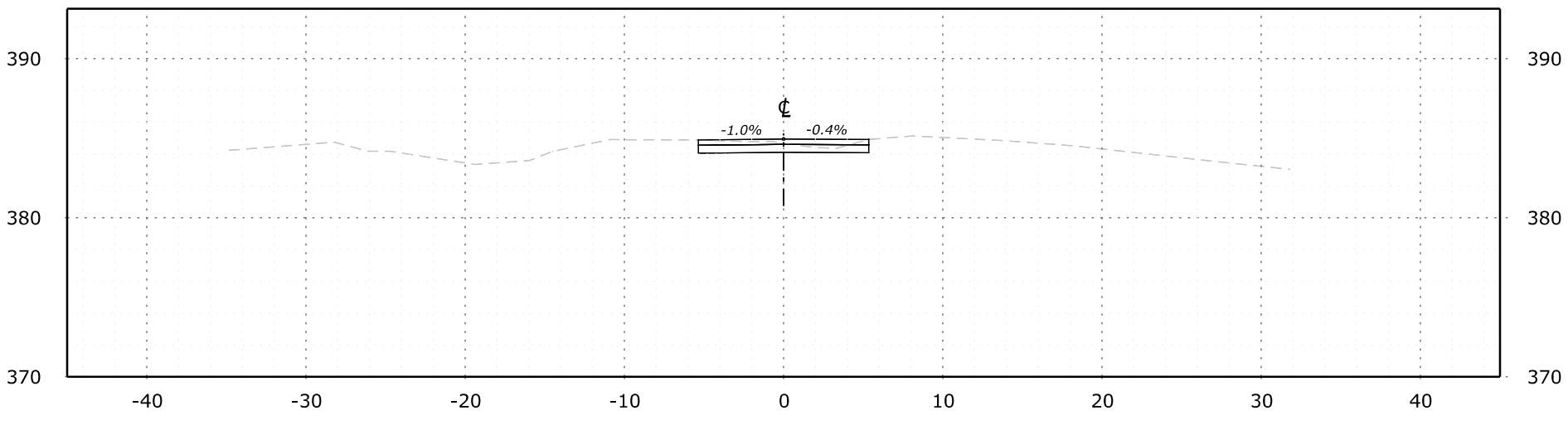
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W02



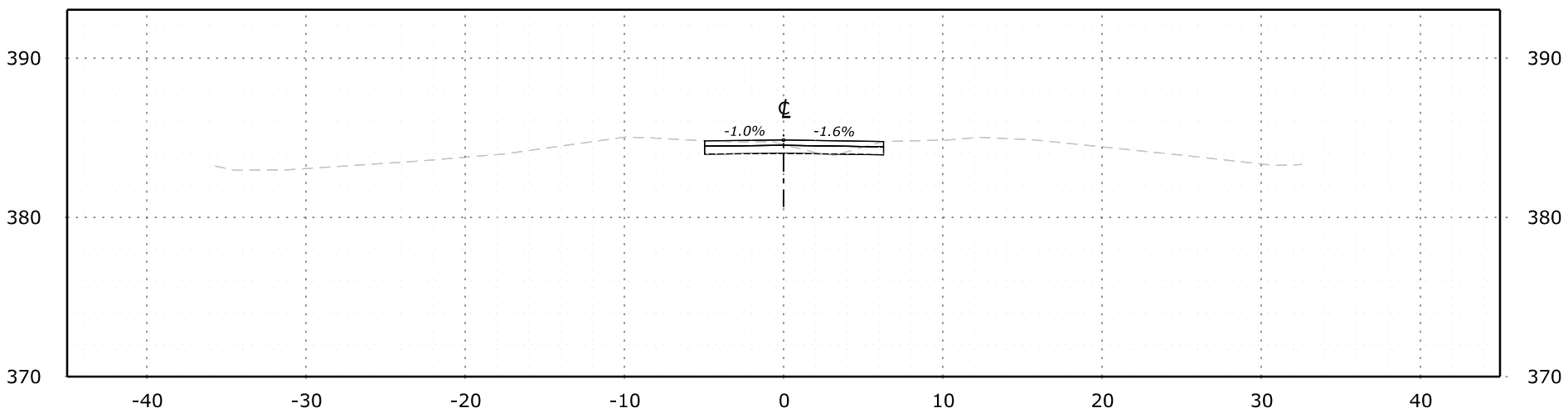
403+50.00
Des Grade = 385.11
Subgrade = 384.77
Original Grnd = 385.10



403+00.00
Des Grade = 385.48
Subgrade = 385.14
Original Grnd = 385.49



402+50.00
Des Grade = 384.95
Subgrade = 384.62
Original Grnd = 384.69



402+00.00
Des Grade = 384.85
Subgrade = 384.52
Original Grnd = 384.57

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

402+00 TO 403+50

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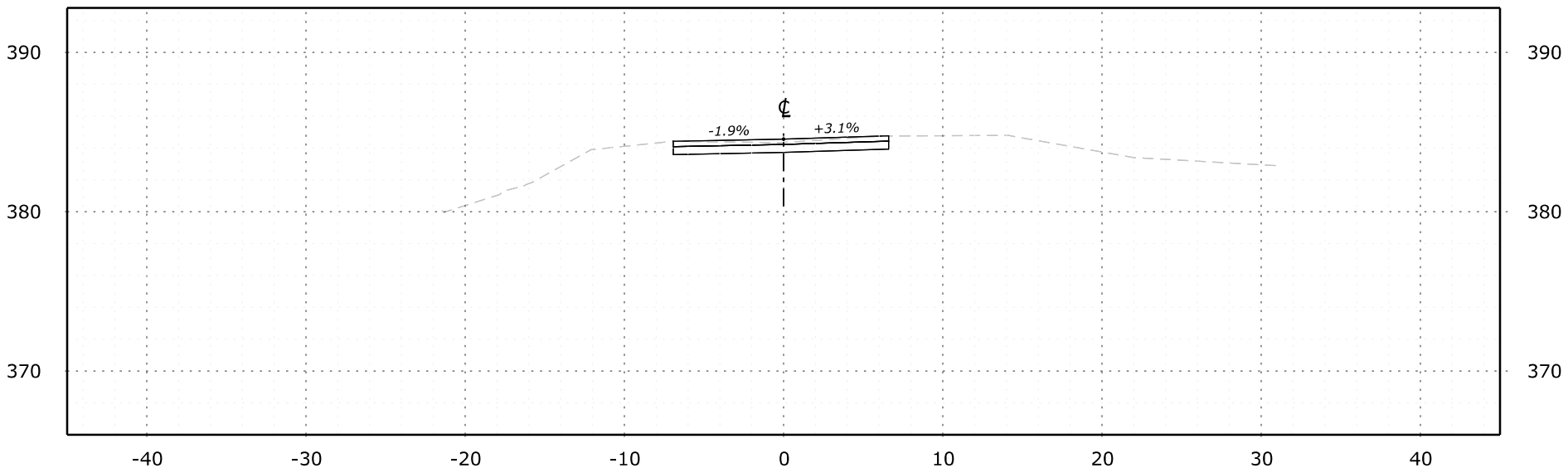
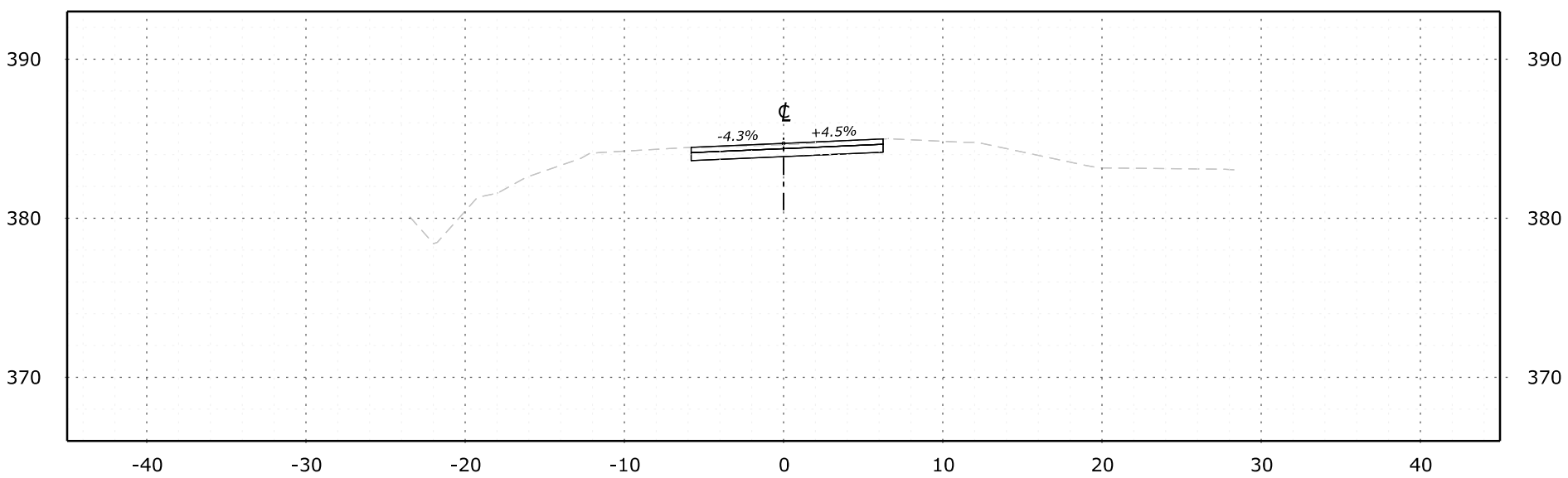
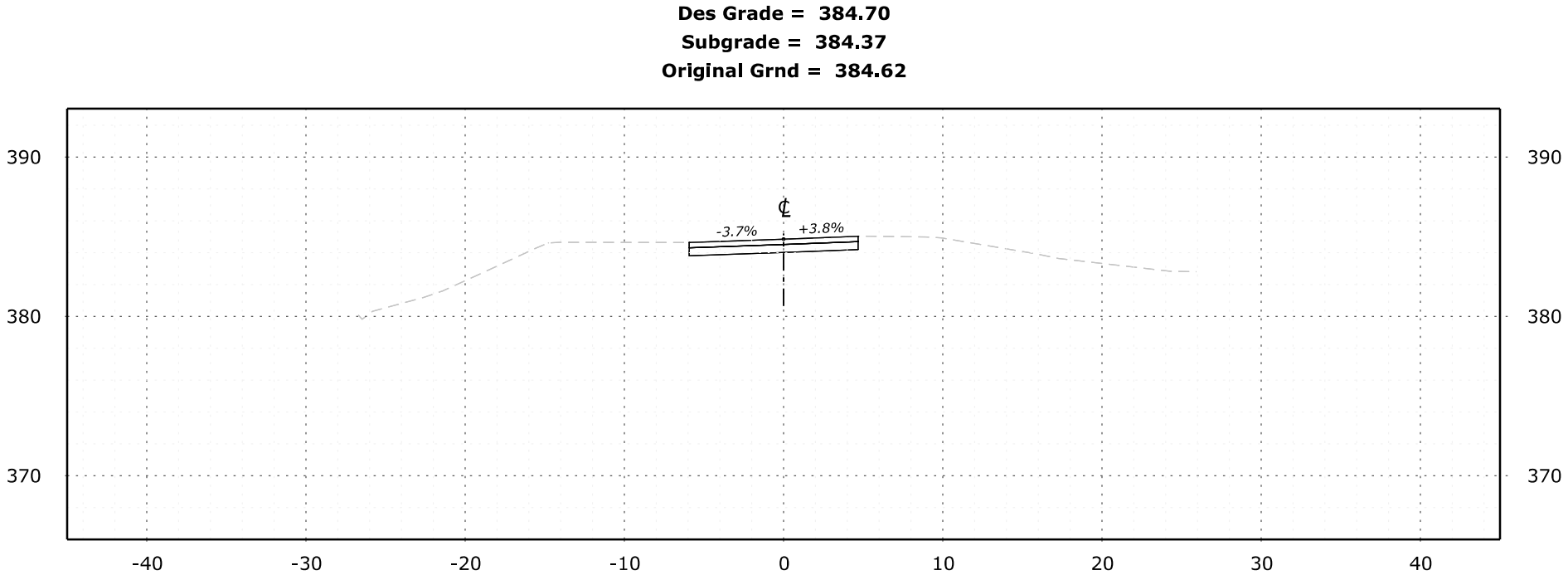
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W03



405+00.00
Des Grade = 384.55
Subgrade = 384.22
Original Grnd = 384.35



404+50.00
Des Grade = 384.70
Subgrade = 384.37
Original Grnd = 384.62



404+00.00
Des Grade = 384.85
Subgrade = 384.52
Original Grnd = 384.82

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

404+00 TO 405+00

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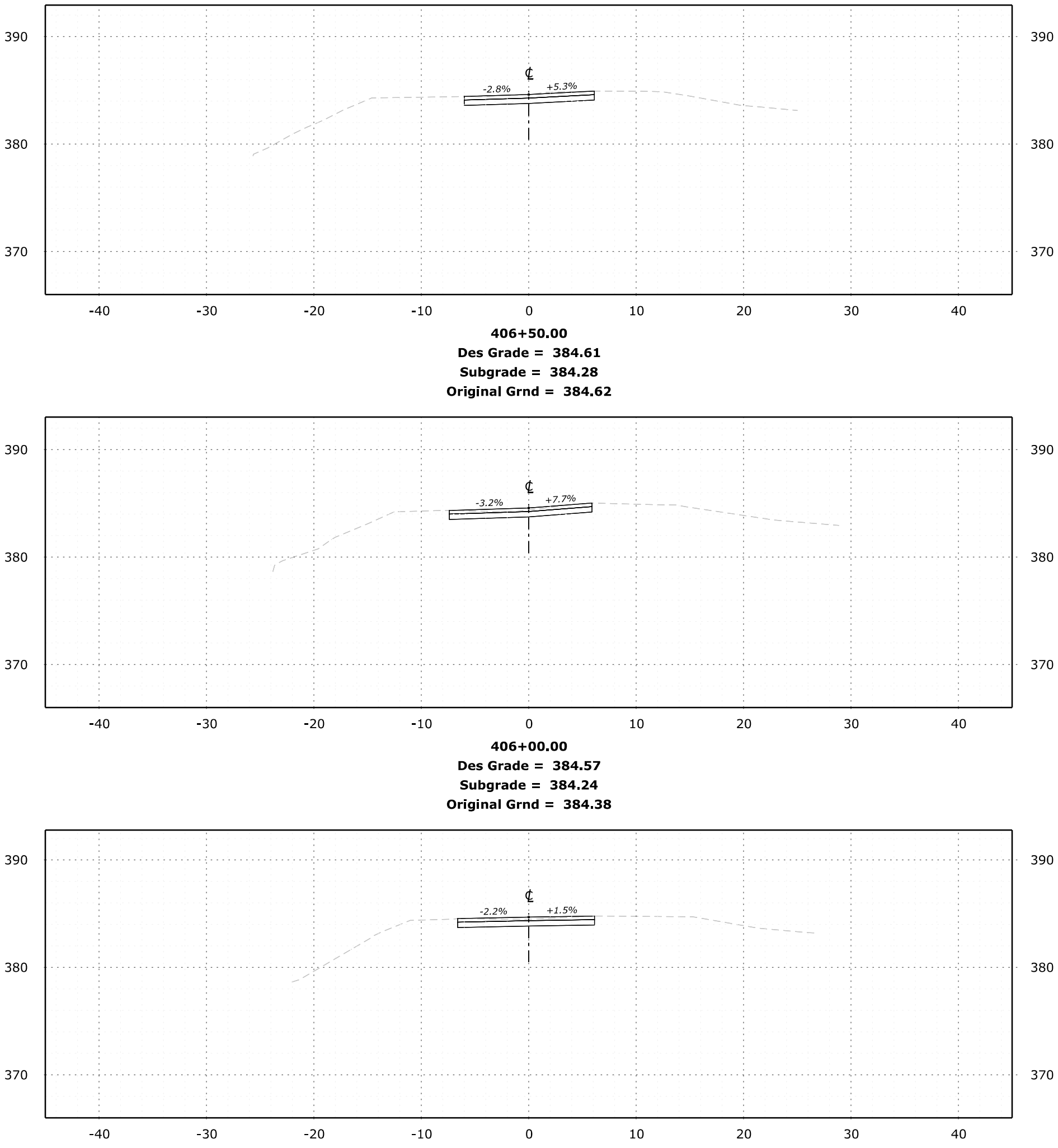
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W04



406+50.00
Des Grade = 384.61
Subgrade = 384.28
Original Grnd = 384.62

406+00.00
Des Grade = 384.57
Subgrade = 384.24
Original Grnd = 384.38

405+50.00
Des Grade = 384.68
Subgrade = 384.35
Original Grnd = 384.47

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

405+50 TO 406+50

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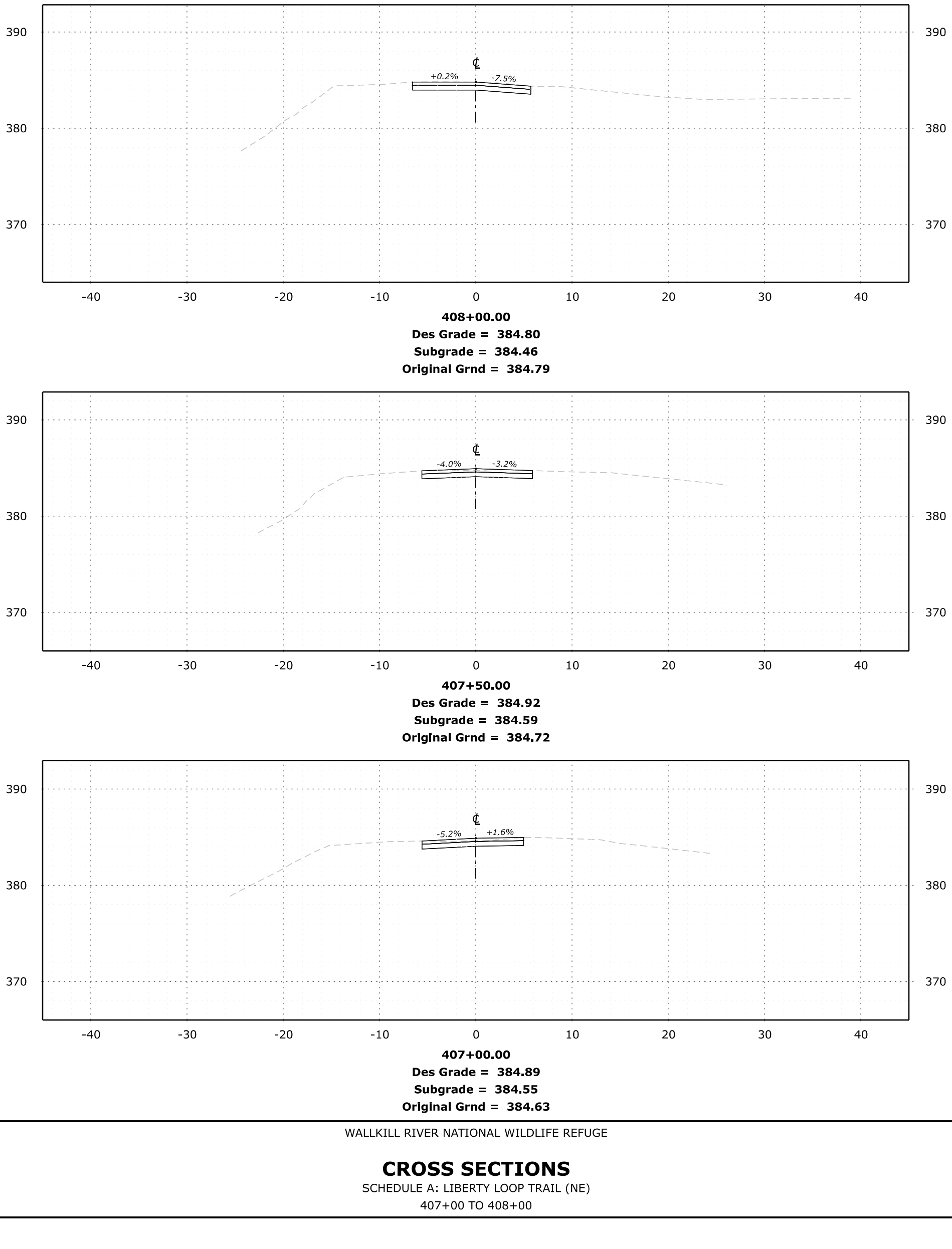
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W05



408+00.00
Des Grade = 384.80
Subgrade = 384.46
Original Grnd = 384.79

407+50.00
Des Grade = 384.92
Subgrade = 384.59
Original Grnd = 384.72

407+00.00
Des Grade = 384.89
Subgrade = 384.55
Original Grnd = 384.63

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

407+00 TO 408+00

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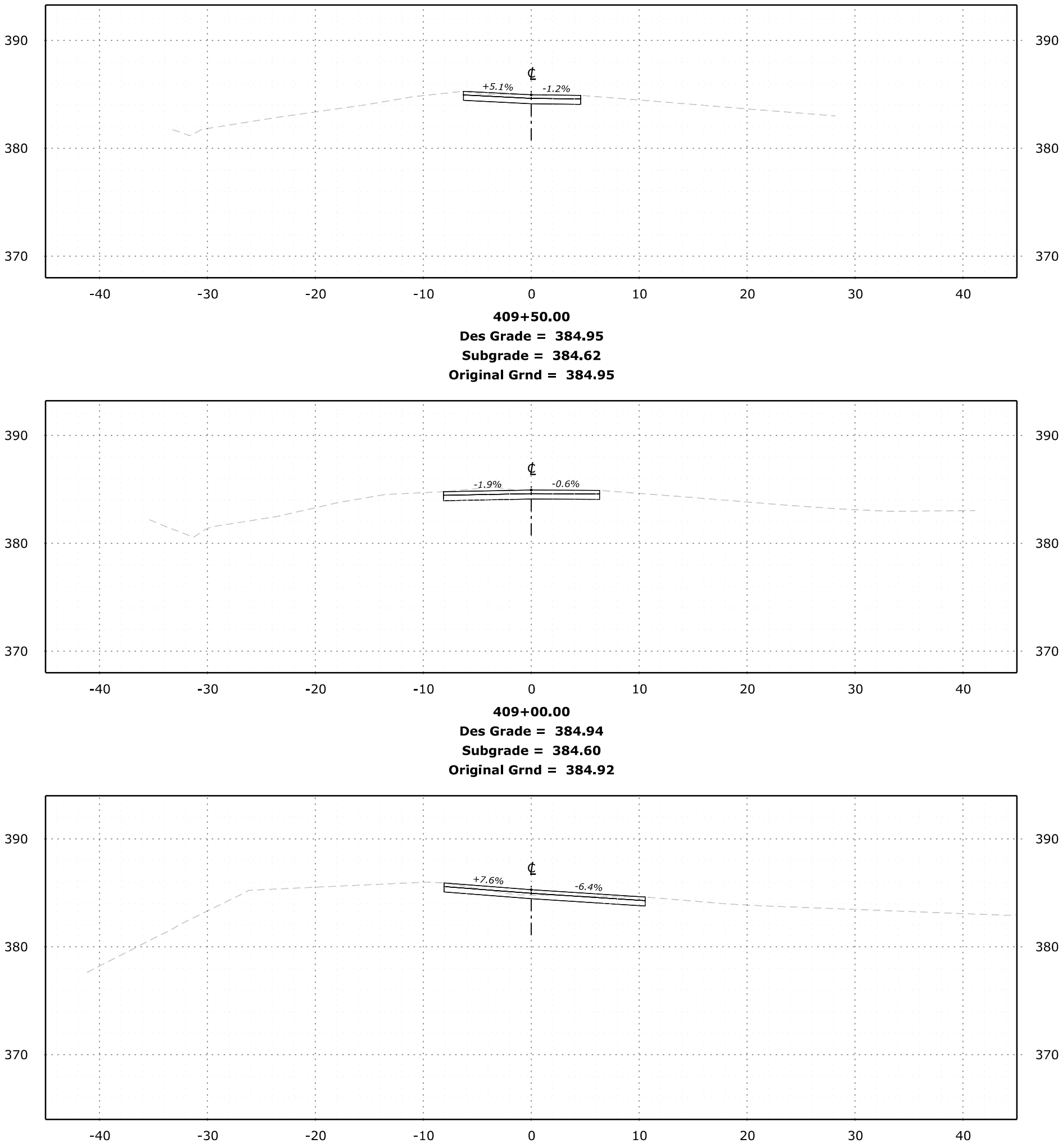
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W06



409+50.00
Des Grade = 384.95
Subgrade = 384.62
Original Grnd = 384.95

409+00.00
Des Grade = 384.94
Subgrade = 384.60
Original Grnd = 384.92

408+50.00
Des Grade = 385.29
Subgrade = 384.96
Original Grnd = 384.79

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

408+50 TO 409+50

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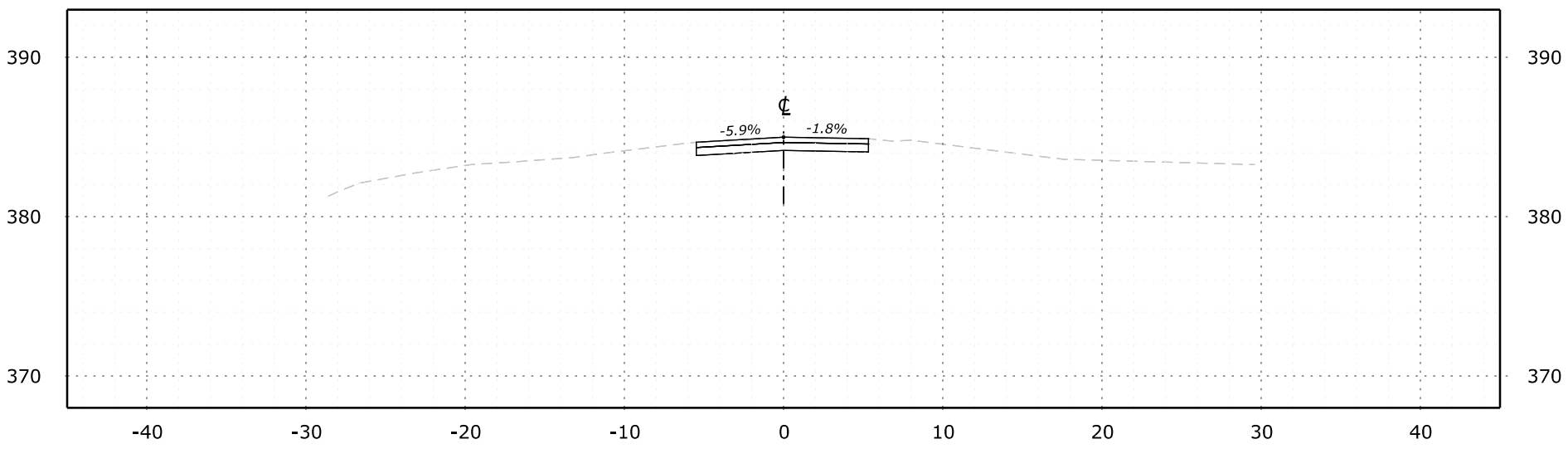
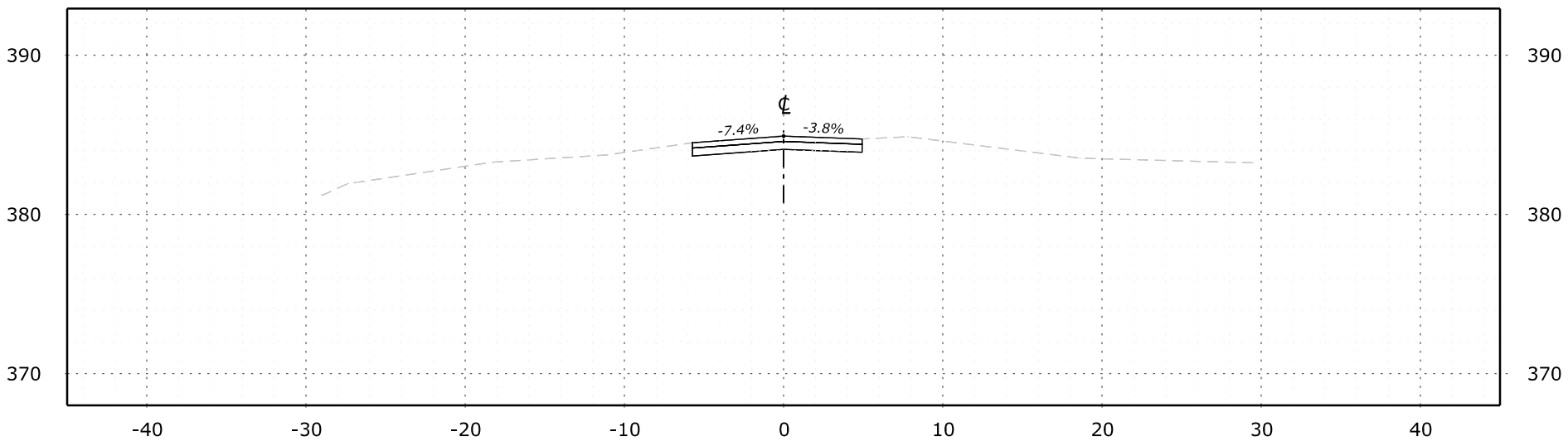
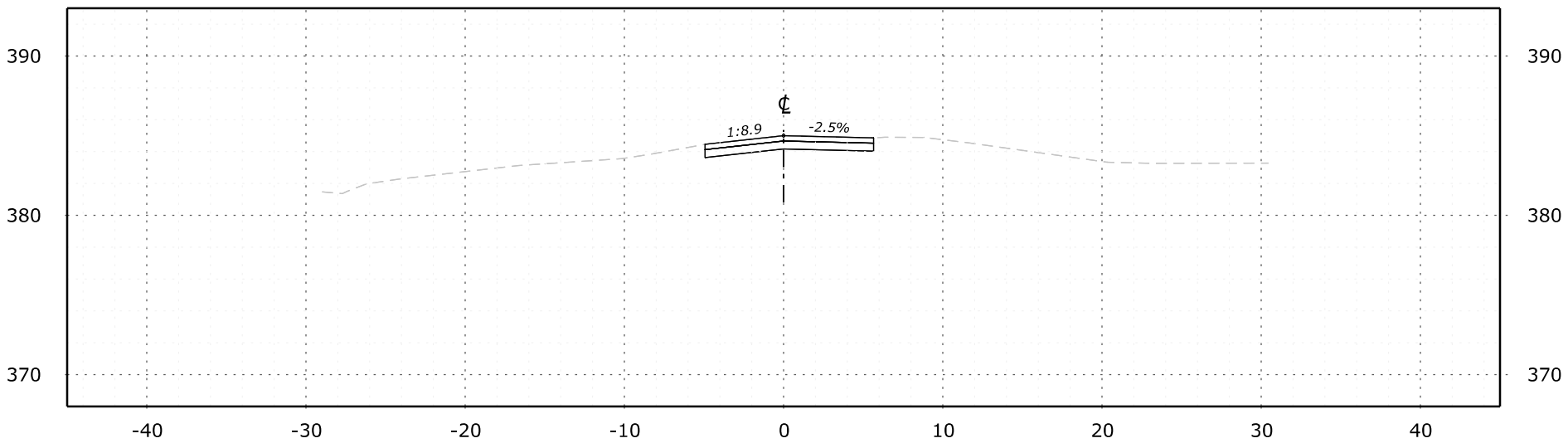
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W07



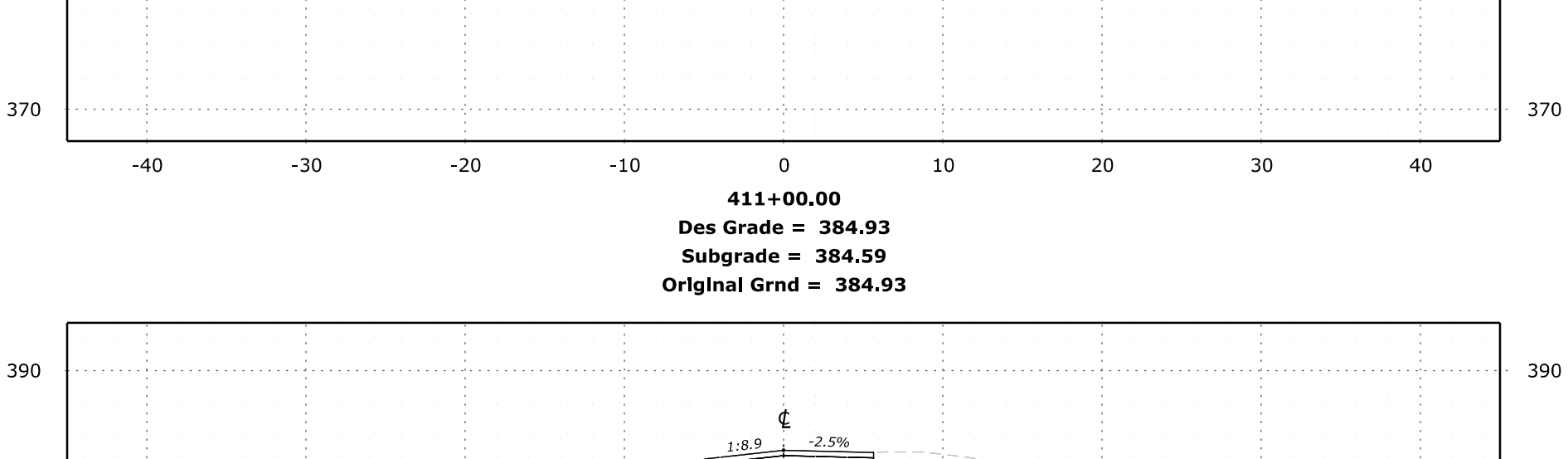
411+50.00
Des Grade = 384.99
Subgrade = 384.66
Original Grnd = 384.99



411+00.00
Des Grade = 384.93
Subgrade = 384.59
Orlginal Grnd = 384.93



410+50.00
Des Grade = 385.00
Subgrade = 384.67
Original Grnd = 384.98



410+00.00
Des Grade = 384.99
Subgrade = 384.66
Original Grnd = 384.99

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

410+00 TO 411+50

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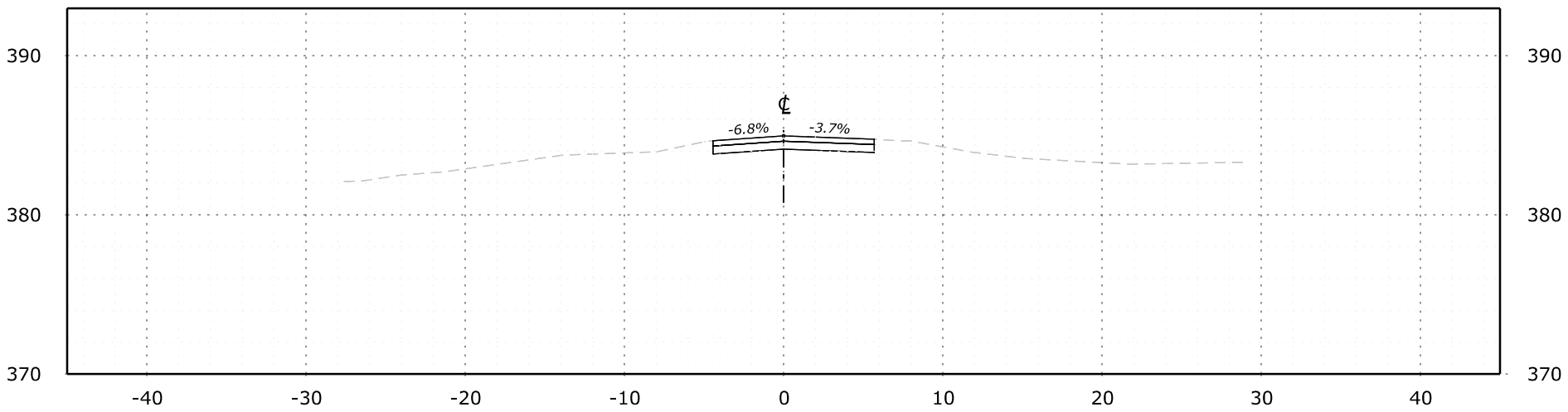
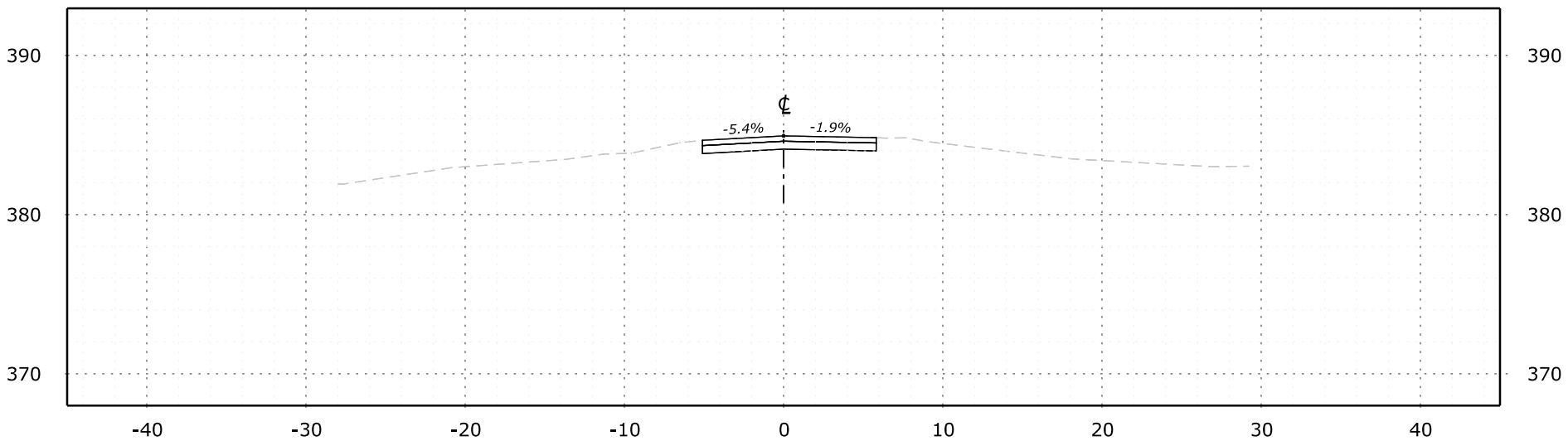
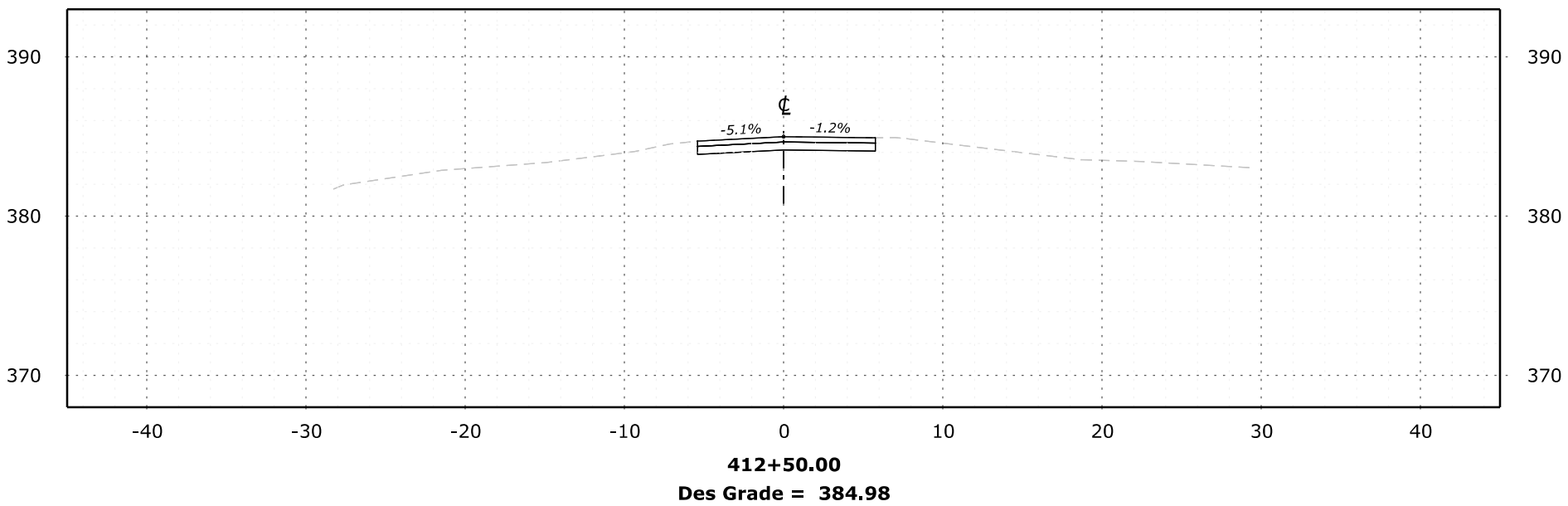
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W08



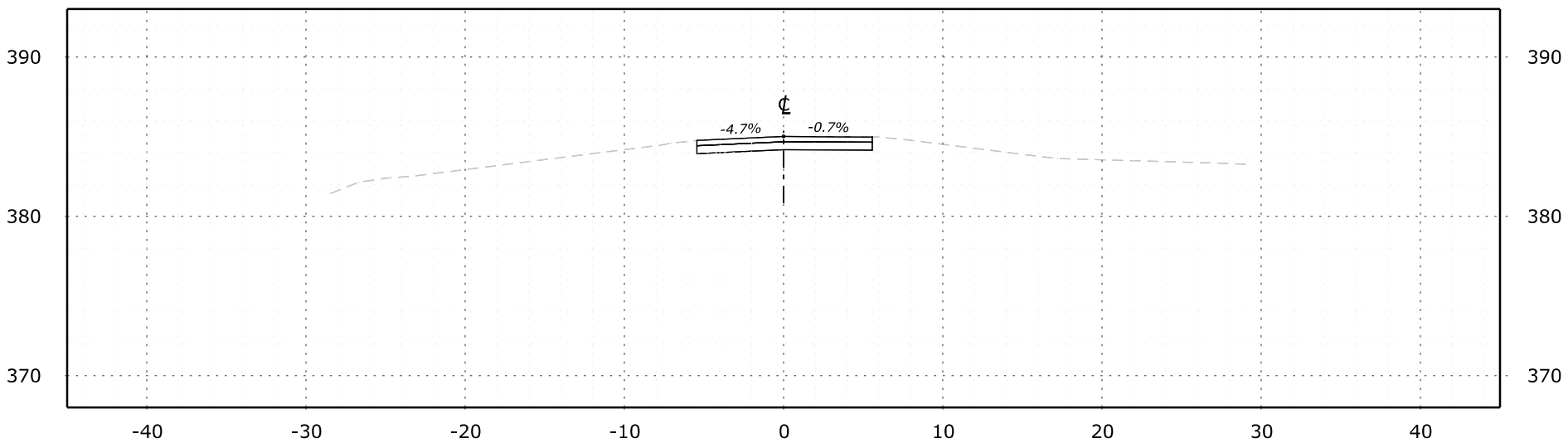
413+50.00
Des Grade = 384.96
Subgrade = 384.62
Original Grnd = 384.95



413+00.00
Des Grade = 384.95
Subgrade = 384.61
Original Grnd = 384.96



412+50.00
Des Grade = 384.98
Subgrade = 384.65
Original Grnd = 384.98



412+00.00
Des Grade = 385.02
Subgrade = 384.68
Original Grnd = 385.02

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

412+00 TO 413+50

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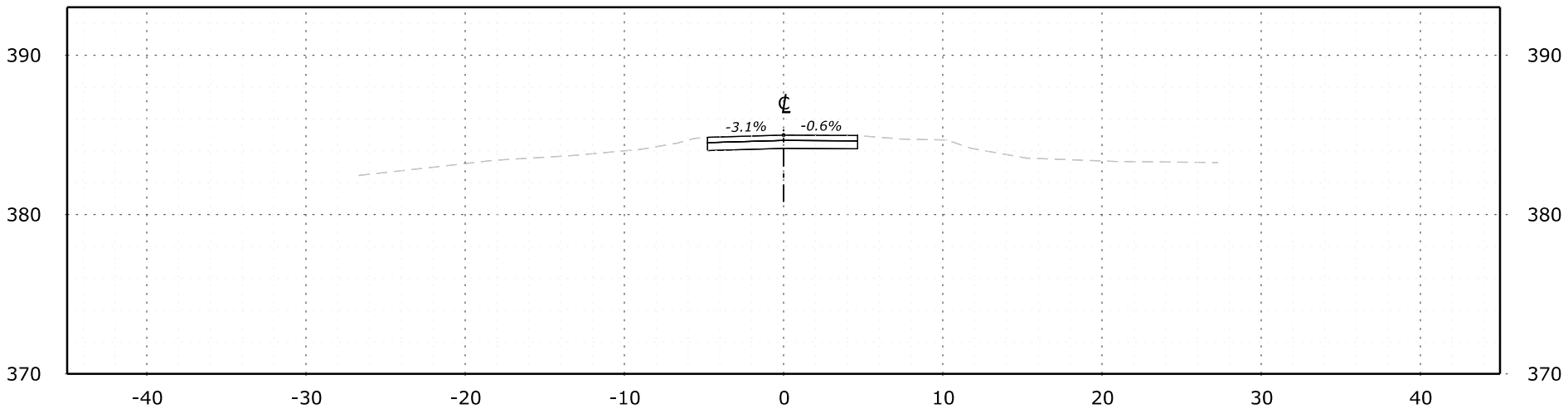
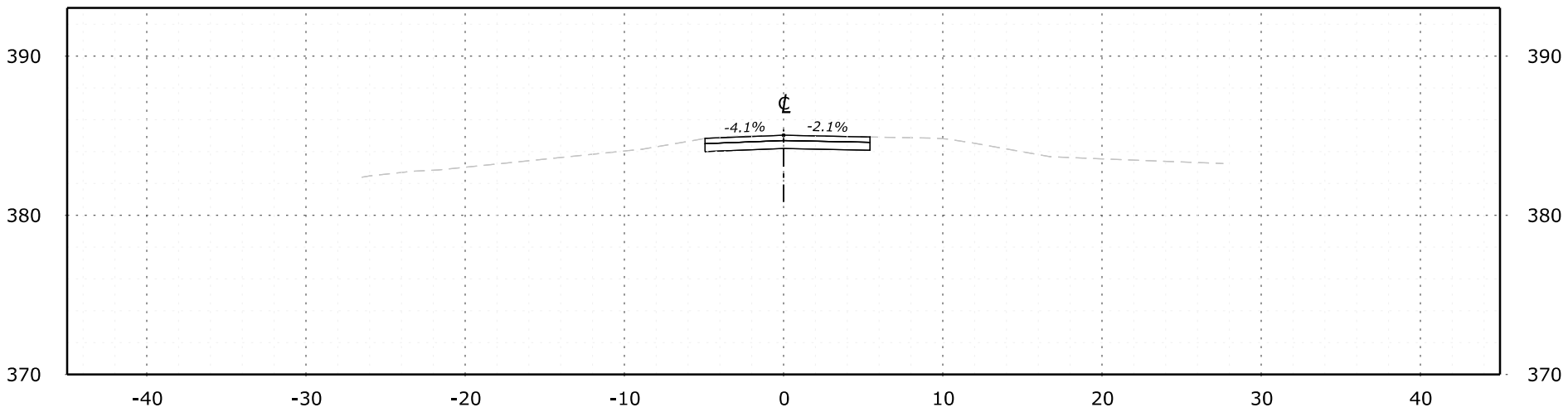
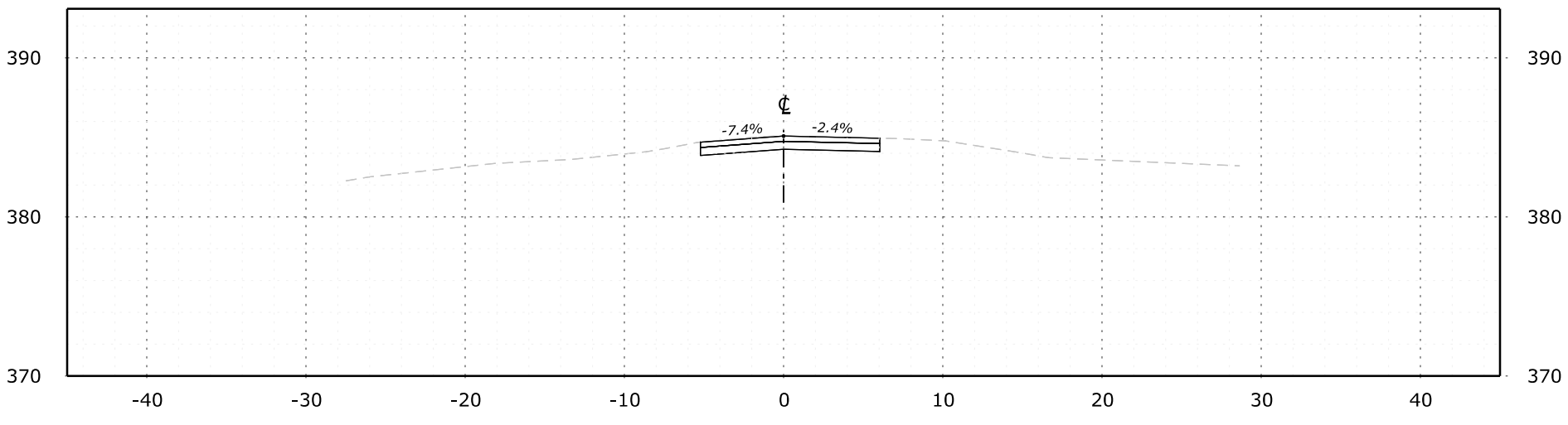
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W09



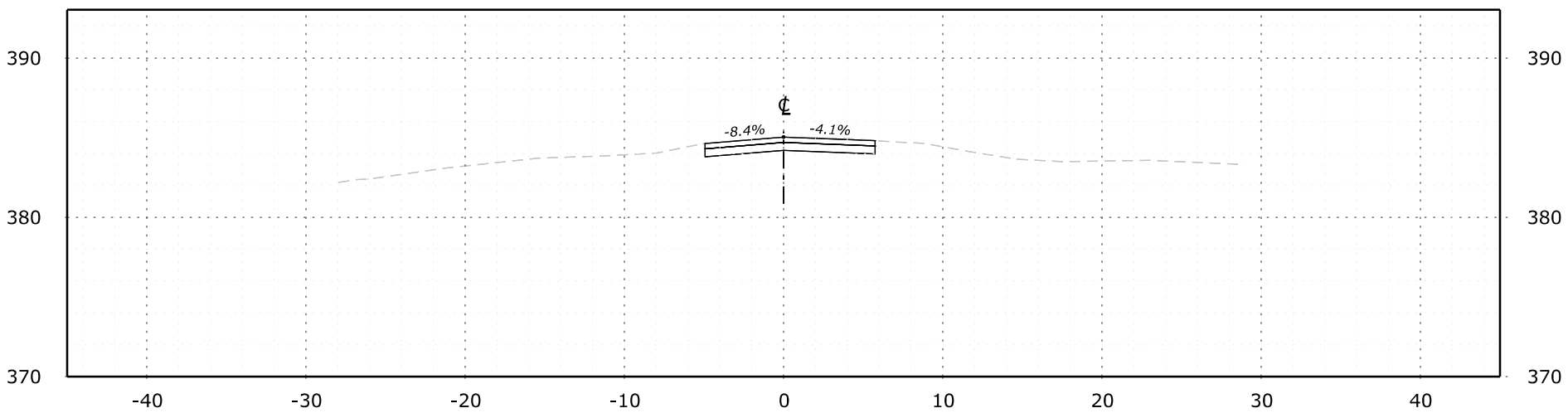
415+50.00
Des Grade = 384.99
Subgrade = 384.65
Original Grnd = 384.99



415+00.00
Des Grade = 385.03
Subgrade = 384.70
Original Grnd = 385.03



414+50.00
Des Grade = 385.08
Subgrade = 384.75
Original Grnd = 385.08



414+00.00
Des Grade = 385.05
Subgrade = 384.71
Original Grnd = 385.04

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

414+00 TO 415+50

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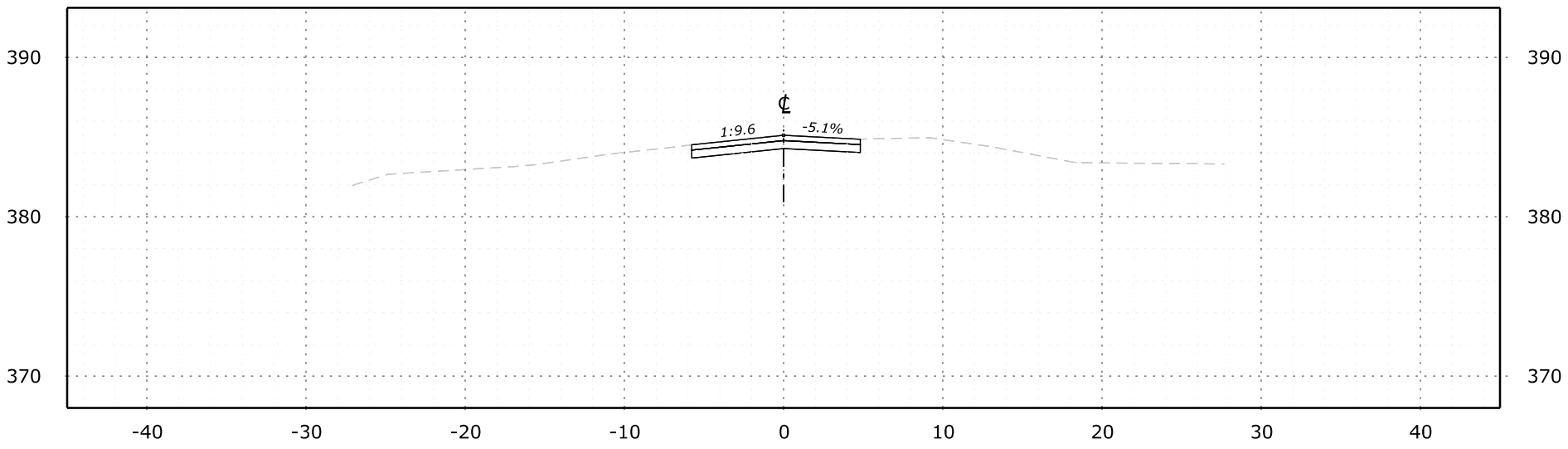
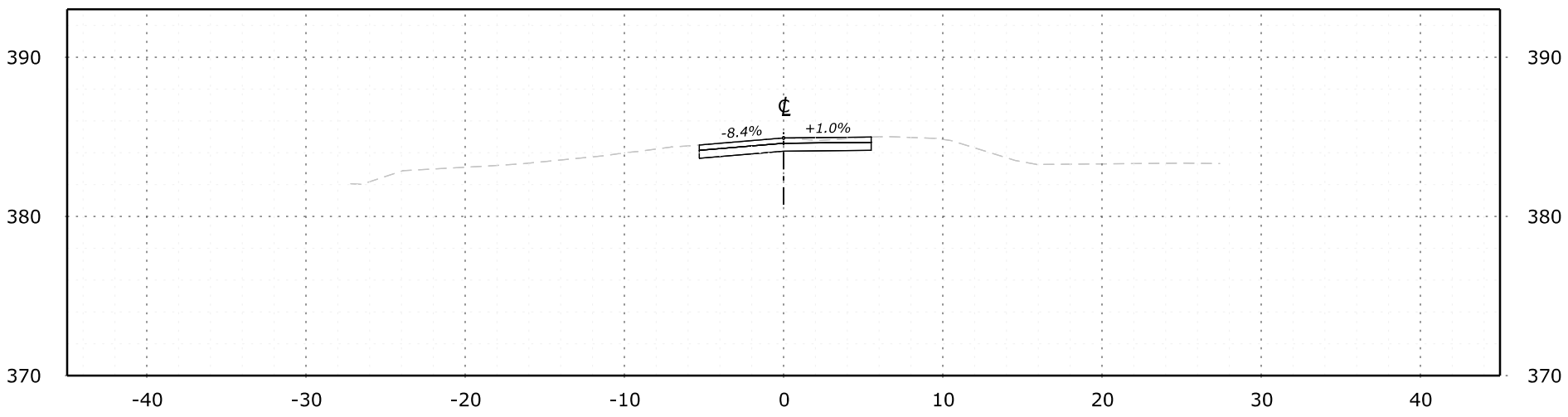
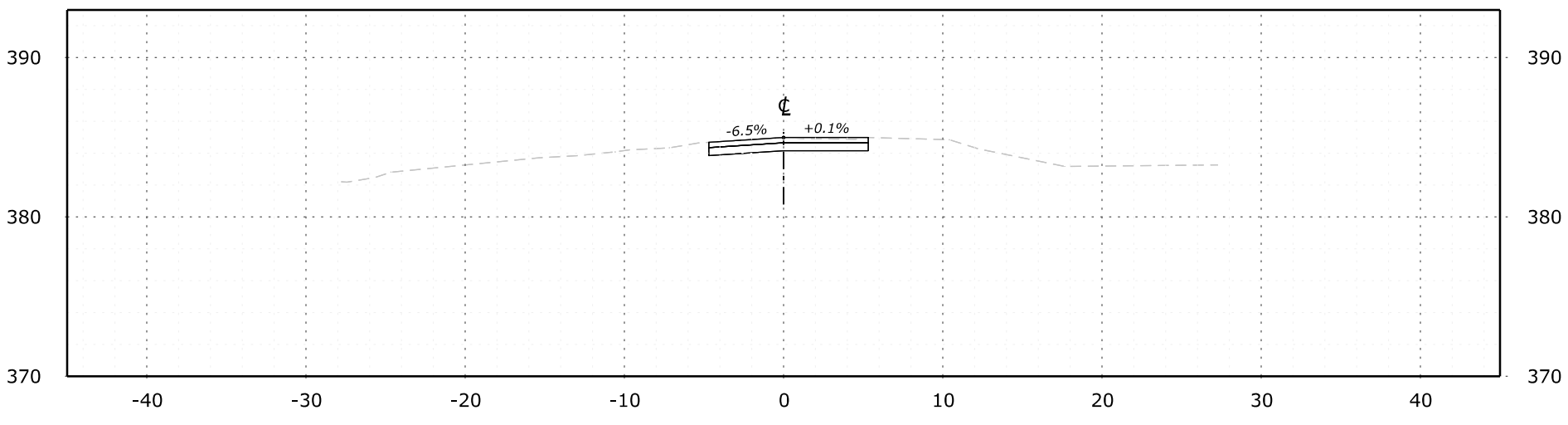
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W10



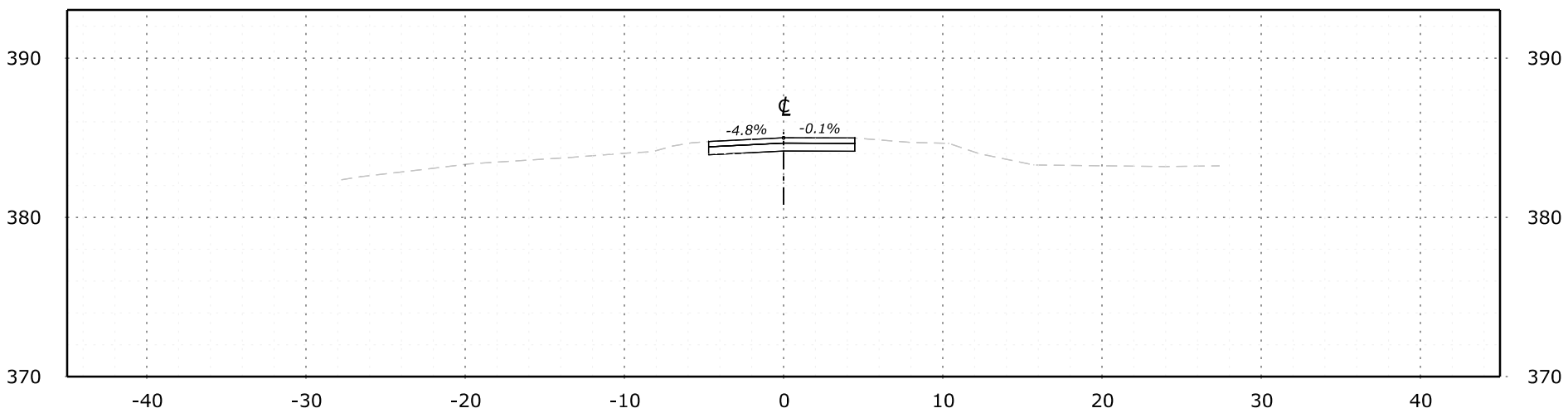
417+50.00
Des Grade = 385.11
Subgrade = 384.77
Original Grnd = 385.09



417+00.00
Des Grade = 384.94
Subgrade = 384.60
Orlglnal Grnd = 384.93



416+50.00
Des Grade = 384.98
Subgrade = 384.64
Original Grnd = 384.98



416+00.00
Des Grade = 384.99
Subgrade = 384.66
Original Grnd = 384.99

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

416+00 TO 417+50

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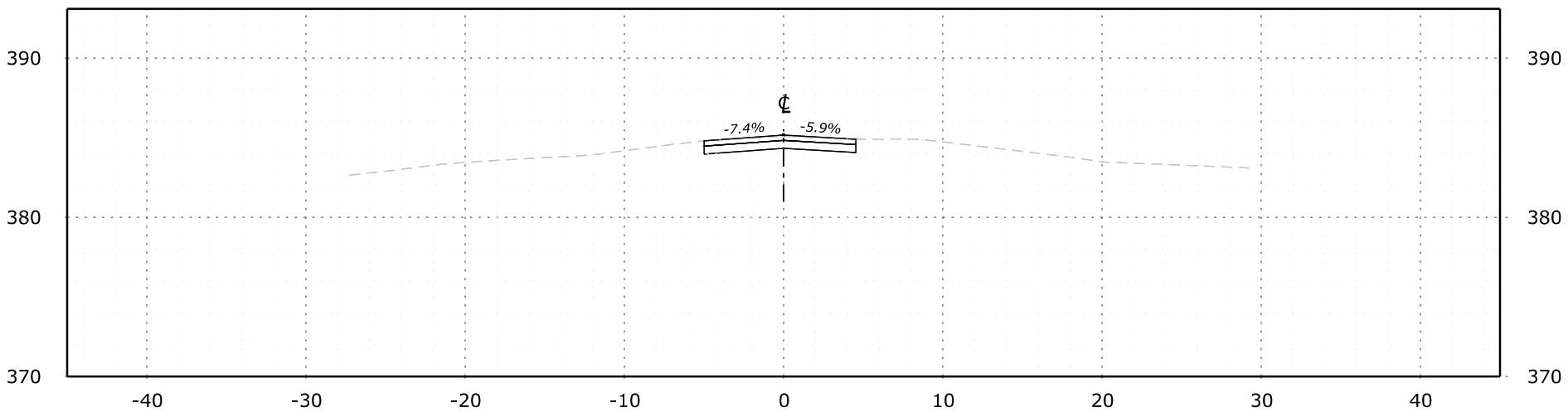
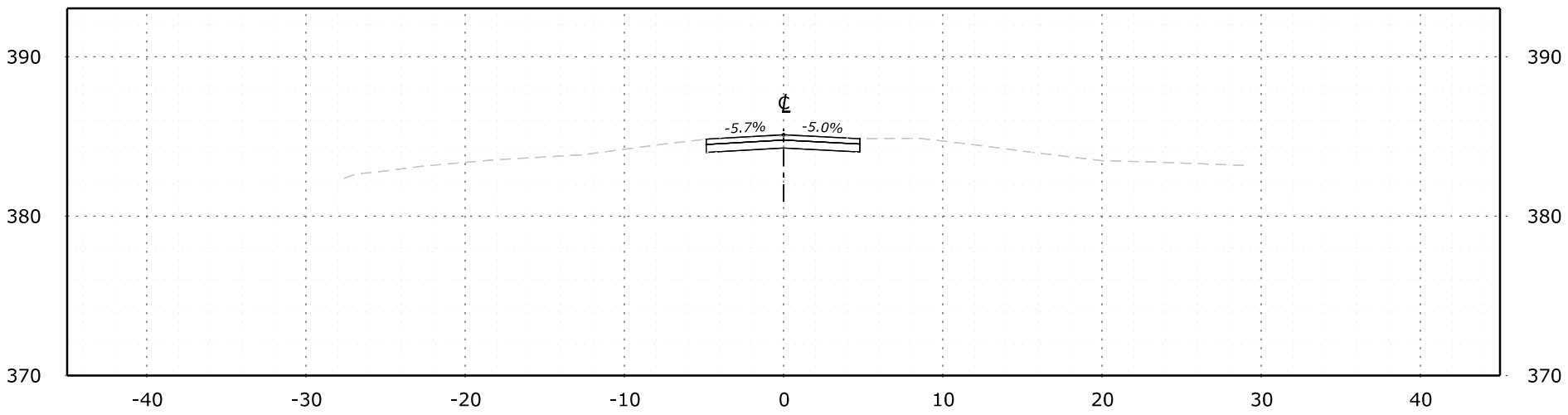
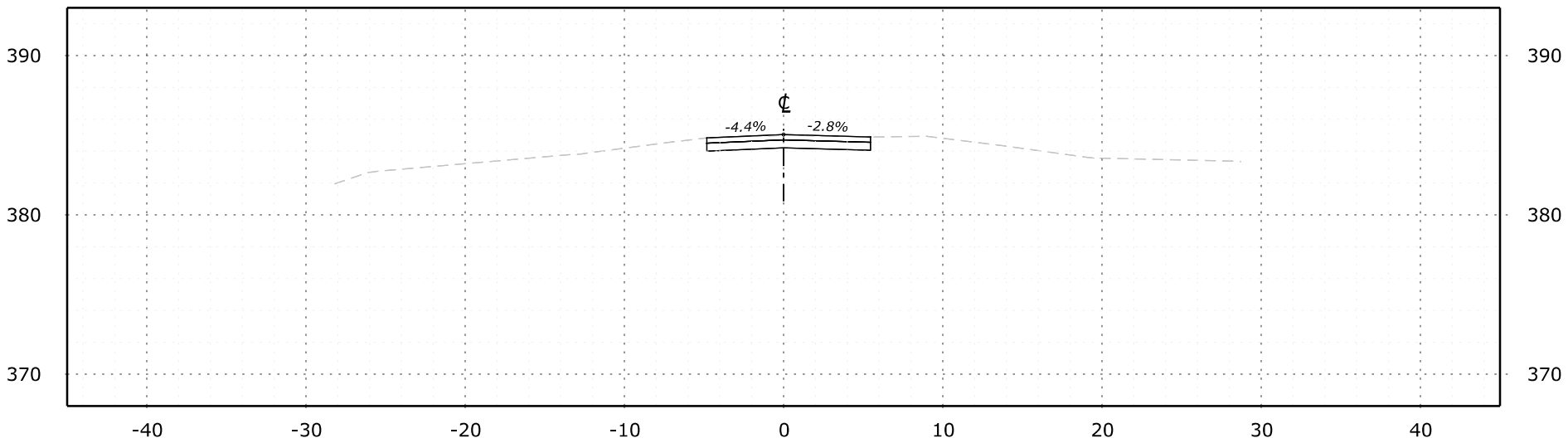
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W11



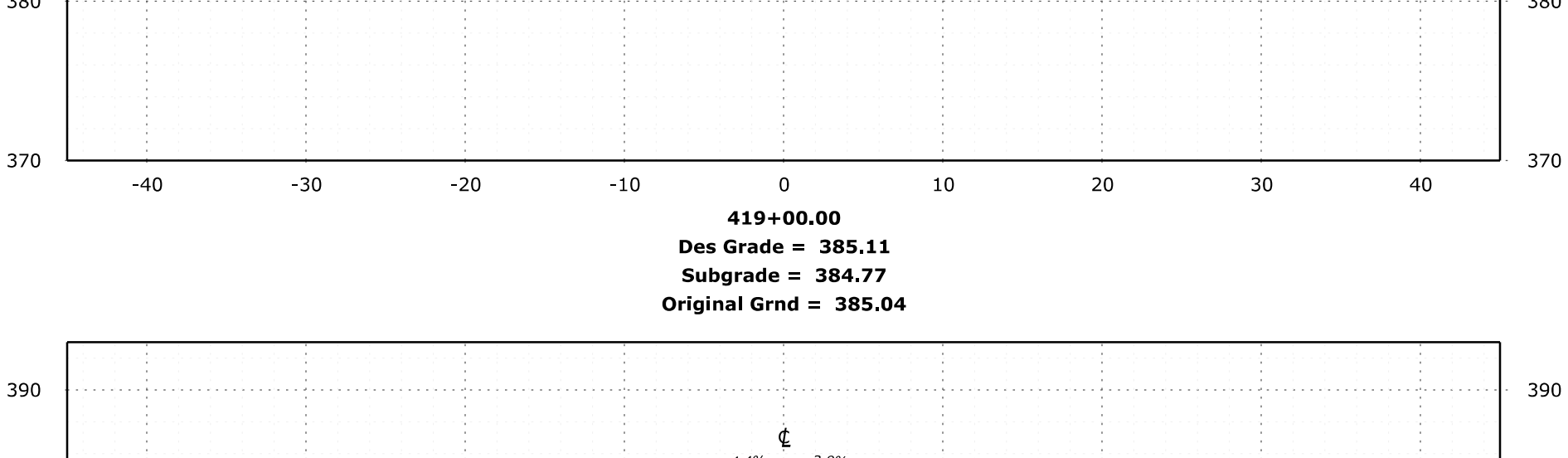
419+50.00
Des Grade = 385.17
Subgrade = 384.84
Original Grnd = 385.09



419+00.00
Des Grade = 385.11
Subgrade = 384.77
Original Grnd = 385.04



418+50.00
Des Grade = 385.04
Subgrade = 384.70
Original Grnd = 384.99



418+00.00
Des Grade = 385.04
Subgrade = 384.71
Original Grnd = 385.04

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

418+00 TO 419+50

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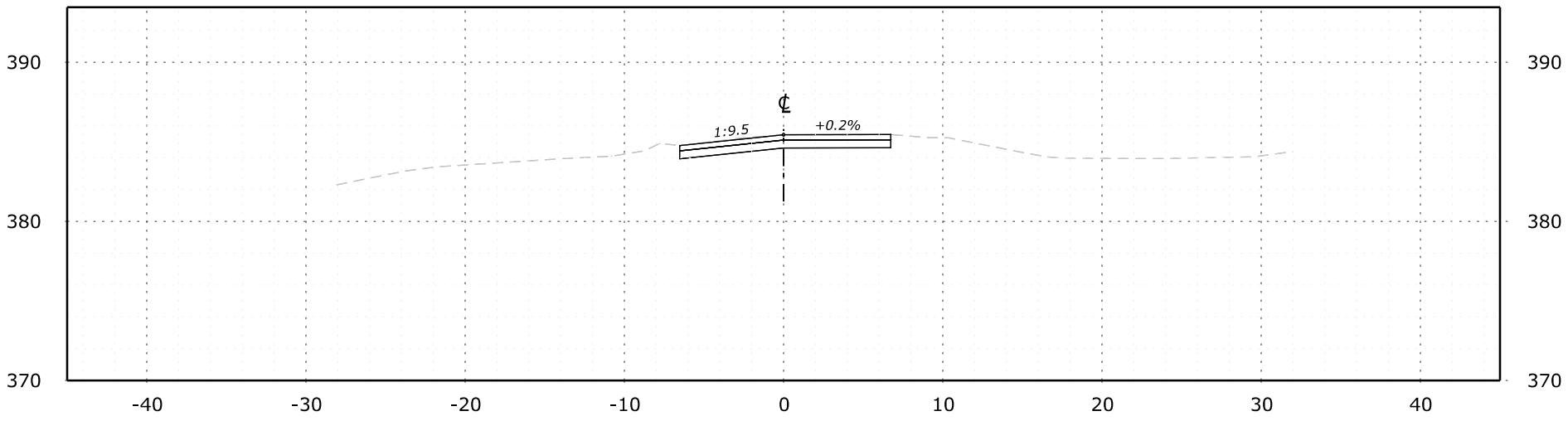
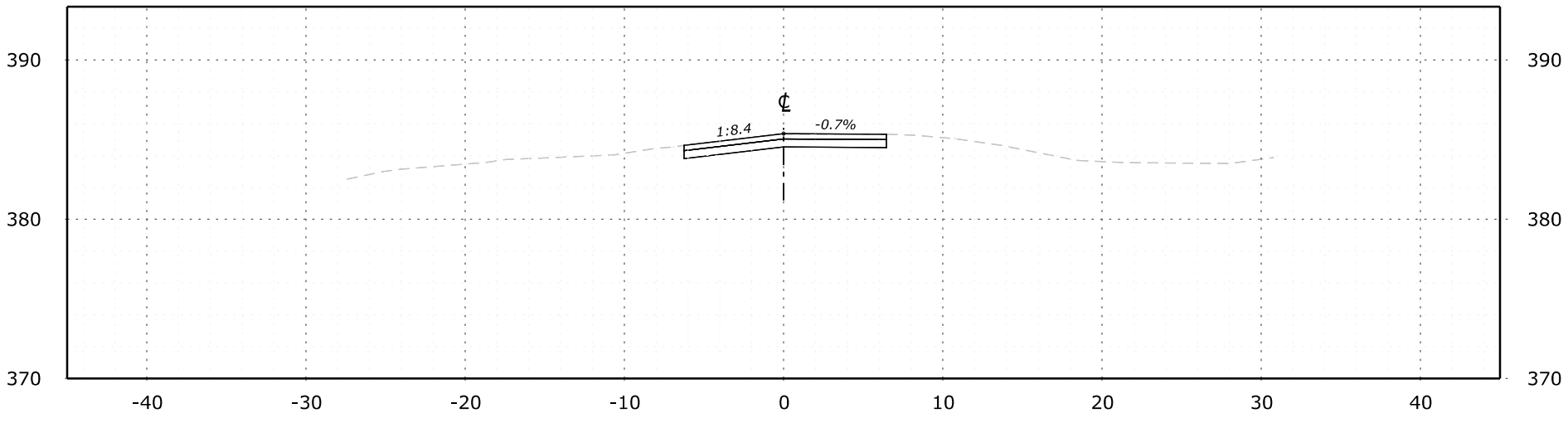
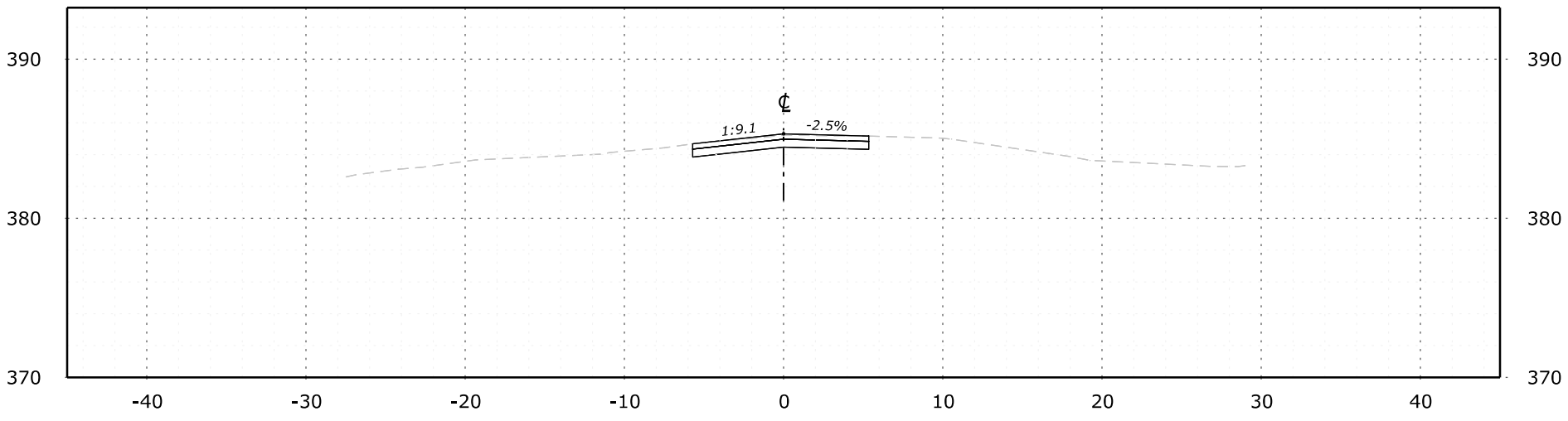
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W12



421+50.00
Des Grade = 385.44
Subgrade = 385.11
Original Grnd = 385.44



421+00.00
Des Grade = 385.37
Subgrade = 385.04
Original Grnd = 385.35



420+50.00
Des Grade = 385.31
Subgrade = 384.97
Original Grnd = 385.24

420+00.00
Des Grade = 385.24
Subgrade = 384.91
Original Grnd = 385.16

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

420+00 TO 421+50

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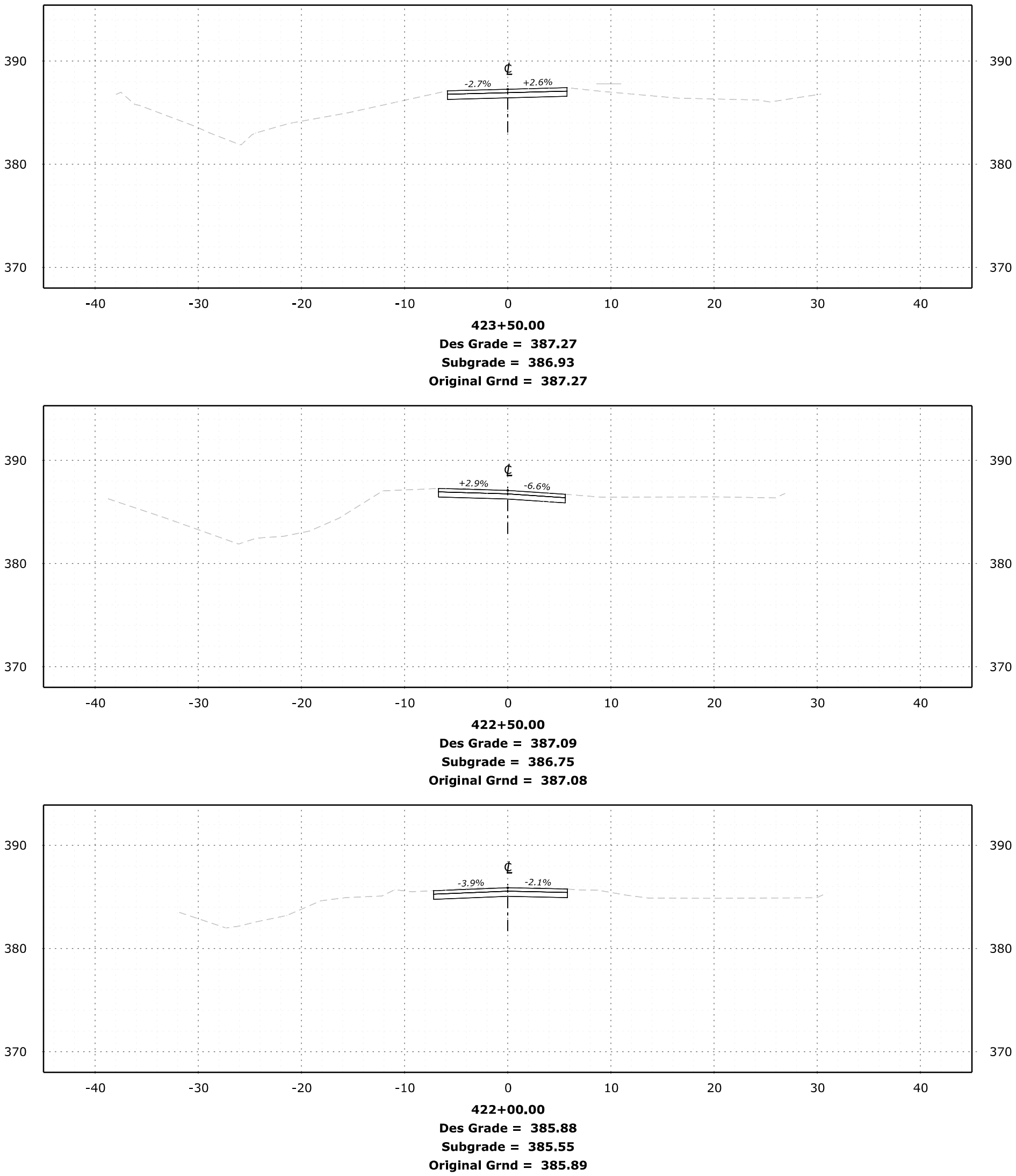
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W13



423+50.00
Des Grade = 387.27
Subgrade = 386.93
Original Grnd = 387.27

422+50.00
Des Grade = 387.09
Subgrade = 386.75
Original Grnd = 387.08

422+00.00
Des Grade = 385.88
Subgrade = 385.55
Original Grnd = 385.89

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

422+00 TO 423+50

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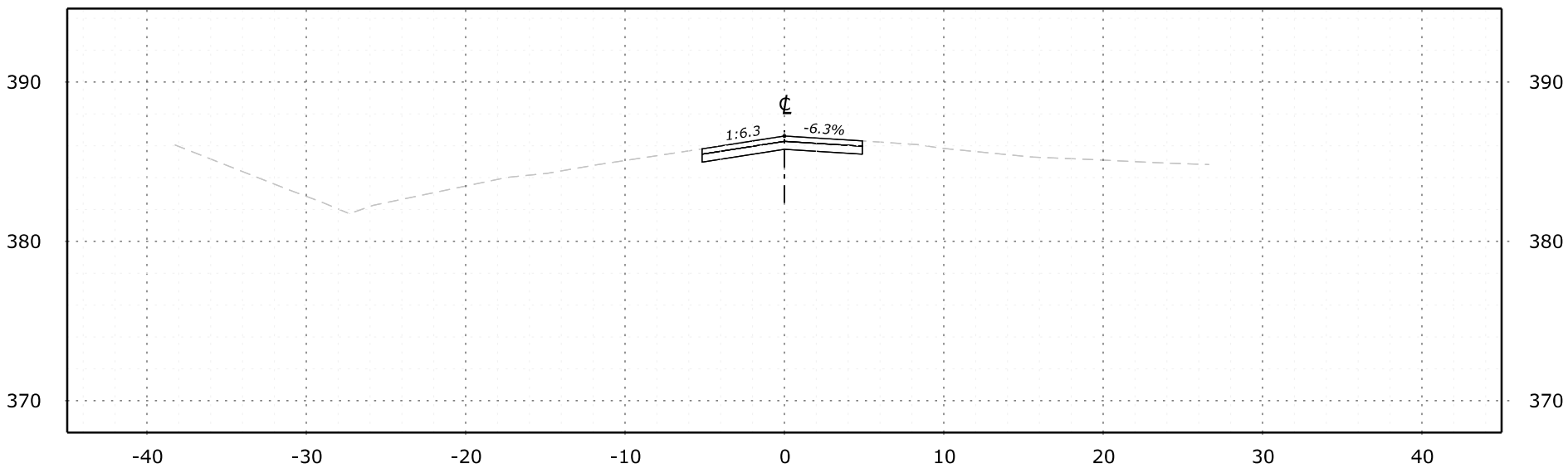
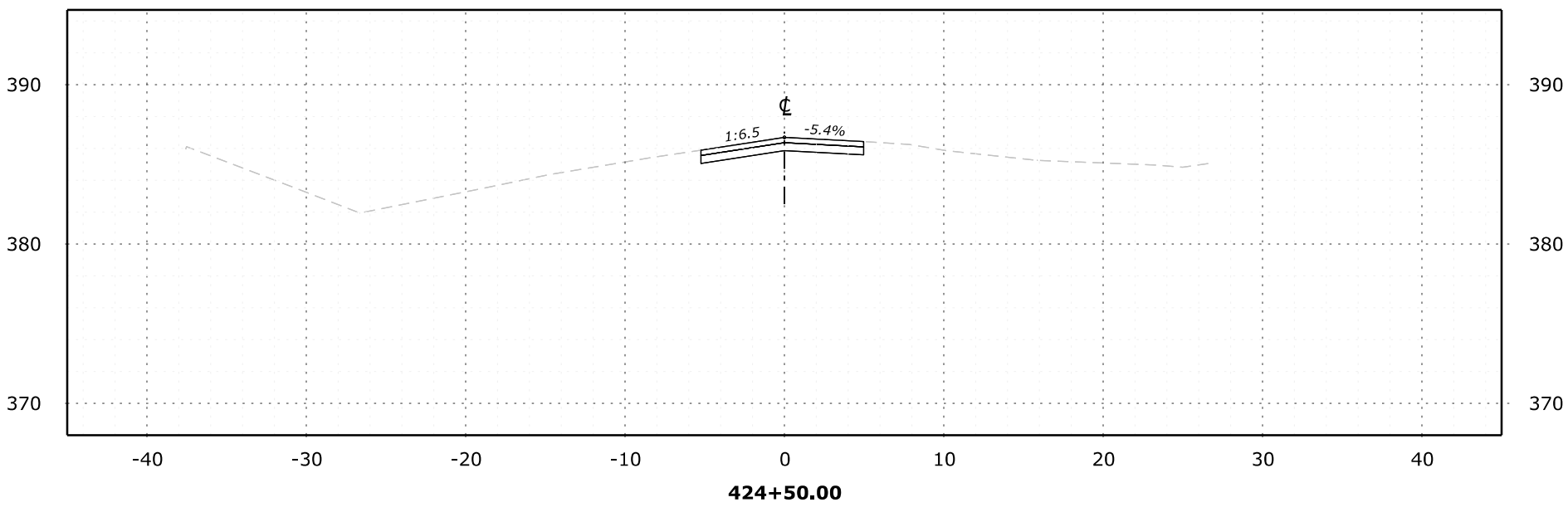
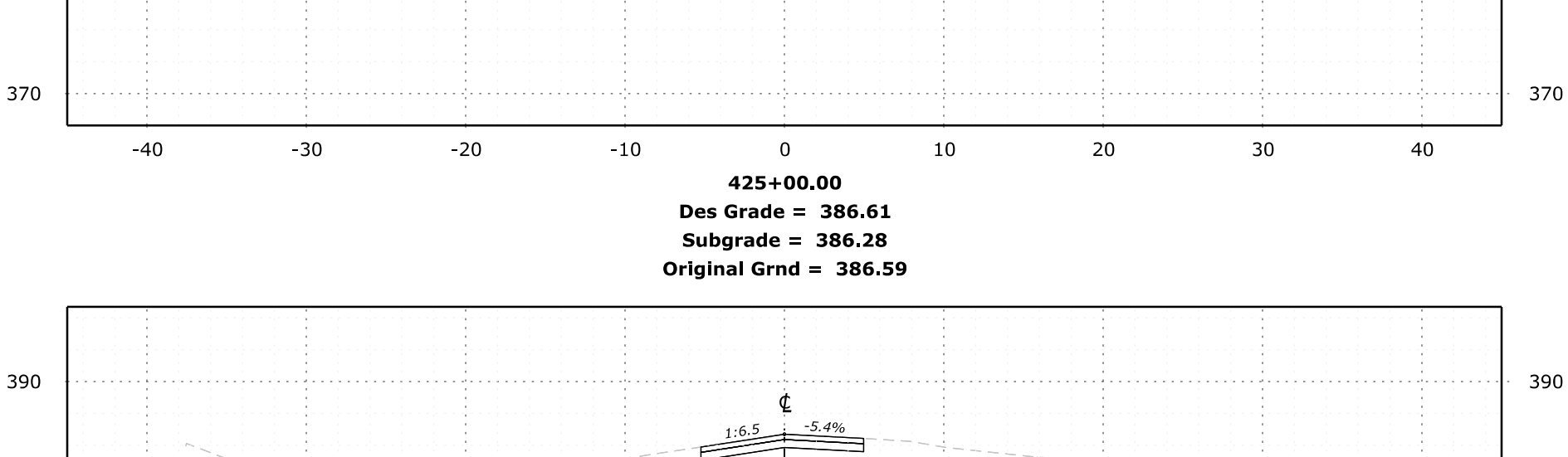
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W14



425+00.00
Des Grade = 386.61
Subgrade = 386.28
Original Grnd = 386.59



424+50.00
Des Grade = 386.69
Subgrade = 386.36
Original Grnd = 386.68



424+00.00
Des Grade = 386.78
Subgrade = 386.44
Original Grnd = 386.78

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

424+00 TO 425+00

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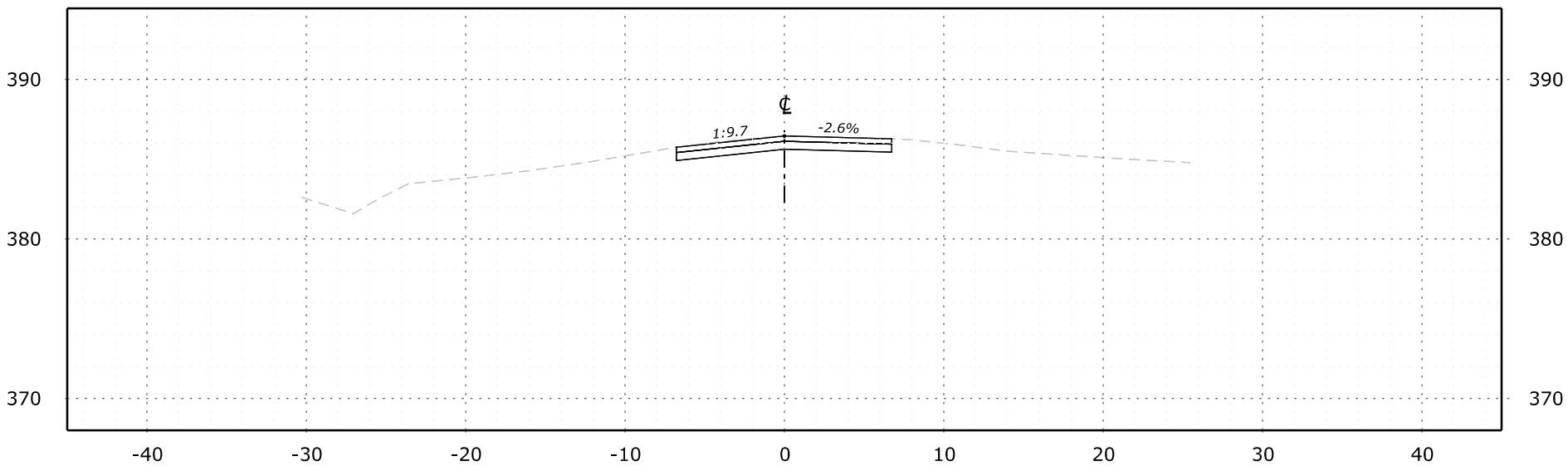
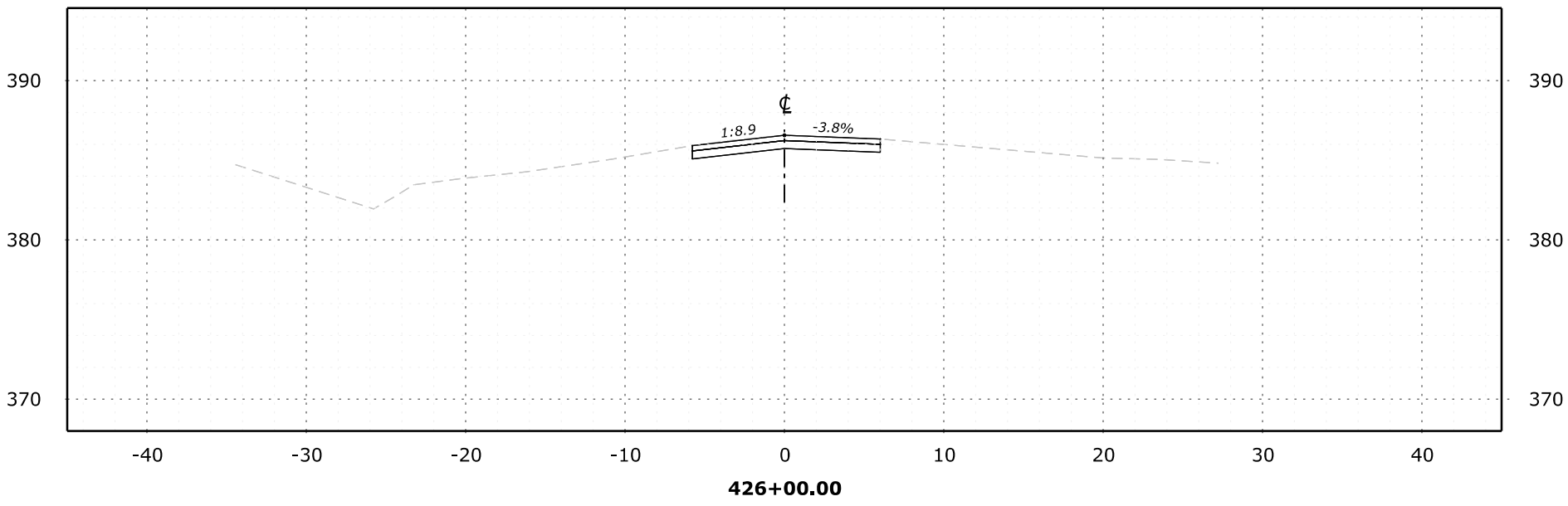
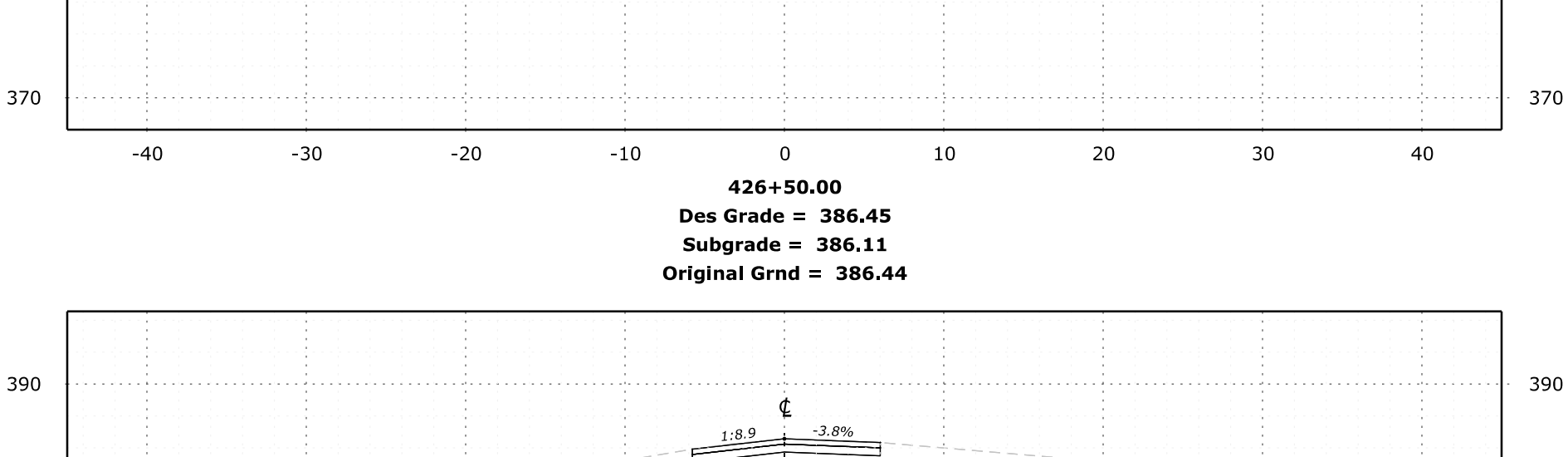
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W15



426+50.00
Des Grade = 386.45
Subgrade = 386.11
Original Grnd = 386.44



426+00.00
Des Grade = 386.56
Subgrade = 386.23
Original Grnd = 386.54



425+50.00
Des Grade = 386.55
Subgrade = 386.21
Original Grnd = 386.55

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

425+50 TO 426+50

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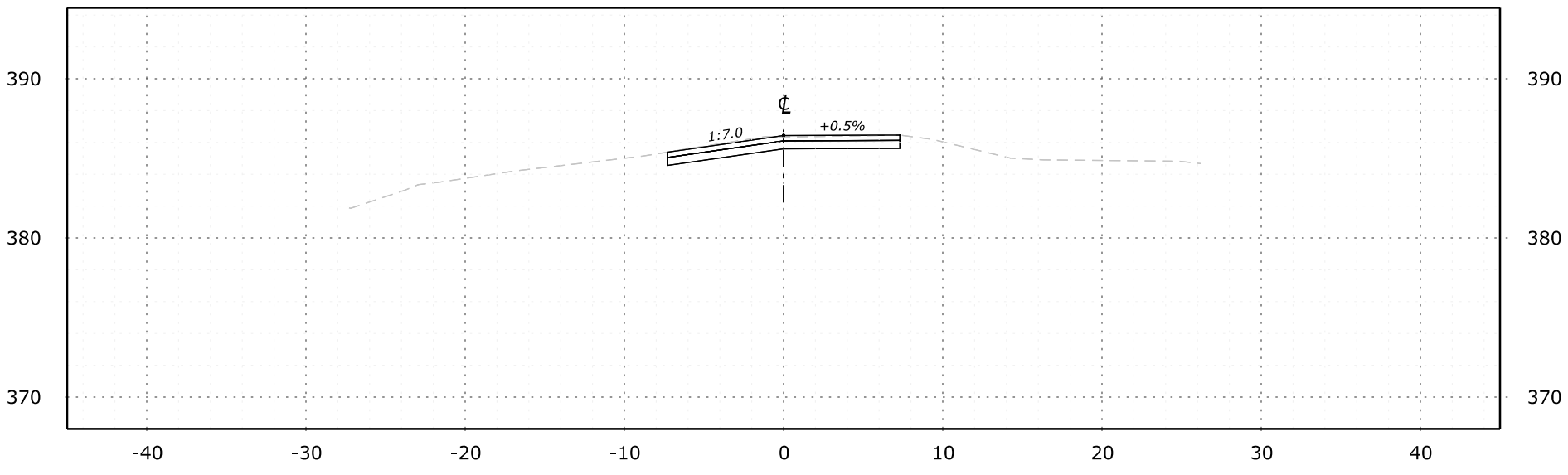
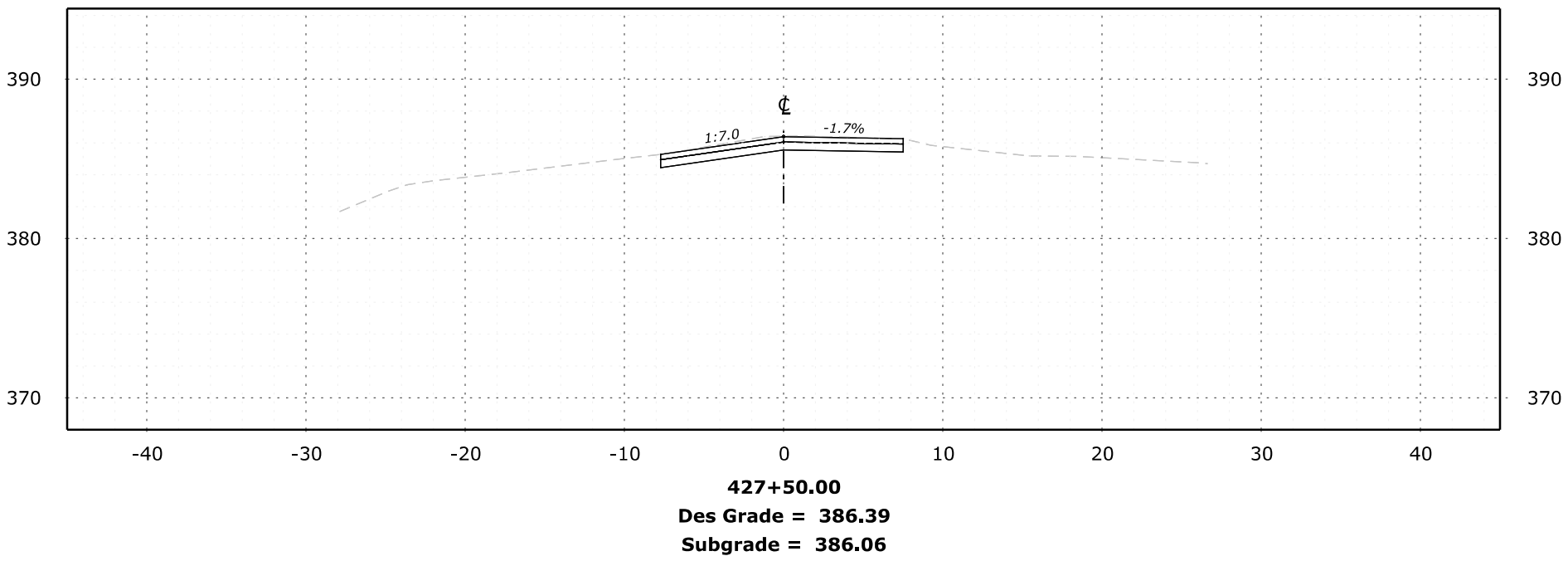
3 August 2026 5:11 PM

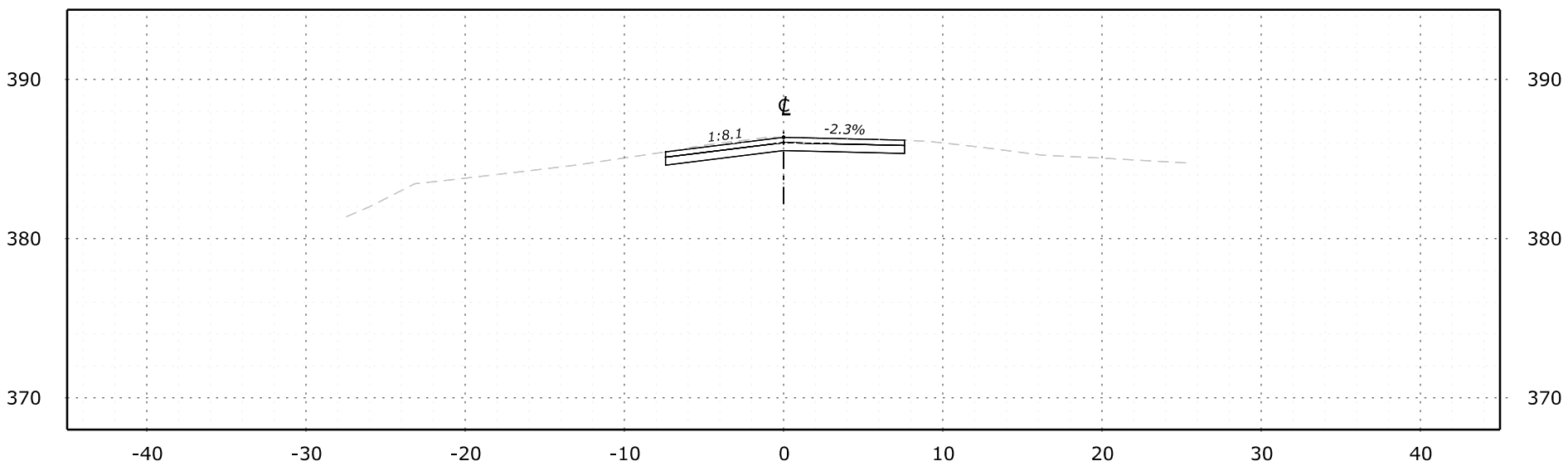
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W16



428+00.00
Des Grade = 386.43
Subgrade = 386.09
Original Grnd = 386.35



427+50.00
Des Grade = 386.39
Subgrade = 386.06
Original Grnd = 386.41



427+00.00
Des Grade = 386.36
Subgrade = 386.02
Original Grnd = 386.36

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

427+00 TO 428+00

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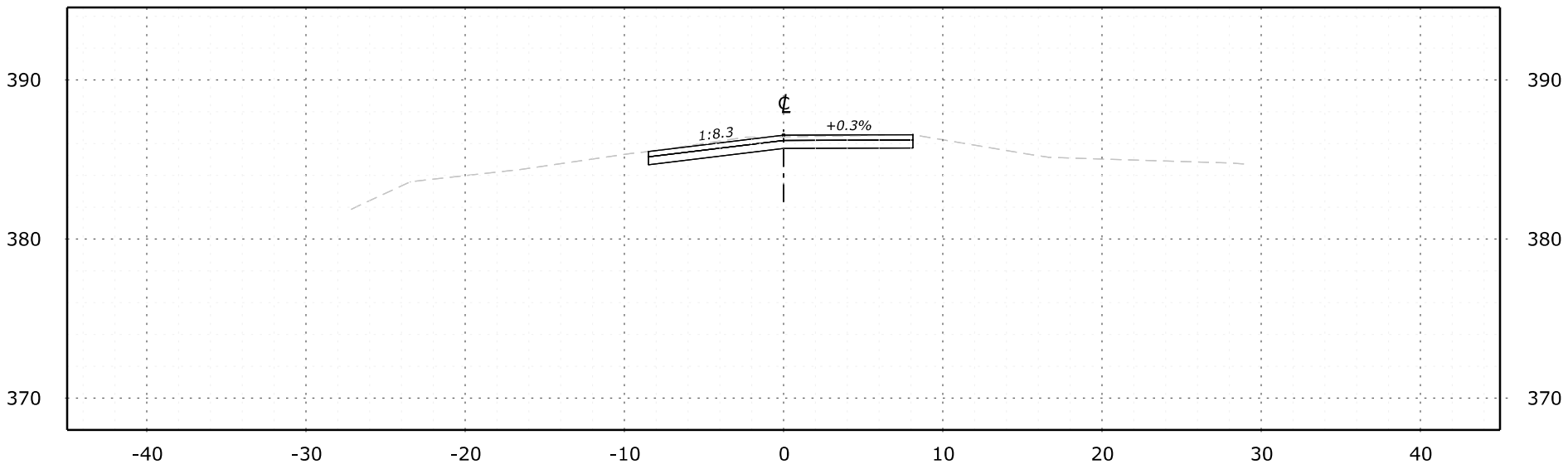
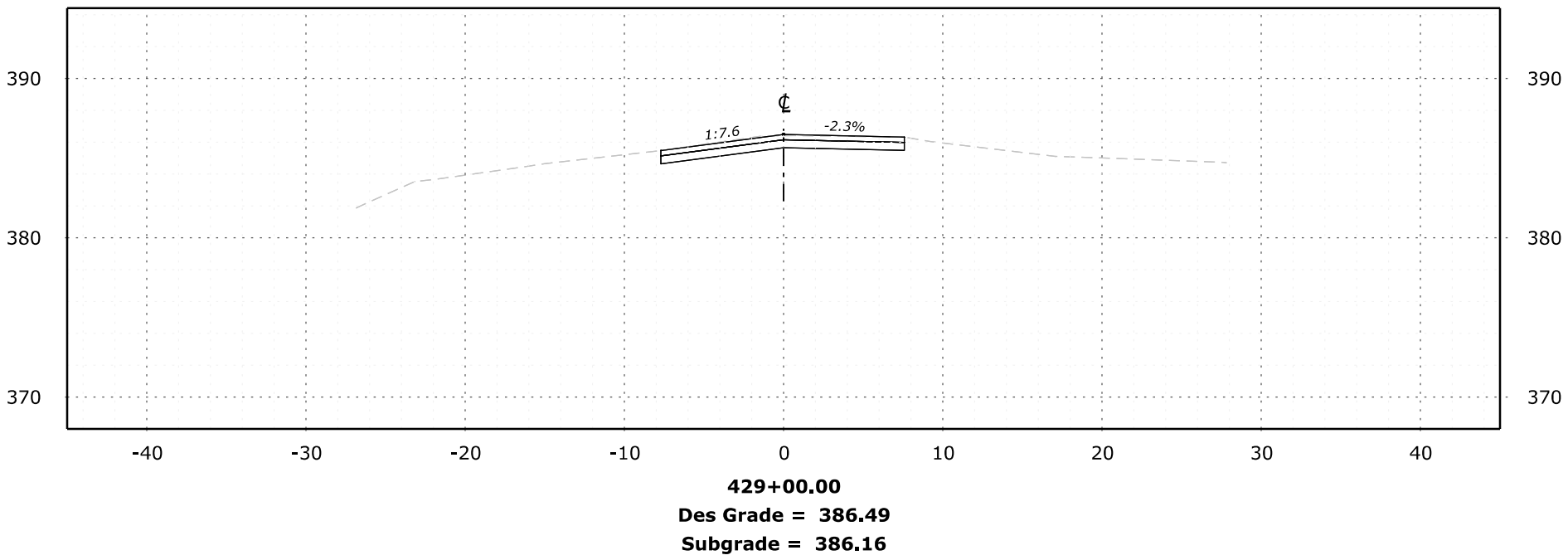
3 August 2026 5:11 PM

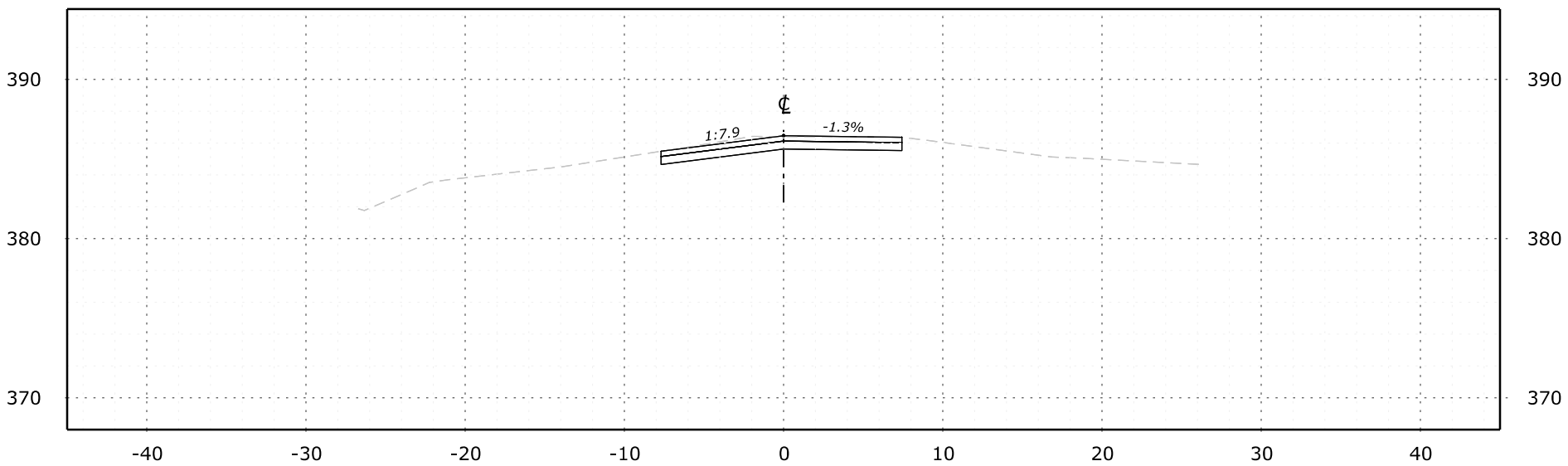
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W17



429+50.00
Des Grade = 386.53
Subgrade = 386.20
Original Grnd = 386.42



429+00.00
Des Grade = 386.49
Subgrade = 386.16
Original Grnd = 386.41



428+50.00
Des Grade = 386.46
Subgrade = 386.13
Original Grnd = 386.41

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

428+50 TO 429+50

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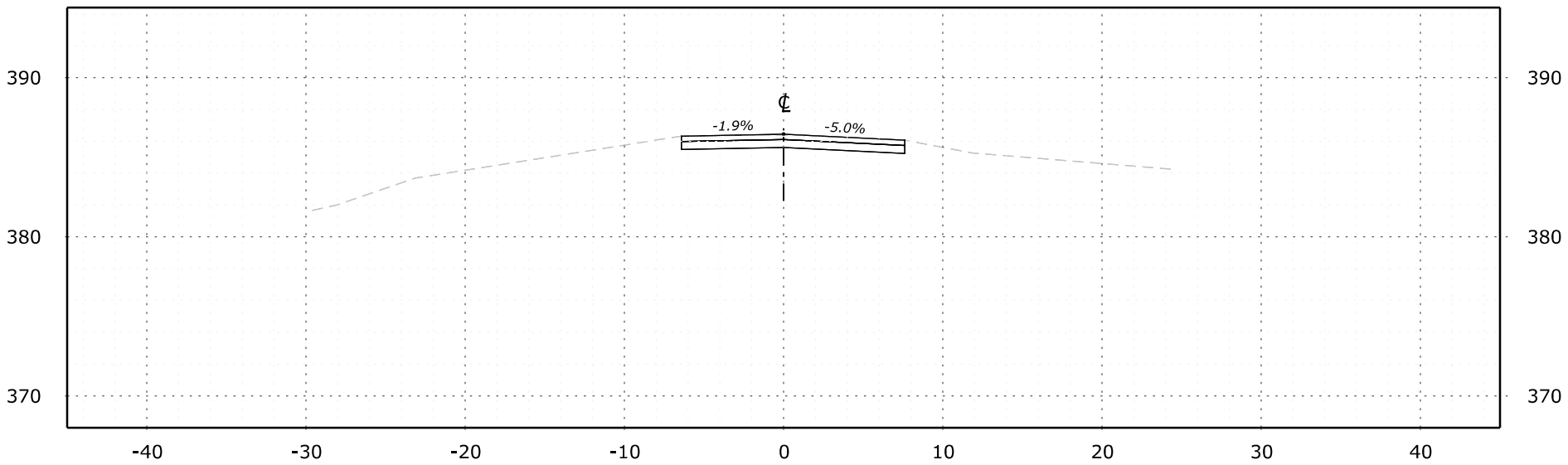
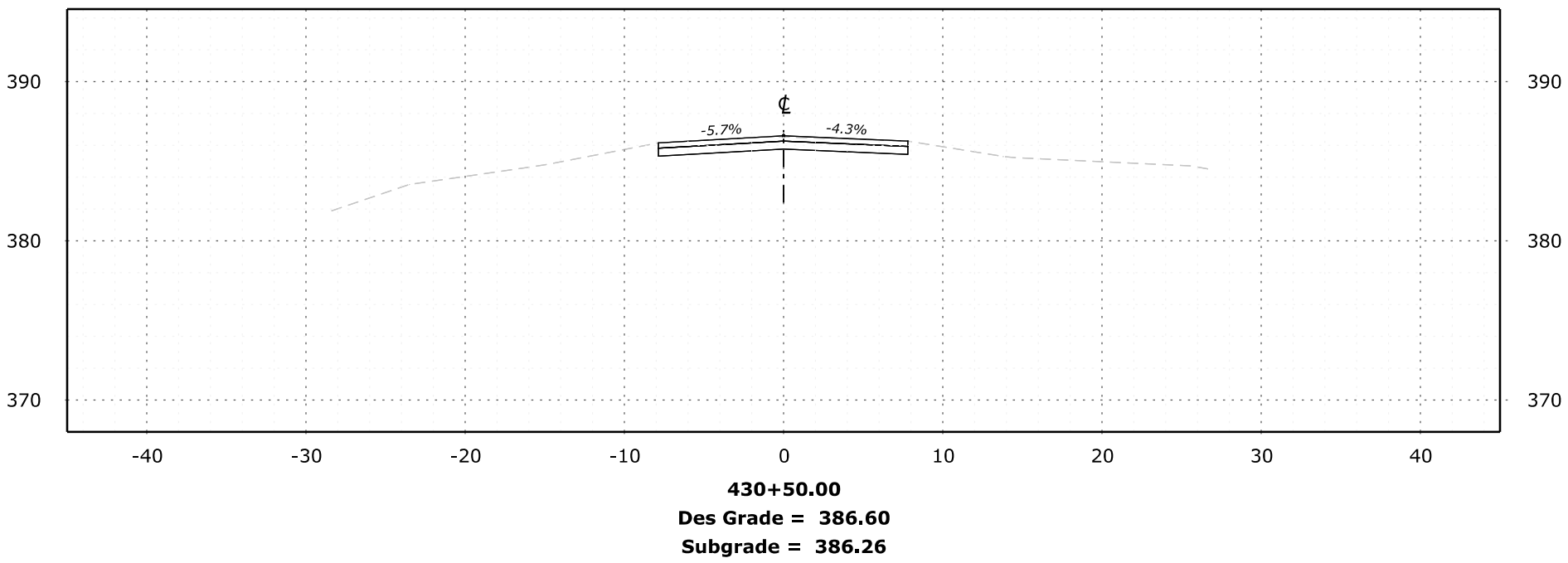
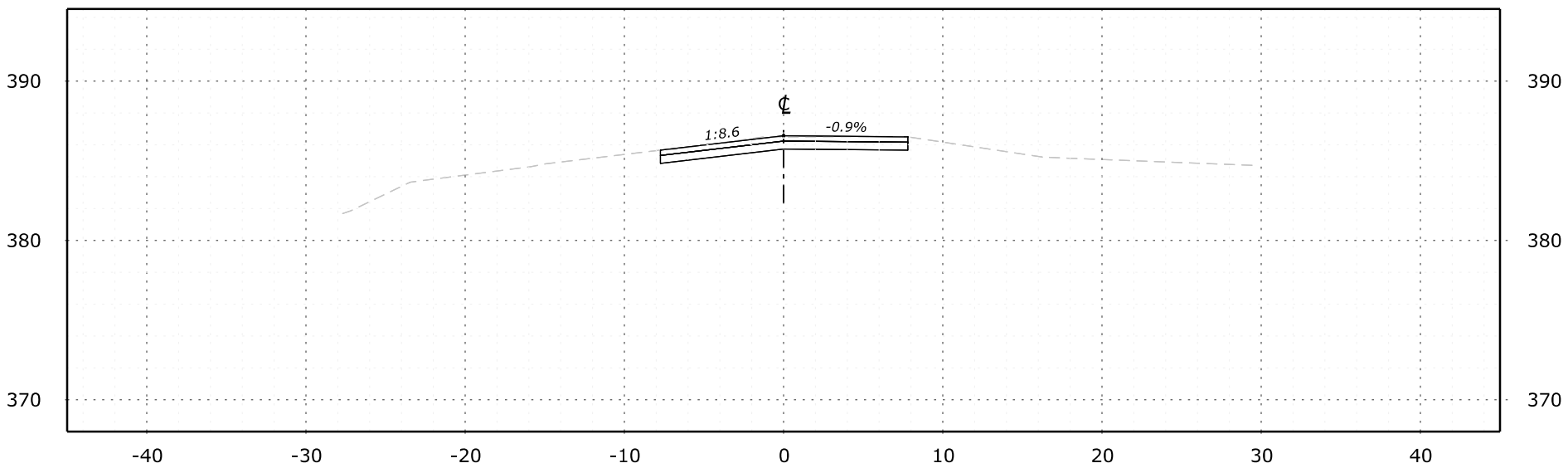
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W18



431+00.00
Des Grade = 386.45
Subgrade = 386.12
Original Grnd = 386.39



430+50.00
Des Grade = 386.60
Subgrade = 386.26
Original Grnd = 386.53



430+00.00
Des Grade = 386.56
Subgrade = 386.23
Original Grnd = 386.49

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

430+00 TO 431+00

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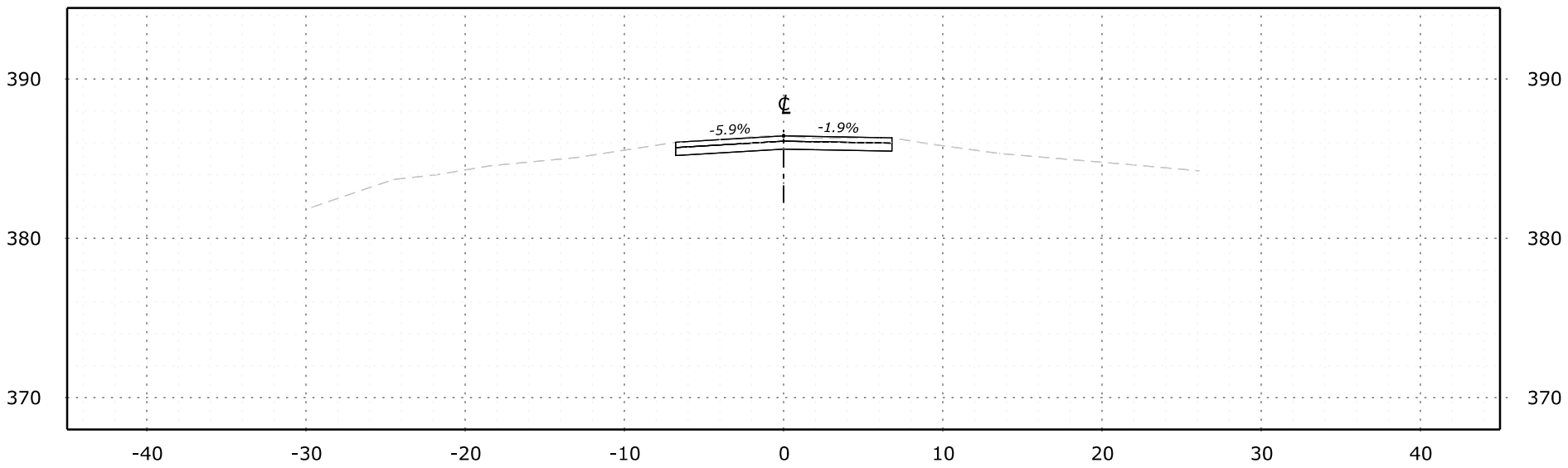
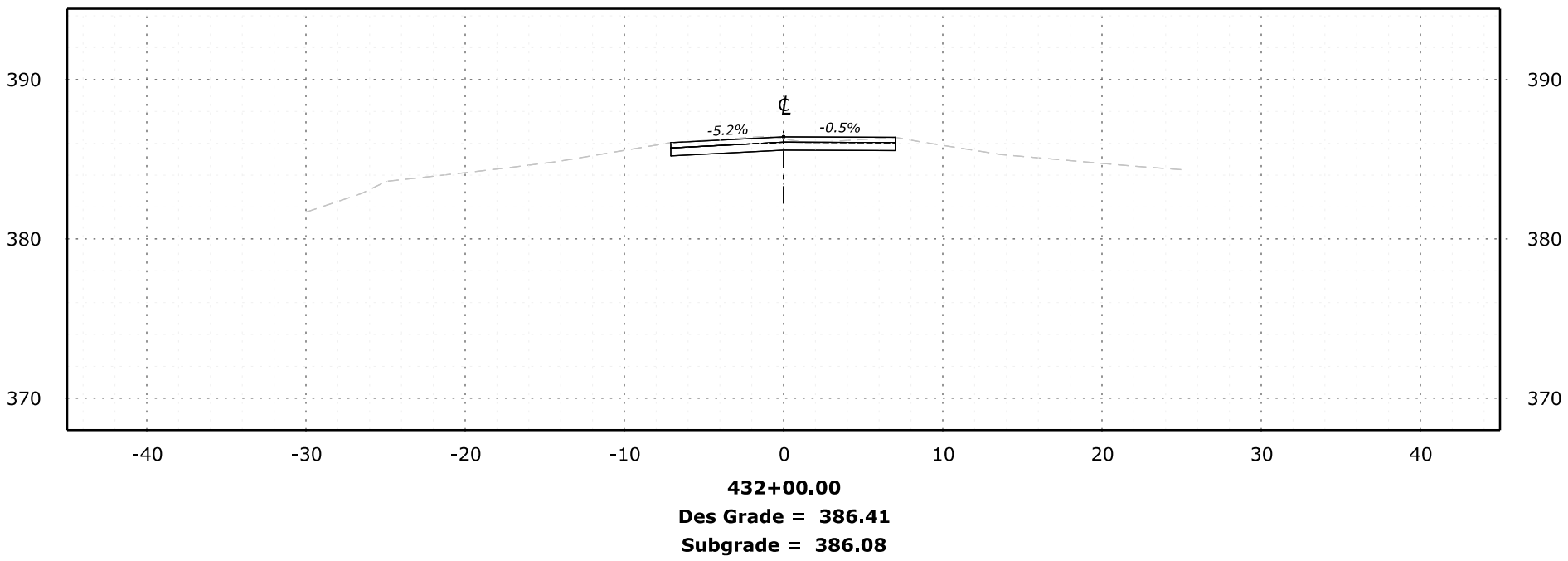
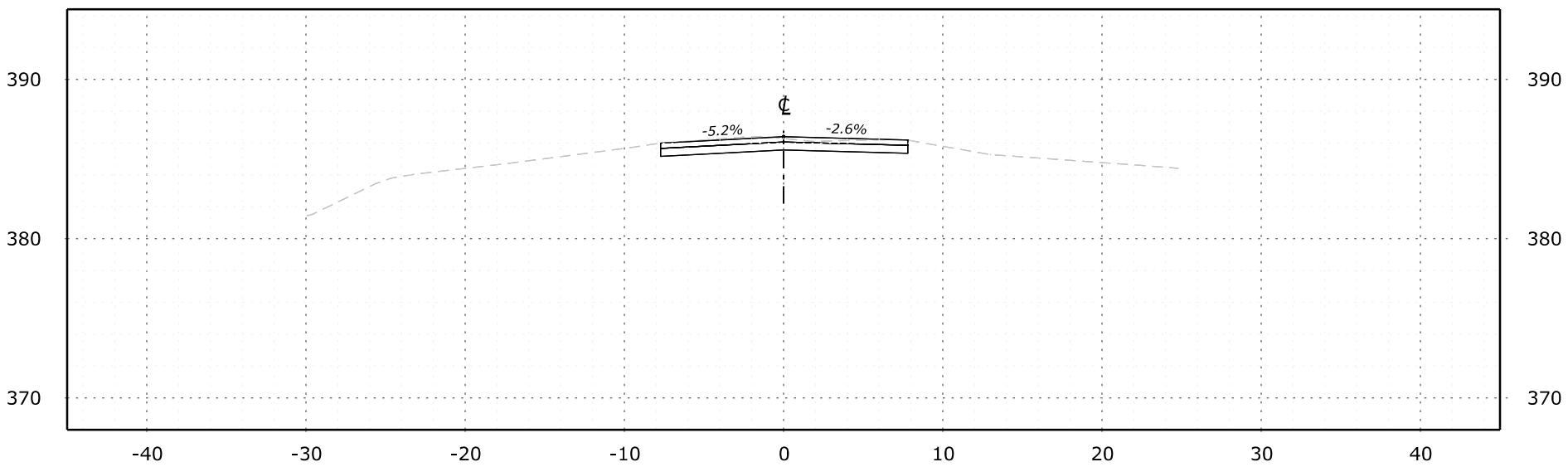
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W19



432+50.00
Des Grade = 386.43
Subgrade = 386.09
Original Grnd = 386.42



432+00.00
Des Grade = 386.41
Subgrade = 386.08
Original Grnd = 386.29



431+50.00
Des Grade = 386.40
Subgrade = 386.06
Original Grnd = 386.29

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

431+50 TO 432+50

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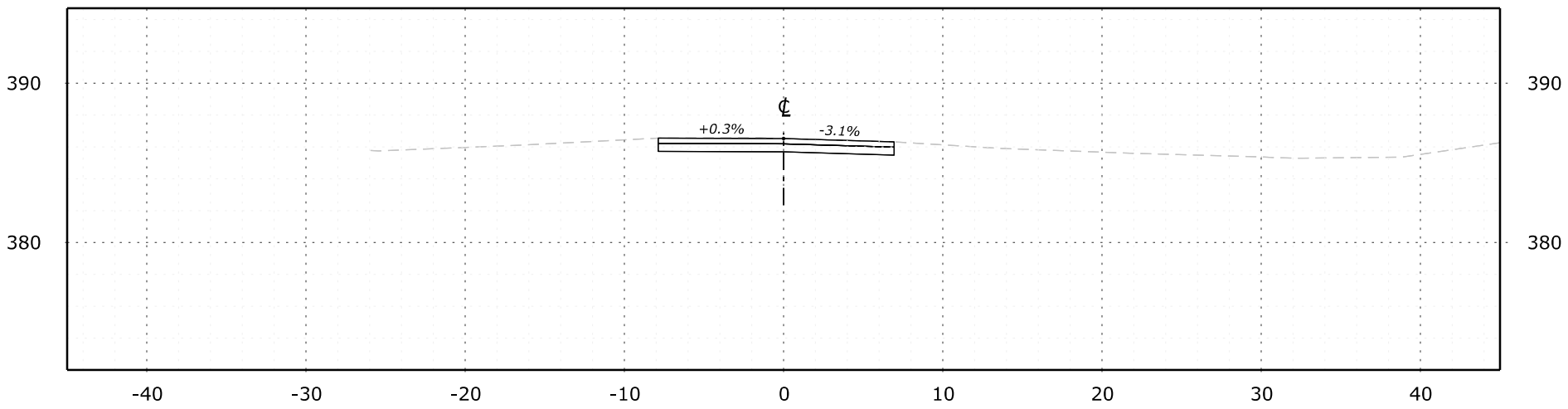
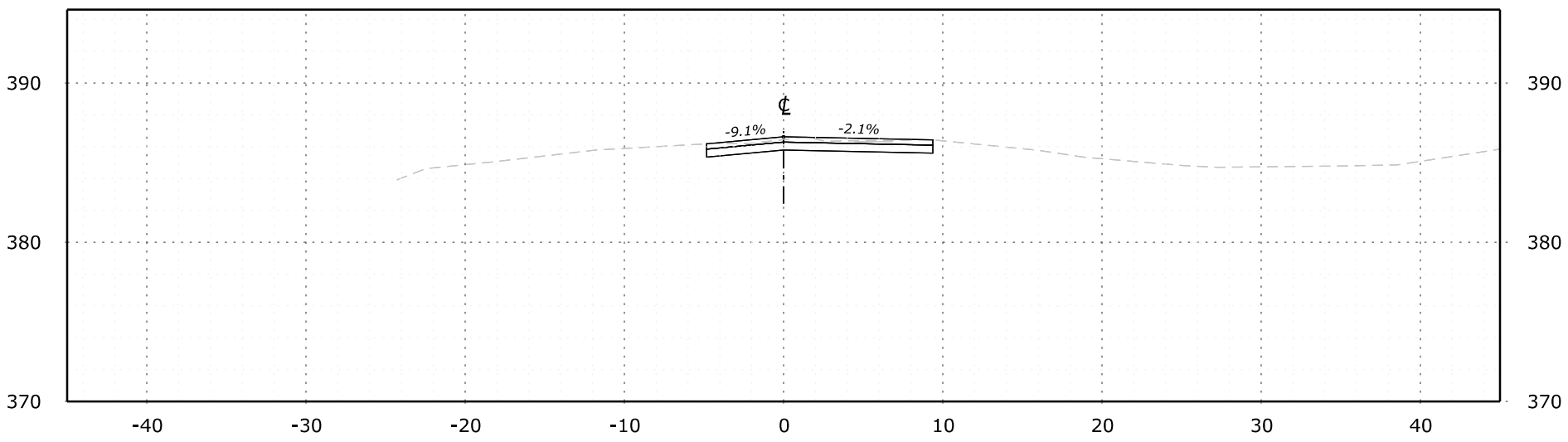
3 August 2026 5:11 PM

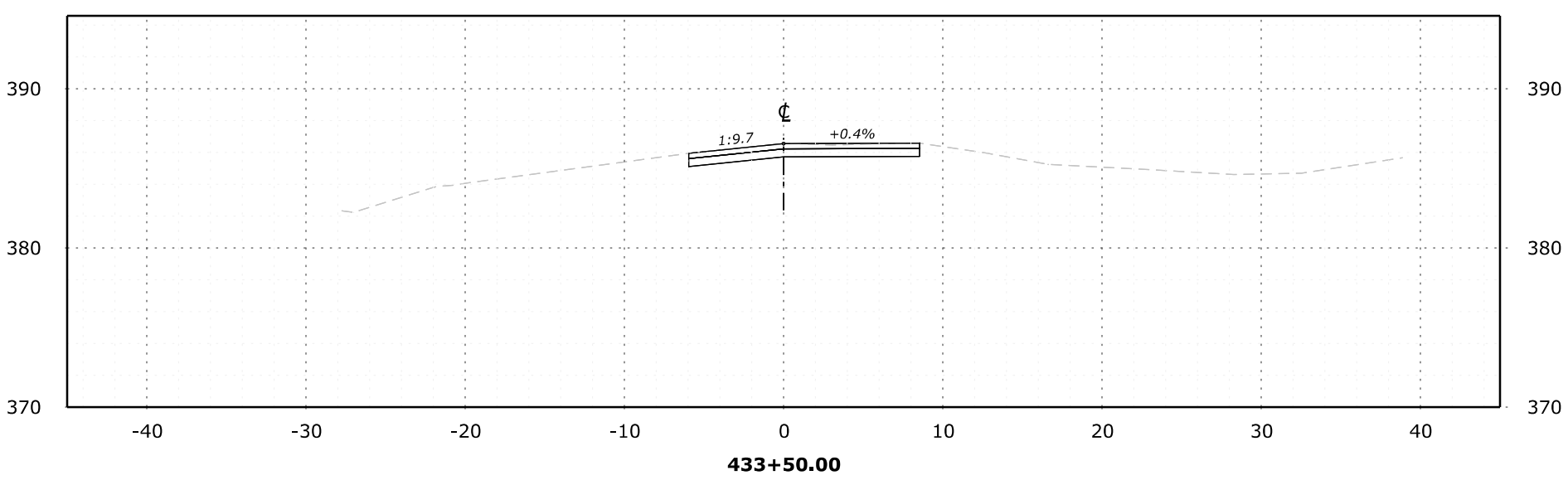
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	W20



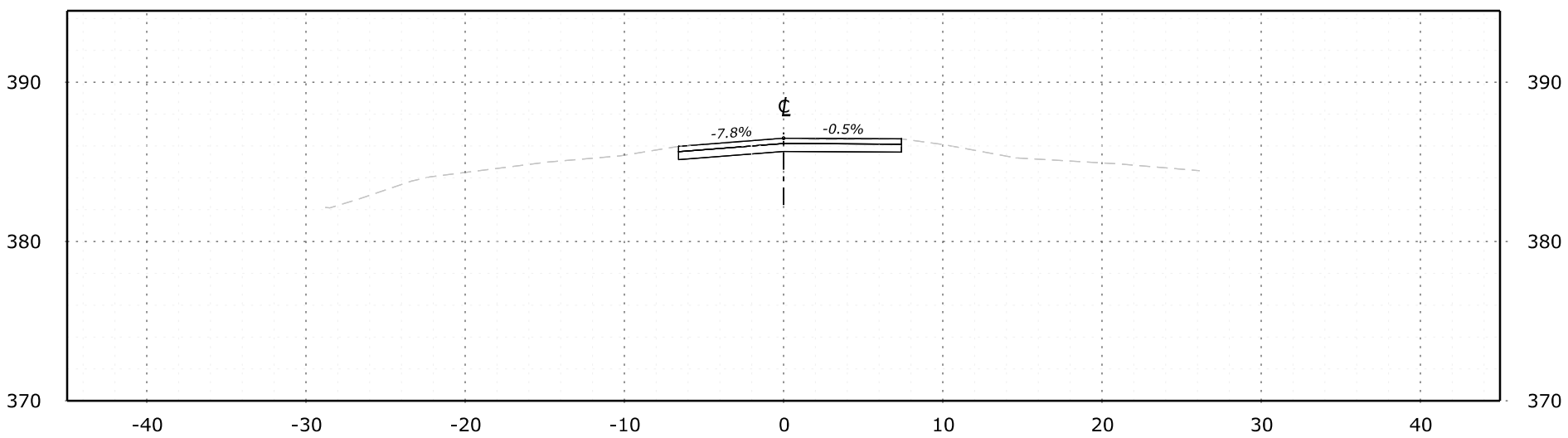
434+23.00
Des Grade = 386.52
Subgrade = 386.19
Original Grnd = 386.52



434+00.00
Des Grade = 386.62
Subgrade = 386.29
Original Grnd = 386.44



433+50.00
Des Grade = 386.55
Subgrade = 386.22
Original Grnd = 386.53



433+00.00
Des Grade = 386.48
Subgrade = 386.15
Original Grnd = 386.48

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE A: LIBERTY LOOP TRAIL (NE)

433+00 TO 434+23

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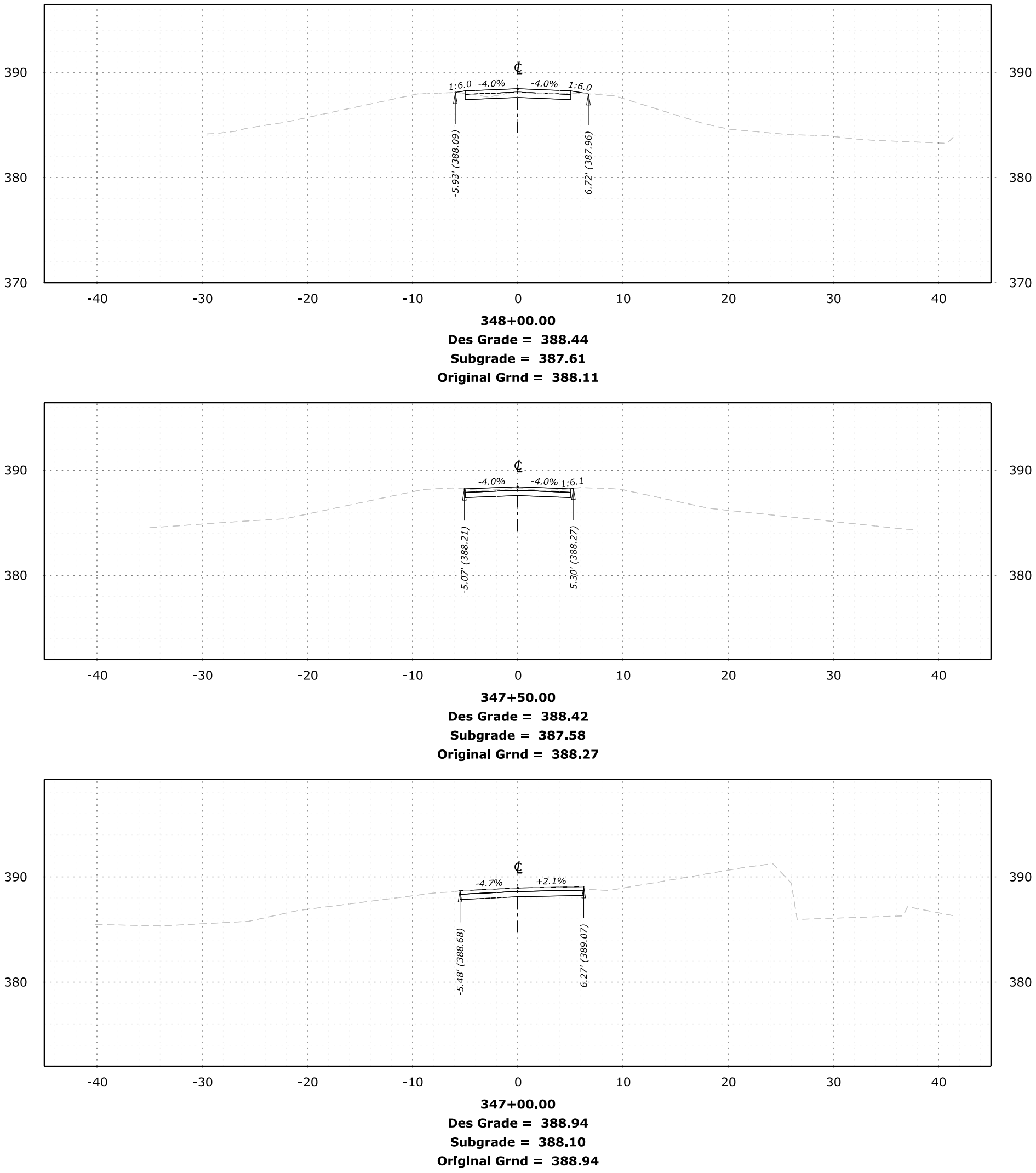
3 August 2026 5:11 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X01



348+00.00
Des Grade = 388.44
Subgrade = 387.61
Original Grnd = 388.11

347+50.00
Des Grade = 388.42
Subgrade = 387.58
Original Grnd = 388.27

347+00.00
Des Grade = 388.94
Subgrade = 388.10
Original Grnd = 388.94

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

347+00 TO 348+00

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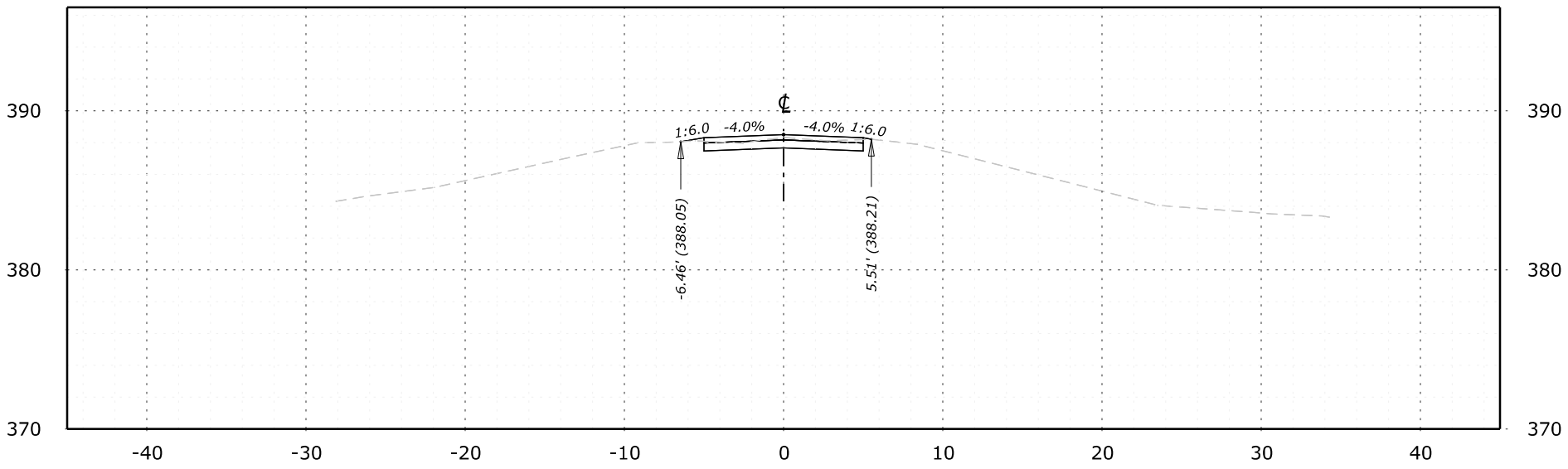
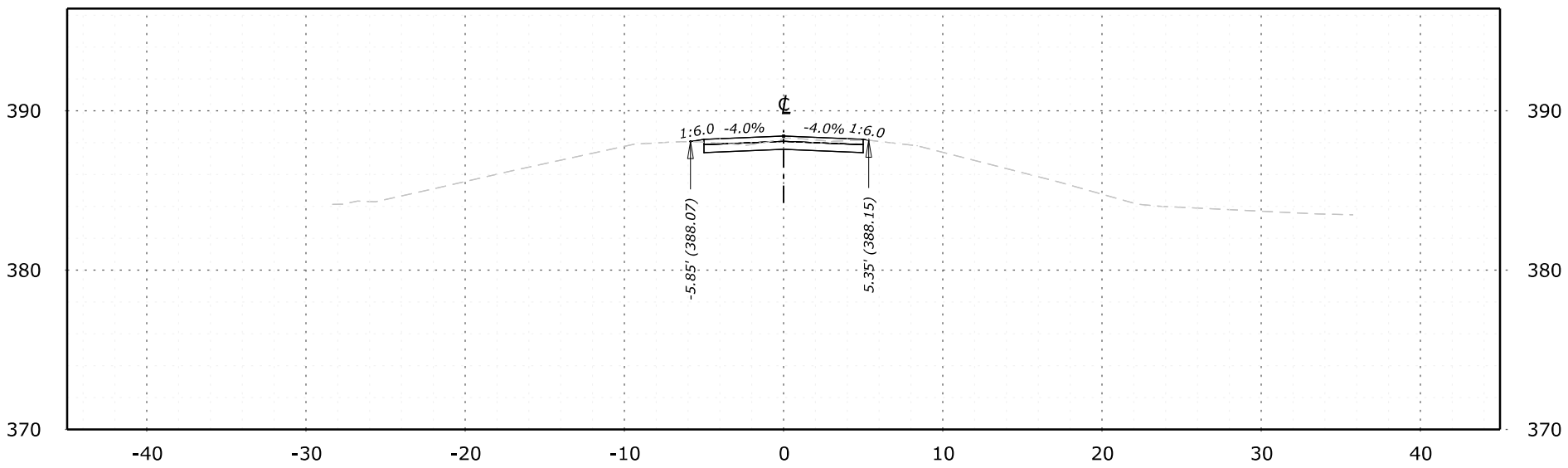
3 August 2026 5:02 PM

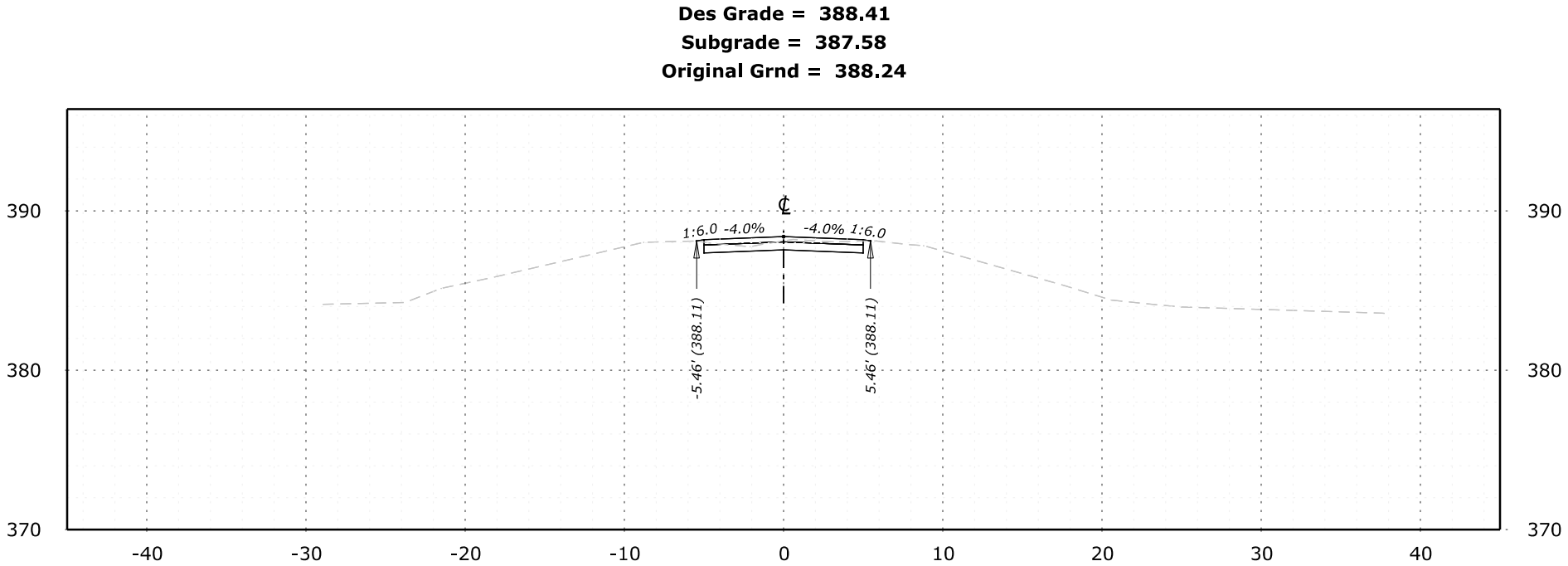
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X02



349+50.00
Des Grade = 388.50
Subgrade = 387.66
Original Grnd = 388.31



349+00.00
Des Grade = 388.41
Subgrade = 387.58
Original Grnd = 388.24



348+50.00
Des Grade = 388.39
Subgrade = 387.55
Original Grnd = 388.11

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

348+50 TO 349+50

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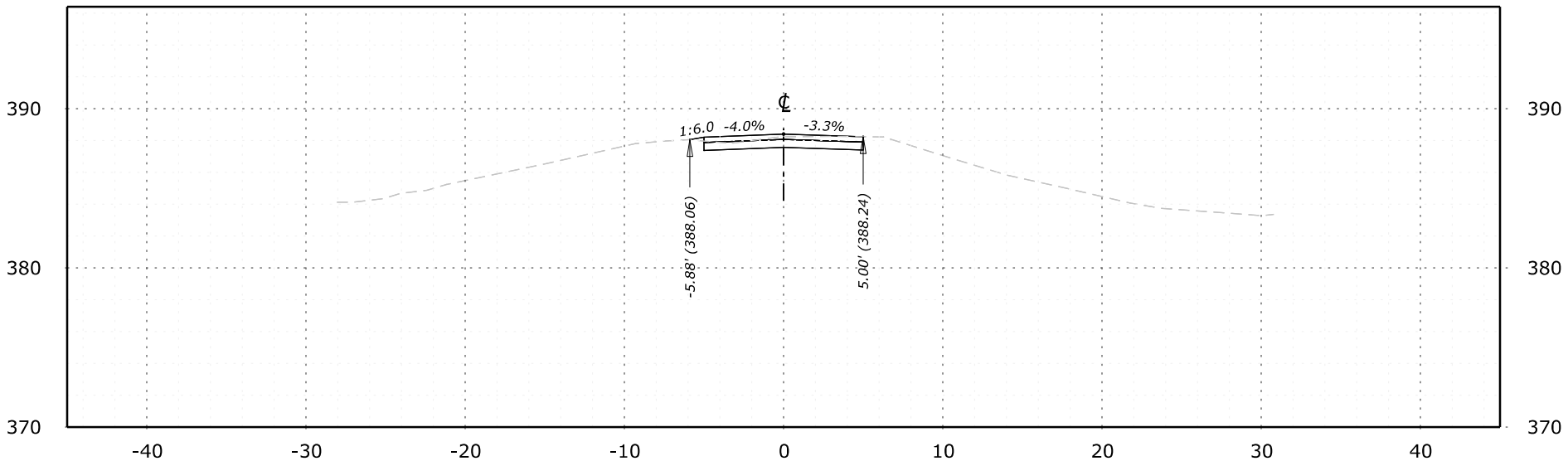
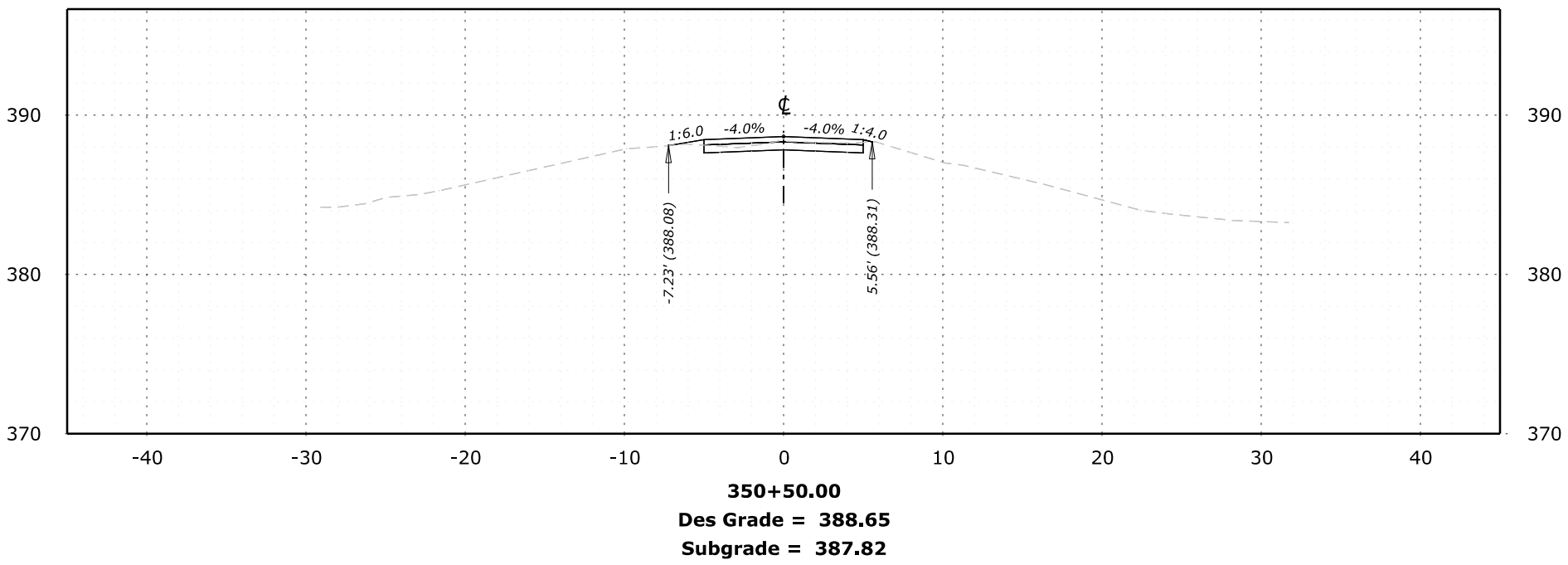
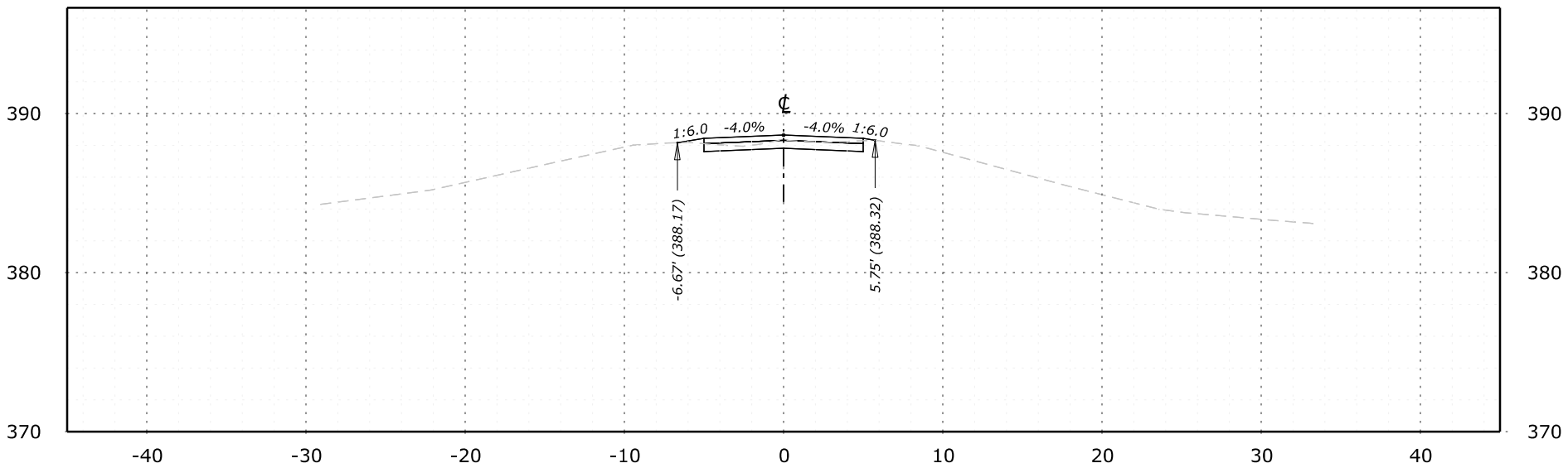
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X03



351+00.00
Des Grade = 388.41
Subgrade = 387.57
Original Grnd = 388.24



350+50.00
Des Grade = 388.65
Subgrade = 387.82
Original Grnd = 388.36



350+00.00
Des Grade = 388.64
Subgrade = 387.81
Original Grnd = 388.34

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

350+00 TO 351+00

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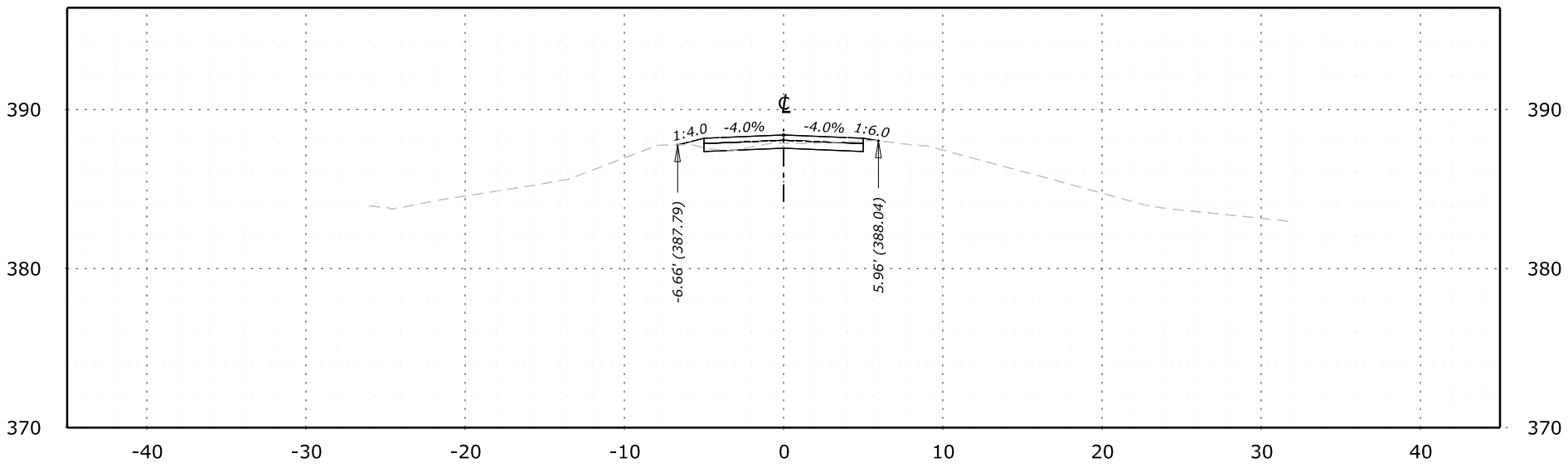
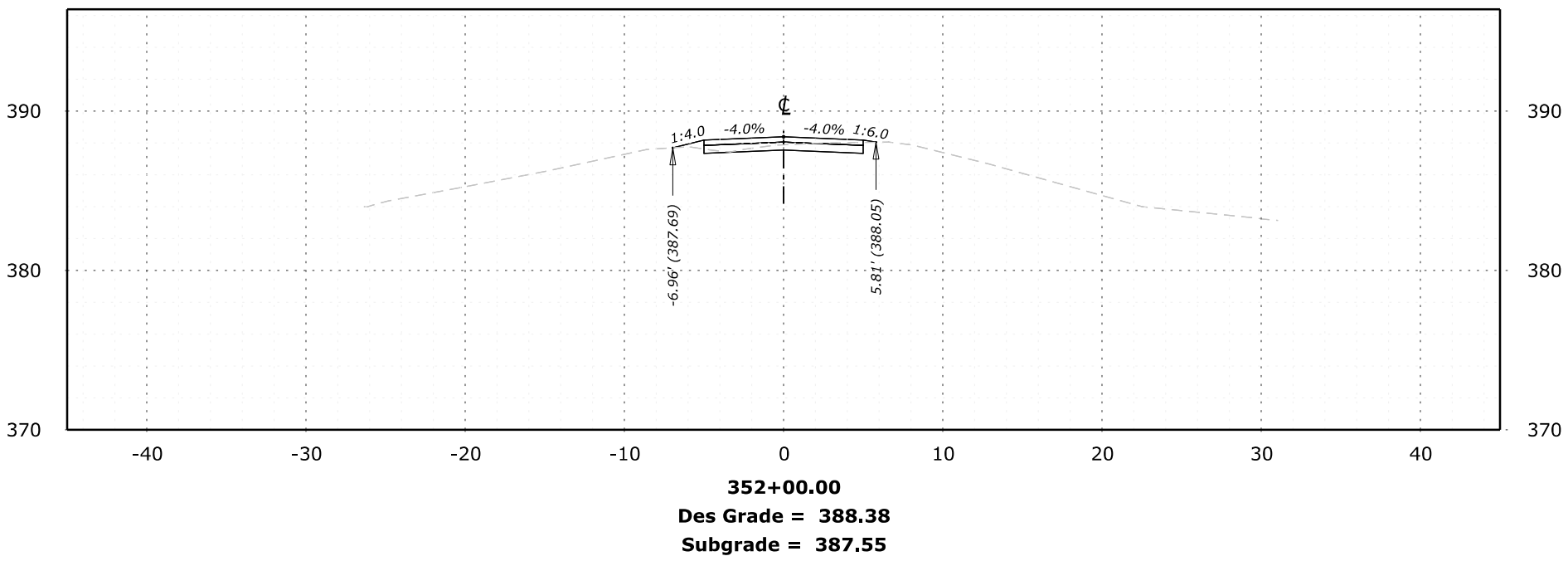
3 August 2026 5:02 PM

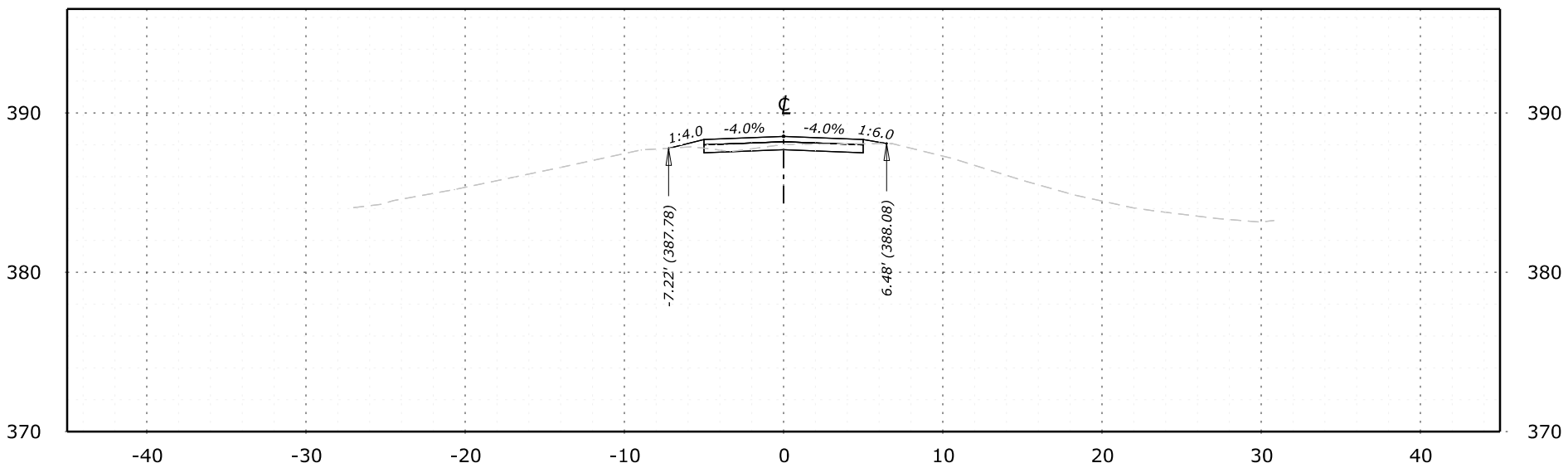
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X04



352+50.00
Des Grade = 388.40
Subgrade = 387.57
Original Grnd = 387.89



352+00.00
Des Grade = 388.38
Subgrade = 387.55
Original Grnd = 387.89



351+50.00
Des Grade = 388.53
Subgrade = 387.70
Original Grnd = 388.01

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

351+50 TO 352+50

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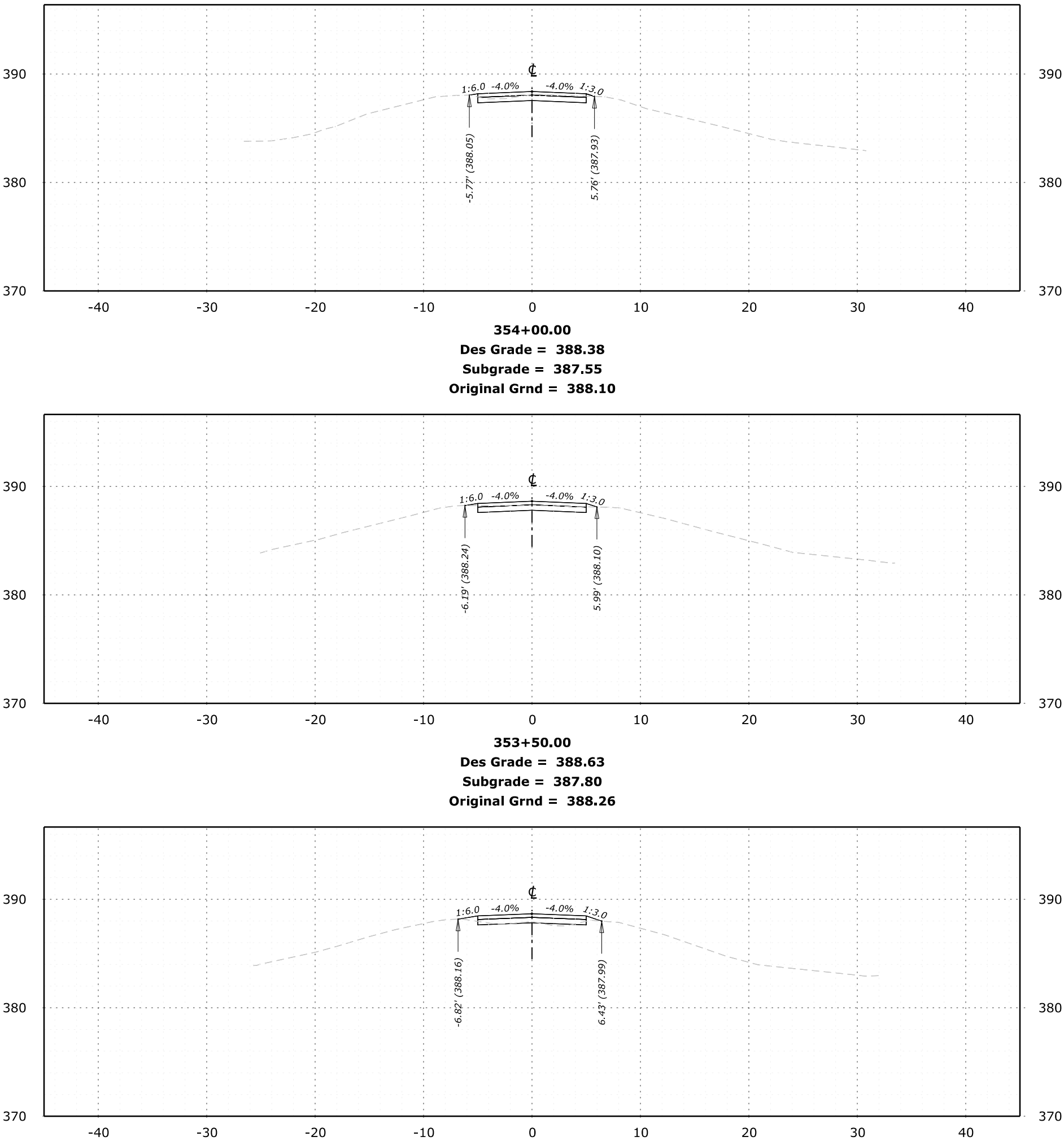
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X05



354+00.00
Des Grade = 388.38
Subgrade = 387.55
Original Grnd = 388.10

353+50.00
Des Grade = 388.63
Subgrade = 387.80
Original Grnd = 388.26

353+00.00
Des Grade = 388.67
Subgrade = 387.83
Original Grnd = 387.93

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

353+00 TO 354+00

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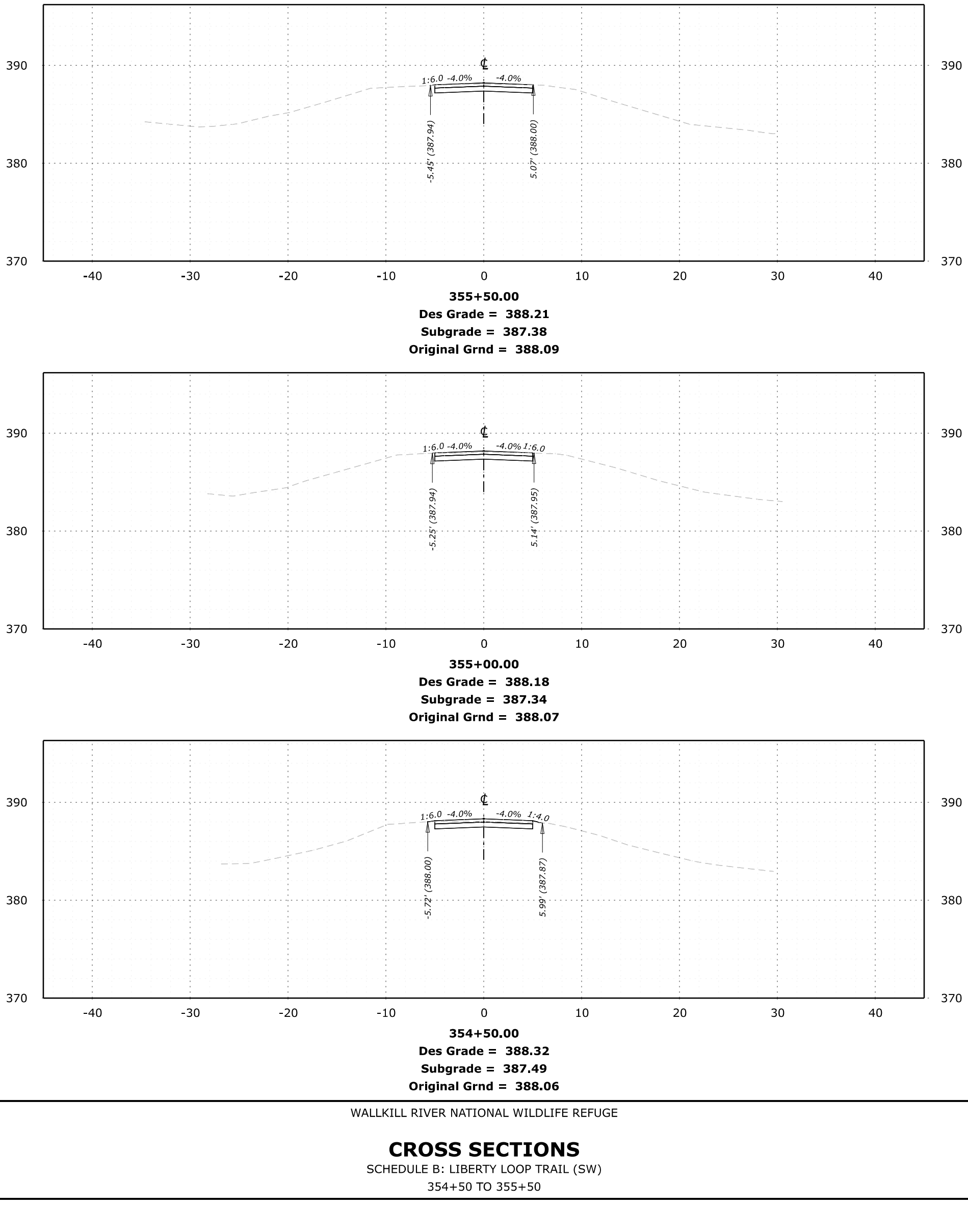
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X06



355+50.00
Des Grade = 388.21
Subgrade = 387.38
Original Grnd = 388.09

355+00.00
Des Grade = 388.18
Subgrade = 387.34
Original Grnd = 388.07

354+50.00
Des Grade = 388.32
Subgrade = 387.49
Original Grnd = 388.06

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

354+50 TO 355+50

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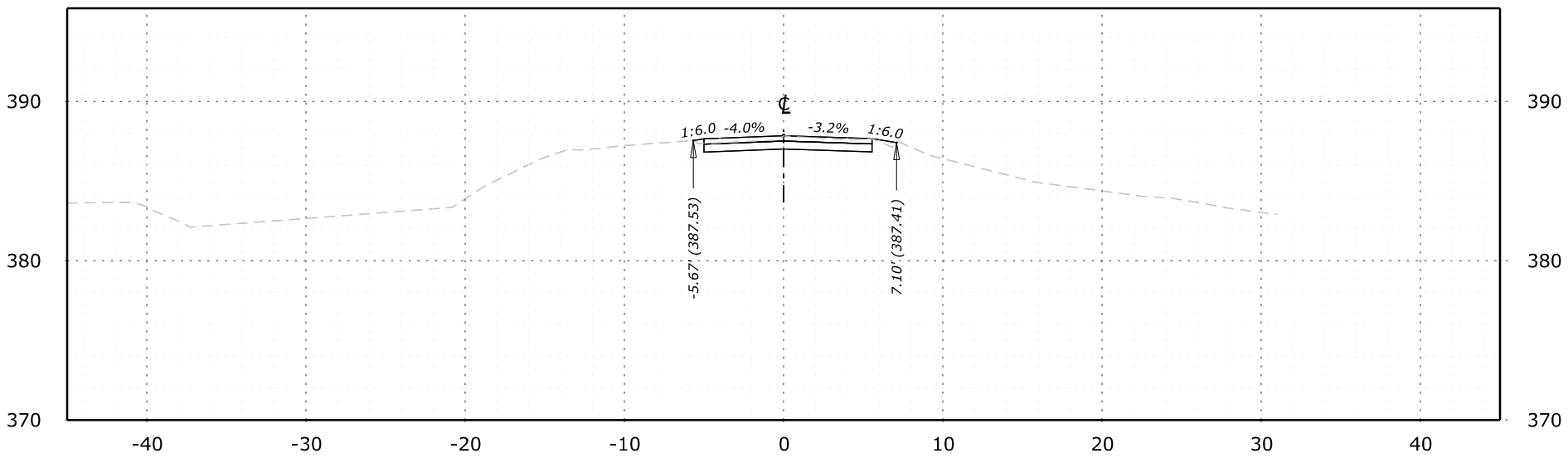
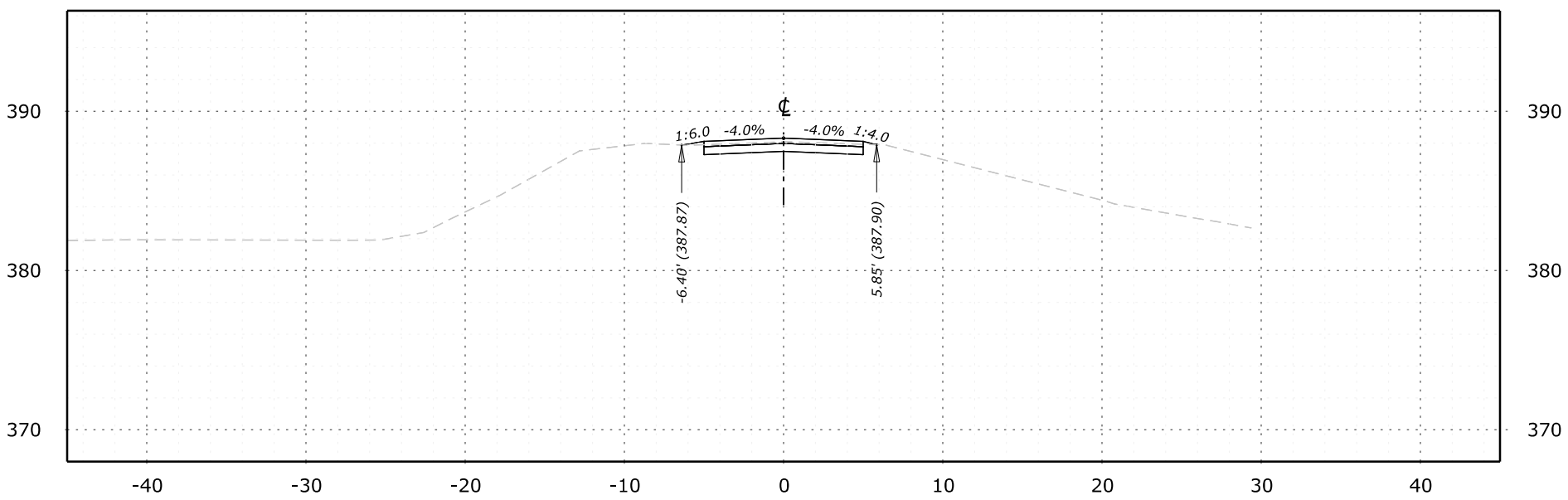
3 August 2026 5:02 PM

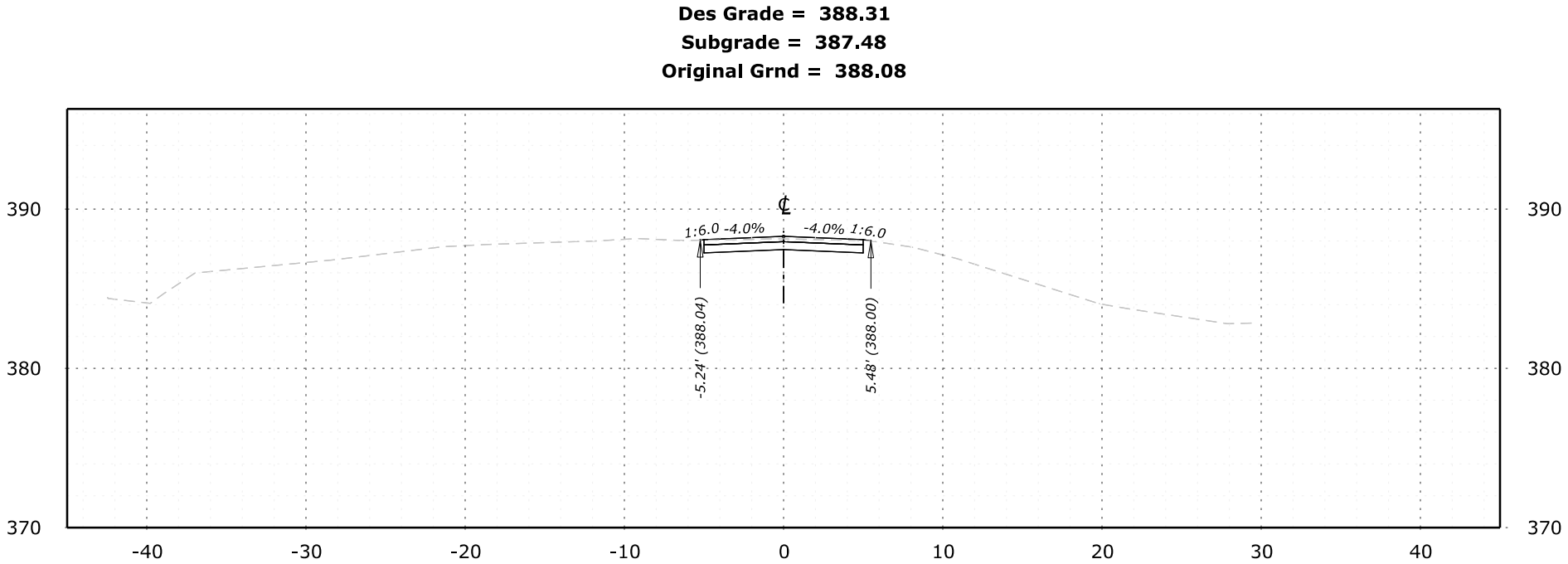
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X07



357+00.00
Des Grade = 387.84
Subgrade = 387.01
Original Grnd = 387.83



356+50.00
Des Grade = 388.31
Subgrade = 387.48
Original Grnd = 388.08



356+00.00
Des Grade = 388.28
Subgrade = 387.45
Original Grnd = 388.12

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

356+00 TO 357+00

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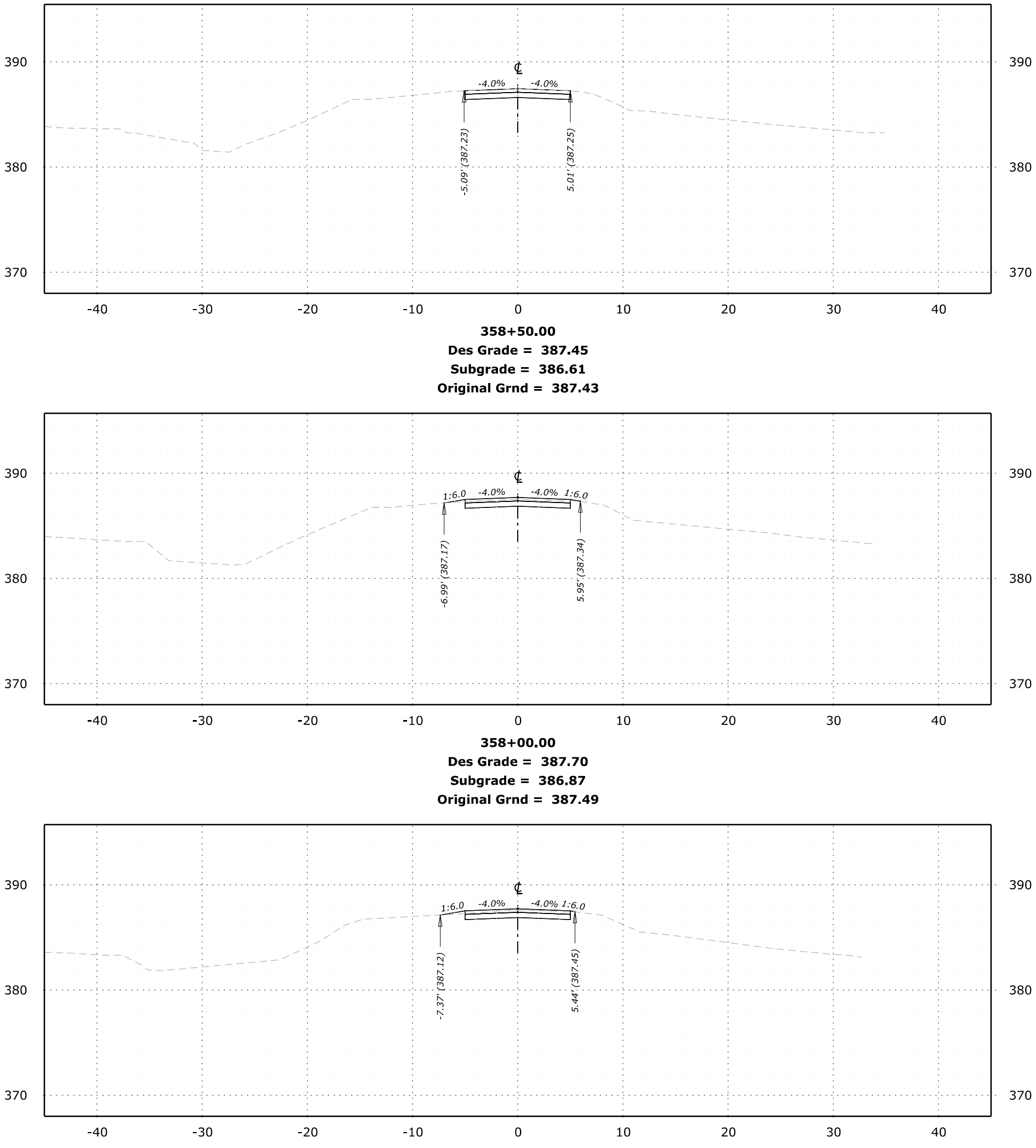
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X08



358+50.00
Des Grade = 387.45
Subgrade = 386.61
Original Grnd = 387.43

358+00.00
Des Grade = 387.70
Subgrade = 386.87
Original Grnd = 387.49

357+50.00
Des Grade = 387.72
Subgrade = 386.89
Original Grnd = 387.59

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

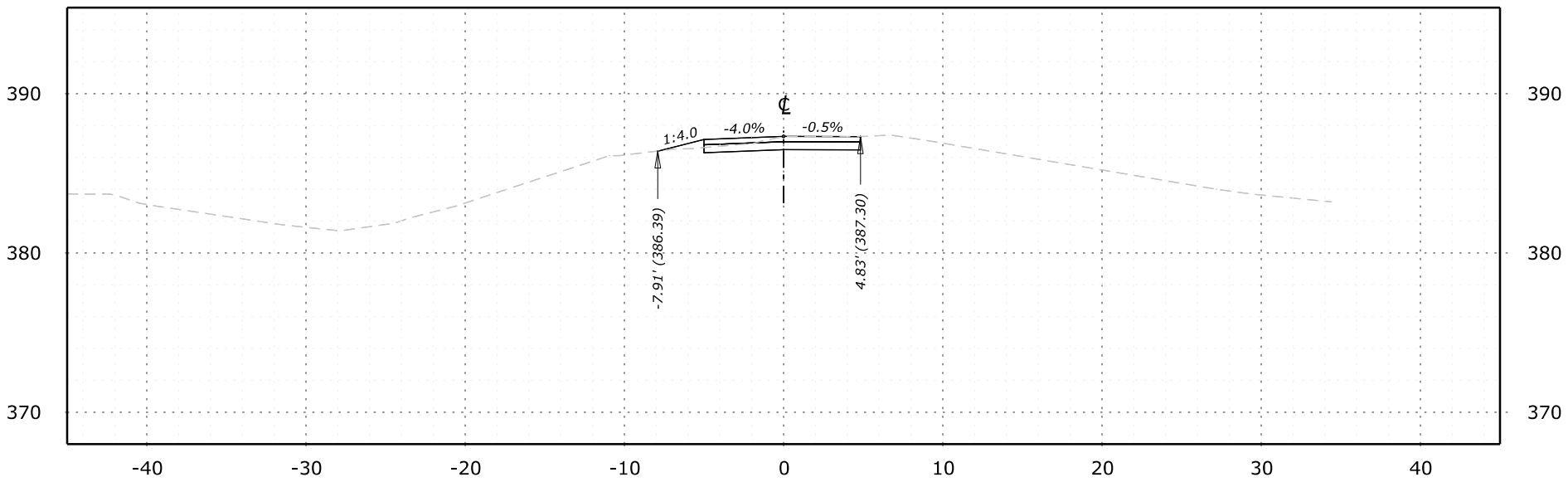
CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

357+50 TO 358+50

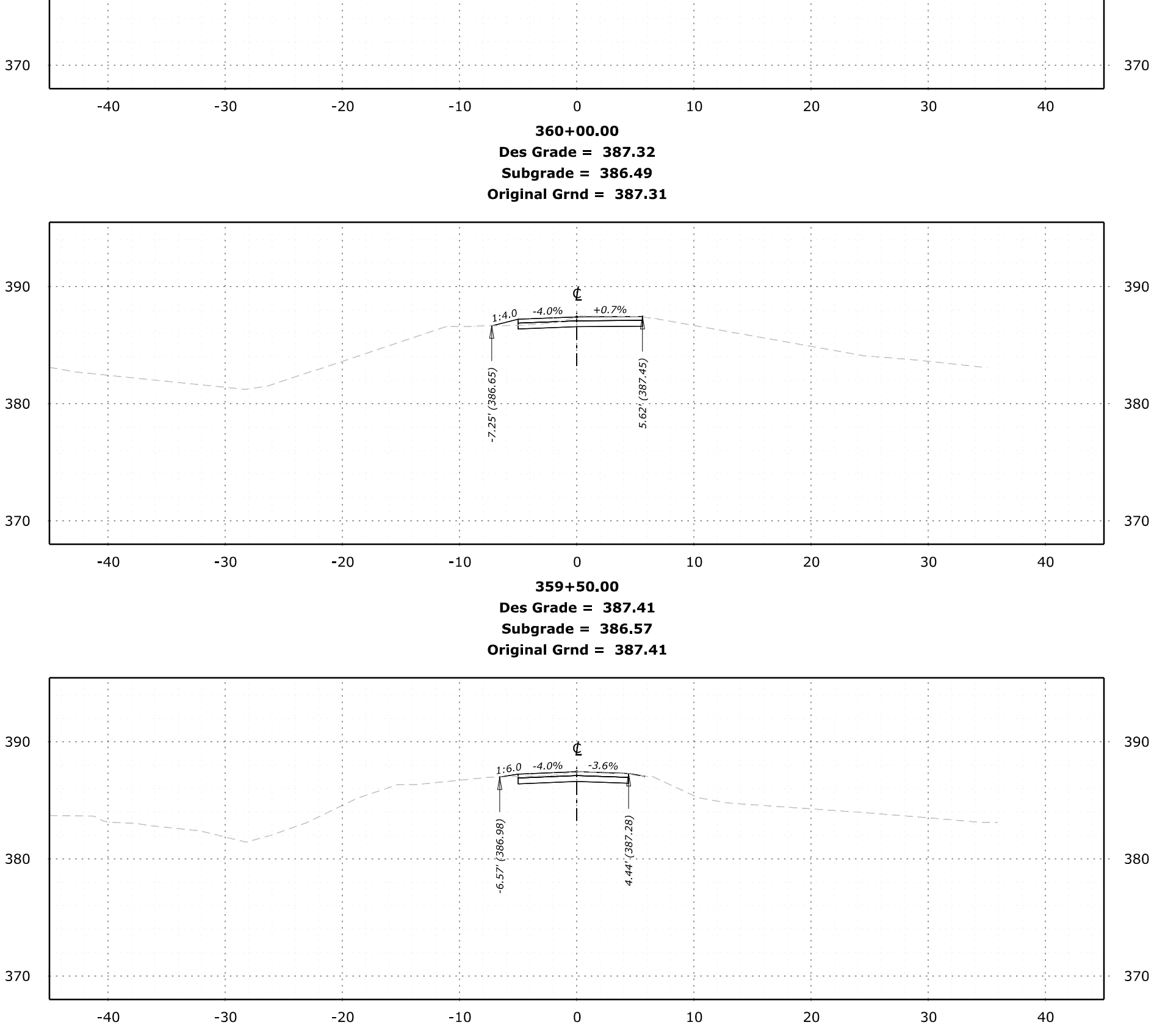
\\fhfl15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\ZXX_NJ-WARI_2019-1(1)_LibertyLoop_xc.dgn [LL - 357+50.00 [Sheet]]
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X09



360+00.00
Des Grade = 387.32
Subgrade = 386.49
Original Grnd = 387.31

359+50.00
Des Grade = 387.41
Subgrade = 386.57
Original Grnd = 387.41



359+00.00
Des Grade = 387.44
Subgrade = 386.61
Original Grnd = 387.44

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

359+00 TO 360+00

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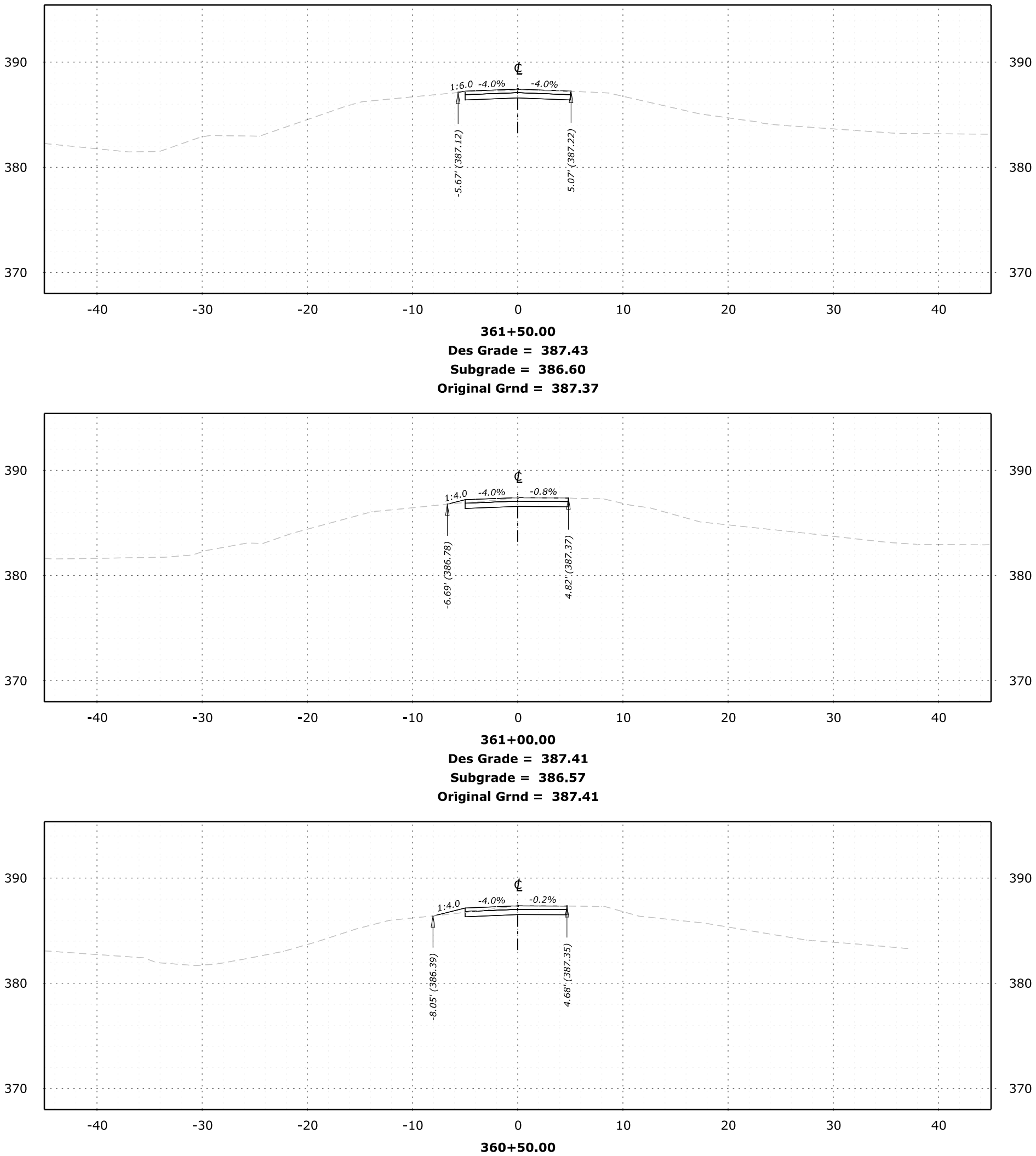
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X10



361+50.00
Des Grade = 387.43
Subgrade = 386.60
Original Grnd = 387.37

361+00.00
Des Grade = 387.41
Subgrade = 386.57
Original Grnd = 387.41

360+50.00
Des Grade = 387.36
Subgrade = 386.52
Original Grnd = 387.36

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

360+50 TO 361+50

\\fhfl15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\ZXX_NJ-WARI_2019-1(1)_LibertyLoop_xc.dgn [LL - 360+50.00 [Sheet]]
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PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X11

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

361+60 TO 362+10

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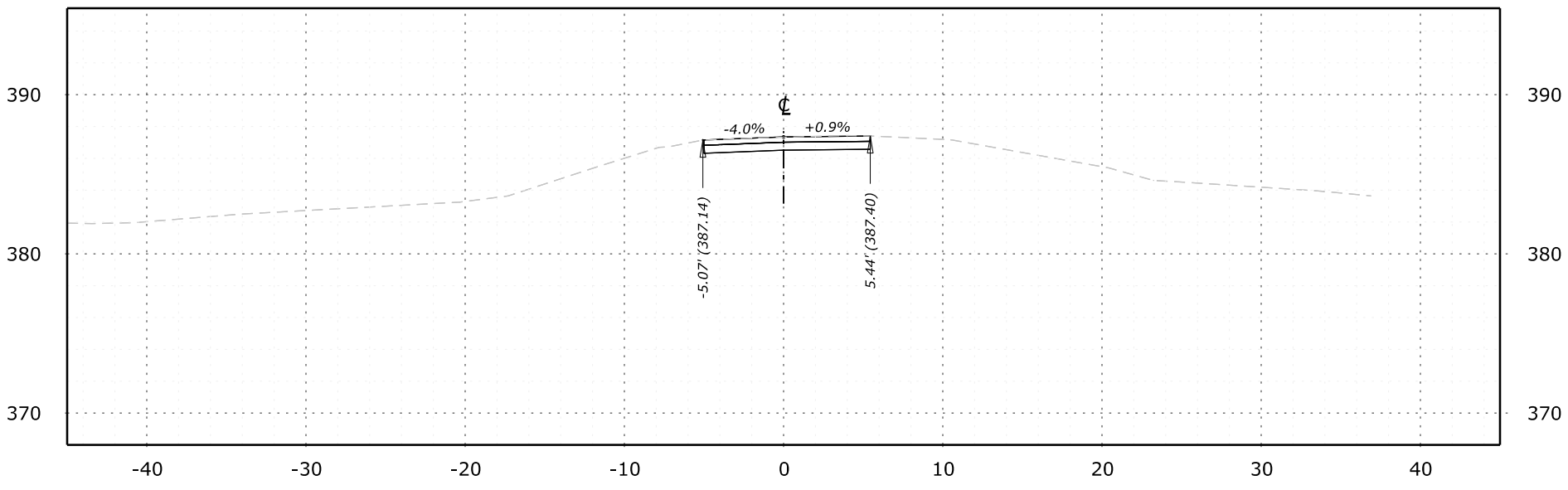
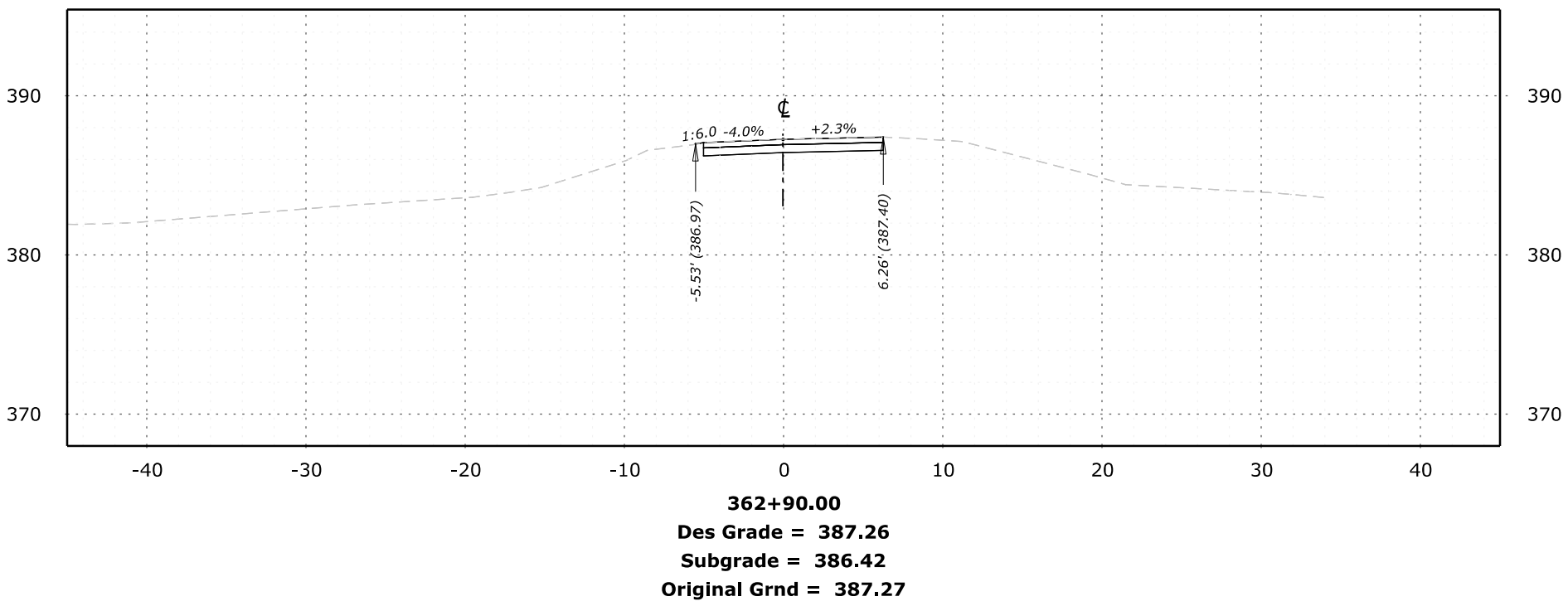
3 August 2026 5:02 PM

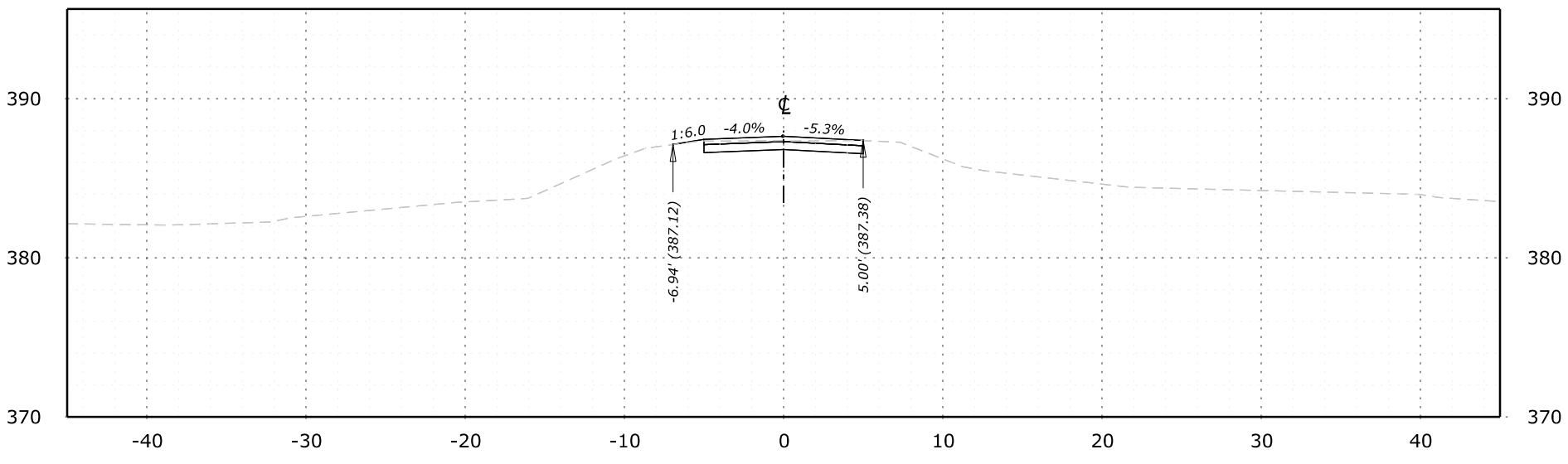
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X12



363+00.00
Des Grade = 387.36
Subgrade = 386.52
Original Grnd = 387.34



362+90.00
Des Grade = 387.26
Subgrade = 386.42
Original Grnd = 387.27



362+50.00
Des Grade = 387.64
Subgrade = 386.81
Original Grnd = 387.35

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

362+50 TO 363+00

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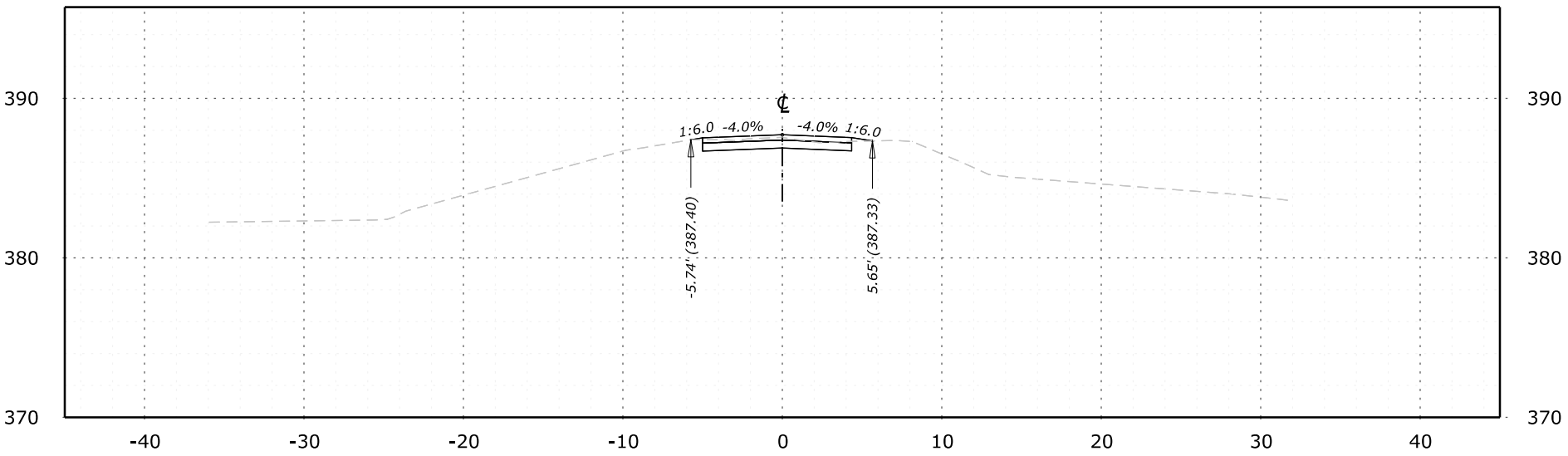
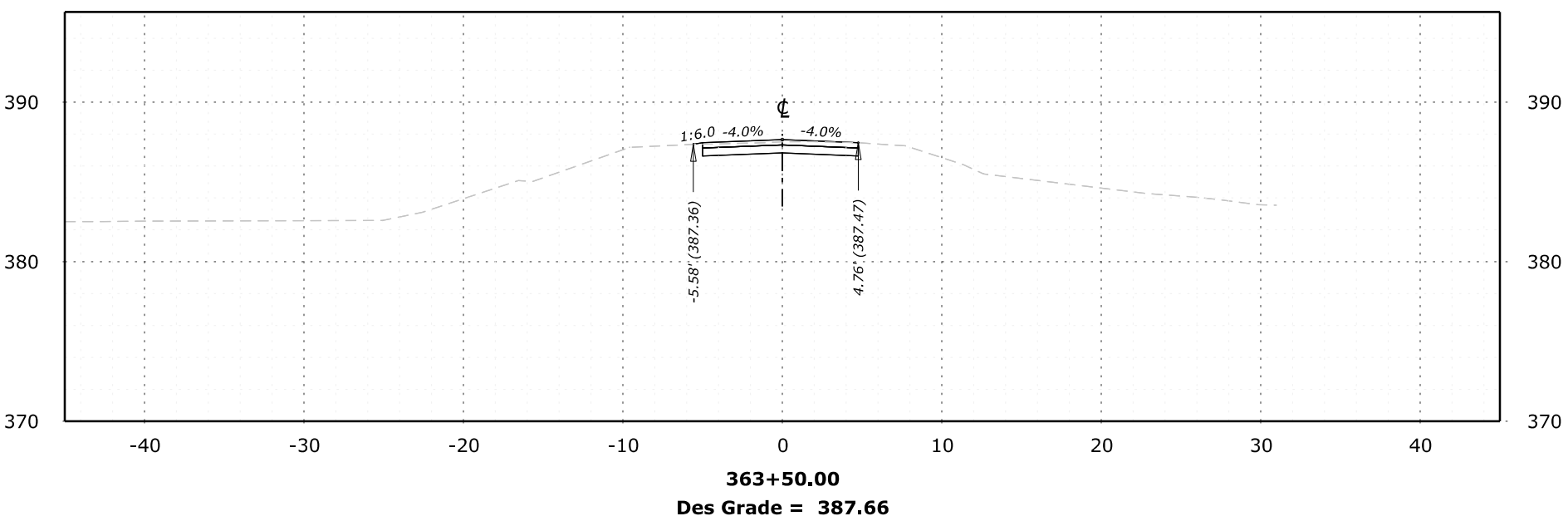
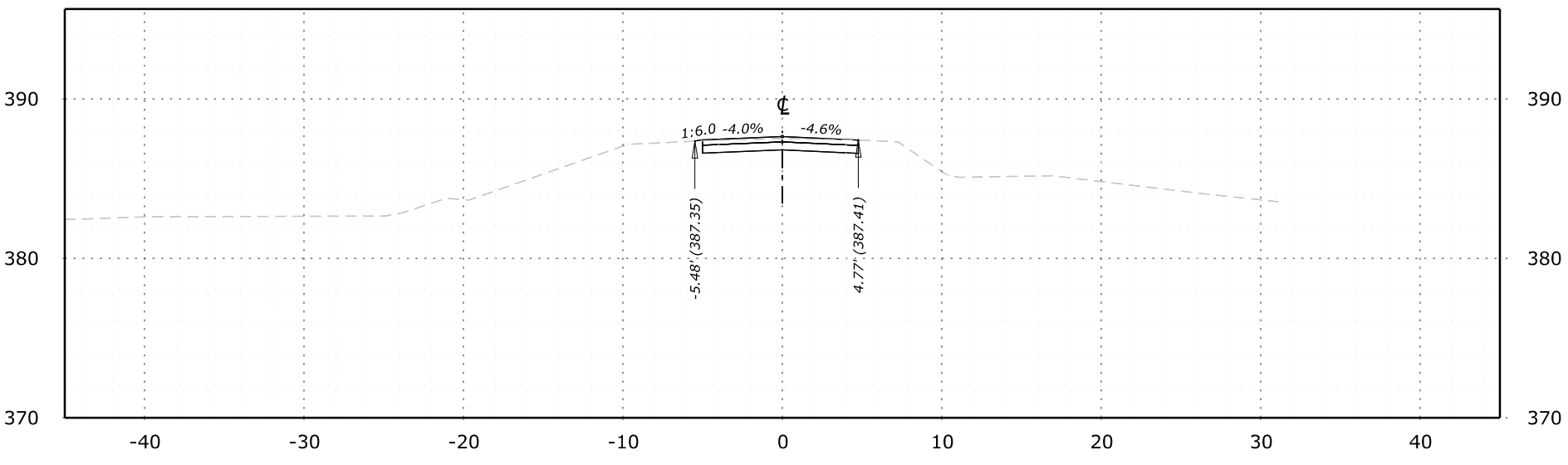
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X13



364+00.00
Des Grade = 387.72
Subgrade = 386.89
Original Grnd = 387.54



363+50.00
Des Grade = 387.66
Subgrade = 386.83
Original Grnd = 387.52



363+28.00
Des Grade = 387.63
Subgrade = 386.80
Original Grnd = 387.51

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

363+28 TO 364+00

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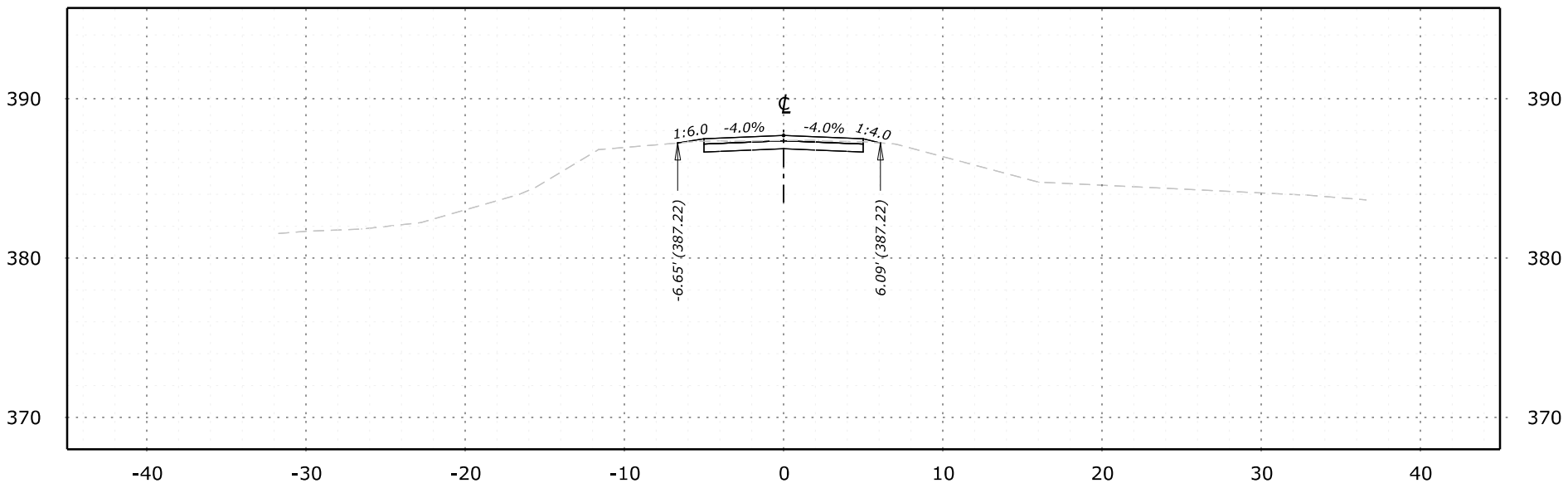
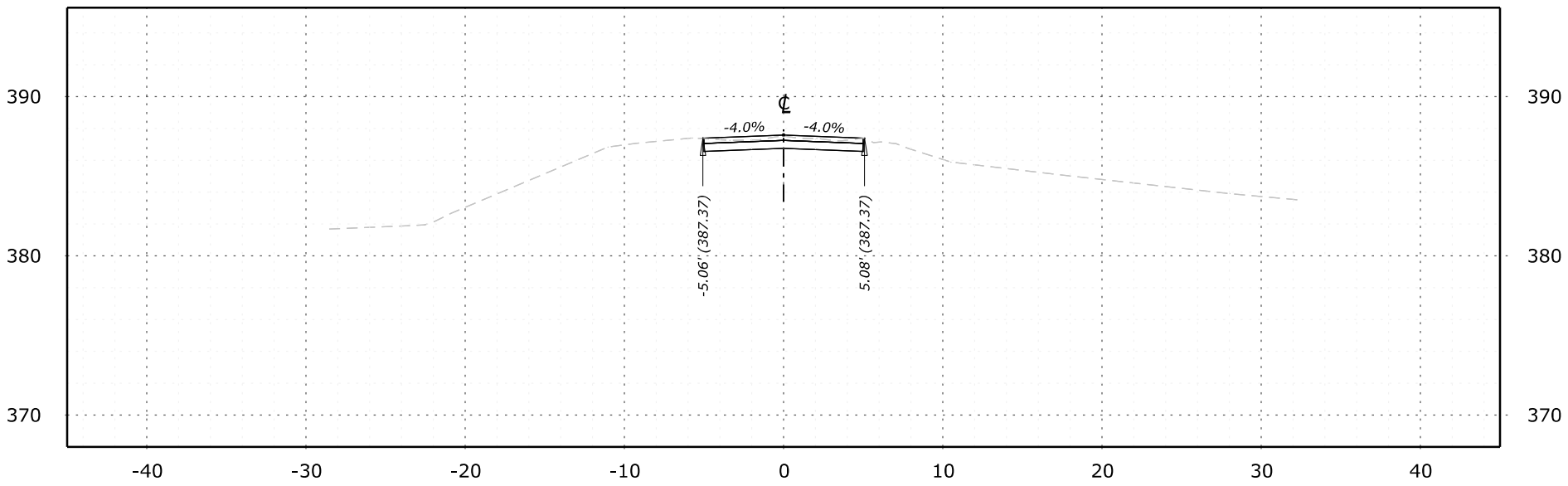
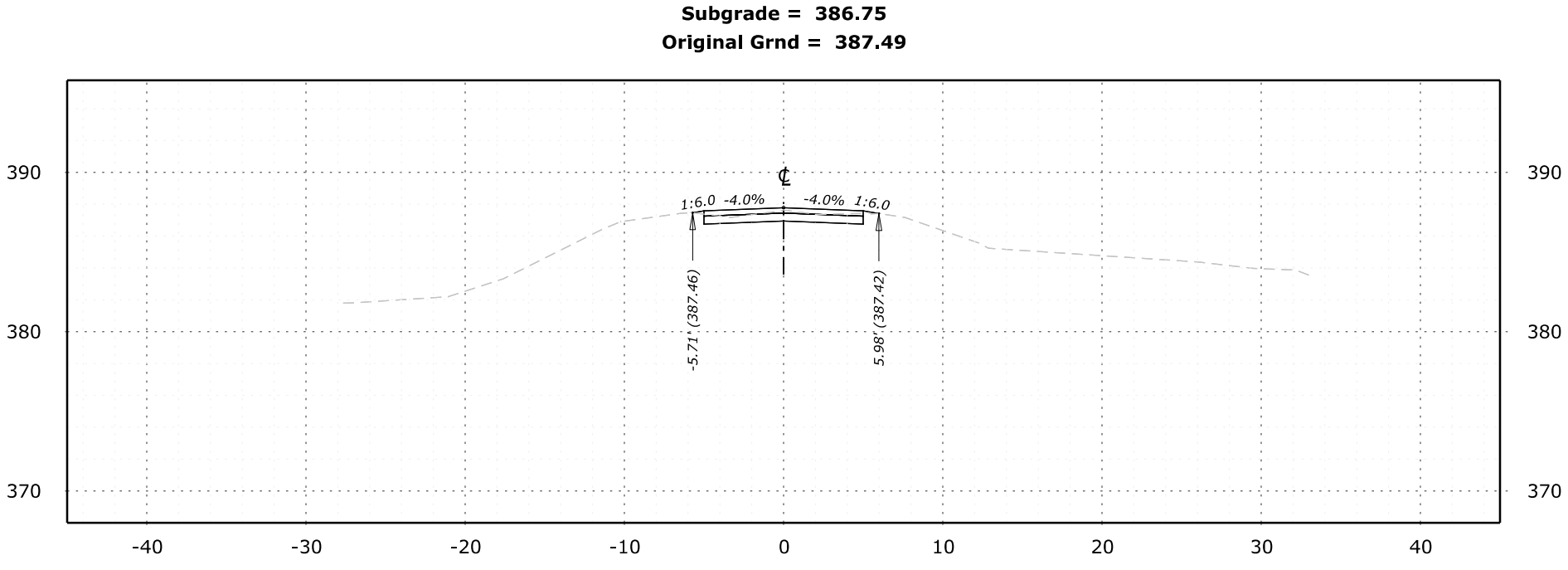
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X14



365+50.00
Des Grade = 387.69
Subgrade = 386.86
Original Grnd = 387.38



365+00.00
Des Grade = 387.58
Subgrade = 386.75
Original Grnd = 387.49



364+50.00
Des Grade = 387.78
Subgrade = 386.95
Original Grnd = 387.60

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

364+50 TO 365+50

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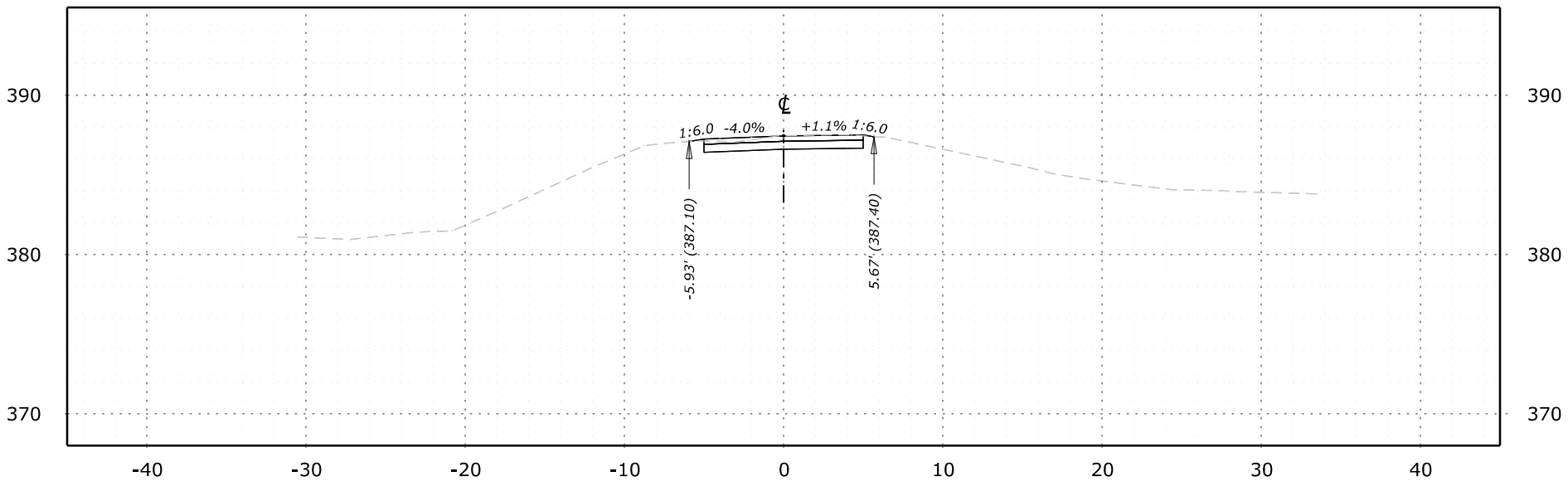
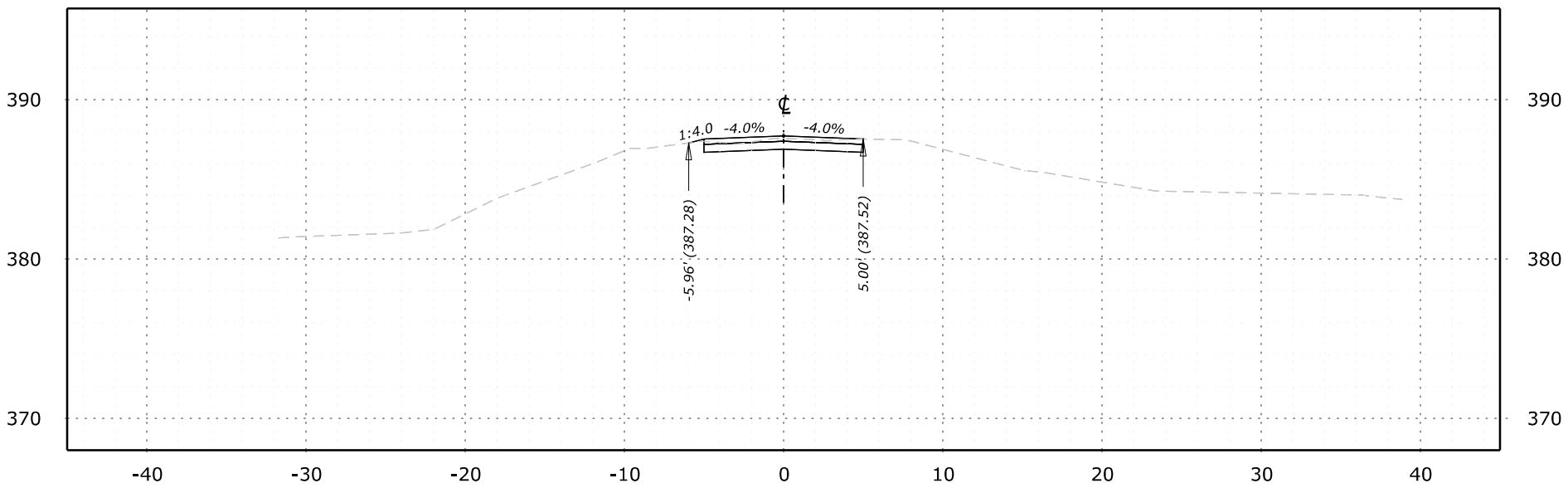
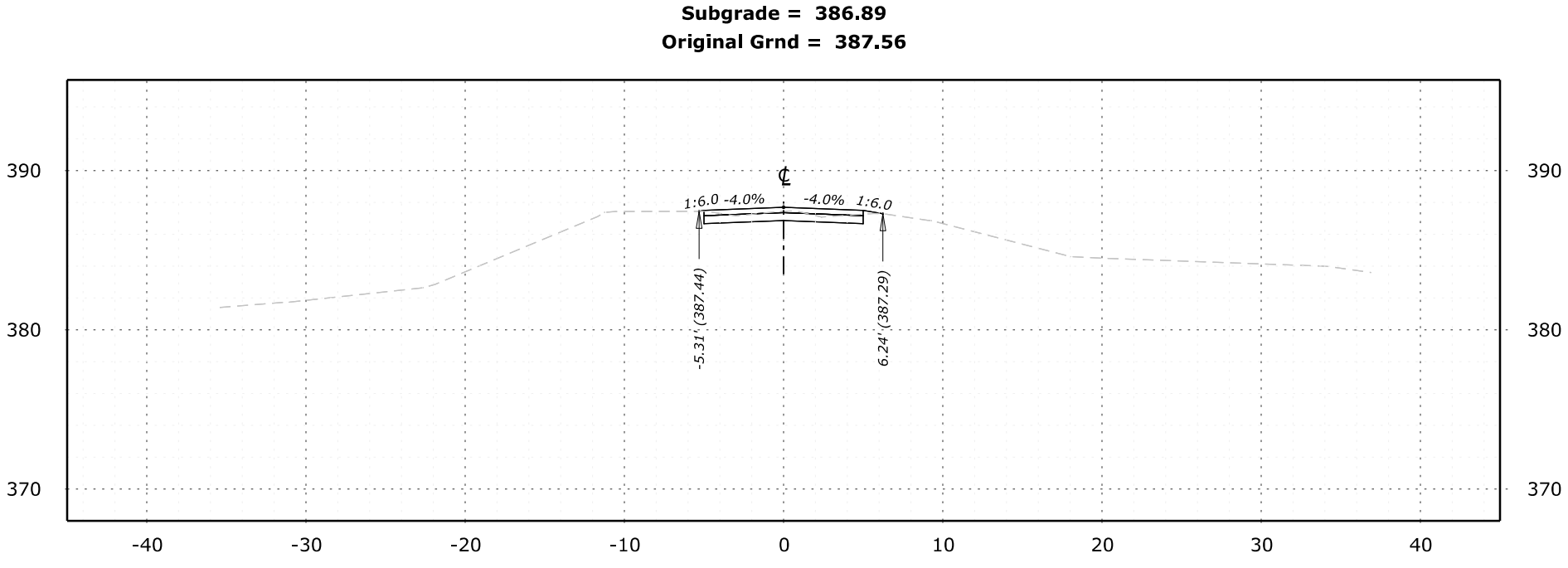
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X15



367+00.00
Des Grade = 387.45
Subgrade = 386.62
Original Grnd = 387.45



366+50.00
Des Grade = 387.72
Subgrade = 386.89
Original Grnd = 387.56



366+00.00
Des Grade = 387.69
Subgrade = 386.86
Original Grnd = 387.45

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

366+00 TO 367+00

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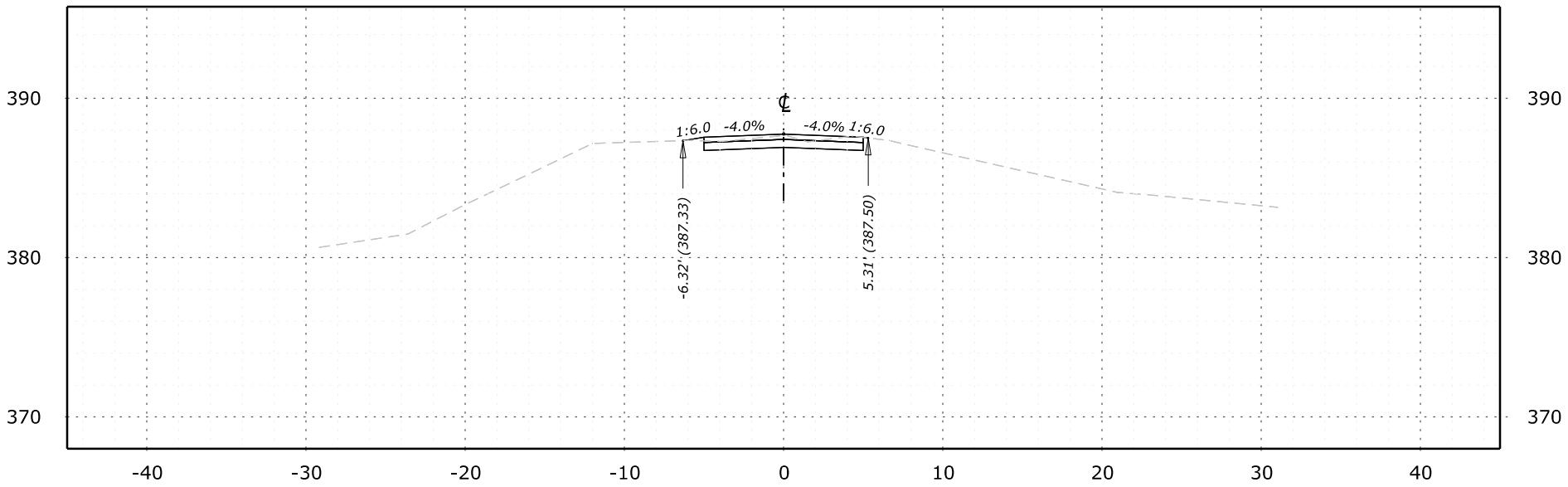
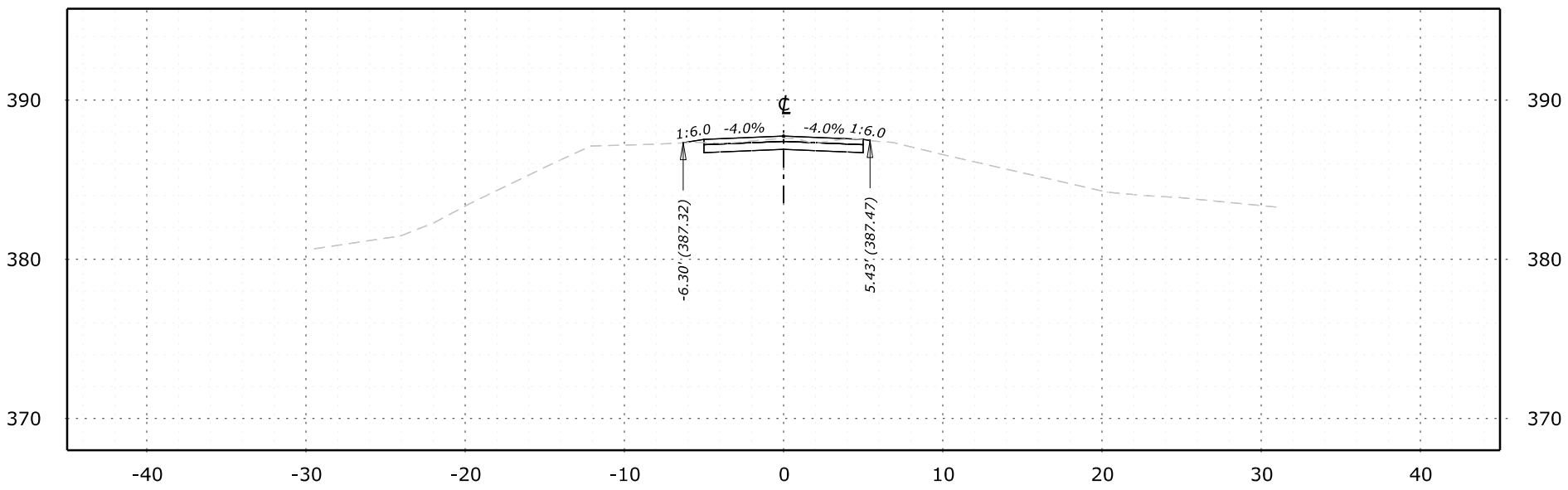
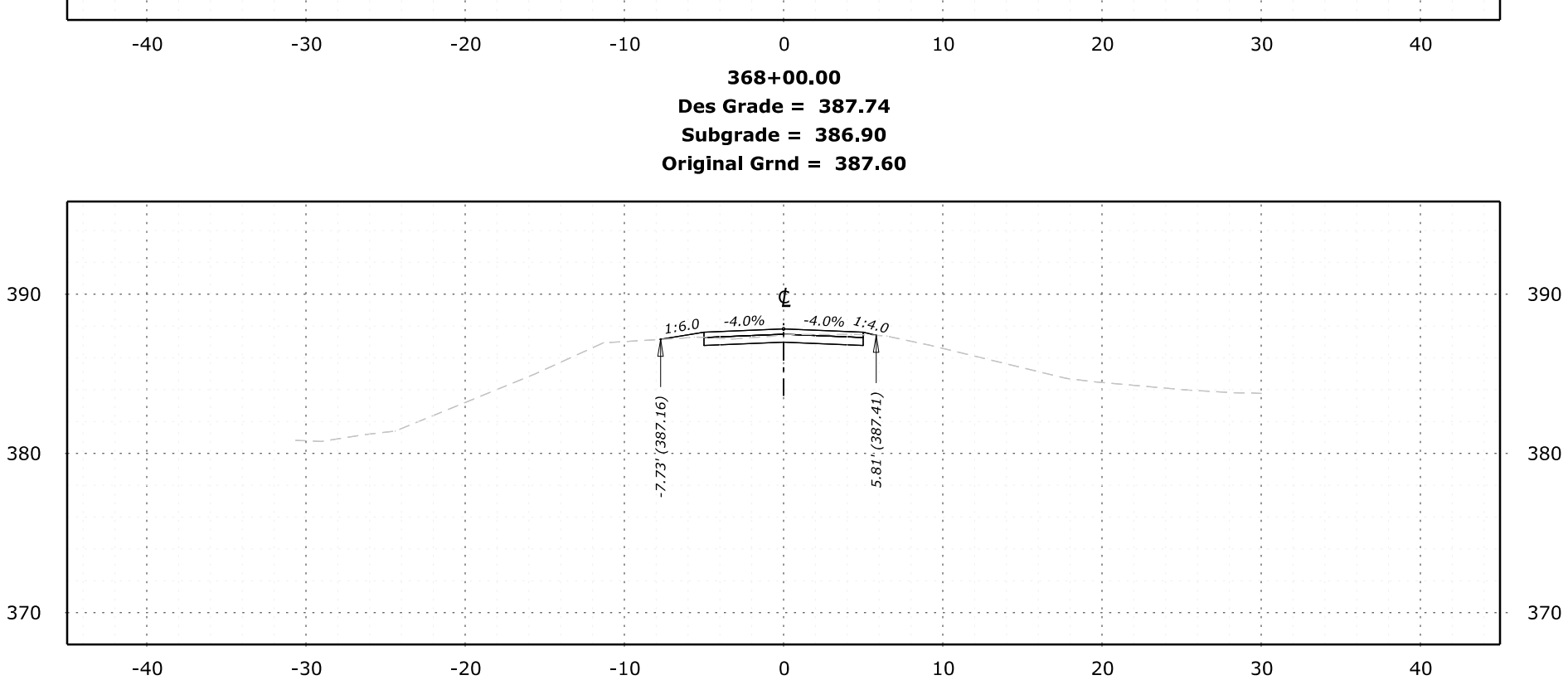
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X16



368+13.00
Des Grade = 387.75
Subgrade = 386.92
Original Grnd = 387.65



368+00.00
Des Grade = 387.74
Subgrade = 386.90
Original Grnd = 387.60



367+50.00
Des Grade = 387.81
Subgrade = 386.98
Original Grnd = 387.46

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

367+50 TO 368+13

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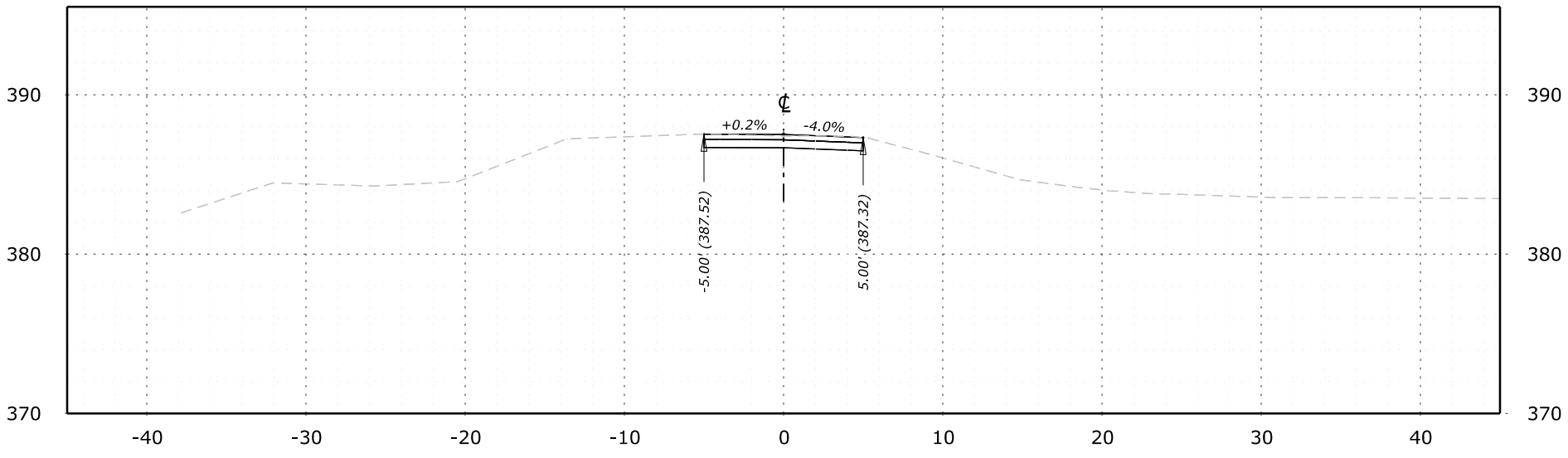
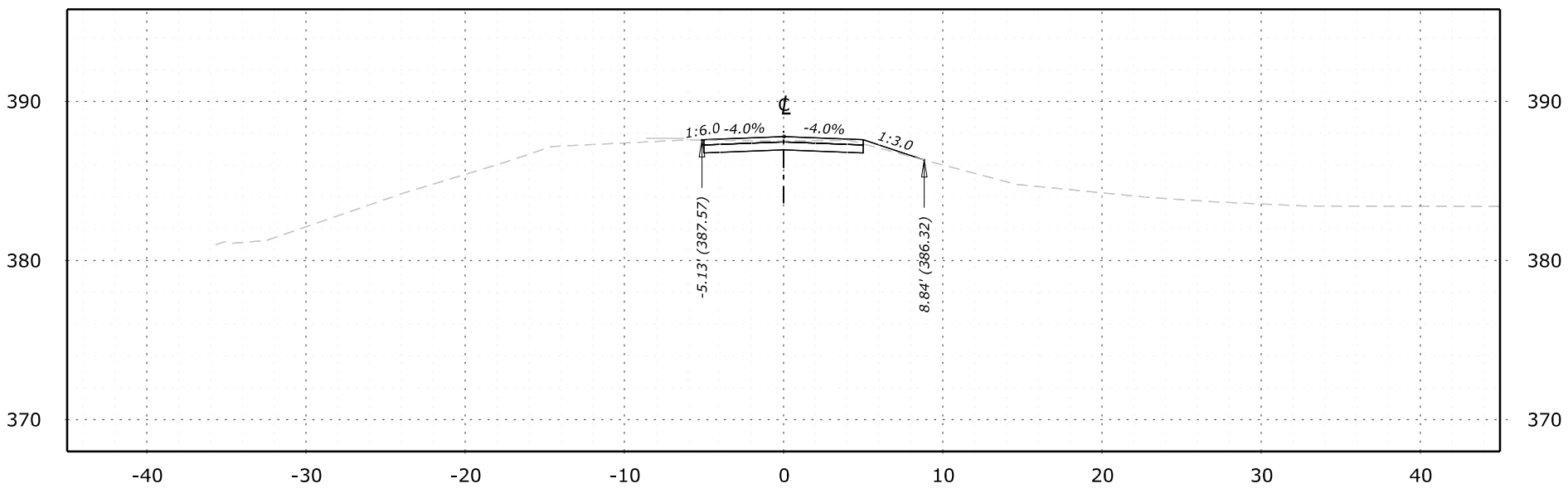
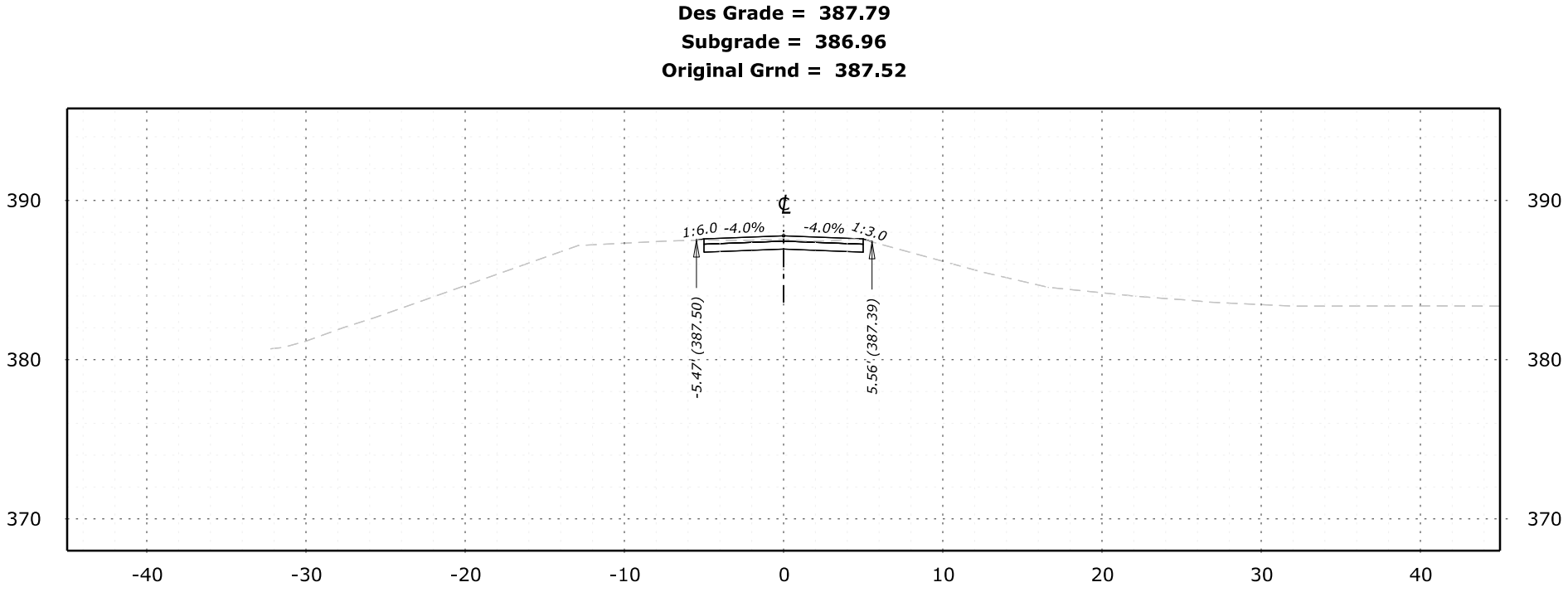
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X17



369+00.00
Des Grade = 387.52
Subgrade = 386.68
Original Grnd = 387.42



368+67.00
Des Grade = 387.79
Subgrade = 386.96
Original Grnd = 387.52



368+50.00
Des Grade = 387.78
Subgrade = 386.95
Original Grnd = 387.57

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

368+50 TO 369+00

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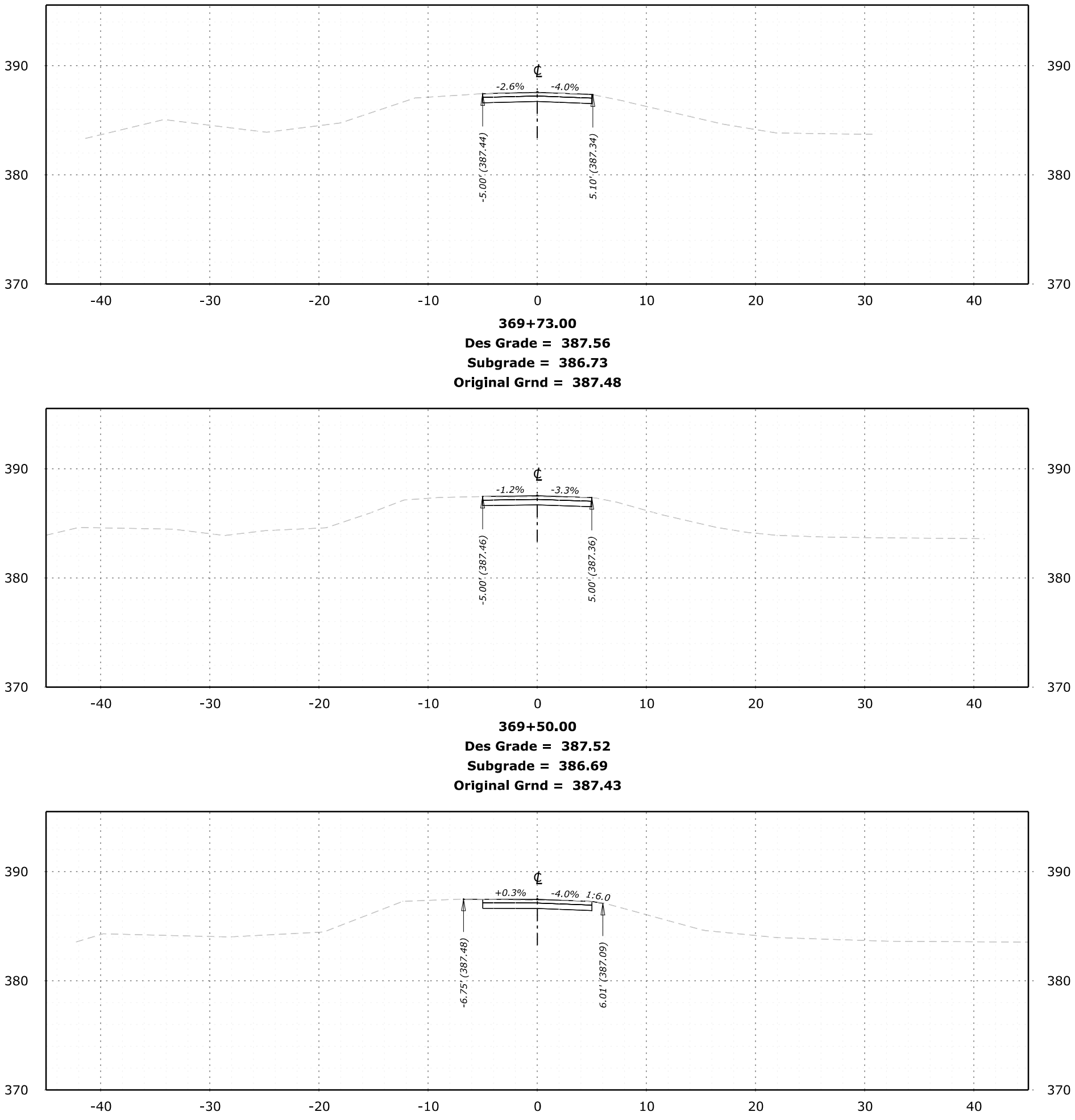
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X18



369+73.00
Des Grade = 387.56
Subgrade = 386.73
Original Grnd = 387.48

369+50.00
Des Grade = 387.52
Subgrade = 386.69
Original Grnd = 387.43

369+18.00
Des Grade = 387.46
Subgrade = 386.63
Original Grnd = 387.36

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

369+18 TO 369+73

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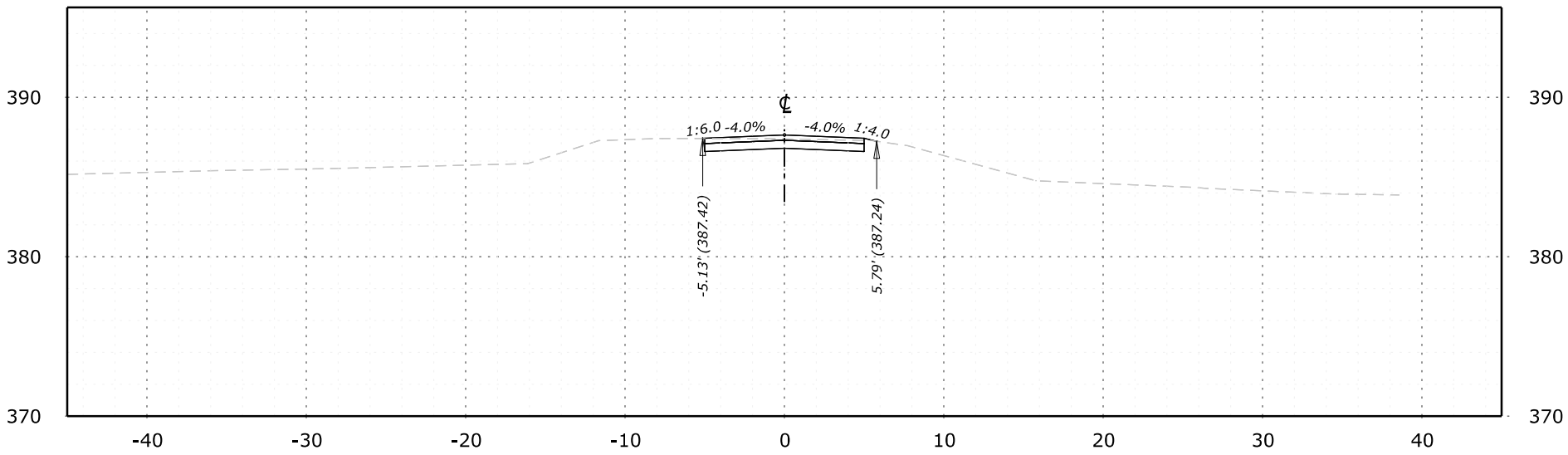
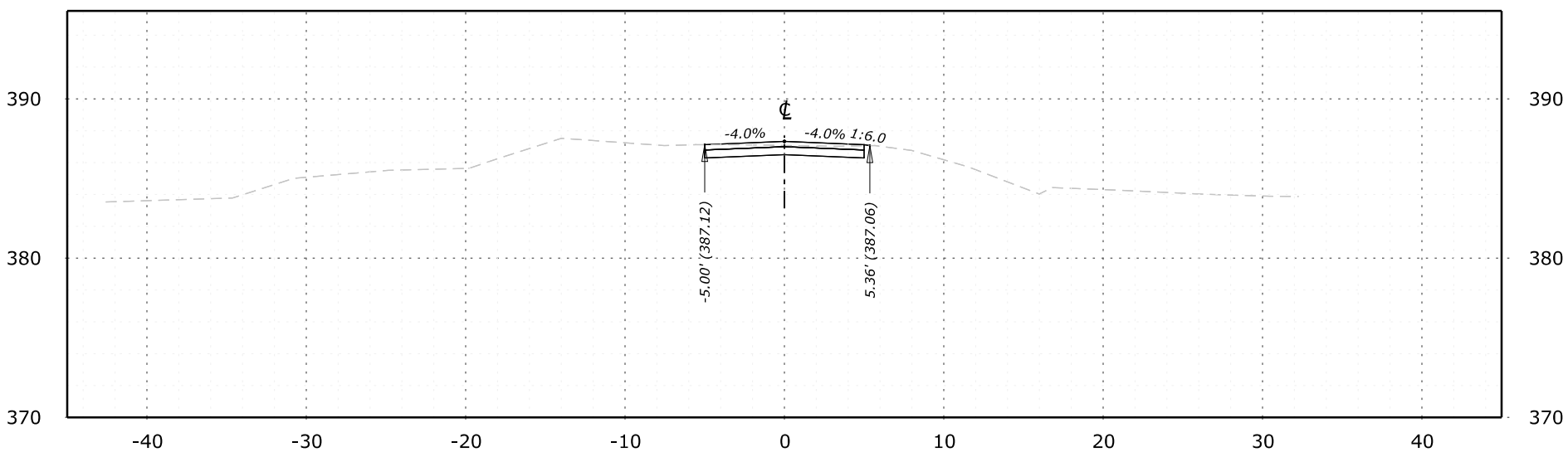
3 August 2026 5:02 PM

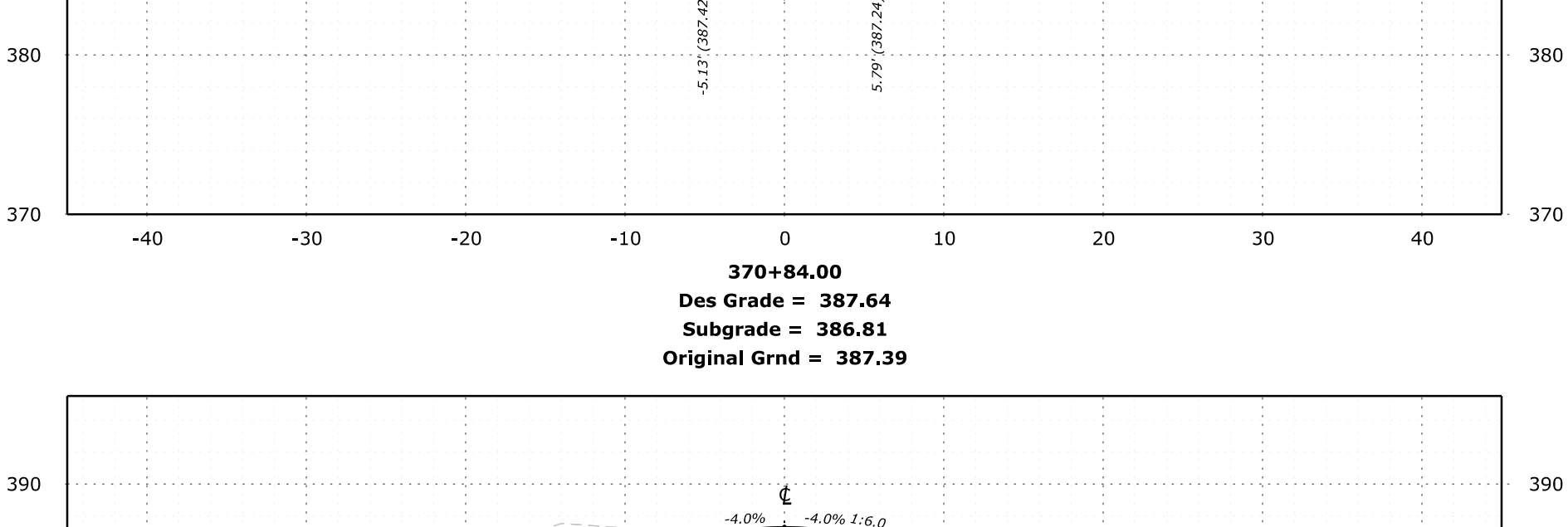
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X19



370+84.00
Des Grade = 387.64
Subgrade = 386.81
Original Grnd = 387.39



370+50.00
Des Grade = 387.32
Subgrade = 386.49
Original Grnd = 387.07



370+00.00
Des Grade = 387.36
Subgrade = 386.53
Original Grnd = 387.13

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

370+00 TO 370+84

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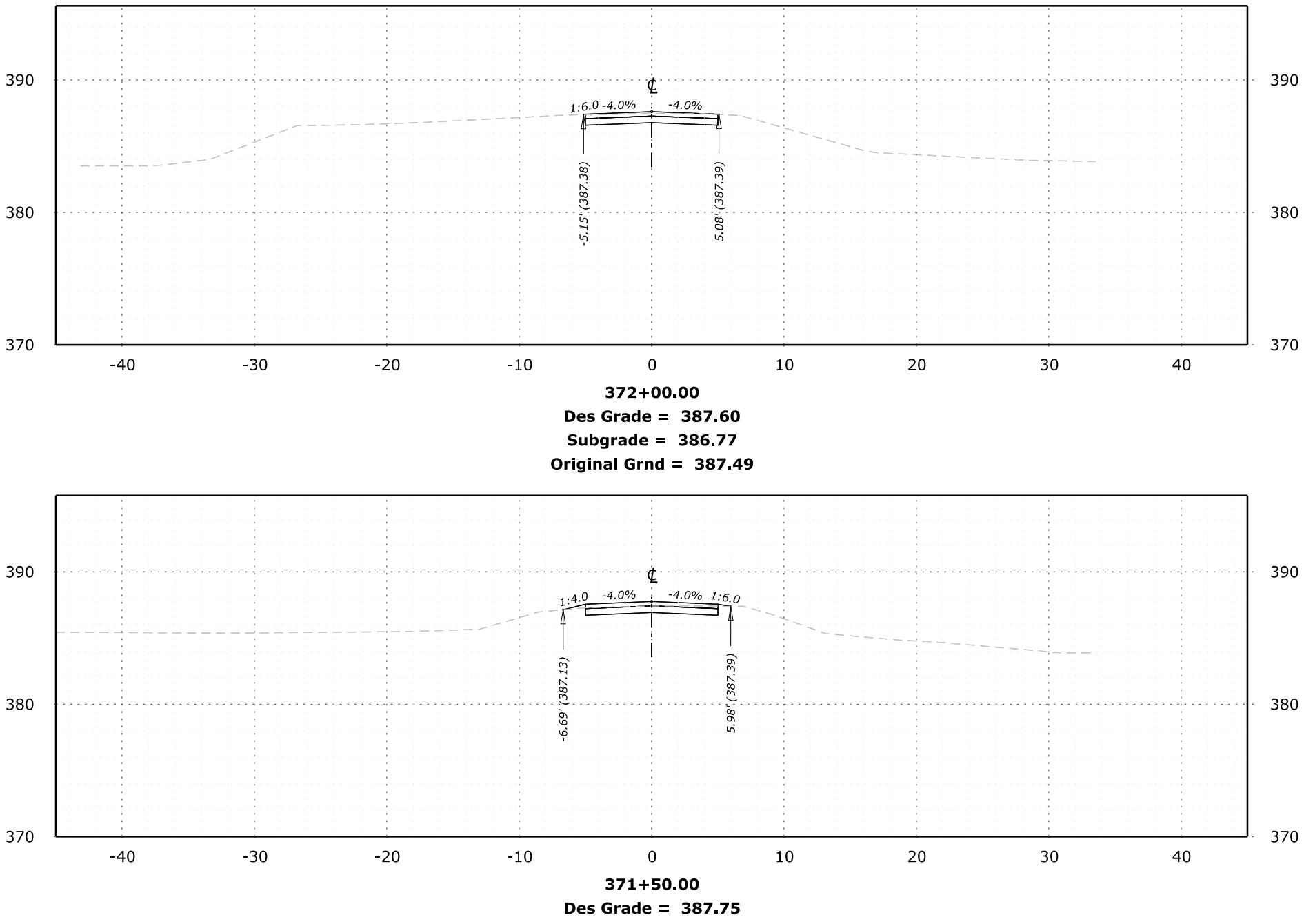
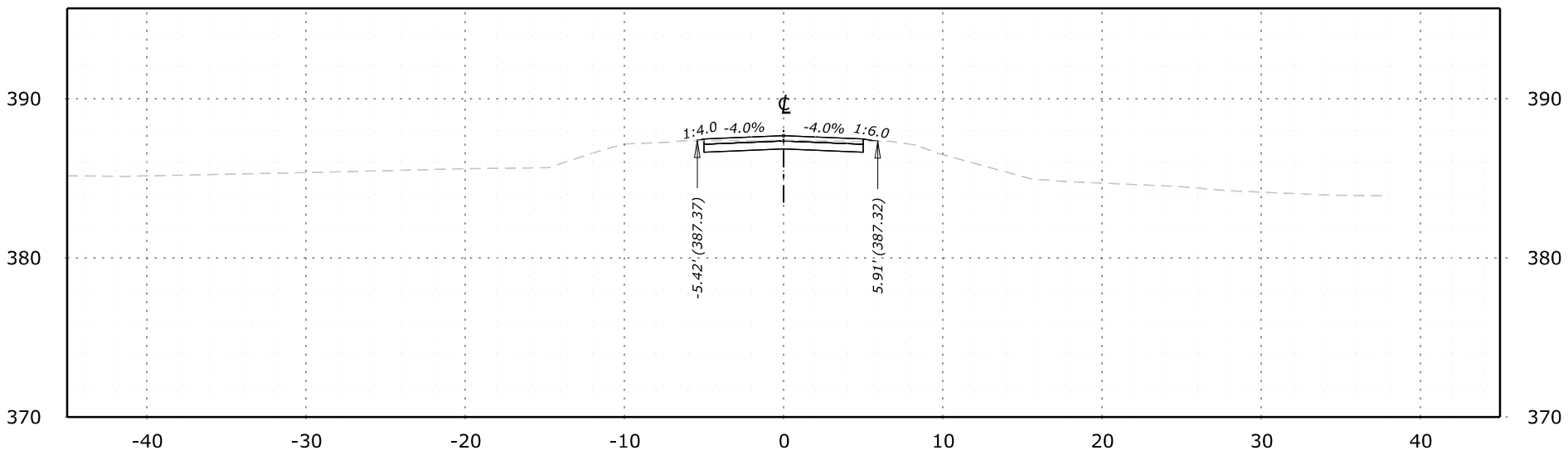
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X20



372+00.00
Des Grade = 387.60
Subgrade = 386.77
Original Grnd = 387.49

371+50.00
Des Grade = 387.75
Subgrade = 386.92
Original Grnd = 387.41



371+00.00
Des Grade = 387.67
Subgrade = 386.84
Original Grnd = 387.40

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

371+00 TO 372+00

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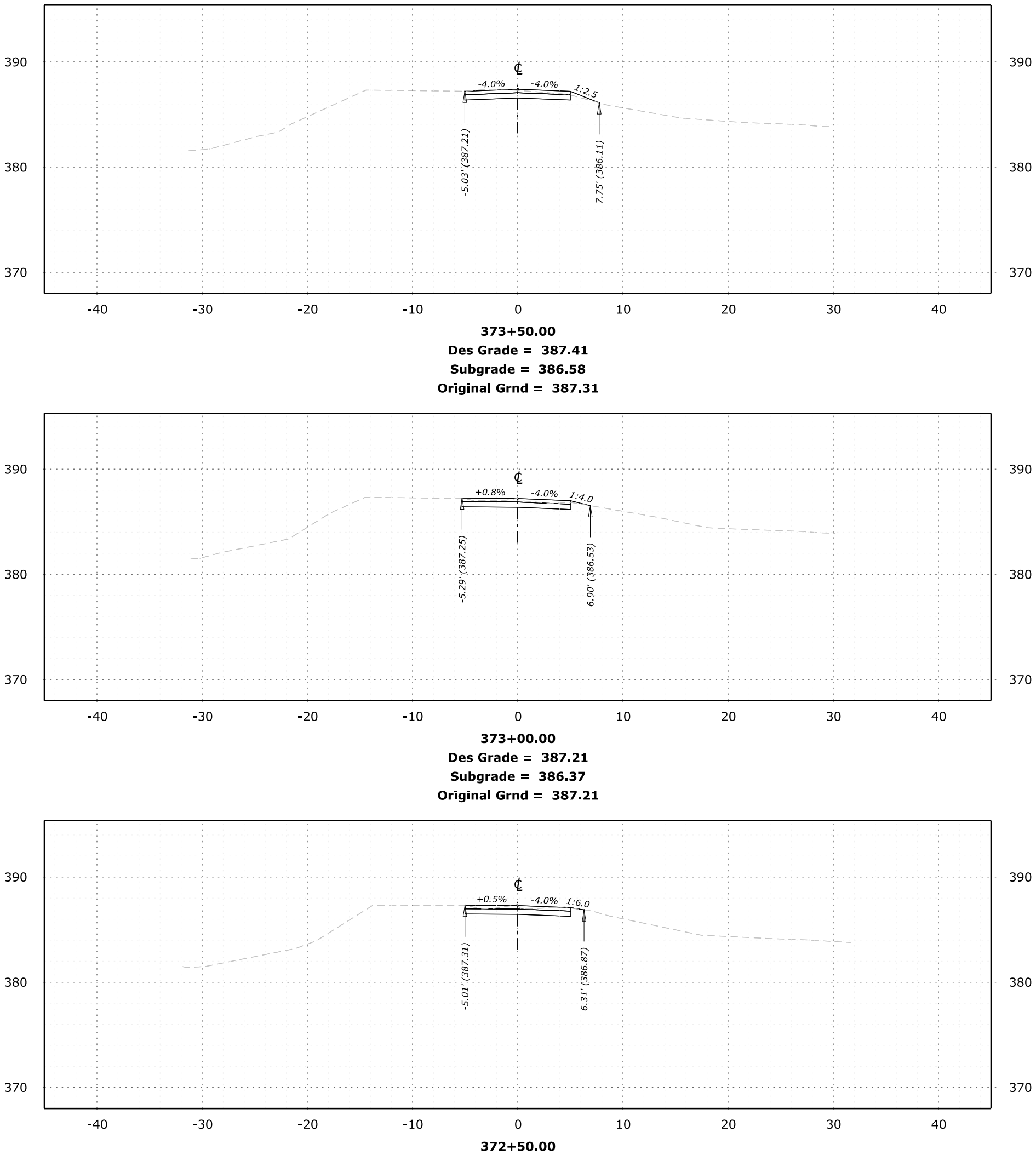
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X21



373+50.00
Des Grade = 387.41
Subgrade = 386.58
Original Grnd = 387.31

373+00.00
Des Grade = 387.21
Subgrade = 386.37
Original Grnd = 387.21

372+50.00
Des Grade = 387.29
Subgrade = 386.46
Original Grnd = 387.29

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

372+50 TO 373+50

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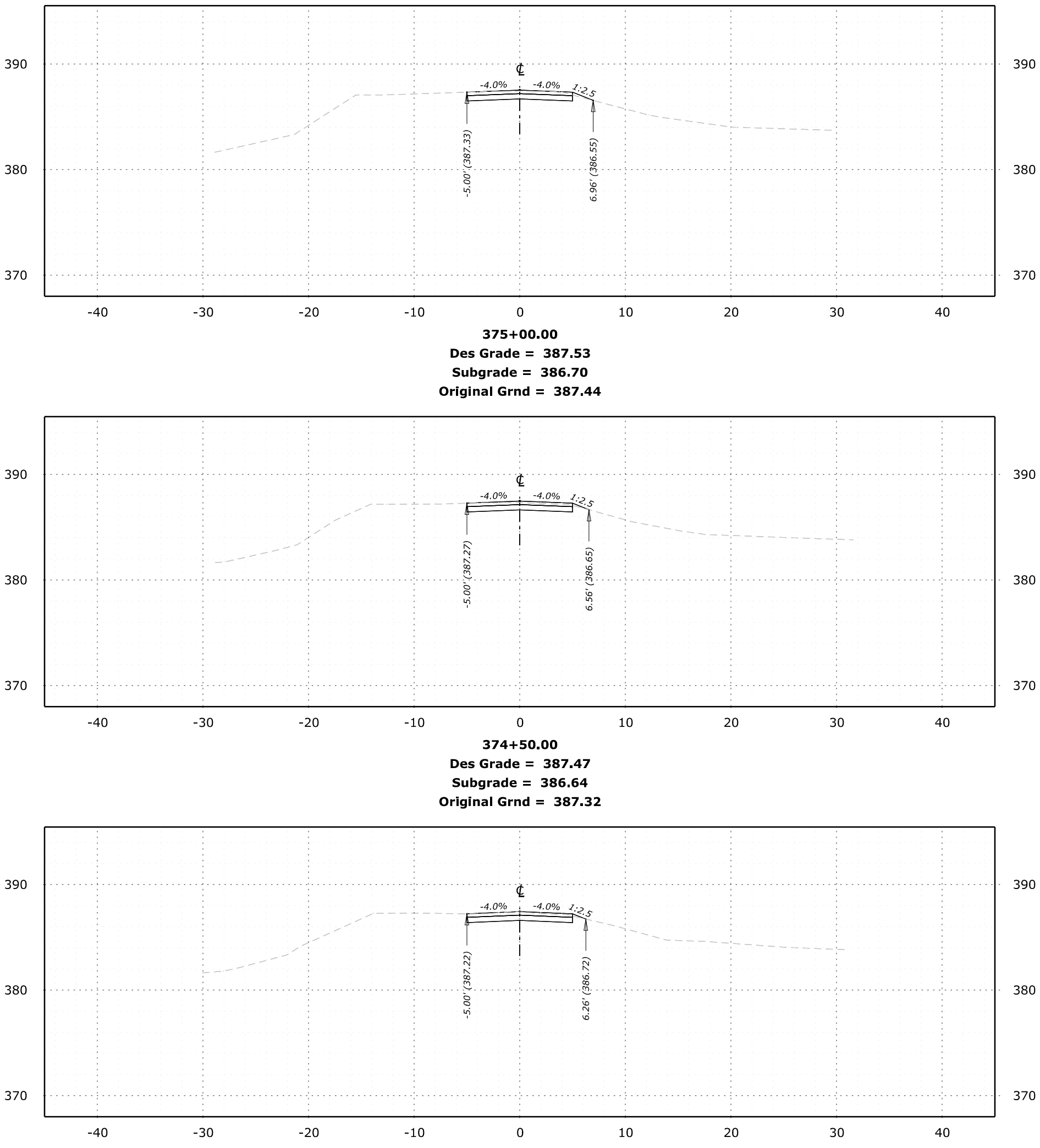
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X22



375+00.00
Des Grade = 387.53
Subgrade = 386.70
Original Grnd = 387.44

374+50.00
Des Grade = 387.47
Subgrade = 386.64
Original Grnd = 387.32

374+00.00
Des Grade = 387.42
Subgrade = 386.59
Original Grnd = 387.40

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

374+00 TO 375+00

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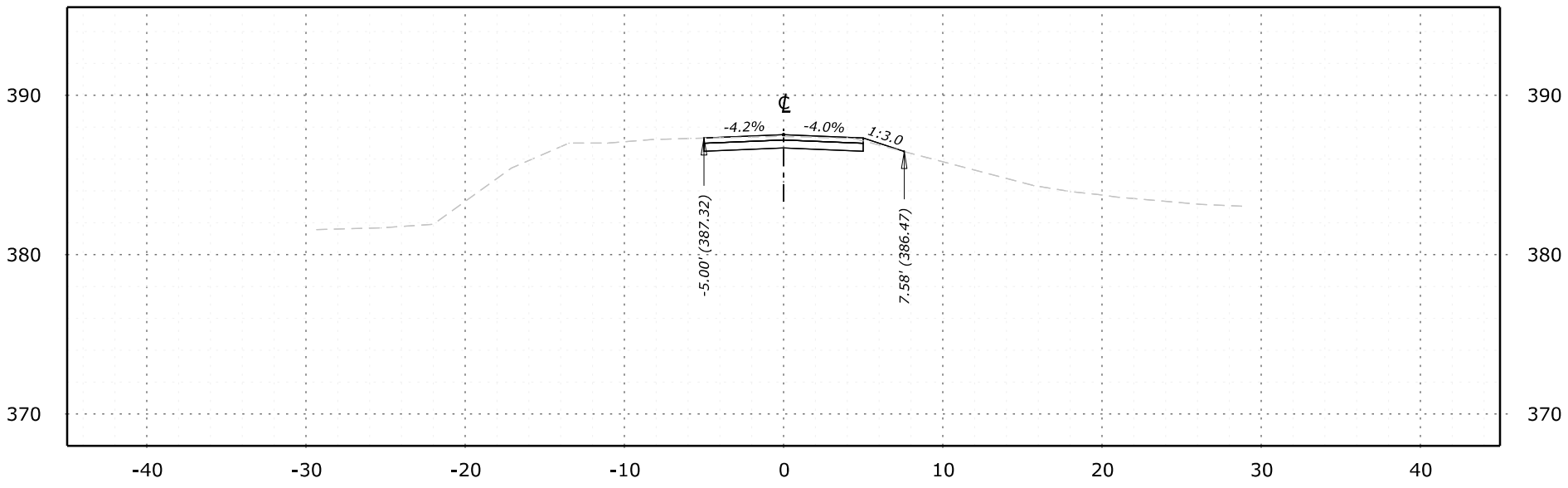
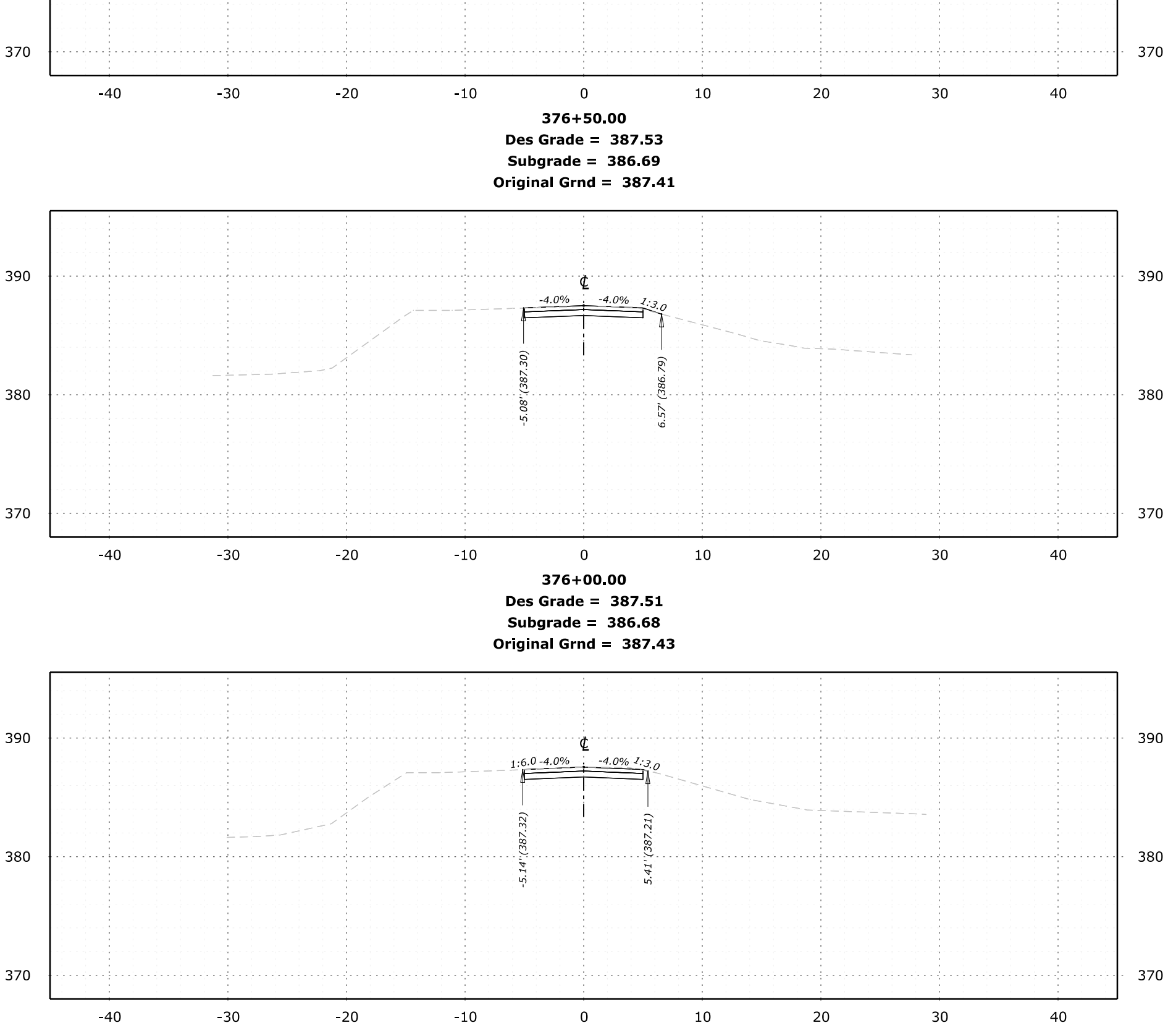
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X23



376+50.00
Des Grade = 387.53
Subgrade = 386.69
Original Grnd = 387.41

376+00.00
Des Grade = 387.51
Subgrade = 386.68
Original Grnd = 387.43



375+50.00
Des Grade = 387.55
Subgrade = 386.71
Original Grnd = 387.53

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

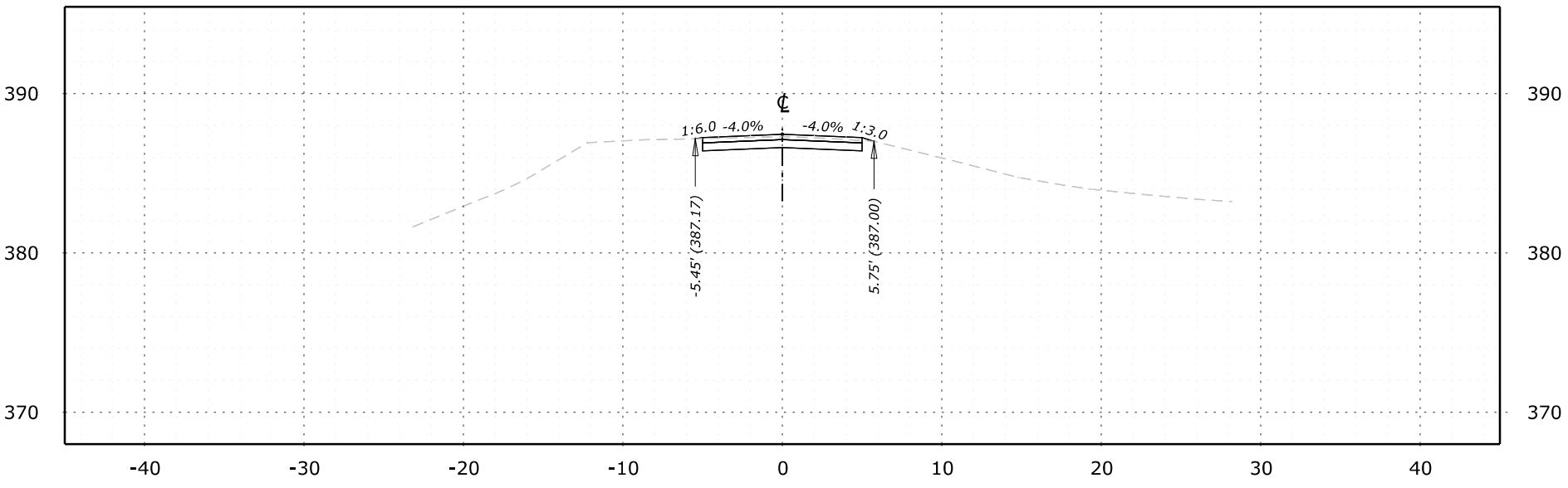
CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

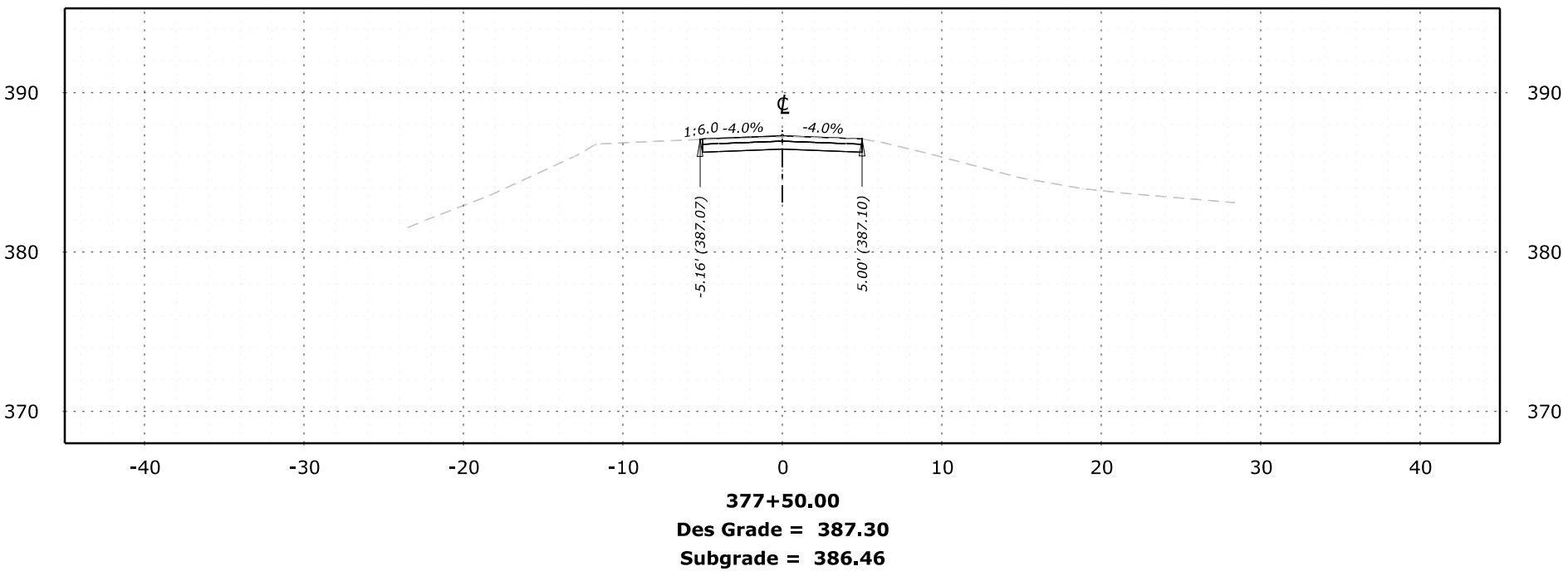
375+50 TO 376+50

\\fhfl15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\ZXX_NJ-WARI_2019-1(1)_LibertyLoop_xc.dgn [LL - 375+50.00 [Sheet]]
3 August 2026 5:02 PM

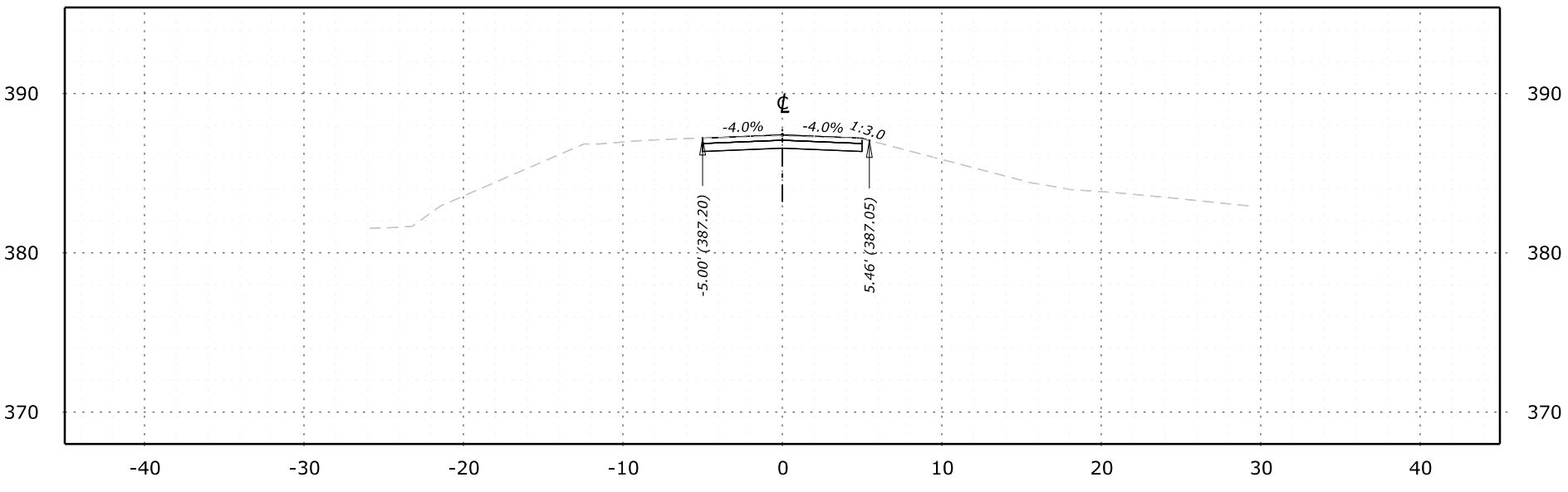
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X24



378+00.00
Des Grade = 387.45
Subgrade = 386.61
Original Grnd = 387.29



377+50.00
Des Grade = 387.30
Subgrade = 386.46
Original Grnd = 387.23



377+00.00
Des Grade = 387.40
Subgrade = 386.56
Original Grnd = 387.34

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

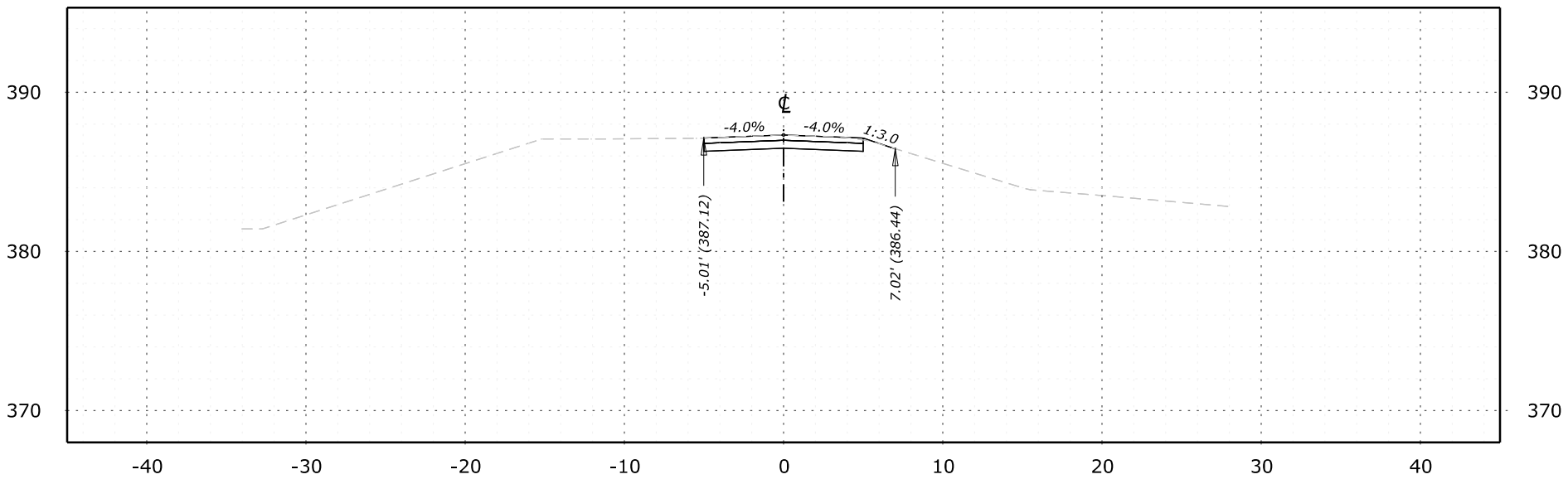
CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

377+00 TO 378+00

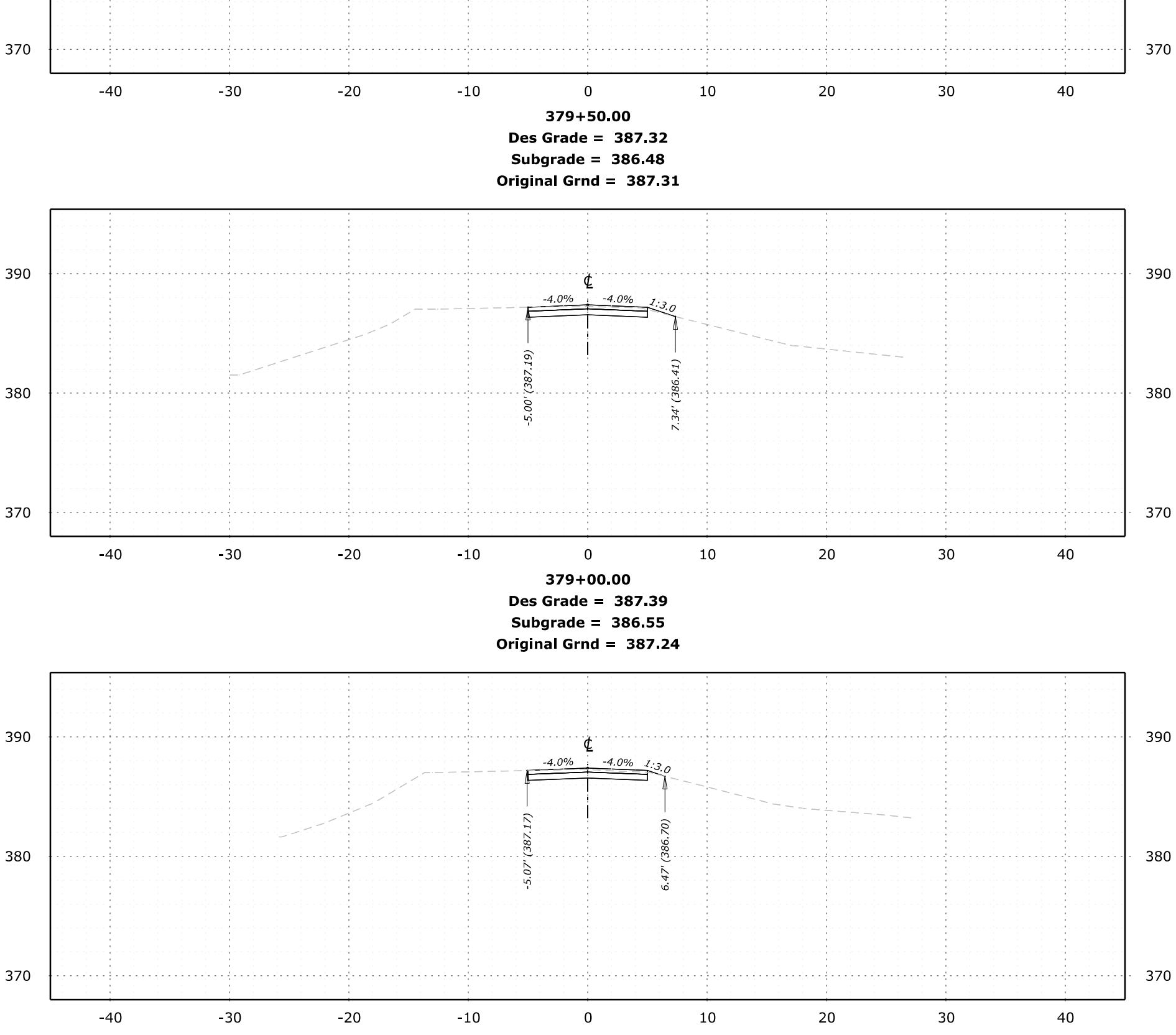
\\fhfl15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\ZXX_NJ-WARI_2019-1(1)_LibertyLoop_xc.dgn [LL - 377+00.00 [Sheet]]
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X25



379+50.00
Des Grade = 387.32
Subgrade = 386.48
Original Grnd = 387.31

379+00.00
Des Grade = 387.39
Subgrade = 386.55
Original Grnd = 387.24



378+50.00
Des Grade = 387.39
Subgrade = 386.55
Original Grnd = 387.27

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

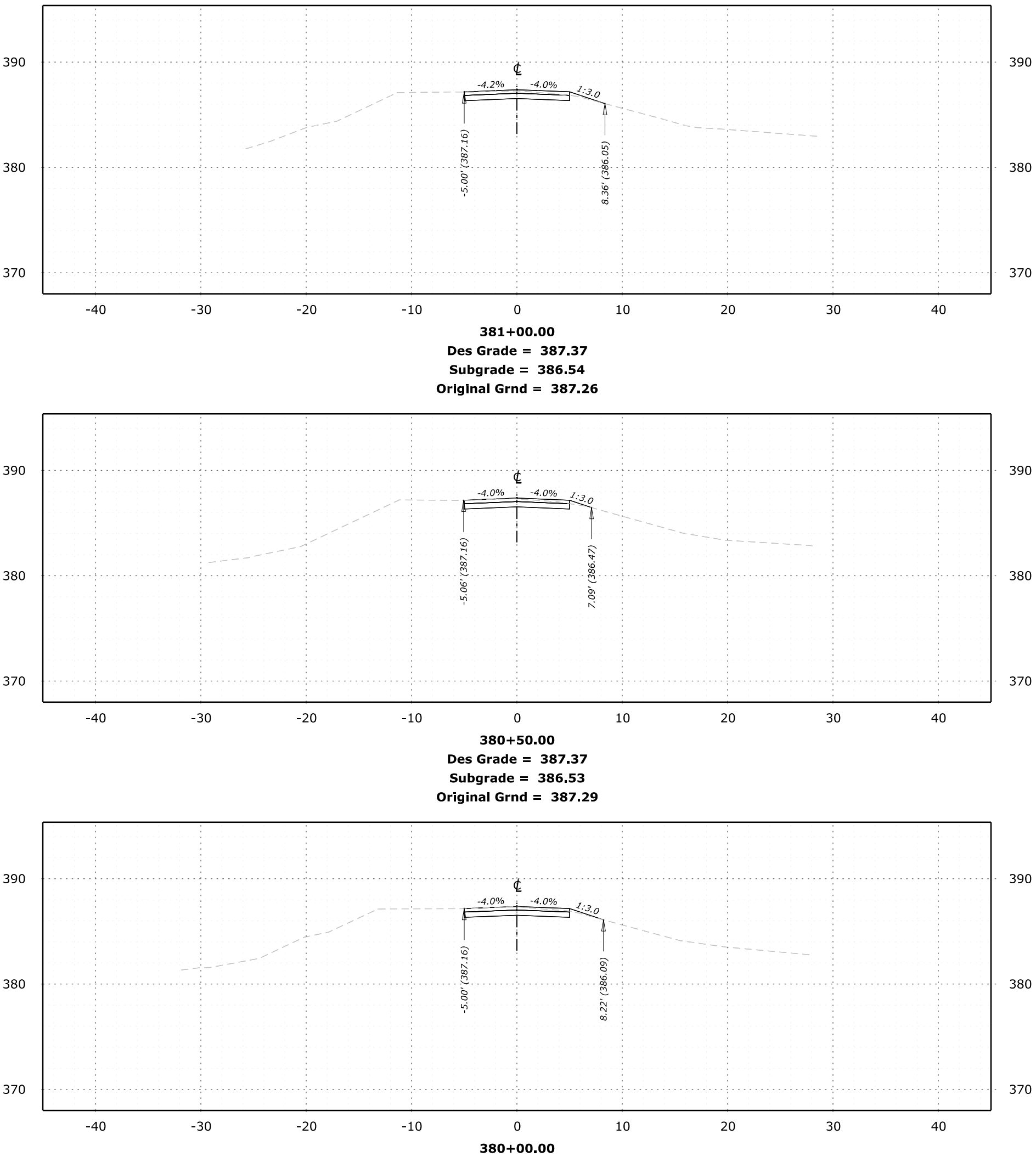
CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

378+50 TO 379+50

\\fhfl15fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\ZXX_NJ-WARI_2019-1(1)_LibertyLoop_xc.dgn [LL - 378+50.00 [Sheet]]
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X26



381+00.00
Des Grade = 387.37
Subgrade = 386.54
Original Grnd = 387.26

380+50.00
Des Grade = 387.37
Subgrade = 386.53
Original Grnd = 387.29

380+00.00
Des Grade = 387.36
Subgrade = 386.53
Original Grnd = 387.33

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

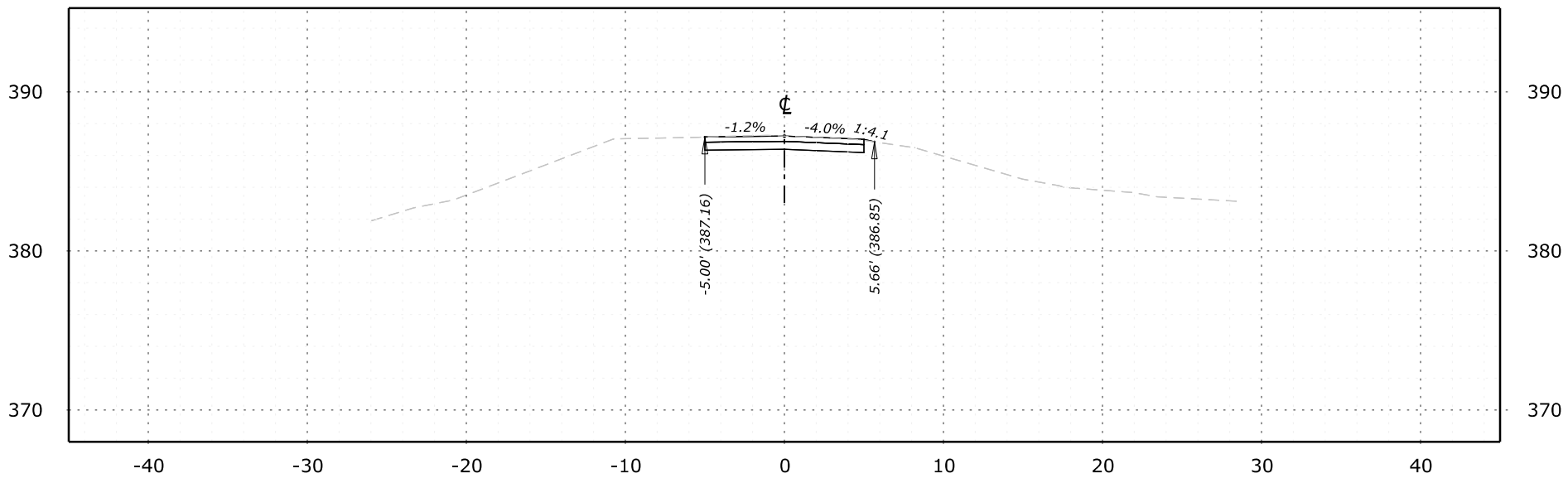
CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

380+00 TO 381+00

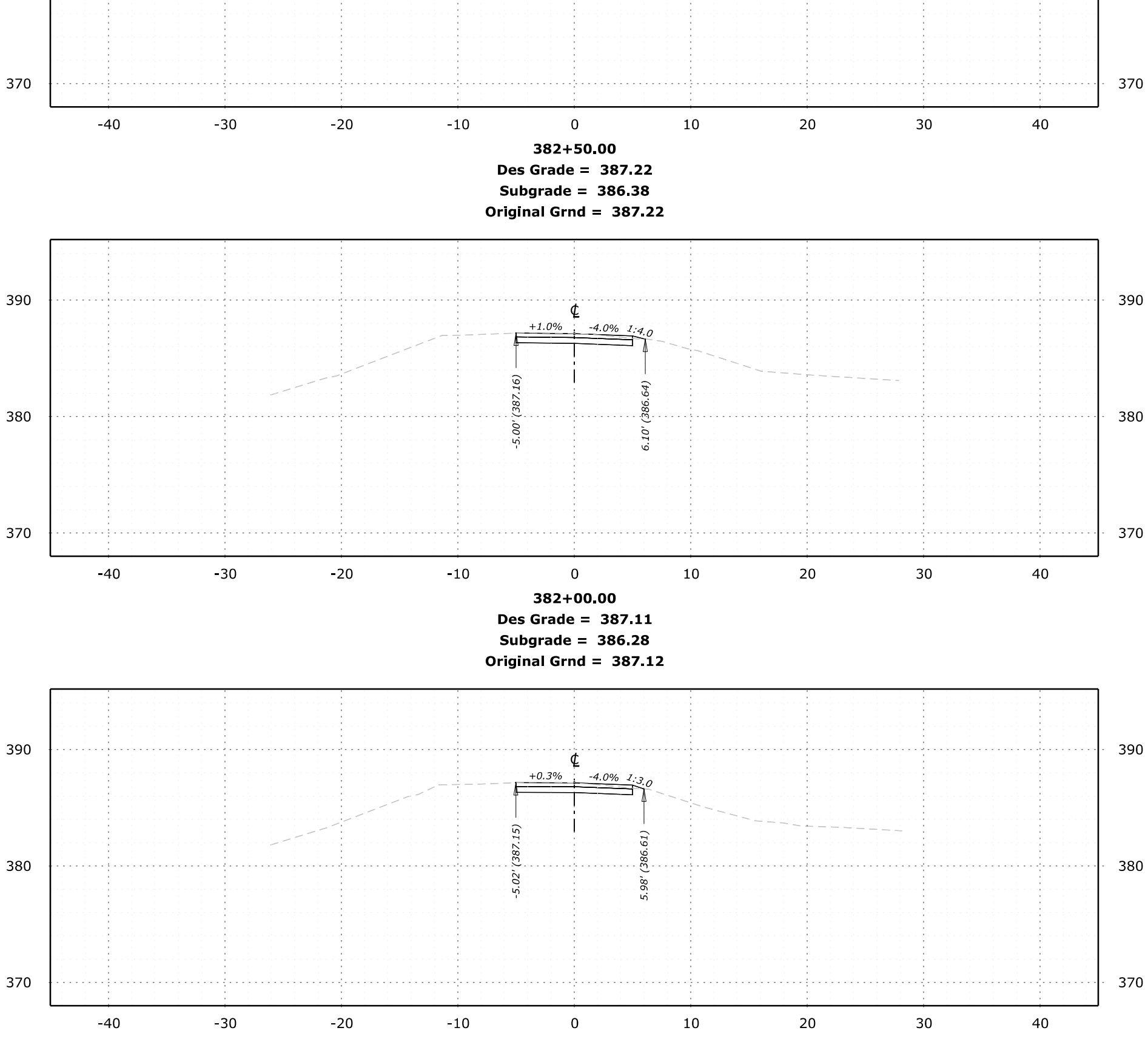
\\fhf115fileserve.fhd.fhwa.dot.gov\data\PROJECTS_FWS\WARI\NJ ERFO FW WARI 2019-1(1)\HWY_DES\CADD\ZXX_NJ-WARI_2019-1(1)_LibertyLoop_xc.dgn [LL - 380+00.00 [Sheet]]
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X27



382+50.00
Des Grade = 387.22
Subgrade = 386.38
Original Grnd = 387.22

382+00.00
Des Grade = 387.11
Subgrade = 386.28
Original Grnd = 387.12



381+50.00
Des Grade = 387.14
Subgrade = 386.30
Original Grnd = 387.14

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

381+50 TO 382+50

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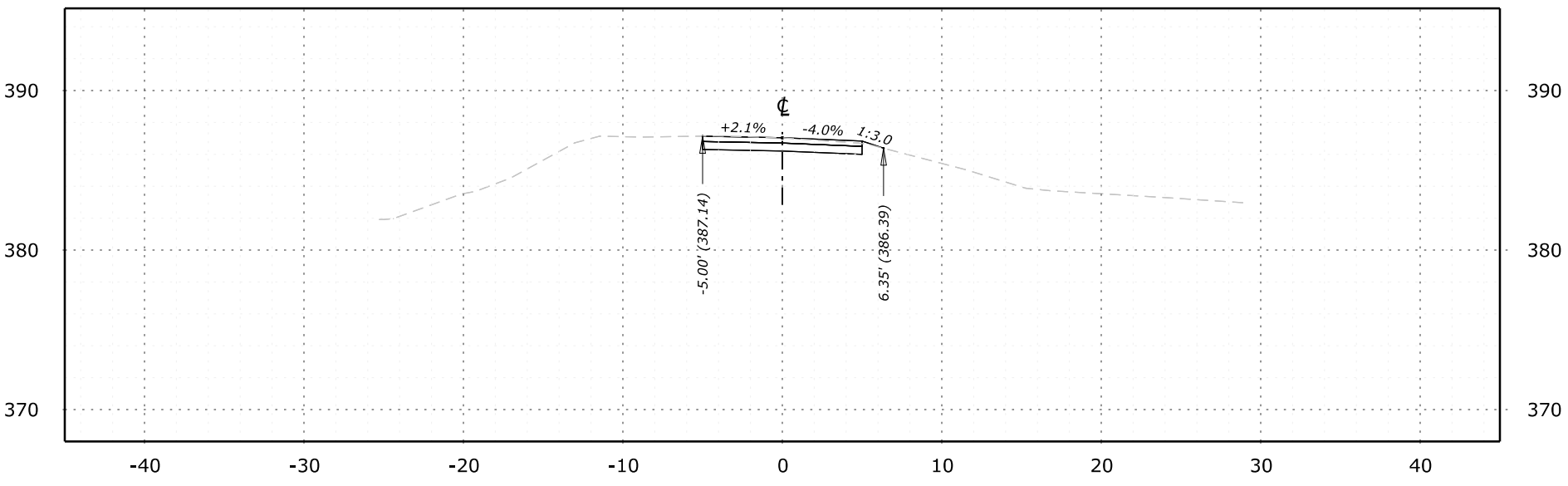
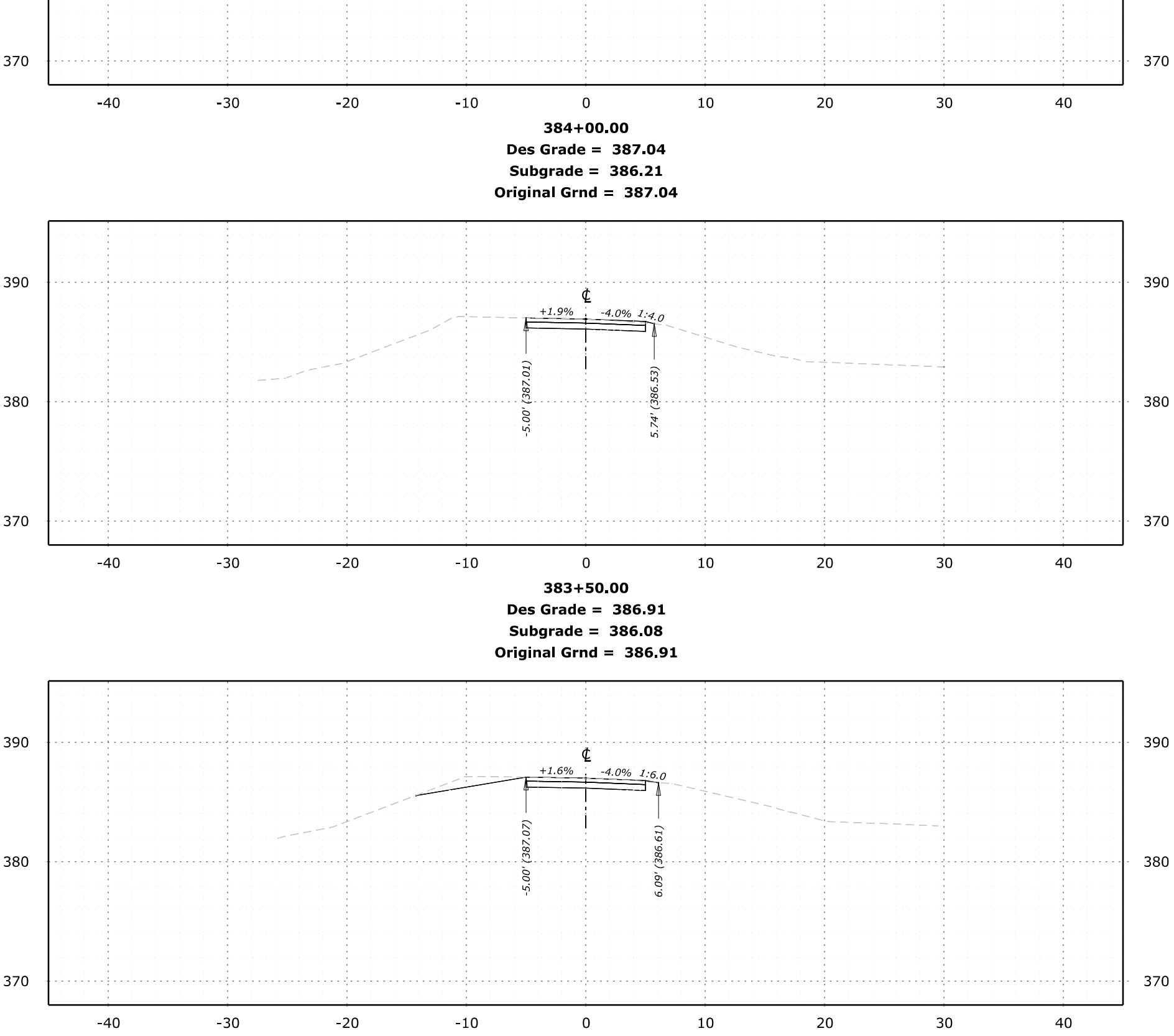
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X28



384+00.00
Des Grade = 387.04
Subgrade = 386.21
Original Grnd = 387.04

383+50.00
Des Grade = 386.91
Subgrade = 386.08
Original Grnd = 386.91



383+00.00
Des Grade = 386.99
Subgrade = 386.16
Original Grnd = 386.99

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

383+00.00 TO 384+00.00

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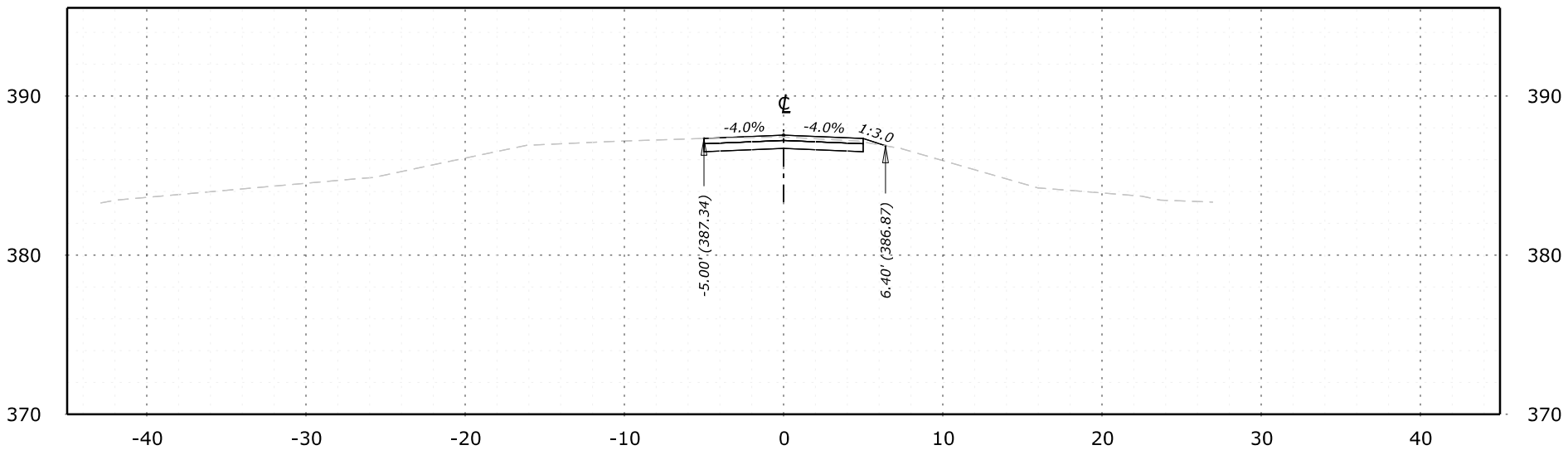
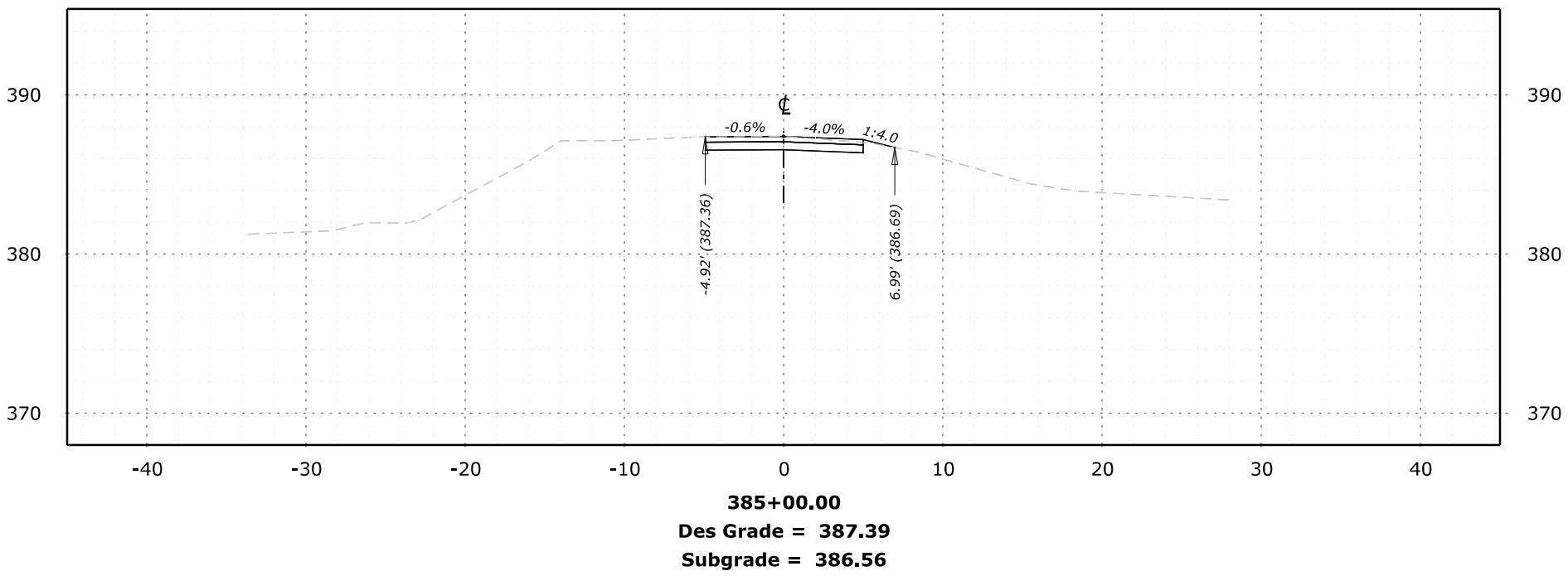
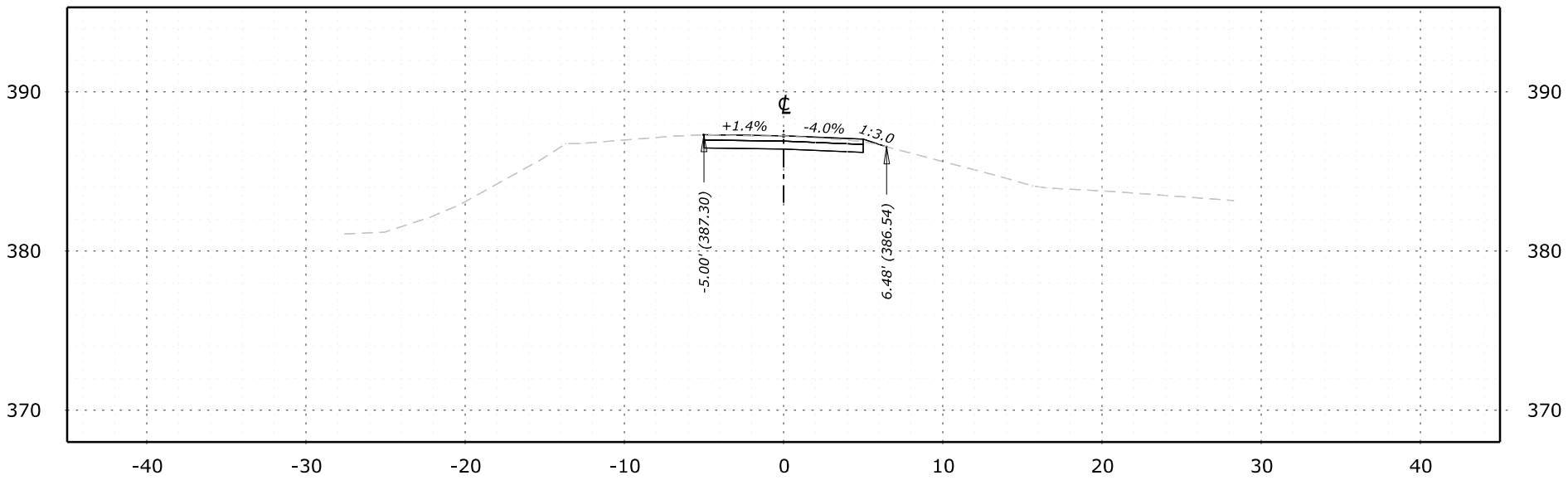
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X29



385+50.00
Des Grade = 387.54
Subgrade = 386.71
Original Grnd = 387.43



385+00.00
Des Grade = 387.39
Subgrade = 386.56
Original Grnd = 387.39



384+50.00
Des Grade = 387.23
Subgrade = 386.39
Original Grnd = 387.23

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

384+50 TO 385+50

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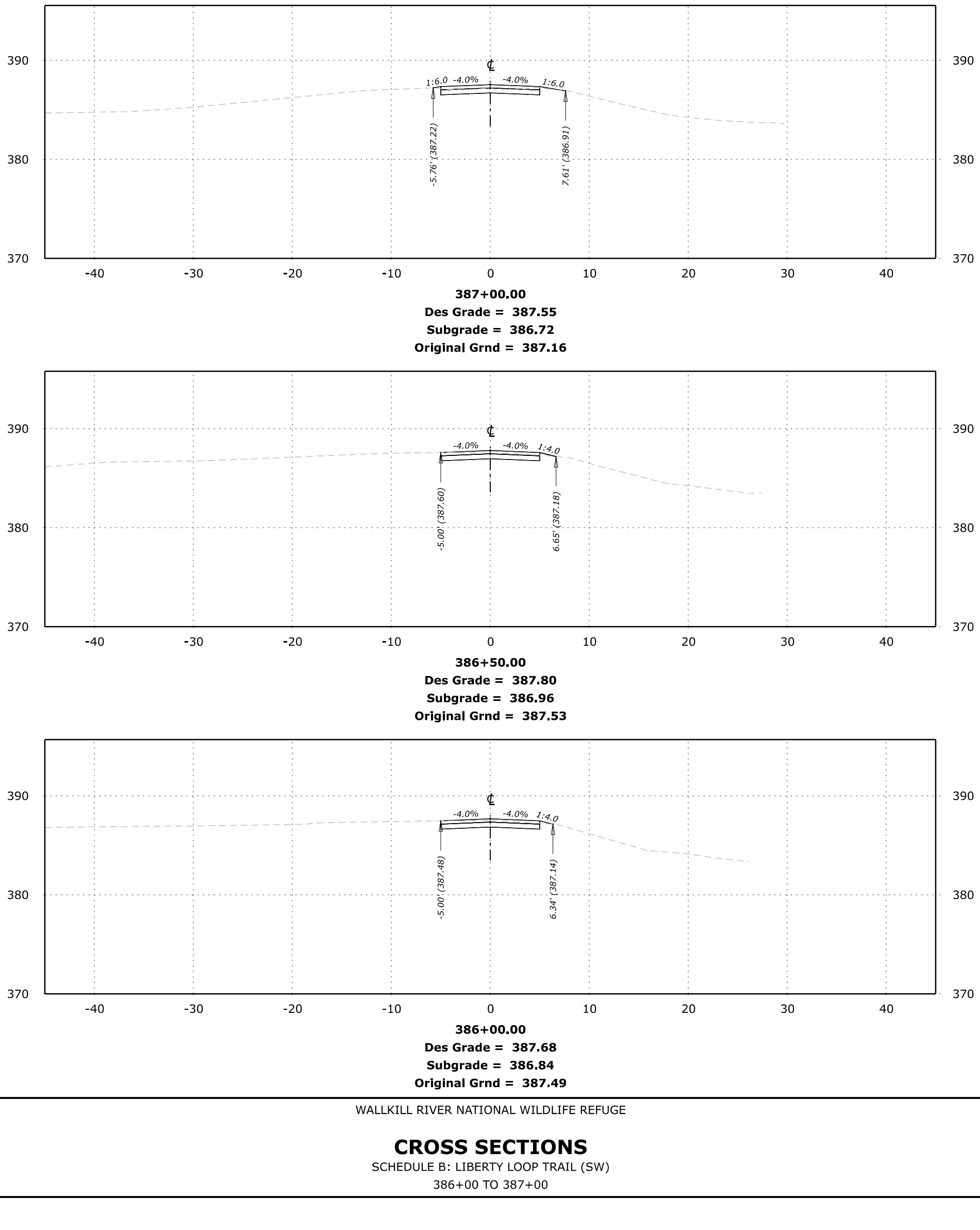
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X30



387+00.00
Des Grade = 387.55
Subgrade = 386.72
Original Grnd = 387.16

386+50.00
Des Grade = 387.80
Subgrade = 386.96
Original Grnd = 387.53

386+00.00
Des Grade = 387.68
Subgrade = 386.84
Original Grnd = 387.49

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

386+00 TO 387+00

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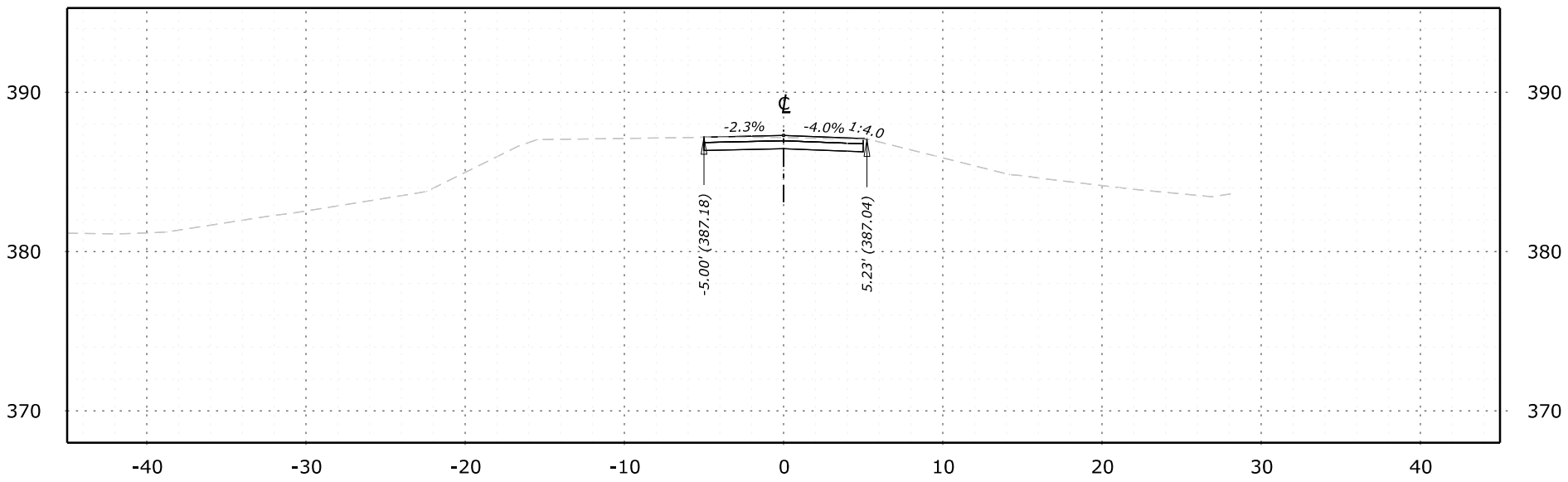
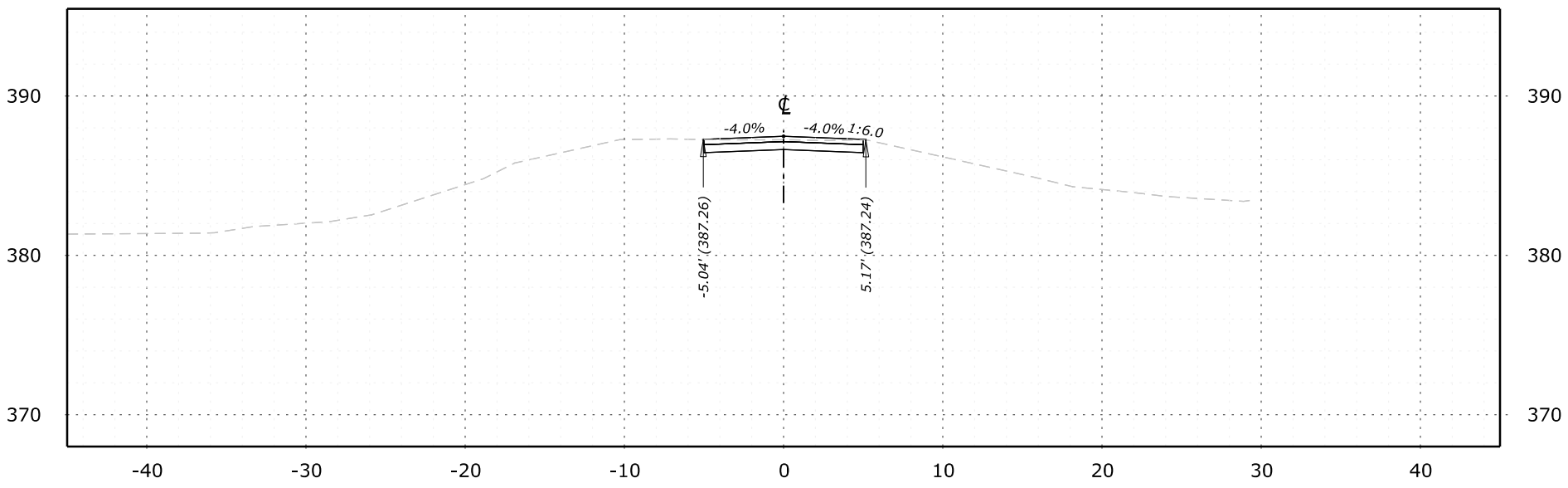
3 August 2026 5:02 PM

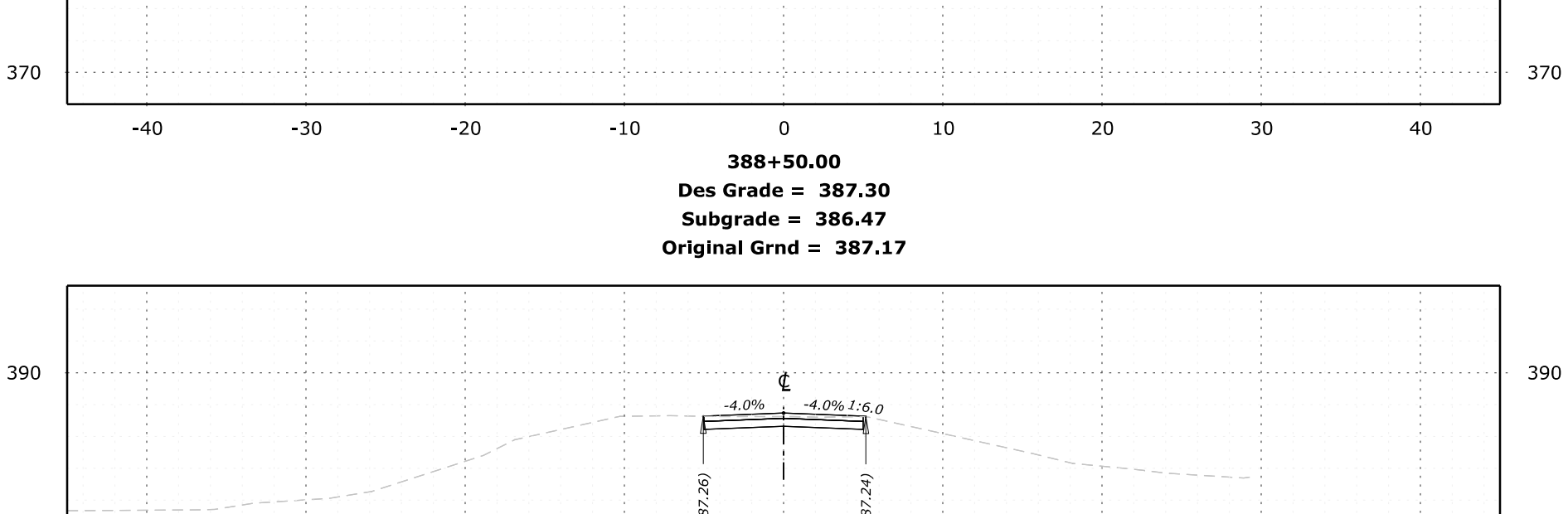
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X31



388+50.00
Des Grade = 387.30
Subgrade = 386.47
Original Grnd = 387.17



388+00.00
Des Grade = 387.47
Subgrade = 386.63
Original Grnd = 387.24



387+50.00
Des Grade = 387.41
Subgrade = 386.58
Original Grnd = 386.96

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

387+50 TO 388+50

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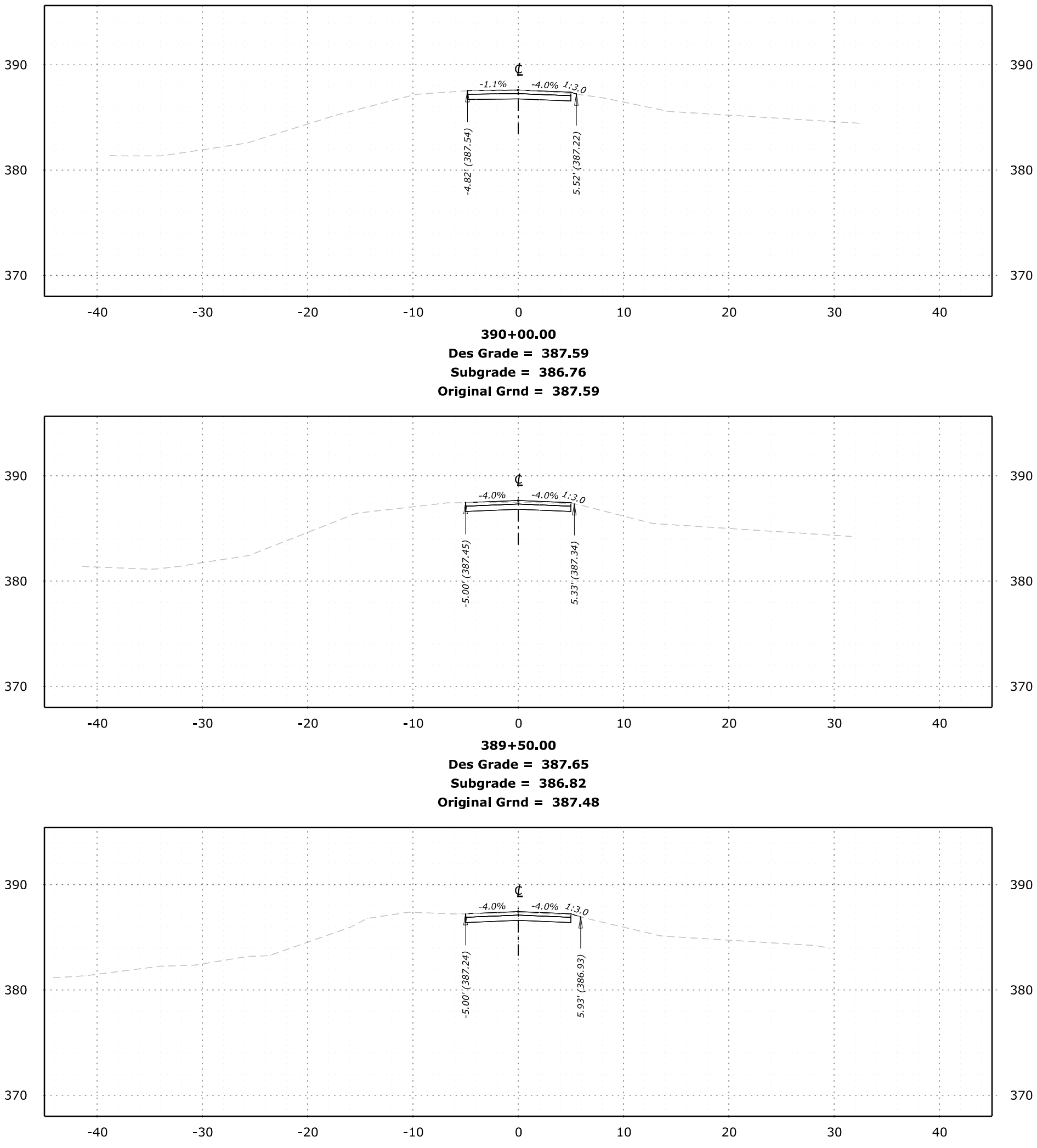
3 August 2026 5:02 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X32



390+00.00
Des Grade = 387.59
Subgrade = 386.76
Original Grnd = 387.59

389+50.00
Des Grade = 387.65
Subgrade = 386.82
Original Grnd = 387.48

389+00.00
Des Grade = 387.44
Subgrade = 386.61
Original Grnd = 387.35

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

389+00 TO 390+00

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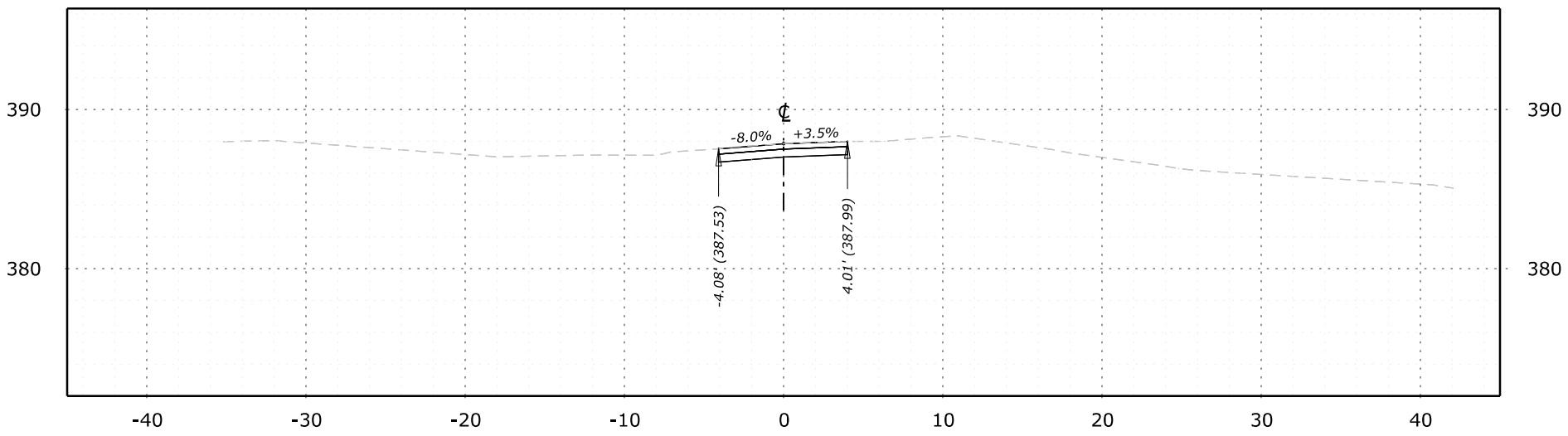
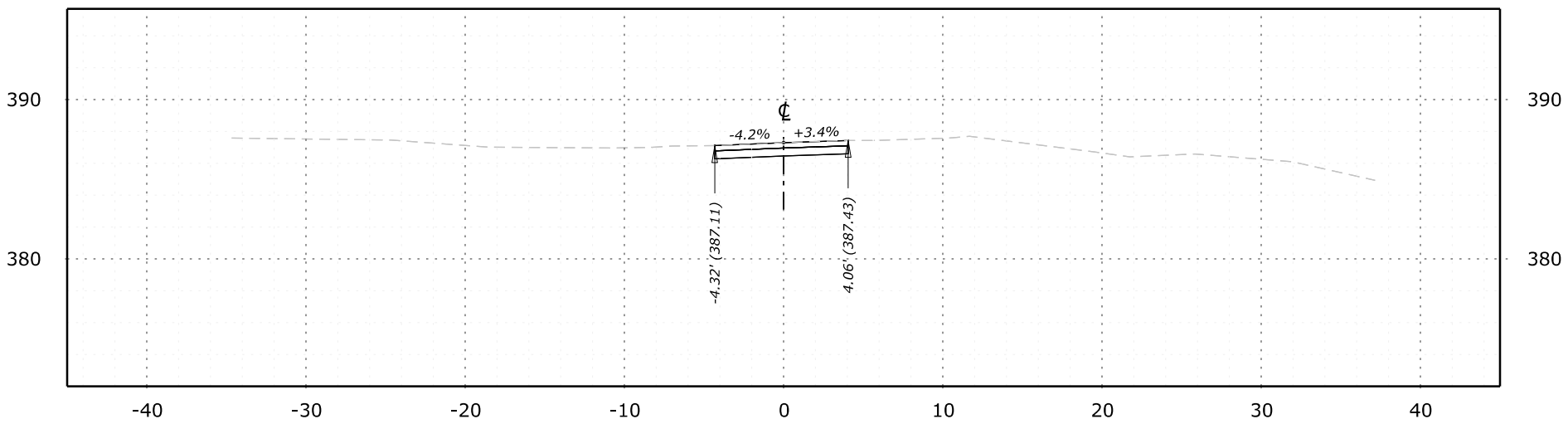
3 August 2026 5:02 PM

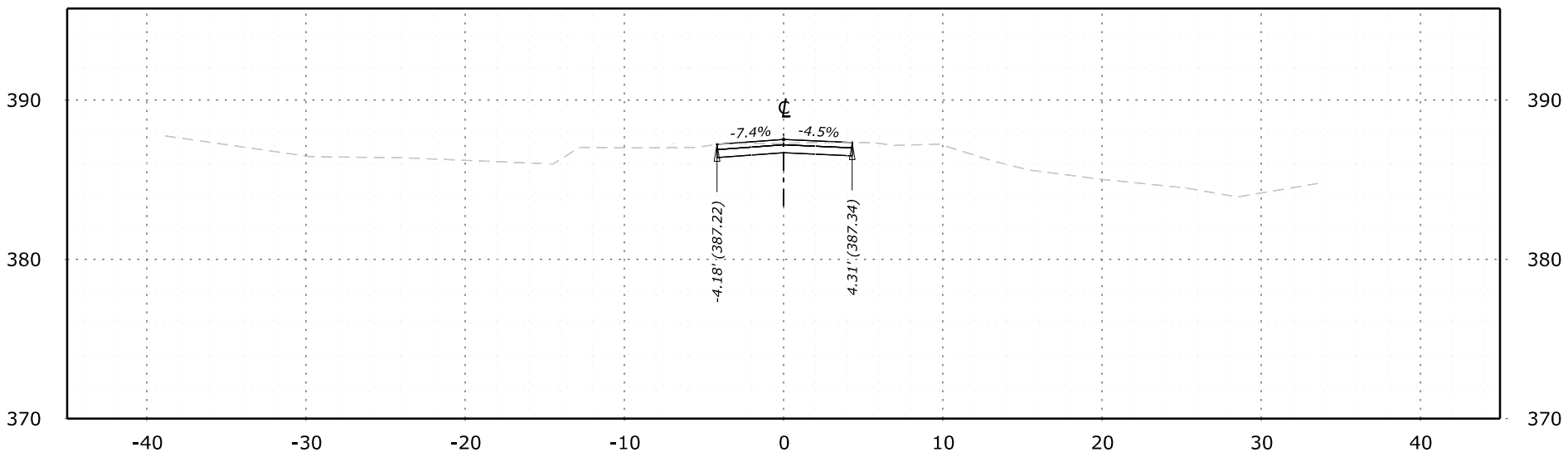
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X33



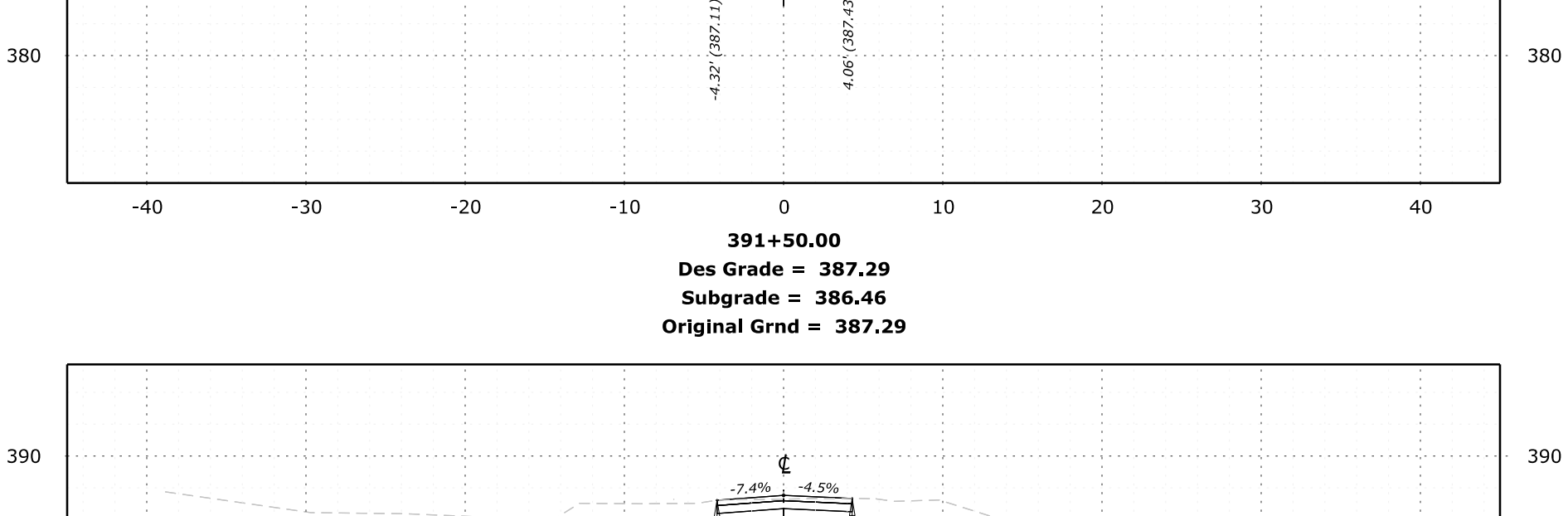
392+00.00
Des Grade = 387.85
Subgrade = 387.02
Original Grnd = 387.85



391+50.00
Des Grade = 387.29
Subgrade = 386.46
Original Grnd = 387.29



391+00.00
Des Grade = 387.53
Subgrade = 386.70
Original Grnd = 387.28



390+50.00
Des Grade = 387.65
Subgrade = 386.82
Original Grnd = 387.40

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

390+50 TO 392+00

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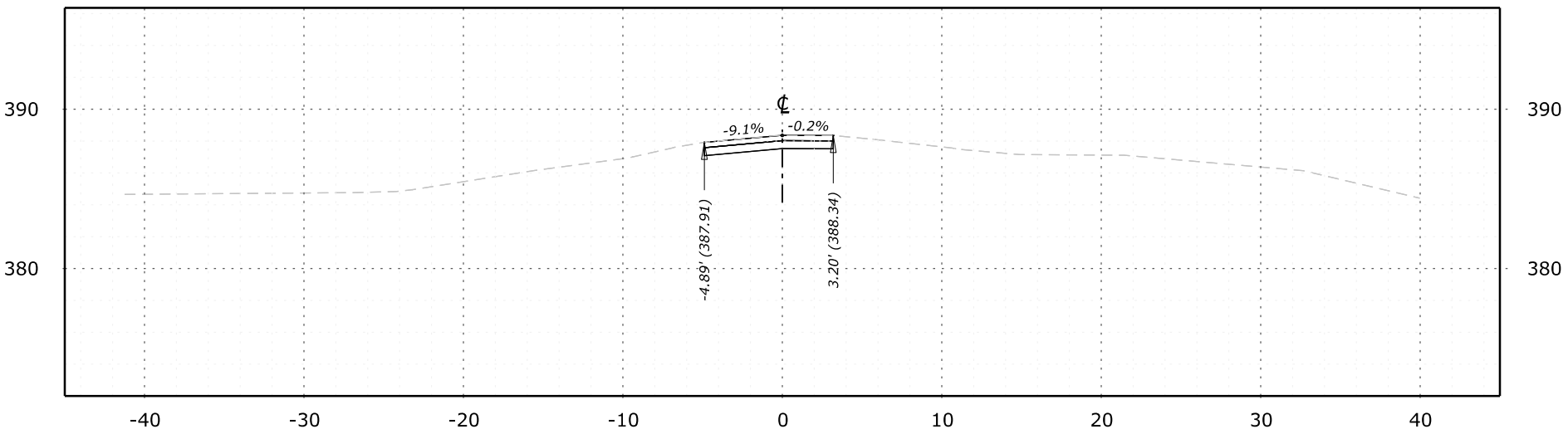
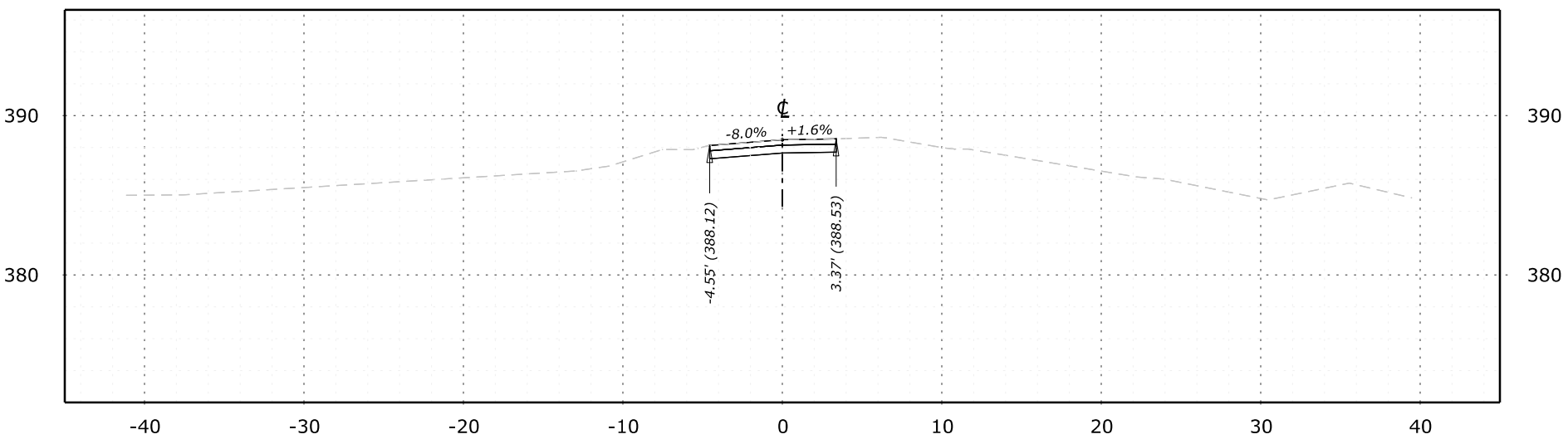
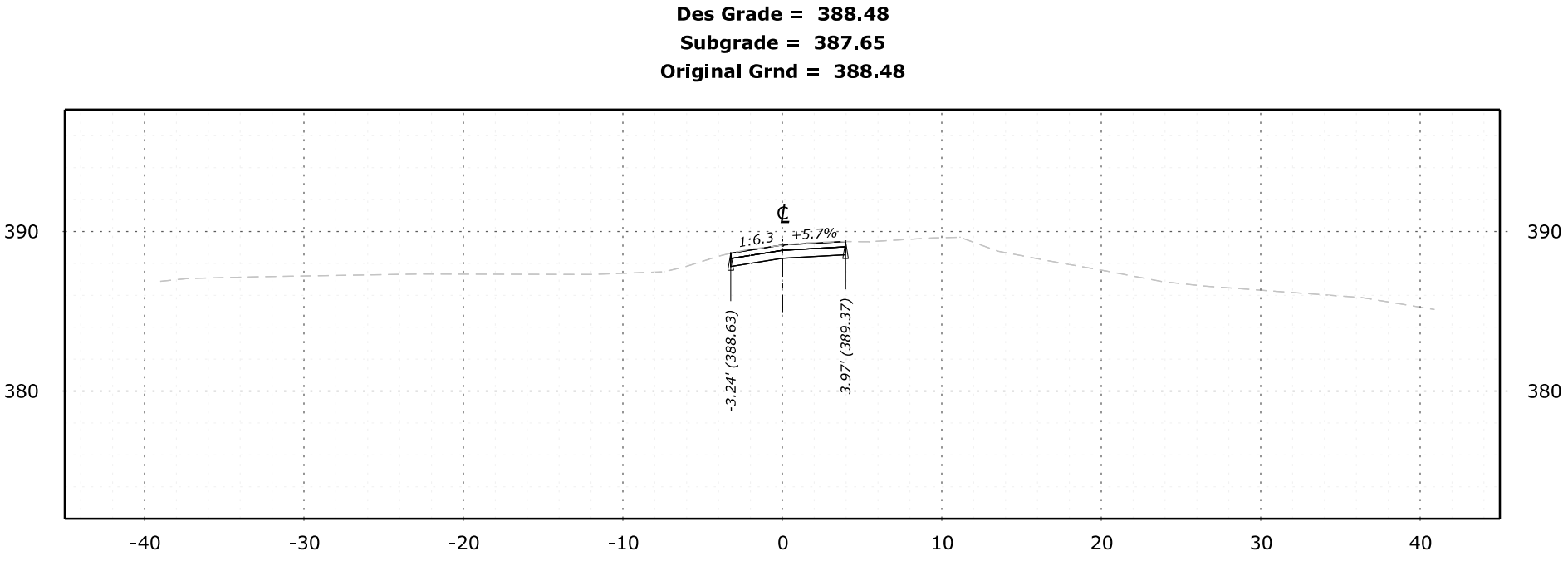
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	X34



393+50.00
Des Grade = 388.35
Subgrade = 387.52
Original Grnd = 388.35



393+00.00
Des Grade = 388.48
Subgrade = 387.65
Original Grnd = 388.48



392+50.00
Des Grade = 389.15
Subgrade = 388.31
Original Grnd = 389.14

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE B: LIBERTY LOOP TRAIL (SW)

392+50 TO 393+50

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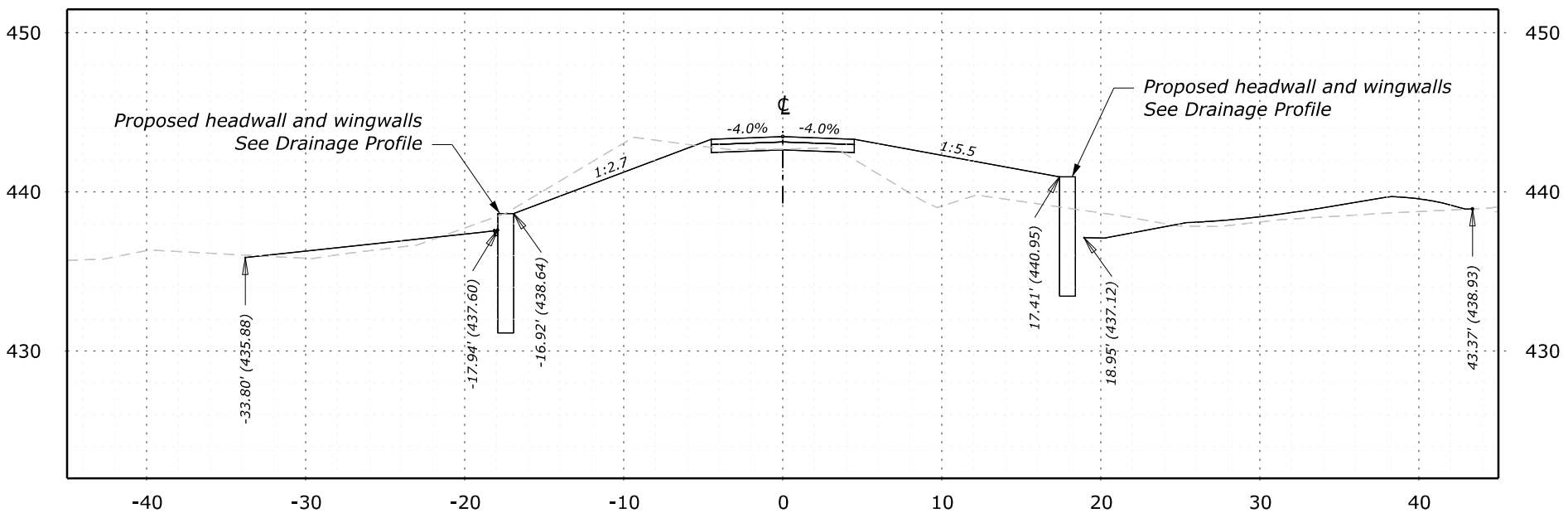
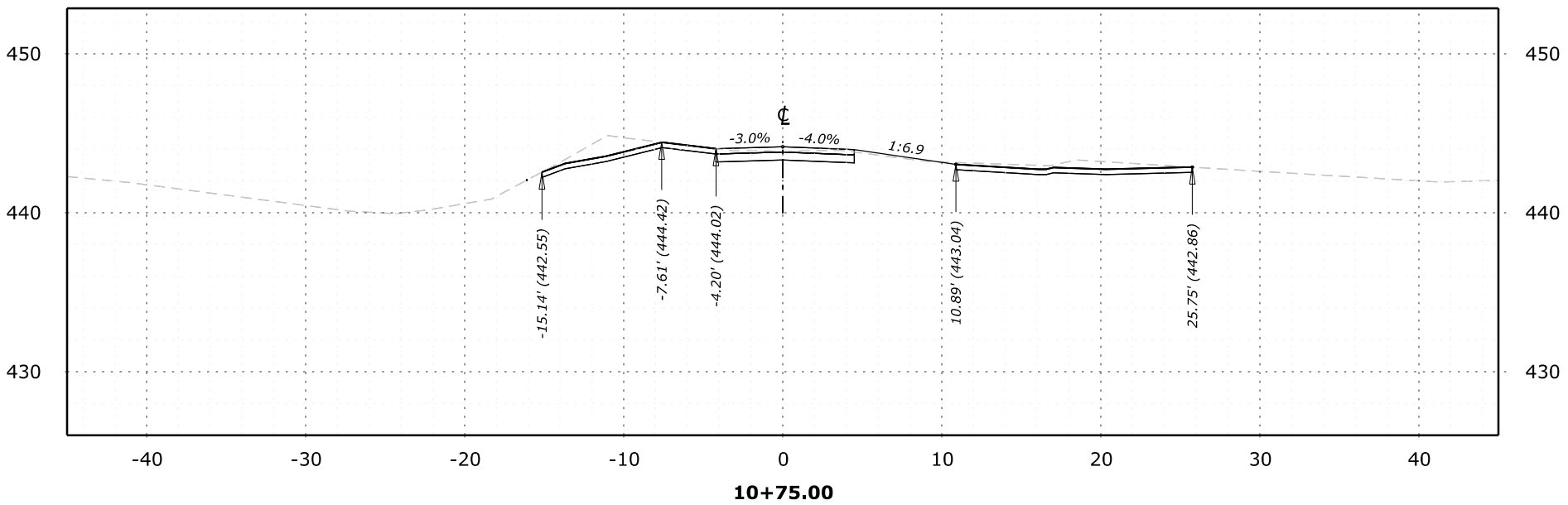
3 August 2026 5:02 PM

Disaster No. NJ2019-1-FWS and NY2019-1-FWS

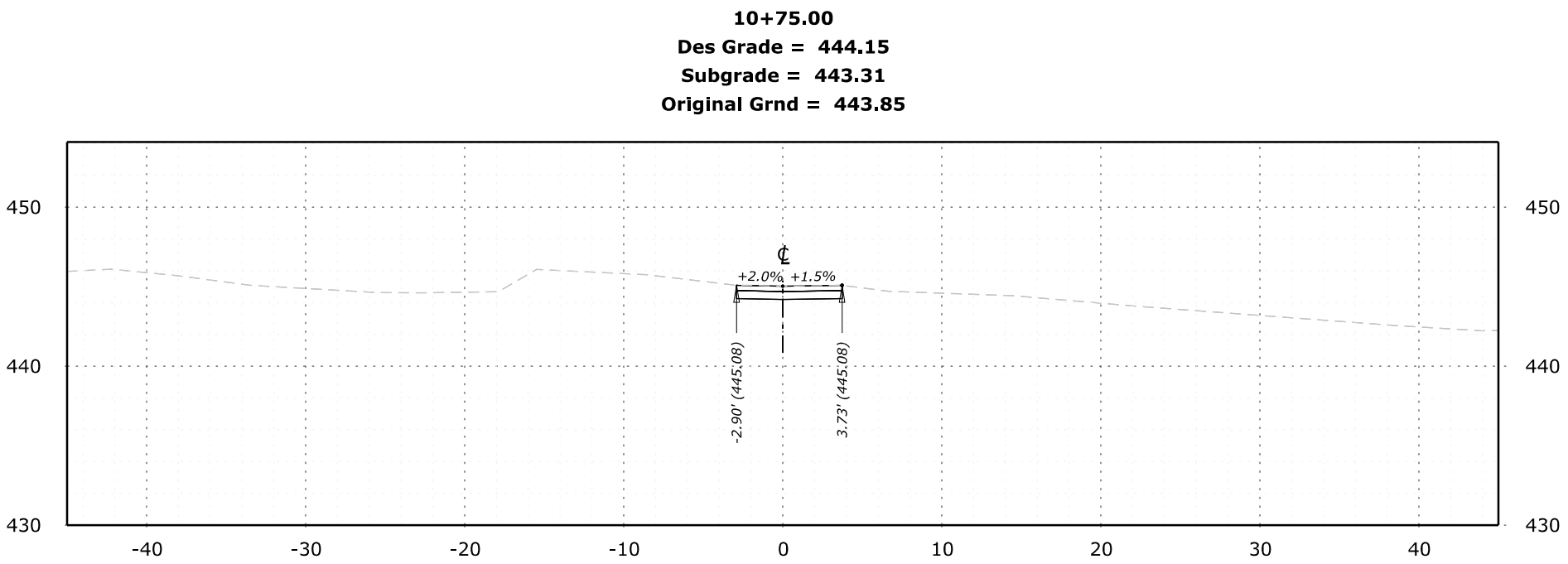
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	Y01



10+90.00
Des Grade = 443.48
Subgrade = 442.65
Original Grnd = 442.68



10+75.00
Des Grade = 444.15
Subgrade = 443.31
Original Grnd = 443.85



10+60.00
Des Grade = 445.02
Subgrade = 444.19
Original Grnd = 445.02

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1

10+60 TO 10+90

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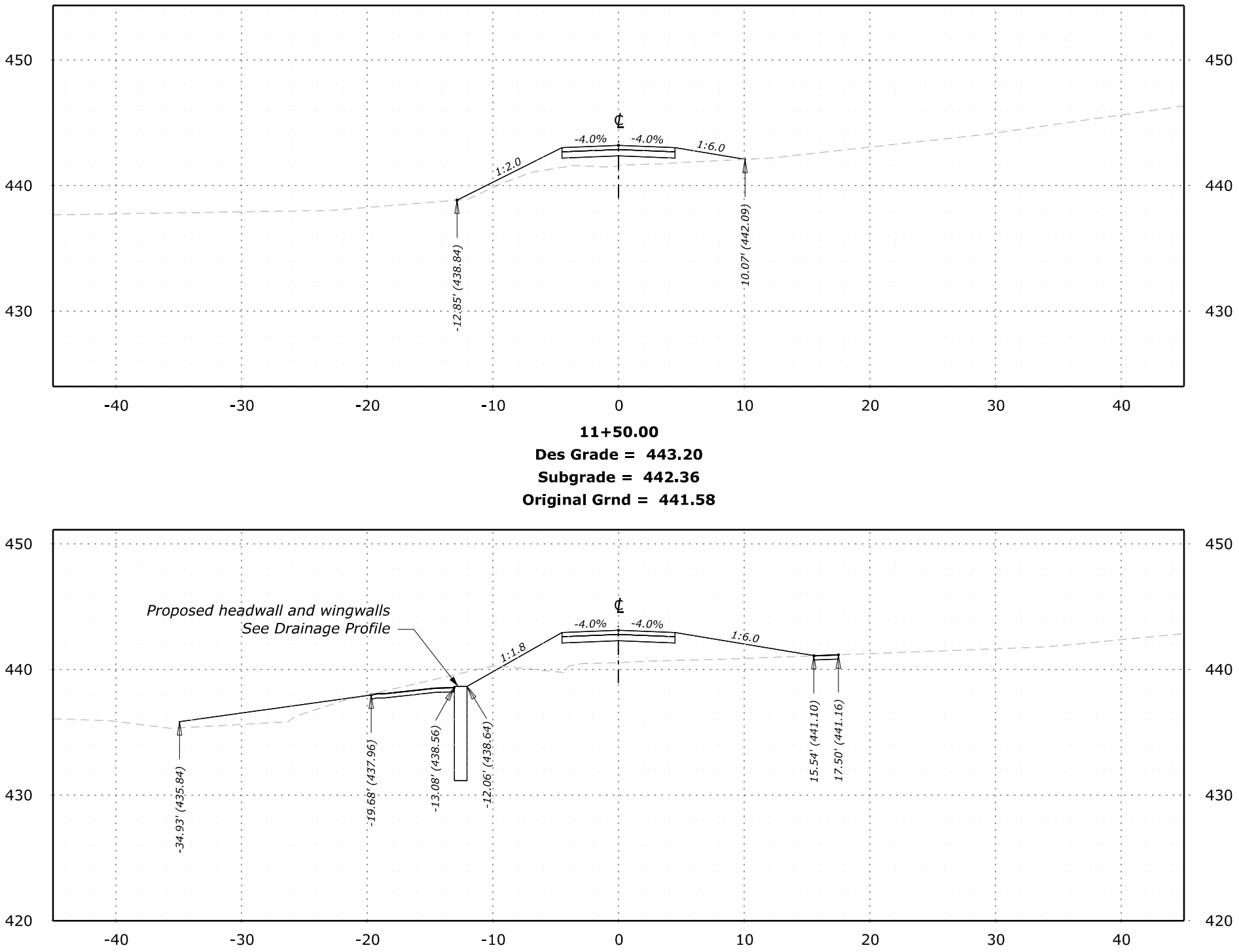
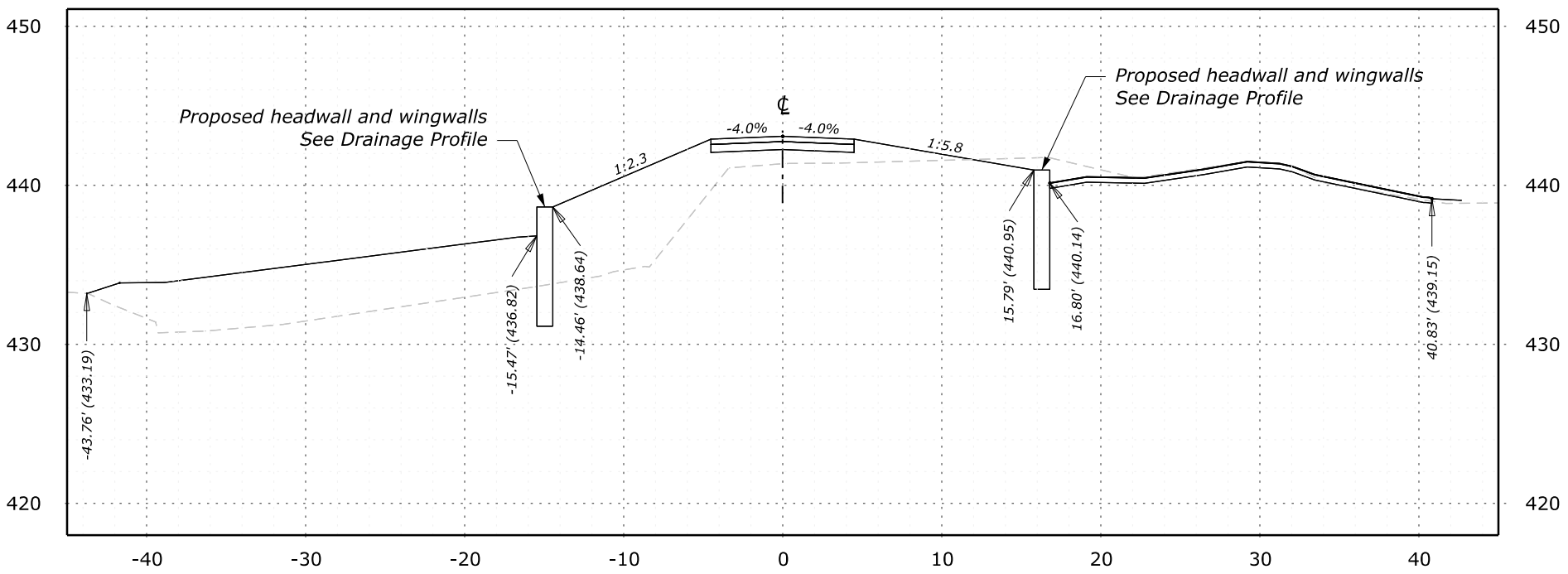
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	Y02



11+50.00
Des Grade = 443.20
Subgrade = 442.36
Original Grnd = 441.58

11+25.00
Des Grade = 443.12
Subgrade = 442.29
Original Grnd = 440.53



11+10.00
Des Grade = 443.08
Subgrade = 442.24
Original Grnd = 439.33

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1

11+10 TO 11+50

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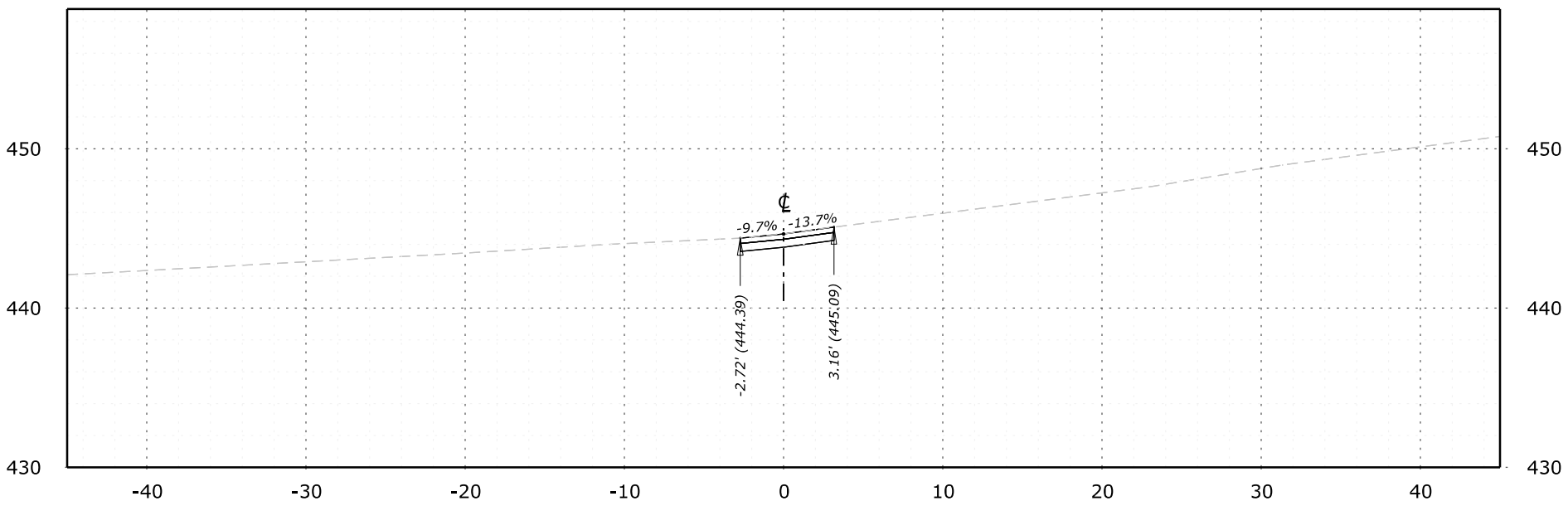
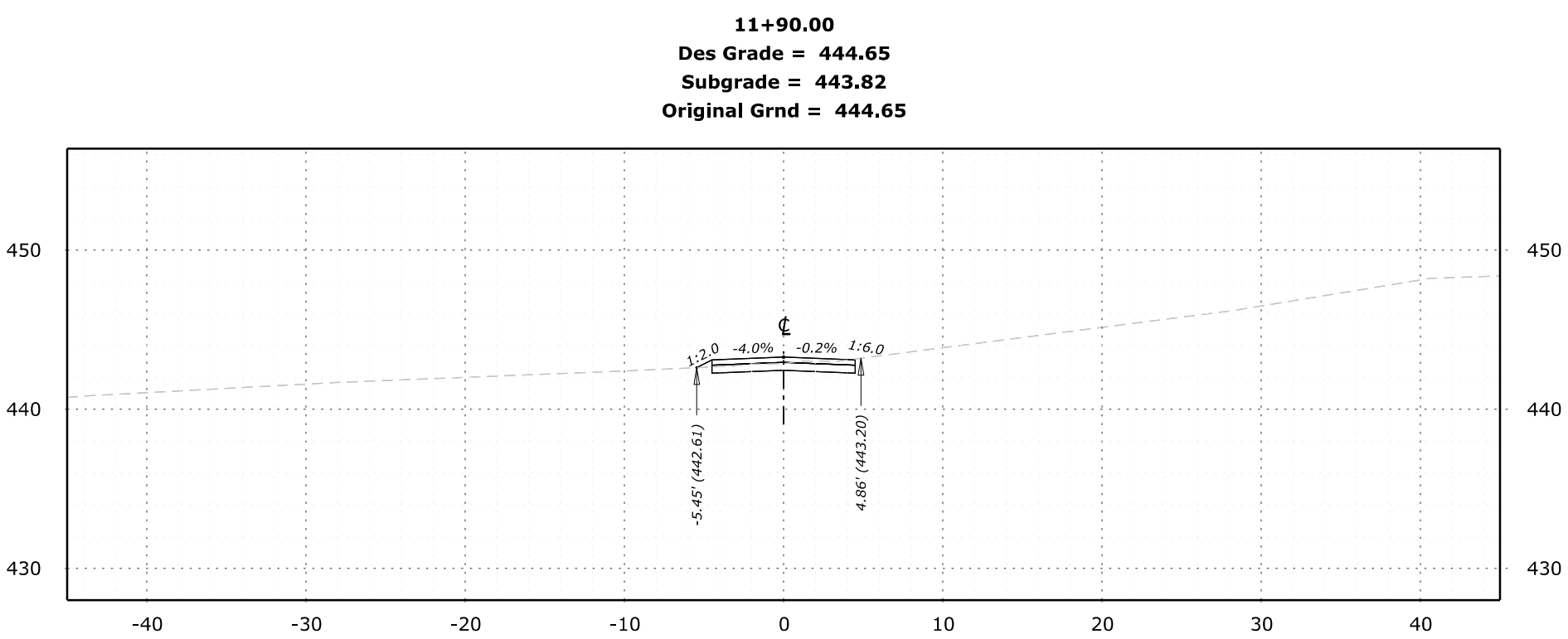
3 August 2026 5:30 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	Y03



11+90.00
Des Grade = 444.65
Subgrade = 443.82
Original Grnd = 444.65



11+75.00
Des Grade = 443.27
Subgrade = 442.44
Original Grnd = 442.86

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1

11+75 TO 11+90

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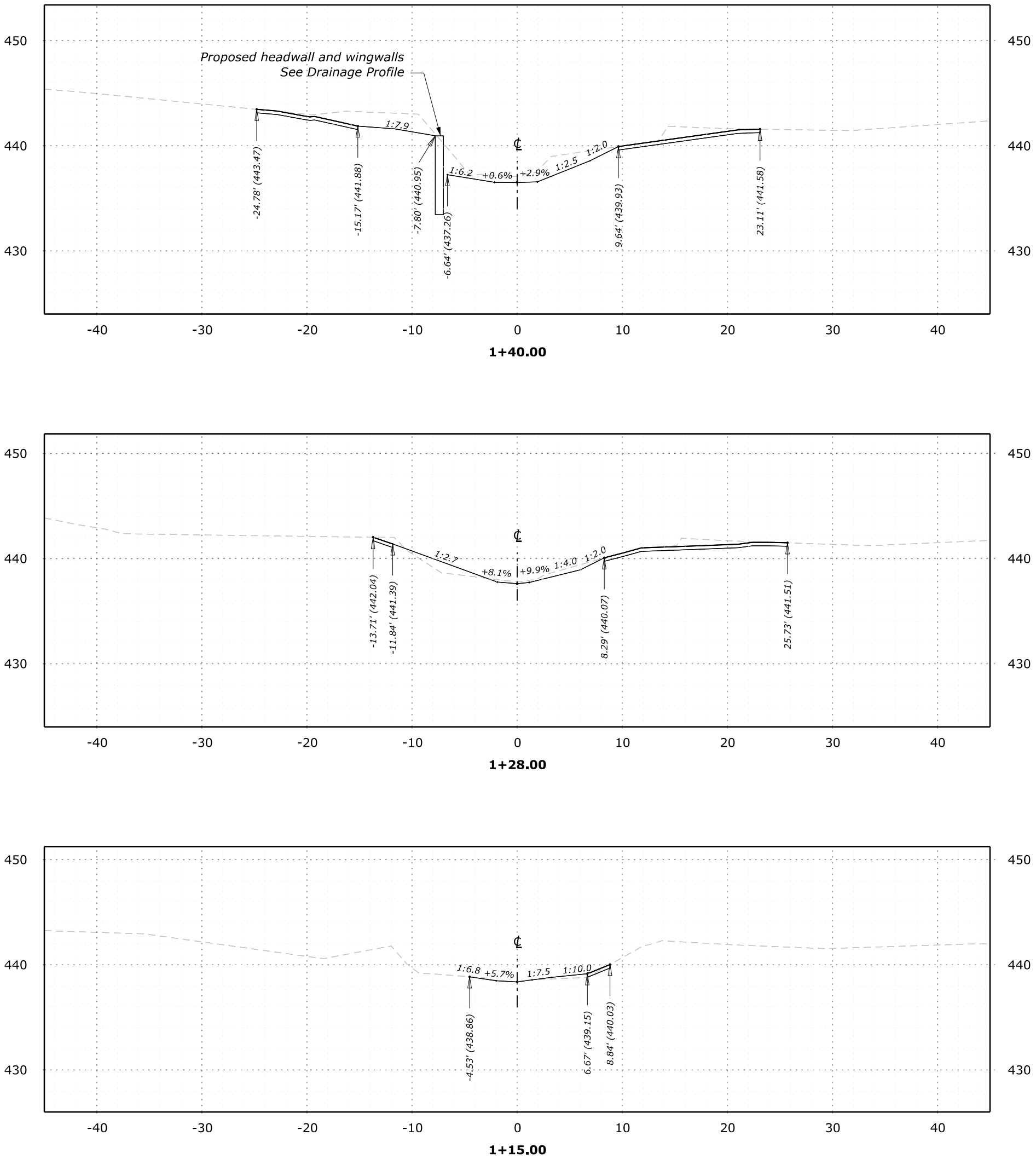
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	Y04



WALLKILL RIVER NATIONAL WILDLIFE REFUGE

STREAM CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1

1+15 TO 1+40

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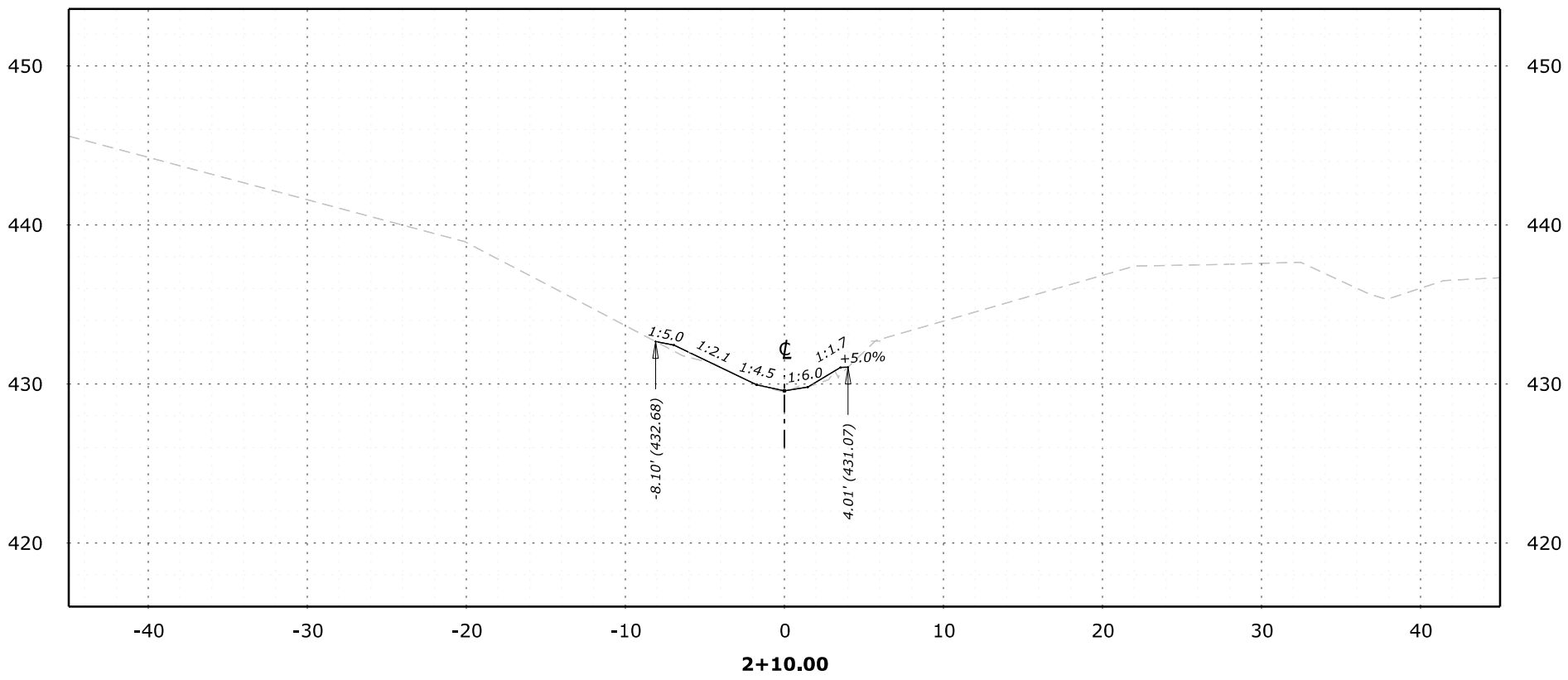
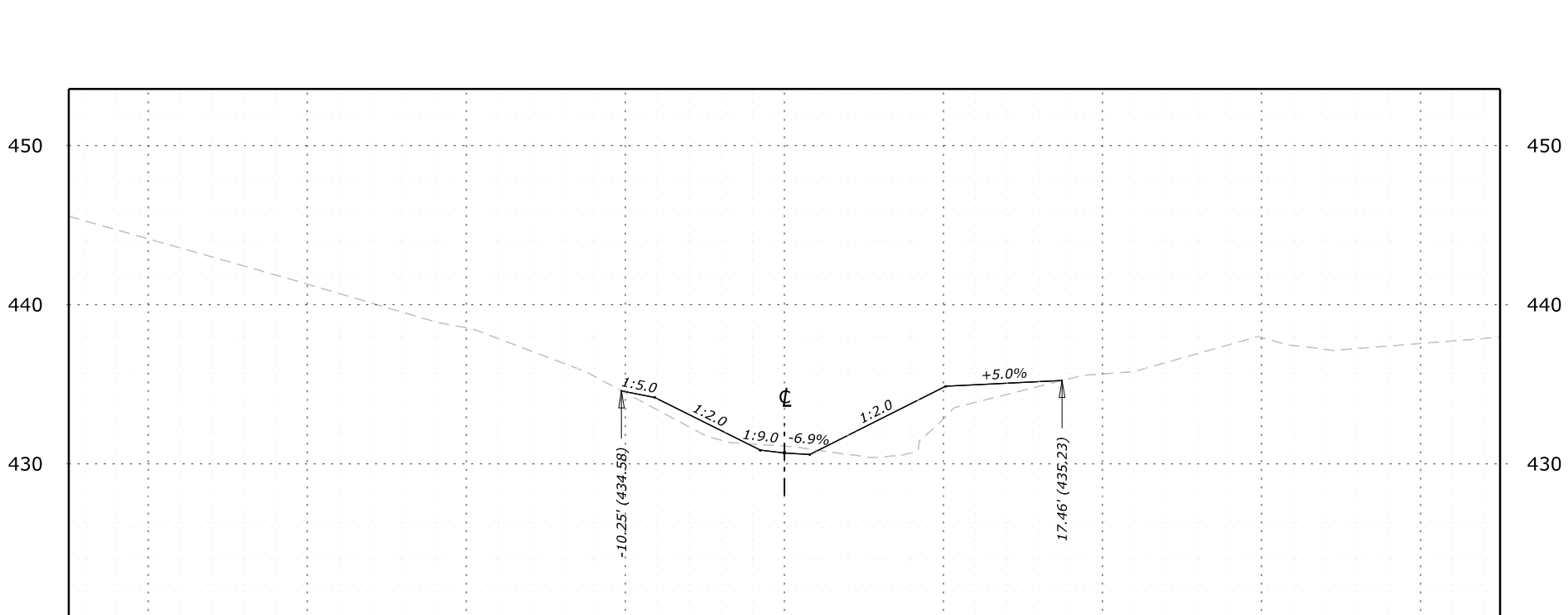
3 August 2026 5:30 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	Y05





WALLKILL RIVER NATIONAL WILDLIFE REFUGE

STREAM CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #1

1+85 TO 2+10

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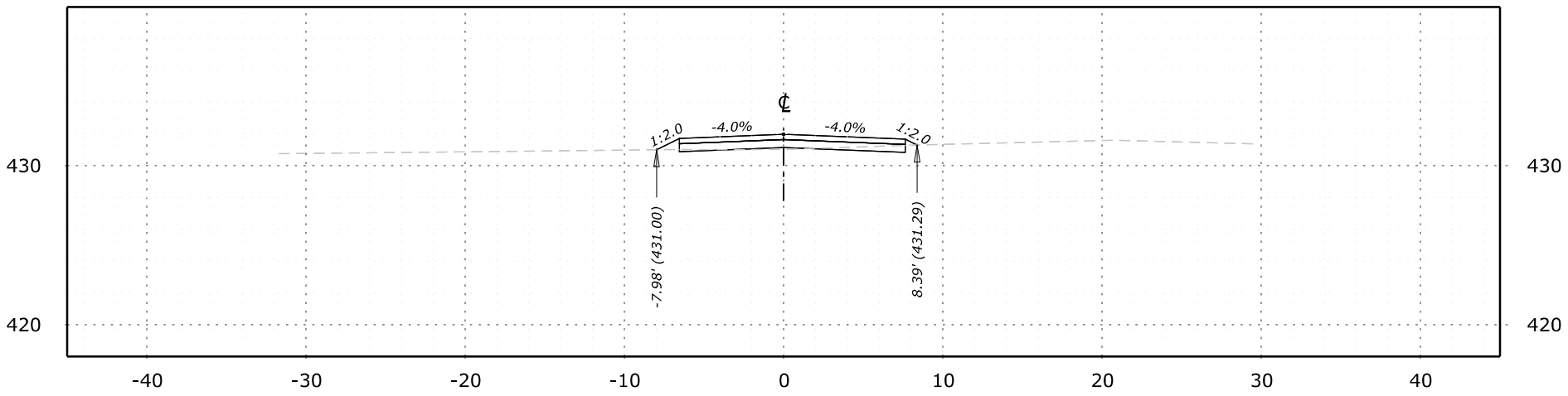
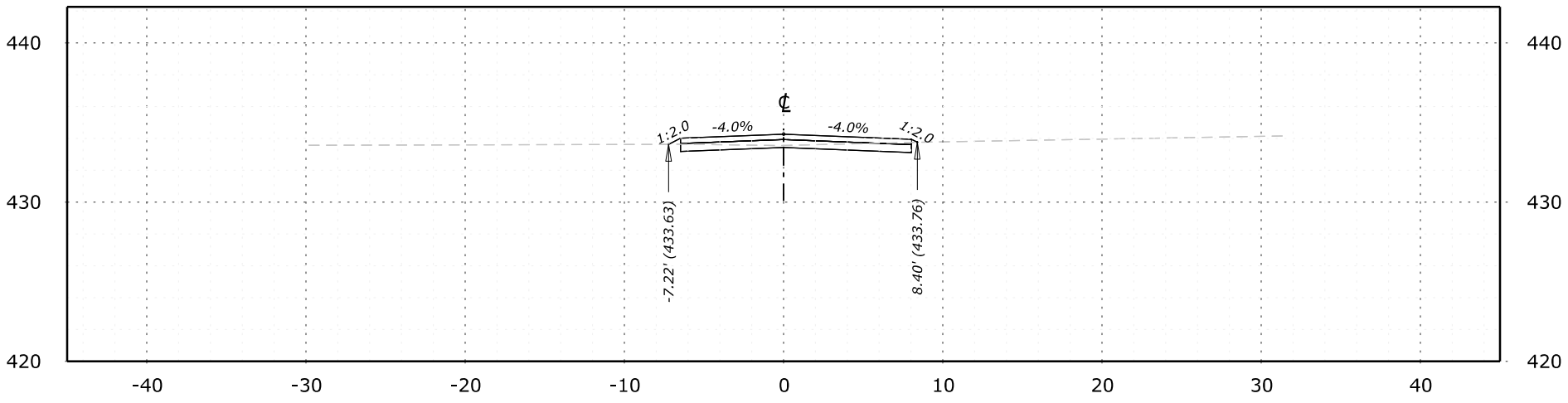
3 August 2026 5:30 PM

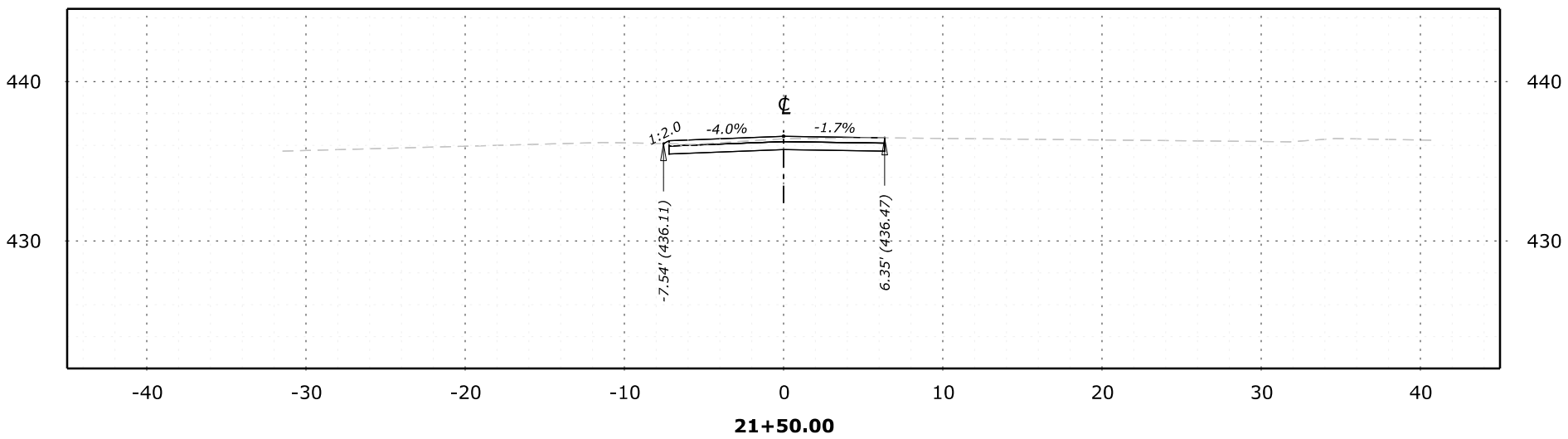
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	Z01



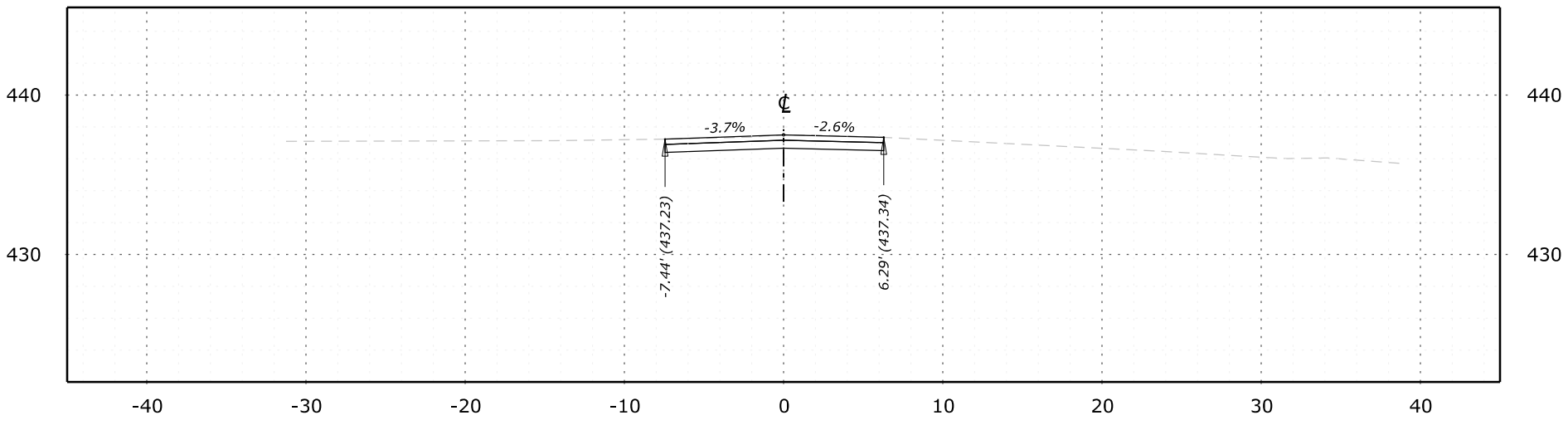
22+00.00
Des Grade = 431.97
Subgrade = 431.14
Original Grnd = 431.02



21+75.00
Des Grade = 434.27
Subgrade = 433.43
Original Grnd = 433.56



21+50.00
Des Grade = 436.57
Subgrade = 435.74
Original Grnd = 436.41



21+37.68
Des Grade = 437.50
Subgrade = 436.67
Original Grnd = 437.50

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #2

21+37.68 TO 22+00

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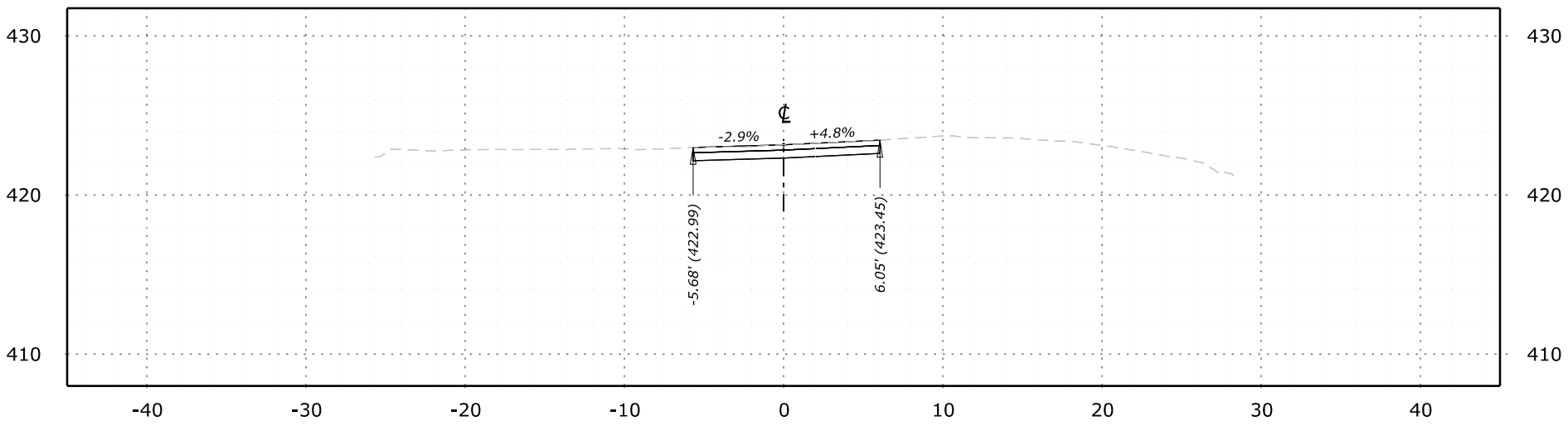
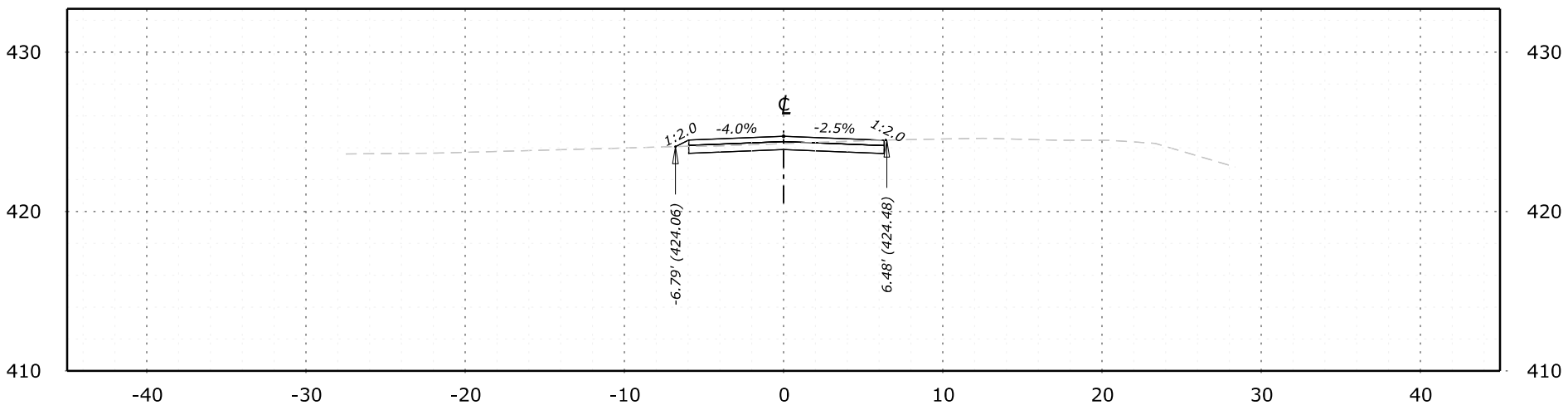
3 August 2026 5:35 PM

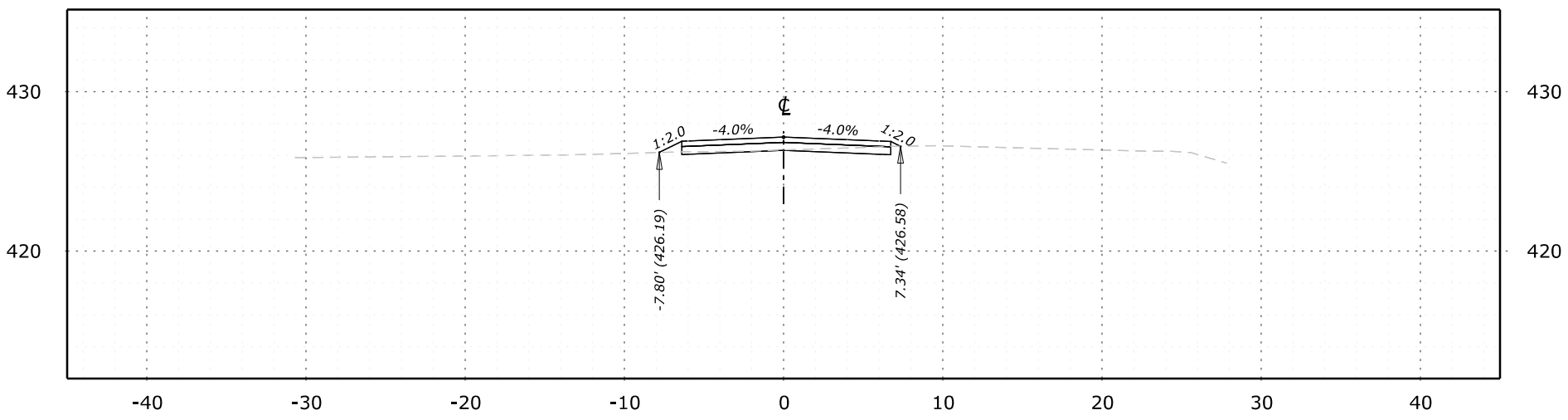
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	Z02



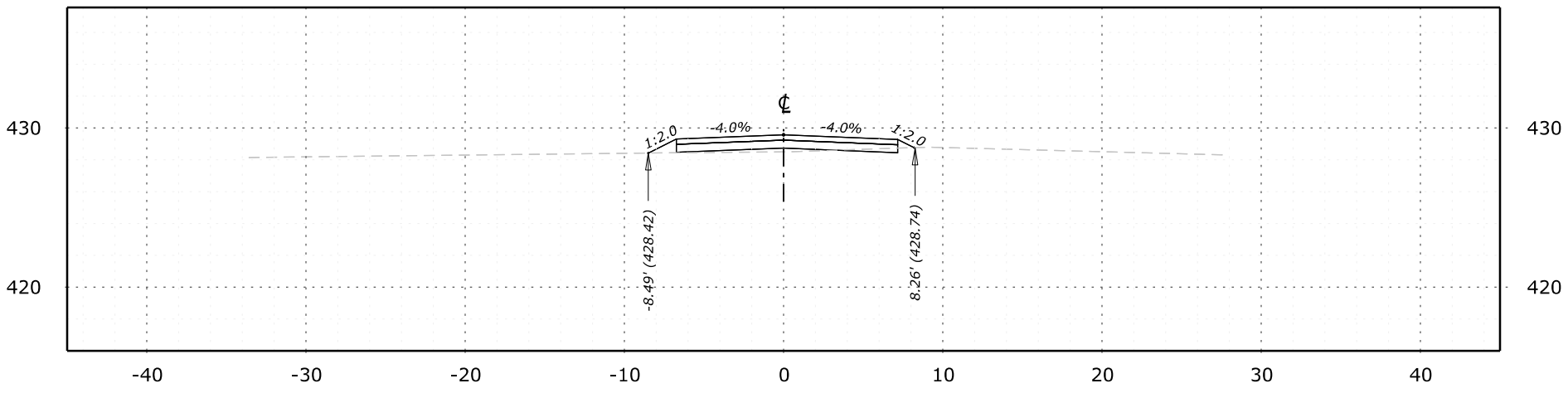
22+91.23
Des Grade = 423.16
Subgrade = 422.32
Original Grnd = 423.16



22+75.00
Des Grade = 424.71
Subgrade = 423.88
Original Grnd = 424.26



22+50.00
Des Grade = 427.15
Subgrade = 426.32
Original Grnd = 426.34



22+25.00
Des Grade = 429.57
Subgrade = 428.73
Original Grnd = 428.50

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #2

22+25 TO 22+91.23

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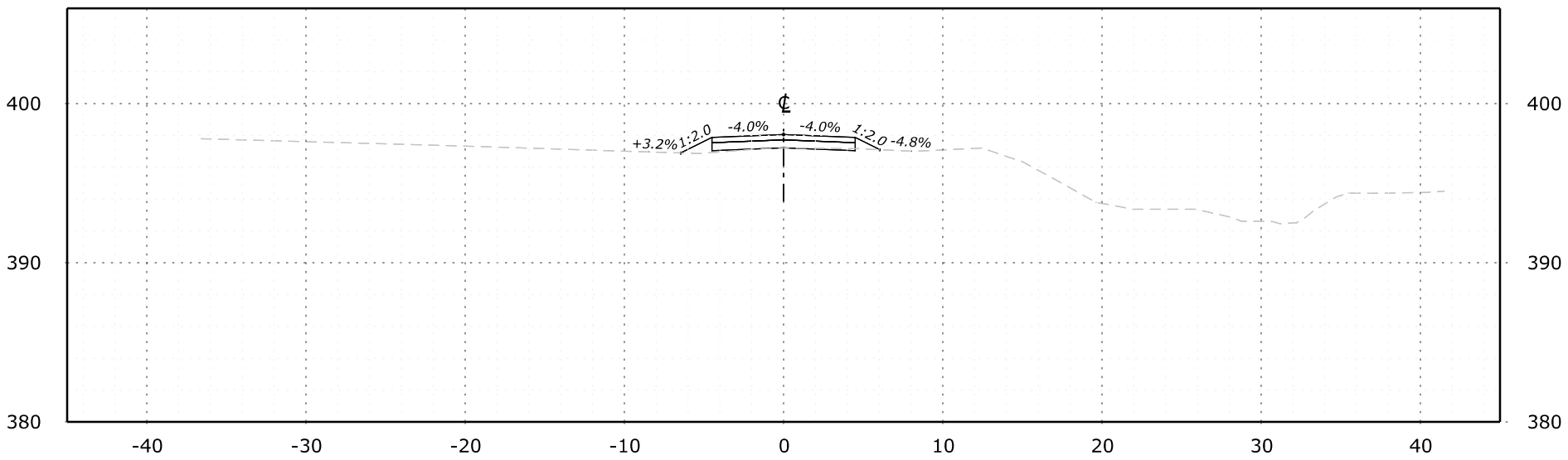
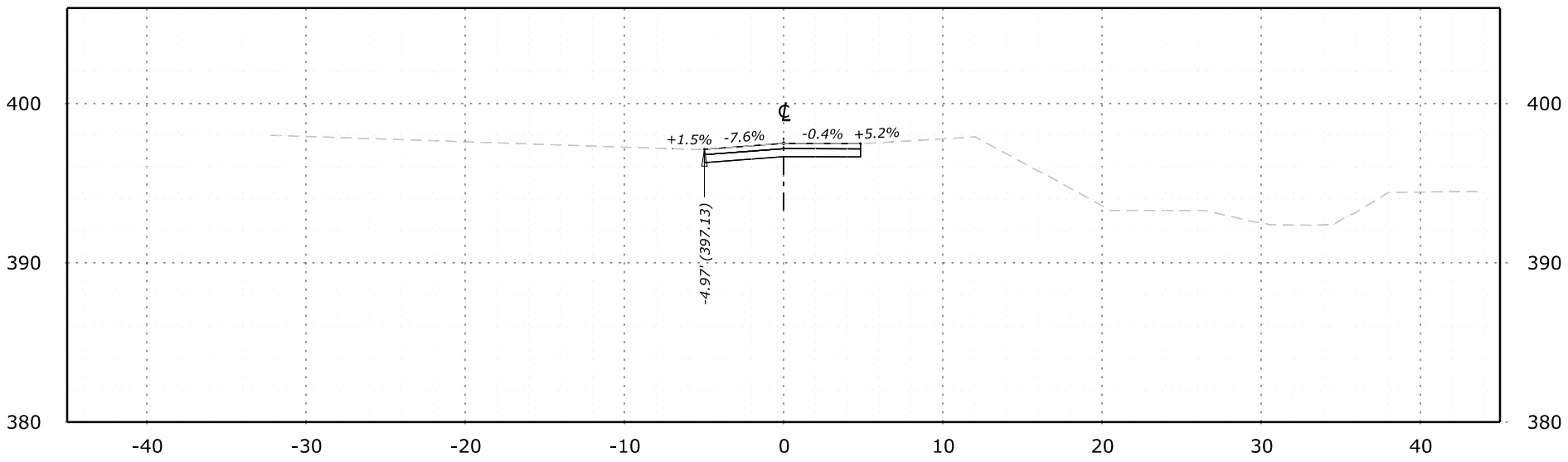
3 August 2026 5:35 PM

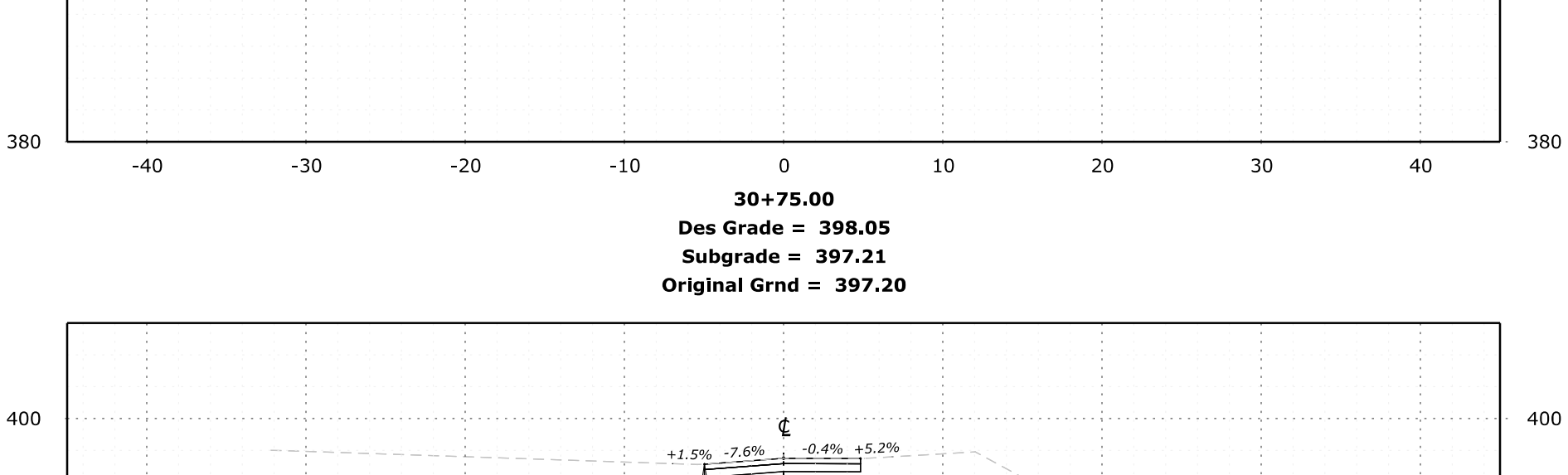
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	AA01



30+75.00
Des Grade = 398.05
Subgrade = 397.21
Original Grnd = 397.20



30+50.00
Des Grade = 397.51
Subgrade = 396.67
Original Grnd = 397.51



30+43.34
Des Grade = 397.46
Subgrade = 396.63
Original Grnd = 397.46

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #3

30+43.34 TO 30+75

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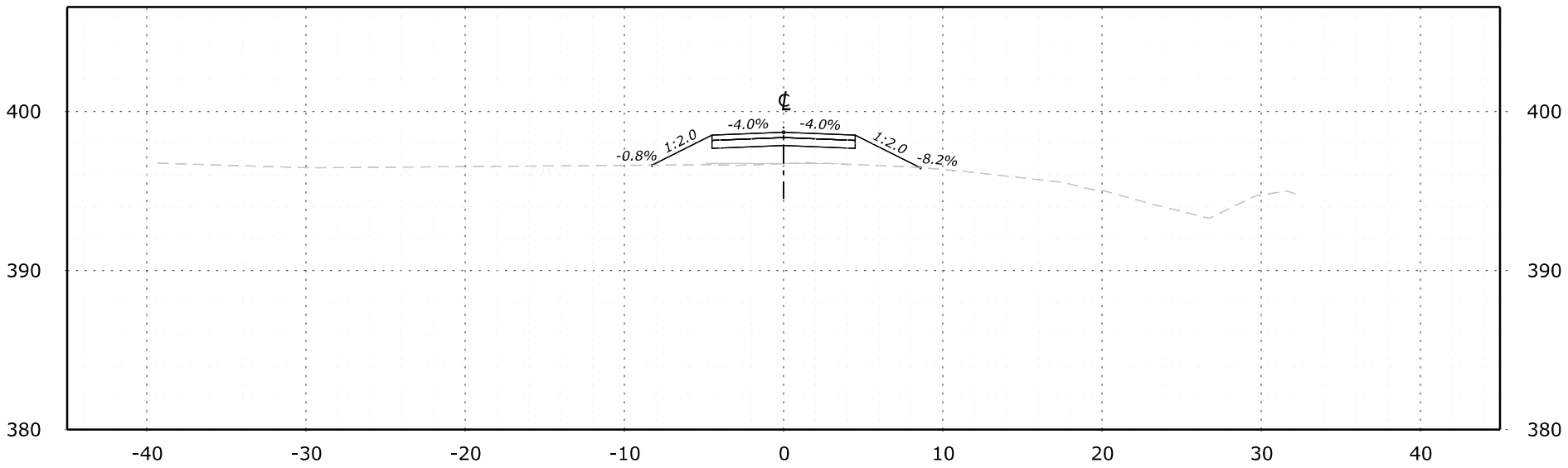
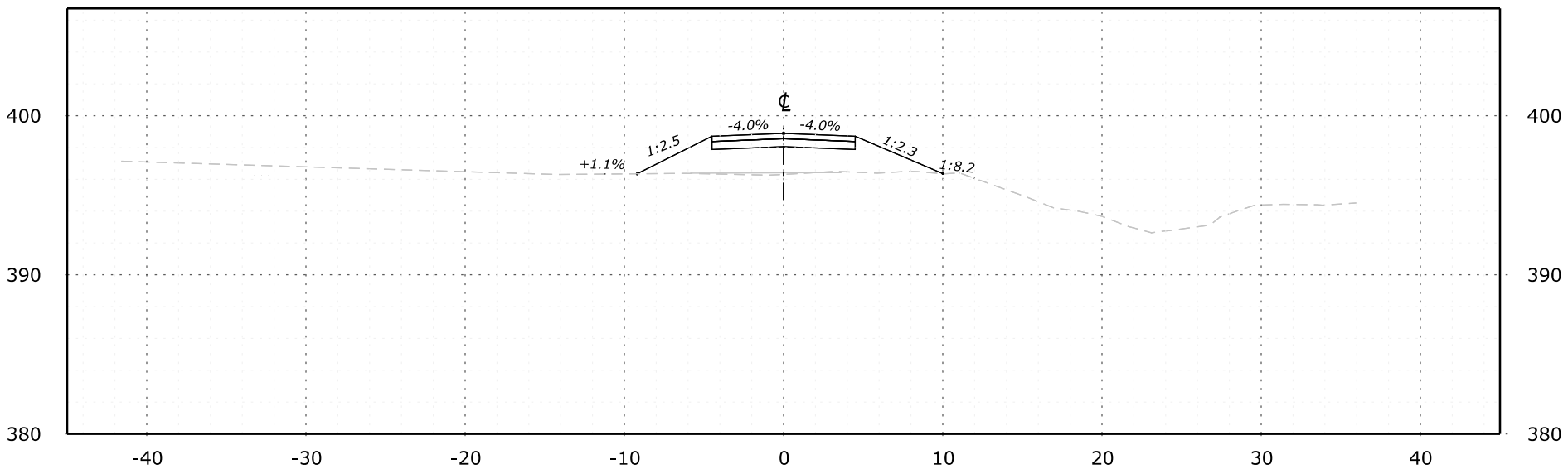
3 August 2026 5:39 PM

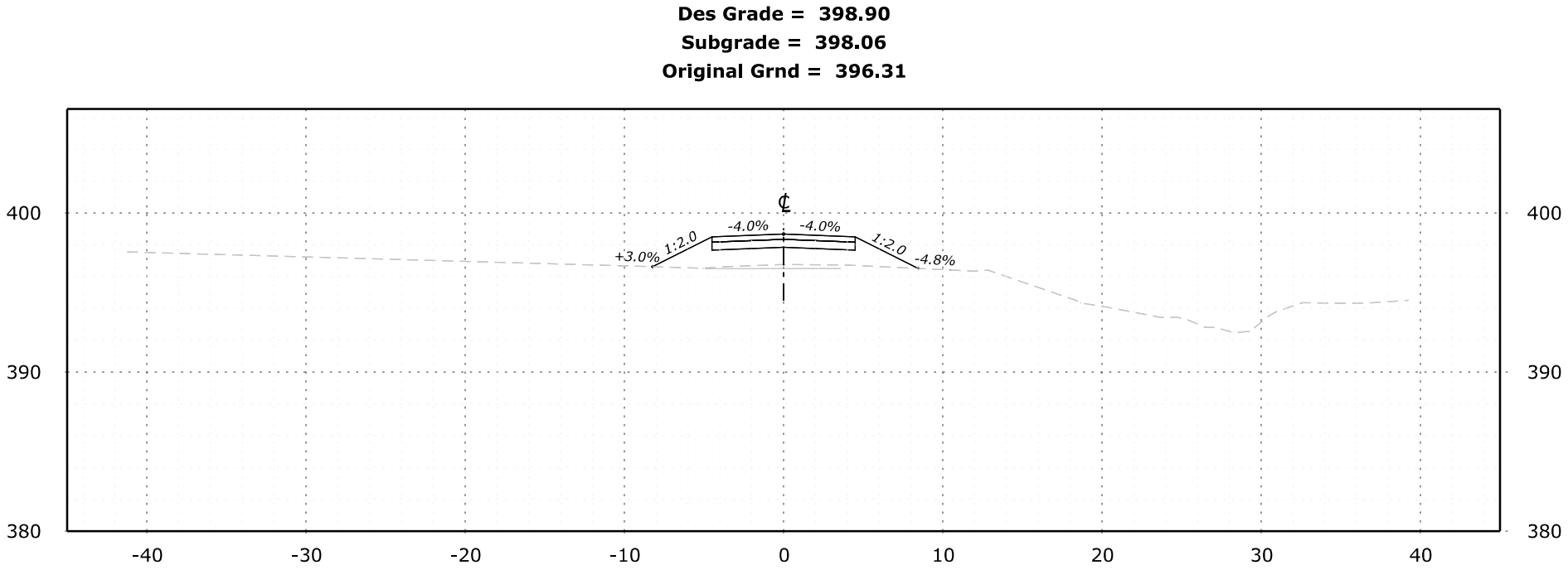
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	AA02



31+50.00
Des Grade = 398.69
Subgrade = 397.86
Original Grnd = 396.71



31+25.00
Des Grade = 398.90
Subgrade = 398.06
Original Grnd = 396.31



31+00.00
Des Grade = 398.68
Subgrade = 397.84
Original Grnd = 396.77

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #3

31+00 TO 31+50

3 August 2026 5:39 PM
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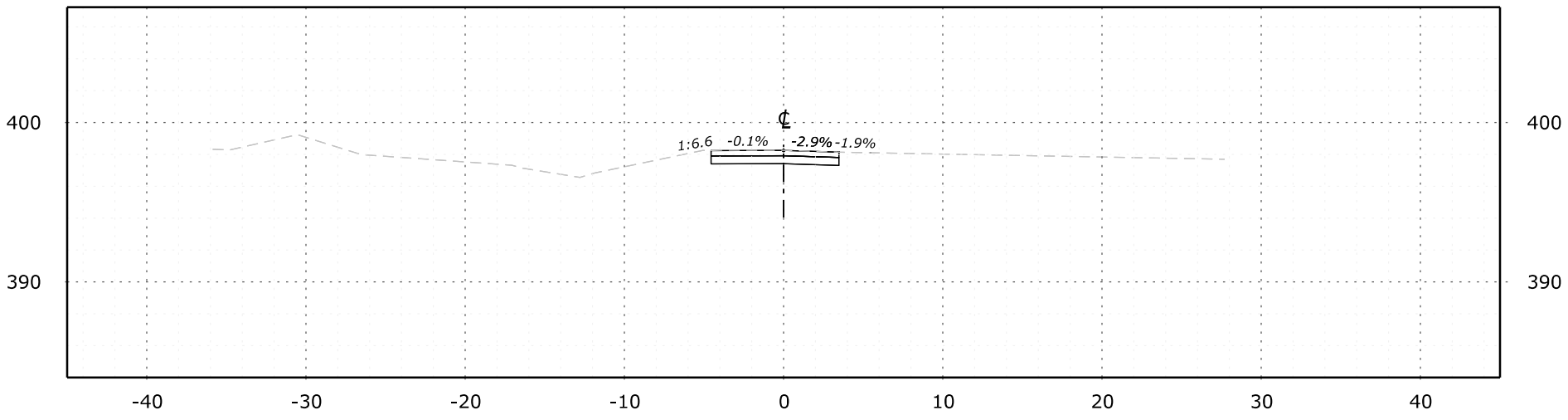
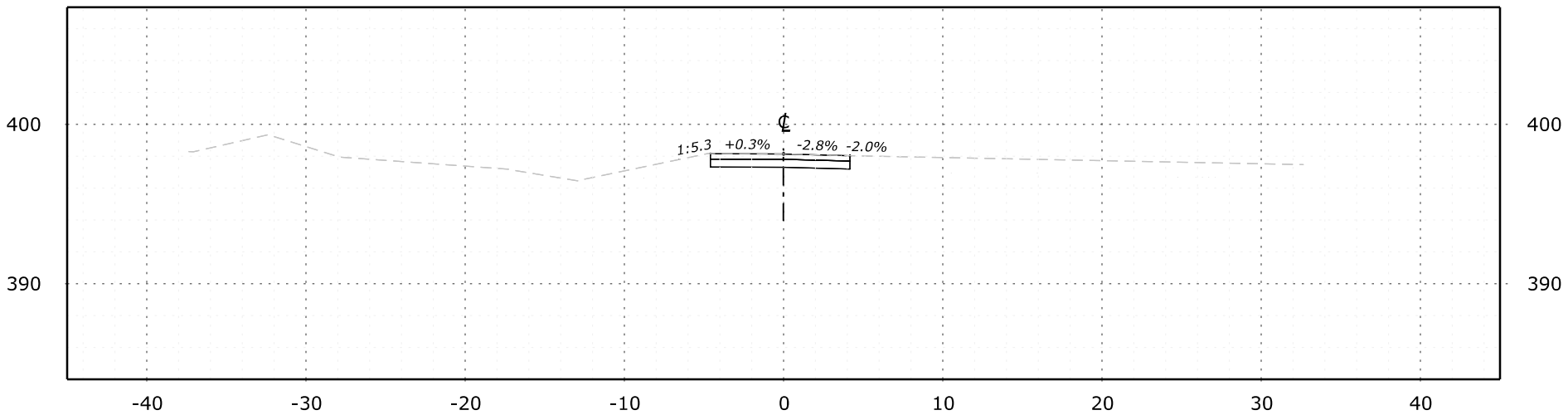
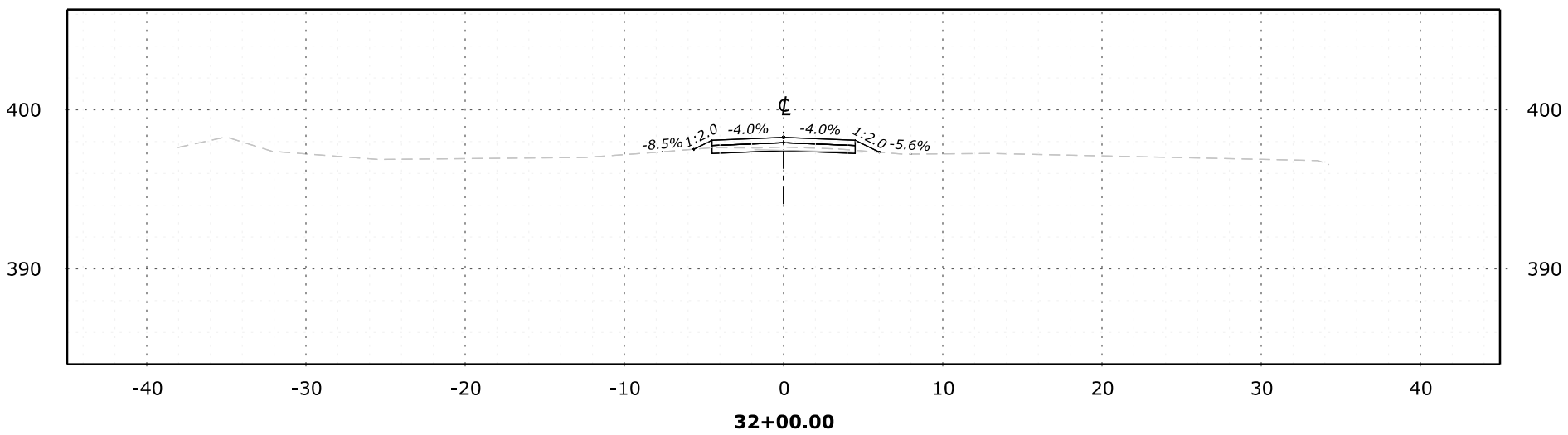
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	AA03



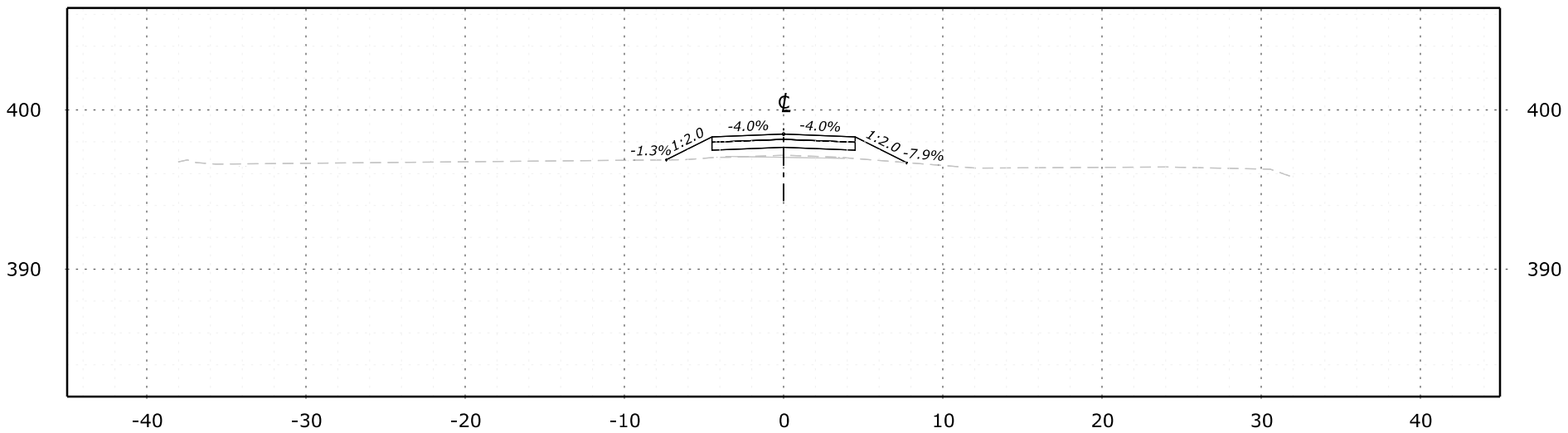
32+32.64
Des Grade = 398.24
Subgrade = 397.41
Original Grnd = 398.24



32+25.00
Des Grade = 398.14
Subgrade = 397.31
Original Grnd = 398.14



32+00.00
Des Grade = 398.27
Subgrade = 397.44
Original Grnd = 397.66



31+75.00
Des Grade = 398.48
Subgrade = 397.65
Original Grnd = 397.15

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #3

31+75 TO 32+32.64

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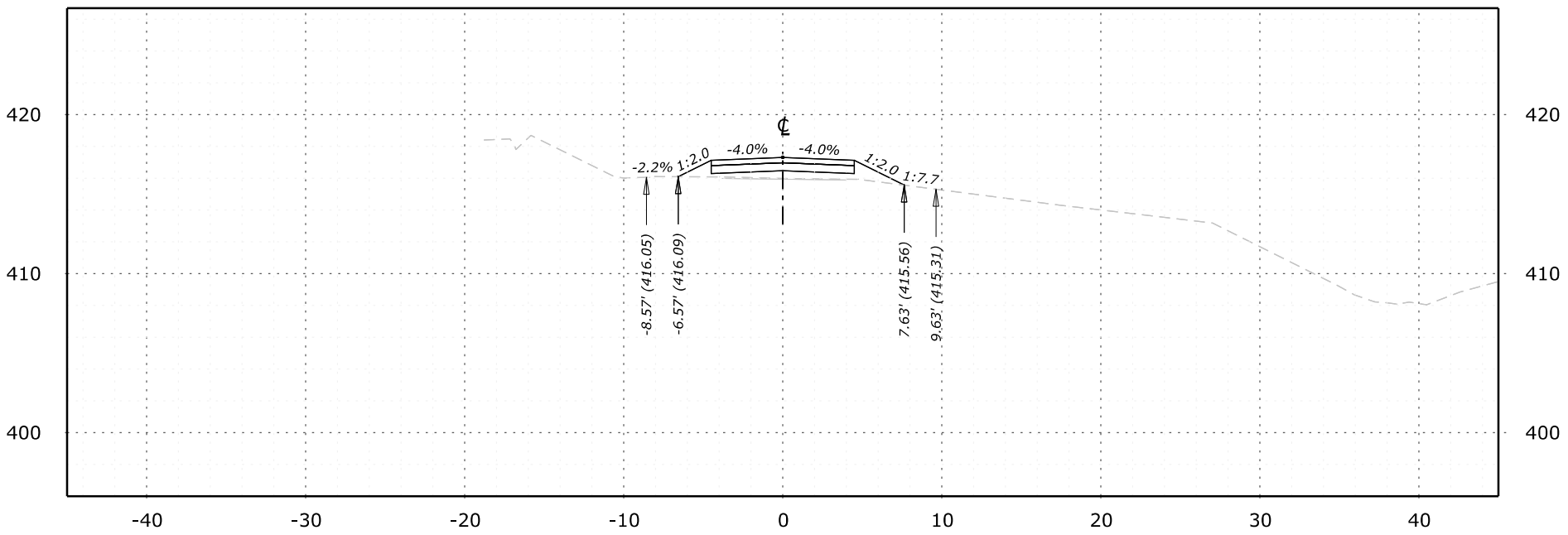
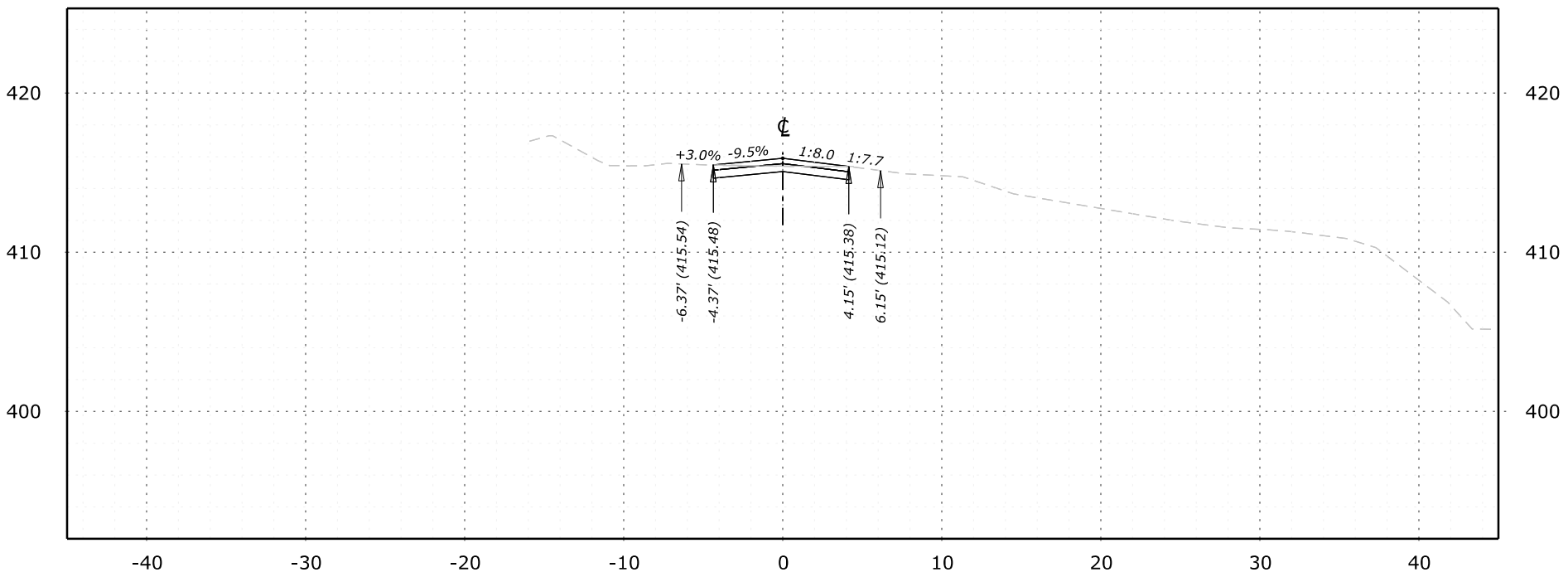
3 August 2026 5:39 PM

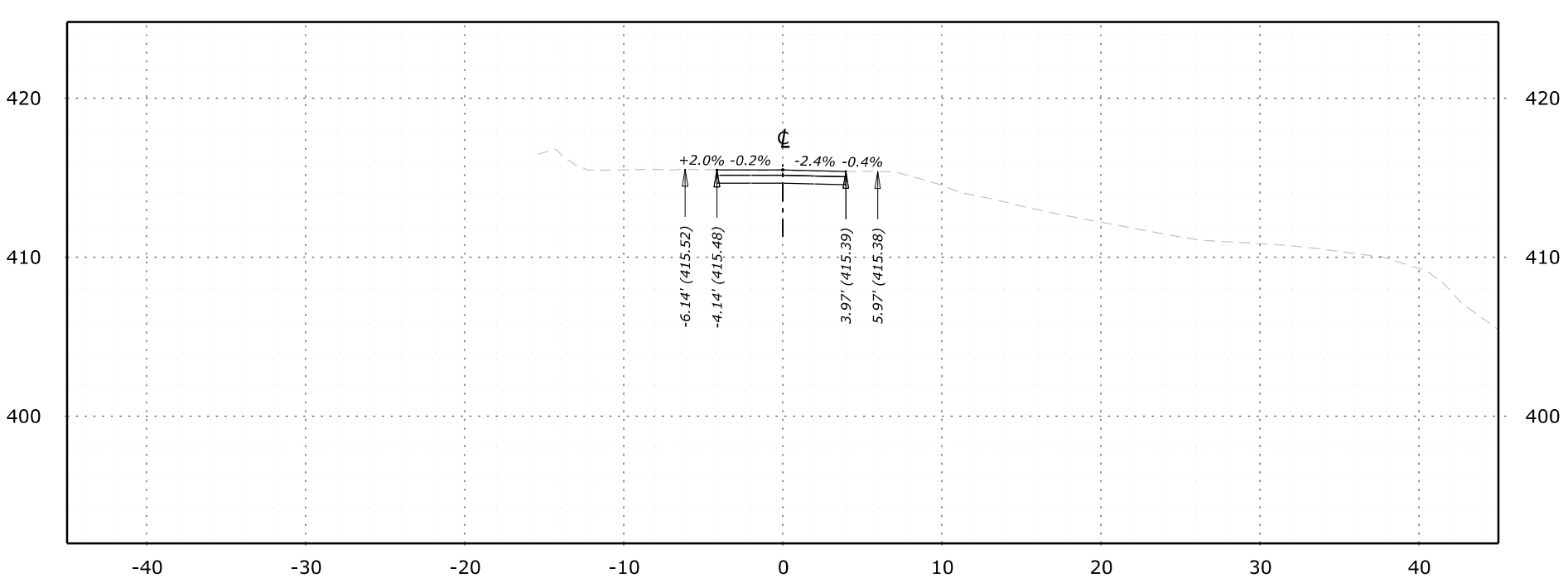
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	BB01



40+50.00
Des Grade = 417.31
Subgrade = 416.48
Original Grnd = 415.99



40+25.00
Des Grade = 415.90
Subgrade = 415.06
Original Grnd = 415.39



40+15.57
Des Grade = 415.49
Subgrade = 414.65
Original Grnd = 415.49

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #4

40+15.57 TO 40+50

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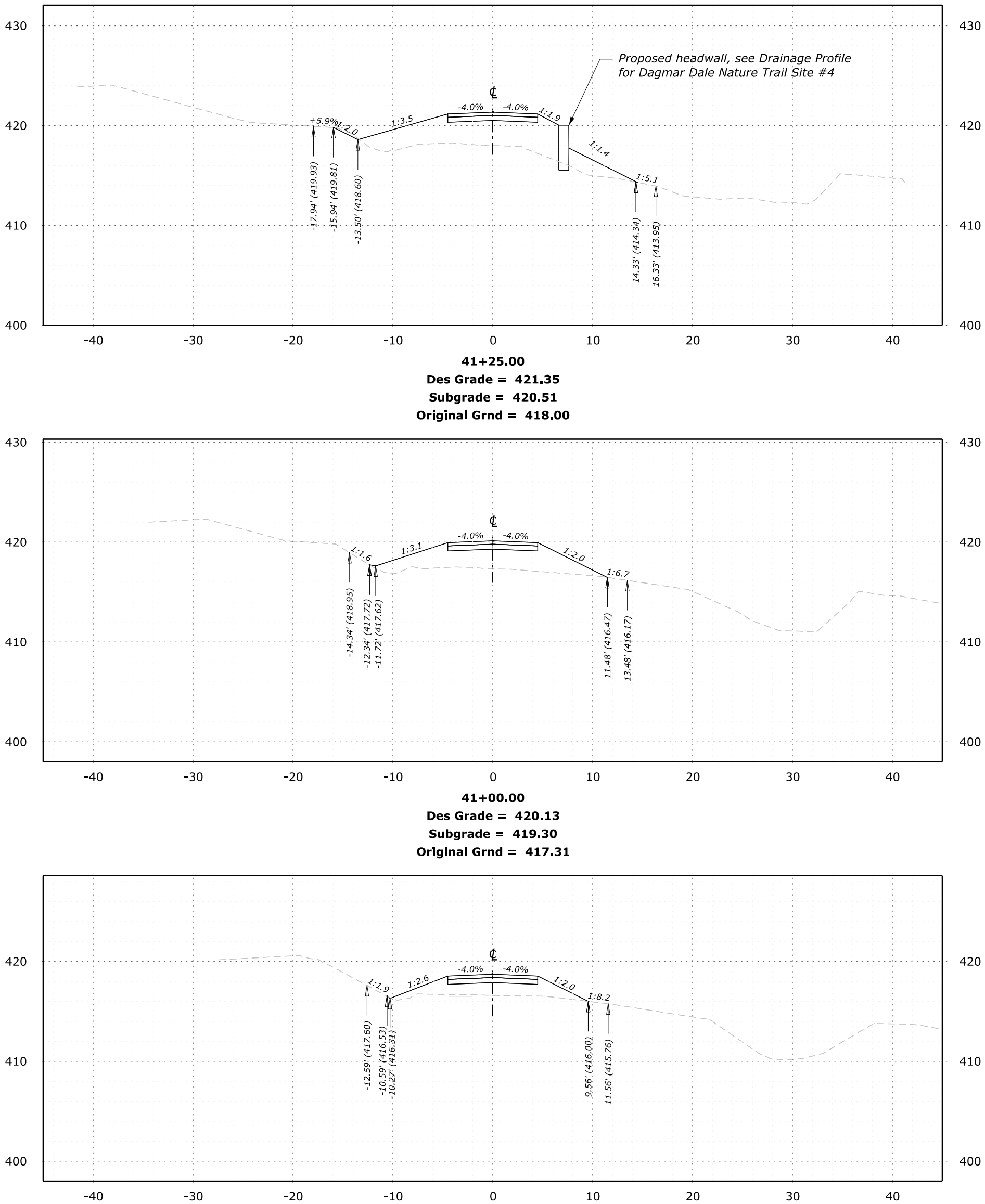
3 August 2026 5:45 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	BB02



41+25.00
Des Grade = 421.35
Subgrade = 420.51
Original Grnd = 418.00

41+00.00
Des Grade = 420.13
Subgrade = 419.30
Original Grnd = 417.31

40+75.00
Des Grade = 418.72
Subgrade = 417.89
Original Grnd = 416.59

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #4

40+75 TO 41+25

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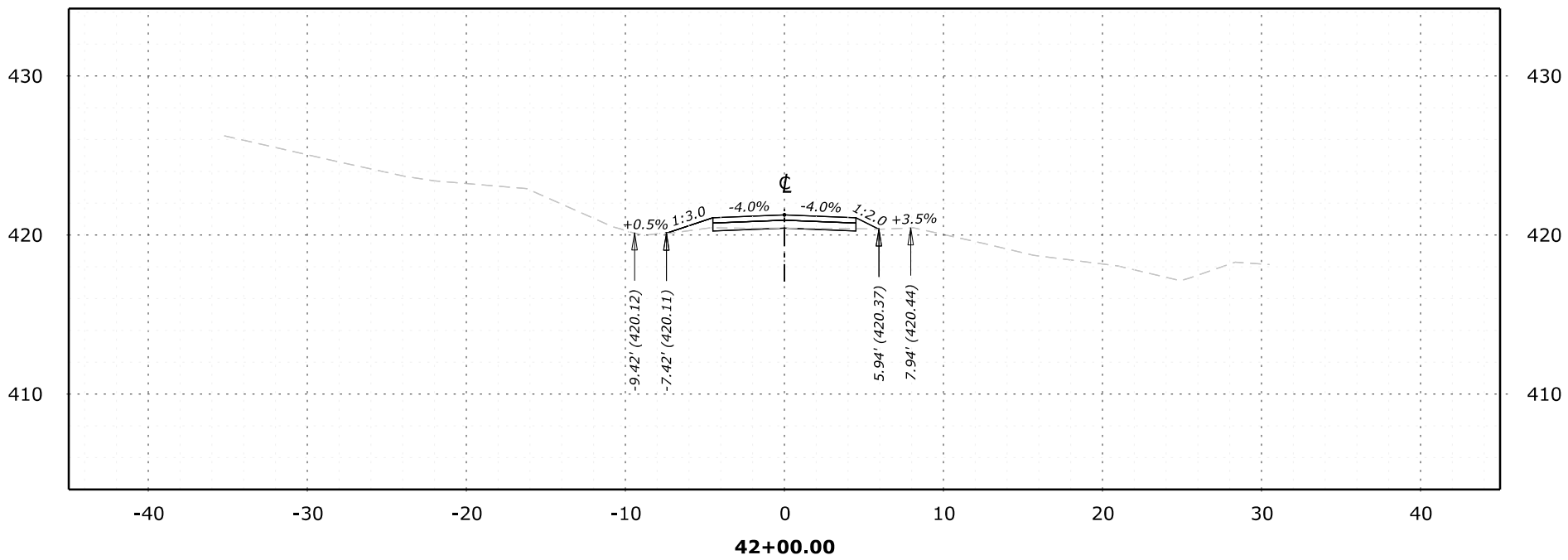
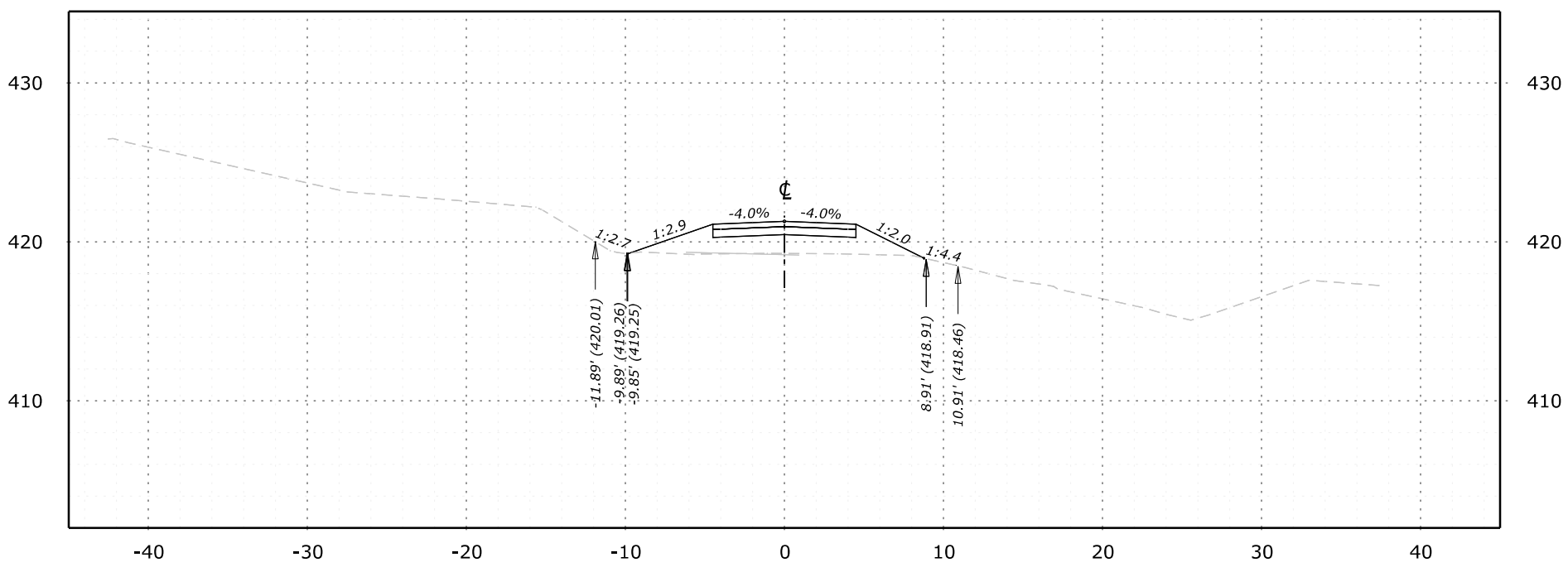
3 August 2026 5:45 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	BB03



42+00.00
Des Grade = 421.27
Subgrade = 420.43
Original Grnd = 420.43



41+75.00
Des Grade = 421.29
Subgrade = 420.46
Original Grnd = 419.29

41+50.00
Des Grade = 421.32
Subgrade = 420.49
Original Grnd = 418.59

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #4

41+50 TO 42+00

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3 August 2026 5:45 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	BB04

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3 August 2026 5:45 PM

42+19.32
Des Grade = 421.32
Subgrade = 420.49
Original Grnd = 421.32

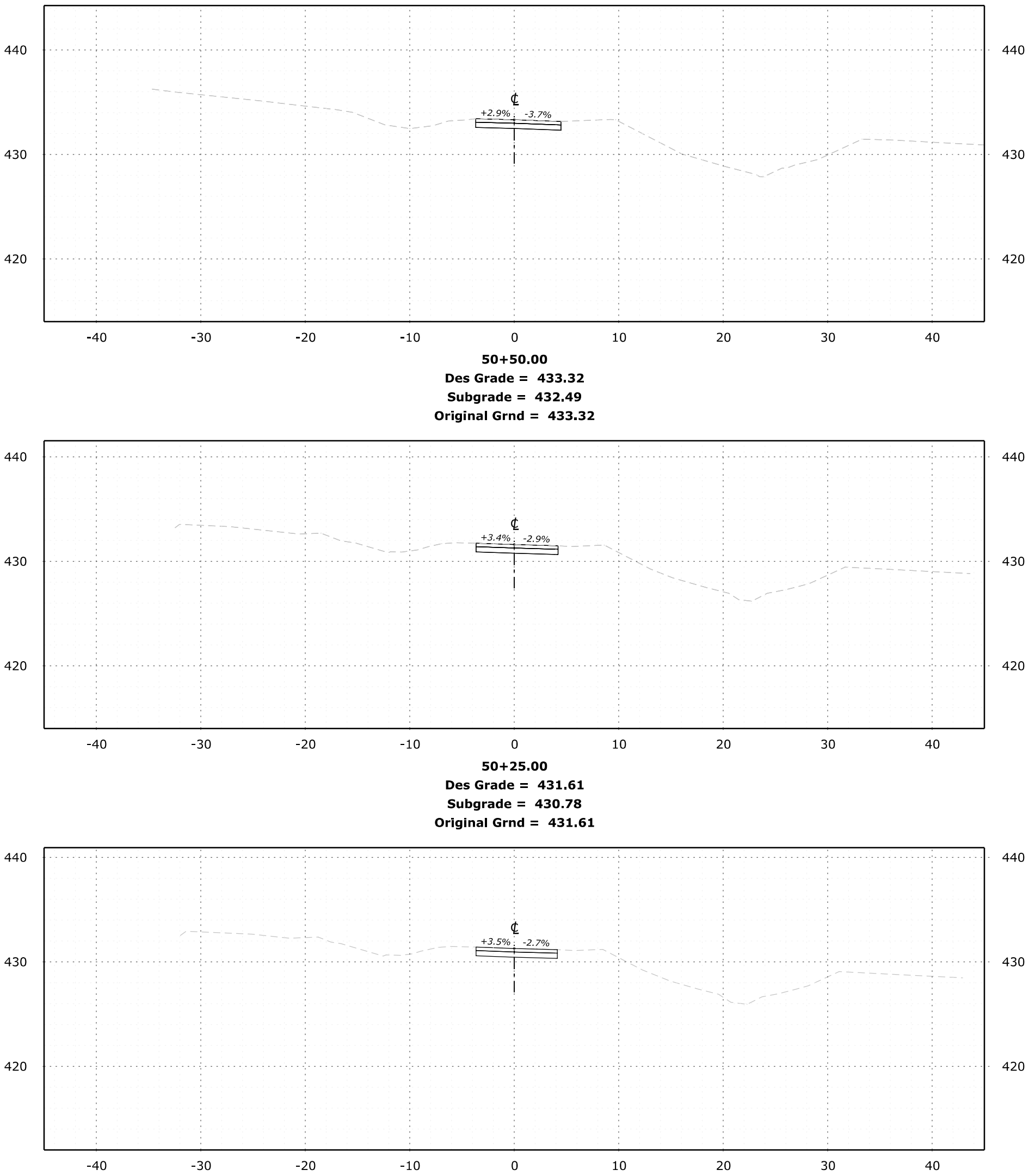
WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #4

42+19.32

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	CC01



50+50.00
Des Grade = 433.32
Subgrade = 432.49
Original Grnd = 433.32

50+25.00
Des Grade = 431.61
Subgrade = 430.78
Original Grnd = 431.61

50+20.00
Des Grade = 431.28
Subgrade = 430.45
Original Grnd = 431.28

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #5

50+20 to 50+50

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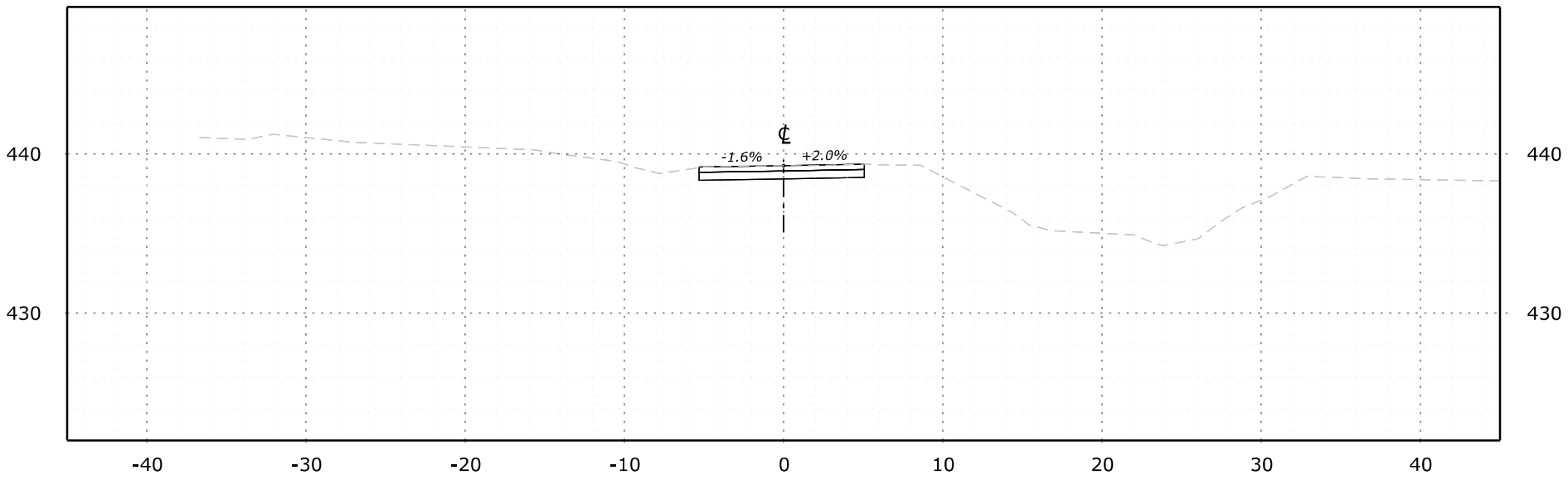
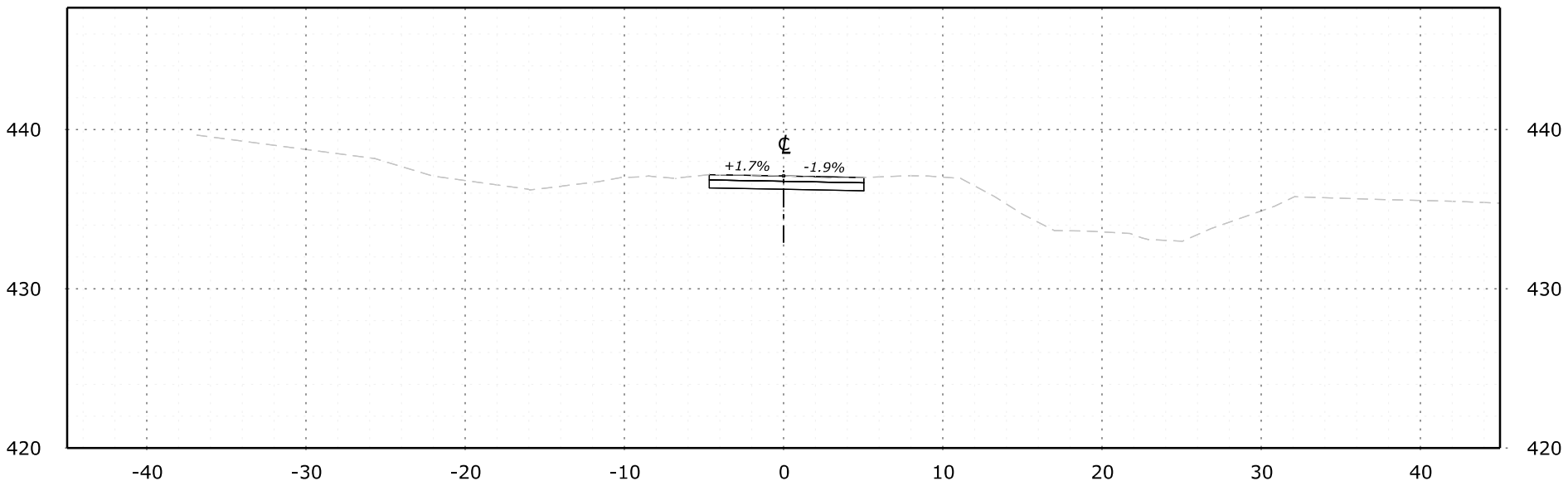
3 August 2026 5:50 PM

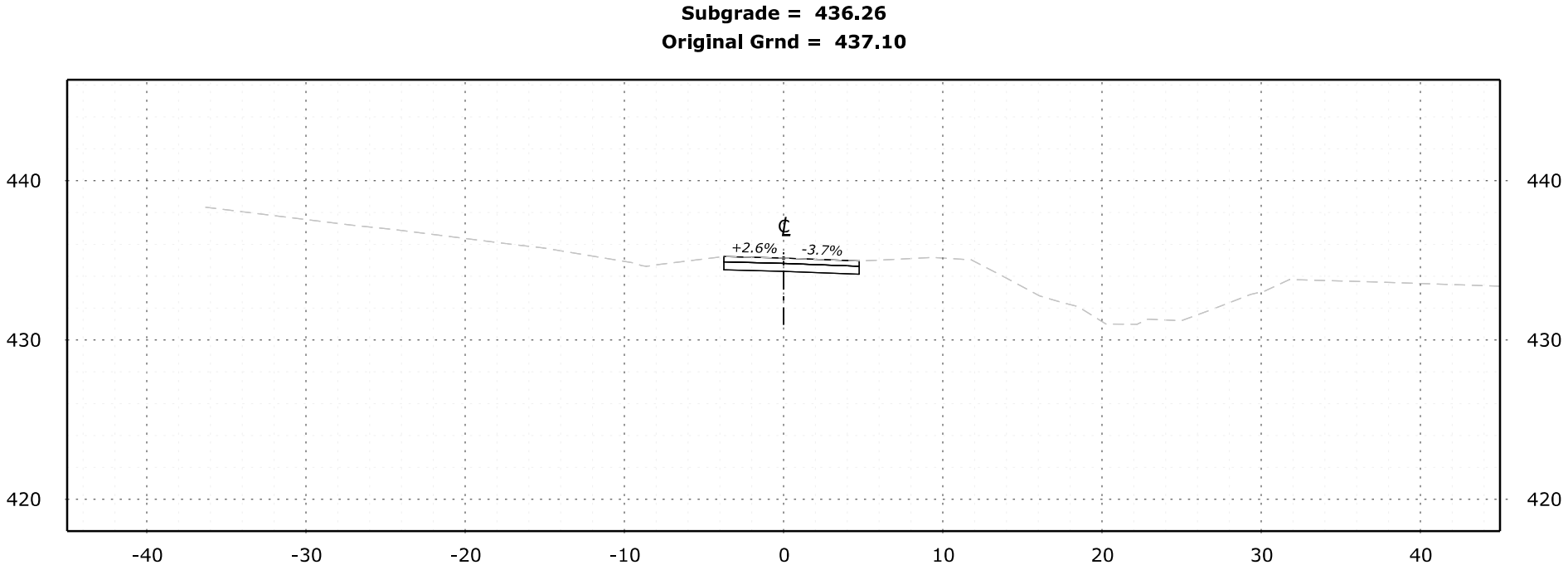
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	CC02



51+25.00
Des Grade = 439.26
Subgrade = 438.43
Original Grnd = 439.27



51+00.00
Des Grade = 437.10
Subgrade = 436.26
Original Grnd = 437.10



50+75.00
Des Grade = 435.14
Subgrade = 434.30
Original Grnd = 435.14

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #5

50+75 to 51+25

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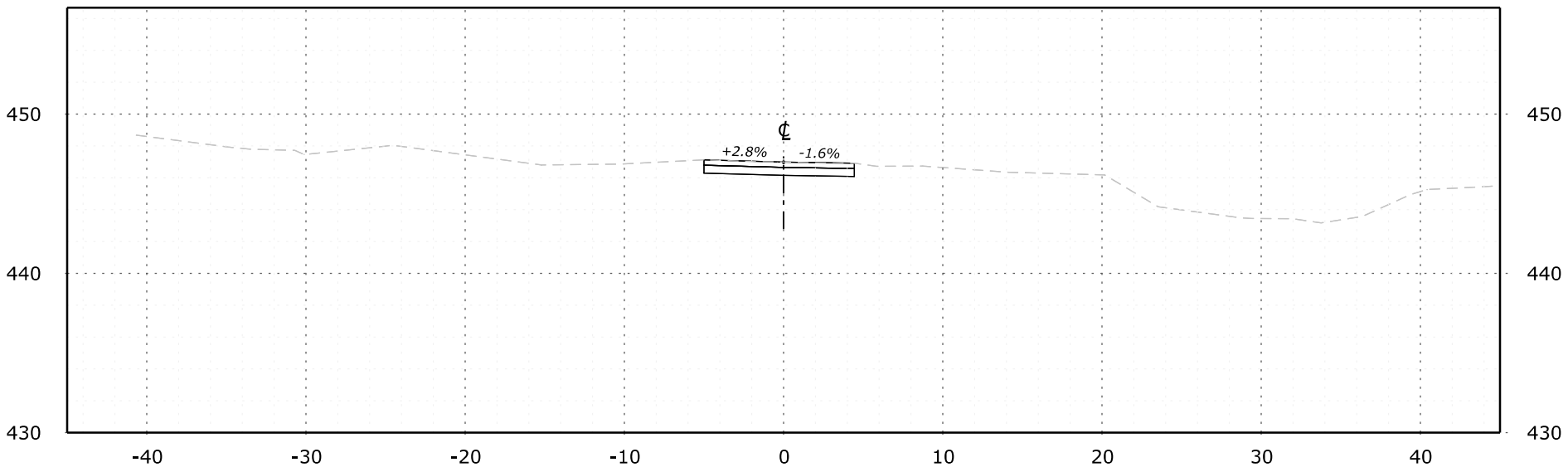
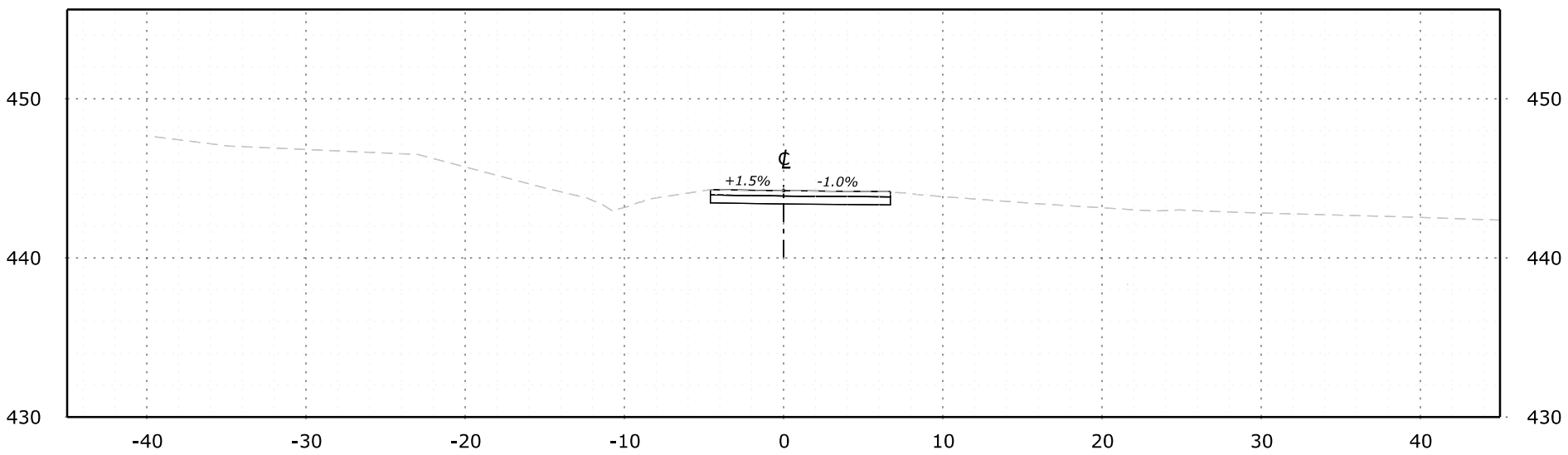
3 August 2026 5:50 PM

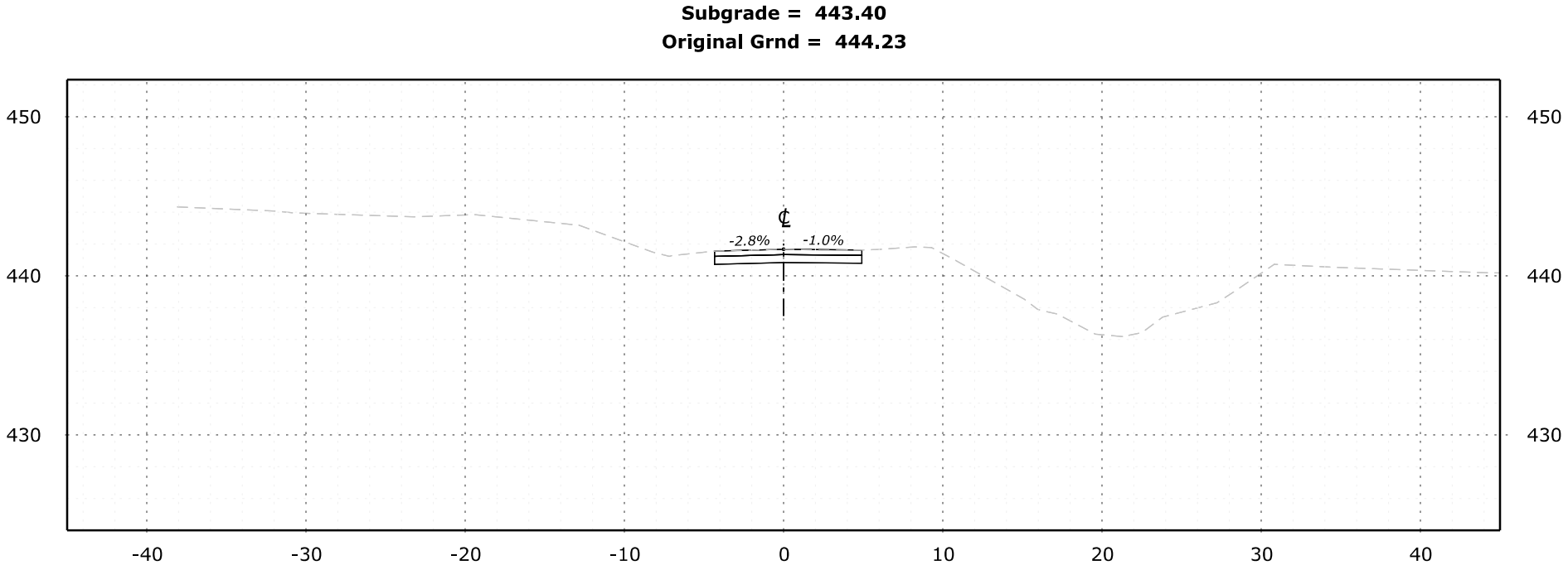
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	CC03



52+00.00
Des Grade = 446.99
Subgrade = 446.16
Original Grnd = 446.99



51+75.00
Des Grade = 444.23
Subgrade = 443.40
Original Grnd = 444.23



51+50.00
Des Grade = 441.67
Subgrade = 440.84
Original Grnd = 441.67

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #5

51+50 to 52+00

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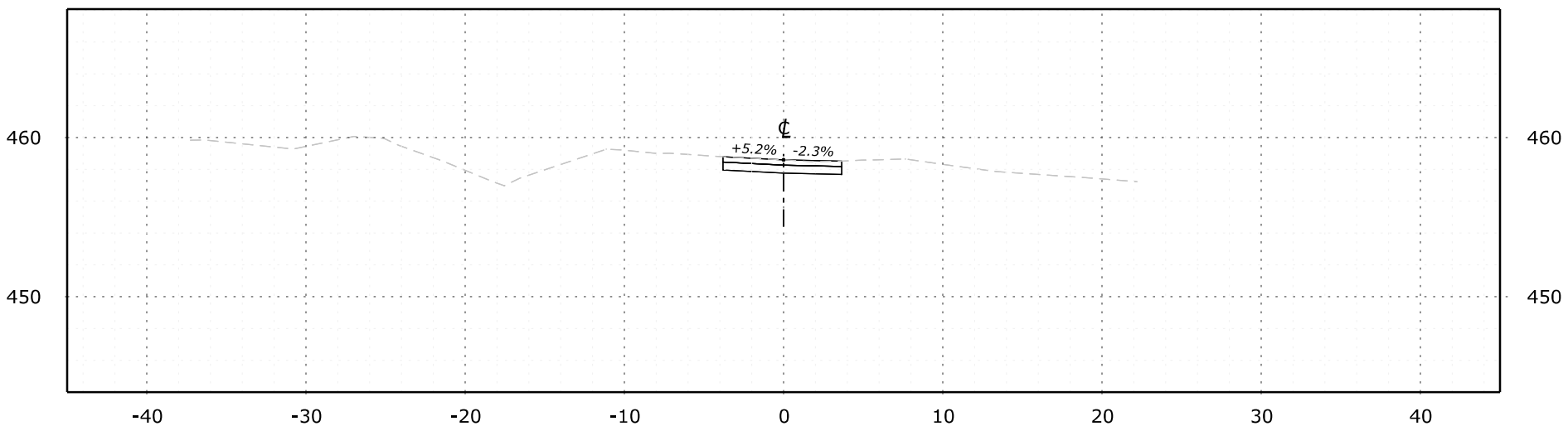
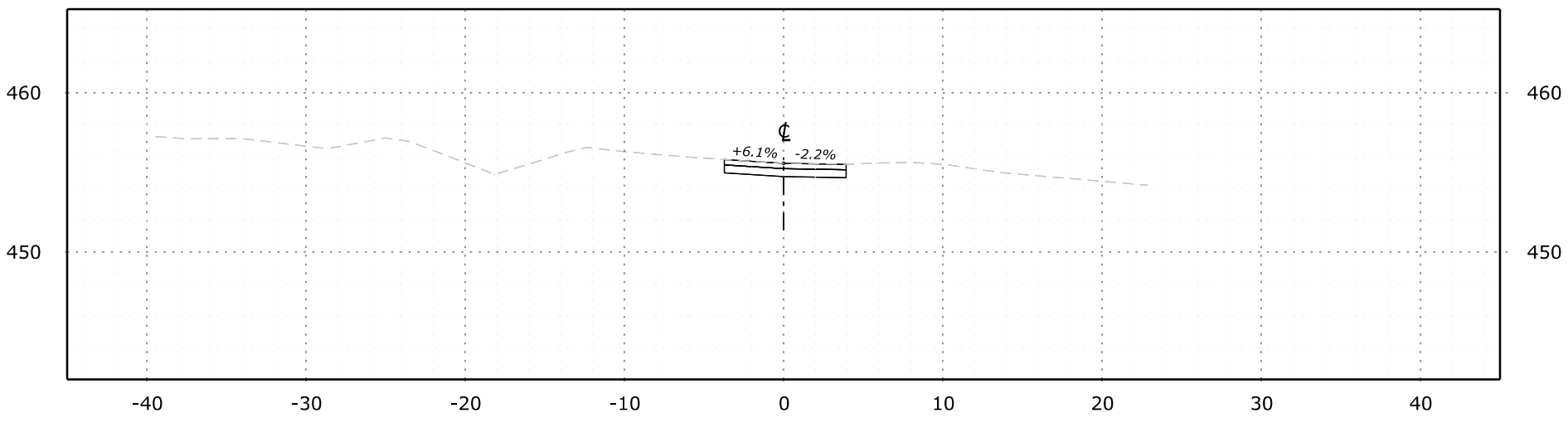
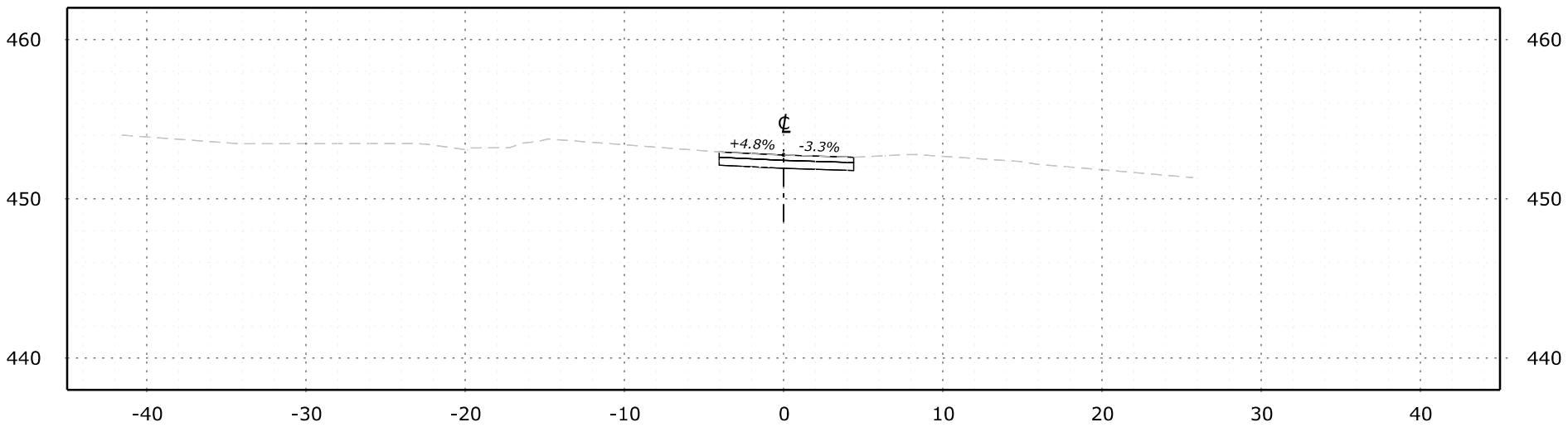
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	CC04



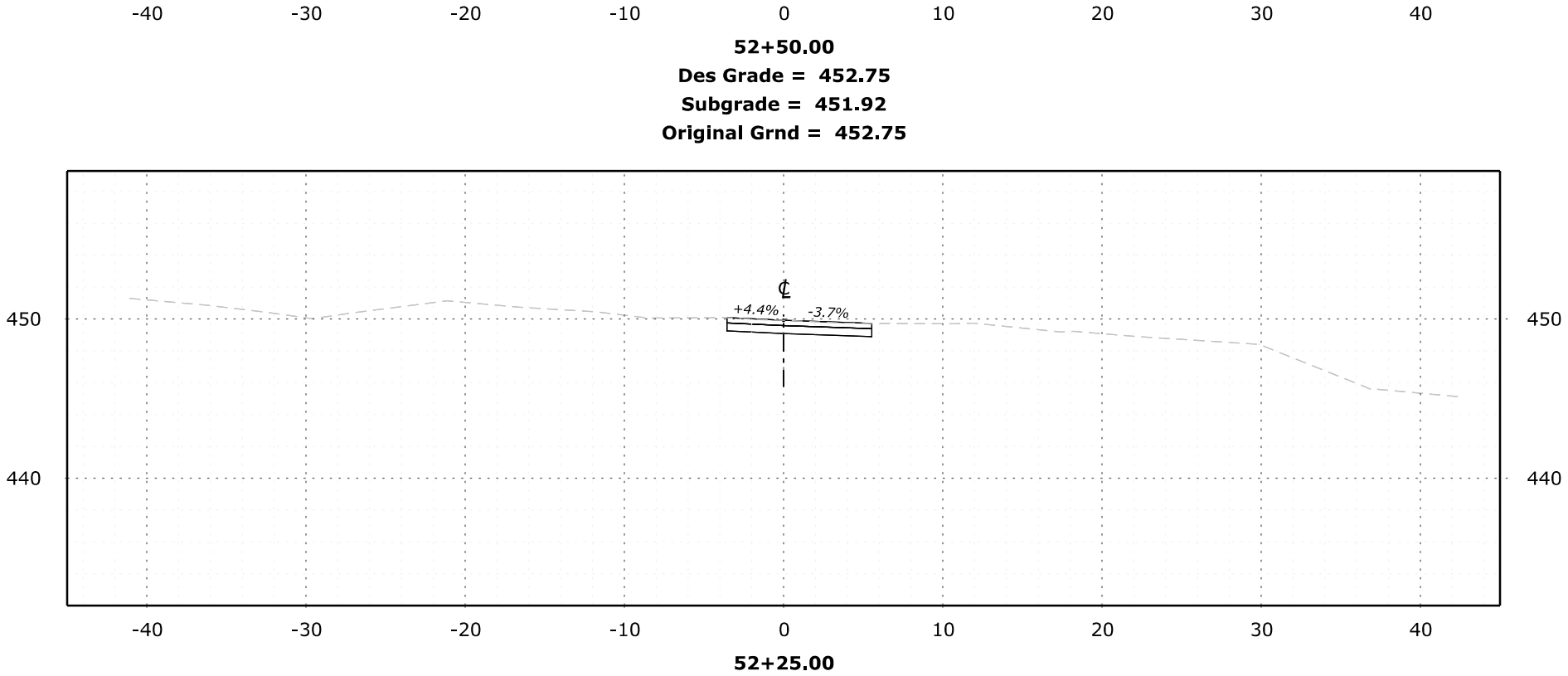
53+00.00
Des Grade = 458.60
Subgrade = 457.76
Original Grnd = 458.60



52+75.00
Des Grade = 455.58
Subgrade = 454.75
Original Grnd = 455.58



52+50.00
Des Grade = 452.75
Subgrade = 451.92
Original Grnd = 452.75



52+25.00
Des Grade = 449.92
Subgrade = 449.08
Original Grnd = 449.92

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #5

52+25 to 53+00

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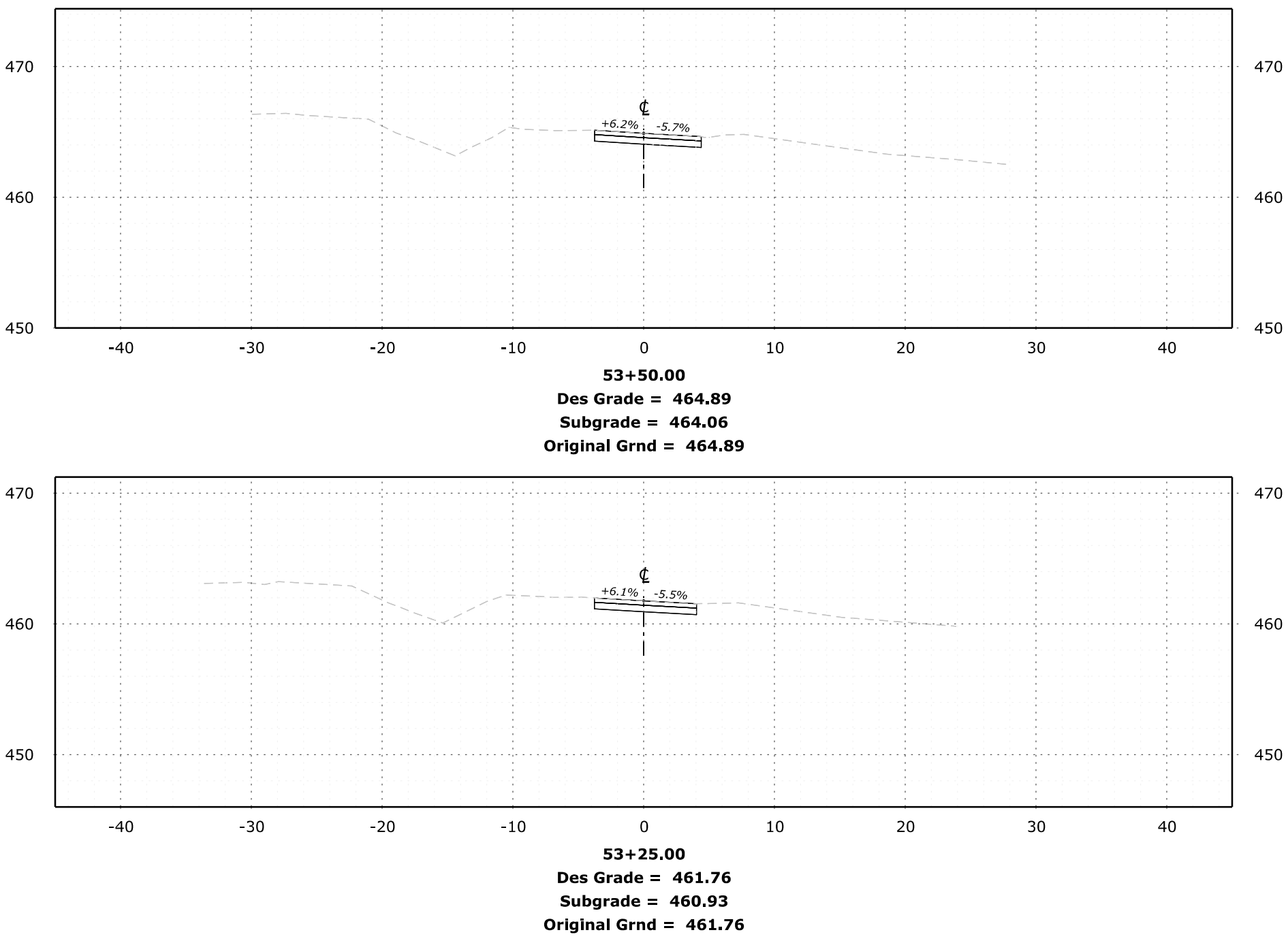
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	CC05



WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #5

53+25 to 53+50

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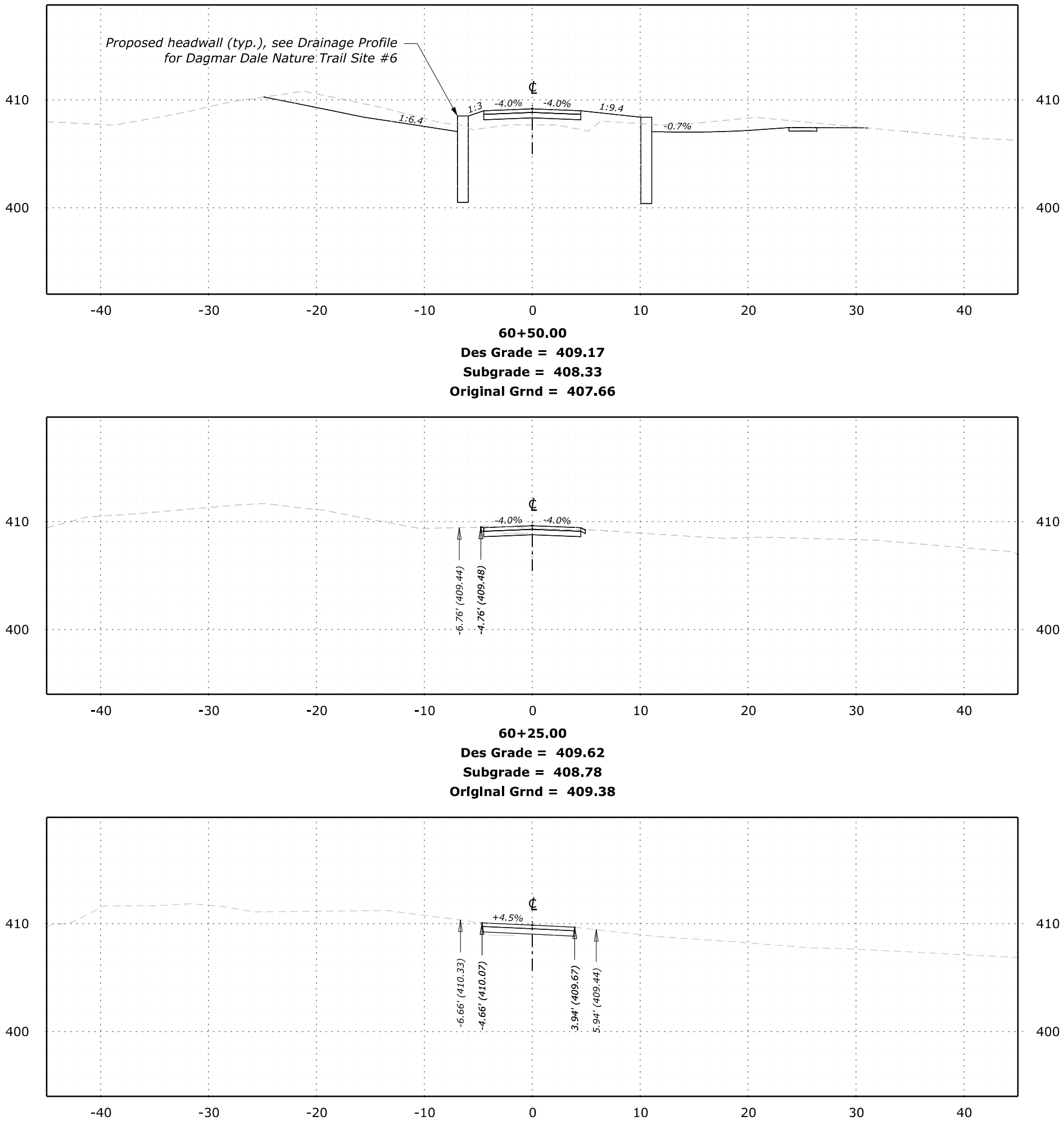
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	DD01



60+50.00
Des Grade = 409.17
Subgrade = 408.33
Original Grnd = 407.66

60+25.00
Des Grade = 409.62
Subgrade = 408.78
Original Grnd = 409.38

60+13.00
Des Grade = 409.86
Subgrade = 409.02
Original Grnd = 409.86

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

60+13 TO 60+50

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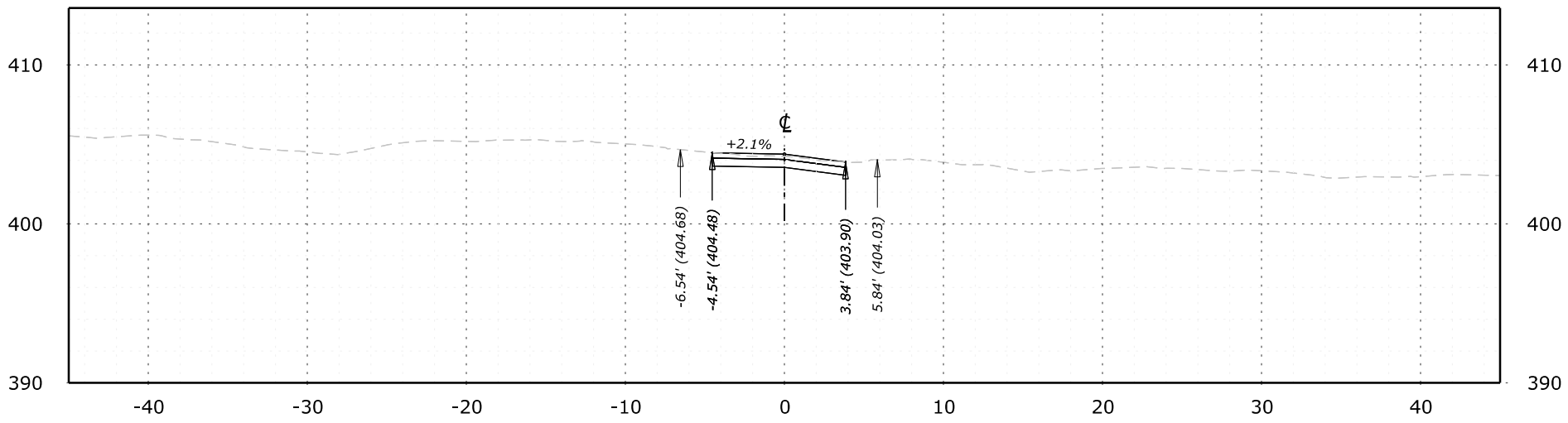
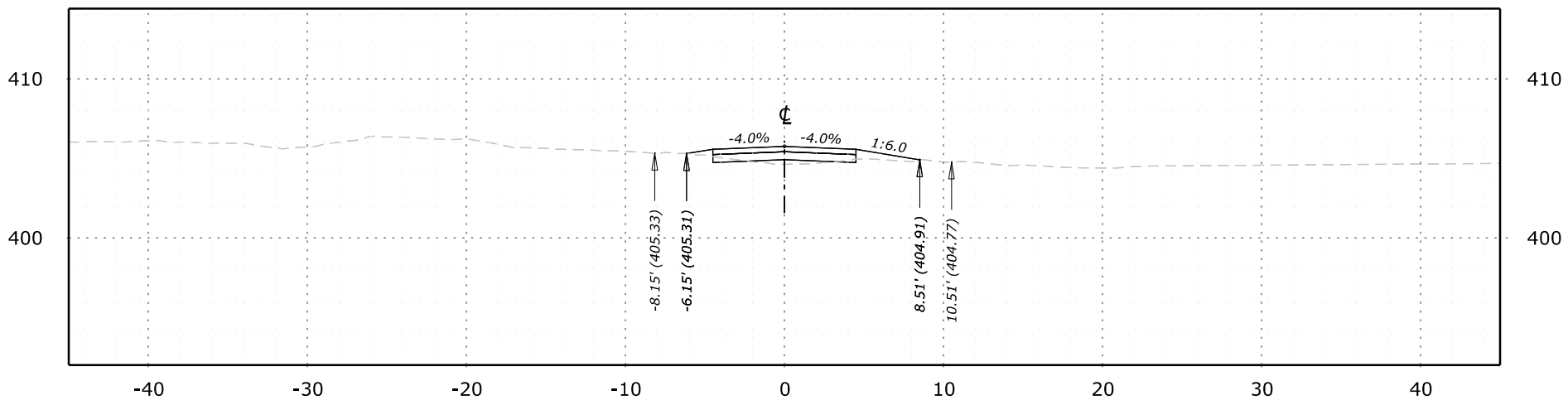
3 August 2026 5:57 PM

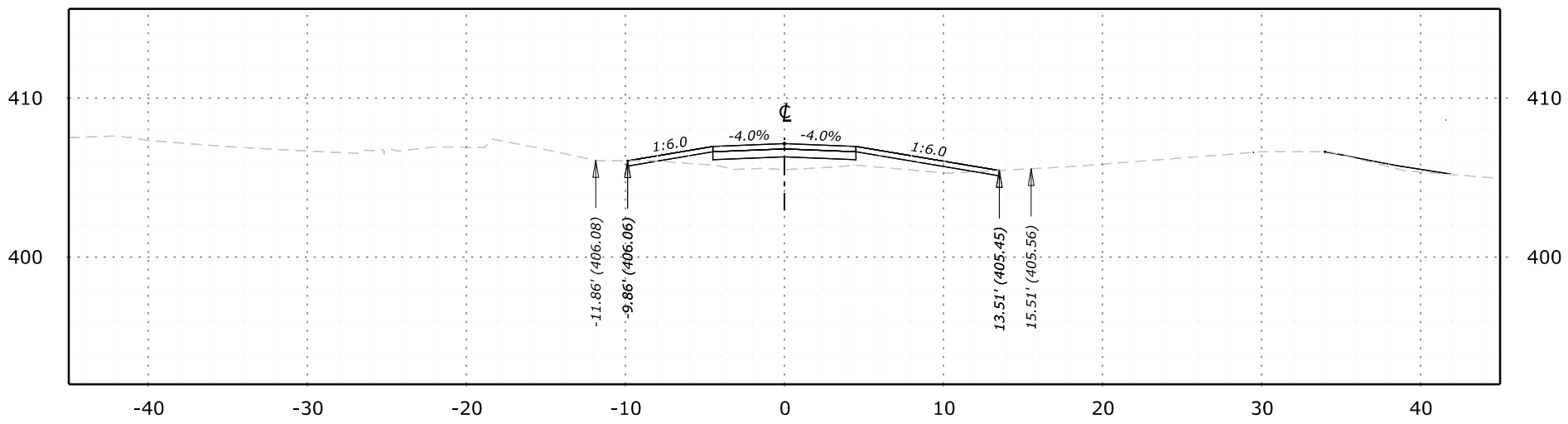
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	DD02



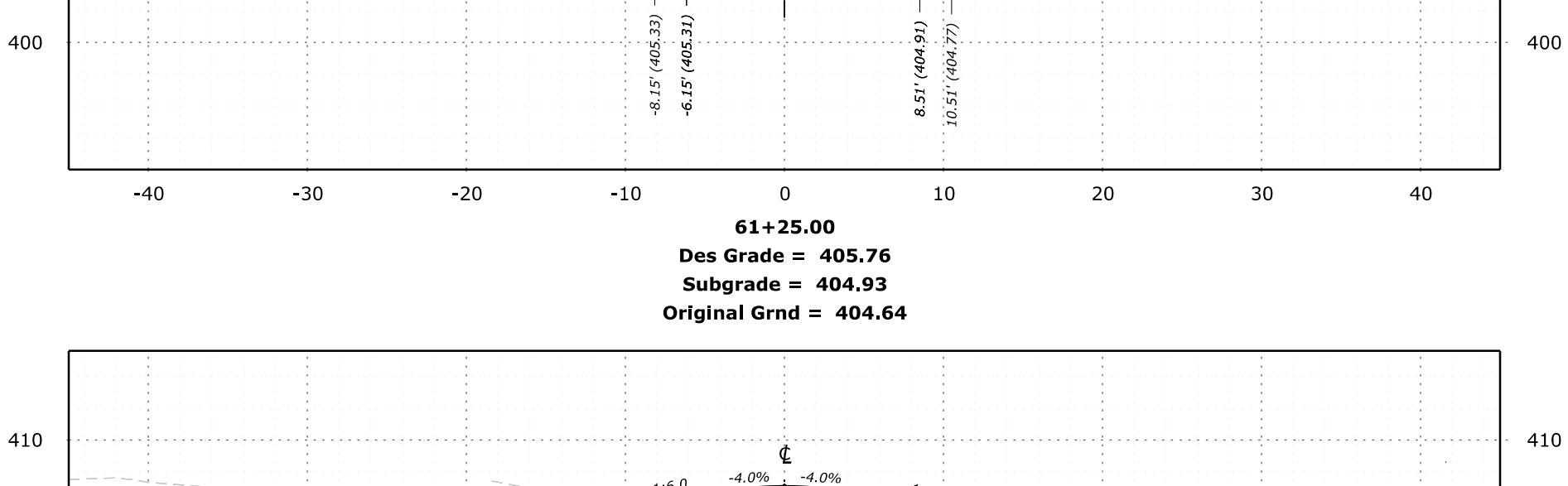
61+50.00
Des Grade = 404.39
Subgrade = 403.56
Original Grnd = 404.26



61+25.00
Des Grade = 405.76
Subgrade = 404.93
Original Grnd = 404.64



61+00.00
Des Grade = 407.13
Subgrade = 406.30
Original Grnd = 405.49



60+75.00
Des Grade = 408.50
Subgrade = 407.66
Original Grnd = 406.34

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

60+75 TO 61+50

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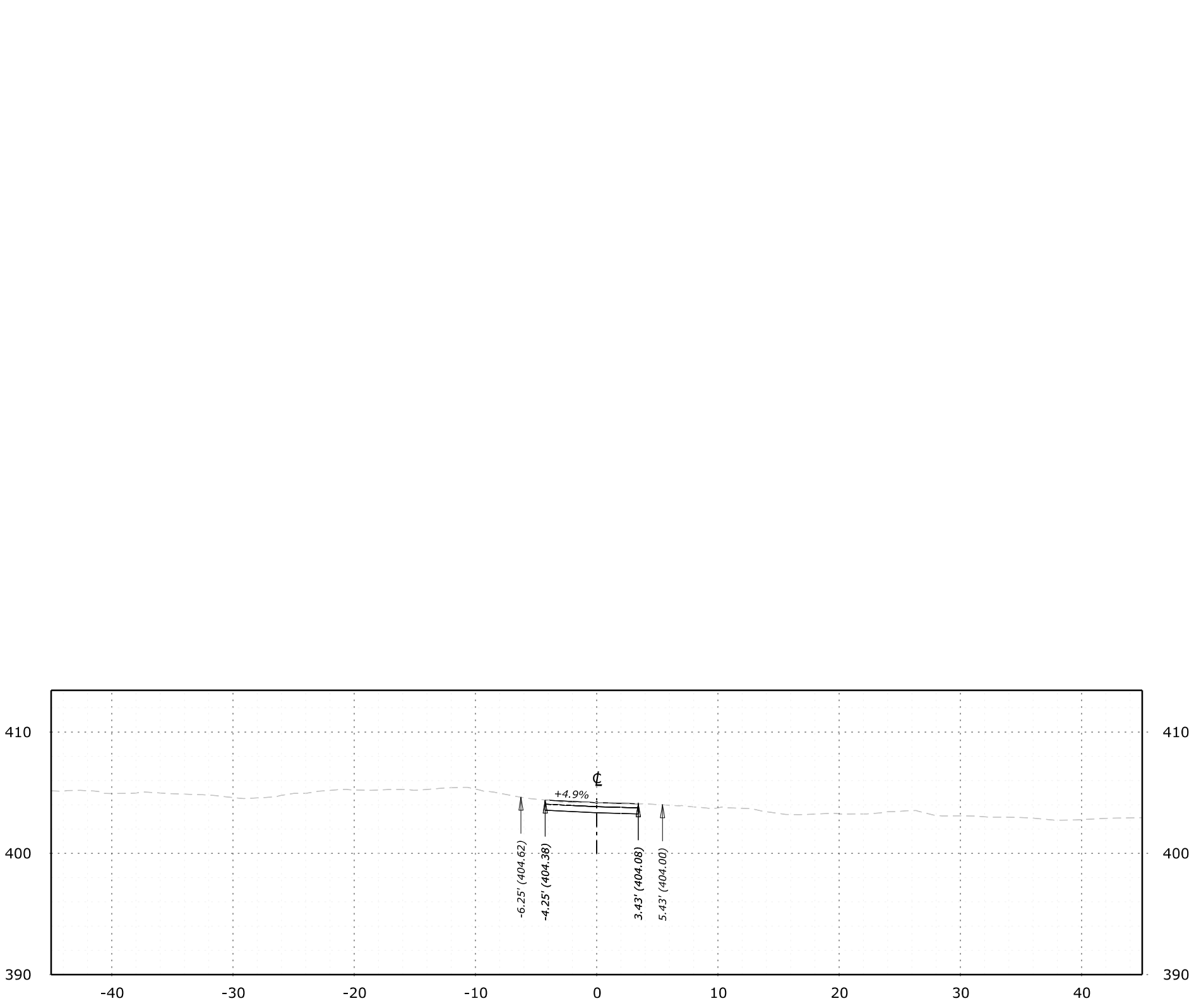
PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	DD03



61+53.99
Des Grade = 404.18
Subgrade = 403.34
Original Grnd = 404.18

WALLKILL RIVER NATIONAL WILDLIFE REFUGE

CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

61+53.99

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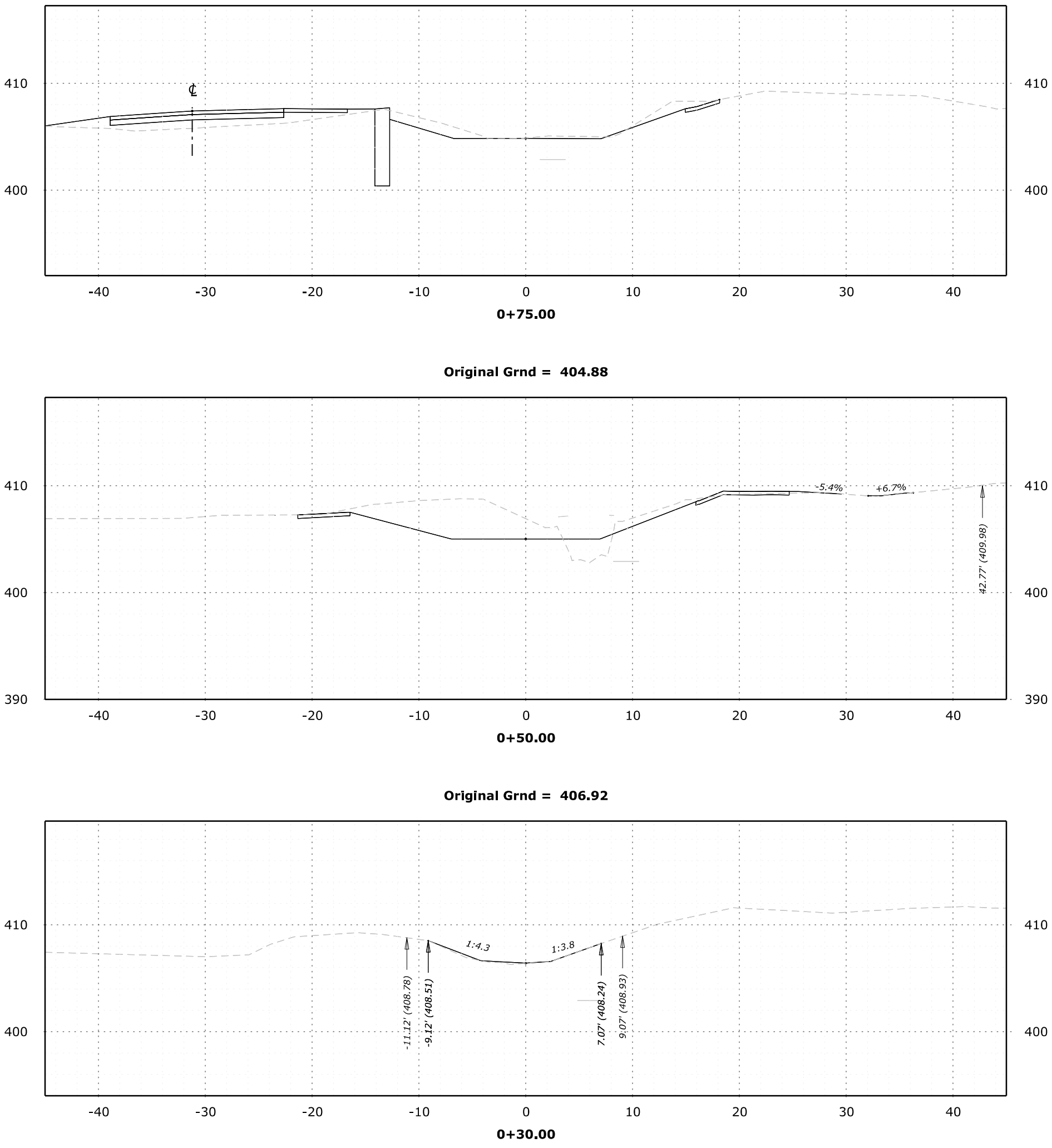
Disaster No. NJ2019-1-FWS and NY2019-1-FWS

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	DD04



WALLKILL RIVER NATIONAL WILDLIFE REFUGE

STREAM CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

0+30 TO 0+75

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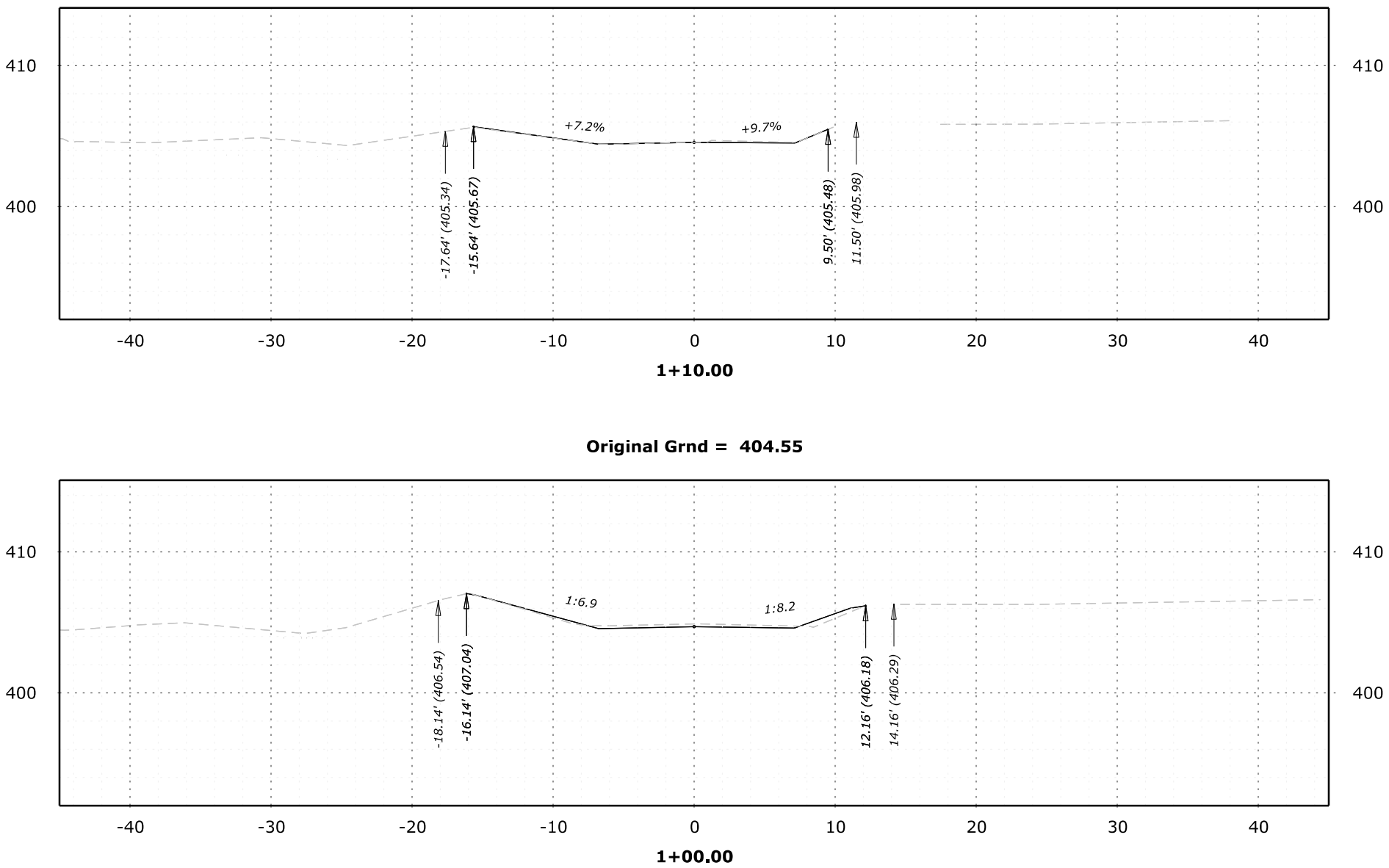
3 August 2026 5:57 PM

PROJECT	SHEET NUMBER
NJ-NY ERFO FW WARI 2019-1(1)	DD05



WALLKILL RIVER NATIONAL WILDLIFE REFUGE

STREAM CROSS SECTIONS

SCHEDULE C: DAGMAR DALE NATURE TRAIL SITE #6

1+00 TO 1+10

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